Readability Studio 2026

User Manual

Readability Analysis Software

ECLIPSE FOUNDATION

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Readability Studio 2026

User Manual

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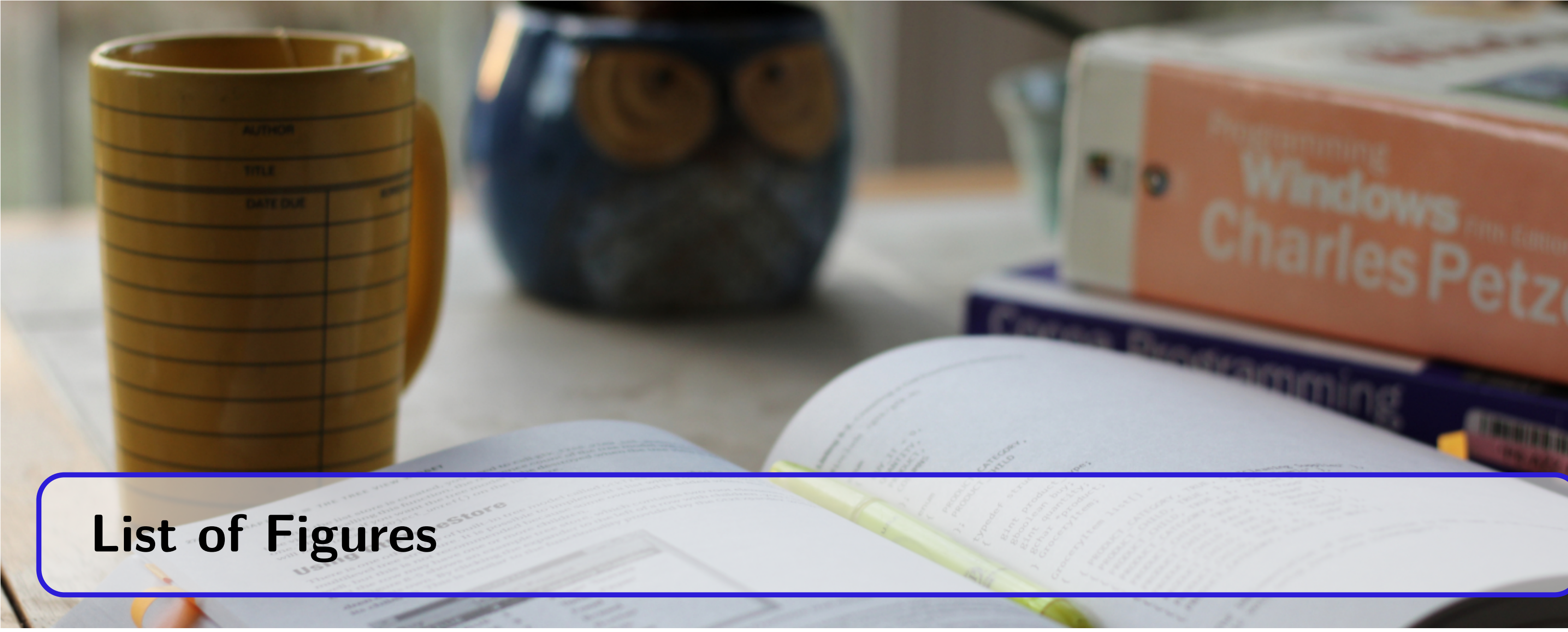

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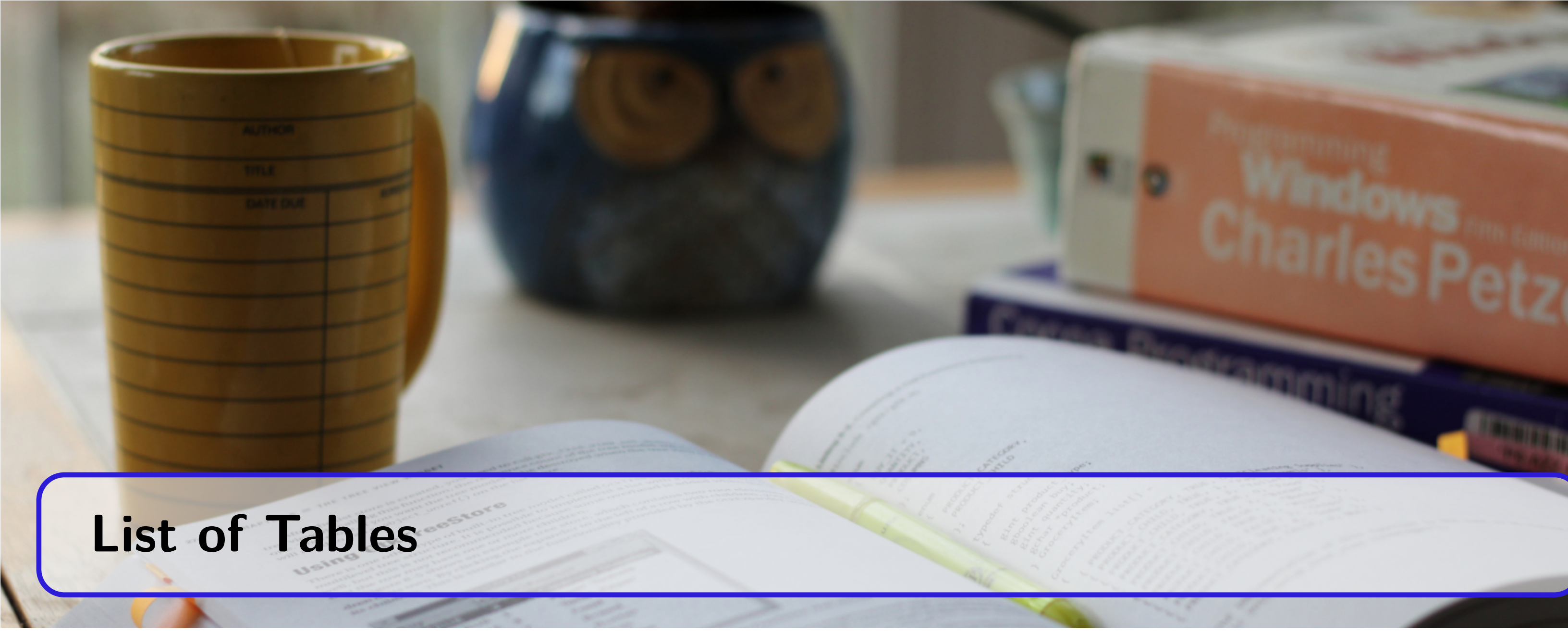

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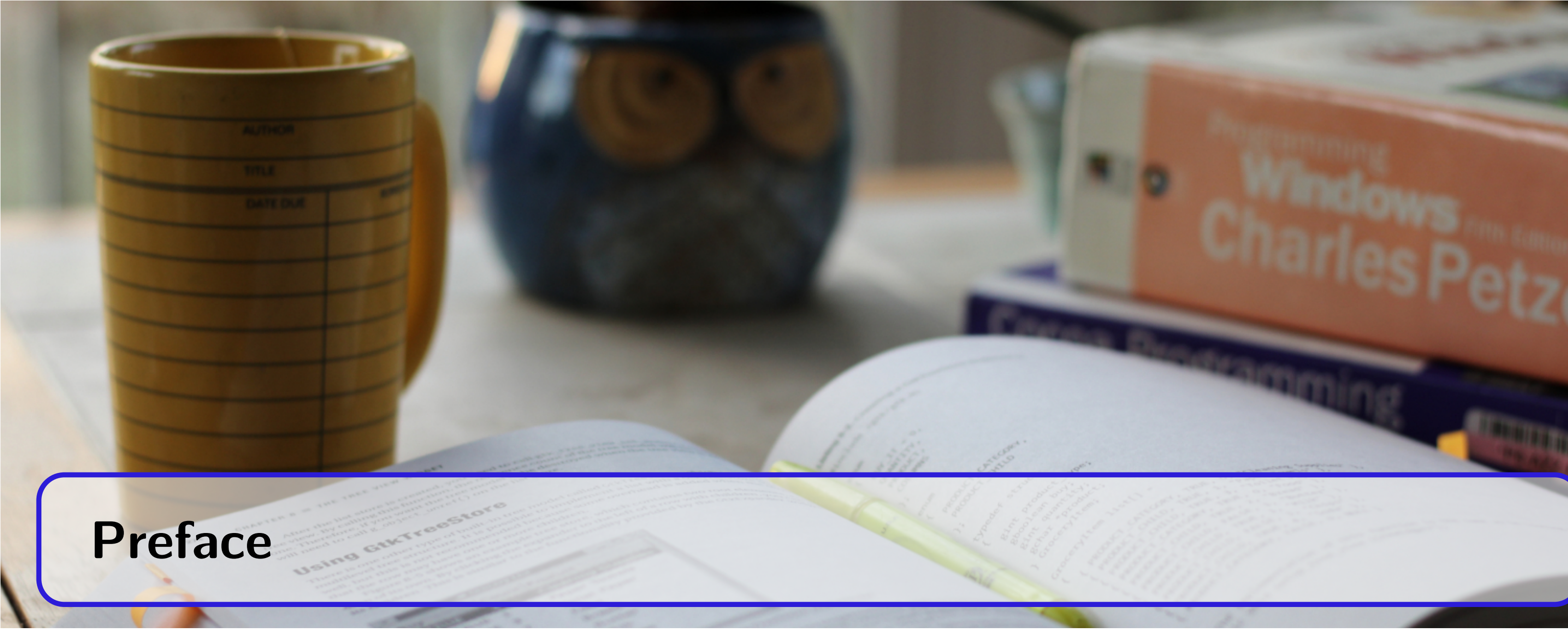


Preface

This book is a guide to the software product *Readability Studio*, as well as a reference for readability formulas.

Chapters 10, 11, and 12 provide overviews for English, Spanish, and German tests, respectively.

All translations are by the author unless otherwise indicated.

Overviews

1. Technical Overviews

When writing for a target audience or classifying documents for specific age groups, readability formulas are invaluable tools. They help determine the minimum age group that can quickly process reading material by using an analytical approach. This approach uses factors such as sentence length, word difficulty, and word familiarity.

Most readability scores represent the grade level of the youngest reader that can understand a document. They can also determine how easily a broader audience can process it. For example, most magazine and newspaper articles should aim for scores around the 14–16 age range. This is not because most readers of newspapers are in this age group. It is because this type of material should be easy to read for most age groups. If an article has numerous complex and uncommon words, then it will take much longer to read and understand. (It will also make it less enjoyable to read.) These are not qualities appropriate for something meant to be read quickly by most readers.

Note that these tests only determine the reading level of material based on how easy it is to process. They do not consider the content or themes of a given work. The age level reported by a test score only reflects the age group that can comprehend a document. It does not reflect whether the topics covered in the material are meant for that age group. If you are classifying books into age groups, you should first review them for any mature topics. Then you should use readability tests to determine their comprehension levels.

1.1 How Tests are Calculated

All readability formulas rely on factors such as sentence length, syllable counts, character counts, or word familiarity. Which factors are used vary from test to test, but most tests use between one to three of these factors to gauge a document's difficulty.

Where readability tests may differ is how they are calculated. Most tests are performed using the following methods:

- linear regression equations
- lookup tables
- graphs

The vast majority of tests use multiple linear regressions. Basically, multiple linear regression is a statistical method for modelling the relationship between a dependent variable and two or more independent variables (i.e., predictors). In the case of readability tests, the dependent variable is human-judged grade levels for a set of samples, while the independents are factors such as the samples' sentence lengths and syllable counts. A multiple regression is performed to determine the strongest predictors (usually the top two), and the prediction equation created by the analysis will be the new readability formula.

Another method is lookup tables, where the intersection of factors (e.g., number of sentences and number of syllables) is found in a table. When located, the position in the table will indicate the grade level. Examples of table-based tests include Wheeler-Smith and New Dale-Chall. (Refer to tbl. 10.29 for an example.)

The final method also relies on finding the intersection of factors, but on a graph instead of a table. Most of these plots are divided into bands or brackets, with each of these regions representing a grade or difficulty level. For example, Fry and Raygor use polygonal bands, while Flesch Reading Ease uses brackets along a "ruler":

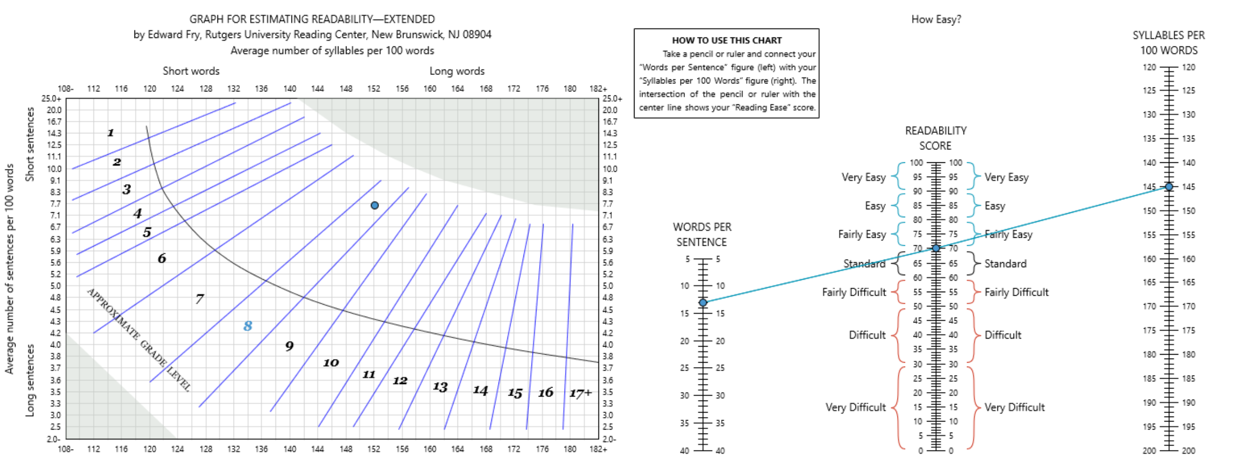


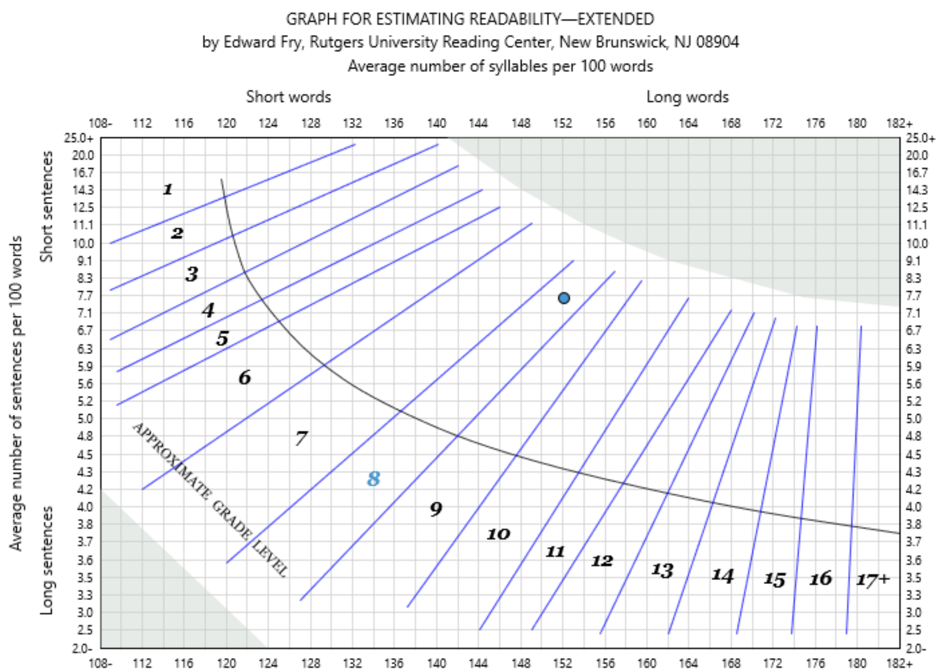
Figure 1.1: Comparison of bands vs. brackets

Once the intersection of two factors is located on the plot, the band or bracket that the intersection falls into will indicate the score.

As a final note, there are a few hybrid tests that can be calculated via a regression equation or graph. These tests were initially constructed as regression equations, and this is the preferred method for performing them. The tests' authors, however, would also offer a graphical version as a quicker and simpler way for performing these tests. Examples of these tests include Flesch Reading Ease and Crawford. Note that the program will include both forms of these tests in the results.

1.2 Grade-level Clamping

A key difference between graph and table-based tests versus regression-based tests is how the former has defined upper limits for their scores. For example, graph-based tests acquire their scores by finding the intersection of certain factors (e.g., sentence length) within a set of bands. These bands represent grades, and there are a specific set of grade bands available. As an example, a Fry graph has bands ranging from 1st grade to 17th (i.e., college):



Likewise, table-based tests take a set of factors and look up the score from a table. These tables have a specific set of grade levels and will also have an upper limit. (Refer to tbl. 10.29 for an example.)

In contrast, tests that use regression equations do not have upper limits. In theory, a document with extreme factors (e.g., abnormally long sentences) could yield scores above grade 19 (i.e., PhD level). To keep results sensible, such scores should be clamped to 19 and be displayed as 19+. This is done for a number of reasons.

First, scores such as 27th grade do not reflect reality. To describe a document as only being understandable by a PhD is relatable. To say that a document can only be processed by someone in the 27th grade, however, makes little sense. Another way to view this is that 19+ implies a document's failure to be readable—similar to an *F* on a test.

Second, because graph and table-based tests are bound to sensical grade scales, it is reasonable to treat regression-based tests the same way. For example, a document scoring 16+ for New Dale-Chall should clamp tests such as Flesch-Kincaid to 19+, thus keeping their scores on a similar scale. In other words, grade and table-based tests were designed for grade scales from 0–19 (sometimes even lower). Likewise, it is reasonable to assume that was the intention of regression-based tests as well.

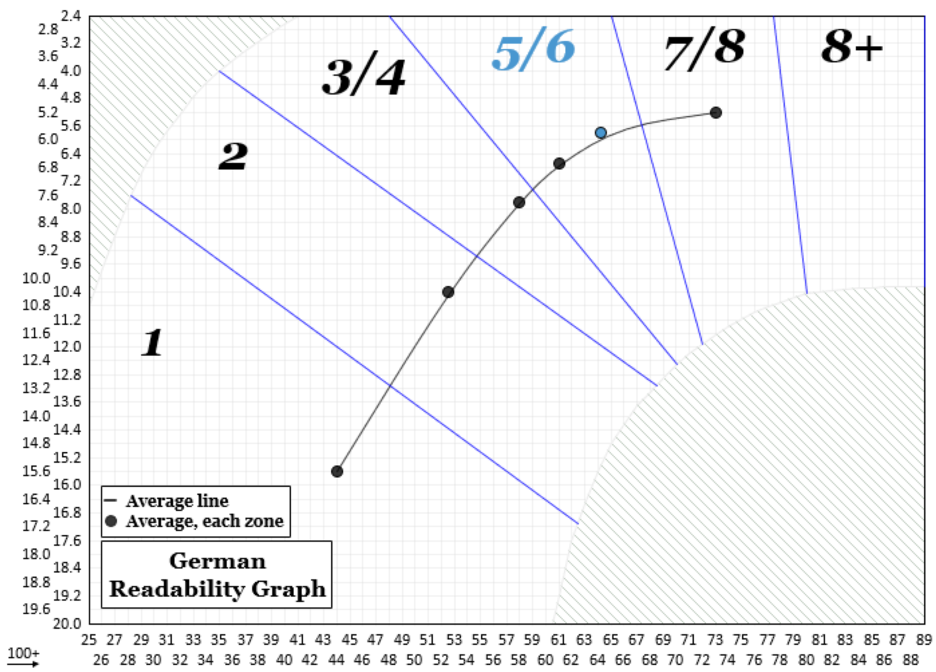
Finally, the source materials for these formulas cite example scores within 0–19. Again, it can be inferred that the authors' intention was for documents to be bound to a reasonable grade scale.

Although not recommended, it is possible to create a copy of any regression-based test and remove this clamping. (Refer to sec. 19.2 for an example.)

1.3 How Tests Can Fail

Most readability tests will return a result, even when the document being reviewed contains extreme difficulty factors. For example, with table-based tests, documents with long sentences and/or numerous unfamiliar words will fall within the top bracket of the lookup table. (Refer to tbl. 10.29 for an example). Likewise, plugging a document's statistics into a regression-based test will also produce a result without issue (although extreme results may need to be clamped, see sec. 1.2).

It is possible, however, for certain graphical tests to not return a meaningful result. Tests such as Fry and Schwartz use plots that are divided into bands (i.e., polygonal strips) which represent different grade levels. To produce a result, these tests find the intersection of a document's statistics along the plot's axes. Once the intersection is found, the band that the point falls within will indicate the grade level of the document.



For most documents this will yield a result, but extreme documents may fall outside the grade-level bands. (This can happen if there are too many difficult words or lengthy sentences.) When this occurs, the program will indicate that the test failed and explain which factor (e.g., high syllable count) caused the failure.

2. Program Overview

2.1 What *Readability Studio* Offers

Readability Studio eases the task of performing these tests. The program quickly imports and analyzes your work using the most popular tests. In addition, it offers insightful explanations and suggestions for improving your documents.

Readability Studio's Unique Features

Numerous insightful tests

Readability Studio offers many popular readability tests, along with in-depth explanations. Each test result includes an explanation of the score and the factors that affect it.

The program also provides many useful features to help review the difficult aspects of your document. This includes lists of various difficult words and views of the document with difficult words and sentences highlighted. These tools can assist you in improving your documents so that they meet their target readers.

Another important feature is that the program analyzes the entire content of your document (rather than samples). This will give you the most accurate results possible.

Finally, *Readability Studio* offers multiple-document analysis. The program can analyze hundreds (or even thousands) of documents at once and aggregate their results. This includes all readability scores, difficult word and sentence statistics, and grammar issues. You can also drill down into a specific document to review it in deeper detail.

The following tests are available:

Table 2.1: English Tests

Automated Readability Index	FORCAST	Spache Revised
Bormuth Cloze Mean Test	Fry Graph	Wheeler-Smith
Bormuth Grade Placement Test	Gunning Fog	
Coleman-Liau	Gunning Fog (Powers-Sumner-Kearl)	
Dale-Chall (Powers-Sumner-Kearl)	Harris-Jacobson	
Danielson-Bryan 1	Läsbarhetsindex (Lix)	
Danielson-Bryan 2	McAlpine EFLAW©	
Degrees of Reading Power	Modified SMOG	
Degrees of Reading Power (GE)	New Automated Readability Index	
Easy Listening Formula	New Dale-Chall	
Farr-Jenkins-Paterson	New Farr-Jenkins-Paterson	
Farr-Jenkins-Paterson (Powers-Sumner-Kearl)	New Fog Count	
Flesch-Kincaid	Rate Index (Rix)	
Flesch Reading Ease	Raygor Estimate Graph	
Flesch Reading Ease (Powers-Sumner-Kearl)	SMOG	

Table 2.2: Spanish Tests

Crawford
FRASE Graph
Gilliam-Peña-Mountain Fry Graph
INFLESZ
Läsbarhetsindex (Lix)
Rate Index (Rix)
SOL

Table 2.3: German Tests

Amstad
Läsbarhetsindex (Lix)
Lix (German children's literature)
Lix (German technical literature)
Neue Wiener Sachtextformel 1
Neue Wiener Sachtextformel 2
Neue Wiener Sachtextformel 3
Quadratwurzelverfahren
Rate Index (Rix)
Rix (German fiction)
Rix (German non-fiction)
Schwartz
SMOG (Bamberger-Vanecek)
Wheeler-Smith (Bamberger-Vanecek)

Helpful assistance for test selection

Readability Studio presents an intuitive interface for selecting readability tests. Experts can manually select the tests that they want to perform, while those less familiar with these tests can be guided through a helpful wizard. This wizard can recommend which tests to include based on the document's type (e.g., text book) or industry (e.g., health care).

Graphics

Readability Studio includes numerous graphs to help visualize your results. Graphical tests such as Fry and Raygor are included, as well as a bar chart showing a breakdown of the difficult word categories. If you are reviewing Dolch Sight Words, then breakdown charts of your category usage and coverage are also included.

Multiple-document graphs are also available. Here, histograms and box plots show the distribution of a document collection's test scores. This is useful for summarizing the general reading level of a batch of documents. It is also useful for finding problematic documents in a batch (i.e., outliers).

Detailed statistics

Numerous sentence and word statistics are provided. This includes a full difficult-words breakdown and word & sentence-length averages. The type of statistics included is also customizable.

Lists of difficult words and sentences

Lists of difficult words and sentences are broken down by category and can easily be searched, sorted, printed, and exported. A separate list for each type of difficult word (e.g., 3+ syllables) is included, as well as a list of overly-long sentences. These lists provide detailed information about each difficult word, such as its syllable count and frequency. This enables you to search for either the most difficult words or the difficult words that occur most frequently.

These lists are also interactive. You can double-click on a word (or sentence) in a list, and the program will switch to the document display with the first occurrence of that word selected. This is useful for quickly finding a difficult word or sentence in its original context.

Document highlighting

Difficult words and sentences are also shown highlighted in their original context. This is useful in finding trouble spots in your document.

Grammar tools

Along with readability analysis, *Readability Studio* also offers basic grammar checking. The program can detect the following problems:

- wording errors and known misspellings (English only)
- possible misspellings (English only)
- repeated words
- article mismatching (English only)
- redundant phrases (English only)
- wordy items (English only)
- clichés (English only)
- overly-long sentences
- lowercased sentences
- sentences that begin with conjunctions
- passive voice (English only)

These issues are shown highlighted in your document for easy review. They are also aggregated into sortable lists to give a general overview of any problems and to assist in finding specific issues.

Dolch Sight Words

Readability Studio also includes a Dolch suite of statistics and graphs, which is useful for educators and writers who work with young readers. The program provides a complete breakdown of Dolch words that you are using (and not using) in your content. This enables writers to see how they can improve their Dolch practice materials. This also enables educators to determine which materials are appropriate for Dolch practice.

Numerous file formats supported

Readability Studio supports importing text from many file formats, including:

- ANSI and Unicode text (*.txt)
- *Microsoft® Word* 97–2003 (*.doc)
- *Microsoft® Word* 2007 (*.docx)
- *Microsoft® PowerPoint* 2007 (*.pptx)
- *Microsoft® Excel* 2007 (cells can be batch analyzed as documents or filepaths to other documents)
- *OpenDocument* Text (*.odt)
- *OpenDocument* Presentation (*.odp)
- HTML (*.htm)
- Markdown (*.md, *.Rmd, *.qmd)
- Rich text (*.rtf)
- Postscript (*.ps)
- ZIP (documents within can be batch analyzed)
- Interface Definition Language (*.idl)
- (Doxygen style) C++ source (*.cpp and *.h)

You do not need the native application for these file formats. *Readability Studio* is capable of importing these files without any additional programs.

Exporting capabilities

Readability Studio provides printing and exporting capabilities for all test scores, word lists, statistics, graphs, etc. This enables you to easily share and publish your results.

- Test results, statistics, and difficult word lists can all be exported as HTML, text, or LaTeX.
- Statistics summary report can be exported as HTML.
- Highlighted document views can be exported as RTF or HTML.
- Graphs can be exported as TIFF, JPG, BMP, PNG, SVG, GIF, TARGA, or WebP.

Flexibility

Finally, *Readability Studio* is highly configurable. Many options, such as how documents are analyzed and how the results are displayed, are customizable. For example, you can specify how numeric words (e.g., *10/31/1974*) are syllabized. Other programs either sound out each digit or treat the entire word as one syllable and you cannot change this. With *Readability Studio*, you can toggle this behavior at any time.

II

Projects

3. Creating a New Project

The first step to analyzing a document is to create a new project. A project is where you select your document, define how to parse it, and choose which tests to perform. Projects organize results (e.g., readability tests, graphs, etc.) and settings into a single file that can be opened later.

In this chapter, we will step through the process of creating a project, explaining all the options available along the way.

3.1 Project Types

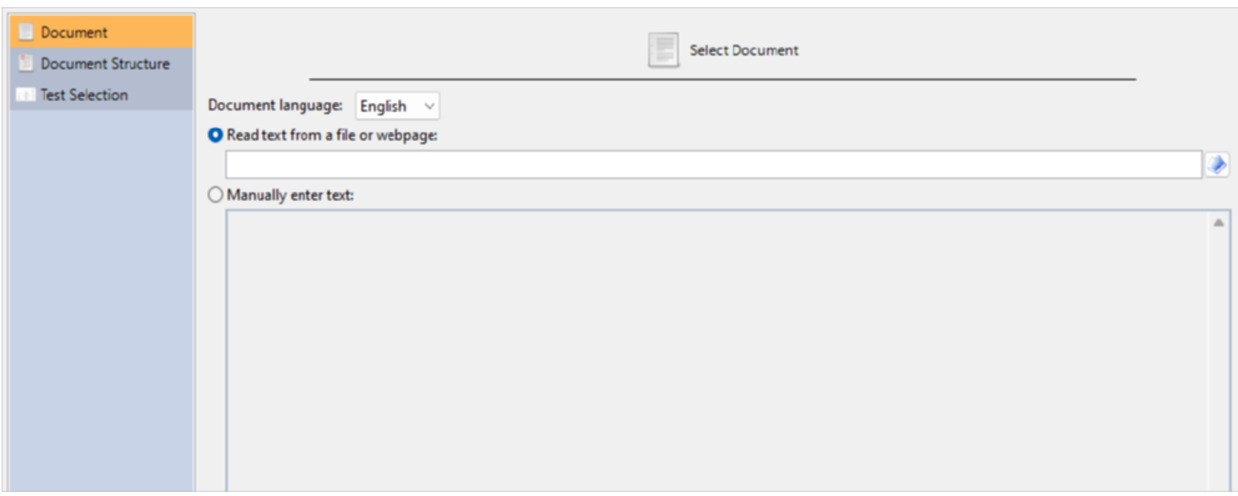
To begin, click the New button on the Home tab to display the New Project dialog. This dialog enables you to create either a standard or batch project.

Standard project: Select this option to create a standard project, which can analyze a file, a webpage, or typed text.

Batch project: Select this option to create a batch project, which can analyze a collection of documents or a website.

3.2 Standard Project

To determine a document's readability you will need to create a new project. First, click the New button on the Home tab to open the New Project wizard. Then select Standard Project and click OK. On the first page of the wizard, the options for specifying the document and its language will be shown:

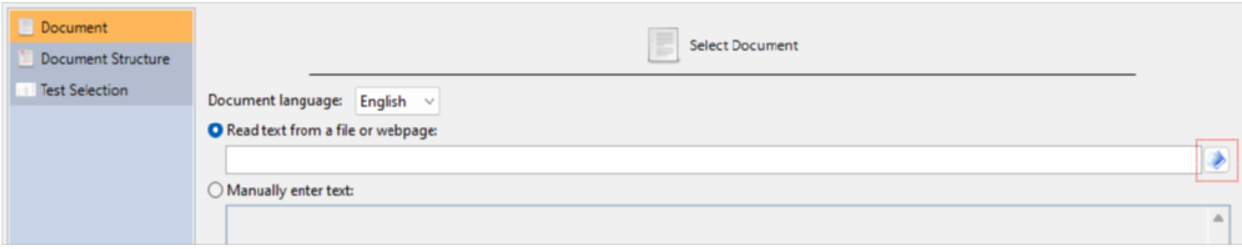


The language option will control how the document is analyzed (e.g., syllable counting). This will also affect which tests, word lists, and grammar features will be available. For example, if we select Spanish, then only Spanish tests will be included, and features related to English word lists (e.g., New Dale-Chall) will not be available. For this example, select English.

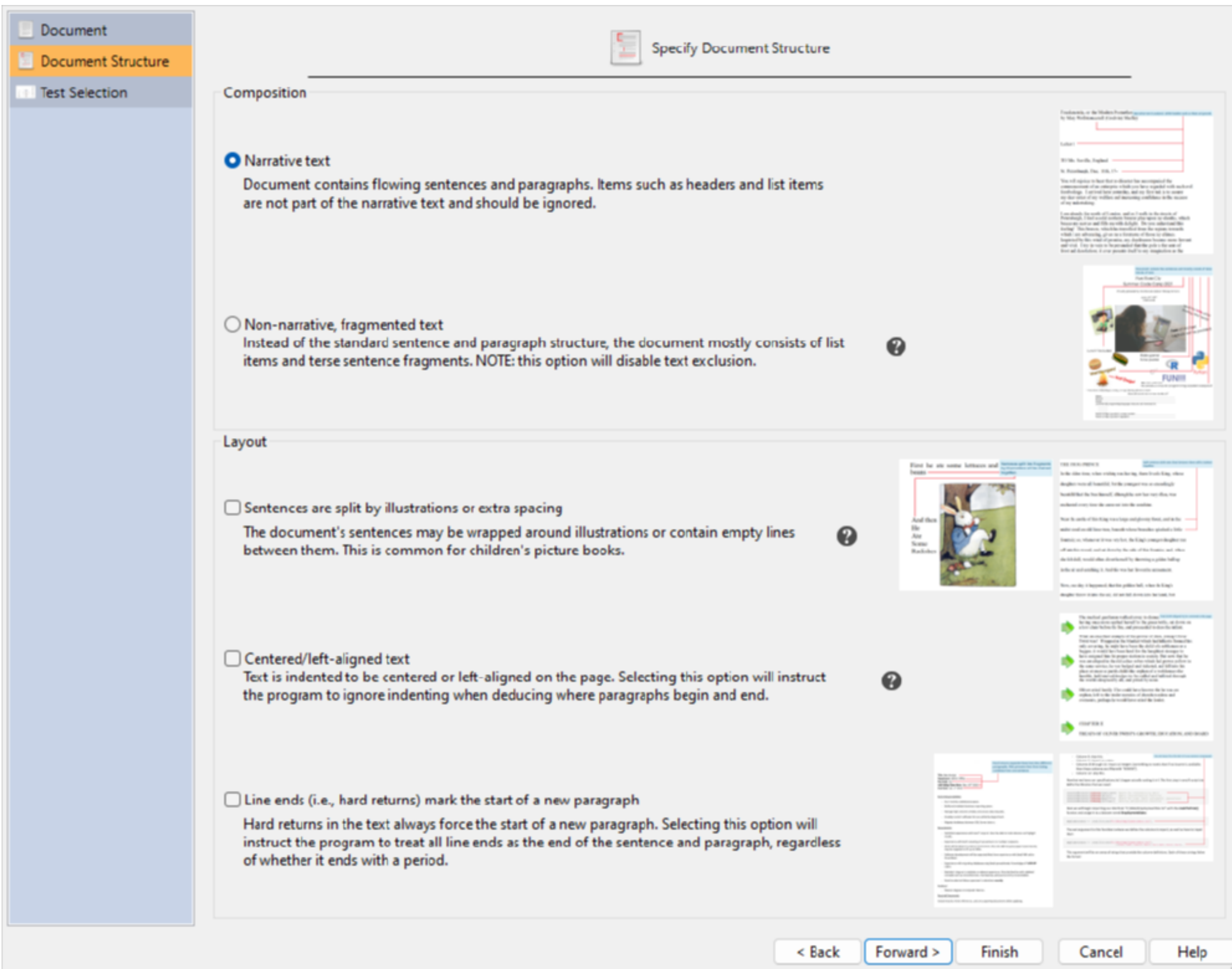
The next options specify whether you will be analyzing a file or manually-entered text.

For example, to analyze an email, copy the text from your email program, select the Manually enter text option, and paste it into the text box.

To analyze a file, select the Read text from a file or webpage option. Then click the button next to the file path field and browse to the document:



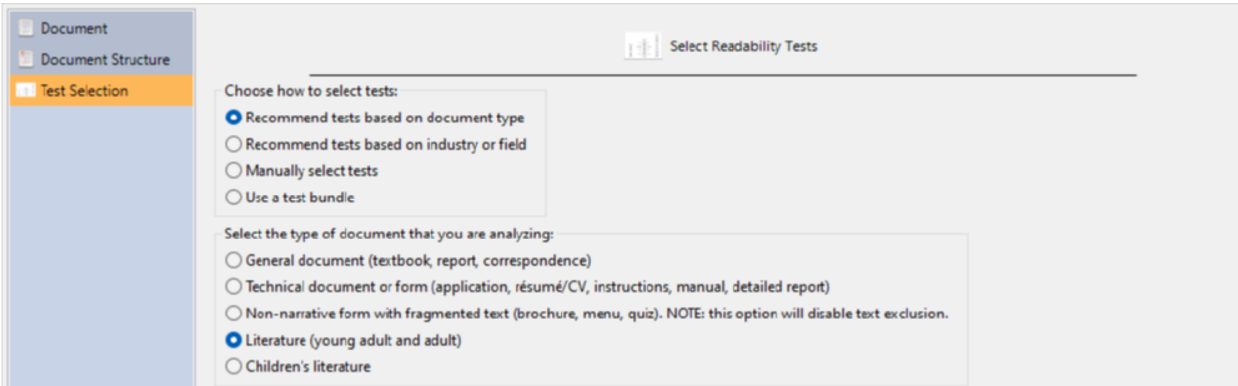
After you have selected your file, click the Forward > button to continue to the Document Structure page, as shown below:



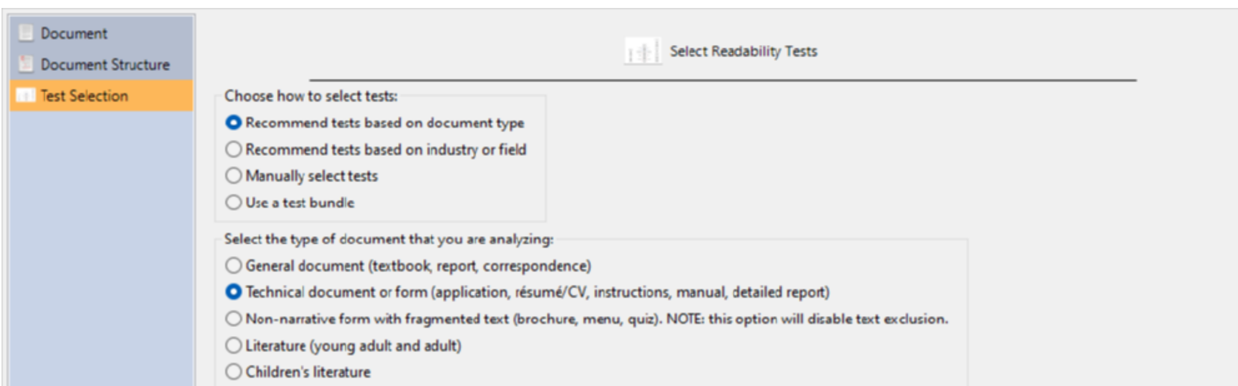
On this page, you will specify the format that best describes your document. The composition types that you can select are either narrative text or non-narrative/fragmented text. These options describe the content of your document and will help the program select the appropriate text exclusion method. The layout options are used to describe the format of the document and will help the program select the appropriate sentence deduction method. For example, if your document is formatted to fit a page and also wraps its text around illustrations, then it is recommended to check the first two Layout options.

> 💡 Tip
> Previews of these document formats are shown next to each option. To view the full-size image of any preview, click on it with your mouse.

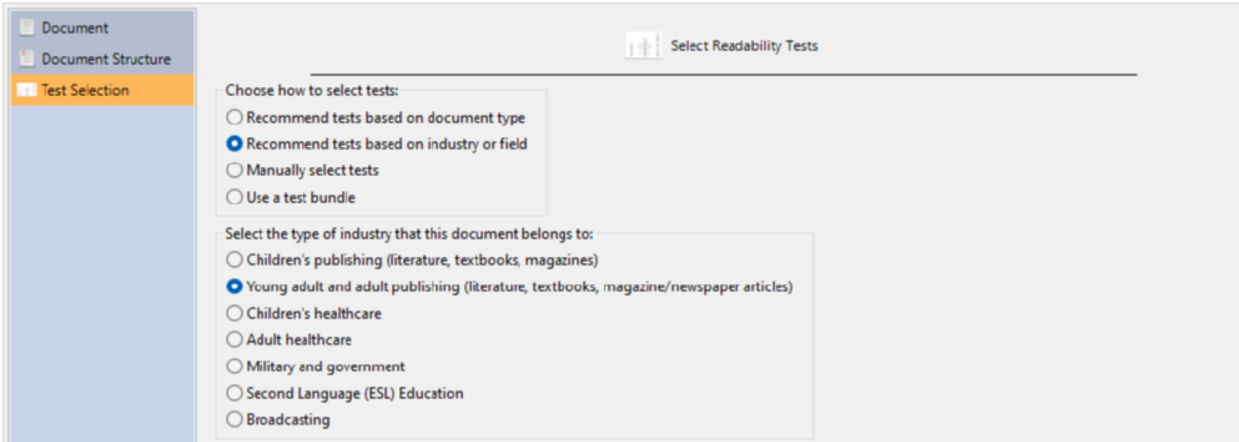
After specifying your document structure, click the Forward > button to continue to the Test Selection page, as shown below:



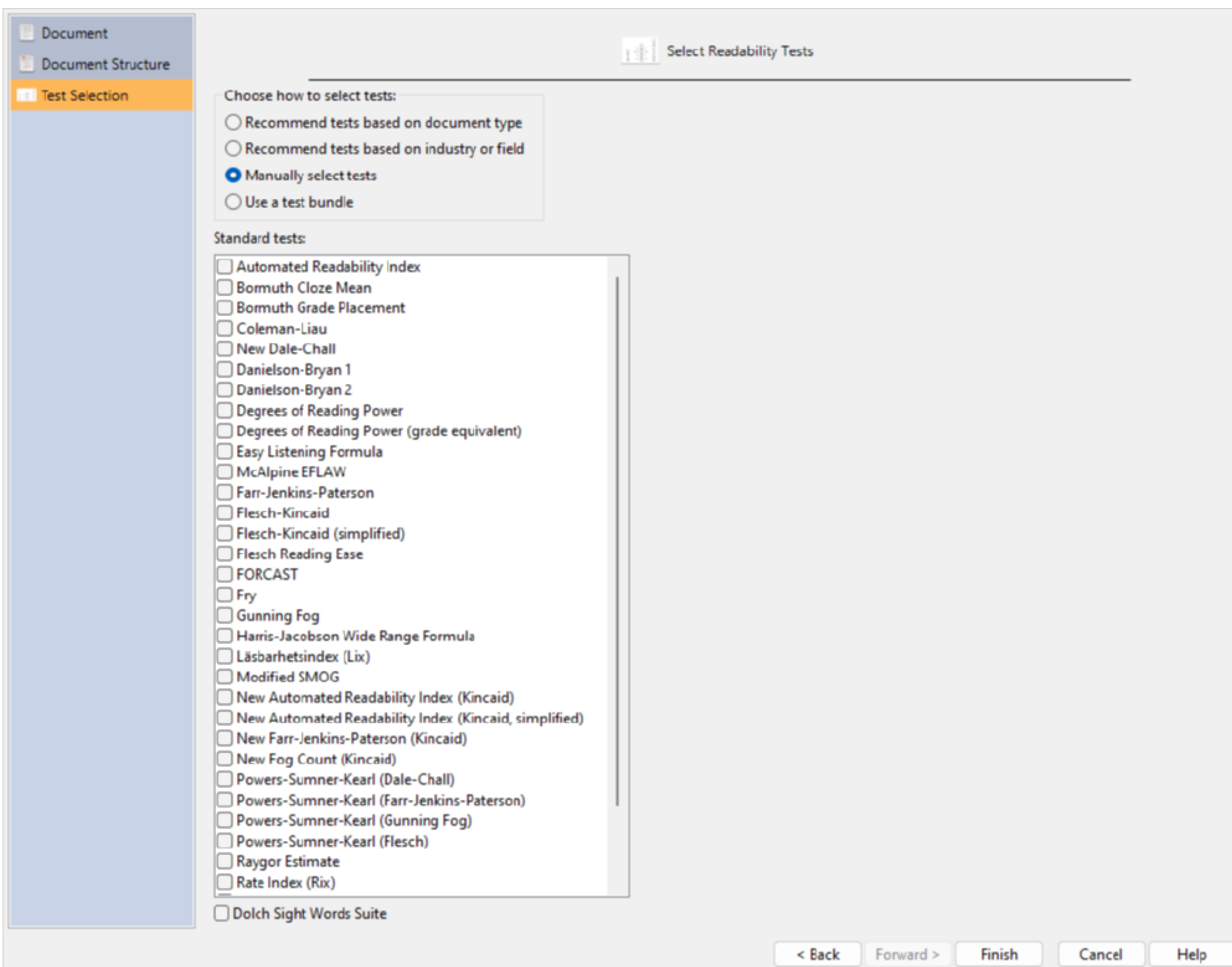
On this page, you will select how you want *Readability Studio* to choose the readability tests to perform. Your first option is to tell the program which type of document you are analyzing and it will select the most appropriate tests based on that. To do this, select the Recommend tests based on document type option. You will then be presented with the Select the type of document... options, where you can select which type of document best describes your file (e.g., a literary work or technical form), as shown below:



Your second option is to tell *Readability Studio* which type of industry your document relates to and it will select the most appropriate tests based on that. To do this, select the Recommend tests based on industry or field option. You will then be presented with the Select the type of industry... options, where you can select which field your file relates to (e.g., magazine publishing or the military), as shown below:

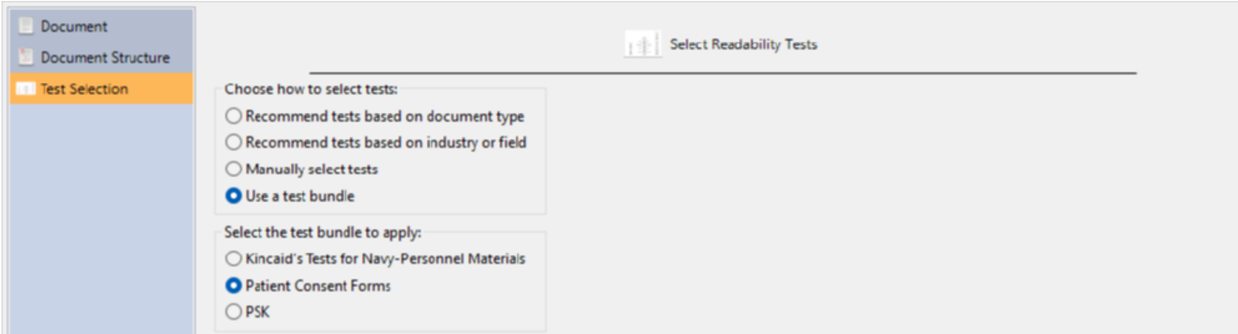


Your third option is to manually select the tests yourself. To do this, select the Manually select tests option. You will then be presented with the Standard tests options, where you can select any of the tests that *Readability Studio* has to offer, as shown below:



The tests available on this page will depend on the project's language. For example, if you are manually selecting Spanish tests, then tests such as Crawford will be available, but English tests (e.g., Spache) will not. Also, some industry and document-type options may not have tests associated with them depending on the selected language.

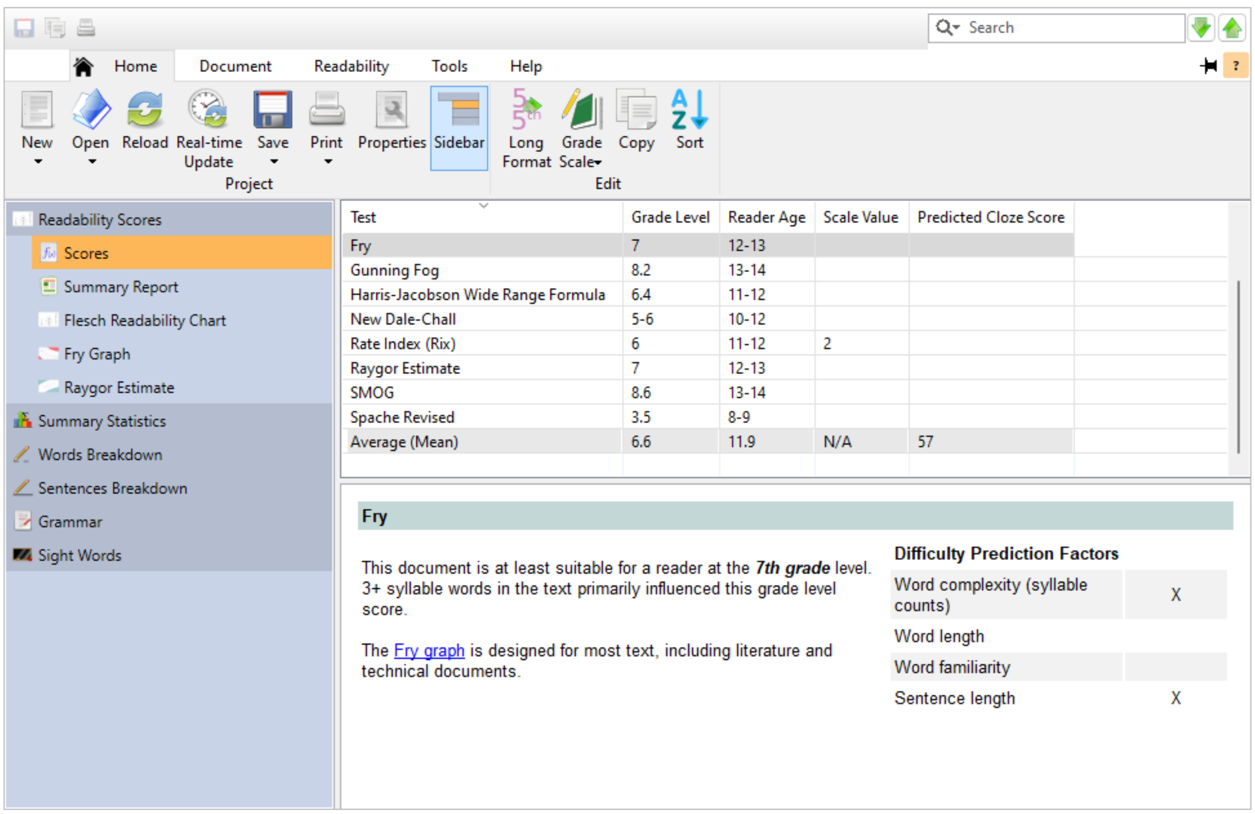
Your final option is to select a test bundle.



Along with adding tests to a project, test bundles can also include goals to help with reviewing your scores.

Note that you can always add or remove tests from your project at any time. If you select either the document type or industry options, you are not bound to the tests that *Readability Studio* recommends.

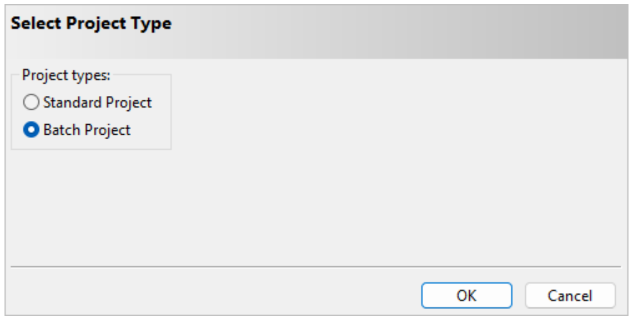
Finally, click the Finish button and the project will be created and will appear like this:



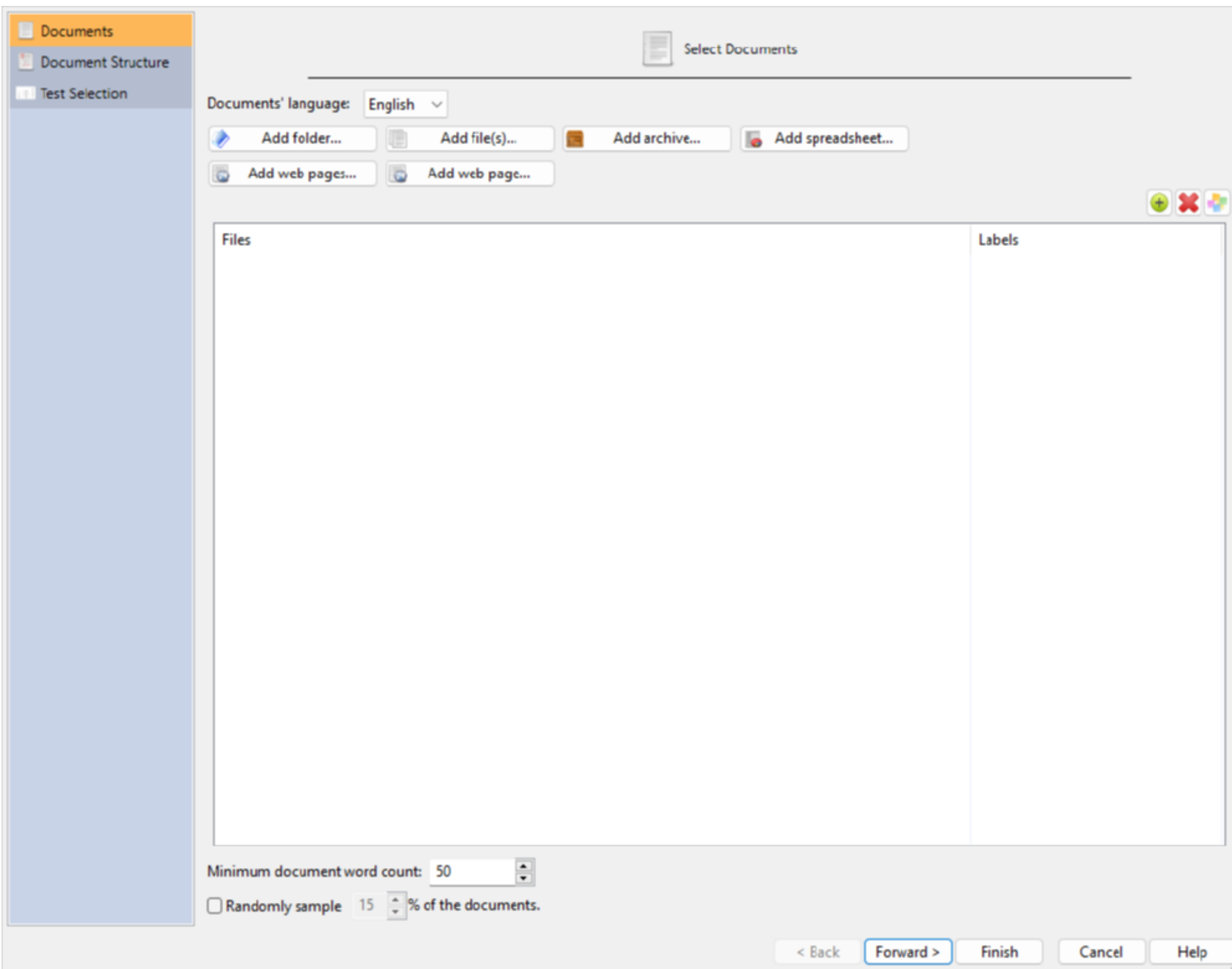
From here, you can review the readability scores and graphs, useful document statistics, and various lists of difficult words. You can also review the document with its difficult words and sentences highlighted.

3.3 Batch Projects

To determine the readability of a document collection, you will need to create a new batch project. First, click the New button on the Home tab to open the Select Project Type dialog, as shown below:



Select Batch Project and click the OK button to proceed to the New Project wizard dialog, as shown below:

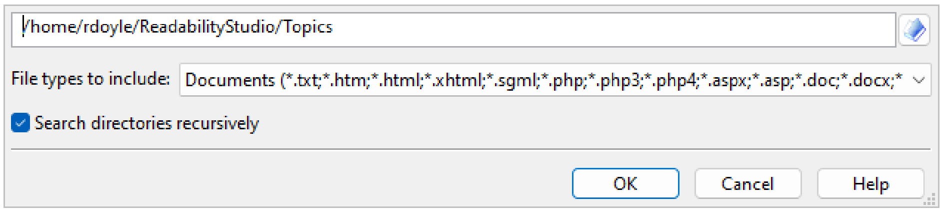


On the first page of this dialog, Documents, are options for selecting your documents and how to specify their language. The language option will control how the documents are analyzed (e.g., syllable counting). This will also affect which tests, word lists, and grammar features will be available.

For example, if we select Spanish, then only Spanish tests will be included, and features related to English word lists (e.g., New Dale-Chall) will not be available. For this example, select English.

Now you can add files from a folder, archive file, *Excel*® 2007 file, or website to the project.

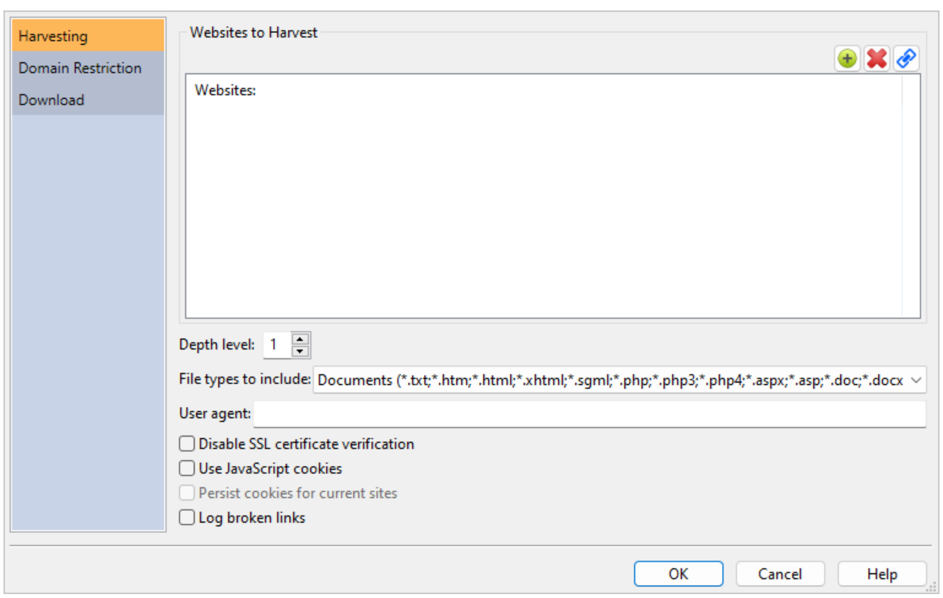
To add files from a folder, click the Add folder button. You will be presented with the Select Directory dialog. On this dialog, you can select the document types to search for, as well as whether to search recursively within the folders.



To add files from an archive (i.e., a *.zip file), click the Add archive button. You will be presented with the Select Archive File dialog. Like the Select Directory dialog, you can specify which documents to search for within the archive. (Note that the entire archive will be searched recursively for documents.)

To add documents from an *Excel* (*.xlsx) file, click the Add spreadsheet button. After selecting an *Excel* file, you will be prompted for which cells to import. After specifying these, the cells will be imported one of two ways. The first way is if a cell's content is a file path. In this case, the document referenced by this file path will be loaded into the project. Otherwise, if a cell is not a file path, then its content will be imported as a document itself. In this case, the cell's name (e.g., *B19*) will be used as the document's name.

To search for files from a website and include them in the project, click the Add web pages button. This will display the Web Harvester dialog, as shown below:



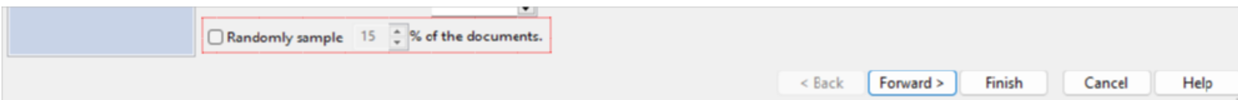
To add an individual web page to the project, click the Add web page button. This will display a text box where you can enter the web page's path.

You can also select local files to include in the project. Click the Add file(s) button and then select the file(s) that you want to include.

As you add files to the project, you can also include a helpful description next to any file. Double-click in the Description column next to any filename, and then type a description to be attached to that file. When the batch project is loaded, this description will appear in any results window that displays filenames. (Note that for some file formats, the program will fill this field with the document's subject or title information if you leave it blank.)

After you have selected your files, you can randomly sample a subset of these documents for your project. Normally, you will want to analyze all the documents that you have selected and ignore this feature. However, some research projects may require random sampling. For these situations, check the % of documents to randomly sample option and enter the percentage of documents to analyze.

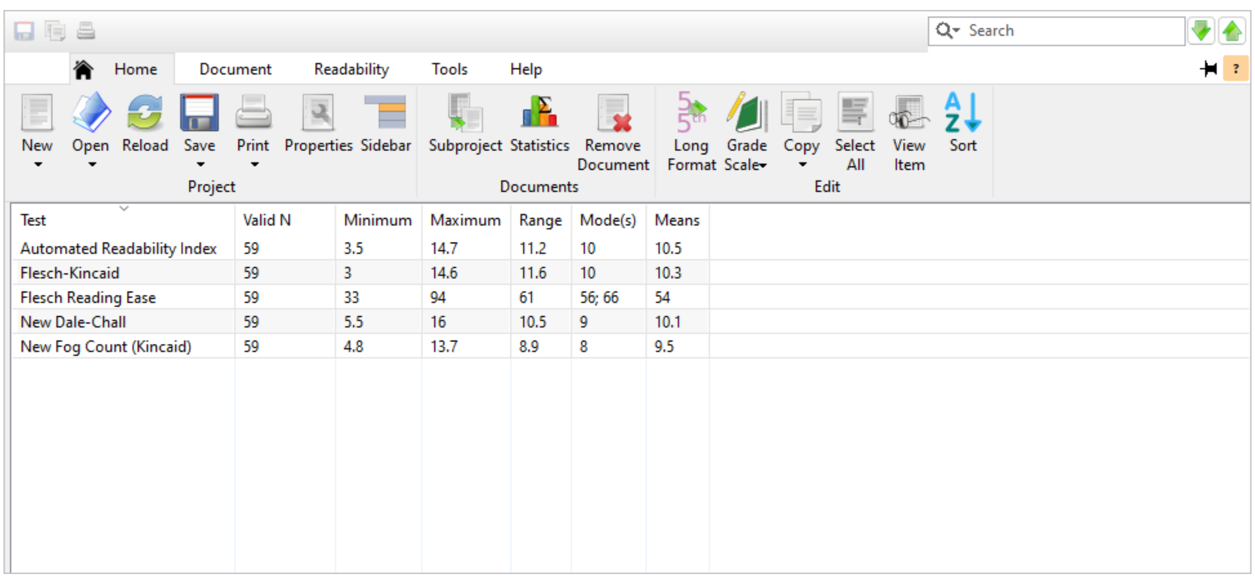
You can also filter documents based on content size. Enter into the field Minimum document word count the minimum number of words a document must have to be included in the batch. Any document that does not meet this threshold will be ignored during import.



Note that when the project is first analyzed, the documents that were not included in the sample will be removed from the project.

Clicking the Forward > button will move to the Document Structure and Test Selection pages. Refer to sec. 3.2 for an explanation of these pages.

After you have specified your document structure and tests, click the Finish button to create the project, as shown below:



From here, you can review the readability scores, box plots, histograms, readability graphs, Dolch statistics, and difficult words. You may also want to review any warnings that were encountered while analyzing the documents.

4. Reviewing Standard Projects

4.1 Test Scores

After opening a project, click on the Readability Scores icon on the project sidebar, and then click on Scores beneath that. Here you can view all the test scores, as well as their respective explanations and recommendations.

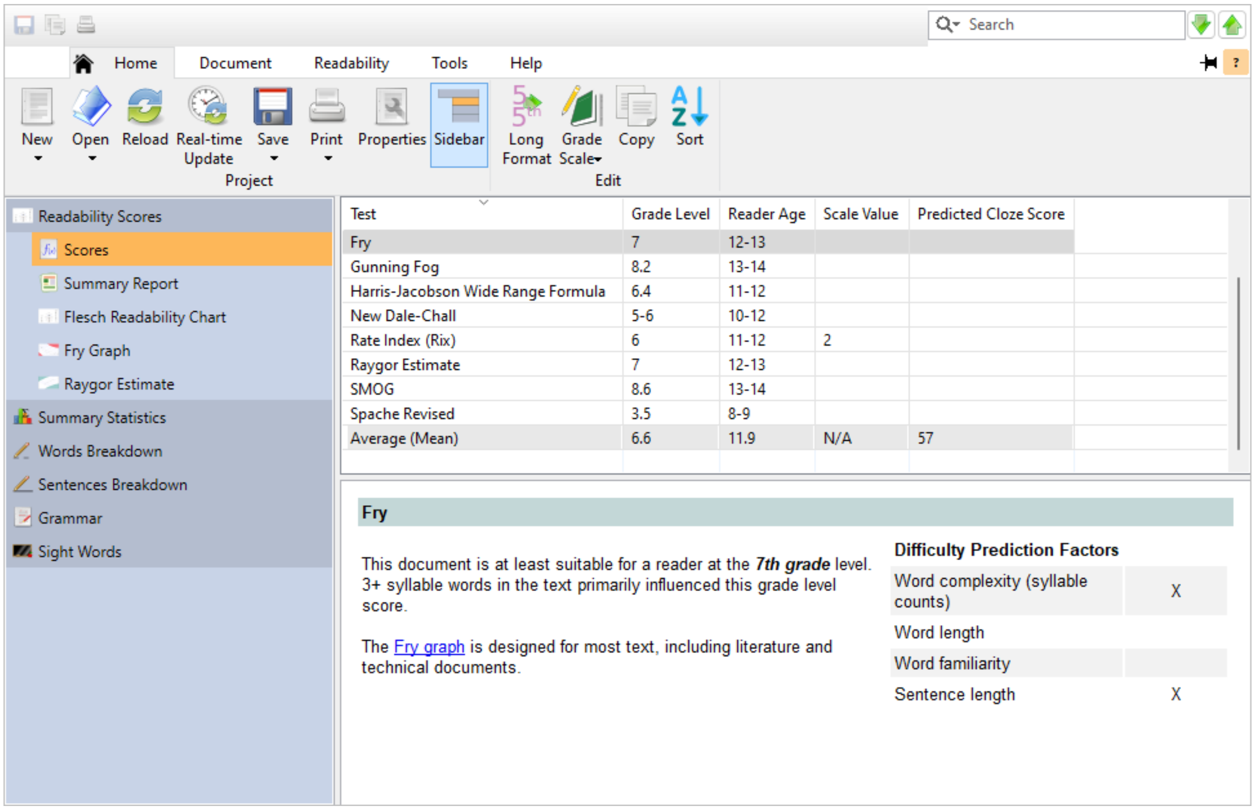
The top area of the Scores window is the score grid. In here there are four columns:

- Test: the name of the readability test.
- Grade Level: the minimum grade level that the test determined the document is suitable for.
- Reader Age: the minimum reader age that the test determined the document is suitable for. The reader age is determined from the test's score. It can be shown as either a range for the entire grade, or be rounded to the semester that the grade score falls into.
- Scale Value: the index value for tests that use special scales.
- Predicted Cloze Score: the estimated score if the document were to be used for a cloze test. Note that cloze scores will be displayed in percentage format (i.e., 0–100) instead of floating-point format (i.e., .0–1.0) for easier reading.

Note that most tests have just one type of result (e.g., grade level), so non-applicable columns are left blank. For example, a Gunning Fog result may be a grade level of 8.3. This translates to an eighth grade level reader (in his or her third month of class). Because this test only returns a grade level and age, the Scale Value and Predicted Cloze Score columns are empty. Another example is a result for the Flesch Reading Ease test, which could be a scale value of 79. This test has its own scale (instead of a grade level) so its age, grade level, and cloze columns are empty.

When a grade-level test scores a document at its maximum value, a + will be appended to the score. Although most grade-level tests can score up to 19, some tests which are designed for primary and secondary education materials will have lower maximum scores. For example, Wheeler-Smith is designed for primary-age materials and its maximum score is the fourth-grade level. If a document's Wheeler-Smith score is 4, then it will be displayed as *4+*.

Select any test in the results grid to display an explanation of its score in the window below.



Along with an explanation of the test's results, the factors that it uses to determine difficulty are also shown. For example, if Sentence Length and Word Length have an *X* next to them, then these variables are used in the test's equation. To lower the score for a given test, try to improve its factors (e.g., lower the document's average sentence length).

Finally, the averages (means) of your test scores and ages are displayed at the bottom of the results window. Their respective modes, medians, and standard deviations are also included if Extended Information is enabled. Note that scores are truncated when searching for the grade-level mode. For example, *6.7* and *6.2* will both be treated as *6*. Mode generally requires discrete data to produce meaningful results; therefore, grade-level scores are converted to their base grades for this statistic.

When calculating these statistics, failed test scores are treated as missing observations and not included. For example, let us say that you included 5 tests, but one of them could not be calculated. In this case, the valid 4 scores will be added and divided by 4 (not 5) to acquire the means. Also, grade range scores–such as *5–6* from New Dale-Chall–will be converted to its own average (e.g., *5–6* will be 5.5).

To remove any test from this list, select it and press Ctrl+Del (,) or ⌘+⌫ () on your keyboard. To edit a custom test, double-click on it in this list. To view the help for a standard test, double-click on it in this list.

There will also be readability graphs in this section if you included their respective tests. Refer to sec. 4.2 for further information.

4.2 Readability Graphs

There are numerous readability tests that are graphically plotted, including Fry, Raygor, Flesch, Lix, Gilliam-Peña-Mountain, FRASE, Crawford, and Schwartz. These graphs consist of either grade-level or reading-level regions that a document may fall within. They also include indicators for the most influential factor for the document's reported level (i.e., word difficulty vs. sentence difficulty).

To view these graphs, click on the Graphs icon in a standard project or the Readability Graphs icon in a batch project.

Let us first review a Fry graph:

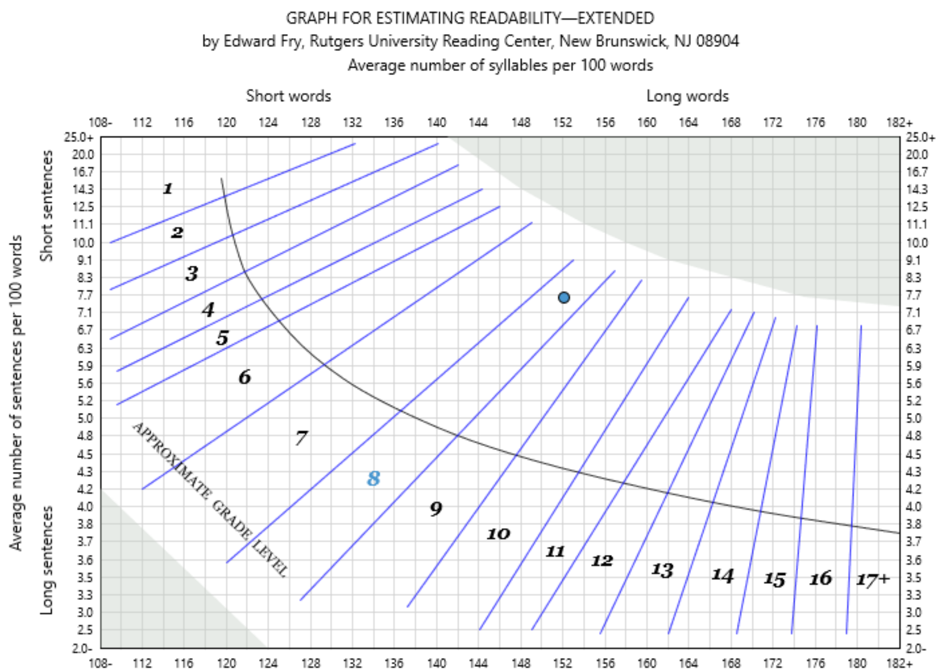


Figure 4.1: Fry graph

To read this graph, search for the red data point–this value represents the document's score. Note that there are numbers 1 through 17+ on this chart, each partitioned into small slices. The band that the data point falls into is the grade level of this document. In the example above, the data point is inside of the 8th grade slice (notice that the 8 is marked in blue).

Running through each slice is a gray line, which is a separator to help show the most predominant cause of the document's difficulty. If the data point is above the gray line, then the strongest influence on the document's difficulty is high-syllable words. If it is below the line, then the strongest influence is long sentences. In the above example, it is above the line, implying that it is slightly more influenced by high-syllable words. However, because it is fairly close to the line, this document is well balanced between difficult words and difficult sentences.

If the data point does not fall inside of a slice then the test will fail. For example, the Raygor graph below indicates that the document's overly-long words caused the test to fail:

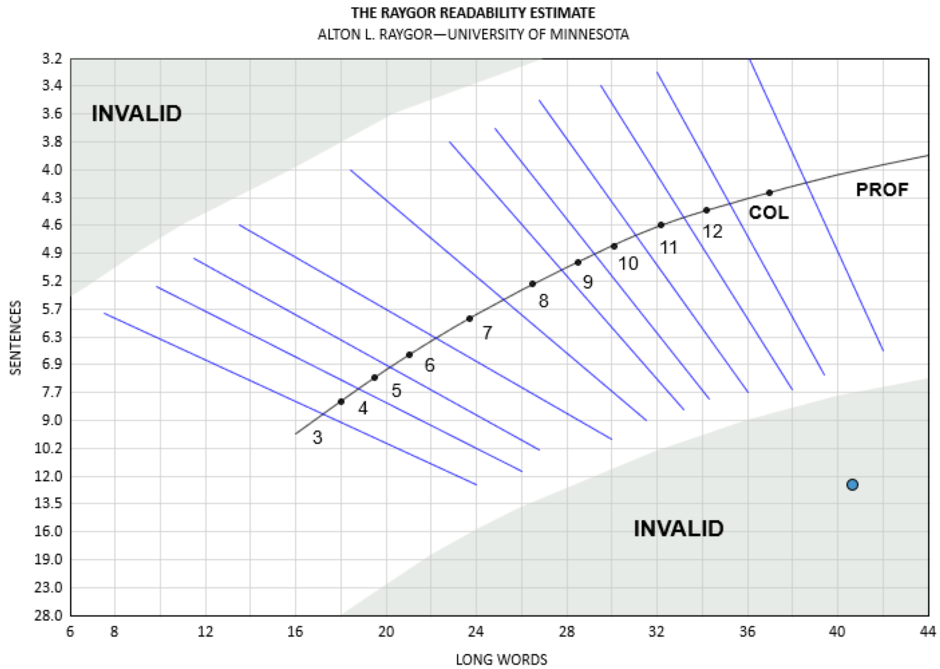


Figure 4.2: Raygor Estimate graph

The document was almost classified as 12th grade level material, but its long words make it too difficult to be assigned to any grade. At this point, you would want to review the document's lengthy sentences (select Highlighted Report under the Grammar sidebar item) and try to shorten some of them.

The Gilliam-Peña-Mountain graph is a variation of Fry designed for Spanish text. It is identical to Fry, except that its x-axis begins at 175 (instead of 108). This enables this graph to plot Spanish text, which usually has a much higher syllable count than its English counterpart. Below is an example of a Gilliam-Peña-Mountain graph:

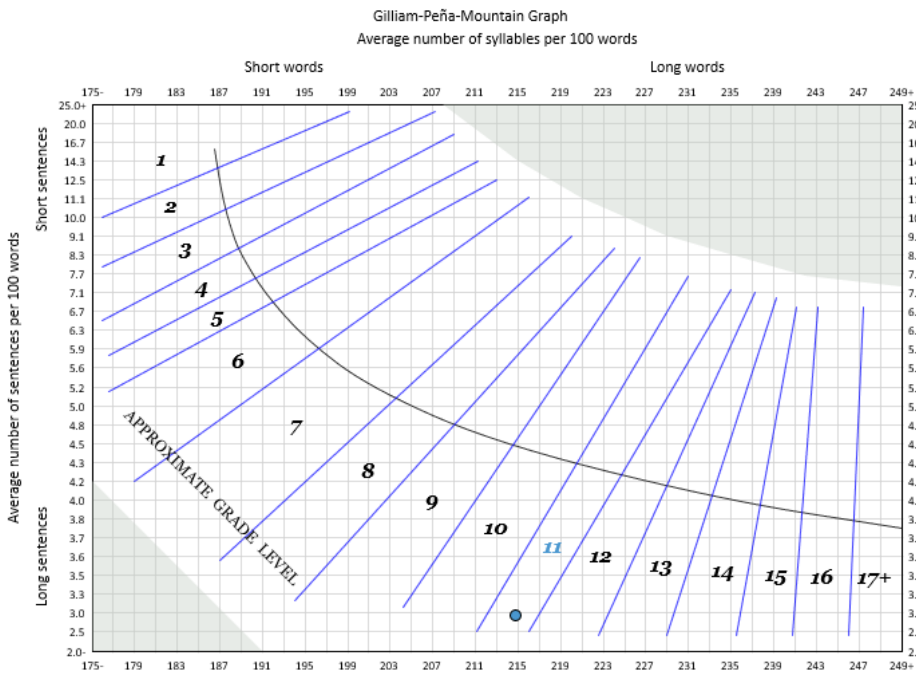


Figure 4.3: Gilliam-Peña-Mountain graph

The Schwartz graph is another Fry-like graph, which is designed for German (primary and secondary-reader) text. Although similar in appearance to Fry, this graph uses its own reading-level regions mapped to the German materials it was trained on. This approach differs from Gilliam-Peña-Mountain, which uses the original Fry template and adjusts its axis ranges to accommodate Spanish text. Below is an example of a Schwartz graph:

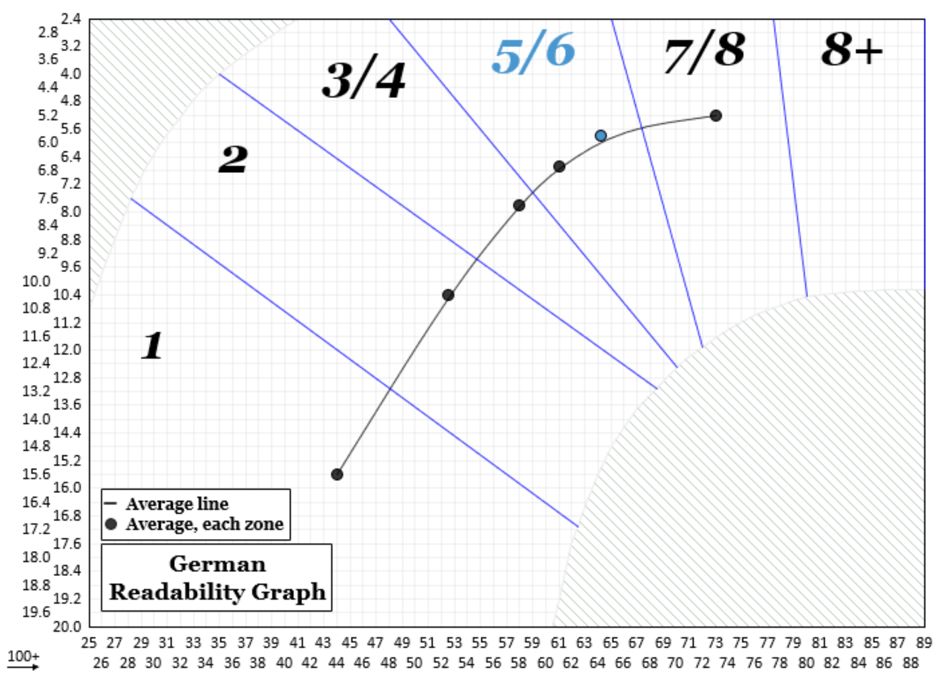


Figure 4.4: Schwartz graph

The Fry and Raygor graphs are very similar and are both interpreted in the same manner; however, there are a few differences between them. The Fry graph classifies documents within the 1st grade to doctorate level. In contrast, Raygor concentrates on the range of 3rd grade to the sophomore collegiate level. It is recommended to only use Raygor graphs for secondary-age materials.

Another difference is that the Fry graph is influenced by high-syllable words, while the Raygor graph is influenced by long words. Long sentences remain a key factor for both of these graphs, though.

The other difference between the two graphs is that they are inverted from one another. In a Fry graph, the long-sentence region is in the bottom left-hand corner (going downward), whereas in a Raygor graph the long-sentence region is in the top left-hand corner (going upward). However, the same rules apply for how to interpret the grade level and influence of difficult words vs. difficult sentences.

The FRASE graph is similar to Fry in terms of factors used and axis orientation. Its most influential factor (i.e., word vs. sentence difficulty) is determined in the same manner, and its Y (sentence) axis is descending. Below is an example of a FRASE graph:

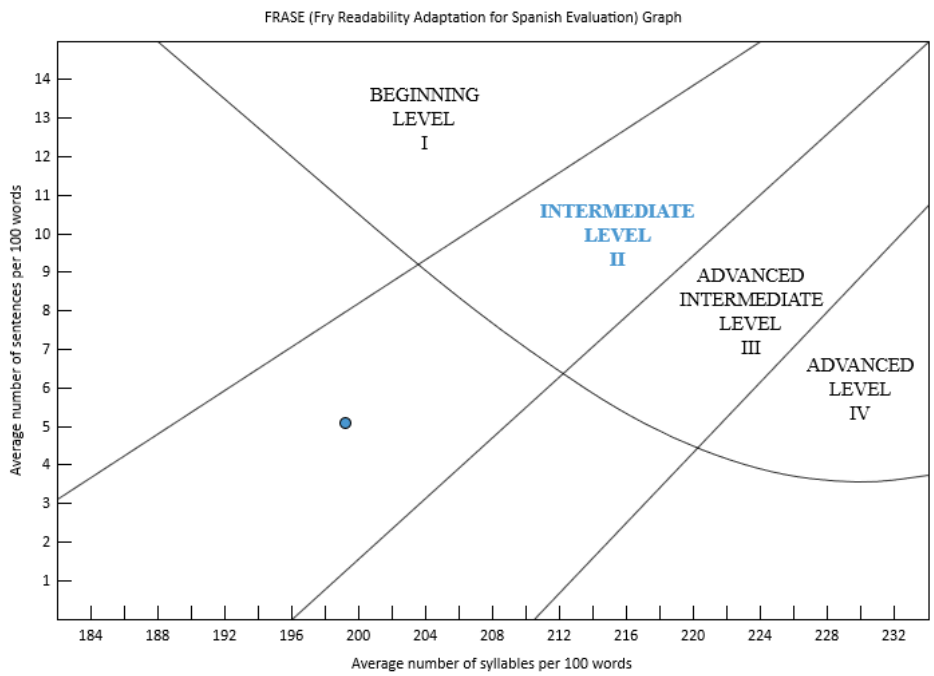


Figure 4.5: FRASE graph

A key difference with FRASE is its use of reading levels (rather than grade levels). In the above example, we can see that the document has been scored at the *Intermediate* level. The other difference is that it is designed for Spanish text—note how its X (syllable-count) axis begins at 182, not 108.

The Crawford graph is also similar to Fry in terms of factors used. However, it differs from most other graphs in that it uses the interior of the plot to find its sentence-count factor (instead of an axis). Below is an example of a Crawford graph:

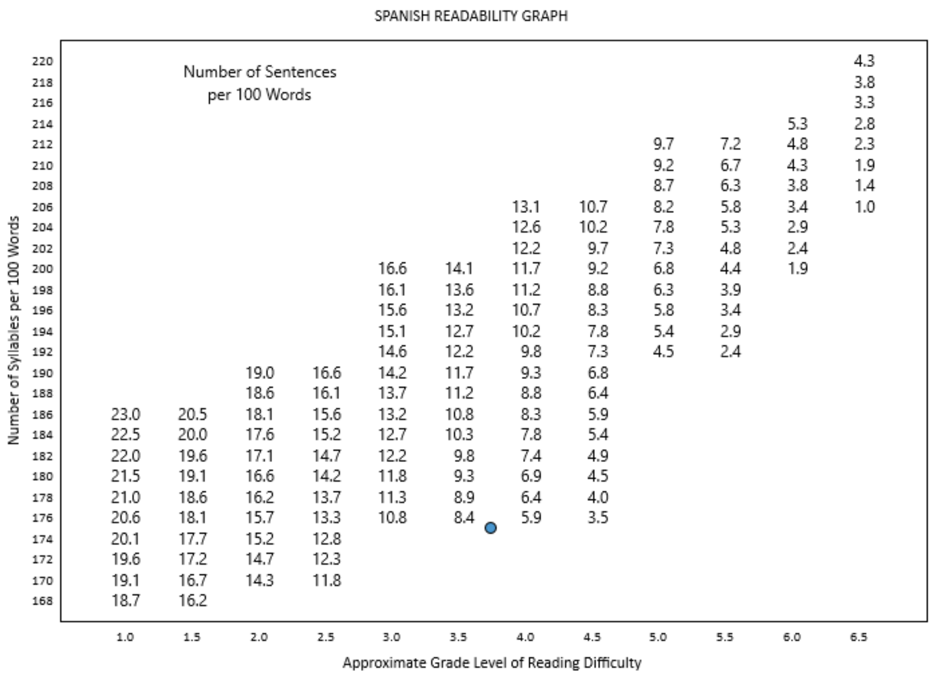


Figure 4.6: Crawford graph

In the above graph, we can see that the intersection of the factors is 175 and 7. The number of syllables is 175 (y-axis). The number of sentences is approximately 7 (between 8.4 and 5.9 on the plot [corresponding to 175 on the y-axis]). From this intersection, the point's x-axis value is 3.7. Hence, 3.7 is the grade-level score for this graph.

The Flesch chart plots a document's factors onto a pair of "rulers" and then connects them to determine the Flesch score. This score ranges from 0 ("very difficult") to 100 ("very easy"). Below is an example of a Flesch chart:

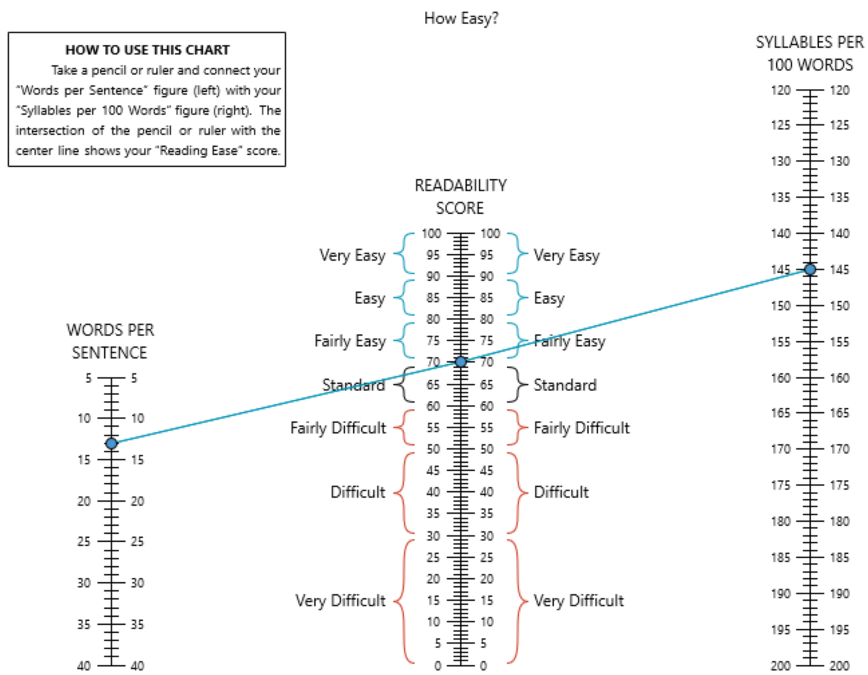


Figure 4.7: Flesch chart

Danielson-Bryan 2—a variation of Flesch Reading Ease— includes a similar graphic, which also includes grade-level information:

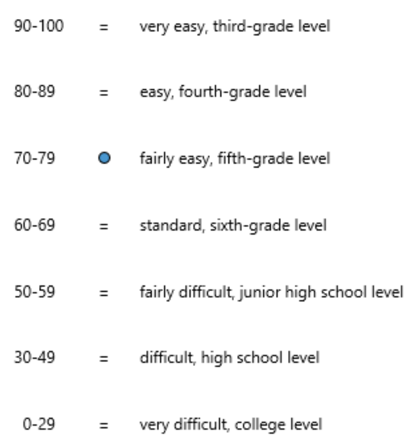


Figure 4.8: Danielson-Bryan 2 plot

The INFLESZ Scale—another variation— plots a score (based on an adjusted FRE calculation) on a 0–100 scale. This score can then be compared between three different classification scales: INFLESZ, Szigriszt, and Flesch Reading Ease. (INFLESZ is preferred over Szigriszt, and Flesch Reading Ease is only included for reference.) Similar to Flesch, it groups its scores from *"MUY DIFÍCIL"* ("very difficult") to *"MUY FÁCIL"* ("very easy").

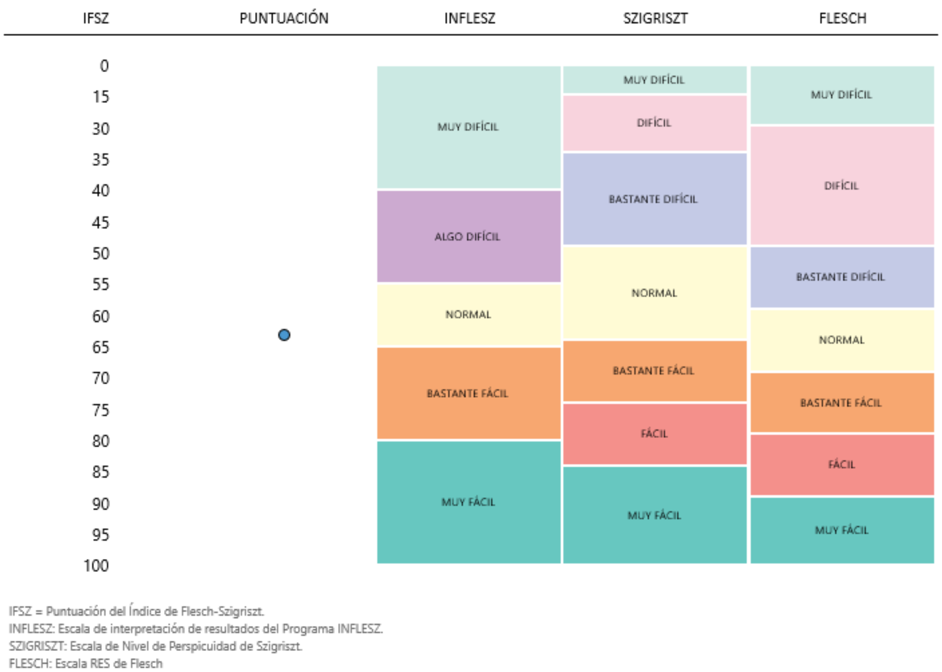


Figure 4.9: INFLESZ

Similar to a Flesch chart, the Lix gauge plots a document's Lix index score on a chart showing its difficulty level:

Figure 4.10: Lix gauge

In the above chart, we can see that Lix scores can range from 0 (“very easy”) to 60+ (“very difficult”). There is also a variation of the Lix gauge that is adjusted for German materials:

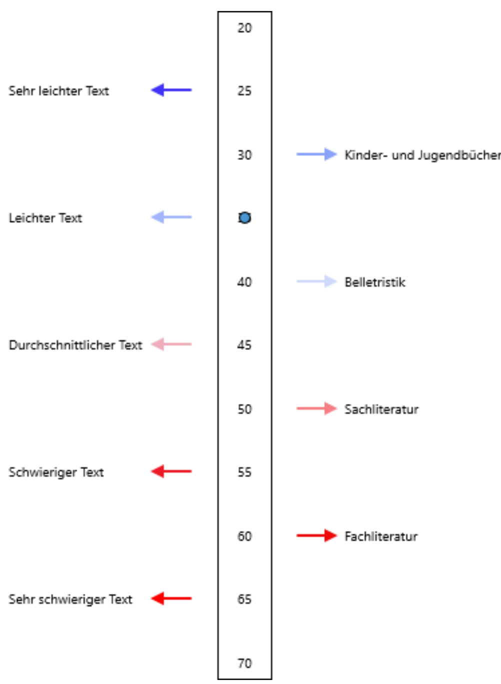


Figure 4.11: German Lix gauge

This version adjusts the scaling of the score, assuming that German text is inherently more difficult than other languages. It also includes additional levels of difficulty (e.g., "children and young adult").

Note that if you plot any of these graphs, its score can also be viewed in the Readability Scores area.

As a final note, batch projects include multi-document versions of all these graphs:

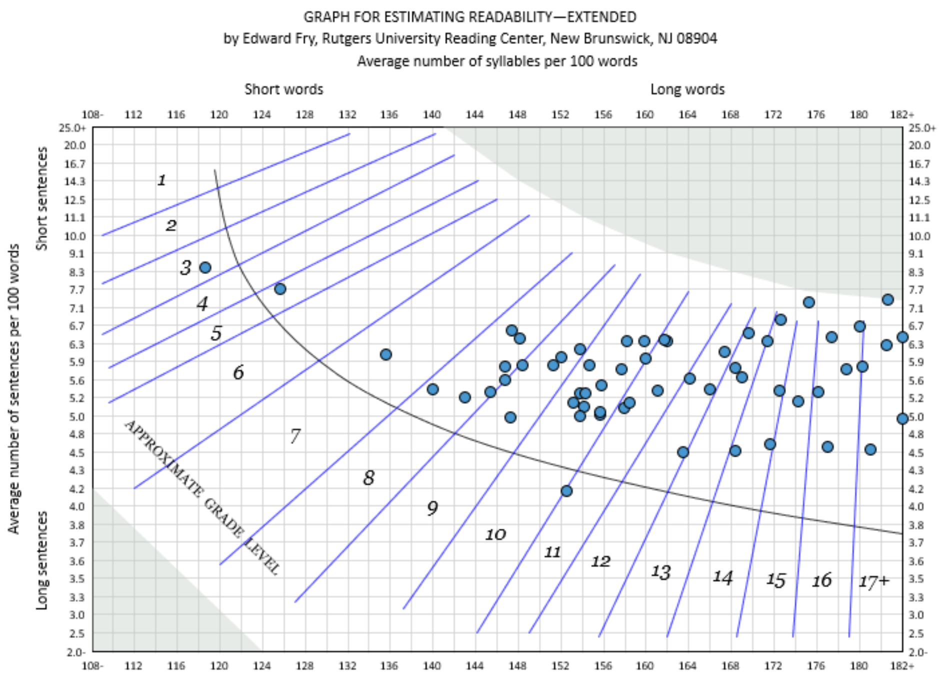


Figure 4.12: Multiple documents on a Fry graph

Here, we can see that an entire collection of documents is plotted onto a single Fry graph. This is useful for visualizing the grade-level layout of a group of documents. In the above example, we can see that the documents are clustering in the high-school region. We can also interpret that complex words are heavily influencing the documents' difficulty.

You can also print or save a blank copy of any of these graphs. This is useful for printing handouts for instructors or students to use. To print a blank graph, go to the Readability tab on the ribbon, click the Blank Graphs button, and select the graph that you wish to save.

4.3 Goals

Goals are recommended score ranges that a document should fall within.

If a project contains goals, then a Goals section will be available to show whether they were met. For example:

Test/Statistic	Minimum Recommended Value	Maximum Recommended Value
⚠ Fry	✔	⚠ 10
⚠ Flesch Reading Ease	⚠ 60	✔
⚠ Easy Listening Formula	✔	⚠ 12

In the above image, we can see that this project is recommending:

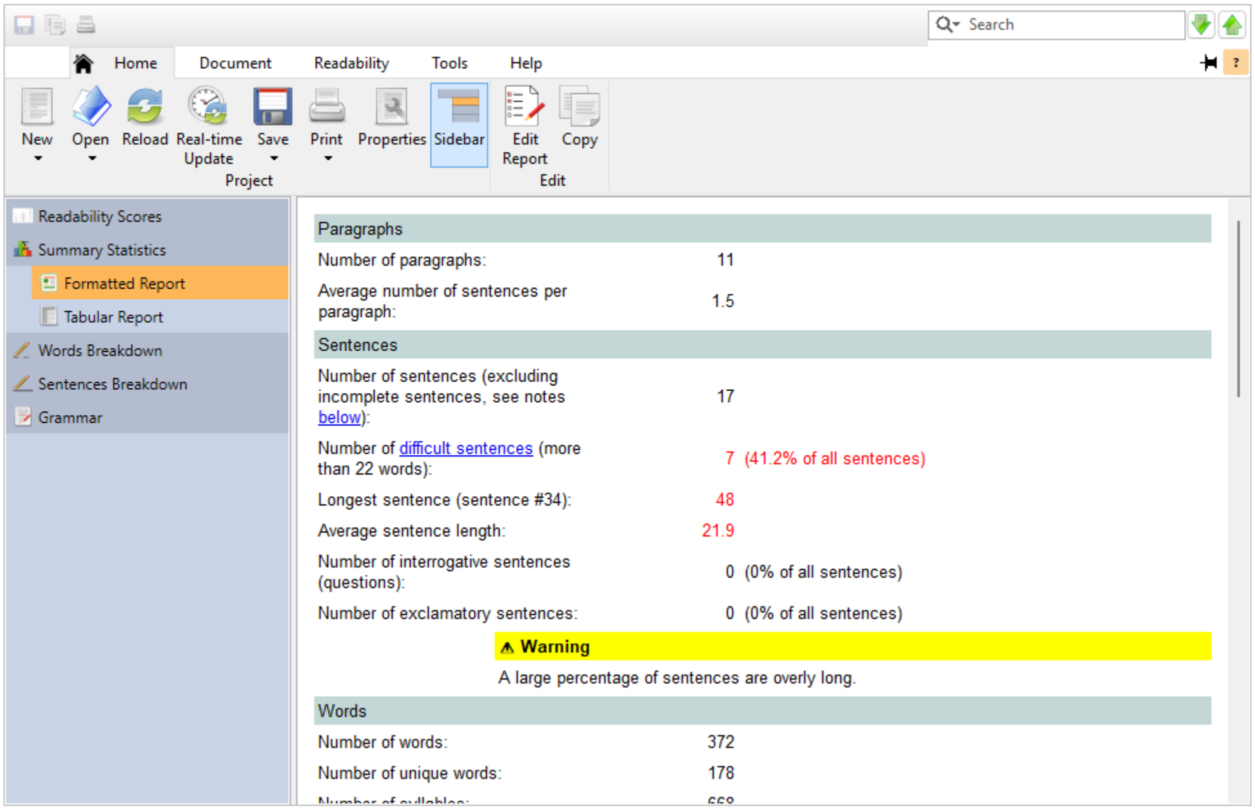
- Fry should not be higher than 10
- Flesch Reading Ease should be higher than 60
- Easy Listening Formula should not be higher than 12

Also, the warning icons indicate that currently the document is failing to meet all these criteria.

Goals can be created through a test bundle. When the bundle is applied, both its tests and goals will be added to the project.

4.4 Statistics

After opening a project, click on the Summary Statistics icon on the project sidebar to view the document's statistics:



The Report and Tabular Report windows display the document's statistics in two formats: a formatted report and a tabular report, respectively. Both of these reports contain the following statistics:

- number of paragraphs and average number of sentences per paragraph
- number of sentences and average number of words per sentence
- number and percentage of difficult sentences
- number of units/independent clauses (This statistic is only available if Extended Information is enabled.)
- number of interrogative sentences (questions)
- number of exclamatory sentences
- number of words
- number of unique words
- number of syllables and average number of syllables per word
- number of characters (with and without punctuation) and average number of characters per word
- number and percentage of numerals
- number and percentage of proper nouns (not available for German projects)
- number and percentage of total (and unique) monosyllabic words
- number and percentage of total (and unique) 3+ syllable words
- number and percentage of total (and unique) hard SMOG words. This statistic is only available if SMOG is included in the project

- number and percentage of total (and unique) hard Fog words. This statistic is only available if Fog is included in the project
- number and percentage of total (and unique) 6+ letter words
- number and percentage of total (and unique) unfamiliar words (This includes word lists such as New Dale-Chall and Spache. These statistics are only available if the respective tests are included in the project.)
- number of grammar errors
- text size (the size of the text in a document, not the actual file size). (This statistic is only available if Extended Information is enabled.)

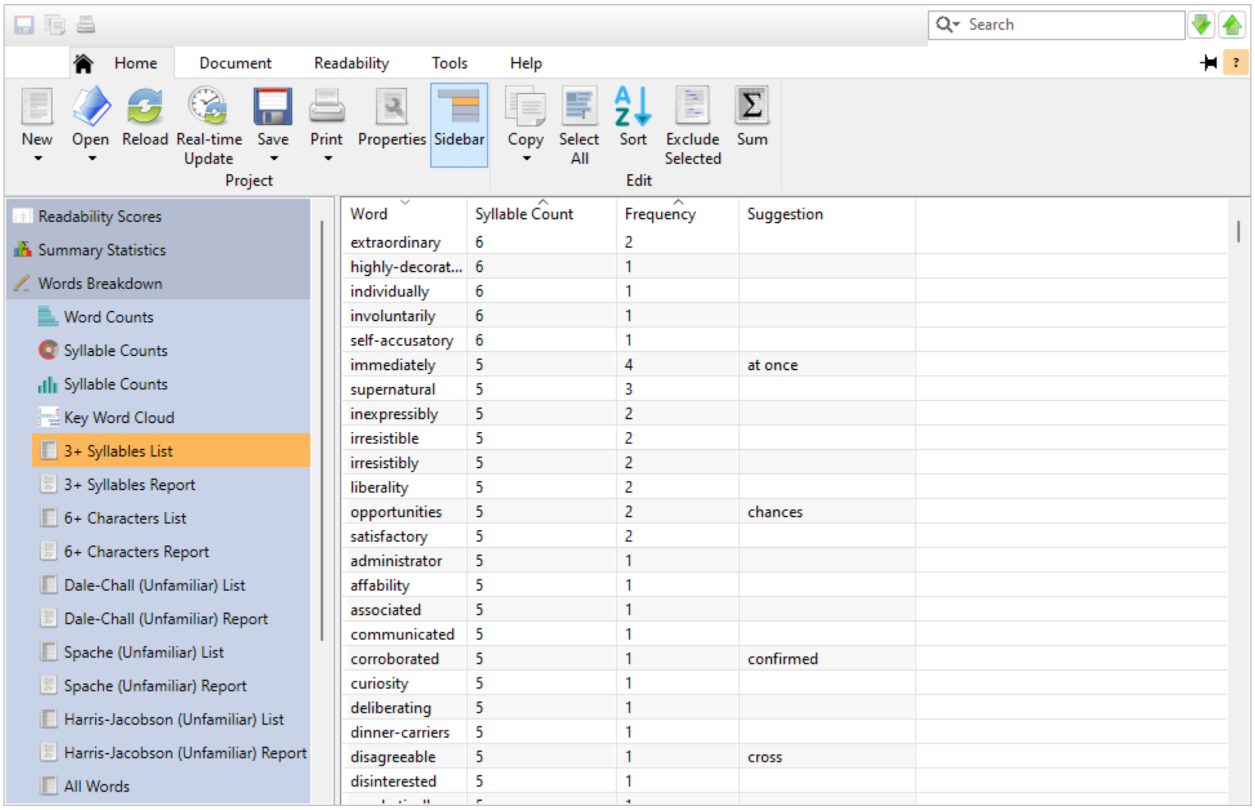
Statistics for any custom familiar-word tests will also be included in this window.

A Notes section may be included at the bottom of the formatted report if there are any items of interest detected (e.g., a high sentence-length average). If your document requires improvement, then this section is usually a good place to begin.

> Note
> 3+ syllable words and Fog and SMOG hard words are all the same except for how numerals (e.g., phone numbers and dates) are handled. SMOG always sounds out each digit of a numeric word, whereas Fog ignores numeric words. You can control how numeric words are handled from the Document Indexing section of the Options dialog (available on the Tools tab).

4.5 Words Breakdown

After opening a project, click on the Words Breakdown icon on the project sidebar to view the lists, highlighted reports, and graphs of difficult words:



Reviewing Word Lists

The types of word lists that may be shown are:

3+ Syllables: This list displays all the document's words that are three or more syllables. It also displays how many syllables each word contains and their respective frequency counts. This window is only available if the document contains 3+ syllable words.

6+ Characters: This list displays all the document's words that are six or more characters. It also displays how many characters each word contains and their respective frequency counts. This window is only available if the document contains 6+ character words.

Dale-Chall (Unfamiliar): This list displays all the document's words that are not in the New Dale-Chall familiar words list and their respective frequency counts. The New Dale-Chall word list is a collection of words that are considered to be familiar to most fourth grade readers. Words in the document that are not found on this list may be less familiar to younger readers and hence more difficult. This window is only available if a Dale-Chall related test is included in the project.

Spache (Unfamiliar): This list displays all the document's words that are not in the Spache familiar words list and their respective frequency counts. The Spache word list is a collection of words that are considered to be familiar to most primary-age readers. Words in the document that are not found on this list may be less familiar to younger readers and hence more difficult. This window is only available if the Spache test is included in the project.

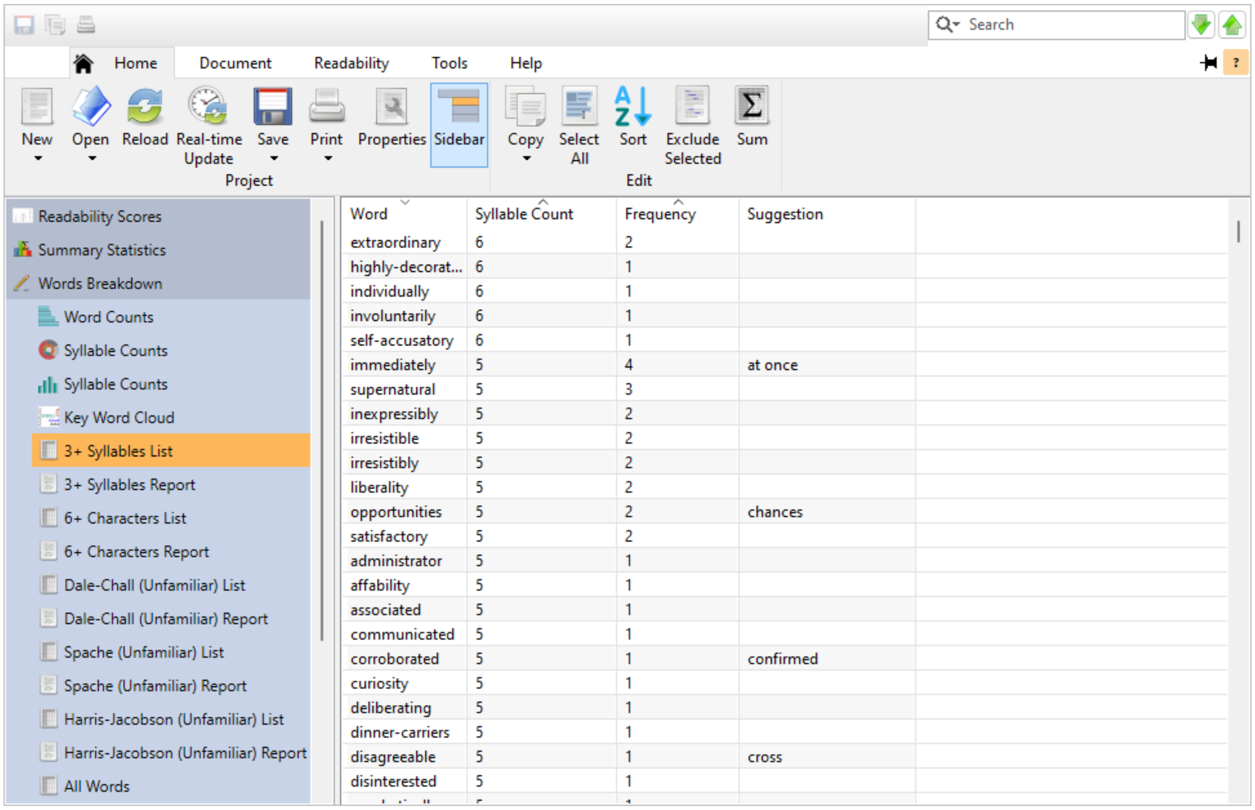
Harris-Jacobson (Unfamiliar): This list displays all the document's words that are not in the Harris-Jacobson familiar words list and their respective frequency counts. The Harris-Jacobson word list is a collection of words that are considered to be familiar to most second-grade readers. Words in the document that are not found on this list may be less familiar to younger readers and hence more difficult. This window is only available if the Harris-Jacobson test is included in the project.

All Words: This list displays all the words from the document and their respective frequency counts.

Key Words: This list is the same as All Words, except that it excludes filenames, numeric words, and common words, and then combines the remaining words. In regards to combining words, words with the same root will be merged into a single row. For example, *report*, *reports*, and *reporting* will be combined into one row, with the frequency counts for all three words also combined. As for excluding words, common words (e.g., *the*), numerals, and filenames are not included in this list. (If no words can be combined and there are no common words to ignore, then this list will not be shown.)

> Note
> If you are ignoring incomplete sentences, then these lists will only include the words from complete sentences. If you need to review all the words from the document, then select the option Do not exclude any text. Refer to sec. 20.1 to learn more about how *Readability Studio* handles incomplete sentences.

To find the most complex words, sort the Syllable Count column of the 3+ Syllables list. You can sort any of the columns by clicking on the column header. To change the sorting order to either ascending or descending, click the column header a second time. As you can see below, the words with the most syllables are now shown at the top of the list.



If your test scores indicate that the document is too difficult for your audience, then you should review overly complex or long words. You may consider seeing if any shorter synonyms could be used in their place. The same type of review can also be performed with the 6+ Characters list by sorting the Character Count column.

To help find the most difficult words for the New Dale-Chall list, select the Dale-Chall (Unfamiliar) subitem and sort the Frequency column. The most frequently occurring words are the ones that should be replaced with more familiar synonyms that most readers would recognize.

To find a specific word in a list, type the word into the Search bar at the top of program and hit the Enter (⊞, 🐧) or ⌅ () button on your keyboard. You can also select the list and begin typing the word. If the word is in the list, then the selection will move to it.

Double-click on any word in these lists to display the respective highlighted text report with the first instance of that word selected.

Reviewing Word Reports

The different types of text reports that may be shown are:

3+ Syllables: This text report contains the document with all the words that are three or more syllables highlighted. This window is only available if the document contains 3+ syllable words.

6+ Characters: This text report contains the document with all the words that are six or more characters long highlighted. This window is only available if the document contains 6+ character words.

Dale-Chall (Unfamiliar): This text report contains the document with all the New Dale-Chall unfamiliar words highlighted. The New Dale-Chall word list is a collection of words that are considered to be familiar to most fourth grade readers. Words in the document that are not found on this list may be less familiar to younger readers and hence more difficult. This window is only available if a Dale-Chall related test is included in the project.

Spache (Unfamiliar): This text report contains the document with all the Spache unfamiliar words highlighted. The Spache word list is a collection of words that are considered to be familiar to most primary-age readers. Words in the document that are not found on this list may be less familiar to younger readers and hence more difficult. This window is only available if the Spache test is included in the project.

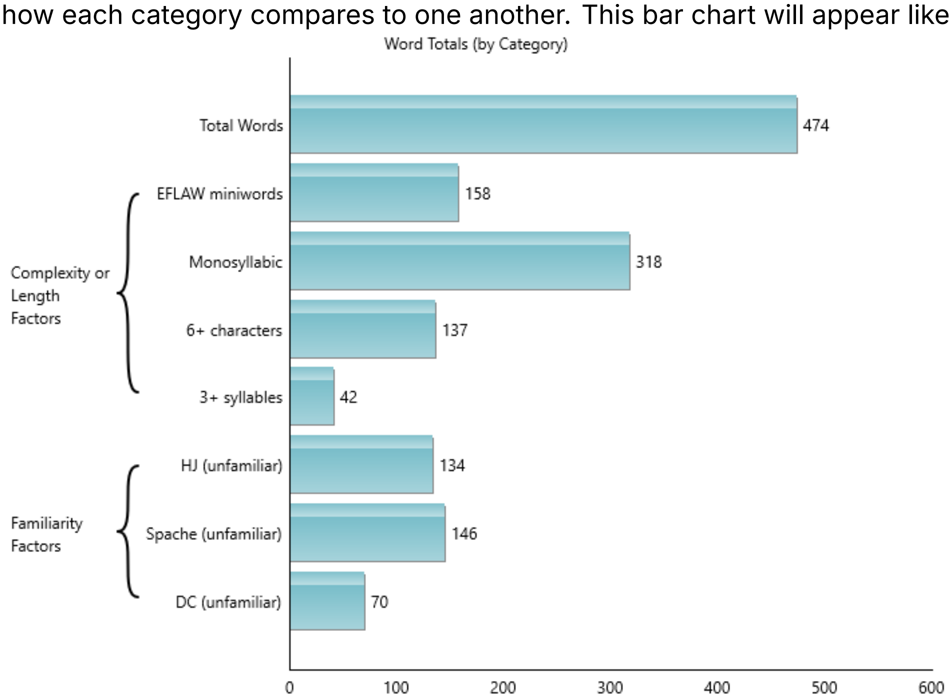
Harris-Jacobson (Unfamiliar): This text report contains the document with all the Harris-Jacobson unfamiliar words highlighted. The Harris-Jacobson word list is a collection of words that are considered to be familiar to most primary-age readers. Words in the document that are not found on this list may be less familiar to younger readers and hence more difficult. This window is only available if the Harris-Jacobson test is included in the project.

You can change both the font and highlighting color by opening the Project Properties dialog and selecting Highlighted Reports. Note that changing these options from the Project Properties dialog will only affect the current project. If you want to make changes to any future projects, then go to the Options dialog from the Tools tab instead.

Reviewing Word-based Graphs

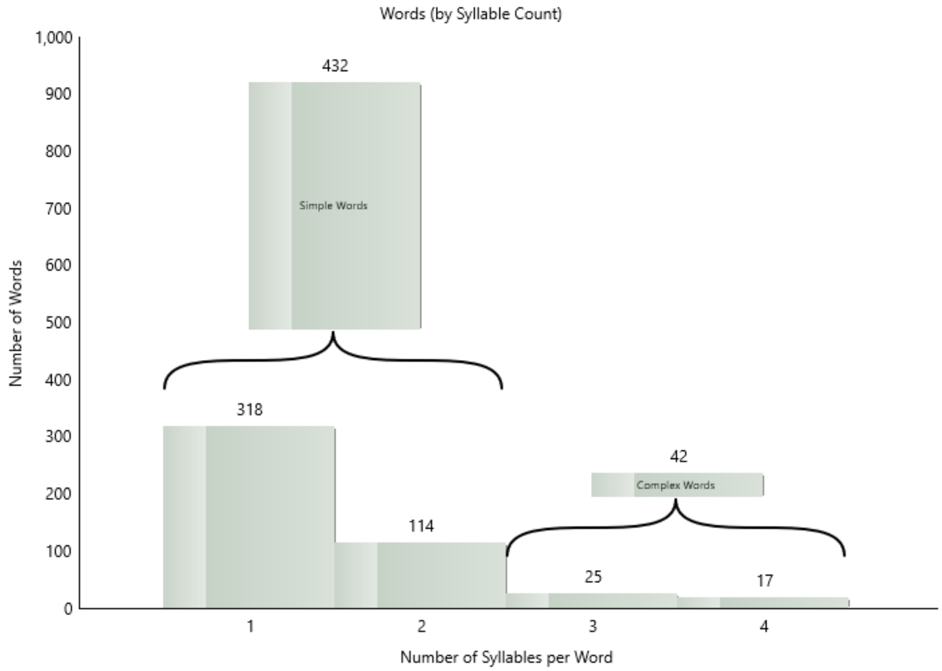
The different types of graphs that may be shown are:

Word Counts bar chart: This graph shows the total counts of each type of word (e.g., 3+ syllable words) and how each category compares to one another. This bar chart will appear like this:



In the above example, we can see that there are 70 unfamiliar New Dale-Chall words, but only 42 three+ syllable words. From this, we could infer that we should first concentrate on replacing these unfamiliar words. Lists of these words can be found by clicking on the Difficult Words icon on the project sidebar.

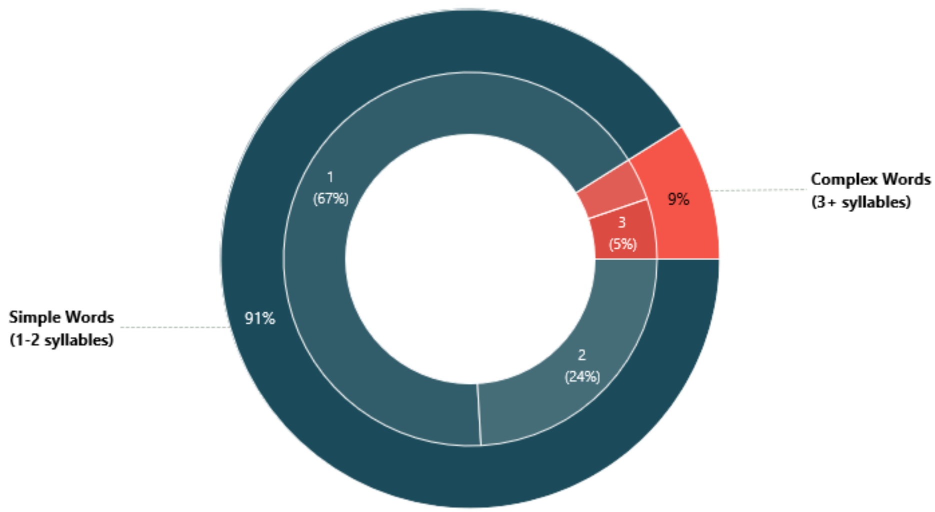
Syllable Counts: This is a histogram of the words, grouped by syllable count.



In the above histogram, we see that most of the words are either 1 or 2 syllables. There are only 25 three-syllable words and 17 four-syllable words in the document. (Note that selecting a bar will display the first 25 words within that syllable group.) Combined, there are 42 complex words, as shown by the cumulative group bar above the 3 and 4 bars.

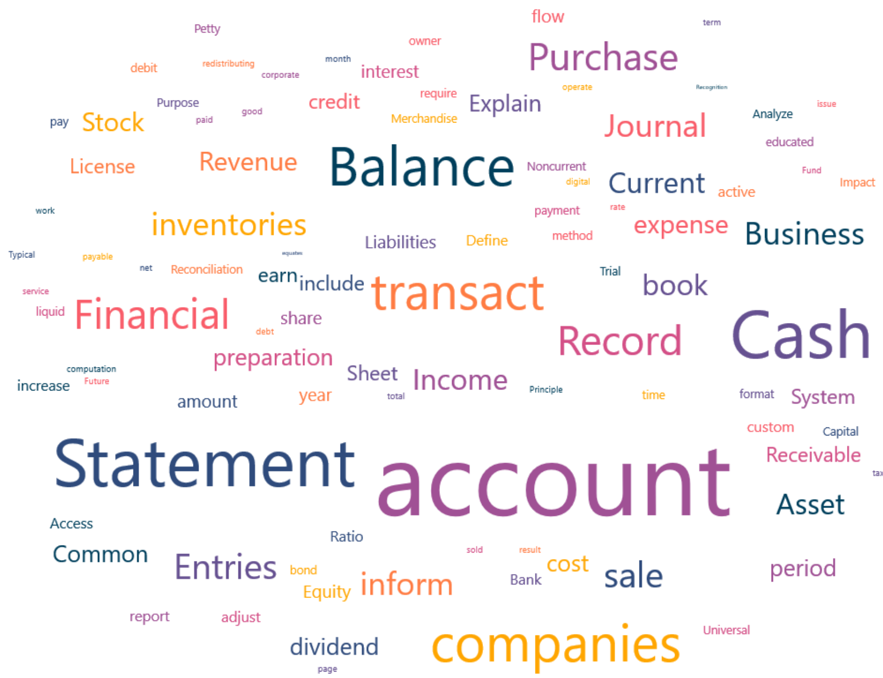
> **Note**
> The total value of a word count reflects the number of words, including repeats of the same word. The unique value of a word count reflects the number of words, excluding repeats. For example, a document contains the word *liquidation* three times. This word will be tallied three times when factored into the total word count, but only once into the unique word count. Most tests use the total word count variable, although there are some that use unique counts (e.g., Spache uses its unique unfamiliar word count).

Syllable Counts: This is a donut chart of the words, grouped by syllable count (along the inner ring). Additionally, an outer ring categorizes these subgroups into simple and complex words. (One- and two-syllable words are considered simple, while three [and higher] syllable words are complex.)



The above donut chart shows similar results to the previous histogram. With this visualization, however, it is easier to compare the overall complex- and simple-word groups. In this case, less than 10% of the document consists of complex words. Also, looking at the inner ring, we can see that ~two-thirds of the words are one-syllable.

Key Word Cloud: In conjunction with the Key Words list, a key word cloud is also included to visualize the main concepts of the document(s).



A word cloud shows words from a document in a random layout with randomly assigned colors. The size of each word is based on its relative frequency within the document. In other words, the larger the word, the more often it is used compared to other words.

This particular word cloud displays the ~100 most frequently occurring key words appearing in the document. Like the Key Words list, these will be combined words with uncommon ones (e.g., *the*) removed. In the case of combined words, the most frequent instance of similar words will be shown. For example, if the word *report* appears more times than *reporting* and *reports*, then *report* will appear in the word cloud. Also, the size of the word *report* will be based on the number of times *report*, *reporting*, and *reports* all appear.

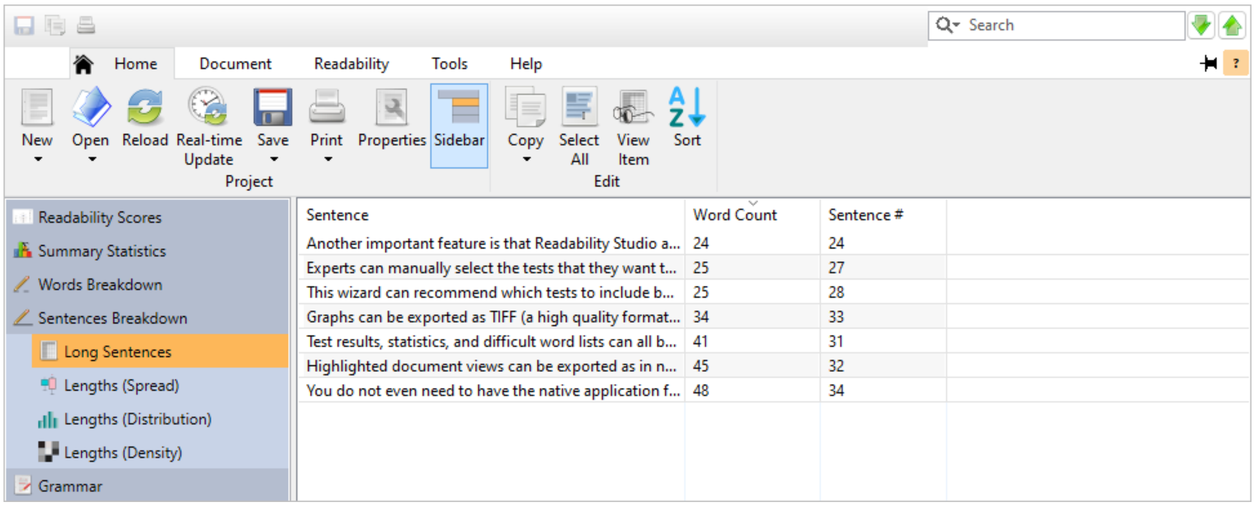
To focus on important topics, only words occurring two or more times will be shown (unless there are fewer than one unique hundred words).

In a batch project, this word cloud will show the key words across all documents.

4.6 Sentences Breakdown

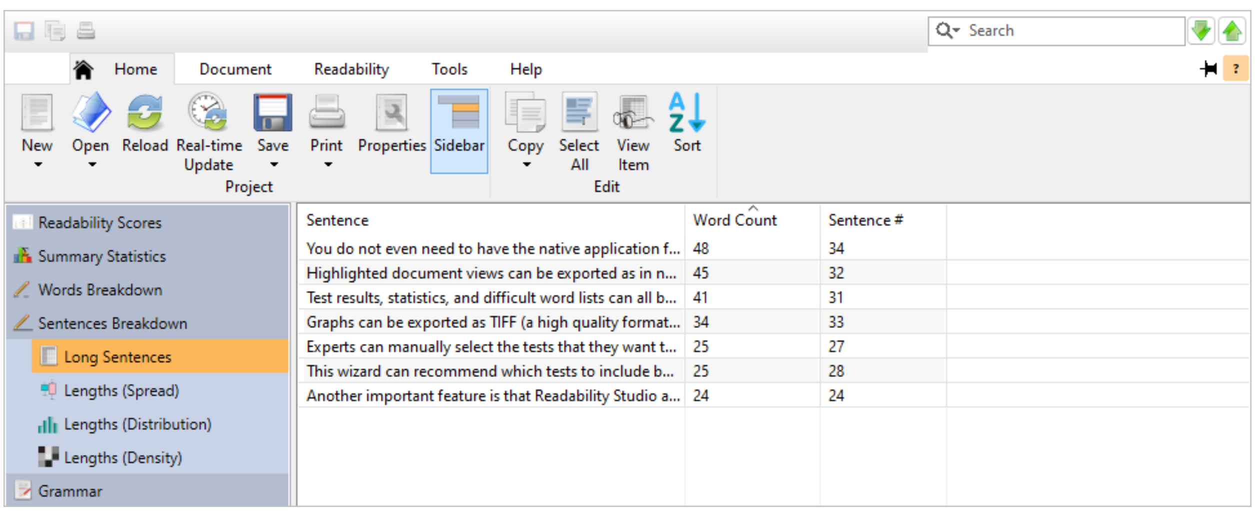
After opening a project, click on the Sentences Breakdown icon on the project sidebar to view a breakdown of the document's sentences. This includes the following:

A list of overly-long sentences:



Note that if you are ignoring incomplete sentences, then this list will only include the complete sentences. If you need to review all the sentences from the document, then select the option Do not exclude any text. Refer to sec. 20.1 to learn more about how *Readability Studio* handles incomplete sentences.

To find the longest sentence, sort the Word Count column into descending order. As you can see below, the longest sentence is now shown at the top of the list.

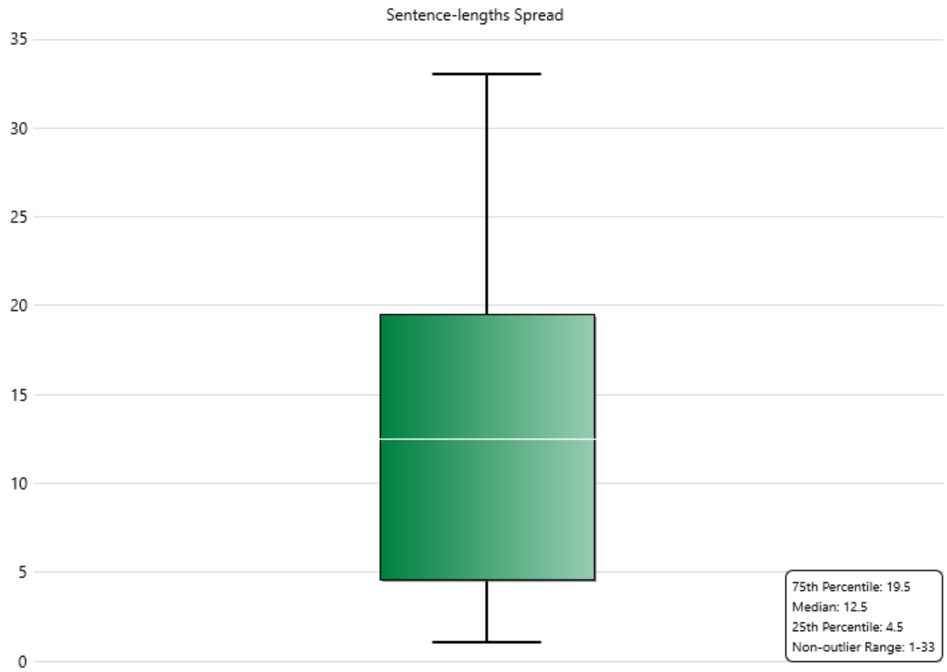


To view any sentence in its original context, double-click on it. This will take you to the Highlighted Report page with that sentence selected.

If the test scores indicate that the document is too difficult for your audience, then review the overly-long sentences. You should consider splitting these sentences into smaller ones and revising any wordy items.

Note that if there are no long sentences in your document, then this window will not be included.

A box plot of the sentence lengths (measured by word count) is also available:



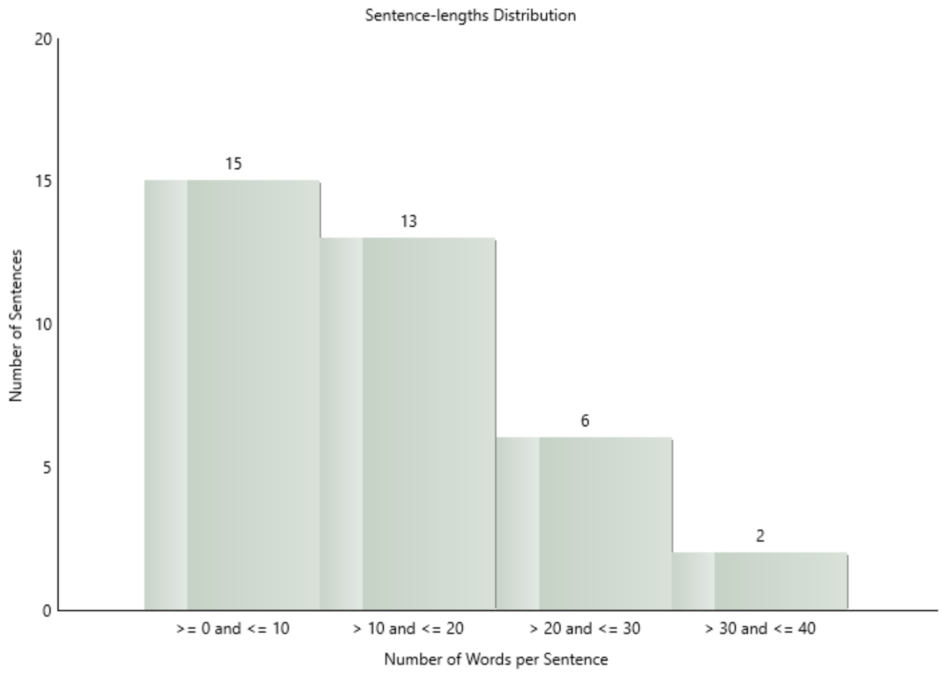
The first item to review is the box, which measures the percentiles range. Above, we see that this range is 4.5–19.5, meaning that half of the sentences are between 4.5–19.5 words.

Next, note the line drawn across the box—this represents the median. In this case, the median is 12.5, which means that the middle sentence is 12.5 words. This median line is almost equidistant, which implies that the sentence lengths in this range are evenly distributed.

The last items to review are the whiskers, which represent the non-outlier range. The datapoints in this range are outside the middle half the data, but still within a reasonable limit. Above, we can see that the remaining sentences are skewed towards having longer sentences, with the largest being 33 words. Although 33 words is reasonable (compared to the box's range), you may want to review some of these longer sentences.

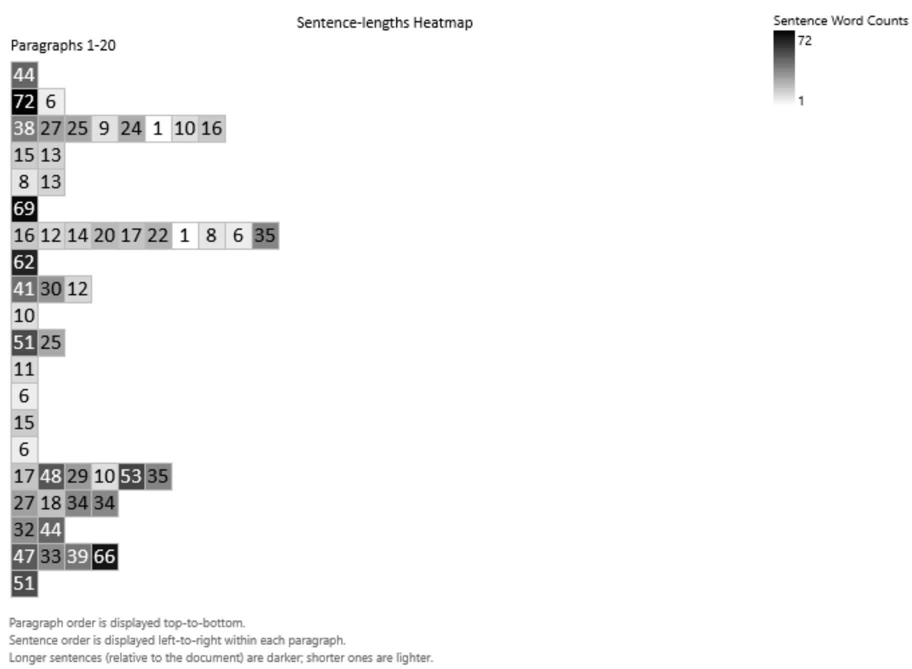
Finally, any sentence lengths outside the non-outlier range will be displayed as red dots. These points represent sentences which are significantly longer (or shorter) than the others. Clicking on an outlier will display the sentence's number, along with a preview of it.

A histogram is also available to visualize how the sentence lengths are distributed:



In the above graph, we can see a large portion of short sentences, with the rest falling into a normal distribution.

A heatmap of these sentence lengths is also included:



This graph displays the document's sentence, going from left-to-right, top-to-bottom. Each square represents a sentence, and the square's color represents its word count. The darker a square, the higher the sentence's word count (relative to the document). Likewise, lighter squares represent the shorter sentences in the document. This is useful for visualizing the trend and density of a document's longer sentences. For example, if sections of the grid are darker than others, then the sentences are longer in these spots and should be reviewed. The example below demonstrates this:

In the above graph, note how the first couple of sentences are the longest. After that, the middle section has relatively shorter sentences. Finally, the last section contains a significant block of the document's longer sentences.

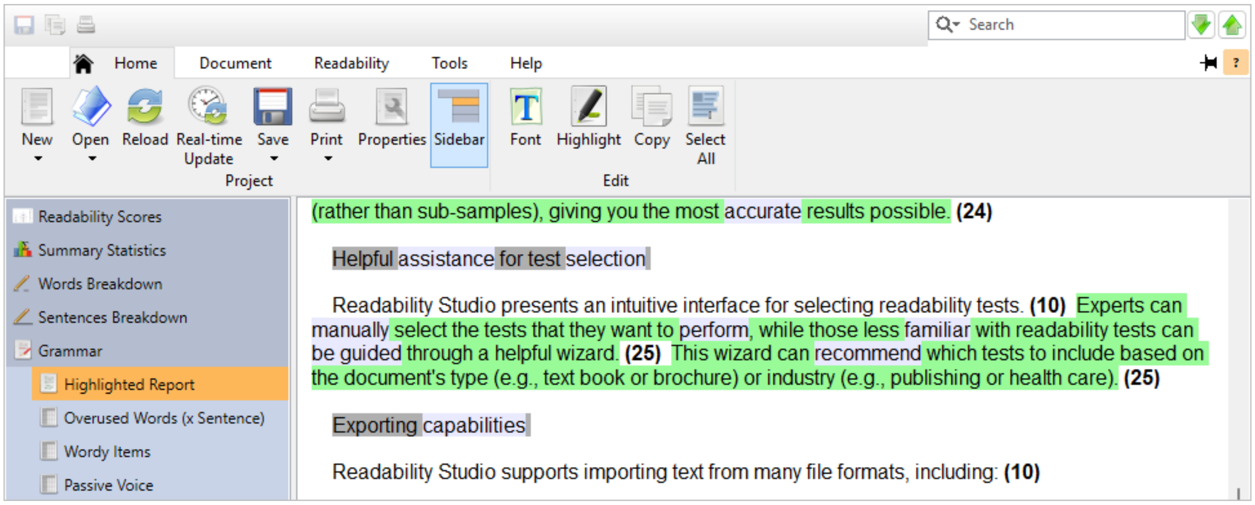
Although black squares refer to the longer sentences, this may not imply that these sentences are overly long. It only means that they are the longest, relative to the rest of the document. With this in mind, this heatmap is useful for locating overly-dense sections of longer sentences. In other words, clusters of black squares indicate problematic areas where lengthy sentences are running together.

To find the location of a sentence, click on the square and its sentence number will be shown. Also, if the sentence's length is an outlier (compared to the other sentences), then a preview of it will also be shown.

Refer to example sec. 14.6 for a more detailed tutorial on sentence heatmaps.

4.7 Grammar

After opening a project, click on the Grammar icon on the project sidebar and select the Highlighted report subitem. This text report will display the document with its difficult sentences and wordy items highlighted.



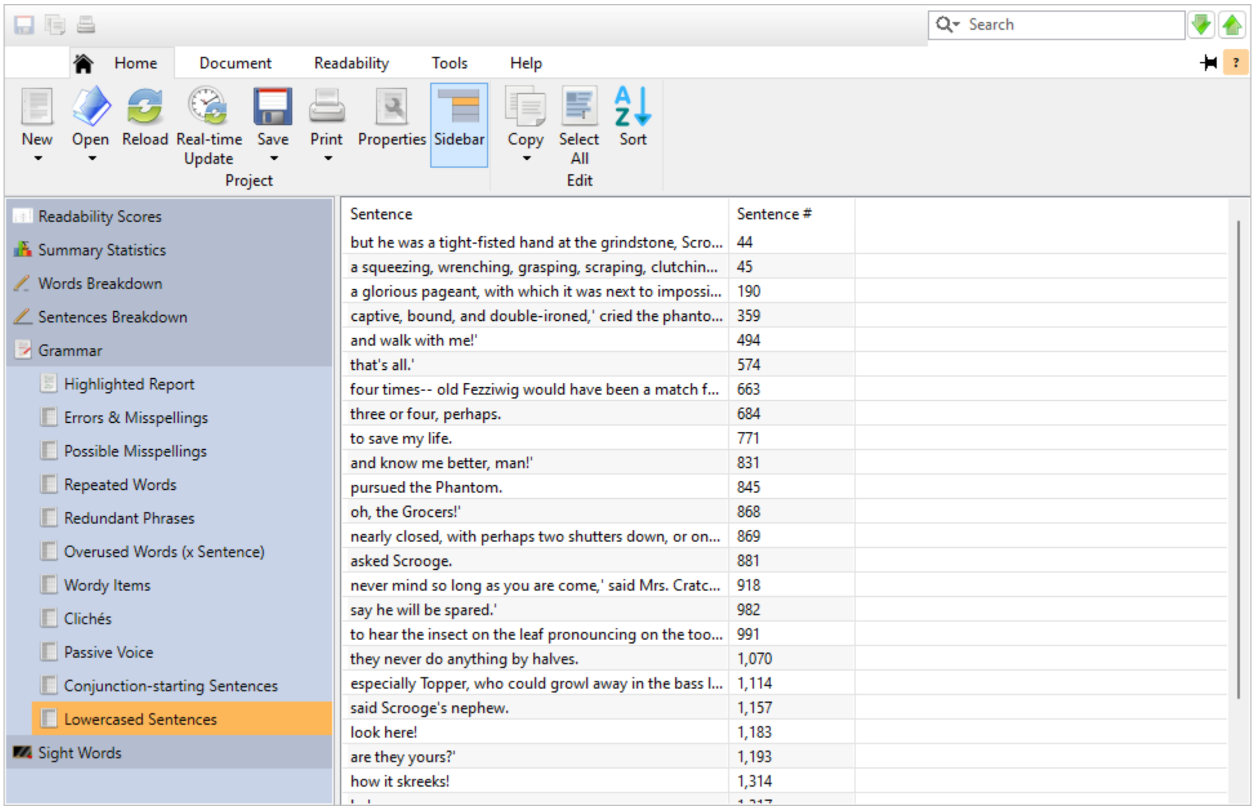
Note that each sentence is followed by its respective word count (in parentheses).

You can change both the font and highlighting color by opening the Project Properties dialog and selecting Highlighted Reports. Note that changing these options from the Project Properties dialog will only affect the current project. If you want to make configuration changes for future projects, then go to the Options dialog from the Tools tab instead.

For documents with numerous sentences over twenty words, it is recommended to use outlier searching. Open the Project Properties dialog, select Document Indexing, and select the option Outside sentence-length outlier range. This will make it so that only the extremely long sentences (compared to the other sentences in the document) are highlighted. This helps pinpoint the most troublesome sentences that need to be shortened.

Reviewing Lowercased Sentences

After opening a project, click on the Grammar icon on the project sidebar and select the Lowercased Sentences subitem. This window will display a list of all sentences that begin with lowercased words (which is usually a typo).



Lowercased sentence searching is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

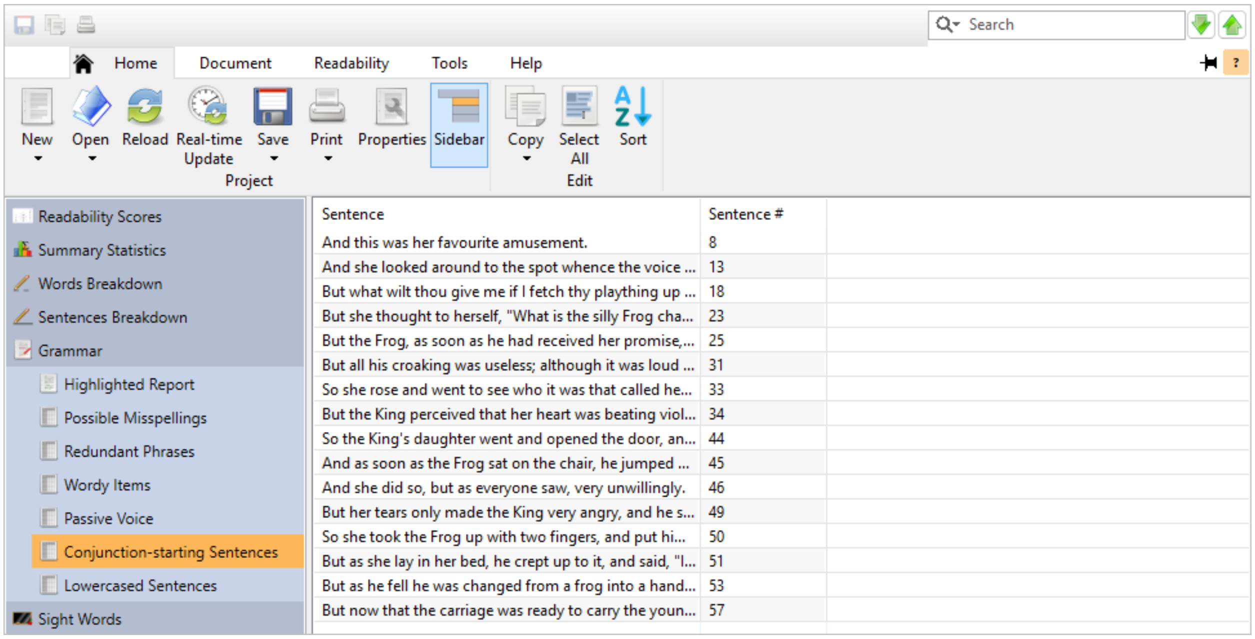
Also note that if the option Sentences must begin with capitalized words is enabled, then this feature will be limited to sentences that begin new paragraphs. If you want to search for any possible sentences that begin with lowercased words, then you should leave this option unchecked.

To view any sentence in its original context, double-click on it. This will take you to the Highlighted Report page with that sentence selected.

Note that if there are no lowercased sentences in your document, then this window will not be included.

Reviewing Conjunction-starting Sentences

After opening a project, click on the Grammar icon on the project sidebar and select the Conjunction-starting Sentences subitem. This window will display a list of all sentences which begin with coordinating conjunctions.



Beginning sentences with conjunctions is not an issue for most writing, but should be avoided with professional publications. This feature is provided to grammatically improve professional documents.

The conjunctions that *Readability Studio* searches for are the most common coordinating conjunctions. These are:

- and
- nor
- but
- or
- yet
- so

Generally for technical writing, only separate clauses should begin with these words, not sentences. Any sentence that begins with these words should be combined with the previous sentence. Another option would be to replace the conjunction. For example, *and* could be replaced with *In addition*.

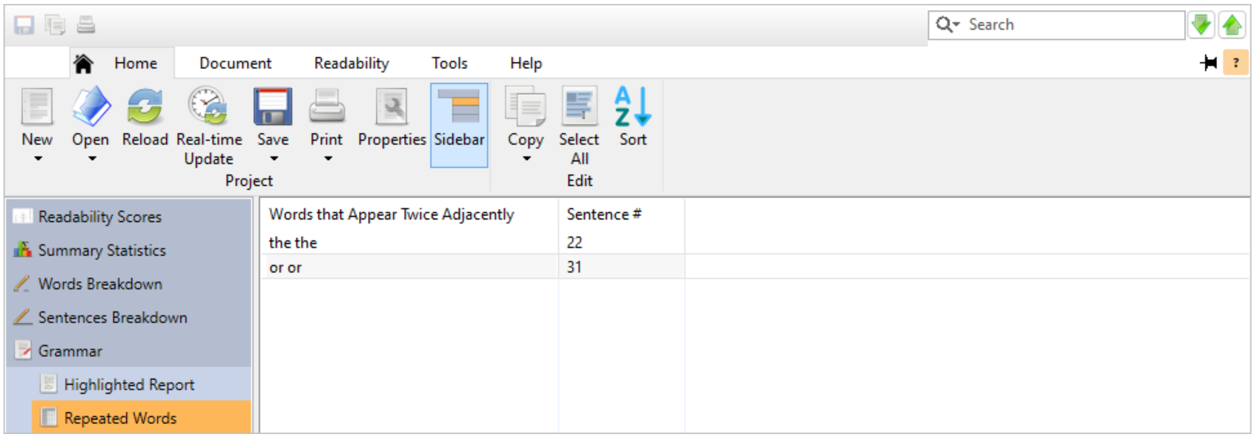
This search is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

To view any sentence in its original context, double-click on it. This will take you to the Highlighted Report page with that sentence selected.

Note that if there are no conjunction-starting sentences in your document, then this window will not be included.

Reviewing Repeated Words

After opening a project, click on the Grammar icon on the project sidebar and select the Repeated Words subitem. This window will display a list of all repeated words from the document.



A repeated word is when the same word appears after itself (which is usually a typo). For example:

> Gabi gave her report *to to* the manager.

In this case, the word *to* was accidentally typed twice. *Readability Studio* will consider this a repeated word and include it in this list.

Repeated words are bound to the sentence level. If the same word ends one sentence and begins the next, then it will not be considered a repeated word. For example, *Blake* would not be considered a repeated word here:

> Gabi gave her sales report to *Blake. Blake* really appreciated Gabi's promptness.

Repeated word searching is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

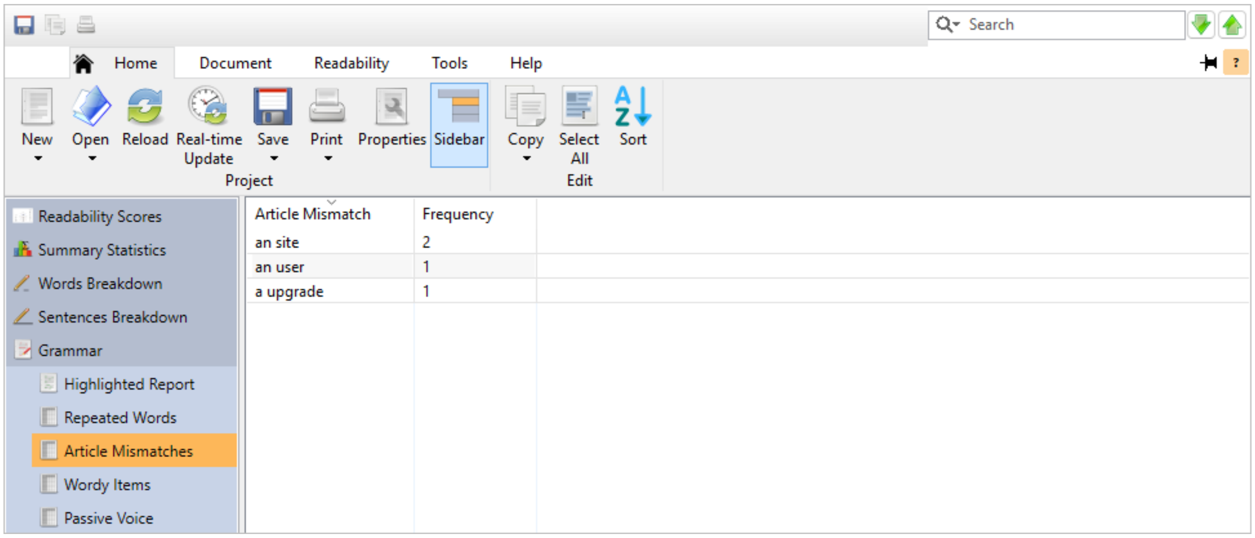
To view any repeated word in its original context, double-click on it. This will take you to the Highlighted Report page with that repeated word selected.

Note that if there are no repeated words in your document, then this window will not be included.

Refer to sec. 20.6 for more information on how repeated words are detected.

Reviewing Article Mismatches

After opening a project, click on the Grammar icon on the project sidebar and select the Article Mismatches subitem. This window will display a list of articles (i.e., *a* and *an*) that do not match their following noun.



For example, the following articles would be marked as mismatching:

- a electronic
- an textbook

This search is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

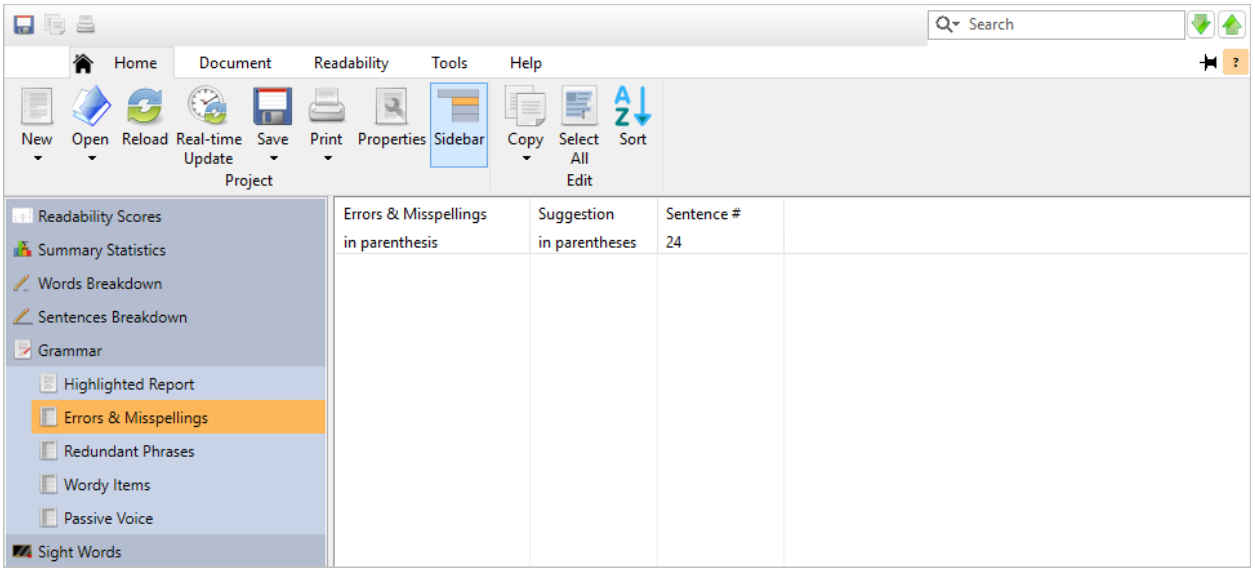
To view any item in its original context, double-click on it. This will take you to the Highlighted Report page with that item selected.

Note that if there are no article mismatches in your document, then this window will not be included.

Refer to sec. 20.7 to learn more.

Reviewing Wording Errors & Misspellings

After opening a project, click on the Grammar icon on the project sidebar and select the Errors & Misspellings subitem. This window will display a list of misused phrases, grammatically incorrect wording, and known misspellings from each document.



For example:

> *We is* offering standard benefits (e.g., 401K matching and health *ensurance*), as well as *French benefits* to all new employees.

In this case, *We is* is not grammatically correct, and *We are* will be offered as a suggested replacement. Also, *ensurance* is a misspelling of *insurance* (which will be the suggested replacement). Finally, *French benefits* is not the correct expression and *fringe benefits* will be offered as a suggested replacement. All of these will be considered wording errors and shown in this list.

Wording errors are bound to the sentence level. If part of a phrase ends one sentence and begins the next, then it will not be considered an error.

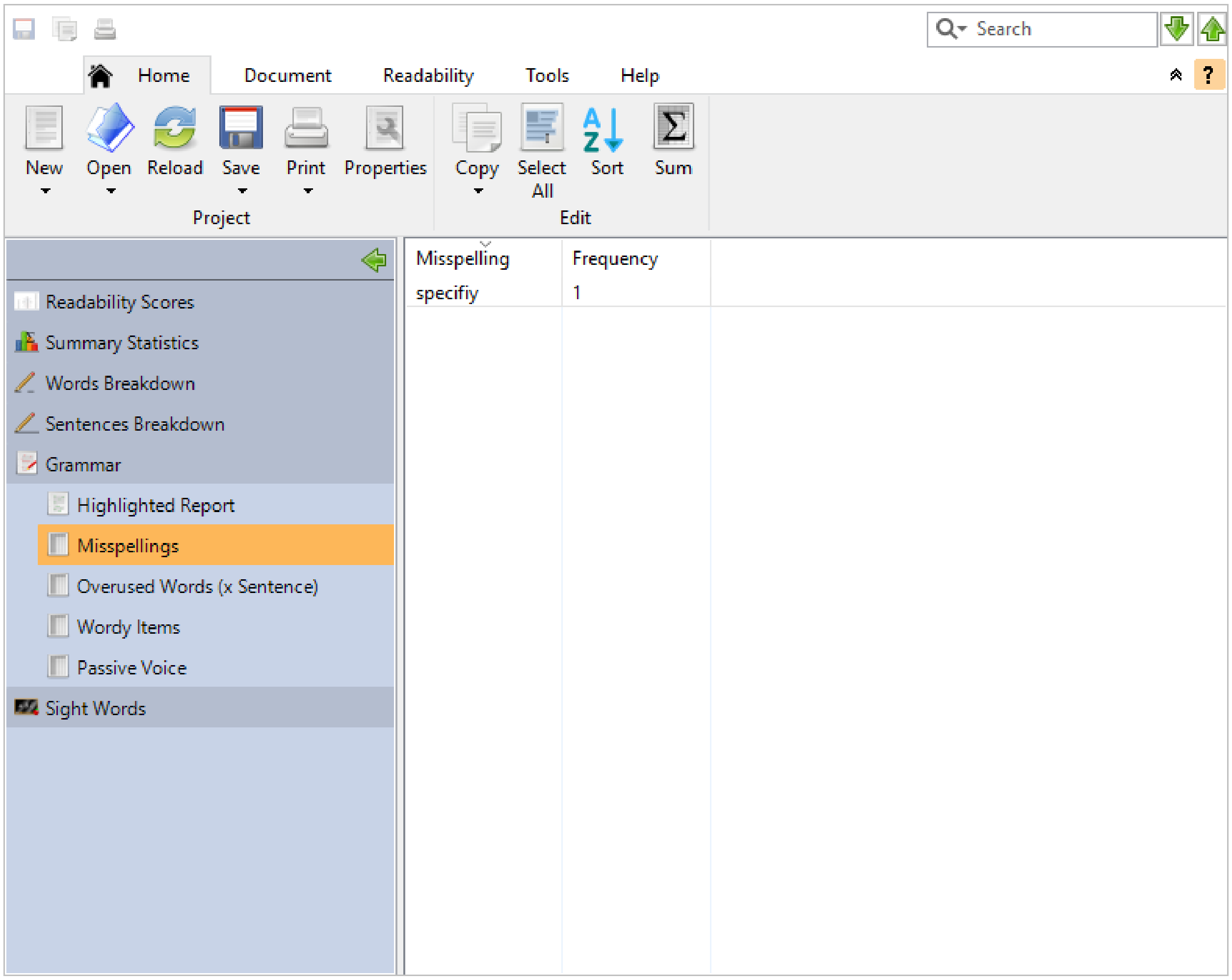
This search is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

To view any wording error in its original context, double-click on it. This will take you to the Highlighted Report page with that phrase selected.

Note that if there are no errors in your document, then this window will not be included.

Reviewing Possible Misspellings

After opening a project, click on the Grammar icon on the project sidebar and select the Possible Misspellings subitem. This window will display a list of all possible misspellings from the document.



Options for controlling how misspellings are searched for are available in the Grammar options.

Misspelling searching is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

To view any misspelling in its original context, double-click on it. This will take you to the Highlighted Report page with that word selected.

To add words to your dictionary, first select them in this list. Then, click on the Document tab, and click the arrow next to the Spell Checker button. Then, select Add Selected Words and the selected words in this list will no longer be reported as misspellings.

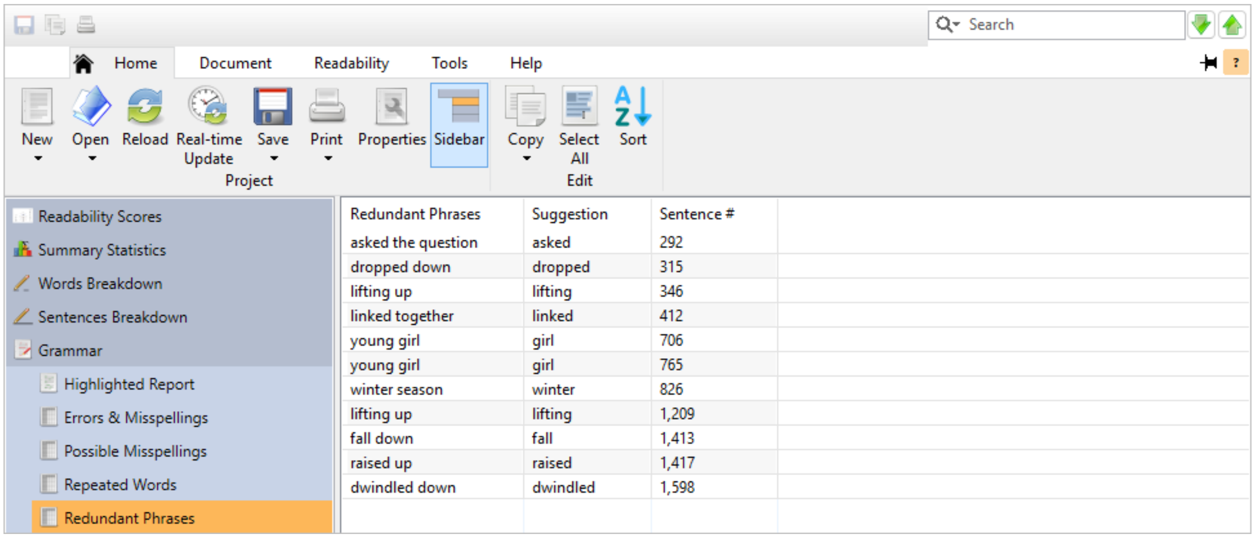
To edit your custom dictionary, click on the arrow next to the Spell Checker button and select Edit... to show the Edit Custom Dictionary dialog. From here, you can add, remove, or edit the words in your custom dictionary.

To change your spell-checking options, click on the arrow next to the Spell Checker button and select Settings... to edit your grammar settings.

Note that if there are no misspellings in your document, then this window will not be included.

Reviewing Redundant Phrases

After opening a project, click on the Grammar icon on the project sidebar and select the Redundant Phrases subitem. This window will display a list of redundant phrases (and suggested replacements) from the document.



A redundant phrase is when a modifier expresses something that is already implied by the noun it is modifying. For example:

> Gabi and Blake *collaborated together* on the report. They should have a *brief summary* ready later today.

In this case, *collaborated together* and *brief summary* are redundant. Collaboration already implies two or more people working together. The *together* in this phrase adds nothing and just makes this sentence wordy. As for *brief summary*, summaries are always brief; hence, the redundancy in that phrase. *Readability Studio* will consider these as redundant phrases and include them in this list.

Redundant phrases are bound to the sentence level. If part of a phrase ends one sentence and begins the next, then it will not be considered a redundant phrase. For example, *brief. Summaries* would not be considered a redundant phrase here because the words are in different sentences:

> Gabi and Blake need to be brief. Summaries are very important before presenting a report.

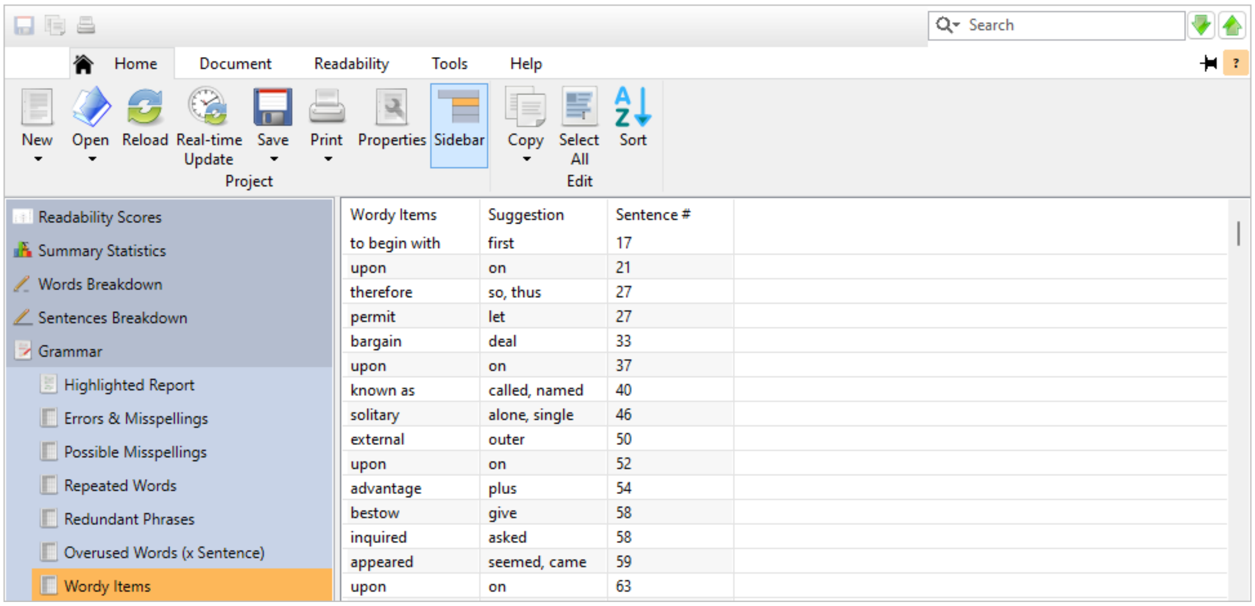
Redundant phrase searching is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

To view any redundant phrase in its original context, double-click on it. This will take you to the Highlighted Report page with that phrase selected.

Note that if there are no redundant phrases in your document, then this window will not be included.

Reviewing Wordy Items

After opening a project, click on the Grammar icon on the project sidebar and select the Wordy Items subitem. This window will display a list of wordy phrases and difficult words (and their suggested replacements) from the document.



A phrase is wordy if it contains too many words or contains complex words that could be replaced by simpler ones. For example:

> *Over the course of* the presentation, Blake amazed the Provost with his data visualizations.

In this case, *Over the course of* is wordy. *Readability Studio* will consider this a wordy phrase and offer *during* or *throughout* as replacements.

Wordy phrases are bound to the sentence level. If part of a phrase ends one sentence and begins the next, then it will not be considered a wordy phrase. For example, *a host of* would not be considered a wordy phrase here because its words are in two different sentences:

> For the party, we need a host. Of course he should be very funny.

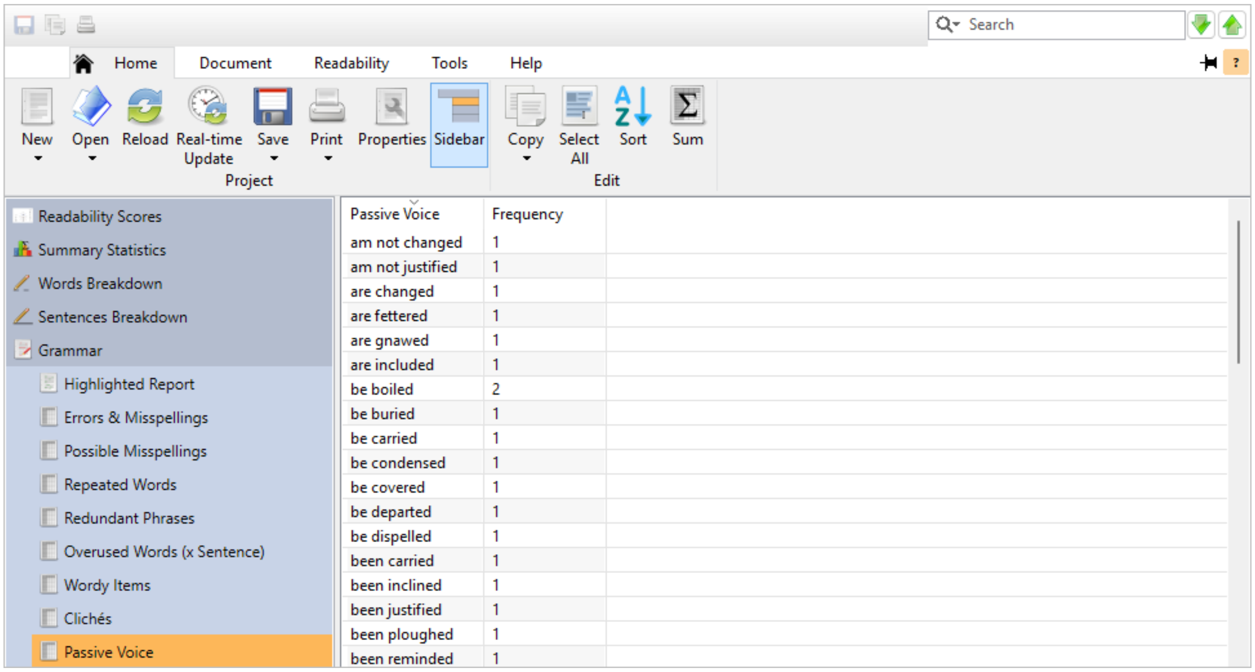
Wordy-item searching does not affect readability calculations, so this analysis is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

To view any wordy item in its original context, double-click on it. This will take you to the Highlighted Report page with that item selected.

Note that if there are no wordy items in your document, then this window will not be included.

Reviewing Passive Voice

After opening a project, click on the Grammar icon on the project sidebar and select the Passive Voice subitem. This window will display a list of passive phrases.



For example, the following would be marked as passive:

> The show had been given many awards by the academy.

Although not a grammatical error, passive voice can cause confusion as to who is doing what to whom. In the above example, the ordering of words makes it difficult for early readers to understand who was giving the awards. The following would be direct and clear:

> The academy gave the show many awards.

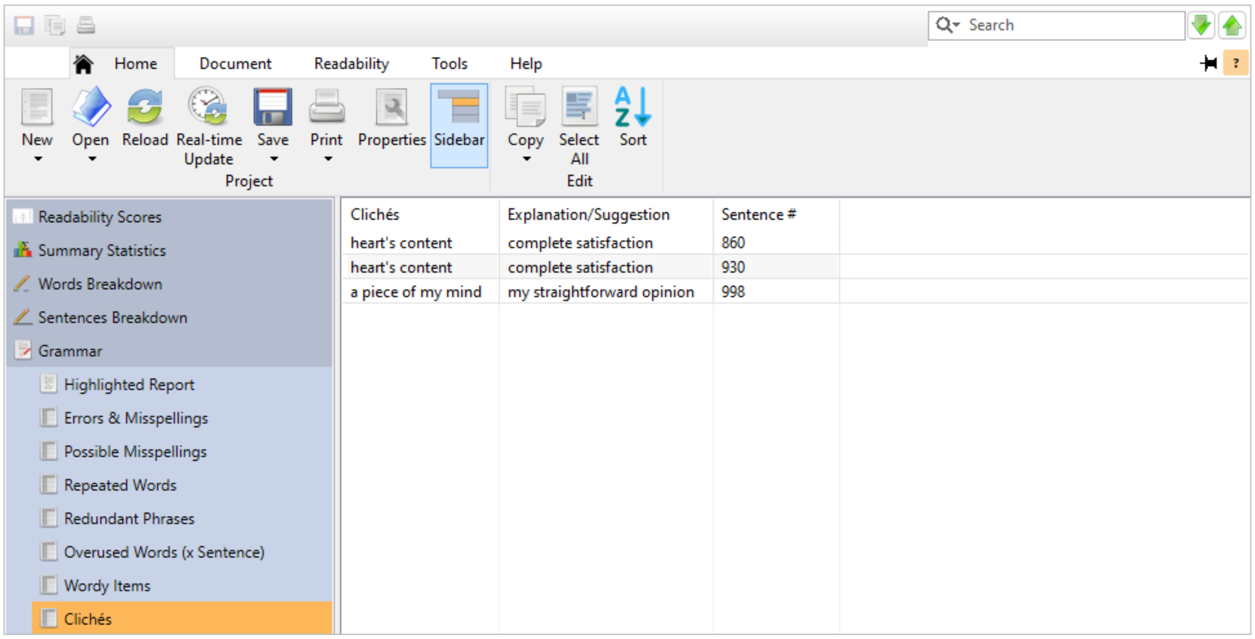
This search is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

To view any item in its original context, double-click on it. This will take you to the Highlighted Report page with that item selected.

Note that if there are no passive phrases in your document, then this window will not be included.

Reviewing Clichés

After opening a project, click on the Grammar icon on the project sidebar and select the Clichés subitem. This window will display a list of clichés (and their suggested replacements) from the document.



A cliché is an overused phrase that should be avoided if you are writing professional documents. In the context of readability, clichés can be unfamiliar to early (young or ESL) readers. For example:

> He is acting very *off the wall*.

In this case, *off the wall* is a cliché and will be listed with *unusual* being offered as a replacement. A phrase such as this is considered *tired* and would not appear professional in a business report or technical document. Also, those new to English may not understand what this expression means. If you are targeting early readers, then you should avoid clichés for clarity.

Clichés are bound to the sentence level. If part of a phrase ends one sentence and begins the next, then it will not be considered a cliché.

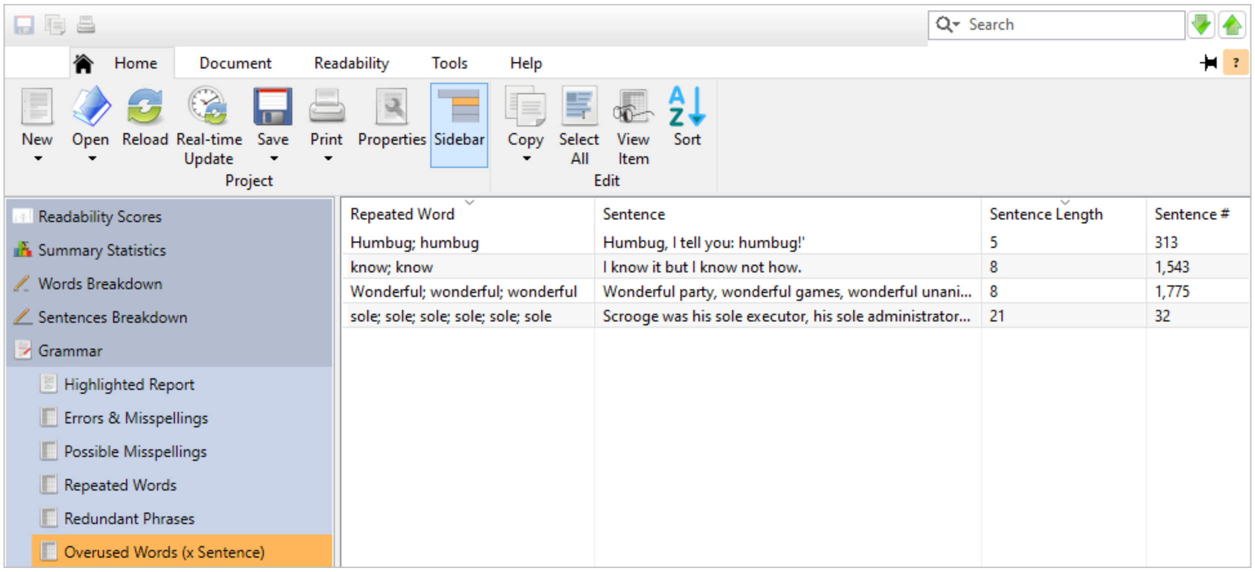
Cliché-searching is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

To view any cliché in its original context, double-click on it. This will take you to the Highlighted Report page with that cliché selected.

Note that if there are no clichés in your document, then this window will not be included.

Reviewing Overused Words (x Sentence)

After opening a project, click on the Grammar icon on the project sidebar and select the Overused Words (x Sentence) subitem. This window will display a list of all sentences that use the same word multiple times. (These are more of style suggestions, rather than grammar issues.) When reviewing these, it is recommended to change some of the repeated words to add more variety to your content.



This check is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

Numeric words, proper nouns, words containing less than three letters, and common words (e.g., *the*) are excluded from this review.

Also, a sentence with a pair of repeated words that may be part of the same phrase are ignored. For example:

> When they asked Blake if he would be willing to work over the weekend, he just *laughed* and *laughed*.

Finally, a word must appear on average once every fifth word in the sentence. For example, a word appearing twice in an eleven-word sentence will not be counted. If the sentence is 10 words, however, it will be counted.

Note that if there are no sentences with overused words in your document, then this window will not be included.

To view any sentence in its original context, double-click on it. This will take you to the Highlighted Report page with that sentence selected.

4.8 Dolch Sight Words

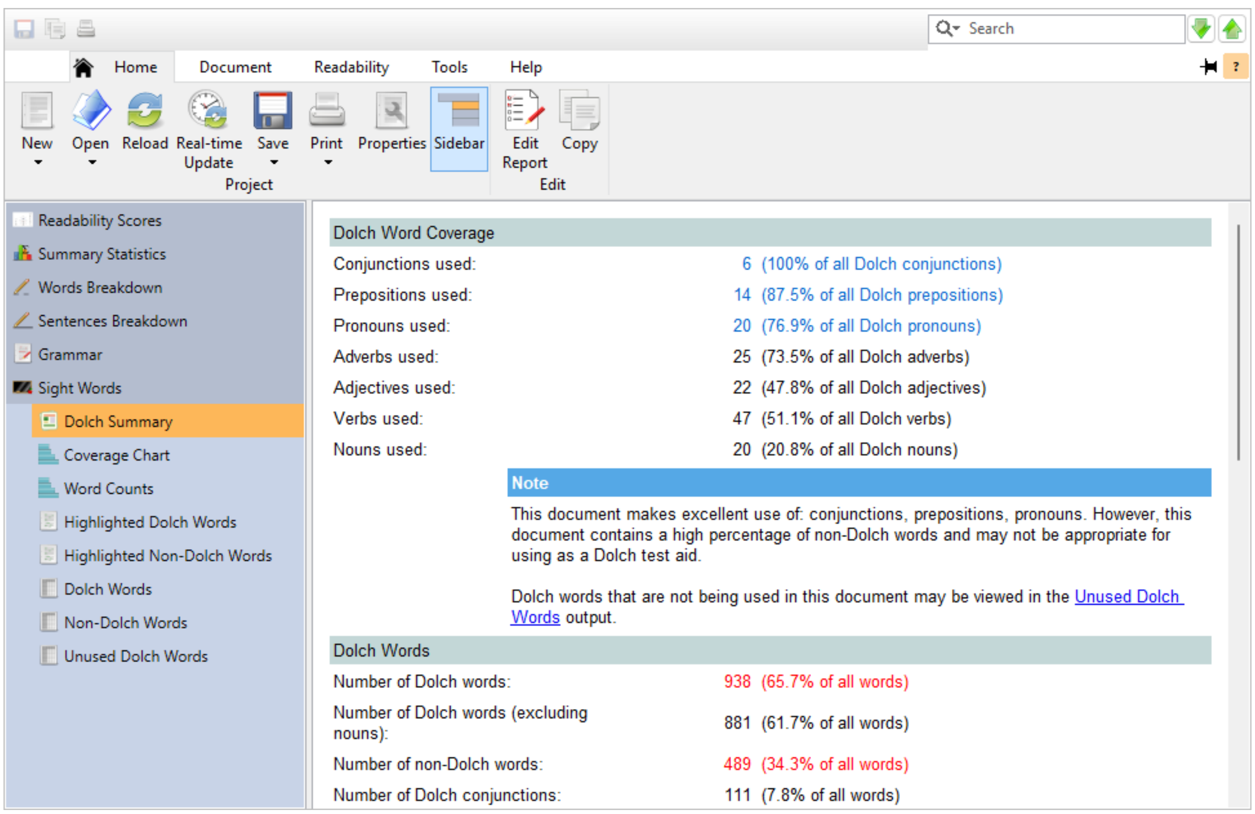
The Dolch Sight Words (456–60) represent the most frequently occurring service words[1] in most text, especially children's literature. Early readers need to learn and recognize these words to attain reading fluency. Many of these words cannot be sounded out or represented by pictures; therefore, they must be learned by sight[2].

If you are writing educational materials for early readers, then use as many of these words as possible to help readers practice them. Also, keep the percentage of non-Dolch words low so that readers can better focus on the sight words.

Readability Studio offers a suite of Dolch statistics and graphs as a tool for educators and writers targeting early readers. If you are creating material intended to help early readers practice Dolch words, then it is recommended to include this in your project. To add Dolch statistics to your project, go to the Readability tab and select either Primary-age Reading or Second Language Reading and then Dolch Sight Words. A Dolch Sight Words icon will appear at the bottom of the project sidebar—clicking on that will display all the Dolch information.

Reviewing the Dolch Statistics

After opening a project, click on the Dolch Sight Words icon on the project sidebar and select the Dolch Summary subitem to view the document's Dolch statistics:



There are two types of statistics included: Dolch coverage and raw Dolch word counts.

[1] Pronouns, adjectives, adverbs, prepositions, conjunctions, and verbs.

[2] A separate list of nouns commonly found in children's literature is also included with the Dolch collection. However, the sight words are generally the focus of most Dolch activities.

The Dolch Word Coverage section displays how much of each category is used in the document. For example, if six of the conjunction Dolch words are used in the document, then 6 will be displayed next to Conjunctions used. The percentage will also be included in parentheses after these counts. In this case, it will be 100% because there are six words on the Dolch conjunction list.

Notes will be shown at the bottom of the Dolch Word Coverage section to indicate which (if any) categories are being used well by your document. This is the best way to see whether or not your document is useful as a Dolch practice tool.

If a document does not provide good coverage for any of the categories and/or contains numerous non-Dolch words, then it will not be an appropriate Dolch training document. This does not necessarily mean that it is not an appropriate document for younger readers. It simply means that the document does not use enough of the Dolch words to help readers learn these words. For example, *The Tale of Peter Rabbit* has decent coverage of the Dolch conjunctions and prepositions, but it also contains a large percentage of non-Dolch words. This book's readability levels indicate that it is proper for young readers. However, its high non-Dolch words percentage will make it inappropriate for readers to use it for practicing Dolch words.

To graphically view the Dolch coverage, click on the Coverage Chart icon. This will display a bar chart of these statistics, listed by category.

Below this is the Dolch Words section, which lists all the raw Dolch word counts, broken down by category. Each category includes its total and unique counts of words found in the document. For example, let us say that four unique words from the pronoun list are found in the document, each appearing twice. In this case, Number of Pronoun Dolch words will be eight (four words each appearing twice) and Number of unique Pronoun Dolch words will be four.

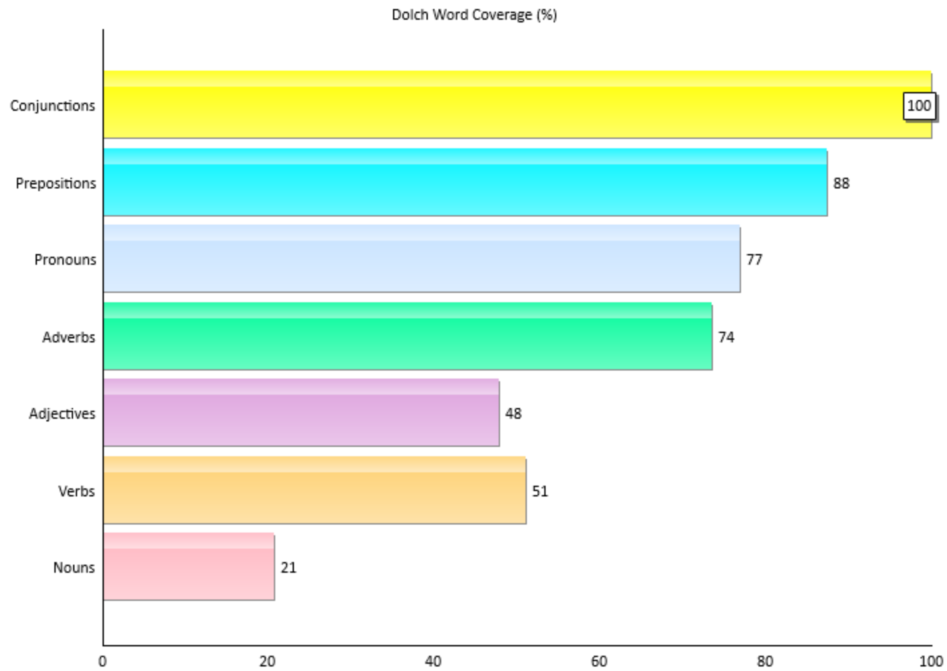
Note that there may be a slight discrepancy between a category's coverage value and its unique word count. This will happen if your document contains variations of certain words, such as *ask* and *asked*. The program will treat morphological variations of a word the same as its root word when calculating coverage statistics. However, when calculating unique word statistics, different forms of words will be counted separately. For example, the words *asked* and *asking* will be counted as separate unique words; however, they both will be seen as "ask" when calculating the verb-list coverage.

To graphically view the Dolch words breakdown, click on the Word Counts subitem. This will display a bar chart of these statistics, listed by category.

Reviewing the Dolch Graphs

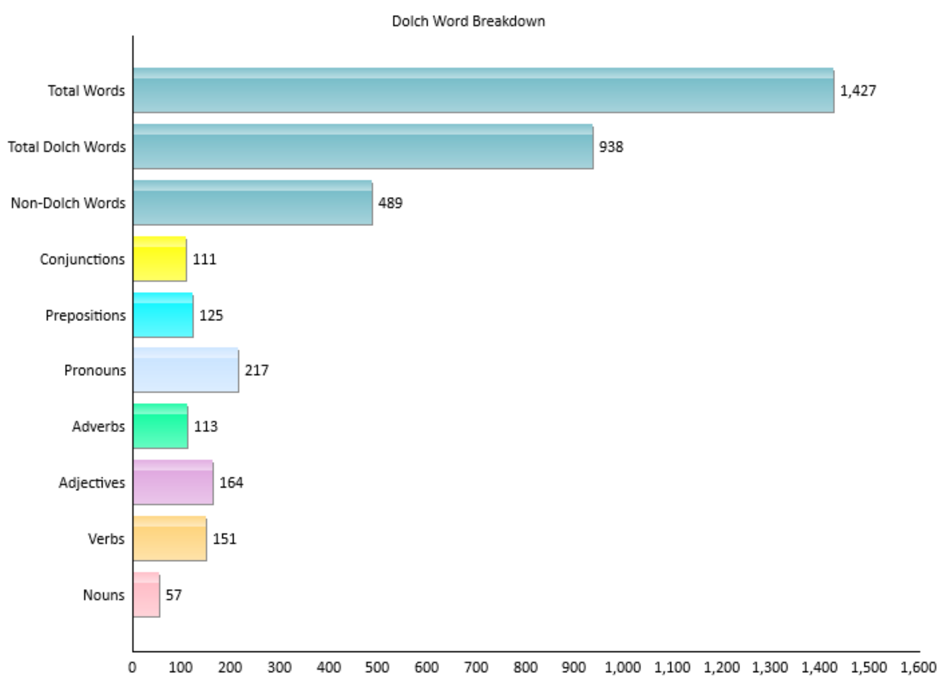
After opening a project, click on the Dolch Sight Words icon on the project sidebar. There are two graphs available for review: Dolch Coverage and Word Counts.

The Dolch Coverage graph displays how much of each category is used in the document. Refer to sec. 4.8 for more information.



As we can see in the above graph, 87.5% of the Dolch prepositions are used in the document. Pronouns and adverbs are also well covered with 76.9% and 73.5%, respectively. However, less than half of the Dolch adjectives and nouns are used in the document. From this, we can see that this document may be useful for some Dolch categories, but not others.

The Words Breakdown graph displays all the raw Dolch word counts, broken down by category. Refer to sec. 4.8 for more information.

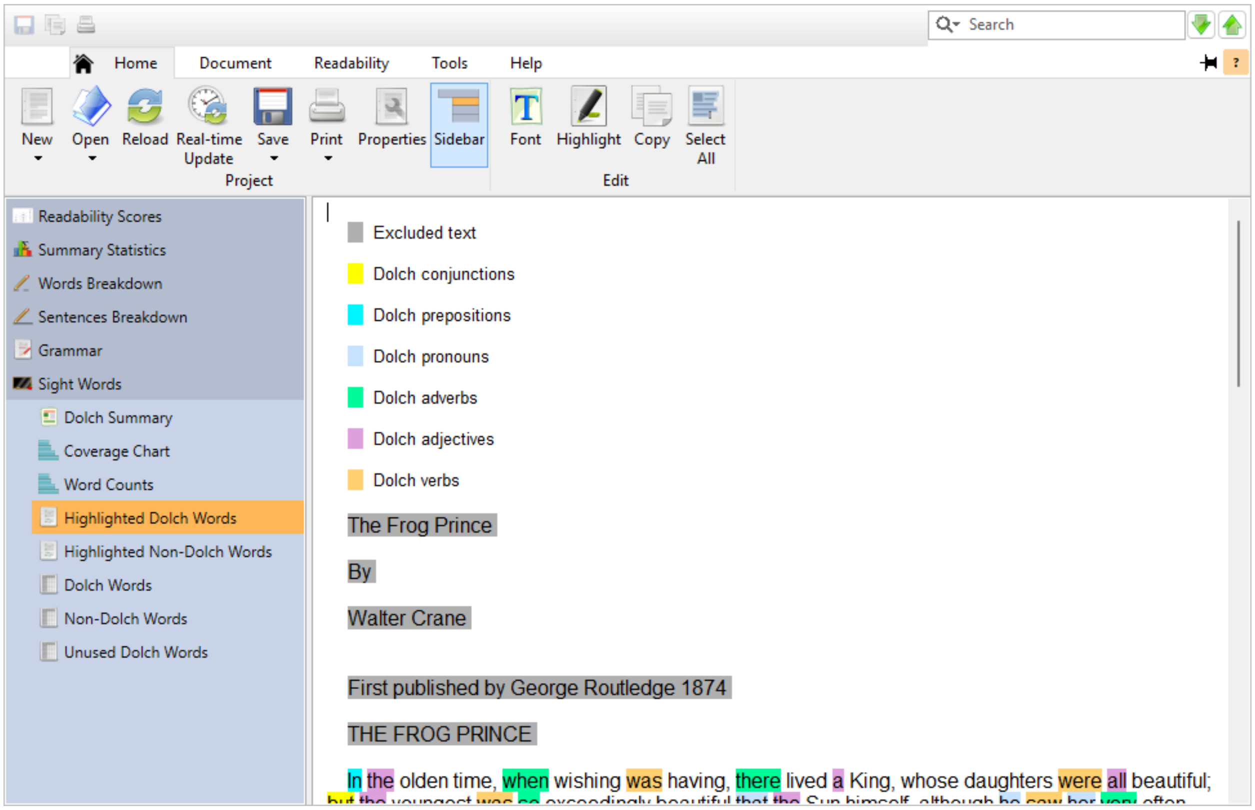


As we can see in the above graph, there are 938 total Dolch words used in this document. Unfortunately, there are also 489 non-Dolch words. Although there is a strong number of Dolch words, the high number of non-Dolch words may make this document a poor candidate for practicing Dolch sight words.

Reviewing the Dolch Highlighted Text

After opening a project, click on the Dolch Sight Words icon on the project sidebar. There will be two types of highlighted text reports available: Dolch Words and Non-Dolch Words.

The Dolch Words subitem will display the document with the Dolch words highlighted:



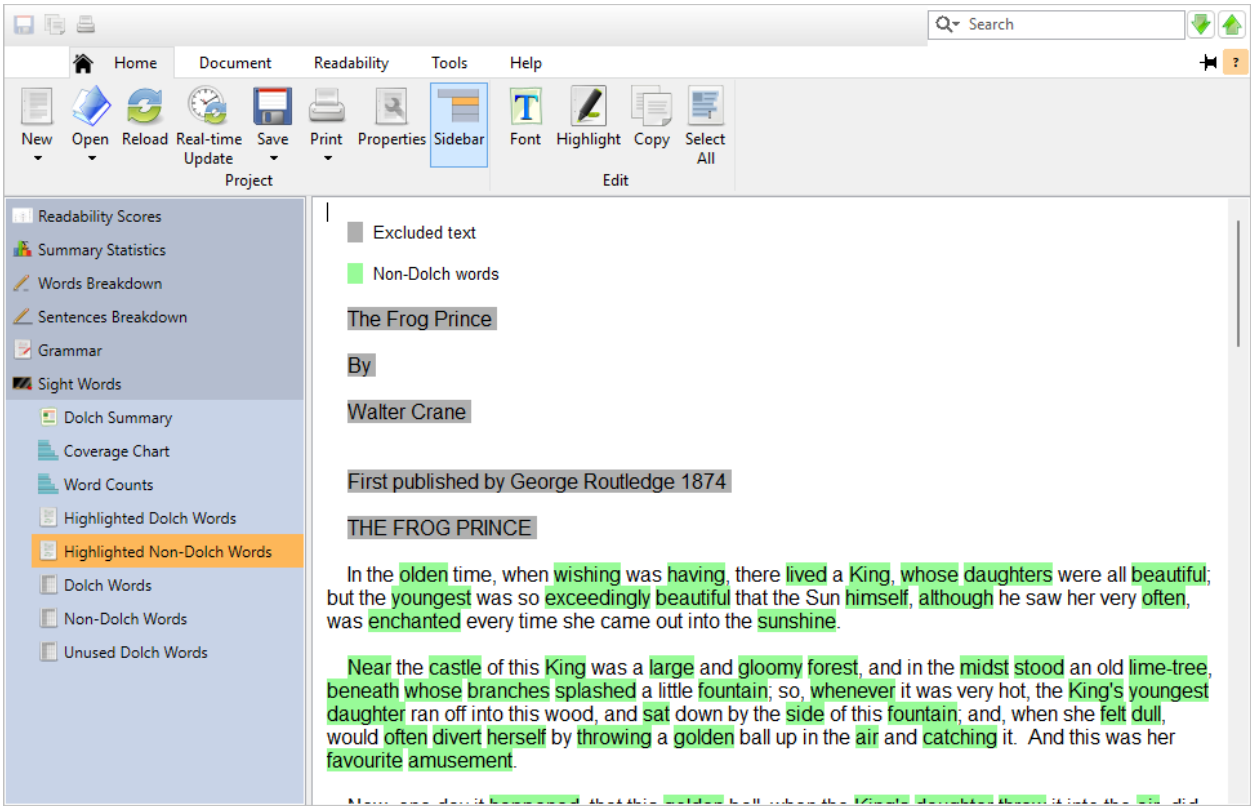
Each category (e.g., nouns) will be highlighted by a different color.

To customize category highlighting, click the Properties button on the Home tab to display the Project Properties dialog. Then select the Highlighted Reports icon and select the Dolch Sight Words subitem.

Home 〉 Project Properties 〉 Highlighted Reports 〉 Dolch Sight Words

Here you can choose which categories to highlight and select their respective highlight colors.

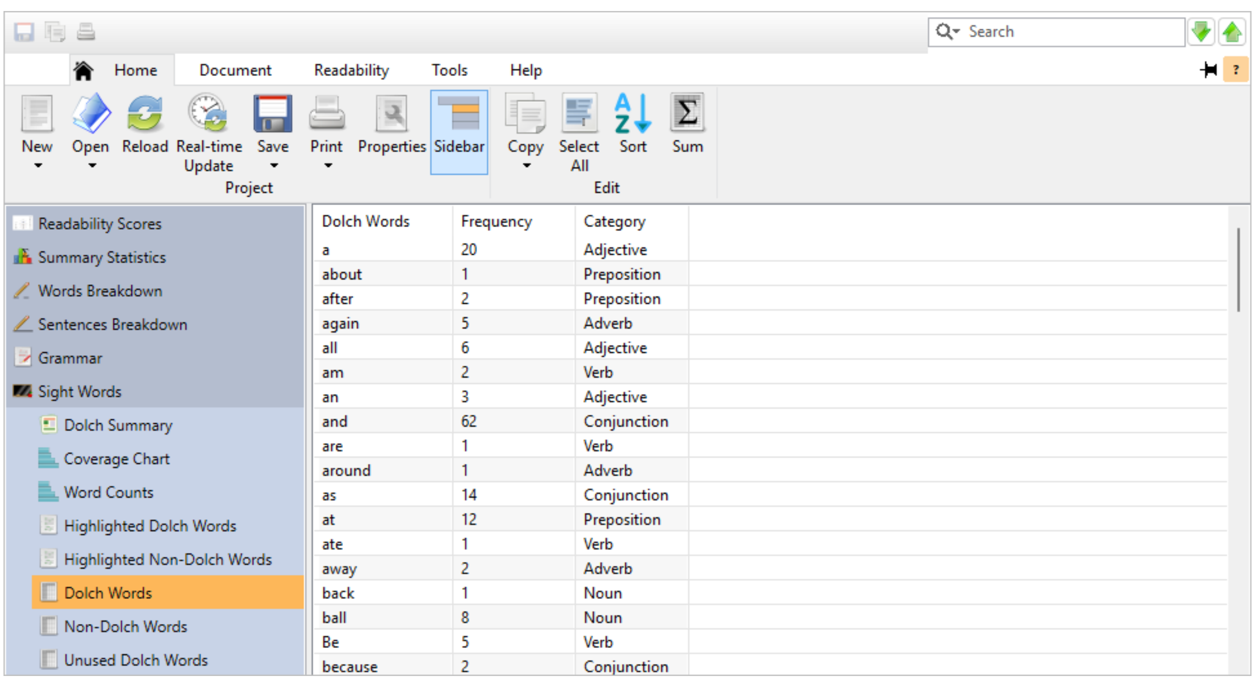
The Non-Dolch Words subitem will display the document with all the non-Dolch words (words that do not appear in any of the Dolch lists) highlighted:



Reviewing the Dolch Word Lists

After opening a project, click on the Dolch Sight Words icon on the project sidebar. There will be three types of word lists available: Dolch Words, Non-Dolch Words, and Unused Dolch Words.

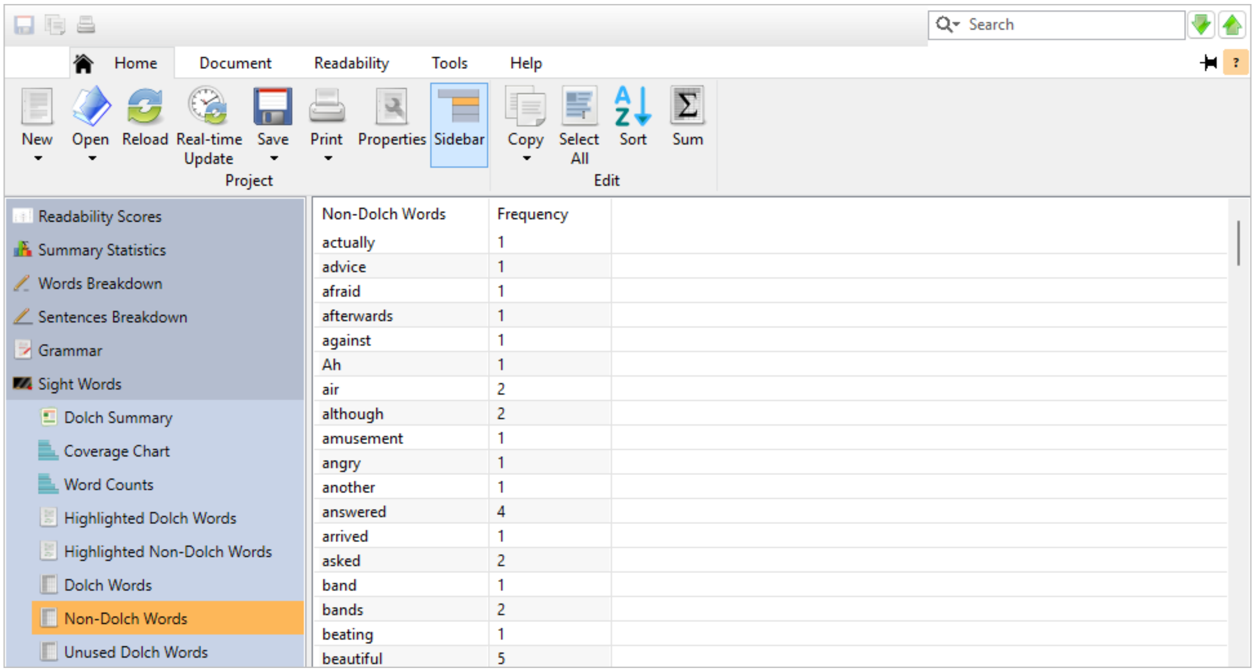
The Dolch Words subitem will display a list of the Dolch words from the document:



Next to each word will be two columns. The first column is Frequency, which will represent the number of times the word appears in the file. The second column is Category, which represents the Dolch category to which the word belongs.

Double-click on any word in this list to display the Dolch Words text report with the first instance of that word selected.

The Non-Dolch Words subitem will display a list of the words from the document that do not appear on any of the Dolch lists:

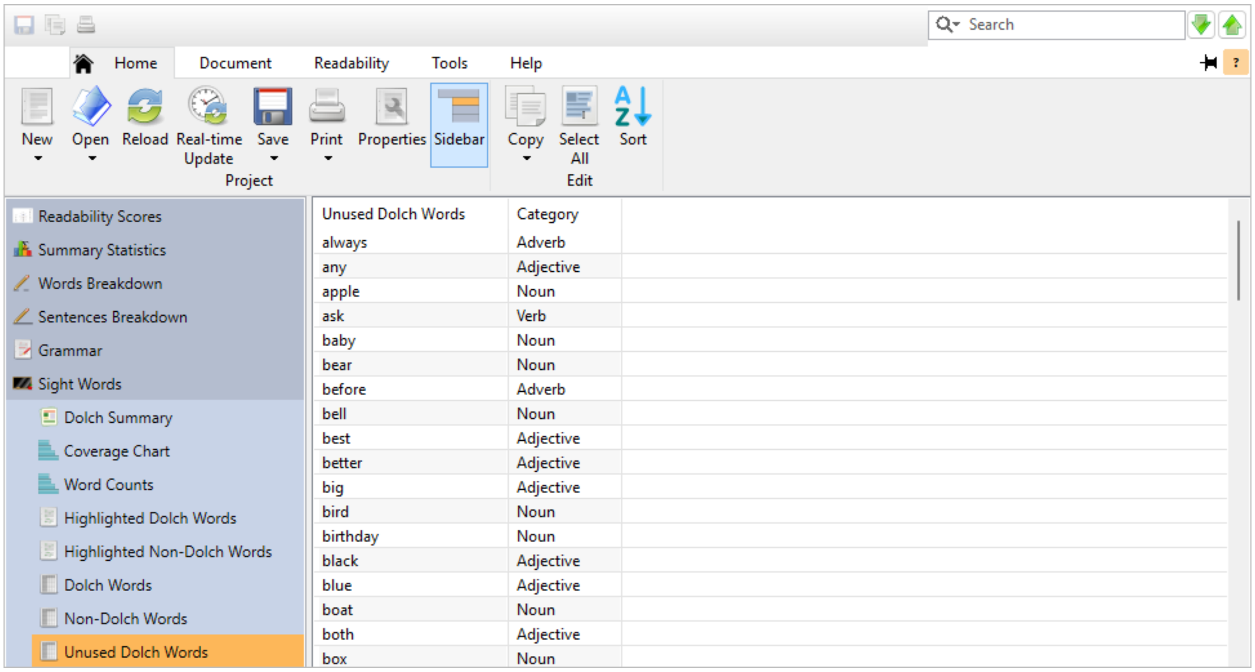


Next to each word will be a Frequency column, which will represent the number of times the word appears in the file.

A high volume of non-Dolch words will adversely affect a document's ability to be a good Dolch practice tool. If the non-Dolch word count is high, consider replacing or removing words from this list that have high frequency counts.

Double-click on any word in this list to display the Non-Dolch Words text report with the first instance of that word selected.

The Unused Dolch Words subitem will display a list of the Dolch words that do not appear in the document:



Next to each word will be a Category column, which represents the Dolch category that the word belongs to. If your document has low coverage of a category that you are targeting, try adding the words listed here to the document.

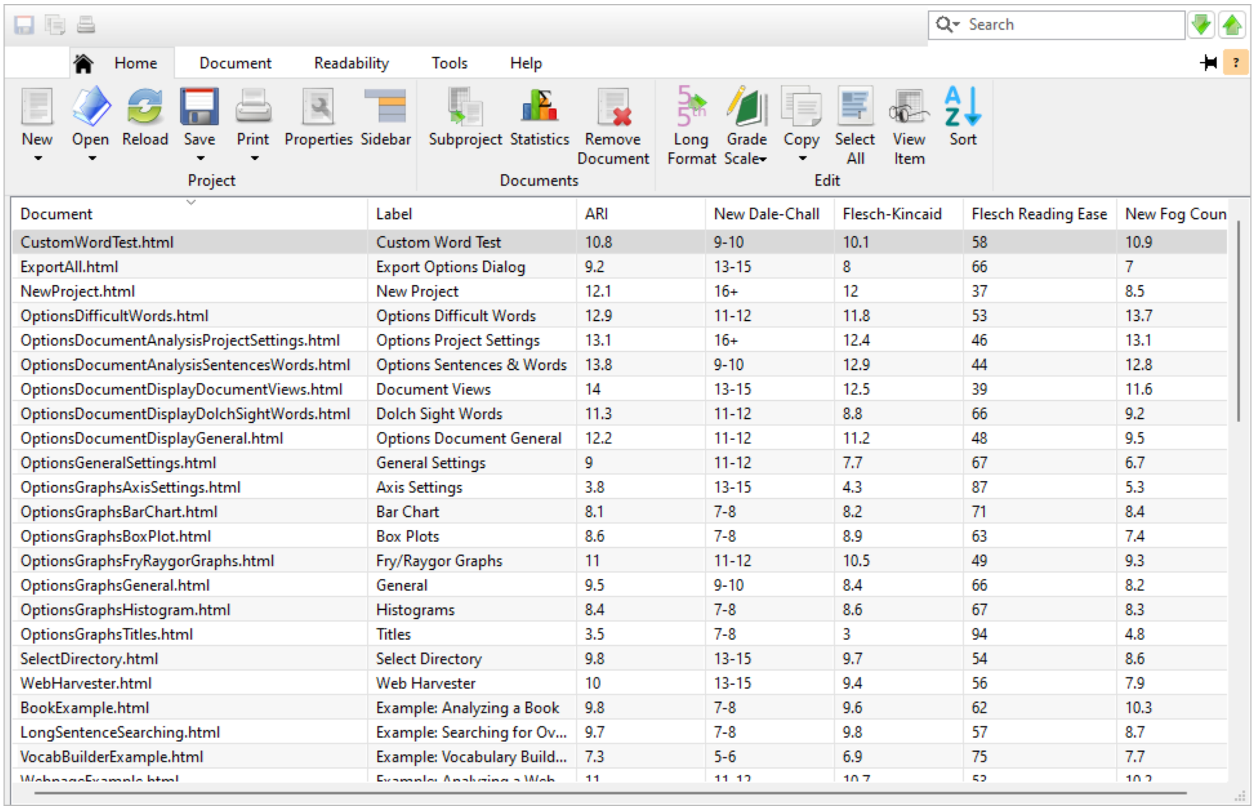
5. Reviewing Batch Projects

> Tip
>
> For any list that displays a document name in the first column, the following features are available:
>
> To create a standard project for any document in the list, select it and click the Subproject button on the Home tab. This is useful for reviewing a particular document in greater detail. Note that when this standard project is created, it will copy all the project settings from the batch project. (These options may differ from your global configurations.)
>
> To remove any document in the list from the project, select it and press Del (⊞, 🐧) or ⌫ () on your keyboard.

5.1 Test Scores

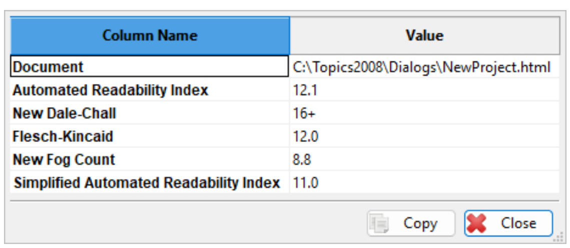
After opening a project, click on the Readability Scores icon on the project sidebar. The following lists may be available: Raw Scores, Goals, Score Summary, Grade Score Summary (x Document), and Cloze Score Summary (x Document).

The Raw Scores subitem will display the scores for each document:



Document	Label	ARI	New Dale-Chall	Flesch-Kincaid	Flesch Reading Ease	New Fog Coun
CustomWordTest.html	Custom Word Test	10.8	9-10	10.1	58	10.9
ExportAll.html	Export Options Dialog	9.2	13-15	8	66	7
NewProject.html	New Project	12.1	16+	12	37	8.5
OptionsDifficultWords.html	Options Difficult Words	12.9	11-12	11.8	53	13.7
OptionsDocumentAnalysisProjectSettings.html	Options Project Settings	13.1	16+	12.4	46	13.1
OptionsDocumentAnalysisSentencesWords.html	Options Sentences & Words	13.8	9-10	12.9	44	12.8
OptionsDocumentDisplayDocumentViews.html	Document Views	14	13-15	12.5	39	11.6
OptionsDocumentDisplayDolchSightWords.html	Dolch Sight Words	11.3	11-12	8.8	66	9.2
OptionsDocumentDisplayGeneral.html	Options Document General	12.2	11-12	11.2	48	9.5
OptionsGeneralSettings.html	General Settings	9	11-12	7.7	67	6.7
OptionsGraphsAxisSettings.html	Axis Settings	3.8	13-15	4.3	87	5.3
OptionsGraphsBarChart.html	Bar Chart	8.1	7-8	8.2	71	8.4
OptionsGraphsBoxPlot.html	Box Plots	8.6	7-8	8.9	63	7.4
OptionsGraphsFryRaygorGraphs.html	Fry/Raygor Graphs	11	11-12	10.5	49	9.3
OptionsGraphsGeneral.html	General	9.5	9-10	8.4	66	8.2
OptionsGraphsHistogram.html	Histograms	8.4	7-8	8.6	67	8.3
OptionsGraphsTitles.html	Titles	3.5	7-8	3	94	4.8
SelectDirectory.html	Select Directory	9.8	13-15	9.7	54	8.6
WebHarvester.html	Web Harvester	10	13-15	9.4	56	7.9
BookExample.html	Example: Analyzing a Book	9.8	7-8	9.6	62	10.3
LongSentenceSearching.html	Example: Searching for Ov...	9.7	7-8	9.8	57	8.7
VocabBuilderExample.html	Example: Vocabulary Build...	7.3	5-6	6.9	75	7.7

Next to each document name, all of its respective test scores are displayed. In the above example, we can see that the file *NewProject.html* has an ARI score of 12.1, a New Dale-Chall score of 16+, and a Flesch-Kincaid score of 12. To view any document's scores in a vertical layout, double-click on it to display the View Item dialog:



Column Name	Value
Document	C:\Topics2008\Dialogs\NewProject.html
Automated Readability Index	12.1
New Dale-Chall	16+
Flesch-Kincaid	12.0
New Fog Count	8.8
Simplified Automated Readability Index	11.0

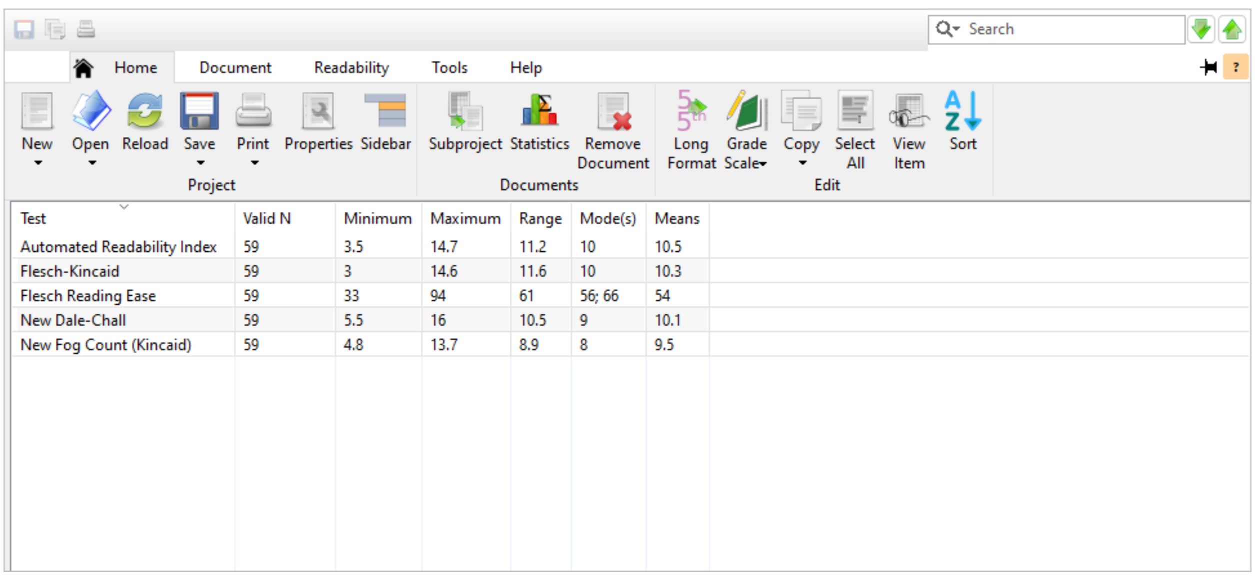
The Goals subitem will display any goals that are currently failing in the project:

Document	Label	Easy Listening Formula Max	Flesch Reading Ease Min	Fry Max
NewProject.html	Topics	✔ 12	⚠ 60	⚠ 10
CustomWordTest.html	Topics	✔ 12	⚠ 60	✔ 10
SentenceDeduction.html	Topics	✔ 12	⚠ 60	✔ 10
ReviewingStatistics.html	Topics	⚠ 12	⚠ 60	⚠ 10
ColemanLiau.html	Topics	✔ 12	⚠ 60	⚠ 10
DaleChall.html	Topics	⚠ 12	⚠ 60	⚠ 10
FORCAST.html	Topics	✔ 12	⚠ 60	⚠ 10
Flesch.html	Topics	✔ 12	⚠ 60	⚠ 10
FleschKincaid.html	Topics	✔ 12	⚠ 60	⚠ 10
Fry.html	Topics	✔ 12	⚠ 60	⚠ 10
GunningFog.html	Topics	✔ 12	⚠ 60	⚠ 10
NewFogCount.html	Topics	✔ 12	⚠ 60	⚠ 10
PSK.html	Topics	✔ 12	⚠ 60	⚠ 10
Raygor.html	Topics	✔ 12	⚠ 60	⚠ 10
SARI.html	Topics	✔ 12	⚠ 60	⚠ 10
SMOG.html	Topics	✔ 12	⚠ 60	⚠ 10
Spache.html	Topics	✔ 12	⚠ 60	⚠ 10

Above, we can see that most of the documents are passing the goal of having an ELF score below 12. However, most are failing to have an FRE score above 60.

> **Note**
> For simplicity, only goals that are failing are shown here. If all goals are passing, then this window will display a message indicating that.

The Score Summary subitem will display the aggregated statistics for each test:



Test	Valid N	Minimum	Maximum	Range	Mode(s)	Means
Automated Readability Index	59	3.5	14.7	11.2	10	10.5
Flesch-Kincaid	59	3	14.6	11.6	10	10.3
Flesch Reading Ease	59	33	94	61	56; 66	54
New Dale-Chall	59	5.5	16	10.5	9	10.1
New Fog Count (Kincaid)	59	4.8	13.7	8.9	8	9.5

By default, the following statistics for each test are included:

- Valid N: the number of valid scores.
- Minimum: the lowest test score.
- Maximum: the highest test score.
- Range: the distance between the lowest and highest test scores.
- Mode(s): the most frequently occurring score. In the case of a tie, all modes will be listed. Note that grade scores are rounded down when searching for the mode.
- Means: the average test score. This is calculated by adding all the scores and dividing the total by the number of scores.

If Extended Information is enabled, then these additional statistics will be included:

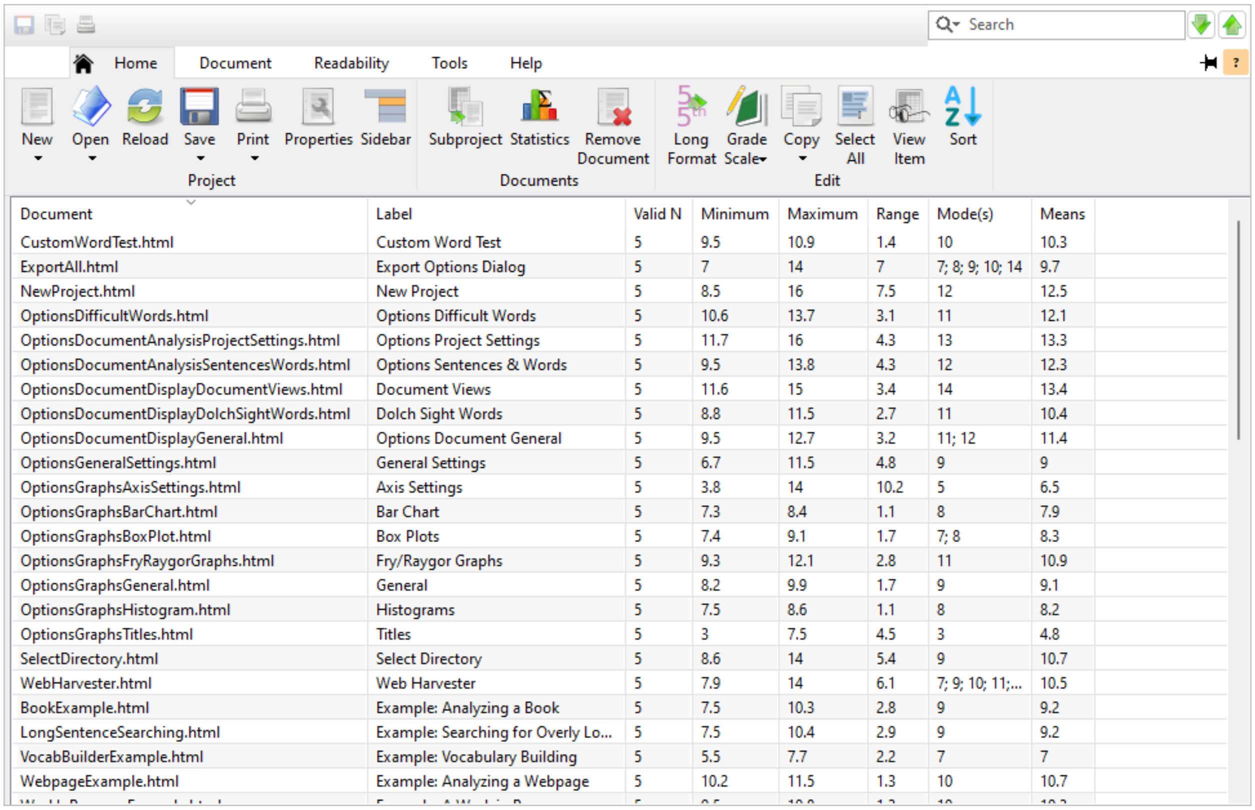
- Median: the middle test score. This represents the middle point separating the lower and upper halves of the scores. This is calculated by sorting the scores and taking the score in middle if the number of scores is odd. If the number of scores is even, then the mean of the two middle points is taken.
- Skewness: the asymmetry of the scores' distribution.
- Kurtosis: the peakedness of the scores' distribution.
- Standard Deviation: the variability (or spread) of the scores. Note that sample std. dev. is used here because it is assumed that the batch contains a sampling of a larger document collection.
- Variance: another measurement of the scores' variability, which is std. dev. squared.
- Lower quartile: the test score at the 25% percentile. This is the score separating the lower quarter of the scores from the rest of the scores.
- Upper quartile: the test score at the 75% percentile. This is the score separating the upper quarter of the scores from the rest of the scores.

Note that when calculating these statistics, failed test scores are treated as missing observations and not included. For example, let us say that you analyze 100 documents, but 3 of them cannot calculate Fry because their average sentence length is too high. In this case, the valid 97 Fry scores will be added and divided by 97 (not 100) to get the means.

Some special scores will be converted when being aggregated into these statistics. For example, grade ranges such as *5–6* (from New Dale-Chall) will be converted to its own average (e.g., *5–6* will be 5.5).

To remove any test in this list from the project, select it and press Ctrl+Del (,) or ⌘+⌫ () on your keyboard.

The Grade Score Summary (x Document) subitem will display aggregated statistics for all grade-level scores, grouped by document:

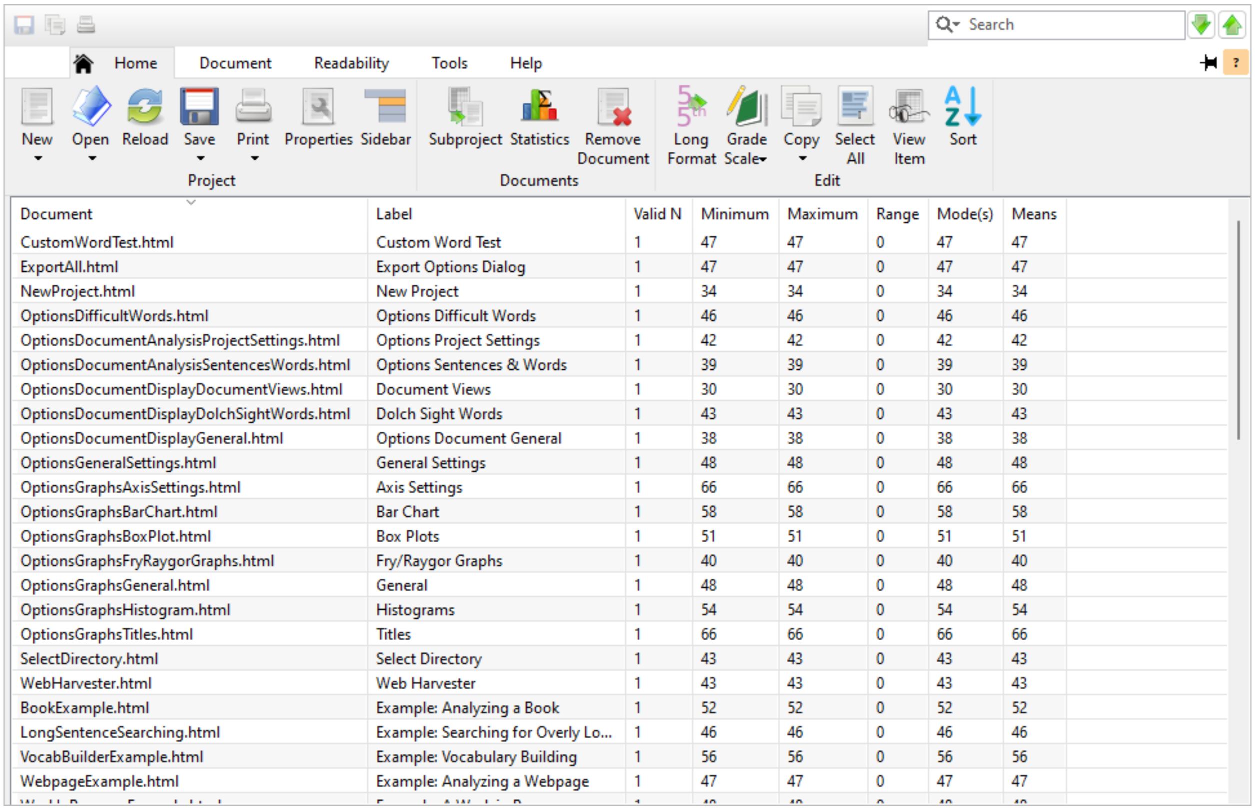


Document	Label	Valid N	Minimum	Maximum	Range	Mode(s)	Means
CustomWordTest.html	Custom Word Test	5	9.5	10.9	1.4	10	10.3
ExportAll.html	Export Options Dialog	5	7	14	7	7; 8; 9; 10; 14	9.7
NewProject.html	New Project	5	8.5	16	7.5	12	12.5
OptionsDifficultWords.html	Options Difficult Words	5	10.6	13.7	3.1	11	12.1
OptionsDocumentAnalysisProjectSettings.html	Options Project Settings	5	11.7	16	4.3	13	13.3
OptionsDocumentAnalysisSentencesWords.html	Options Sentences & Words	5	9.5	13.8	4.3	12	12.3
OptionsDocumentDisplayDocumentViews.html	Document Views	5	11.6	15	3.4	14	13.4
OptionsDocumentDisplayDolchSightWords.html	Dolch Sight Words	5	8.8	11.5	2.7	11	10.4
OptionsDocumentDisplayGeneral.html	Options Document General	5	9.5	12.7	3.2	11; 12	11.4
OptionsGeneralSettings.html	General Settings	5	6.7	11.5	4.8	9	9
OptionsGraphsAxisSettings.html	Axis Settings	5	3.8	14	10.2	5	6.5
OptionsGraphsBarChart.html	Bar Chart	5	7.3	8.4	1.1	8	7.9
OptionsGraphsBoxPlot.html	Box Plots	5	7.4	9.1	1.7	7; 8	8.3
OptionsGraphsFryRaygorGraphs.html	Fry/Raygor Graphs	5	9.3	12.1	2.8	11	10.9
OptionsGraphsGeneral.html	General	5	8.2	9.9	1.7	9	9.1
OptionsGraphsHistogram.html	Histograms	5	7.5	8.6	1.1	8	8.2
OptionsGraphsTitles.html	Titles	5	3	7.5	4.5	3	4.8
SelectDirectory.html	Select Directory	5	8.6	14	5.4	9	10.7
WebHarvester.html	Web Harvester	5	7.9	14	6.1	7; 9; 10; 11;...	10.5
BookExample.html	Example: Analyzing a Book	5	7.5	10.3	2.8	9	9.2
LongSentenceSearching.html	Example: Searching for Overly Lo...	5	7.5	10.4	2.9	9	9.2
VocabBuilderExample.html	Example: Vocabulary Building	5	5.5	7.7	2.2	7	7
WebpageExample.html	Example: Analyzing a Webpage	5	10.2	11.5	1.3	10	10.7

The Valid N column represents the number of grade-level tests that were successfully run against each document. After this column are additional columns displaying grade-level score statistics, document-by-document. In the above example, we can see that the document *CustomWordTest.html* had 5 different grade tests ran against it, with 9.5 being the lowest score and 10.9 being the highest.

Note that this subitem will only be available if at least one grade-level test is included in the project.

The Cloze Score Summary (x Document) subitem will display aggregated statistics for all predicted cloze scores, grouped by document:



Document	Label	Valid N	Minimum	Maximum	Range	Mode(s)	Means
CustomWordTest.html	Custom Word Test	1	47	47	0	47	47
ExportAll.html	Export Options Dialog	1	47	47	0	47	47
NewProject.html	New Project	1	34	34	0	34	34
OptionsDifficultWords.html	Options Difficult Words	1	46	46	0	46	46
OptionsDocumentAnalysisProjectSettings.html	Options Project Settings	1	42	42	0	42	42
OptionsDocumentAnalysisSentencesWords.html	Options Sentences & Words	1	39	39	0	39	39
OptionsDocumentDisplayDocumentViews.html	Document Views	1	30	30	0	30	30
OptionsDocumentDisplayDolchSightWords.html	Dolch Sight Words	1	43	43	0	43	43
OptionsDocumentDisplayGeneral.html	Options Document General	1	38	38	0	38	38
OptionsGeneralSettings.html	General Settings	1	48	48	0	48	48
OptionsGraphsAxisSettings.html	Axis Settings	1	66	66	0	66	66
OptionsGraphsBarChart.html	Bar Chart	1	58	58	0	58	58
OptionsGraphsBoxPlot.html	Box Plots	1	51	51	0	51	51
OptionsGraphsFryRaygorGraphs.html	Fry/Raygor Graphs	1	40	40	0	40	40
OptionsGraphsGeneral.html	General	1	48	48	0	48	48
OptionsGraphsHistogram.html	Histograms	1	54	54	0	54	54
OptionsGraphsTitles.html	Titles	1	66	66	0	66	66
SelectDirectory.html	Select Directory	1	43	43	0	43	43
WebHarvester.html	Web Harvester	1	43	43	0	43	43
BookExample.html	Example: Analyzing a Book	1	52	52	0	52	52
LongSentenceSearching.html	Example: Searching for Overly Lo...	1	46	46	0	46	46
VocabBuilderExample.html	Example: Vocabulary Building	1	56	56	0	56	56
WebpageExample.html	Example: Analyzing a Webpage	1	47	47	0	47	47

This subitem is interpreted in the same manner as the Grade Score Summary (x Document) page.

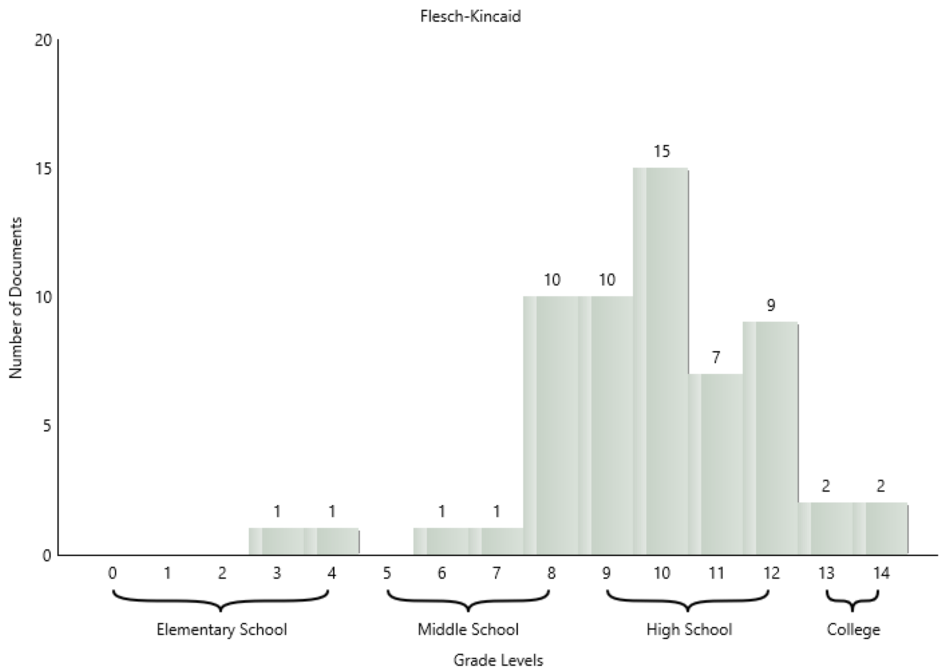
Note that this subitem will only be available if at least one predicted cloze test is included in the project.

Both of these tabs will include the same statistics as the Score Summary page.

5.2 Histograms

After opening a project, click on the Histograms icon on the project sidebar to view your documents' test scores graphically plotted as histograms.

A histogram will be created for each test. For grade level tests, the bins (i.e., bars) will represent either a grade or range of grades. The height of each bar will reflect the number of documents that scored within the bar's range. In the histogram below, we can see that 15 documents scored within the 10th grade range for the Flesch-Kincaid test:

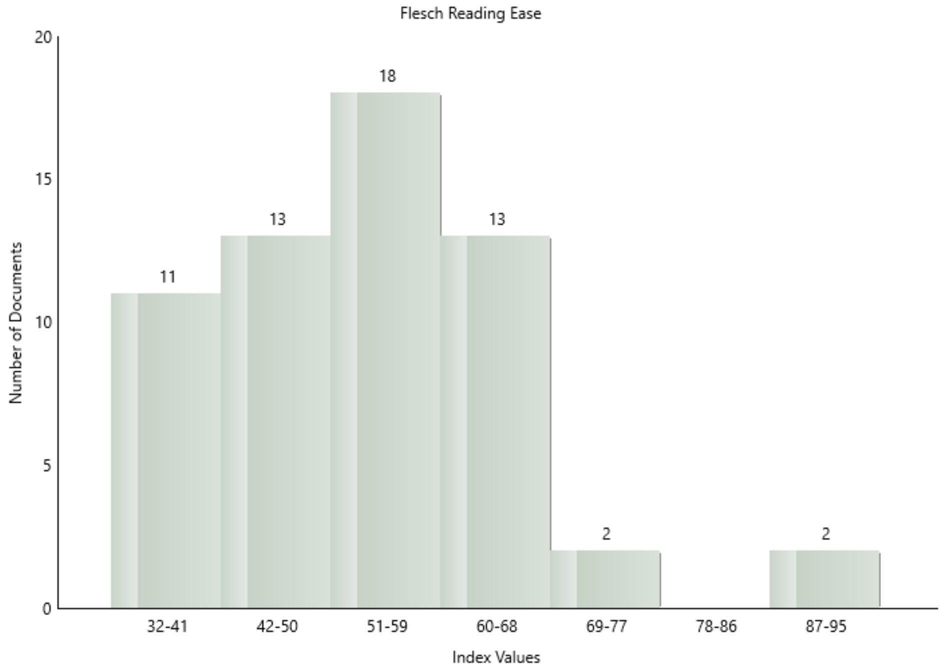


We interpret this first by looking at the highest bar. The label above it displays *15*, which is the number of documents in that category. Now, to see the category, we look at the x-axis labels below the bar. The *10* on the middle of the bar indicates the bar's range (in this case, the 10th grade). Because we are rounding our scores down (the default), scores such as 9.7 will be rounded down and fall into the previous bin (i.e., bar). Scores between 10.0 and 10.9 will fall into this bin; therefore, this bar will represent the 10th grade.

> 💡 Tip
> Selecting a bar will display the first 25 documents within that bin as a tooltip.

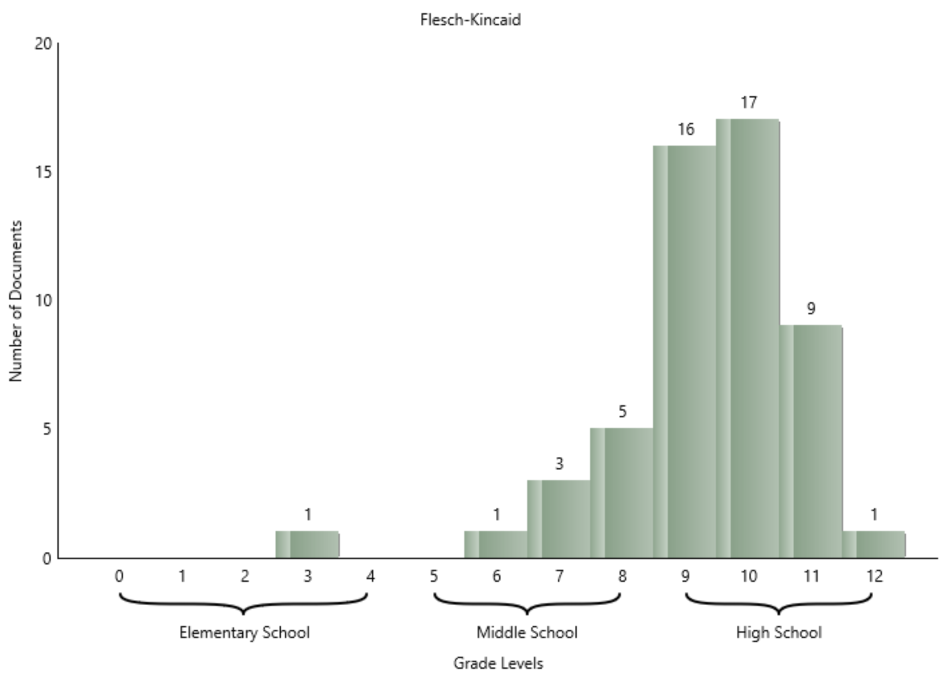
Continuing on, this batch has quite a few difficult documents. There are 15 documents in the 10th grade range, 10 documents in the 9th grade range, etc. In contrast, there are few easy documents, with only 2 in the 6th–7th grade ranges. From this, we can surmise that the readability of this batch needs improving.

Index tests (such as Flesch) will also have histograms, where the bins are ranges in the index. For example, in the histogram below we can see how most of the documents are in the lower range of the Flesch scale:



From this histogram we can also conclude that this batch has too many difficult documents.

If we revise these documents and reload the project (F5), then these graphs will show improvement:



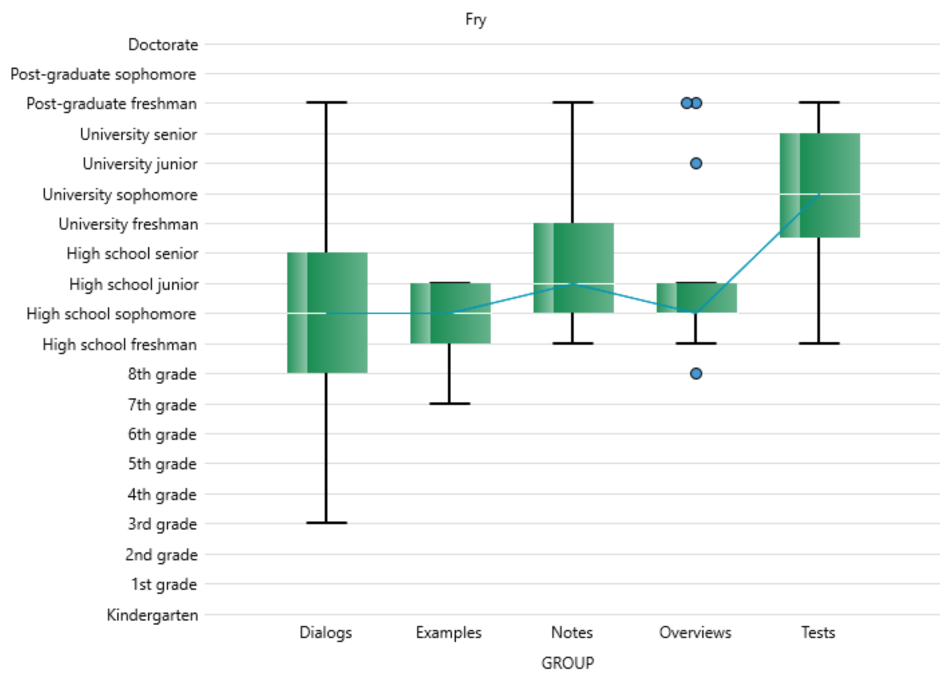
The appearance of these histograms can be customized from the Histogram page of either the Options or Project Properties dialog.

> **Note**
> The binning method for a project is only applied to index and predicted-cloze score histograms. Grade-level histograms will always show one bar for each grade level, and the bar axis will start at 0 (i.e., Kindergarten).

5.3 Box Plots

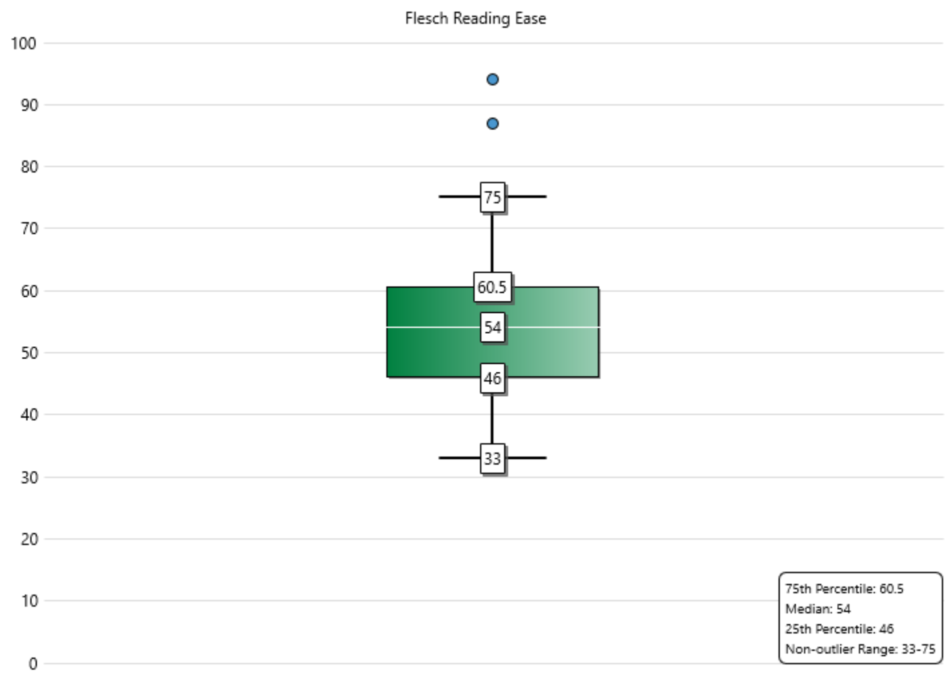
After opening a project, click on the Score Box Plots icon on the project sidebar to view the documents' test scores graphically shown as box plots.

If the project contains groups, then the boxes will be categorized into these groups:



In the above example, note that the documents in the *Overviews* group mostly score in the high-school range. There are, however, a few outliers in this group that may need reviewing.

Along with the grade-level box plots, a separate box plot for each index test will also be included. Below is a box plot of Flesch scores from a batch:



As we can see above, the middle half of the documents fall within a marginally acceptable range of 46–60.5 (60–80 is normally preferred). The upper whisker goes up to 75, meaning that there are a decent number of documents in the 60.5–75 range. Unfortunately, the lower whisker extends all the way to 33—this indicates a fair amount of difficult documents. As with the grade-level plot, this graph helps us conclude that some documents in this batch should be revised to lower their scores.

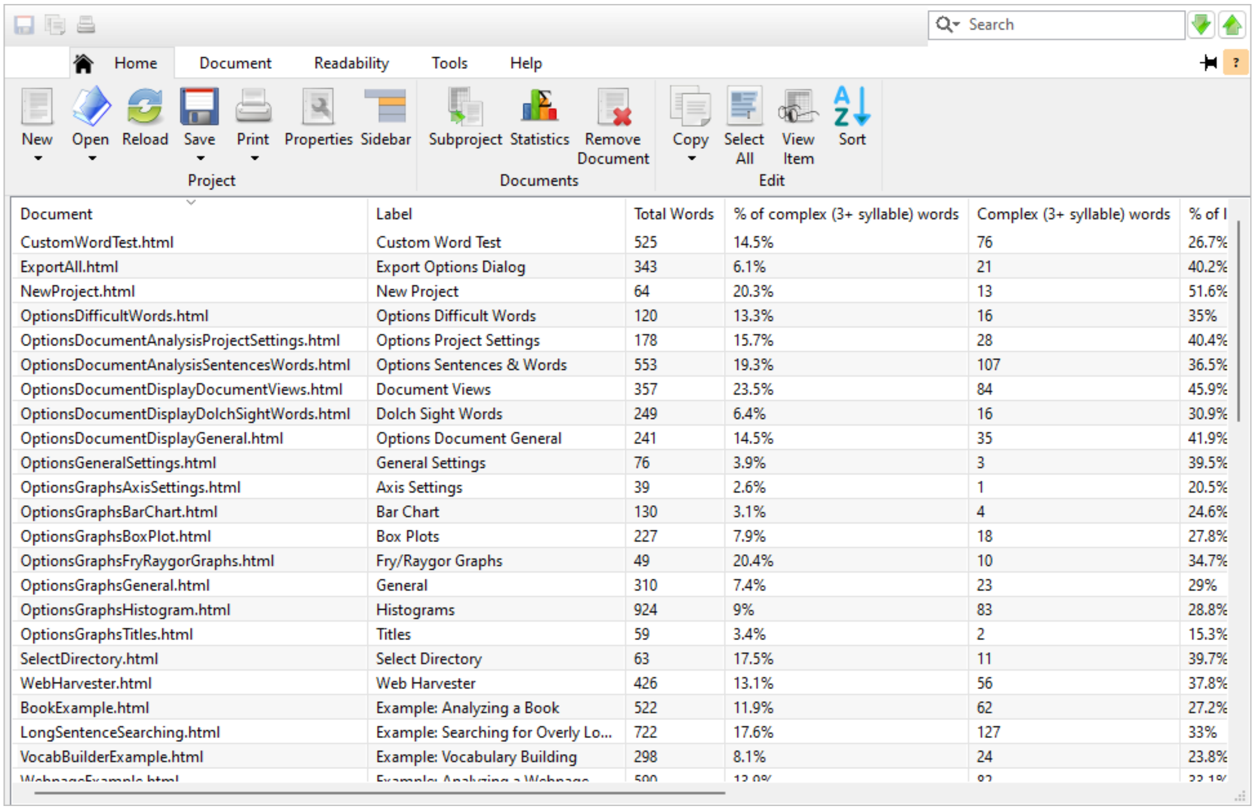
5.4 Readability Graphs

Refer to sec. 4.2.

5.5 Words Breakdown

After opening a project, click on the Words Breakdown icon on the project sidebar, then click any of the following subitems:

Difficult Words: This list displays percentages of difficult words throughout the documents.

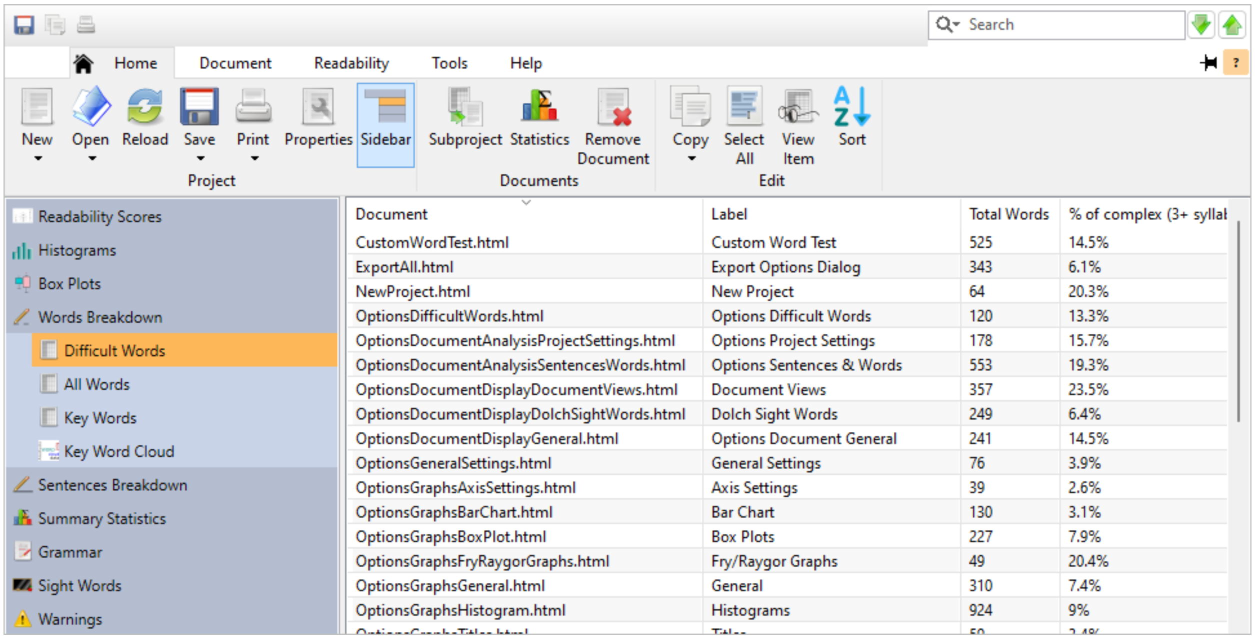


Document	Label	Total Words	% of complex (3+ syllable) words	Complex (3+ syllable) words	% of l
CustomWordTest.html	Custom Word Test	525	14.5%	76	26.7%
ExportAll.html	Export Options Dialog	343	6.1%	21	40.2%
NewProject.html	New Project	64	20.3%	13	51.6%
OptionsDifficultWords.html	Options Difficult Words	120	13.3%	16	35%
OptionsDocumentAnalysisProjectSettings.html	Options Project Settings	178	15.7%	28	40.4%
OptionsDocumentAnalysisSentencesWords.html	Options Sentences & Words	553	19.3%	107	36.5%
OptionsDocumentDisplayDocumentViews.html	Document Views	357	23.5%	84	45.9%
OptionsDocumentDisplayDolchSightWords.html	Dolch Sight Words	249	6.4%	16	30.9%
OptionsDocumentDisplayGeneral.html	Options Document General	241	14.5%	35	41.9%
OptionsGeneralSettings.html	General Settings	76	3.9%	3	39.5%
OptionsGraphsAxisSettings.html	Axis Settings	39	2.6%	1	20.5%
OptionsGraphsBarChart.html	Bar Chart	130	3.1%	4	24.6%
OptionsGraphsBoxPlot.html	Box Plots	227	7.9%	18	27.8%
OptionsGraphsFryRaygorGraphs.html	Fry/Raygor Graphs	49	20.4%	10	34.7%
OptionsGraphsGeneral.html	General	310	7.4%	23	29%
OptionsGraphsHistogram.html	Histograms	924	9%	83	28.8%
OptionsGraphsTitles.html	Titles	59	3.4%	2	15.3%
SelectDirectory.html	Select Directory	63	17.5%	11	39.7%
WebHarvester.html	Web Harvester	426	13.1%	56	37.8%
BookExample.html	Example: Analyzing a Book	522	11.9%	62	27.2%
LongSentenceSearching.html	Example: Searching for Overly Lo...	722	17.6%	127	33%
VocabBuilderExample.html	Example: Vocabulary Building	298	8.1%	24	23.8%

Next to each document name, the respective percentages and raw word counts of the following are displayed:

- Complex (3+ syllable) words
- Long (6+ character) words
- SMOG hard words
- Fog hard words
- Dale-Chall unfamiliar words
- Spache unfamiliar words
- Harris-Jacobson unfamiliar words
- Unfamiliar words for any custom tests

Note that test-specific word counts/percentages (e.g., Dale-Chall unfamiliar words) are only included if the respective test is included in the project.



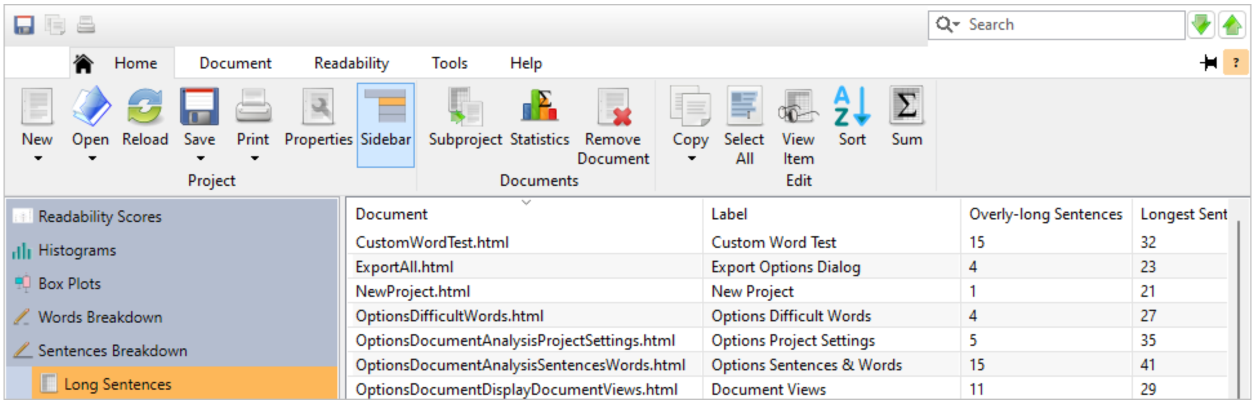
All Words: This list displays all the words from the documents and their respective frequency counts.

Key Words: This list is the same as All Words, except that it excludes filenames, numeric words, and common words, and then combines the remaining words. In regards to combining words, words with the same root will be merged into a single row. For example, *report*, *reports*, and *reporting* will be combined into one row, with the frequency counts for all three words also combined. As for excluding words, common words (e.g., *the*), numerals, and filenames are not included in this list. (If no words can be combined and there are no common words to ignore, then this list will not be shown.)

Key Word Cloud: Refer to sec. 4.5.

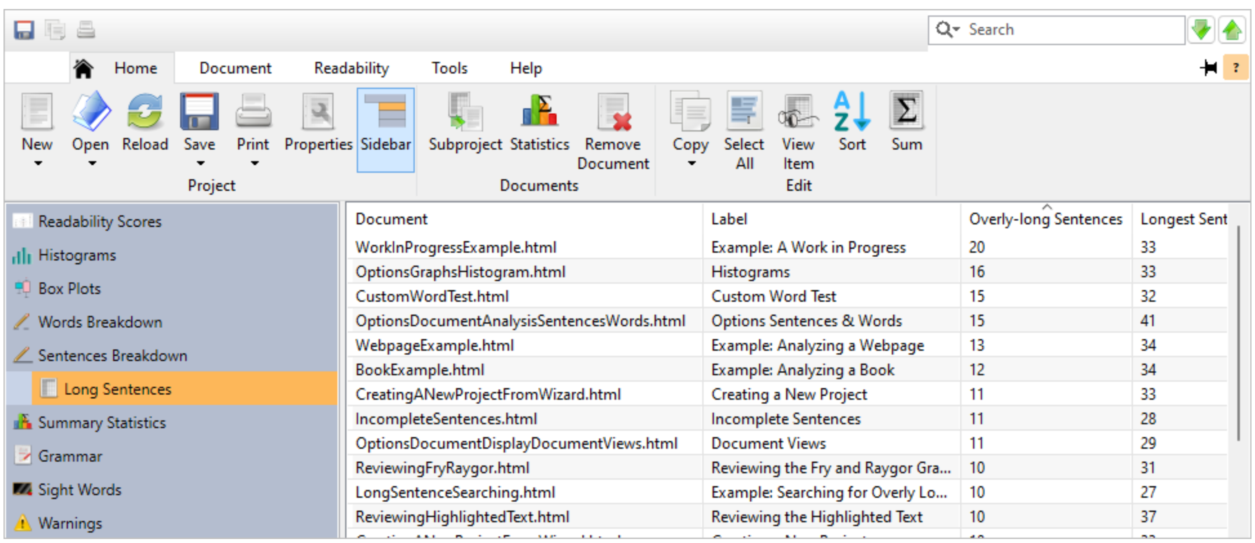
5.6 Long Sentences

After opening a project, click on the Sentences Breakdown icon on the project sidebar and click on the Long Sentences subitem. This window will display a list of the difficult sentences from your documents.



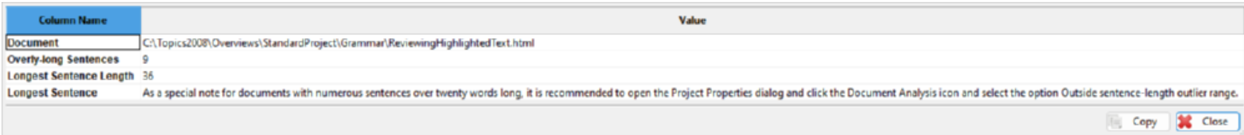
Note that if you are ignoring incomplete sentences, then this will only include the complete sentences. If you need to review all sentences from the documents, then select the option Include in analysis and treat as full sentences. Refer to sec. 20.1 to learn more about how *Readability Studio* handles incomplete sentences.

To find the longest sentence, sort the Longest Sentence Length column into descending order. As we can see below, the longest sentence is now shown at the top of the list.



If the test scores indicate that the documents are too difficult for your audience, then review the overly-long sentences. You should consider splitting these sentences into smaller ones and revising any wordy items.

To view a specific document's longest sentence, double-click on its row in this list. A vertical display of that document's longest sentence will be shown:



Column Name	Value
Document	C:\Topics2008\Overviews\StandardProject\Grammar\ReviewingHighlightedText.html
Overly-long Sentences	9
Longest Sentence Length	36
Longest Sentence	As a special note for documents with numerous sentences over twenty words long, it is recommended to open the Project Properties dialog and click the Document Analysis icon and select the option Outside sentence-length outlier range.

Note that if there are no long sentences in your documents, then this window will not be included.

5.7 Summary Statistics

To display a list containing statistics of all your documents, click on the Summary Statistics icon on the project sidebar. To control which statistics are included in this list, click the Edit Statistics button on the Home tab of the ribbon.

5.8 Grammar

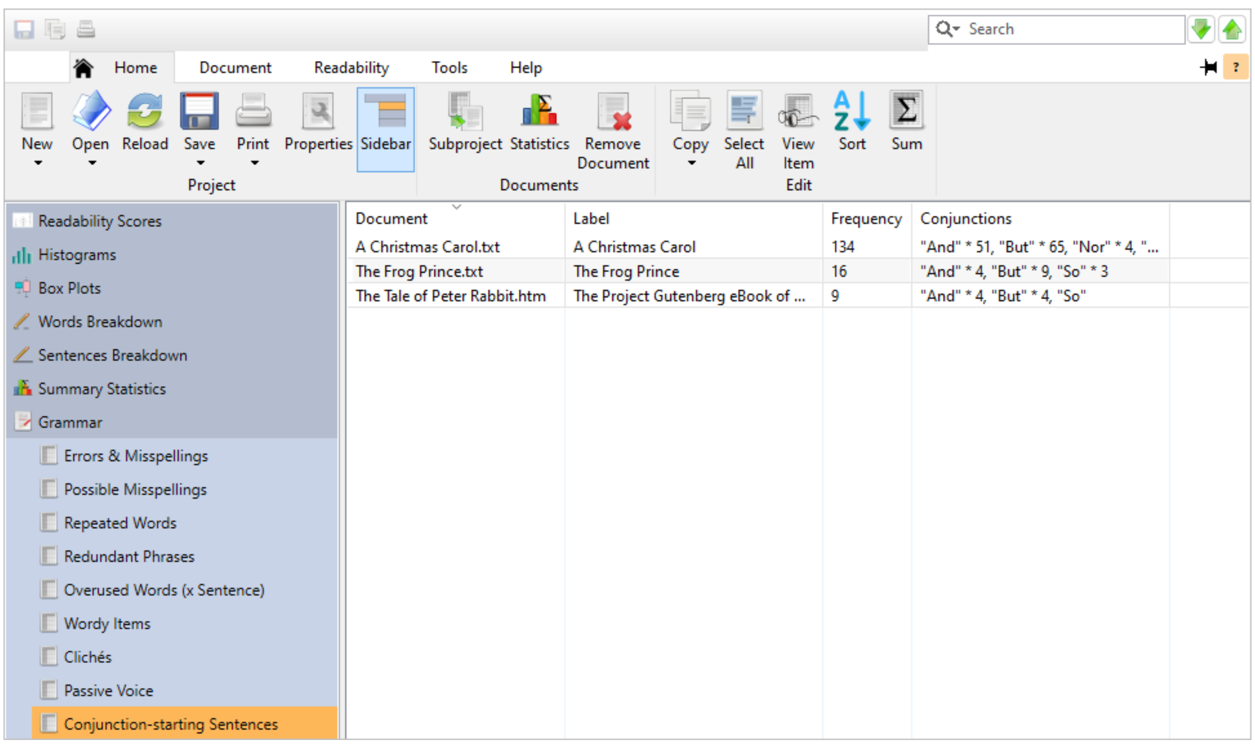
Note that the following windows are only included if their respective grammar issues are detected in the documents.

Also, from any of these windows you can create a standard project. Within a given window, select a document in its list and click the Subproject button on the Home tab.

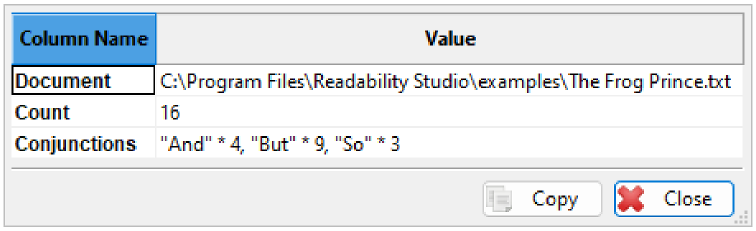
This is useful for reviewing a particular document in greater detail. Note that when this standard project is created, it will copy all the project settings from the batch project. These options may differ from your global configurations.

Reviewing Conjunction-Starting Sentences

After opening a project, click on the Grammar icon on the project sidebar and select the Conjunction-Starting Sentences subitem. This window will display a list of all sentences that begin with coordinating conjunctions.



To view a specific document's conjunction-starting sentences, double-click on its row in this list. A vertical display of that document's conjunction-starting sentences will be shown:

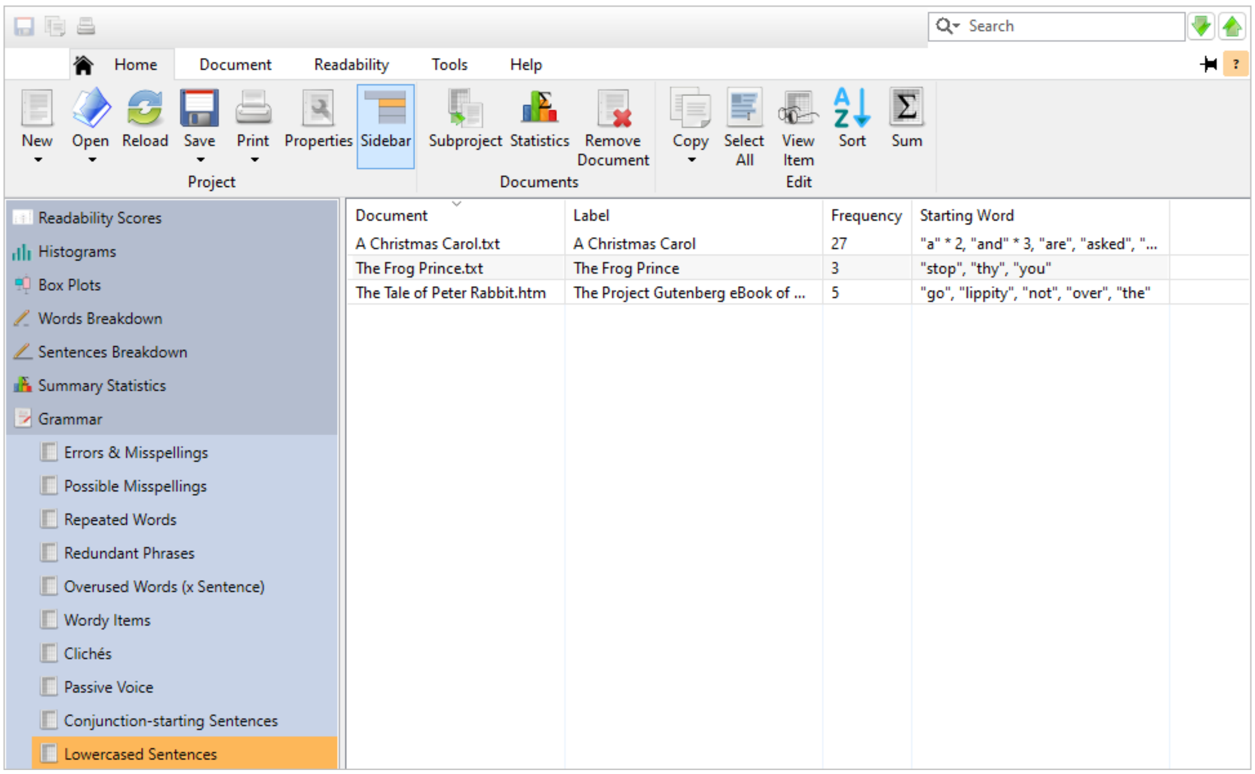


Column Name	Value
Document	C:\Program Files\Readability Studio\examples\The Frog Prince.txt
Count	16
Conjunctions	"And" * 4, "But" * 9, "So" * 3

Refer to sec. 4.7 to learn more about conjunction-starting sentences.

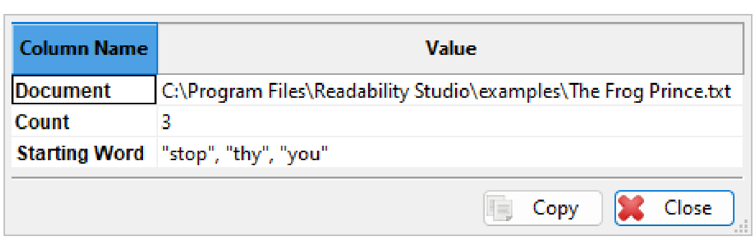
Reviewing Lowercased Sentences

After opening a project, click on the Grammar icon on the project sidebar and select the Lowercased Sentences subitem. This window will display a list of all sentences that begin with lowercased words (which is usually a typo).



Note that if the option Sentences must begin with capitalized words is enabled, then this feature will be limited to sentences that begin new paragraphs. If you want to search for any possible sentences that begin with lowercased words, then you should leave this option unchecked.

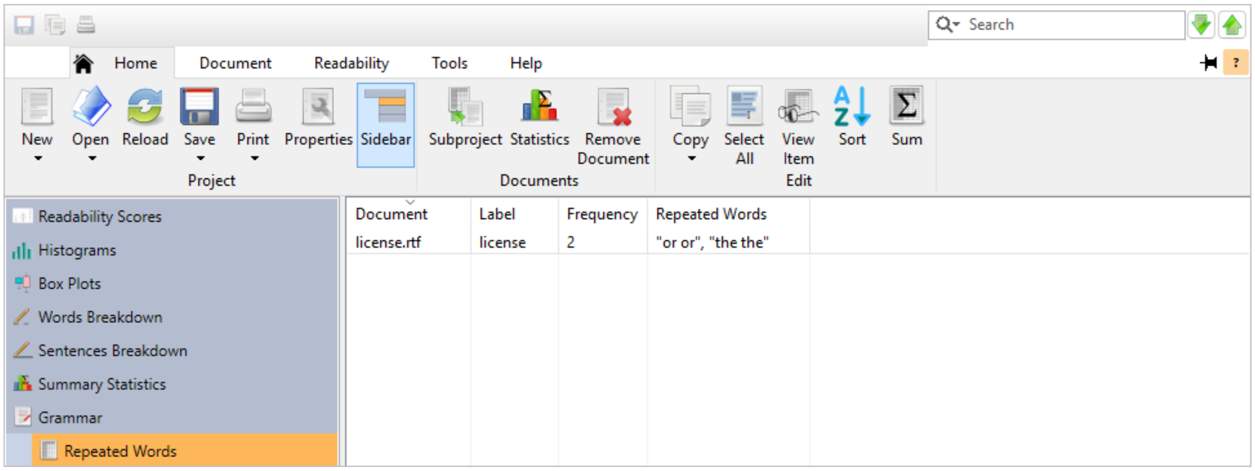
To view a specific document's lowercased sentences, double-click on its row in this list. A vertical display of that document's lowercased sentences will be shown:



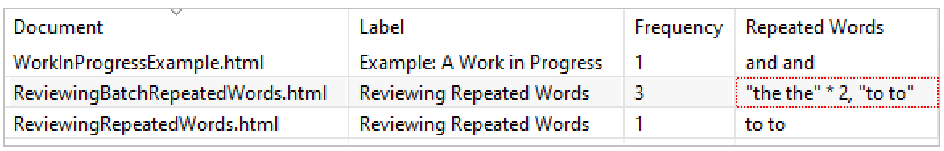
Column Name	Value
Document	C:\Program Files\Readability Studio\examples\The Frog Prince.txt
Count	3
Starting Word	"stop", "thy", "you"

Reviewing Repeated Words

After opening a project, click on the Grammar icon on the project sidebar and select the Repeated Words subitem. This window will display a list of all repeated words from each document.

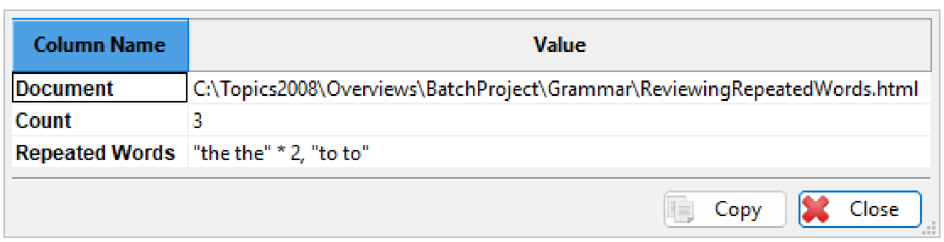


If a repeated word occurs more than once in a document, then a multiplication factor will be shown next to it. For example, if *the the* occurs two separate times in a given document, then this will be displayed as *the the*2*:



Document	Label	Frequency	Repeated Words
WorkInProgressExample.html	Example: A Work in Progress	1	and and
ReviewingBatchRepeatedWords.html	Reviewing Repeated Words	3	"the the" * 2, "to to"
ReviewingRepeatedWords.html	Reviewing Repeated Words	1	to to

To view a specific document's repeated words, double-click on its row in this list. A vertical display of that document's repeated words will be shown:

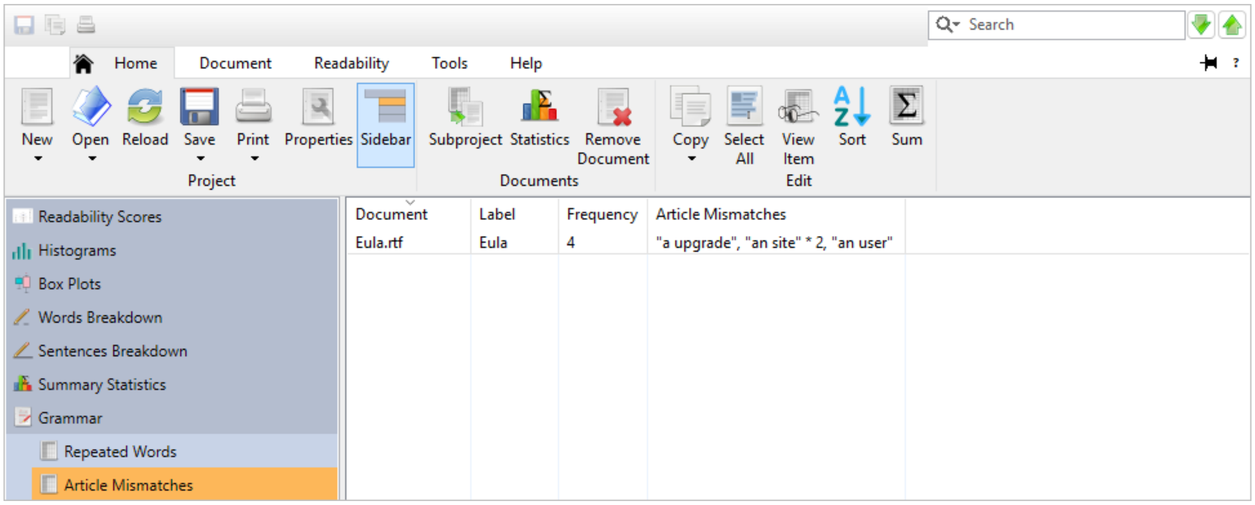


Column Name	Value
Document	C:\Topics2008\Overviews\BatchProject\Grammar\ReviewingRepeatedWords.html
Count	3
Repeated Words	"the the" * 2, "to to"

Refer to sec. 4.7 and sec. 20.6 to learn how repeated words are detected.

Reviewing Article Mismatches

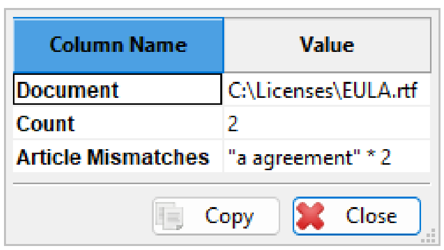
After opening a project, click on the Grammar icon on the project sidebar and select the Article Mismatches subitem. This window will display a list of articles (i.e., *a* and *an*) that do not match their following noun.



Article mismatch searching does not affect readability calculations, so this analysis is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

If an article mismatch occurs more than once in a document, then a multiplication factor will be shown next to it. For example, if *a electronic* occurs twice in a given document, then this will be displayed as *a electronic*2*.

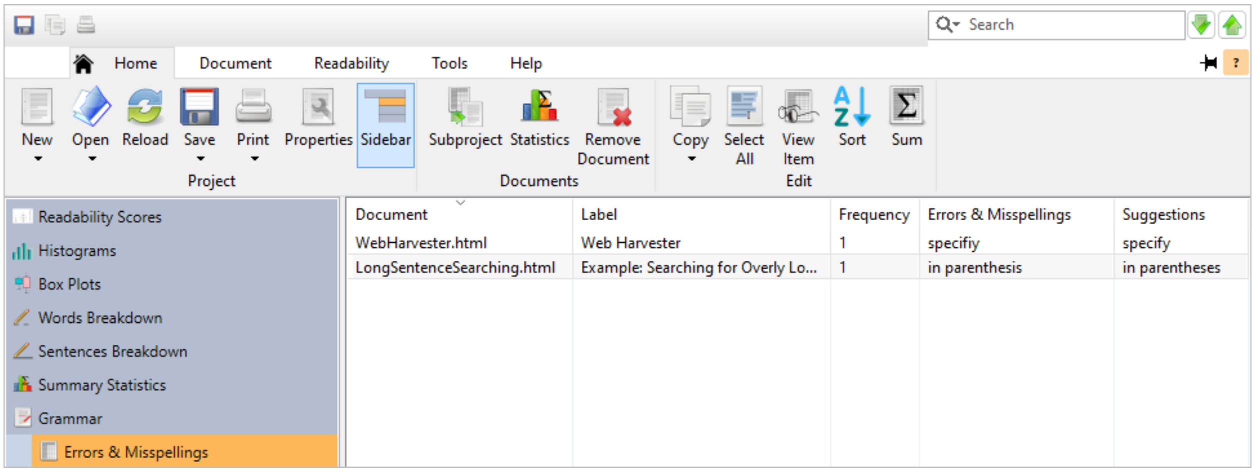
To view a specific document's article mismatches, double-click on its row in this list. A vertical display of that item will be shown:



Refer to sec. 20.7 to learn more.

Reviewing Wording Errors & Misspellings

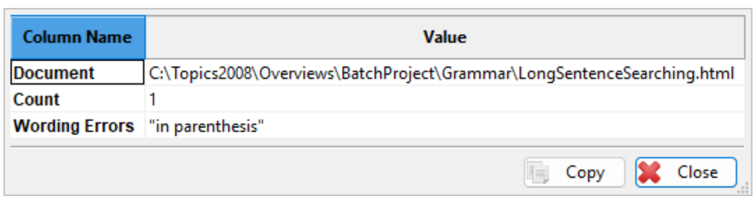
After opening a project, click on the Grammar icon on the project sidebar and select the Errors & Misspellings subitem. This window will display a list of misused phrases, grammatically incorrect wording, and known misspellings from each document.



This search is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

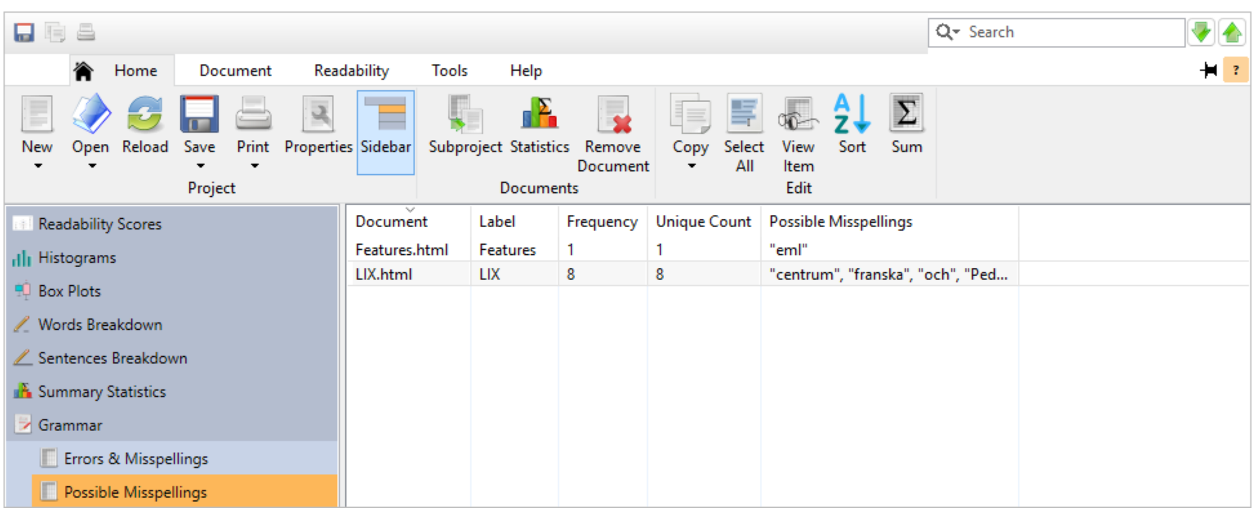
If a wording error occurs more than once in a document, then a multiplication factor will be shown next to it. For example, if *other then* occurs two separate times in a given document, then this will be displayed as *other then*2*.

To view a specific document's errors, double-click on its row. A vertical display of that document's wording errors will be shown:



Reviewing Possible Misspellings

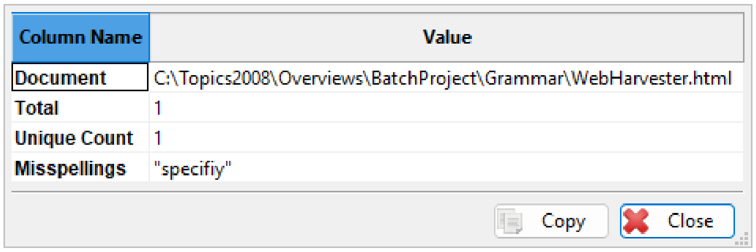
After opening a project, click on the Grammar icon on the project sidebar and select the Possible Misspellings subitem. This window will display a list of all possible misspellings from each document.



This search is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

If a misspelling occurs more than once in a document, then a multiplication factor will be shown next to it. For example, if *specfy* occurs two separate times in a given document, then this will be displayed as *specfy*2*.

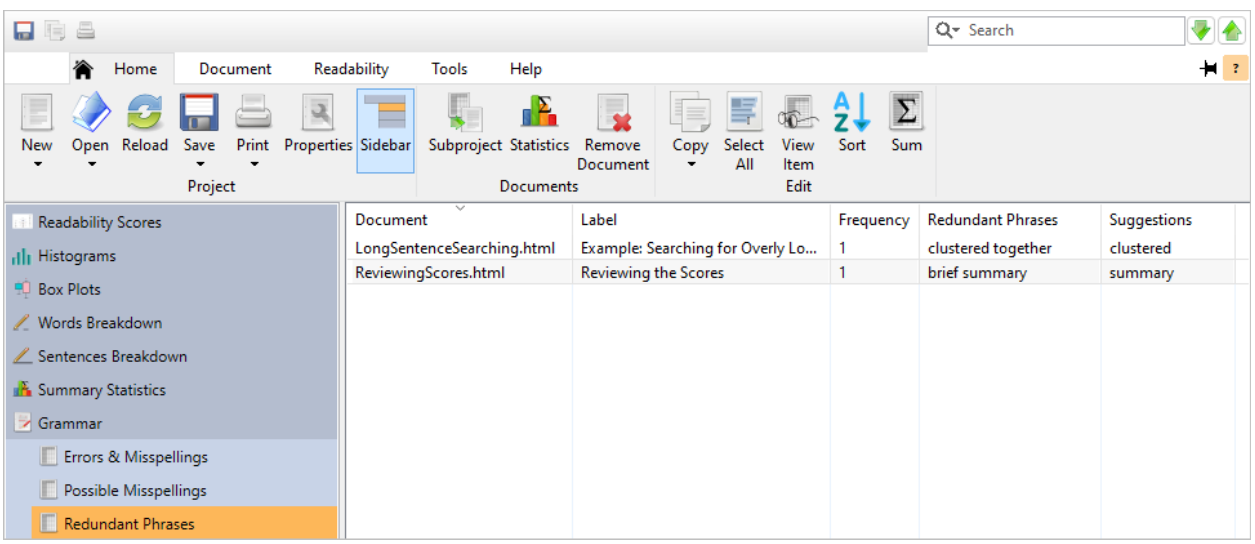
To view a specific document's misspellings, right-click on its row and select View Item. A vertical display of that document's misspellings will be shown:



To add any document's misspellings to your dictionary, double-click on its row to show the Add to Dictionary dialog. From here, you can add any (or all) reported misspellings to your dictionary. Afterwards, these words will no longer be reported as misspelled.

Reviewing Redundant Phrases

After opening a project, click on the Grammar icon on the project sidebar and select the Redundant Phrases subitem. This window will display a list of redundant phrases from the documents. Suggested replacements will also be shown next to each phrase in this list.



Redundant-phrase searching does not affect readability calculations, so this analysis is applied to all sentences. Incomplete sentences will be reviewed for this, even if they are excluded from the analysis.

If a redundant phrase occurs more than once in a document, then a multiplication factor will be shown next to it. For example, if *brief summary* occurs three separate times in a given document, then this will be displayed as *brief summary*3*:

Document	Label	Frequency	Redundant Phrases	Suggestions
WorkInProgressExample.html	Example: A Work in Progress	1	mixed together	mixed
ThirdPartyLicenses.html	Third Party Licenses	2	"each and every", "exactly identic...	"all, each, every", "identical"
ReviewingBatchRedundantPhrases.html	Reviewing Redundant Phrases	7	"brief summary" * 5, "collaborate...	"summary", "collaborated"
ReviewingRedundantPhrases.html	Reviewing Redundant Phrases	5	"brief summary" * 3, "collaborate...	"summary", "collaborated"

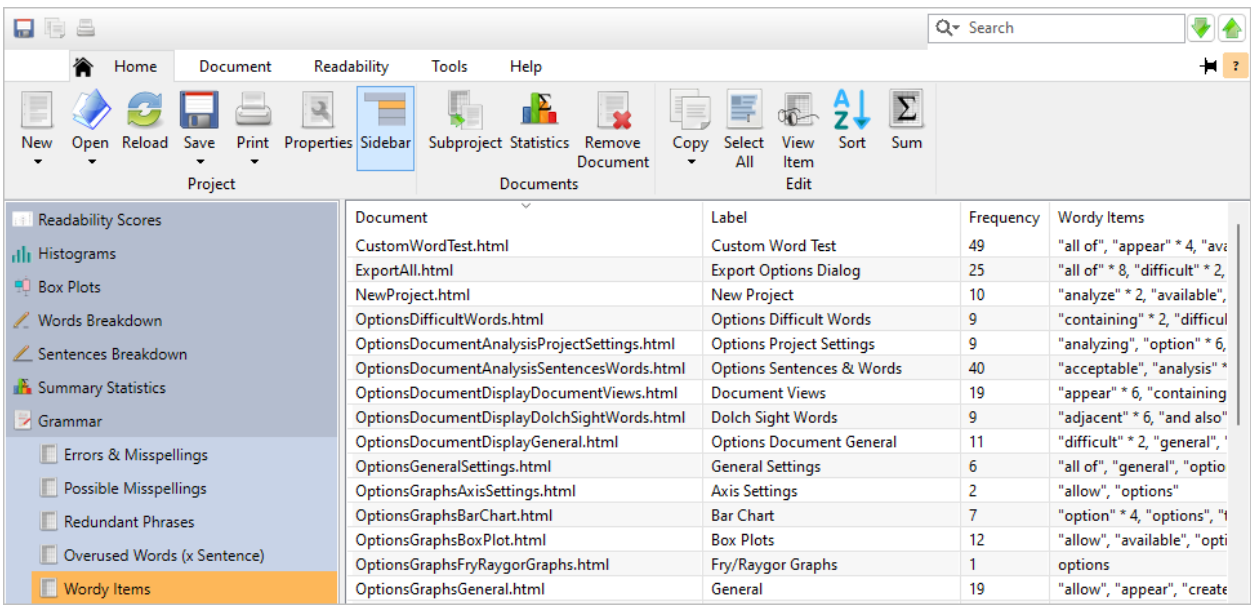
To view a specific document's redundant phrases, double-click on its row in this list. A vertical display of that document's redundant phrases will be shown:

Column Name	Value
Document	C:\Topics2008\Overviews\BatchProject\Grammar\ReviewingRedundantPhrases.html
Count	7
Redundant Phrases	"brief summary" * 3, "collaborated together" * 2, "join together" * 2
Suggestions	"summary", "collaborated", "join, combine"

Refer to sec. 4.7 for an explanation of what redundant phrases are.

Reviewing Wordy Items

After opening a project, click on the Grammar icon on the project sidebar and select the Wordy Items subitem. This window will display a list of wordy phrases and difficult words from the documents. Suggested replacements will also be shown next to each phrase in this list.

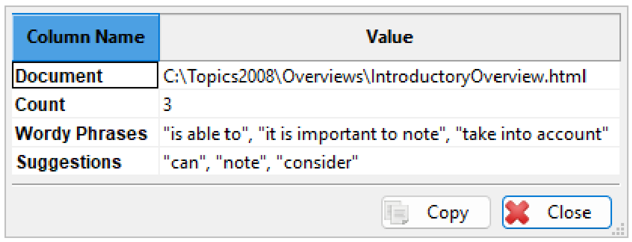


Document	Label	Frequency	Wordy Items
CustomWordTest.html	Custom Word Test	49	"all of", "appear" * 4, "ava
ExportAll.html	Export Options Dialog	25	"all of" * 8, "difficult" * 2,
NewProject.html	New Project	10	"analyze" * 2, "available",
OptionsDifficultWords.html	Options Difficult Words	9	"containing" * 2, "difficul
OptionsDocumentAnalysisProjectSettings.html	Options Project Settings	9	"analyzing", "option" * 6,
OptionsDocumentAnalysisSentencesWords.html	Options Sentences & Words	40	"acceptable", "analysis" *
OptionsDocumentDisplayDocumentViews.html	Document Views	19	"appear" * 6, "containing
OptionsDocumentDisplayDolchSightWords.html	Dolch Sight Words	9	"adjacent" * 6, "and also"
OptionsDocumentDisplayGeneral.html	Options Document General	11	"difficult" * 2, "general",
OptionsGeneralSettings.html	General Settings	6	"all of", "general", "optio
OptionsGraphsAxisSettings.html	Axis Settings	2	"allow", "options"
OptionsGraphsBarChart.html	Bar Chart	7	"option" * 4, "options", "
OptionsGraphsBoxPlot.html	Box Plots	12	"allow", "available", "opti
OptionsGraphsFryRaygorGraphs.html	Fry/Raygor Graphs	1	options
OptionsGraphsGeneral.html	General	19	"allow", "appear", "create

If a wordy item occurs more than once in a document, then a multiplication factor will be shown next to it. For example, if *result in* occurs two separate times in a given document, then this will be displayed as *result in*2*:

Document	Label	Frequency	Wordy Items	Suggestions
AddCustomWordTest.html	Add Custom Word Test	8	"available", "containing", "create"...	"offered, ready", "having, holding.
AdditionalFilteringOptions.html	Additional Filtering Options	24	"additional" * 3, "appear", "availa...	"added, more, other, extra", "see...
CustomTestClassification.html	Classification	2	"creating" * 2	making
CustomTestGeneralSettings.html	General Settings	23	"appear" * 2, "available" * 4, "cont...	"seem, come", "offered, ready", "..
CustomTestProperAndNumerals.html	Numerals	15	"appear" * 2, "familiar" * 4, "occur...	"seem, come", "known", "event, i..
CustomTestWordLists.html	Word Lists	43	"appear" * 3, "available", "benefit...	"seem, come", "offered, ready", "..
EditWordsDialog.html	Edit Word/Phrase List	3	"available", "containing", "delete"	"offered, ready", "having, holding.
ExportAll.html	Export All Options Dialog	25	"available", "option" * 17, "option...	"offered, ready", "choice, way", "...
ExportList.html	List Export Options	10	"additionally", "appear", "option"...	"added, more", "seem, come", "c..
ImageExport.html	Image Export Options	13	"available", "compress", "generat...	"offered, ready", "sqeeze", "create.

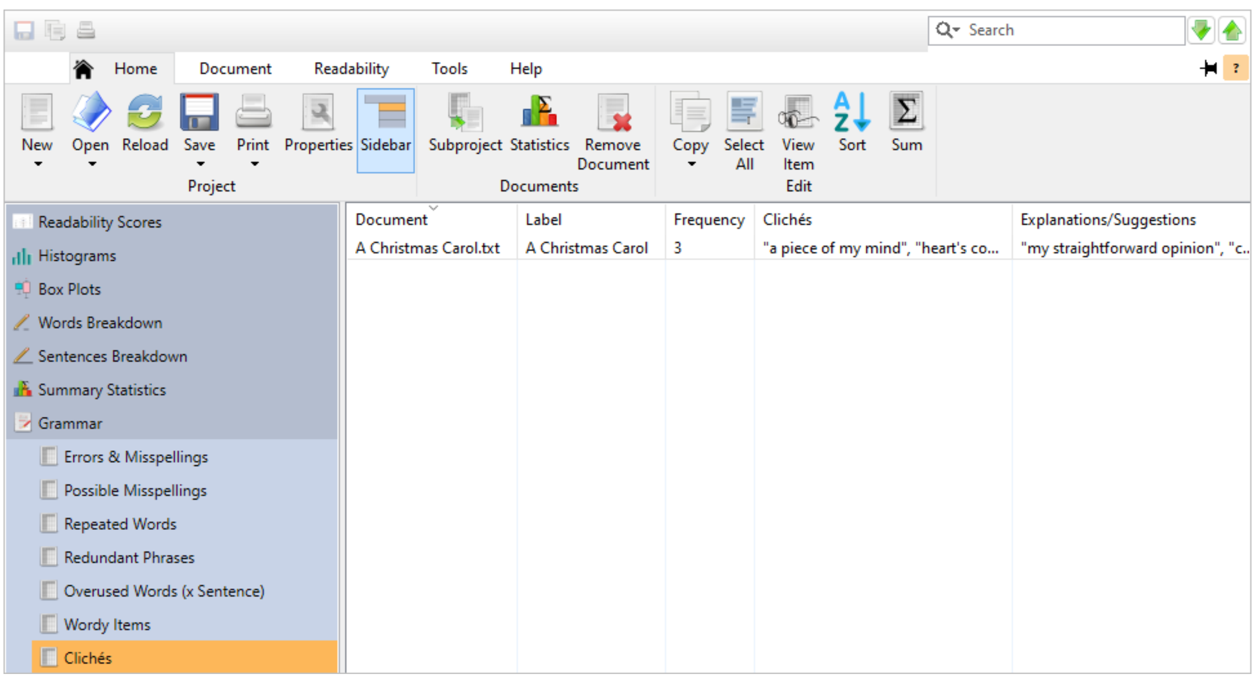
To view a specific document's wordy item, double-click on its row in this list. A vertical display of that item will be shown:



Refer to sec. 4.7 for an explanation about wordy phrases.

Reviewing Clichés

After opening a project, click on the Grammar icon on the project sidebar and select the Clichés subitem. This window will display a list of clichés from the documents. Suggested replacements will be shown next to each cliché in this list.



If a cliché occurs more than once in a document, then a multiplication factor will be shown next to it. For example, if *off the wall* occurs four separate times in a given document, then this will be displayed as *off the wall*4*:

Document	Label	Frequency	Clichés	Explanations/Suggestions
ReviewingBatchCliches.html	Reviewing Clichés	4	"off the wall" * 4	unusual
ReviewingCliches.html	Reviewing Clichés	2	"off the wall" * 2	unusual
eflaw-test.html	McAlpine EFLAW	1	tip of the iceberg	small part of the problem, just th...

To view a specific document's clichés, double-click on its row in this list. A vertical display of that document's clichés will be shown:

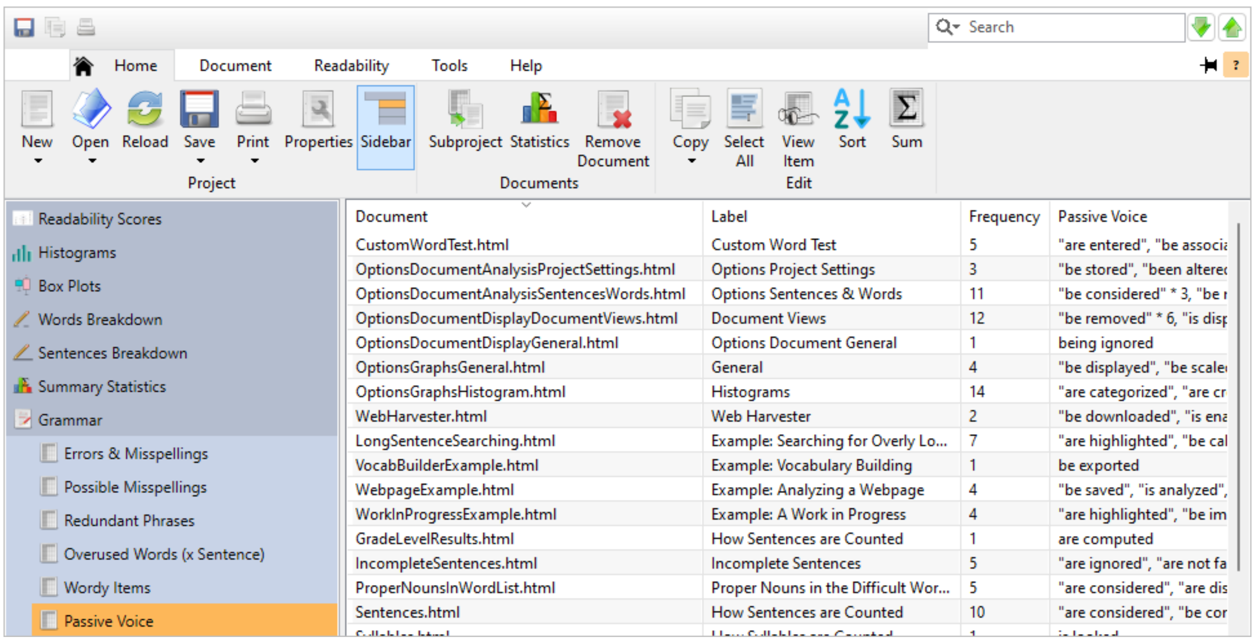
Column Name	Value
Document	C:\Topics2008\Overviews\BatchProject\Grammar\ReviewingBatchCliches.html
Count	4
Clichés	"off the wall" * 4
Explanations/Suggestions	"unusual"

Copy Close

Refer to sec. 4.7 for an explanation about clichés.

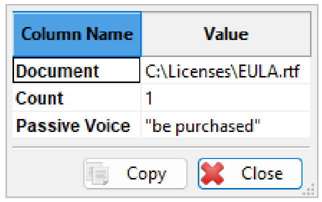
Reviewing Passive Voice

After opening a project, click on the Grammar icon on the project sidebar and select the Passive Voice subitem. This window will display a list of all passive phrases from each document.



If a passive phrase occurs more than once in a document, then a multiplication factor will be shown next to it. For example, if *be chosen* occurs twice in a given document, then this will be displayed as *be chosen*2*.

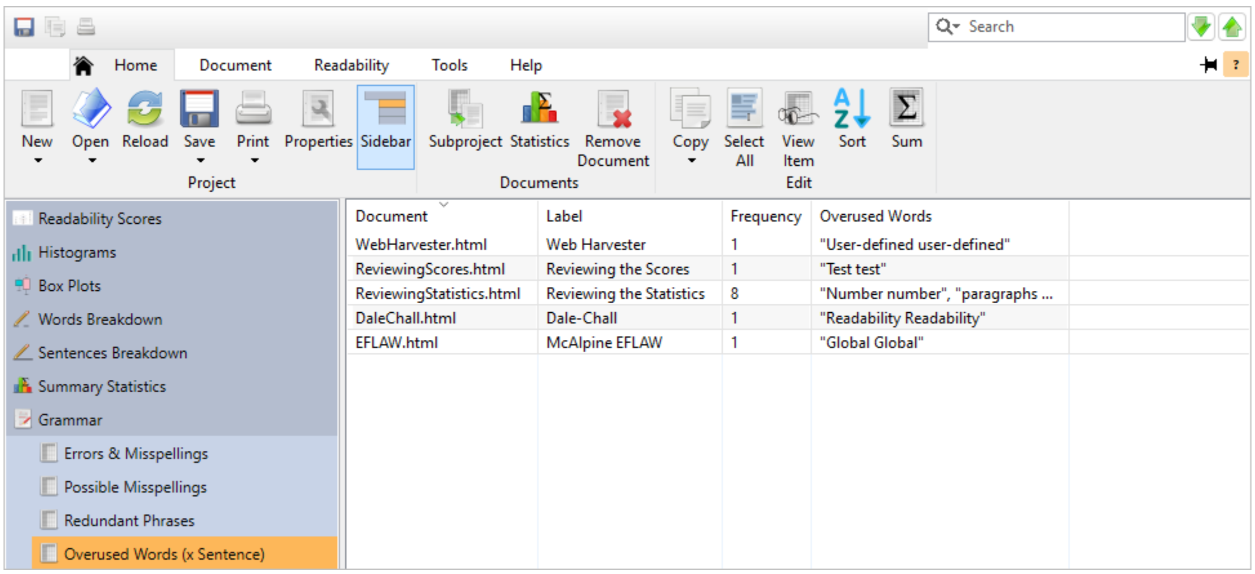
To view a specific document's passive phrases, right-click on its row and select View Item. A vertical display of that document's passive phrases will be shown:



Refer to sec. 4.7 for an explanation about passive phrases.

Reviewing Overused Words (x Sentence)

After opening a project, click on the Grammar icon on the project sidebar and select the Overused Words (x Sentence) subitem. This window will display a list of all sentences that use the same word multiple times from each document. (These are more of style suggestions, rather than grammar issues.) When reviewing these, it is recommended to change some of the repeated words to add more variety to your content.



Numeric words, proper nouns, words containing less than three letters, and common words (e.g., *the*) are excluded from this review.

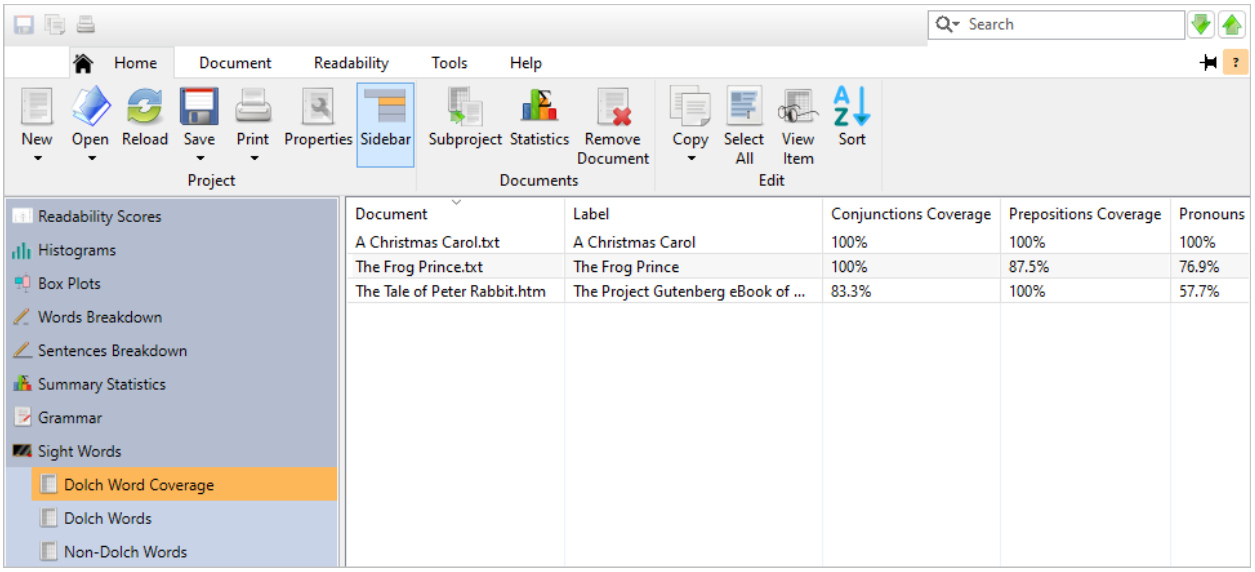
To view a specific document's overused words, right-click on its row and select View Item. A vertical display of that document's overused words will be shown.

Refer to sec. 4.7 for an explanation about overused words.

5.9 Dolch Sight Words

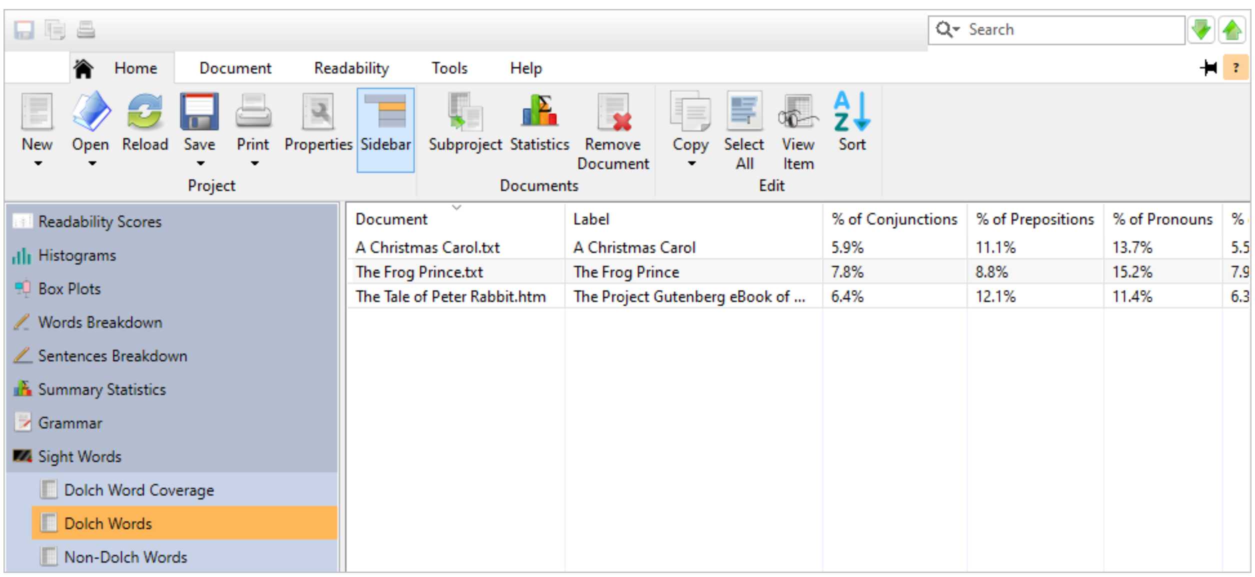
After opening a project, click on the Dolch Sight Words icon on the project sidebar. There will be three lists available: Dolch Word Coverage, Dolch Words, and Non-Dolch Words.

The Dolch Word Coverage subitem will display the Dolch word coverage for each document:



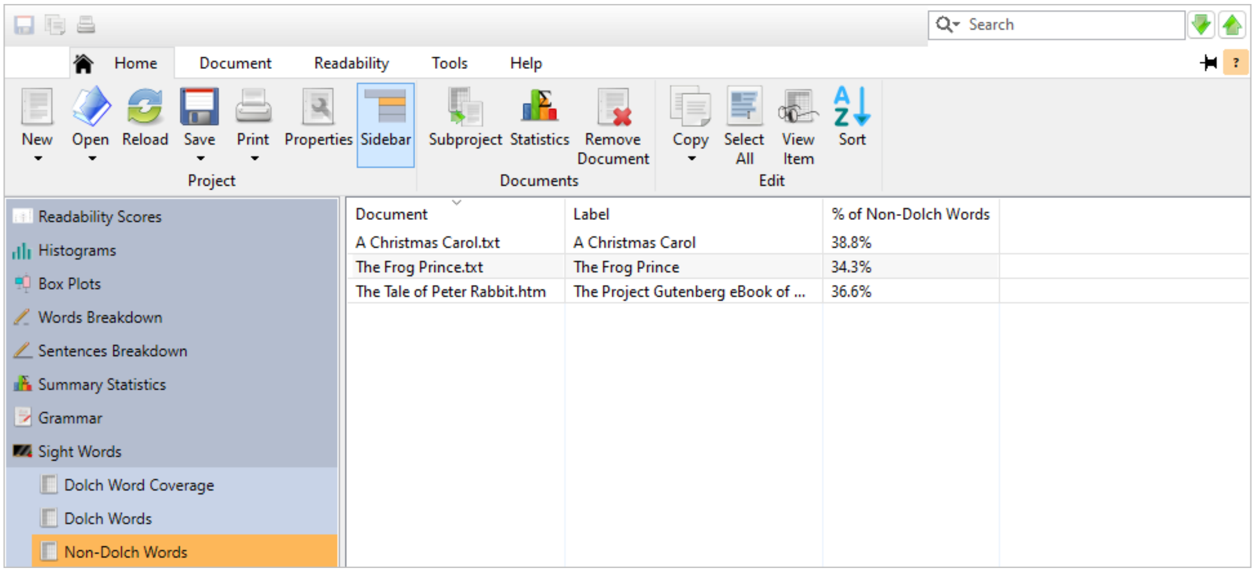
Next to each document name, the respective coverage percentage for each category is displayed.

The Dolch Words subitem will display the percentages of Dolch words used in each document:



Note that the raw counts of these words will also be included if Extended Information is enabled.

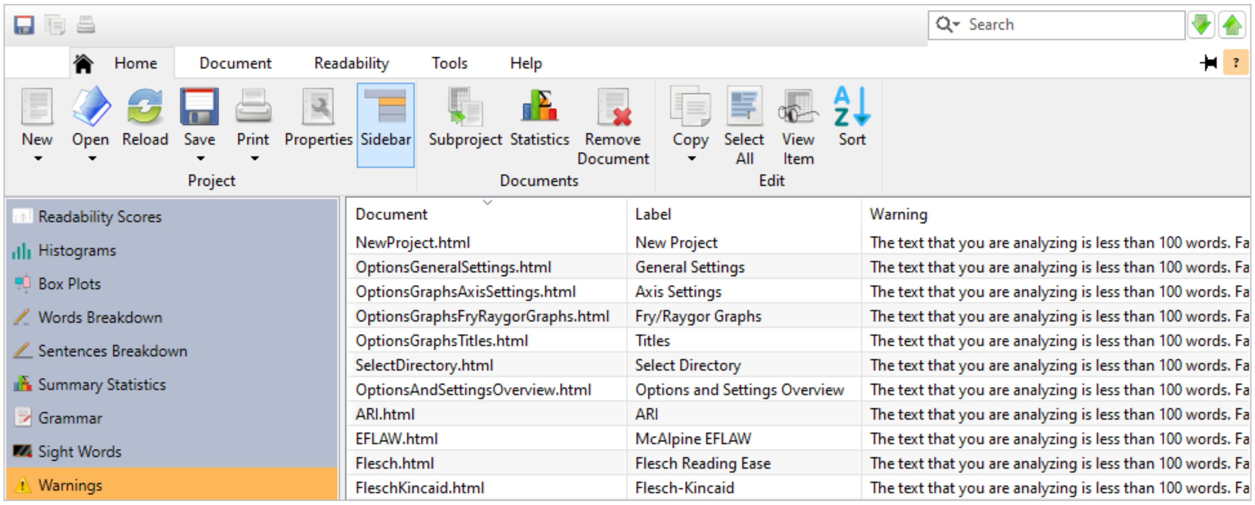
The Non-Dolch Words subitem will display the percentages of words from each document that do not appear on any of the Dolch lists:



Note that the raw counts of these words will also be included if Extended Information is enabled.

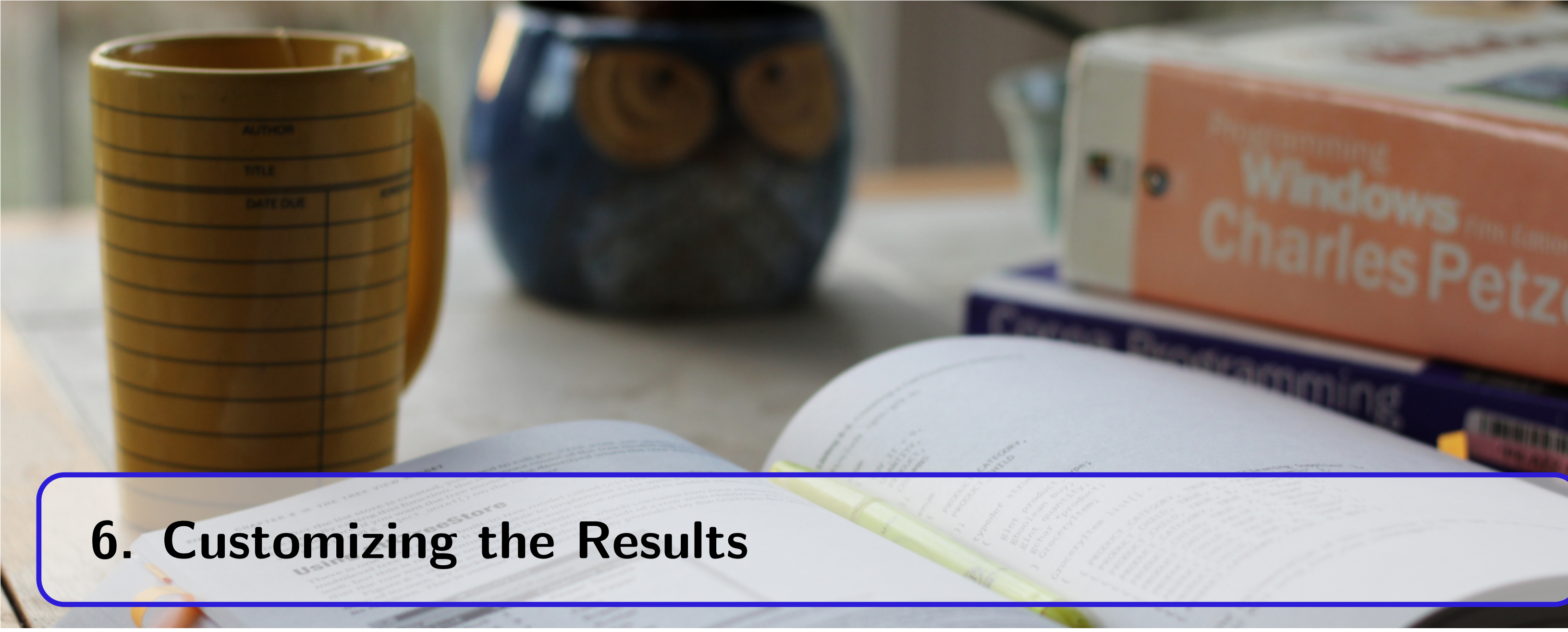
5.10 Warnings

After opening a project, click on the Warnings icon on the project sidebar to view any warnings issued while loading the documents:



Warnings that may occur are:

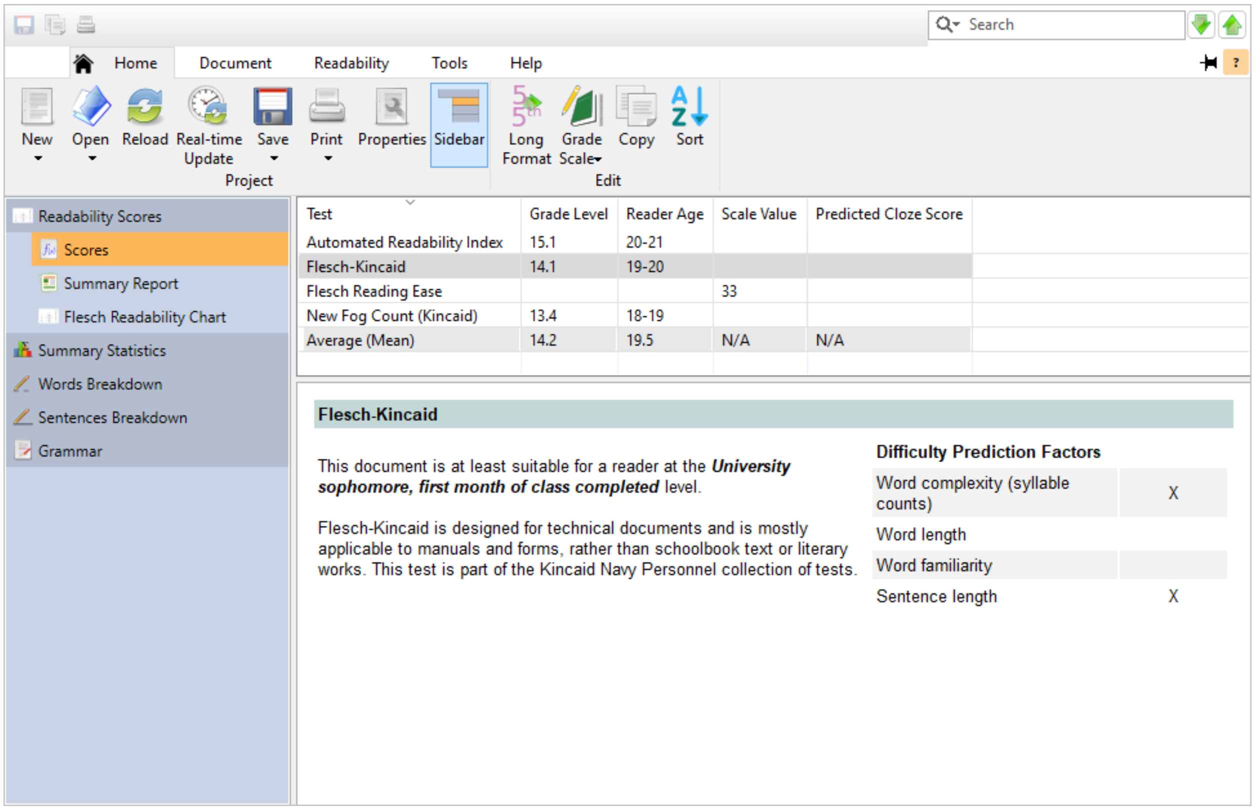
- A document could not be found.
- A document was not analyzed because it did not contain enough text.
- A test could not be run for a particular document.



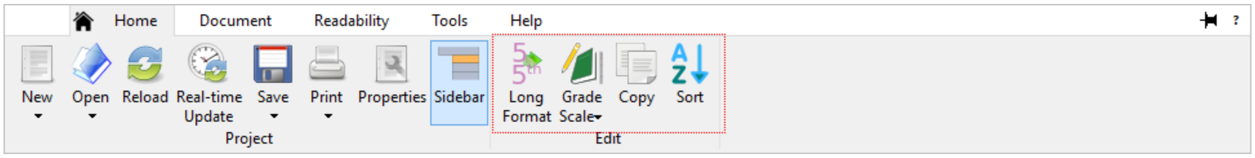
6. Customizing the Results

6.1 Scores

When the Scores section is selected:



Options for customizing their display are made available on the Edit section of the ribbon:



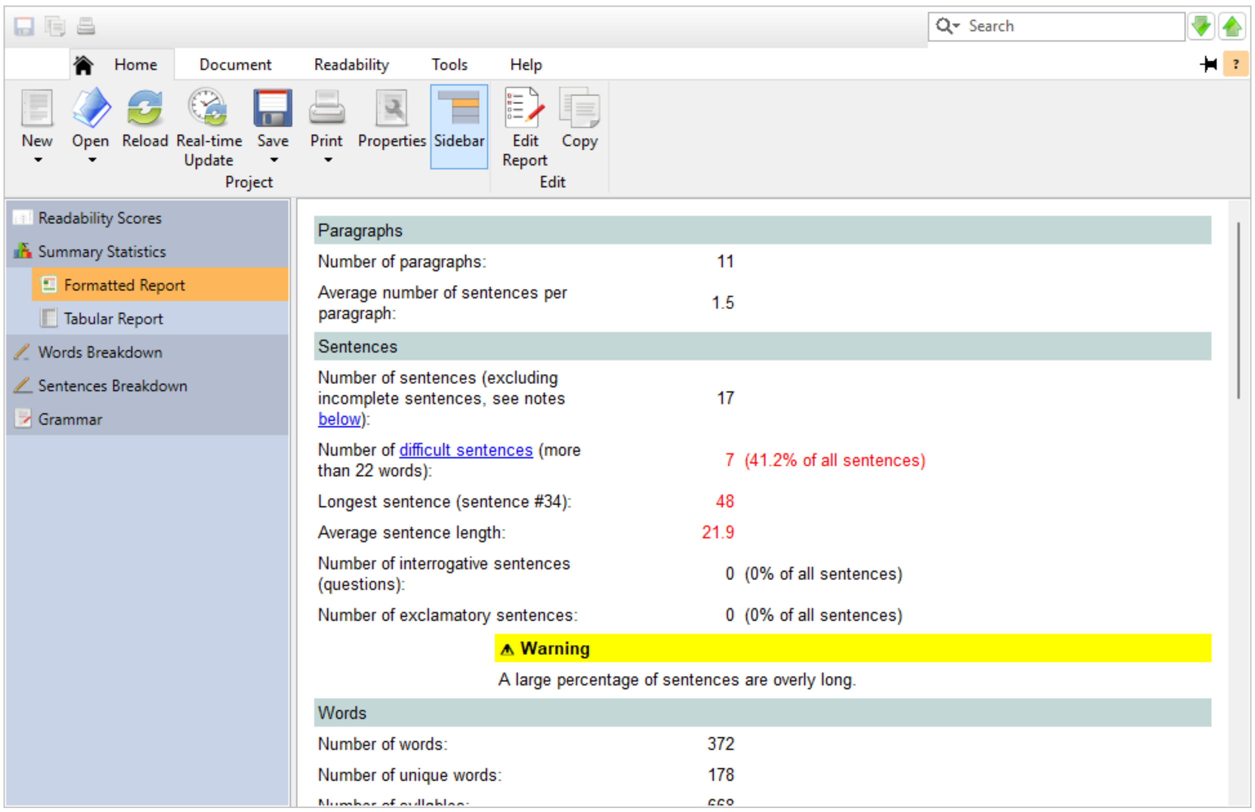
Long format: Click this button to toggle readability scores in long (e.g., *2nd grade*) or short (e.g., *2*) format within the score grid.

Grade scale: Click this button to choose from a list of grade scales to display your scores in.

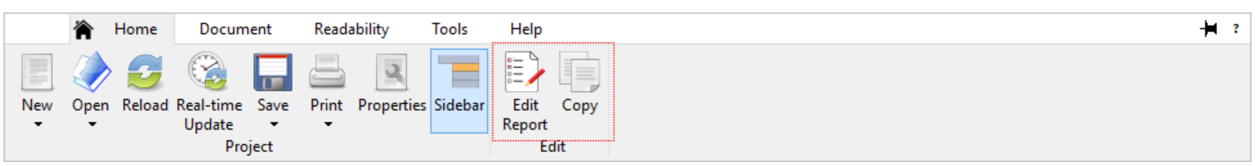
Refer to sec. 21.2 for a list of available scales.

6.2 Summary Statistics

When the Summary Statistics report is selected:



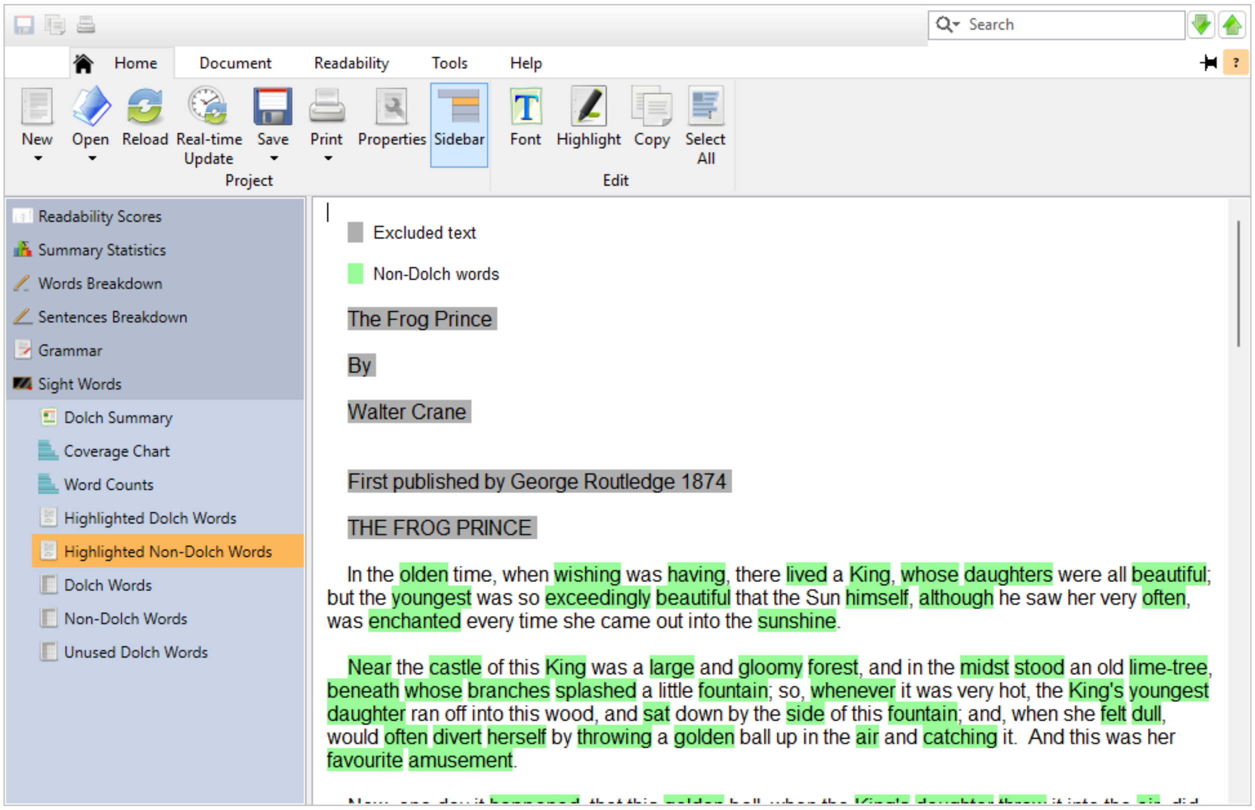
Options for customizing its display are made available on the Edit section of the ribbon:



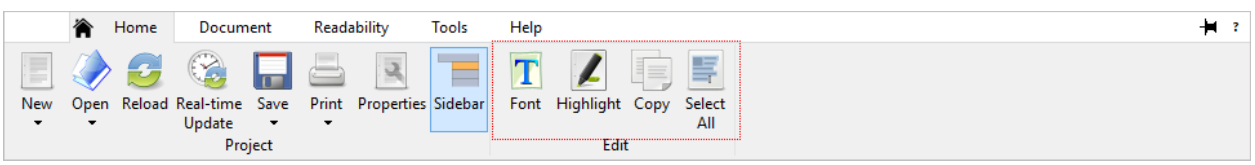
Click the Edit Report button to view the Properties dialog. Here, you can specify which sections to include in your summary report.

6.3 Highlighted Text Reports

When any highlighted text report is selected:



Options for customizing its display are made available on the Edit section of the ribbon:

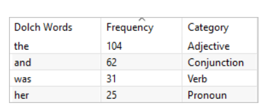


Font: Click this button to change the font.

Highlight: Click this button to open the Properties dialog. Here, you can customize the highlight colors and whether the background or foreground of the text should be highlighted.

6.4 Column Sorting

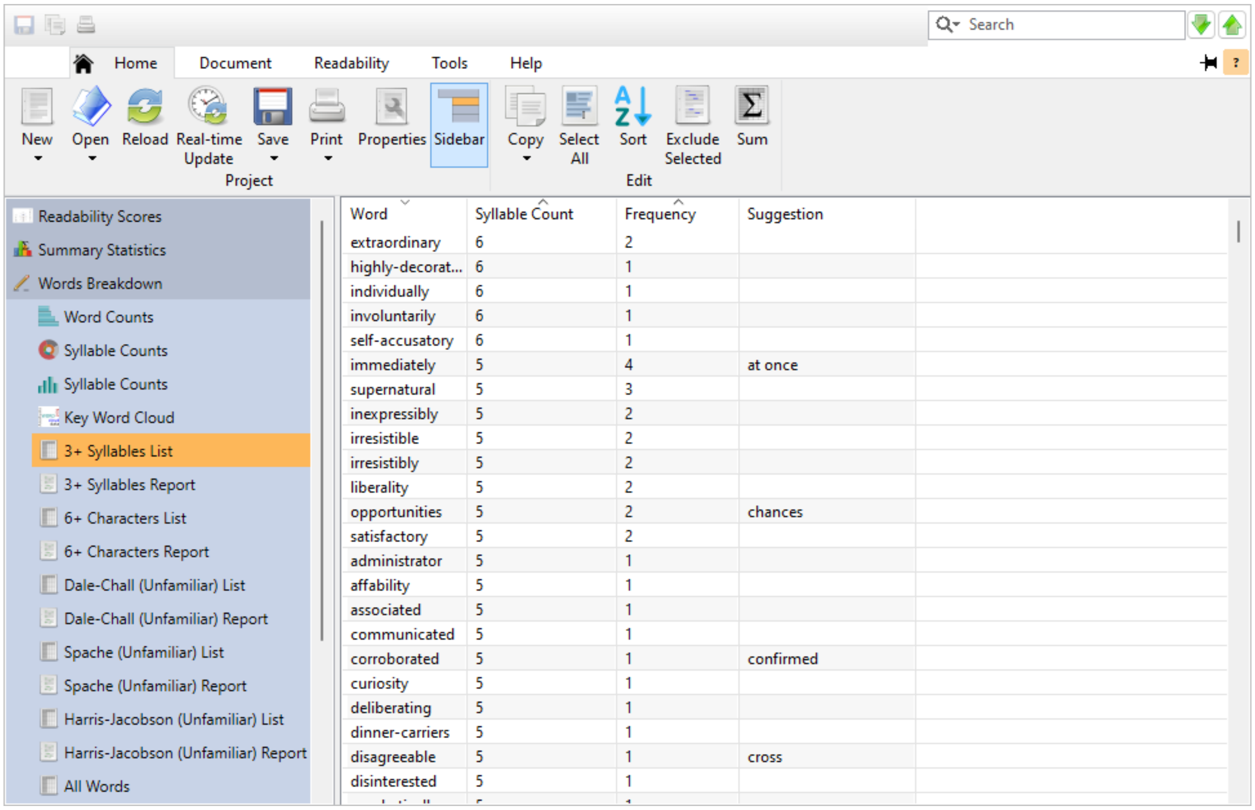
You can sort any list column by clicking on the column header.



Dolch Words	Frequency	Category
the	104	Adjective
and	62	Conjunction
was	31	Verb
her	25	Pronoun

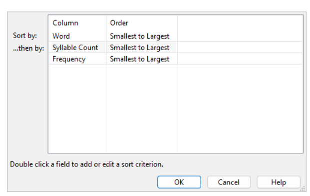
The first time you click a column's header, that column will be sorted into ascending order. This means that the column's values will go from smallest to largest as you move downward. For numeric columns, numbers will be sorted 1, 2, 3, etc. as you go downward. For text columns, the column will be sorted alphabetically, going from A–Z as you move downward.

Clicking the column a second time will sort that column into descending order. As we can see below, the Syllable Count column is sorted in descending order (largest to smallest):



To toggle the sorting order, click the column header again.

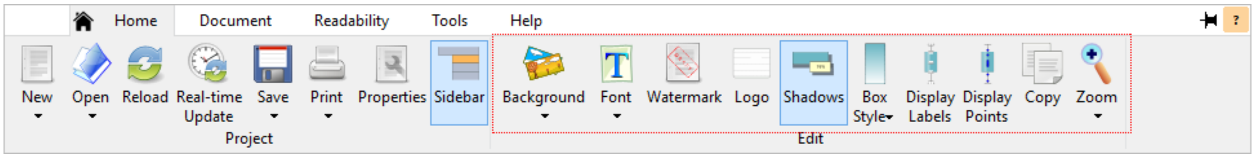
To sort a list by multiple columns, click the Sort button in the Edit section on the Home tab of the ribbon and the Sort Columns dialog will appear.



On this dialog, you will specify which columns to sort by and their order. Going from top to bottom in this grid, the columns specified here will act as the sorting criteria for the list. The list will be sorted by the first specified column (and its respective sorting order). In the case of a tie between two or more rows, the next specified column will become the sort criterion between those rows, and so on. To add a column as a sort criterion, double-click a row in this grid and select the column name from the drop-down box. Once you have selected the columns to sort by, click the OK button to sort the list.

6.5 Graphs

When any graph is selected, options for customizing it are made available on the Edit section of the ribbon:

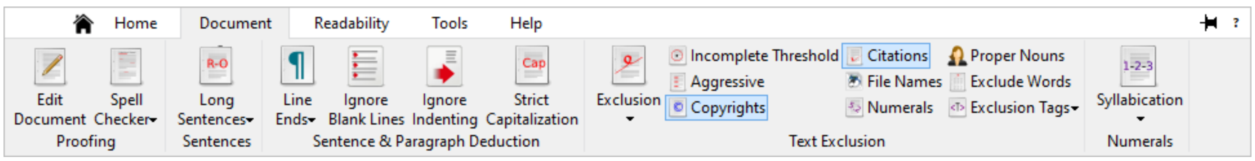


The options made available will depend on the graph type. Refer to sec. 8.10 for further discussions of these options.

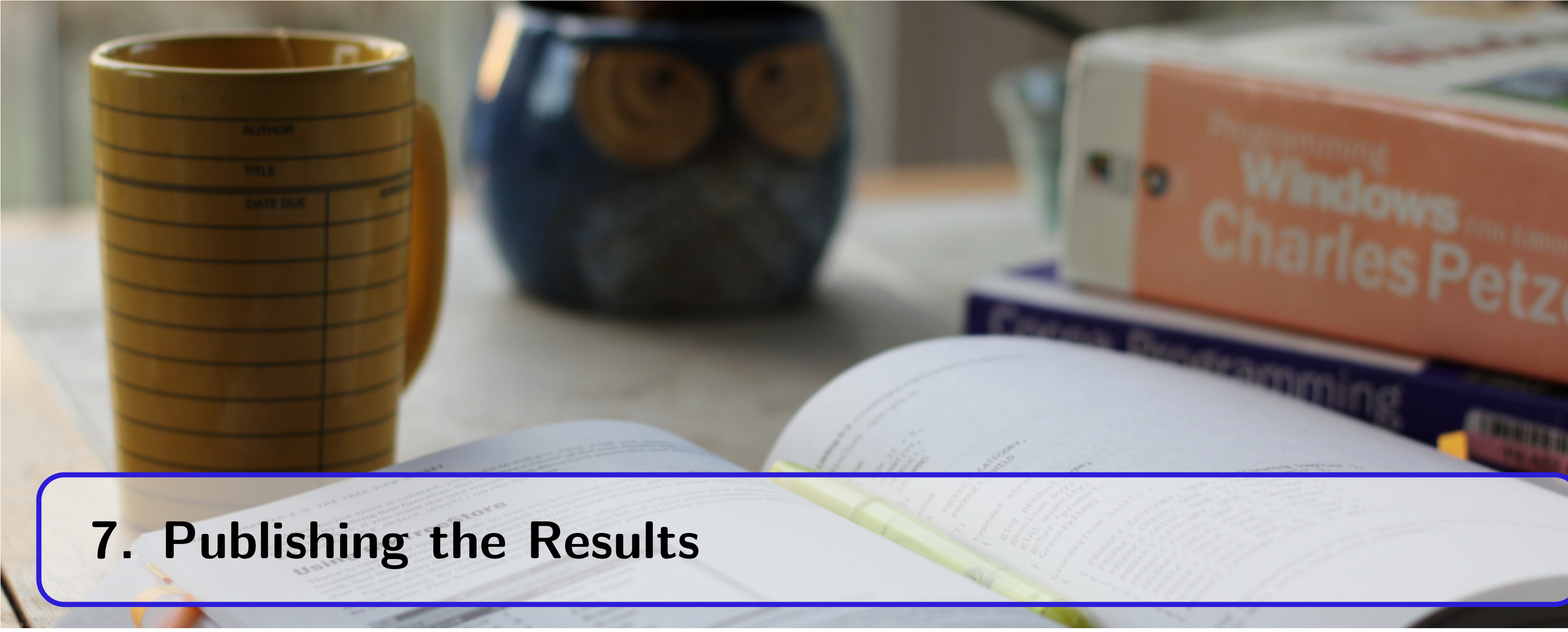
Refer to Chapter 17 for step-by-step tutorials on editing graphs.

6.6 Customizing How Documents are Analyzed

After creating or opening a project, you can customize how the program analyzes the source document(s). For example, you can change how numerals are syllabized or how sentence endings are detected. To edit these options, click on the Document tab of the ribbon:



> **Tip**
> With a standard project, you can review these changes in real time by selecting any highlighted text report in the Words Breakdown or Grammar sections. As you make changes, the program will reanalyze your document and these changes will be visible in this window. For example, changing how words are being excluded or how sentence endings are deduced will become apparent as you make changes from the ribbon.



7. Publishing the Results

All graphs, lists, statistical summaries, and highlighted text reports can be exported and printed from *Readability Studio*. This is useful for reviewing the results outside the program, as well as publishing and sharing them with colleagues.

7.1 Exporting

To export any window, go to the Home tab and click the Save button, or right-click on the window and select Save.

Numerous formats are available for each type of window:

- Highlighted text: HTML, RTF, and Pango (Pango is only available on Linux.)
- Statistical summaries: HTML
- Test results: HTML
- Word, phrase, and sentence lists: HTML, Text, and LaTeX
- Graphs: PNG, BMP, TIFF, JPEG, GIF, TARGA, and SVG

All windows can also be exported at once by clicking the arrow beneath Save and selecting Export All....

7.2 Scores Export Options

When exporting readability scores from a standard project, the following export options are available:

Select which section to save

Grid: Saves the grid of readability scores.

Explanations: Saves a report of the explanations associated with the scores.

Both: Saves the scores grid and explanations.

7.3 Text Window Export Options

When exporting a text report, the following export formats are available:

HTML: A format that can be displayed in Internet browsers and most word-processing programs.

RTF: A format that can be displayed in most word-processing programs.

Pango: A format that can be rendered within libraries such as Cairo or FreeType. (This option is only available on Linux.)

> Tip
> If you plan to edit this output, then RTF is recommended. (Most word processors support editing this format.)

7.4 Image Export Options

When exporting a graph, you will first be prompted for an image format to export as. The following formats are available to choose from:

PNG: A lossless-compressed image format. The image can be compressed to a smaller size without any loss of quality.

JPG: A lossy-compressed image format. Some image quality may be lost, compared to other formats such as PNG.

BMP: An uncompressed raster (i.e., pixel-based) image format.

TIFF: This format can either be compressed or uncompressed and offers both lossy and lossless compression. This format is generally preferred for desktop publishing.

TARGA: A raster (i.e., pixel-based) image format.

GIF: A raster (i.e., pixel-based) image format. Note that this image format is limited to 256 colors.

SVG: A format that uses vector (rather than raster) drawing. Vector-based images can be scaled to much larger sizes, without the loss of quality that raster images would experience.

After selecting the image format, these additional options will be provided:

Image Size

Width: Enter the width of the exported image into this field.

Height: Enter the height of the exported image into this field.

> Note
> When you change the width or height, the other measurement will be adjusted automatically to maintain the aspect ratio.

Color Mode

RGB (Color): Select this option to save the image in color.

Grayscale: Select this option to save the image in shades of gray.

TIFF options

This option is available only when exporting as a TIFF file.

Compression

This option specifies the compression level of the TIFF file and includes the following options:

None: do not compress the image

Lempel-Ziv & Welch: lossless compression, but may generate larger images than other compression methods.

JPEG: lossy compression method. Suitable for high-color images, but some quality may be lost.

Deflate: lossless, high-compression method.

7.5 List Export Options

When exporting a list, you will be provided with the following formats to export as:

HTML: A format that includes formatting and can be opened in browsers and spreadsheet programs.

TXT: A tab-delimited file with no formatting (i.e., colors will be lost).

LaTeX: A LaTeX file with the list formatted into a `longtable{}` environment that can be included in a larger LaTeX document.

After selecting the file format, these additional options will be provided.

Include column headers: Check this option to include the column headers in the output.

Export all rows: Select this option to export all rows and columns from the list.

Export selected rows: Select this option to only export the selected rows in the list.

Export a range of rows: Select this option to specify the rows (and columns) to export.

Paginate using printer settings: Check this option to split the list into smaller tables with page breaks between them. These tables will be sized to fit your current printer paper size and orientation. Additionally, a page break will be inserted between each table, which will appear when you print this output from a browser. To change either your paper size or orientation, select Page Setup from the Print button's menu.

This option only applies to HTML exporting and when exporting either all rows or a range.

Range

The options here specify the range of columns and rows to export.

Rows: from: Enter the starting row to export into this field.

Columns: from: Enter the starting column to export into this field.

Rows: to: Enter the ending row to export into this field.

Columns: to: Enter the ending column to export into this field.

7.6 Export All Options

Select Export All from the Save button's menu to display the Export All Options dialog.

This dialog provides options for exporting an entire project into a single report or folder of separate files.

Folder to export to: Enter the folder path where you want to export the project to into this field. (Only applies if you are exporting as separate files.)

Sections to export

Test scores: Select this option to export all the windows from the Readability Scores section.

Summary statistics: Select this option to export all the windows from the Summary Statistics section.

Histograms/box plots: Select this option to export all histograms and box plots. This option only applies to batch projects.

Words breakdown: Select this option to export all the windows from the Words Breakdown section.

Sentences breakdown: Select this option to export all the windows from the Sentences Breakdown section.

Grammar section: Select this option to export all the windows from the Grammar section.

Dolch sight words section: Select this option to export all the windows from the Dolch Sight Words section. This option only applies if Dolch Sight Words is included in the project.

Warnings section: Select this option to export all the project's warnings. This option only applies to batch projects.

Export file types

Export lists as: Select the file type that you want to save list windows as from this list. This option only applies when exporting as separate files.

Export text reports as: Select the file type that you want to save text reports as from this list. This option only applies to standard projects, and when exporting as separate files.

Export graphs as: Select the image type that you want to save graphs as from this list.

Image Options: Click this button to further customize how to save the graphs as images. This includes options such as color mode and image size. Note that the options available on this dialog will depend on which image type is selected.

Include lists: Select this option to include lists in the output. It is recommended to unselect this option for larger documents. This option only applies to standard projects.

Include text reports: Select this option to include highlighted text reports in the output. It is recommended to unselect this option for larger documents. This option only applies to standard projects.

7.7 Exporting Filtered Documents

Document filtering (or "cleaning") is the removal or conversion of items from a document to assist a textual analysis. This includes items that are not meant to be analyzed, such as:

- Headers
- Footers
- Captions
- Copyright notices
- Citations
- (Short) list items

Also, the romanization of items that may cause issues with legacy programs:

- Non-English letters (e.g., *é*)
- Advanced punctuation (e.g., —)
- Non-sentence-ending periods (e.g., ellipses and abbreviations)

The first set of items are normally excluded from analysis because they a.) are not part of the narrative text and b.) skew the average sentence length.

By default, *Readability Studio* will automatically ignore these items, rather than requiring you to manually remove them from your document. This is helpful by saving time and by not having to maintain two copies of your document: the original and the "cleansed" version used for analysis.

The second set of items do not cause issue with *Readability Studio*. It can recognize extended characters (e.g., *é*, —, etc.) and can properly deduce sentence endings. For these reasons, document cleaning is generally not necessary within *Readability Studio*.

Refer to sec. 20.1 to learn how *Readability Studio* automatically filters documents.

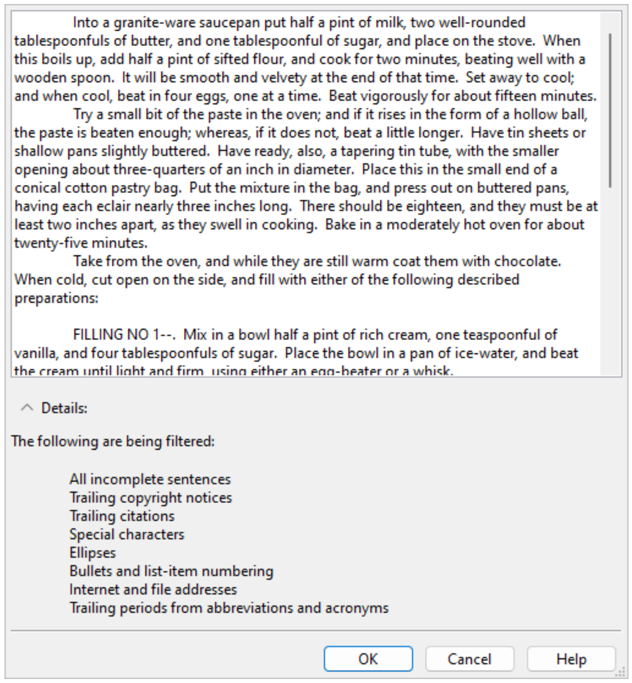
Exporting Filtered Documents

There may be times when you would need a filtered copy of your document. For example, you might need to process your document through another program which does not have auto-filtering. In this situation, you can export a copy of your document with these aforementioned items removed. Then you can analyze your document with other software without needing to manually filter it.

To do this, click on the Home tab of the ribbon, click on the arrow below the Save button, and select Export Filtered Document.

Home > Save > Export Filtered Document

After excluding incomplete sentences from your document, you will be presented with additional options (see below). Once you have specified your filtering options, the Preview Filtered Document dialog (shown on the right) will be displayed. This dialog will show a preview of the filtered document, along with listing the filtering options being used.



Finally, click OK to save the file.

> Tip
> Which incomplete-sentence removal method used is controlled by your project's text exclusion options.

7.7.1 Additional Filtering Options

When exporting a filtered version of your document, the program will first remove incomplete sentences (if applicable). This removal will be based on your project's text exclusion settings. After this text exclusion is applied, additional conversion options will be available from the Additional Filtering Options dialog. From this dialog, you can apply the following filters to the exported copy of your document.

Romanize text (replace accented and special characters): Check this option to simplify text for compatibility with other text analysis tools. This will convert accented characters, smart punctuation, bullets, and similar formatting to simpler equivalents. The following Romanization replacements are used:

Extended ASCII Character	Replacement	Extended ASCII Character	Replacement
ö	oe	ý	y
ü	ue	ÿ	y
ä	ae	•	.
ß	ss	‚	'
ñ	nn	‹	'
å	aa	›	'
æ	ae	‘	'
œ	oe	’	'
à	a	′	'
á	a	„	"
â	a	“	"
ã	a	”	"
å	a	«	"
ç	c	»	"
è	e	…	...
é	e	−	--
ê	e	–	--
ë	e	—	--
ì	i	−	--
í	i	©	(c)
î	i	®	(r)
ï	i	™	(tm)
ò	o	¼	1/4
ó	o	½	1/2
ô	o	¾	3/4
õ	o	±	+-
ø	o		
ù	u		
ú	u		
û	u		

Any extended ASCII character not listed above will be removed from the document.

Remove ellipses: Check this option to remove ellipses (if in the middle of a sentence) or replace with a . (if at the end of a sentence).

Remove bullets and list-item numbering: Check this option to remove bullet symbols and numbers that appear in front of list items.

Remove Internet and file addresses: Check this option to remove Internet URLs and file paths.

Remove trailing periods from abbreviations and acronyms: Check this option to remove the last period from abbreviations and acronyms. For example:

> Dr. Jones has requested a copy of the S.W.O.T. report.

will be exported as:

> Dr Jones has requested a copy of the S.W.O.T report.

This option is recommended for other programs that may have difficulty with deducing sentences properly because of these periods.

Narrow full-width characters: Check this option to convert full-width characters to their narrow counterparts. For example:

Ｈｅｌｌｏ

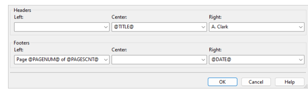
will be exported as:

Hello

7.8 Printing

To print any window, go to the Home tab and click the Print button, or right-click on the window and select Print.

Clicking the arrow beneath the Print button will open a menu containing additional printing options. For example, to change the paper size or orientation of your printouts, select Page Setup from this menu.



To add headers or footers to your printouts, select Headers & Footers... from this menu.

Headers and footers support placeholders for items such as page, date, and time information. These tags are expanded dynamically when printing takes place. These include:

- `@TITLE@`: The title of the printed document.
- `@DATE@`: The date when it was printed.
- `@TIME@`: The time when it was printed.
- `@DATETIME@`: The date and time when it was printed.
- `@PAGENUM@`: The current page number.
- `@PAGESCNT@`: The number of printed pages.
- `@USER@`: The user name currently logged into the computer.

For example, printing:

> *@PAGENUM@ (of @PAGESCNT@ pages)*

may expand to:

> *1 (of 7 pages)*

Custom values can also be entered here (e.g., the editor's name).

Finally, a watermark can be stamped across each page when you print (e.g., *CONFIDENTIAL* or your company name).

To do this, select Watermark... from the menu, enter a label, and click OK.

Note that watermarks support the following placeholders:

- `@DATE@`: The date when it was printed.
- `@TIME@`: The time when it was printed.
- `@DATETIME@`: The date and time when it was printed.
- `@USER@`: The user name currently logged into the computer.

III

Options & Features

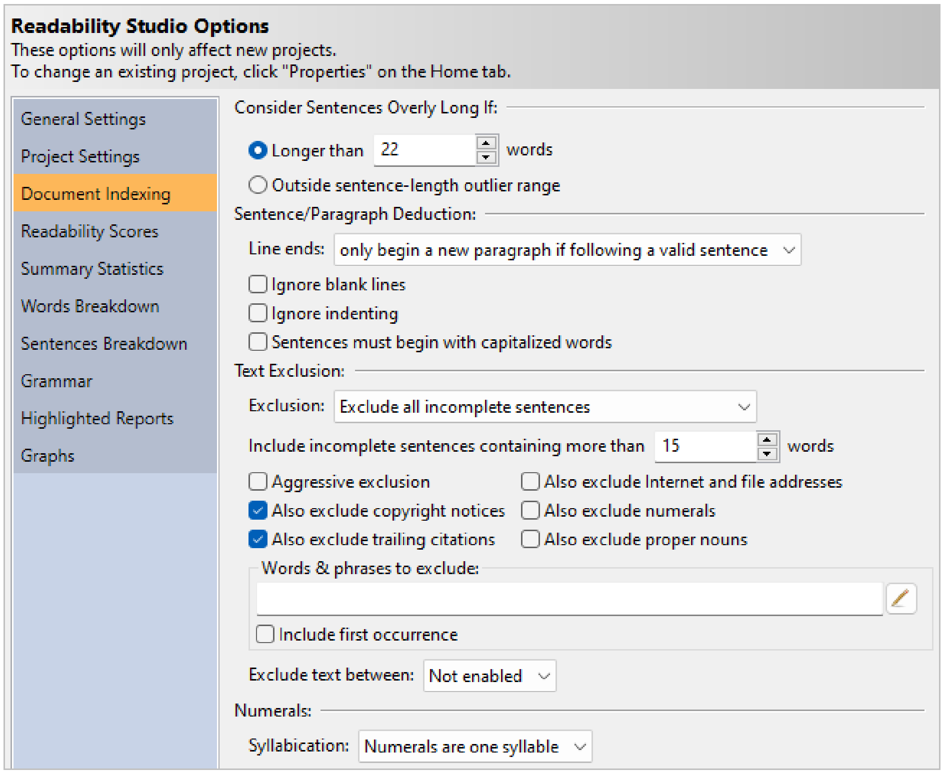
8. Program & Project Options

Readability Studio offers many customizable features, including:

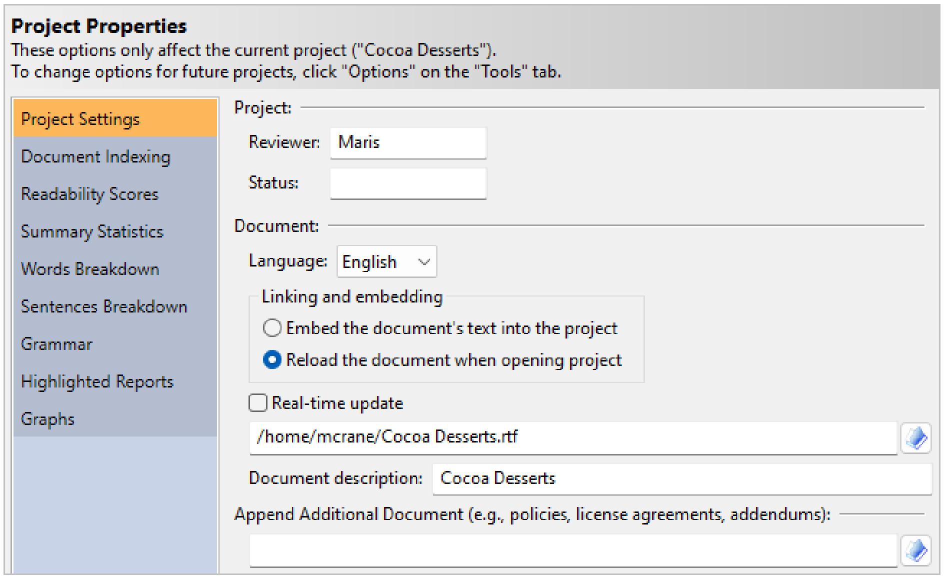
- How documents are parsed and analyzed.
- How results (e.g., statistics and graphs) are displayed.
- How projects connect to the documents they are analyzing.

These options can be configured both globally and for individual projects. Note that changing options globally will affect all future projects, but will not affect currently existing projects.

To set these options globally, click the Options button on the Tools tab to open the Options dialog:



If you wish to change options for an individual project, then click the Properties button on the Home tab to open the Project Properties dialog:



8.1 General Settings

The options on this page manage the program's general settings.

Settings

Import: Click this button to import program settings and defaults from a file.

Export: Click this button to export the current program settings to a file. This includes all your settings on the Options dialog, your defaults for the New Project wizard, and your printer headers & footers.

Reset: Click this button to reset the program settings to the defaults.

Internet

User agent: Enter into this field your user agent. The user agent is the name of the client program that is sent to websites when you try to read and download pages from them.

Disable SSL certificate verification: Check this option to disable SSL certificate verification when reading and downloading webpages. This can be used to connect to self-signed servers or other invalid SSL connections.

> ⚠ Warning
> Disabling SSL verification makes the communication insecure.

Use JavaScript cookies: Check this option to scan for any cookies meant to be sent via JavaScript when reading or downloading a page. If any cookies are found, then the page will be re-read with the cookies being sent. This is useful when connecting to pages that won't load as expected unless certain cookies are sent back to the server.

> ⚠ Warning
> This will result in an additional call to read each webpage and is only recommended if JavaScript is being used to block non-browser connections.

Persist cookies during each harvest: If using JavaScript cookies, check this option to save and resend cookies from each site. As each site is harvested, any cookies from that site and previous ones will be sent. This is useful if harvesting pages from the different portions of a website that may expect the same cookies.

Note that cookies will only persist for the pages being harvested from one instance of the Web Harvester dialog. All cookies will be cleared once the dialog's session is complete.

Warnings & Prompts

Customize: Click this button to display the Warnings & Prompts Display dialog. This dialog enables you to display or suppress various warnings and prompts, including:

- Warn about projects being opened as read-only. This warning is shown when a project will be opened as read-only. If a project is read-only, then you cannot save any changes to it.
- Prompt when removing a test from a project. When removing a readability test from a project, a prompt will be shown to confirm this action.
- Warn about documents that do not contain any words. If you attempt to analyze a document containing no valid words, then you will be warned that no calculations will be performed.
- Warn about documents containing less than 20 words. If you attempt to analyze a document containing less than 20 valid words, then you will be prompted whether you wish to continue. If a document contains such a low amount of text, then most tests will produce questionable results.
- Warn about documents containing less than 100 words. This warning is shown when a document being analyzed contains less than 100 valid words. If a document contains a low amount of text, then some factors (e.g., word and syllable counts) will be standardized for some tests.
- Warn about documents that contain sentences split by paragraph breaks. This warning is shown when a document being analyzed contains what appears to be sentences with paragraph breaks in them. In this situation, the sentences will not be indexed properly, and it is recommended to fix the document.
- Warn about documents that contain long incomplete sentences that will be included in the analysis. This warning is shown when a document being analyzed contains sentences that, although missing terminating punctuation, will be included in the analysis because of their length. To change this behavior, adjust the option Include incomplete sentences containing more than [15] words.
- Prompt if a document should switch to include sentences in the analysis. If a document is excluding a high volume of incomplete sentences, then you will be prompted about switching to include those sentences in the analysis.
- Prompt if a custom NDC test's proper noun settings differ from the standard NDC test. By default, New Dale-Chall requires that the first instance of each proper noun be seen as unfamiliar. If you are creating a custom New Dale-Chall test which uses different proper-noun logic, then you will be prompted whether this is your intent.

- Prompt about New Dale-Chall using a different text exclusion method from the system default. By default, New Dale-Chall uses its own method for excluding text, which may differ from the system default's method. This message will be shown to warn about this.
- Prompt about Harris-Jacobson using a different text exclusion method from the system default. By default, Harris-Jacobson uses its own method for excluding text, which may differ from the system default's method. This message will be shown to warn about this.
- Warn when a custom test's numeral settings will be adjusted. Some tests (e.g., Harris-Jacobson) require numerals to be excluded from the analysis. This warning will be shown if a custom test needs its numeral settings adjusted to take such rules into account.
- Warn about German stemming not supporting proper noun detection. This warning is shown when creating a custom test that uses German stemming and treats proper nouns as familiar.
- Warn about unique-value histograms requiring midpoint axis labels. If you specify unique values as your histogram bin sorting method, then midpoint interval display will need to be used. Because each bin (i.e., bar) will represent only one value, the histograms will not have ranges of values for its intervals. Therefore, cutpoints will not be applicable and midpoint axis labels will be used instead. If you have cutpoints as the interval type and choose unique-value bin sorting, then this warning will be shown before adjusting the interval display.
- Prompt about auto-searching for missing files. By default, projects link to the documents that they analyze. When you open a project, the document will be reanalyzed to include any recent changes. If the document cannot be found, then the program will ask if it should search for a document by the same name. This search will be relative to the project file's location. Unchecking this option will suppress this prompt and the program will automatically search for and load the document.
- Prompt about re-linking to a document that has been embedded. By default, a project will link to the document it is analyzing. When you click the Edit Document button (on the Home tab), the document will be opened in its default editor. However, if you have chosen to embed the document into the project, then this prompt will appear. It will ask you whether you want to change this setting to link to the document and then edit it, or continue embedding the document and then edit the embedded text. Note that if you continue to embed the document and edit the embedded text, then your changes will only be reflected in your project—your changes will not appear in the original document.
- Prompt about failing to load a project that is missing its embedded text. When opening a project which embeds the document's text, then an error message will be shown if the embedded text is not found within the project.
- Check if ClearType is turned on. If ClearType is turned off, then the program will ask you about enabling it. ClearType makes fonts smoother and easier to read. This applies to the Windows® edition of *Readability Studio* only.
- Prompt about whether to set the application's word exclusion list from a project. When adding a word exclusion list from the ribbon for the first time, you will be prompted about setting this list to be included in all future projects.

The following are informational messages shown interactively in the program:

- Prompt about how windows can be exported from the Save button.
- Prompt about how double-clicking a test can show its help.
- Prompt about how background images will not be upscaled beyond their original size when zooming into a graph.
- Prompt about how settings are embedded in projects and how to edit them.

Log

Verbose logging: Select this option to include extended information in the log report.

Append daily log: Select this option to append to the previous log report (when running the program on the same day). The default behavior is to clear the log file every time the program starts.

Developer

Show Developer tab: Select this option to show the Developer tab on the ribbon. The Developer tab provides access to the Lua script editor, which can be used to automate tasks and run custom analyses. Uncheck this to hide the tab if you do not use these features.

Allow Lua scripts to access system commands: Select this option to enable the `io`, os, and debug libraries in the Lua scripting environment. By default, these libraries are disabled for security reasons.

> ⚠ Warning
> Enabling this allows the use of Lua's built-in file I/O, system commands, and debugging features. Only enable this if you trust the scripts you are running.

> ⚠ Warning
> This setting only affects Lua's own standard libraries. Lua scripts can still read or write files through functions provided by this application, regardless of this option.

> Note
> Changing this will take effect the next time the program is started.

8.2 Project

The options on this page customize your general project settings.

Project

Reviewer: Enter your name in this field as it should appear in exported reports.

Status: Enter in this field the status of the document(s) that this project is reviewing (e.g., *DRAFT*). This information will appear in reports when you export all.

Batch options

Minimum document word count: Enter in this field the minimum number of words a document must have to be included in a batch. Any document which does not meet this threshold will be removed from a batch during the analysis.

File path display:

Partially truncate the file path: Select this option to display the longer file paths with their folder names replaced by ellipses in all of the lists.

Show only the file name: Select this option to just display the file names in all of the lists.

Show the full file path: Select this option to display the full paths of the files in all of the lists.

Document

Language: Select from here the language of the document(s) that you will be analyzing. The language selected here will affect how syllables are counted and control which tests, word lists, and grammar features will be made available. For example, if Spanish is selected, then only Spanish tests will be available and the document will be analyzed using the Spanish syllable counter.

> Linking and embedding
>
> Embed the document's text into the project: Select this option to save the source document's text directly into the project. When you open this project, it will not reanalyze your document. If your document has been edited, then those changes will not be reflected in this project.
>
> This is useful for analyzing a snapshot of your work. For example, you could analyze an early draft of your work and have it embedded in the project. As you edit your work, opening this project will still show the results from the original draft. From here, you can create a new project for your revised work and compare its results against your early draft.
>
> Reload the document when opening project: Select this option to reload and reanalyze the source document the next time you open the project. Note that the source document's content will not be stored in the project. If the document cannot be found, then only the results from the previous analysis will be shown.

Real-time update: Select this option to automatically reload the project as the source document is edited externally. This allows for editing the document in another program (e.g., *Word*) and having the results update in *Readability Studio* in real time. (This is only relevant for standard projects and for when the project is linked to a local document.)

> Note
> The changes will take effect in *Readability Studio* only after you save the document from the other program. Real-time update works by checking the modified time on the document every five seconds; if the document's timestamp has changed, then it will be reloaded.

Document description: Enter into this field a description of the document. Note that for some file formats, the program will pre-fill this field with the document's subject or title information.

Append Additional Document (e.g., policies, license agreements, addendums)

Enter into this field an additional document to append to the regular document. This document will be appended to the original document's text (or embedded text) during the project's analysis. This is useful if your project is analyzing a document, but also needs to include a generic template (e.g., license agreement). For example, you could specify a generic insurance policy in this field, and a customized entitlement as the main document. When the project analyzes the entitlement, the policy will be appended to the entitlement (in memory) and both will be analyzed as one document.

Note that for batch projects, this document will be appended to each document in the batch.

Also note that your original document is not affected by this process. The original and additional documents are only combined within *Readability Studio* while it analyzes them.

8.3 Document Indexing

The options here specify how your documents should be parsed and analyzed.

Consider Sentences Overly Long If

Longer than [22] words: Select this option to consider sentences overly long if they exceed the word count specified in this field. The default word count is 22, but can be changed as needed.

Considering sentences with more than 22 words to be overly long is based on research from Kincaid (CRES 19).

Outside sentence outlier range: Select this option to consider sentences overly long if they are comparatively longer than other sentences in the document. This option is only recommended for situations where most of the sentences have an acceptable length, but a few (very) long sentences skew the average. This option is useful for finding these extreme sentences.

These sentences can be viewed in the Grammar and Long Sentences windows in the Grammar section.

Sentence/Paragraph Deduction

Line ends

The following options control how line ends affect how paragraphs are counted.

only begin a new paragraph if following a valid sentence: Select this option to only consider a line feed as the end of the current sentence and paragraph if it follows a valid sentence. This option is recommended for text that is formatted to a specific width (e.g., scanned documents).

Two exceptions are made here. The first exception is for consecutive line feeds–these will be considered the start of a new paragraph. Consecutive line feeds normally indicate a title or chapter header that should be separated from the following paragraph.

The second exception is if the line following a line feed is tabbed or bulleted–this will also indicate the start of a new paragraph. In narrative text, the first sentence of a paragraph is usually tabbed or spaced over. Also, items in a list are usually tabbed or bulleted. In both cases, the program will recognize these as new paragraphs.

Both of these exceptions can be overridden by the options Ignore blank lines and Ignore indenting.

always begin a new paragraph: Select this option to always consider line feeds in the text as the end of the current sentence and paragraph. This should only be used for special cases where a document does not have paragraph indenting and contains numerous lists or headers that are not followed by blank lines.

Refer to sec. 20.3 for more information about these options.

Ignore blank lines: Select this option to not consider incomplete sentences followed by blank lines as the end of the current sentence and paragraph. If an incomplete sentence is encountered, any blank lines after it will be skipped and the sentence will be merged with the following sentence.

This option is not recommended for documents that contain headers or lists that rely on blank lines to separate them from the narrative text. Refer to sec. 20.3 for more information.

Ignore indenting: Select this option to not consider indented lines following incomplete sentences as the start of a new paragraph. This option is recommended for documents that center or left align their text.

> Note
> The options Ignore blank lines and Ignore indenting are disabled if Line ends always begin a new paragraph is selected. In this case, line feeds will always indicate the end of a paragraph, whether or not it is followed by blank or indented lines.

Sentences must begin with capitalized words: Select this option to have sentence deduction assume that all sentences begin with capitalized words. By default, *Readability Studio* takes into consideration writing that does not capitalize the first word of every sentence. To change this behavior, select this option and the program will be more grammatically strict when deducing sentences.

8.3.1 Text Exclusion

> Exclusion
>
> These options control how text is excluded from your documents.
>
> Do not exclude any text: Select this option to treat all incomplete sentences as regular sentences when calculating statistics. This is only recommended for forms that may contain numerous titles and list items and little narrative text. Refer to sec. 20.8 for more information.
>
> Exclude all incomplete sentences: Select this option to ignore all incomplete sentences when calculating statistics. This option is recommended by most tests.
>
> Exclude all incomplete sentences, except headings: Select this option to ignore all tables and list items, but include headers and subheaders as full sentences.

Include incomplete sentences containing more than [15] words: If a sentence is missing terminating punctuation (e.g., . or ?) then it will be considered incomplete. Lines of text missing this punctuation are generally headers or list items that should be excluded from the analysis. However, there are times when a legitimate sentence is simply missing this punctuation (due to writer or formatting error) and should be included in the analysis. To override this, enter into this field the maximum words that an incomplete sentence can have. Any sentence containing more than this number of words will be considered complete, even if its terminating punctuation is absent.

> Note
> Lines of text that resemble equations will continue to be marked as incomplete. Also, if Aggressively exclude is enabled, then list items with more words than this threshold will also continue to be excluded.

Note that if you are not excluding any text, this option will still take effect for tests which explicitly exclude headers (e.g., Harris-Jacobson).

Aggressively exclude: Select this option to mark any valid sentence as incomplete if a). it is between at least two incomplete sentences and b). is less than 10 words. For example:

> - Products by Industry
> - Which Product is Right for You?
> - Products Reviews

By default, the question mark would cause the second sentence to be included in an analysis. Checking this checkbox will instruct the program to consider this sentence as part of the surrounding list items and exclude it.

Finally, checking this will also affect sentences that end with colons or semicolons at the end of a line. Ones that end with semicolons will be excluded. Ones that end with colons that only contain one word will be excluded.

If Also exclude trailing citations is enabled, then checking this option will also cause citation deduction to be more aggressive. By default, only citation blocks in the lower half of the document are excluded. Enabling this option will tell the program to search for citations in the lower three-fourths of the document.

If Also exclude copyright notices is enabled, then checking this option will also cause that feature to include larger paragraphs in its analysis.

Also exclude copyright notices: Select this option to ignore copyright and trademark notices when calculating statistics. Throughout the document, any paragraph consisting of one or two sentences which has a © symbol near the beginning is considered a copyright notice. (Three-sentence paragraphs will also be examined if Aggressively exclude is enabled.)

Near the end of the document, the final paragraphs will be more aggressively reviewed. If the last two paragraphs contain one or two sentences and contain phrases such as *All rights reserved* or *copyright* or symbols such as © or ®, then they will be considered copyright notices.

Also exclude trailing citations: Select this option to ignore a citation section at the end of the document when calculating statistics. If this option is selected, then the program will search for a citation header near the end of the document and exclude any paragraphs after it. A citation header will be any header which contains 5 (or less) words and any of the following labels:

- *Anmerkungen*
- *Bibliographie*
- Bibliography
- *Bibliografía*
- Citations
- *Einzelnachweise*
- For More Information
- *Literatur*
- *Literaturverzeichnis*
- *Para Mas Informacion*
- Reference
- *References*
- *Referencias*

- Related Publications
- *Relación de los Trabajos Publicados*
- Sources
- Textbooks and Curricula
- Works Cited

> Note
> The program will stop excluding text after a *References* header if it detects a paragraph that does not appear to be a citation. This is useful for when a references section is in the middle of a document, followed by other content.

Also exclude Internet and file addresses: Select this option to ignore URLs (i.e., Internet addresses) and filepaths. Note that this applies to words inside of complete sentences, not just incomplete. For example:

> Readability Studio is available from *www.readabilitystudio.com*.

If this option is selected, then *www.readabilitystudio.com* will be excluded from every analysis. This includes sentence-length, word-length, and syllable-count calculations.

Also exclude numerals: Select this option to ignore all numbers. For example:

> Upon arrival at *5:30PM*, please be ready to cover the *$75* boarding fee.

would be analyzed as this:

> Upon arrival at ____, please be ready to cover the ____ boarding fee.

This option may be recommended for mathematical works which contain equations and numbers that may skew the results.

Also exclude proper nouns: Select this option to ignore all proper names of people, places, and things. A useful example would be medical documents that contain drug names that the readers will be familiar with. Rather than penalizing the document for containing these complex product names (that cannot be substituted with simpler words), this option will entirely remove them from the analysis.

Words & phrases to exclude: Enter the path to a text file containing words and phrases to ignore into this field. Any words or phrases from this file will be excluded from all calculations. The format for this text file is a single column, where each line is a separate word or phrase. For example:

```
ex nihilo
quantum fluctuations
particles
```

With the above exclusion list, this text:

> *Quantum Fluctuations* are fluctuations of *particles* within the quantum foam that appear *ex nihilo* and annihilate each other.

would be analyzed as this:

> ____ are fluctuations of ____ within the quantum foam that appear ____ and annihilate each other.

This would be seen as a 13-word sentence, with all the phrases from the exclusion list removed. Note that the first instance of *fluctuations* is removed from the analysis because it is part of the phrase *Quantum Fluctuations* (which is on the exclusion list). However, the second instance of *fluctuations* will be included because it is not part of the phrase that we requested to ignore. To exclude all instances of *fluctuations*, include it in the list on a separate line.

This phrase list can be edited by clicking the button next to this field, which will open the Edit Word/Phrase List dialog.

Include first occurrence: Select this option to not exclude the first occurrence of any word or phrase from your exclusion list. All remaining occurrences of these words in the document will be excluded. This is useful if you wish to only analyze the first instance of certain difficult words.

Exclude text between: Selects whether to exclude all text in between a certain pair of tags. For example, select [and] and any text inside a pair of left and right square brackets will be ignored in the analysis. Along with excluding the bracketed words, any full paragraphs and sentences wrapped inside of these tags will also be excluded (which will be reflected in the statistics). Refer to sec. 15.4 for an example.

> ⚠ Warning
> These options are not available if Do not exclude any text is selected.

8.3.2 Numerals

Syllabication

The following options control how numeric words are syllabized.

Numerals are one syllable: Select this option to have the program consider numeric words (e.g., numbers, dates, phone numbers) as monosyllabic.

Sound out each digit: Select this option to have the program sound out each digit when counting the syllables of numeric words. For example, the word *555-4778* will be considered nine syllables–read as *five-five-five-four-sev-en-sev-en-eight*. As another example, *$52* will be considered four syllables (read as *five-two-doll-ars*).

Note
These options are not applicable if Also exclude numerals is checked.

8.3.3 Edit Word/Phrase List

This dialog can edit either a word or phrase list.

File containing list: Enter the path to the list into this field.

After selecting your list file, its words/phrases will be displayed in the grid below the file-path field. In this grid, you can edit your list with the following keyboard commands:

- Del (Windows, Linux) or ⌫ (Mac)—removes the selected item(s) from the list.
- Ins—edits the selected item in the list. Double-clicking an item with the mouse will also edit it.
- Ctrl+Ins (Windows, Linux) or ⌘+Ins (Mac)—adds a new item to the list.

Buttons to add, edit, and remove items are also available above the grid.

8.4 Readability Scores

Test Options

The options on this page customize the behavior of certain tests.

Gunning Fog

Count independent clauses: Check this option to count independent clauses as separate sentences when calculating a Gunning Fog score. Independent clauses are detected by dashes, colons, and semicolons within a sentence.

Although this behavior is recommended in *The Technique of Clear Writing*, it will produce lower scores than counting sentences with the traditional method. This is especially true if your document contains run-on sentences. For example, if this option is checked, then an 80-word sentence with four clauses will be counted as four smaller sentences and will not adversely affect the score. If this option is unchecked, then this same sentence will be counted as a single 80-word sentence and will adversely affect the score. If you prefer to penalize run-on sentences, then leave this option unchecked.

This option affects the Gunning Fog, (PSK) Gunning Fog, and New Fog Count tests.

Flesch

Numeral syllabication

Select one of the following options to control how numerals are syllabized for Flesch related tests.

Count numerals as one syllable: Select this option to consider all numerals as monosyllabic.

System default: Select this option to use the system default numeral syllabication method.

This option affects the Flesch, PSK Flesch, and Amstad tests.

Note that these options are not applicable if Also exclude numerals is checked.

Flesch-Kincaid

Numeral syllabication

Select from the following options to control how numerals are syllabized for Flesch-Kincaid.

Sound out each digit: Select this option to have the program sound out each digit when counting the syllables of numeric words. For example, the word *555-4778* will be considered nine syllables–read as *five-five-five-four-seven-seven-eight*.

System default: Select this option to use the system default numeral syllabication method.

Note that these options are not applicable if Also exclude numerals is checked.

Dale-Chall

Include Catholic Supplement: Check this option to include Stocker's supplementary word list for Catholic students.

Proper nouns

Select one of the following options to control how proper nouns are treated by Dale-Chall related tests:

Count as unfamiliar: Select this option to consider all proper nouns (that do not appear on the familiar-word list) as unfamiliar.

Count as familiar: Select this option to consider all proper nouns as familiar.

First occurrence of each is unfamiliar: Select this option to consider only the first occurrence of each proper noun (that does not appear on the familiar-word list) as unfamiliar.

Text exclusion

Select one of the following options to control how text is excluded in Dale-Chall related tests:

Exclude incomplete sentences, except headings: Select this option to have New Dale-Chall include headers and subheaders in its analysis, but exclude lists and tables.

Use system default: Select this option to have New Dale-Chall use the default text exclusion method used by other tests.

Note

These options affect both the New Dale-Chall and PSK Dale-Chall tests. The Text exclusion option will also affect any custom test using the following functions:

- `CustomNewDaleChall()`
- `UnfamiliarDaleChallWordCount()`
- `UniqueUnfamiliarDaleChallWordCount()`
- `FamiliarDaleChallWordCount()`

Harris-Jacobson

Text exclusion

Select one of the following options to control how text is excluded in Harris-Jacobson:

Exclude incomplete sentences, except headings: Select this option to have Harris-Jacobson include headers and subheaders in its analysis, but exclude lists and tables.

Use system default: Select this option to have Harris-Jacobson use the default text exclusion method used by other tests.

Note

The text exclusion option will affect any custom test using the following functions:

- `CustomHarrisJacobson()`
- `UnfamiliarHarrisJacobsonWordCount()`
- `UniqueUnfamiliarHarrisJacobsonWordCount()`
- `FamiliarHarrisJacobsonWordCount()`

Scores

The options on this page customize which results to include in the Readability Scores section and how grade-level scores and reading ages are displayed. Most readability tests use U.S. K–12 (+college) grade levels to represent their scores; however, *Readability Studio* can display these scores in other formats as well. For example, a score of 4 could be shown as *Key stage 2 (year 5)* if using the Key Stages grade scale.

Results

Include test-summary report: Check this option to include a summary report of the test scores in the results.

Grade Display

Grade scale: Select from this list the grade scale that you wish to use. This will affect:

- The score explanations shown in the Readability Scores window.
- The score display shown in the Readability Scores window (if long format is selected).
- Selection labels on the Fry, Raygor, and Gilliam-Peña-Mountain graphs.
- The raw and aggregated score display in batch projects (if long format is selected).
- The labels on the grade-level box plot.
- The bin labels on the grade-level histograms.

Refer to sec. 21.2 for a list of available scales.

Display scores in long format: Check this to display readability scores in long format (e.g., *2nd grade*) within the score grid.

Reading Age Display

Calculation

Reading ages are included in the Readability Scores section within Standard projects next to each grade level score. Although most tests do not include reading ages, the program can calculate them from the grade-level scores. The following options are available for customizing how these ages are calculated from their respective grade-level score:

School-year range: Select this option to display the full age range from a given grade level. For example, if a test scores at *5.3* (fifth grade), then the age range for fifth graders (i.e., *10–11*) will be used.

Note that when calculating the reading-age aggregate statistics (e.g., the average reading age), the middle point of each range is used. For example, the average for two ranges such as *10–11* and *11–12* would be 11. The middle point of *10–11* is 10.5, the middle point of *11–12* is 11.5; therefore, (10.5+11.5)/2 = 11.

Round to semester: Select this option to calculate a reading age from a grade level by rounding the grade level. For example, a grade level score of *5.3* will be seen as the first semester of fifth grade and the starting age of most fifth graders (i.e., *10*) will be used as the reading age. As another example, a grade level score of *5.6* will be seen as the second semester of fifth grade and the graduating age of most fifth graders (i.e., *11*) will be used as the reading age.

Note that reading ages do not contain floating-point precision, even if their respective grade-level scores do. This is because students' ages within the same grade can widely vary; therefore, a grade-level score's mantissa cannot accurately be carried over to a group's reading age. A grade-level score can homogeneously represent a group of students in the same grade; however, a reading age must be more generalized to accurately apply to the same group (whose ages differ).

8.5 Statistics

The options on this page customize which statistics to include in your reports and results.

Results

Formatted Report: Check this to include the statistics report in the results.

Tabular Report: Check this to include the statistics (tabular format) in the results.

Summary Report

Paragraphs: Check this to include the paragraph statistics in the report.

Sentences: Check this to include the sentence statistics in the report.

Words: Check this to include the word statistics in the report.

Extended Words: Check this to include extended word statistics (e.g., numerals) in the report.

Grammar: Check this to include grammar statistics in the report.

Notes: Check this to include the notes in the report.

Extended Information: Check this to include more verbose statistics and notes in the results. These additional results will appear in the summary report, the scores grid, and batch statistics.

Dolch Report

Coverage: Check this to include the word coverage statistics in the report.

Words: Check this to include the word statistics in the report.

Explanation: Check this to include an explanation of Dolch in the report.

8.6 Words Breakdown

The options on this page customize which results are included in the Words Breakdown section.

Results

Word Counts: Check this to include the word barchart in the results.

Syllable Counts: Check this to include the syllable histogram and pie chart in the results.

3+ Syllables: Check this to include the 3+ syllable words list in the results.

6+ Characters: Check this to include the 6+ character words in the results.

Dale-Chall: Check this to include the New Dale-Chall unfamiliar words list in the results. (If a Dale-Chall-based test is included in the project).

Spache: Check this to include the Spache unfamiliar words list in the results. (If Spache is included in the project).

Harris-Jacobson: Check this to include the Harris-Jacobson unfamiliar words list in the results. (If Harris-Jacobson is included in the project).

Custom Tests: Check this to include any custom tests' unfamiliar words lists in the results.

All Words: Check this to include a list of all words in the results.

Key Words: Check this to include a list of all uncommon and non-numeric words (in condensed form) in the results. Words with the same root (e.g., *run* and *running*) will be combined into one row.

8.7 Sentences Breakdown

The options on this page customize which results are included in the Sentences Breakdown section.

Results

Long Sentences: Check this to include the overly-long sentences list in the results.

Lengths (Spread): Check this to include a sentence-lengths box plot in the results.

Lengths (Distribution): Check this to include a sentence-lengths histogram in the results.

Lengths (Density): Check this to include a sentence-lengths heatmap in the results.

8.8 Grammar

The options on this page customize grammar analysis.

Results

The options in this section customize which features are included in the grammar analysis.

Highlighted Report: Check this to include the highlighted grammar report in the results.

Wording Errors & Misspellings: Check this to include wording errors and known misspellings in the results.

Possible Misspellings: Check this to include possible misspellings in the results.

Repeated Words: Check this to include repeated words in the results.

Article Mismatches: Check this to include article mismatches in the results.

Redundant Phrases: Check this to include redundant phrases in the results.

Overused Words (by Sentence): Check this to include overused words (by sentence) in the results.

Wordy Items: Check this to include wordy phrases in the results.

Clichés: Check this to include clichés in the results.

Passive Voice: Check this to include passive voice in the results.

Conjunction-starting Sentences: Check this to include conjunction-starting sentences in the results.

Lowercased Sentences: Check this to include lowercased sentences in the results.

Spell Checker

The options in this section select what the spell checker should ignore.

Ignore proper nouns: Check this to not spell check proper nouns. This is useful for excluding names and places that the spell checker may not recognize.

Ignore UPPERCASED words: Check this to not spell check uppercased words (e.g., *FORTRAN* or *S.C.U.B.A.*).

Ignore numerals: Check this to not spell check numerals. Numerals are words that consist of 50% (or more) numbers.

Ignore Internet and file addresses: Check this to not spell check file paths and website addresses.

Ignore programmer code: Check this to not spell check words with mixed upper and lower case words. This is useful for excluding programming syntax which may appear in computer-related documents. For example:

> Projects can be exported via Lua like this:
> a = StandardProject("CrawfordValidation.rsp")
> a:ExportAll("Results")

Checking this option will instruct the program to not spell check words such as *StandardProject* and *ExportAll*.

Allow colloquialisms: Check this to allow colloquialisms, such as *ing* at the end of a word being replaced with *in'*. For example:

> He's been takin' that test for five hours. He ain't never goin' to finish.

Checking this option will instruct the program to not consider words such as *takin'* and *goin'* as misspellings.

Ignore social media hashtags: Check this to ignore hashtags, such as *#ReadabilityFormulasRock* or *@AskLibreOffice*.

8.9 Highlighted Reports

General

The options on this page customize the display of the text views.

Formatting

Font: Click this button to select the font used in the document viewing area.

Highlighting

Excluded text color: Click this button to select the color to highlight excluded text in the document viewing area. This is only applicable when incomplete sentences are being ignored.

Difficult words and sentences color: Click this button to select the color to highlight words and sentences considered difficult in the document viewing area.

Grammar errors color: Click this button to select the color to highlight grammar errors (e.g., repeated words and misspellings). This highlighting will appear in the Grammar document view.

Wordy items color: Click this button to select the color to highlight wordy phrases and difficult words. This highlighting will appear in the Grammar document view.

Highlight text by changing its

Background color: Select this option to highlight text by changing its background color.

Font color: Select this option to highlight text by changing its foreground (font) color.

Dolch Sight Words

The options on this page select which categories of Dolch words to highlight and how to highlight them.

Highlighting

Dolch conjunctions color: Select this checkbox to highlight Dolch conjunctions in the text report. Click the adjacent button to select the color to highlight these words.

Dolch prepositions color: Select this checkbox to highlight Dolch prepositions in the text report. Click the adjacent button to select the color to highlight these words.

Dolch pronouns color: Select this checkbox to highlight Dolch pronouns in the text report. Click the adjacent button to select the color to highlight these words.

Dolch adverbs color: Select this checkbox to highlight Dolch adverbs in the text report. Click the adjacent button to select the color to highlight these words.

Dolch adjectives color: Select this checkbox to highlight Dolch adjectives in the text report. Click the adjacent button to select the color to highlight these words.

Dolch verbs color: Select this checkbox to highlight Dolch verbs in the text report. Click the adjacent button to select the color to highlight these words.

Dolch nouns color: Select this checkbox to highlight Dolch nouns in the text report. Click the adjacent button to select the color to highlight these words.

8.10 Graphs

General

The options on this page customize your graphs.

Color Scheme: Select from here the color scheme to apply to pie charts, word clouds, and grouped histograms (in batch projects).

Background

Color: Selects the color for the graphs' background. Note that if you are displaying an image, then the image will be shown on top of this color.

Fade: Check this to apply a downward fade to the background color of your graphs. The background of your graphs will fade (top-to-bottom) starting with the color that you have selected into white.

Plot Background

Color: Selects the color for the plot area background of the graphs.

Opacity: Sets the transparency of the plot background color. A value of 255 will set the background to be fully opaque, whereas 0 will set the background to be transparent.

Image: Selects the image for the graphs' plot background. This image will be drawn within the area between x- and y-axes of the graphs.

> Effect
>
> The following options can be applied to the background image:
>
> No effect: Does not apply any effect.
>
> Grayscale: Converts the image to shades of gray (i.e., black & white).
>
> Blur horizontally: Applies a horizontal blur across the image.
>
> Blur vertically: Applies a vertical blur across the image.
>
> Sepia: Applies a sepia tone (i.e., faded photograph).
>
> Frosted glass: Applies a frosted glass window effect. In other words, the image as it may appear when viewed through frosted glass.
>
> Oil painting: Applies an oil painting effect.

Opacity: Sets the transparency of the background image. A value of 255 will set the background to be fully opaque, whereas 0 will set the background to be transparent.

Fit

The following options can be used to fit the image to the plot background:

Crop & center: Crop the image to the dimensions of the plot area, with the center of the image as the origin.

Shrink to fit: Shrink the image to the plot area, maintaining its aspect ratio.

Logos

Logo image: Enter the path to the image file used as your graphs' logo into this field. The logo will be displayed in the lower right-hand corner of each graph.

The logo will also have a light transparency added so that the underlying graph slightly shows through it. If the image already has transparent pixels, then those pixels will remain transparent. For example, PNG files support transparency and can be used to create logos with rounded edges.

The logo will also be scaled down to a maximum width and height of 100×100 pixels.

Note that logos will also appear when you print or save a graph.

Effects

Stipple image: Enter into this field the file path to the image used for a stipple brush. A stipple brush can be used to draw stacked images across bars and boxes.

Stipple shape: Select from here which shape to use for a stipple brush. A stipple brush can be used to draw stacked shapes across bars and boxes.

Common image: Enter into this field the file path to the image used for drawing underneath all the bars or boxes on the graphs.

Color: Selects the color used for certain stipple shapes (e.g., *Book*). Note that this color will not be applied to shapes which have their own color scheme (e.g., *Sun*).

Display drop shadows: Check this to include shadows underneath items such as boxes, bars, and labels. Unchecking this option will give your graphs a flat look.

Showcase key items: Check this to draw attention to the complex (i.e., 3+ syllable) word groups on syllable histograms and pie charts (standard projects). This is done by lowering the opacity of the bars and slices for 1- and 2-syllable word groups. Also, the labels for these bars and slices will be removed. Finally, the cumulative bar for the simple-words bar group will be removed on the histogram.

Axes

The options on this page customize the axis settings of your graphs.

Fonts

x-axis: Sets the font for the x-axis labels in your graphs.

y-axis: Sets the font for the y-axis labels in your graphs.

Titles

The options on this page customize the title fonts of your graphs.

Fonts

Top titles: Click this button to select the font for the top title in your graphs.

Bottom titles: Click this button to select the font for the bottom title in your graphs.

Left titles: Click this button to select the font for the left title in your graphs.

Right titles: Click this button to select the font for the right title in your graphs.

Readability Graphs

The options on this page customize various readability graph settings.

Fry/GPM/Raygor/Schwartz Invalid Regions

Color: Selects the color for the invalid sentence/word regions.

> Raygor style
>
> Selects the layout style of the Raygor graph. These options include:
>
> Original: As the graph appeared in the original presentation (Raygor 261). This will show grade labels being closer together, abbreviated, and with a sans-serif font. Also, invalid region polygons are not included.
>
> Baldwin-Kaufman: As the graph appeared in the Baldwin-Kaufman article (149). This is the same as the original, except that it includes invalid region polygons.
>
> Modern: This is a modernized version of Baldwin-Kaufman, where the labels are serif, not abbreviated, and spread out further. The axis labels and titles are also more descriptive.

Flesch Chart

Connect points: Check this to connect the factor and score points on the Flesch chart.

Display grouped documents on syllable ruler: Check this to display document names grouped along the syllable ruler.

This will display brackets along the right-most ruler, divided into ranges of twenty. Next to these brackets, the documents that fall into them (based on syllables per hundred words) will be shown. The documents in the upper brackets will be easier, the lower ones more difficult.

This option only applies to batch projects, and only for batches containing fifty or less documents. Also, this feature will remove the legend (if grouping is in use).

Lix Gauge

Use English translations for German Lix gauge: Check this to display Schulz's (53) English translated bracket labels on the German Lix gauge.

Bar Charts

The options on this page customize your bar charts' settings.

Bar Appearance

Color: Click this button to select the color used for your bars.

Effect

Solid: Select this option to use a solid color for your bars.

Glass effect: Select this option to add a glassy look to your bars.

Color fade, bottom to top: Select this option to apply a color fade across your bars (starting from the bottom).

Color fade, top to bottom: Select this option to apply a color fade across your bars (starting from the top).

Stipple image: Select this option to draw a stack of images across your bars. The image for this brush is specified on the General page.

Stipple shape: Select this option to draw a shape across your bars. The shape for this brush is specified on the General page.

Watercolor: Select this option to draw a watercolor-like effect across the bars. This will appear as if bars are warped and look like they were filled in with watercolor paint (or a marker).

Thick watercolor: This is the same as Watercolor, but will use a more opaque brush and apply a second coat of paint.

Common image: Select this option to draw an image under the graph, with it only showing through where the bars are at.

Opacity: Sets the transparency of the bars. A value of 255 will set the bars to be fully opaque, whereas 0 will set the bars to be transparent.

Orientation

Horizontal: Select this option to display the bars horizontally.

Vertical: Select this option to display the bars vertically.

Display labels on bars: Check this to display labels on the bars.

Histograms

The options on this page customize the test score histograms included in batch projects and the Syllable Counts in standard projects.

Bar Appearance

Color: Click this button to select the color used for your bars.

Effect

Solid: Select this option to use a solid color for your bars.

Glass effect: Select this option to add a glassy look to your bars.

Color fade, bottom to top: Select this option to apply a color fade across your bars (starting from the bottom).

Color fade, top to bottom: Select this option to apply a color fade across your bars (starting from the top).

Stipple image: Select this option to draw a stack of images across your bars. The image for this brush is specified on the General page.

Stipple shape: Select this option to draw a shape across your bars. The shape for this brush is specified on the General page.

Watercolor: Select this option to draw a watercolor-like effect across the bars. This will appear as if bars are warped and look like they were filled in with watercolor paint (or a marker).

Common image: Select this option to draw an image under the graph, with it only showing through where the bars are at.

Opacity: Sets the transparency of the bars. A value of 255 will set the bars to be fully opaque, whereas 0 will set the bars to be transparent.

Binning Options

Bin sorting refers to how values (e.g., index and grade scores) are categorized into separate classes. Each class (or bin) is displayed on a histogram as a bar. The options listed here control how these bins are created and how your data are sorted into them.

Sorting

Create a bin for each unique value: Select this option to create a bin (i.e., a bar) for every unique value. If you have floating-point grade scores and are not rounding your values, then a bin for every unique floating-point value will be created. If you select this option, then it is recommended to set the rounding option to Round down.

Sort by interval: Select this option to have the program select the best bin sizes based on the range of your data. Rather than having a bin for each unique value, bins will be created for ranges of values. For example, you may have a bin representing values for grades 2–4, rather than having separate bins for 2, 3, and 4.

Sort by neat interval: Select this option to have the program select the best integral bin sizes based on the range of your data. This option is the same as Sort by interval, except that it will force the bin ranges and bin sizes to be integer values. This will prevent having ranges beginning at values such as 2.4 and will also force the bin widths into sizes such as 2 or 3 (rather than 3.2). By selecting this option, your intervals will precisely represent grade ranges (e.g., grades 2–4) and prevent any overlap between the grades in your bins (if rounding the values).

Note that if you are not rounding down your values, then scores with no mantissa (fractional part) may fall into a bin different from the same values with fractional parts. For example, if you have a bin ranging from 1–2 and another ranging from 2–3, then:

- The first bin represents values > 1 and ≤ 2
- The second bin represents values > 2 and ≤ 3

If you are not rounding down your scores, then a score of 2 will fall into the 1–2 bin, whereas a score of 2.1 will fall into the 2–3 bin. If you select this option, then it is recommended to set the rounding option to Round down. This will ensure that values such as 2 and 2.1 fall into the same bin.

Also note that if data for a particular histogram contain four or fewer unique values, then unique value binning will be used.

Grade level/index value rounding

This option controls how floating-point values are rounded when being sorted into bins. You may choose from the following options:

Round: Select this option to round grade-level scores up or down (based on their fractional part) when sorting them into the histogram's bins. Values such as 4 and 4.3 will be sorted as 4, whereas 4.5 and 4.9 will be sorted as 5.

Round down: Select this option to round grade-level scores down when sorting them into the histogram's bins. Values such as 4, 4.3, and 4.9 will be sorted as 4. This selection is recommended.

Round up: Select this option to round grade-level scores up when sorting them into the histogram's bins. Values such as 4 will be sorted as 4, whereas 4.1, 4.3, and 4.9 will be sorted as 5. This selection is not recommended and should only be used for special cases.

Do not round: Select this option to not round any grade-level scores. Scores will be sorted into the bins exactly as they appear with their fractional parts. This selection is not recommended and should only be used for special cases.

Bin Display

These options control the display of the axis labels, bar labels, and side title of the histogram.

Interval Display

Cutpoints: Select this option to display the axis bin intervals on the sides of the bars. The label on the left side of a bar will be the ending value of the previous bin. The label on the right side of the bar will be the end of the current bin.

Midpoints: Select this option to display the axis bin intervals centered under the bars. For neat intervals, the label will be the range of the bin; otherwise, it will be the midpoint. This option is recommended if using neat intervals because it may be easier to interpret.

Labels

Counts: Select this option to display the number of items in the bin as its label. This label will appear at the top of each bar.

Percentages: Select this option to display the percentage of items that fall into the bin as its label. This label will appear at the top of each bar.

Counts & percentages: Select this option to display the number of items and percentage of items that fall into the bin as its label. This label will appear at the top of each bar.

No labels: Select this option to not display bin labels.

Box Plots

The options on this page customize the test-score box plots included in batch projects and the Sentence Lengths box plot in standard projects.

Box Appearance

Color: Click this button to select the color used for the boxes.

> **Effect**
>
> Solid: Select this option to use a solid color for your boxes.
>
> Glass effect: Select this option to add a glassy look to your boxes.
>
> Color fade, left to right: Select this option to apply a color fade across your boxes (starting from the left).
>
> Color fade, right to left: Select this option to apply a color fade across your boxes (starting from the right).
>
> Stipple image: Select this option to draw a stack of images across your boxes. The image for this brush is specified on the General page.
>
> Stipple shape: Select this option to draw a shape across your boxes. The shape for this brush is specified on the General page.
>
> Watercolor: Select this option to draw a watercolor-like effect across the boxes. This will appear as if boxes are warped and look like they were filled in with watercolor paint (or a marker).
>
> Common image: Select this option to draw an image under the graph, with it only showing through where the boxes are at.

Opacity: Sets the transparency of the box. A value of 255 will set the box to be fully opaque, whereas 0 will set the box to be transparent.

Box Options:

Display box & whisker labels: Check this to display labels on the middle points, upper/lower control limits, and whiskers.

Connect middle points: Check this to display a line connecting the middle points of each box. This only applies to plots with multiple boxes.

Show all data points: Check this to display all data points on the box and whiskers. If this is unchecked, then only outliers will be shown.

Box coefficient: Enter the coefficient here to specify the percentiles range used for the upper and lower control limits. For example, a coefficient of 25 will use the 25th and 75th percentiles range (i.e., the quartiles range).

9. Additional Features

This chapter discusses additional features not directly related to readability. This includes features such as crawling and downloading documents from a website, as well as logging and command line options.

9.1 Web Harvester

The Web Harvester dialog enables you to gather (and optionally download) links from a website. To access this dialog, select the Tools tab on the ribbon and click the Web Harvest button under Tools & Settings.

Harvesting

Websites to Harvest: Enter the base webpage(s) to begin your search into this list.

Depth Level: Enter the depth of the website that you want to crawl. A level of 1 will gather only the links from the main page, whereas a level of 2 will gather all the links from the main page and then gather all links from those pages. A level of 0 will only download the base webpage(s), not crawling it or gathering any other links from it.

File types to include: Select from this list the types of documents to include from the website.

User agent: Enter into this field your user agent. The user agent is the name of the client program that is sent to websites when you try to read and download pages from them.

Disable SSL certificate verification: Check this option to disable SSL certificate verification when reading and downloading webpages. This can be used to connect to self-signed servers or other invalid SSL connections.

> ⚠ Warning
> Disabling SSL verification makes the communication insecure.

Use JavaScript cookies: Check this option to scan for any cookies meant to be sent via JavaScript when reading or downloading a page. If any cookies are found, then the page will be re-read with the cookies being sent. This is useful when connecting to pages that won't load as expected unless certain cookies are sent back to the server.

> ⚠ Warning
> This will result in an additional call to read each webpage and is only recommended if JavaScript is being used to block non-browser connections.

Persist cookies for current sites: If using JavaScript cookies, check this option to save and resend cookies from each site. As each site is harvested, any cookies from that site and previous ones will be sent. This is useful if harvesting pages from the different portions of a website that may expect the same cookies.

Note that cookies will only persist for the pages currently being harvested from this dialog. All cookies will be cleared once this harvesting session is complete.

Log broken links: Check this option to check for broken links while harvesting and save them to the log report.

> ⚠ Warning
> Checking this option will degrade performance as each web link must be connected to and verified.

Domain Restriction

Domain Restriction

Select from this list the domain restriction that you want to use.

Not restricted to any domain: Select this option to harvest all links from the website, regardless of domain.

Domain restricted: Select this option to only harvest links with the same domain as the webpage you specify (e.g., *company.com*).

Subdomain restricted: Select this option to only harvest links with the same subdomain as the base webpage (e.g., *sales.company.com*).

Restricted to user-defined domain(s): Select this option to only harvest links from a provided list of domains. If folders are included in a specified domain, then the harvesting will be restricted to that folder. For example, if *www.news.mybusiness.com/articles* is specified, then only files from the folder *articles* in the subdomain *www.news.mybusiness.com* will be harvested.

Restricted to external domains: Select this option to only harvest links from domains other than the webpage that you specify.

Restricted to same folder: Select this option to only harvest links from the same web folder as the base URL. For example, if your base webpage is *www.mybusiness.com/history/aboutus.html*, then links will only be harvested from *www.mybusiness.com/history/* (and any of its subdirectories).

User-defined Domain(s): Enter all user-defined domains in this list. This option is only applicable if domain restriction is set to user-defined domains.

Download

These options enable you to download the files from the website prior to analyzing them. Rather than analyzing the files directly from the website, the application will analyze the local files. If file linking is enabled, then the project will link to these local files instead of the website.

This option is recommended if you need to make edits to the files before analyzing them. It may also be recommended to improve performance, as analyzing local files will be faster than reading them from a website.

If you intend to review a website and need to always analyze its latest changes, then it is not recommended to download the website. Rather, you should analyze the website directly and enable file linking. To directly analyze the website, uncheck the option Download files locally.

Download files locally: Select this option to download the files from the website locally.

Folder to download files to: Enter in this field the folder path where the website's files should be downloaded.

Minimum file size to download (in Kbs.): Enter into this field the minimum size that a file must be for it to be downloaded.

Use website's folder structure: Select this option to organize the downloaded files into a folder structure similar to the website's folder structure. Deselecting this option will download the files into a flat folder structure.

> ⚠ Warning
> If you deselect this option, then it is recommended that you also deselect Replace existing files. Downloading all files into the same folder will risk having conflicting file names; deselecting Replace existing files will help with not overwriting these files.

Replace existing files: Select this option to overwrite existing files as they are downloaded. Deselecting this option will rename files as they are downloaded (if necessary) to avoid overwriting existing files.

9.2 Command Line Options

The following command line options are available (⊞, 🐧):

Command	Description
`"filename"`	Opens *filename*, which should be a path to a *Readability Studio* project.
`-logging=n`	This parameter specifies whether to enable logging log while the program is running, where *n* is 0 or 1 (0 = disable, 1 = enable).

The following example would open the document *PatientAgreement.rsp*:

```
readstudio "C:\\users\\kdaly\\Documents\\PatientAgreement.rsp"
```

This example would tell the program to not log any information while running:

```
readstudio -logging=0
```

Note that logging can also be customized from the General page of the Options dialog or the Log Report dialog.

9.3 Log Report

While it is running, *Readability Studio* logs various diagnostic information. This information can provide details such as:

- The location of where the program is loading its license and settings files
- Parsing errors encountered while reading documents (e.g., malformed HTML files)
- Warnings and errors encountered while performing analyses
- The paths of documents being analyzed (verbose mode only)
- Information about the operating system (e.g., OS version, default font, etc.)
- Version information about various libraries used by the program

To view this diagnostic information, go to the Tools tab on the ribbon, then click Log Report under the Tools & Settings section. From the Log Report dialog, you can review, save, and print this information.

The messages in this report are displayed such that:

- General messages have a plain background
- Warnings have a yellow background
- Errors have a red background

This information can be useful if you are experiencing issues with loading a document.

9.4 Batch Importing

Select Labeling

When adding documents to a batch project, you will be prompted about how to label them. Labeling is a useful way of either grouping your documents or applying a description to them.

After selecting a group of documents, you will be presented with this dialog to specify how labeling should be done. On this dialog, you can select whether to use your documents' descriptions as their labels, or to use a single label to group them by.

> Use documents' descriptions (will be loaded during import): Select this to use each document's own description as its label. This is useful when document names may not be descriptive enough; this will show the document's description next to its name.
>
> Use a grouping label: Select this to use your provided label as a grouping label for the selected documents. This is useful when you want to cluster the documents in you batch into subgroups. (These subgroups are visualized on various graphs, such as Fry and the histograms.)
>
> Use the last common folder between files: Select this to group documents by folder. The paths between each pair of files will be compared, and the last folder that they have in common will be their group.

Refer to sec. 18.2 and sec. 18.3 for examples of this feature.

Select Directory

This dialog enables you to select a folder of documents for batch analysis.

Folder path: Enter the path to the folder of documents that you want to include.

File types to include: Select from this list the types of documents to include from the folder.

Search directories recursively: Check this option to include files from all subfolders inside of the selected folder.

Select Archive File

This dialog enables you to select an archive file (e.g., a compressed *.zip file) that contains your documents for batch analysis.

Folder path: Enter the path to the archive file that you want to include.

File types to include: Select from this list the types of documents to include from the archive. All files in the archive matching these document types will be included in the analysis.

Worksheet Preview

When importing a spreadsheet into a batch, this dialog enables you to specify which worksheets and cells to import. Note that only text cells can be imported as documents, so numeric cells will be blank here.

Import:

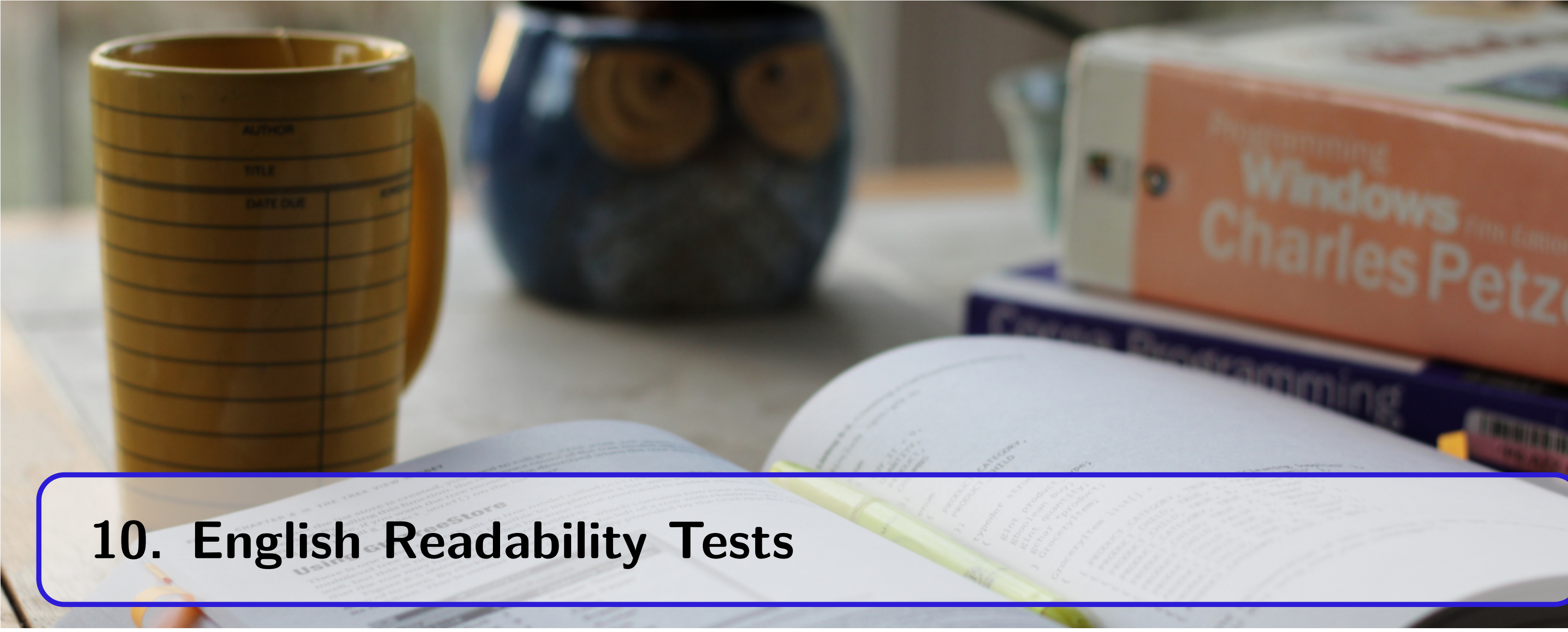
All text cells: Select this option to import the entire worksheet.

Only highlighted cells: Select this option to only import the selected cells (which will be highlighted blue). A block of cells can be selected by holding down the left mouse button and dragging the cursor around the block. Individual cells can be selected by holding down the Ctrl (⊞, 🐧) or ⌘ () button on the keyboard and clicking the cells. Also, entire columns and rows can be selected by clicking the column header or row label, respectively.

Note that selections can also be discontiguous. For example, if you hold down the Ctrl (⊞, 🐧) or ⌘ () button and click the column headers *A* and then *D*, then these two columns will be imported.

IV

Readability Tests



10. English Readability Tests

10.1 Automated Readability Index

The Automated Readability Index–also called "ARI" or "auto"–(E. A. Smith and Senter) was originally created for U.S. Air Force materials. As such, it was designed for technical reports and manuals. This test calculates the grade level of a document based on sentence length and character count.

Kincaid et al. (3, 6–15) also used this test for evaluating peer-prepared reading materials for three studies involving:

- 33 remedial high school students
- 22 graduate students from Southern Georgia College
- 14 trainees participating in a Public Service Careers project at Southern Georgia College

They noted that for the high school study, the materials with lower ARI scores yielded better comprehension rates "as reflected in cloze scores" (3). The graduate students and trainees from the other studies stated that the materials may not be appropriate for remedial high school readers (13, 15), but did agree that the stories were "authentic" and "interesting."

ARI formula

$$G = (4.71 * (RP/W)) + (0.5 * (W/S)) - 21.43$$

Where:

G	Grade level
W	Number of words
RP	Number of strokes[a]
S	Number of sentences

[a] Strokes are characters and punctuation (except for sentence terminating punctuation)

10.2 Bormuth Cloze Mean

The Bormuth Cloze Mean (Machine Computation for Passage Readability) Formula ("Readability: A New Approach" 79–132) calculates a predicted cloze score for a document.

This test is influenced by three variables (word length, sentence length, and word familiarity). Because it uses both word familiarity and syntactic complexity for its analysis, it is often recommended for student documents.

For this formula, a study involving 2,600 students ranging from 4th-12th grade (Development of Readability Analyses 97) was conducted. The students completed 330 110-word cloze passages, and their scores were correlated against numerous linguistic factors from the passages. After a series of factor analyses and multiple regressions, a series of formulas were created to predict cloze and grade-level scores for a passage. These formulas were then cross-validated against 20 250–300 word passages (105). These formulas correlated with between $r^2 = 0.81$ to 0.83 with the original test data and between $r^2 = 0.92$ to 0.93 with the cross validation data.

Of these formulas, the predicted cloze formula is as follows:

Bormuth Cloze Mean formula

$$\begin{aligned} C =& .886593 - .083640 * (R/W) + .161911 * (D/W)^3 - \\ & .021401 * (W/S) + .000577 * (W/S)^2 - .000005 * (W/S)^3 \end{aligned}$$

Where:

C	Estimated cloze score
R	Number of characters
W	Number of words
D	Number of familiar Dale-Chall words
S	Number of sentences

Note
Cloze scores returned by this test should be in percentage format (e.g., 75) rather than fractional format (e.g., .75) to be consistent with other tests.

10.3 Bormuth Grade Placement (GP35)

The Bormuth Grade Placement (Machine Computation for Passage Readability) Formula ("Readability: A New Approach" 79–132) calculates a grade level for a document.

This test is influenced by three variables (word length, sentence length, and word familiarity). Because it uses both word familiarity and syntactic complexity for its analysis, it is often recommended for student documents.

For this formula, a study involving 2,600 students ranging from 4th-12th grade (Development of Readability Analyses 97) was conducted. The students completed 330 110-word cloze passages, and their scores were correlated against numerous linguistic factors from the passages. After a series of factor analyses and multiple regressions, a series of formulas were created to predict cloze and grade-level scores for a passage. These formulas were then cross-validated against 20 250–300 word passages (105). These formulas correlated with between $r^2 = 0.81$ to 0.83 with the original test data and between $r^2 = 0.92$ to 0.93 with the cross validation data.

Of these formulas, the grade-placement formula is as follows:

Bormuth Grade Placement formula

$$\begin{aligned} G =& 3.761864 + 1.053153 * (R/W) - 2.138595 * (D/W)^3 + \\ & .152832 * (W/S) - .002077 * (W/S)^2 \end{aligned}$$

Where:

G	Grade level
R	Number of characters
W	Number of words
D	Number of familiar Dale-Chall words
S	Number of sentences

Note that this particular grade-placement test uses the regression equation that best fit the passages with cloze scores averaging ~35%. There are also 45% and 55% cloze-score fitting Bormuth equations, but as Bormuth noted "35 per cent on a cloze readability test tentatively seemed to represent a criterion for deciding whether or not students are able to exhibit a maximum of information gain as a consequence of reading the passage" (102).

10.4 Coleman-Liau

The Coleman-Liau formula calculates the grade level and estimated cloze score of a document based on sentence length and character count.

Coleman and Liau noted the difficulties of using computers to count syllables (283) for readability formulas. As a solution, they presented a new formula that used character counts as an alternative to syllable counting. This formula was derived from a regression equation trained on the 36 150-word passages from the Miller and Coleman cloze study (851). During validation, the factors of sentences and characters per 100 words yielded a strong correlation of $r^2 = .92$ against Miller and Coleman's cloze scores.

Coleman-Liau calculates the predicted cloze percentage with the following formula:

Coleman-Liau formula

$$E = (141.8401 - (.214590 * R)) + (1.079812 * S)$$

Then convert the cloze percentage into a grade level with the following formula:

$$G = (-27.4004 * (E/100)) + 23.06395$$

Where:

E	Estimated cloze %
G	Grade level
R	Number of characters (per 100 words)
S	Number of sentences (per 100 words)

Though not explicitly mentioned in their article (M. Coleman and Liau 284), the estimated cloze % from the first equation must be converted to a fraction (i.e., divided by 100) to produce the expected results. For this reason, "$(E/100)$" is shown in the second equation, rather than just "estimated cloze %" as it appears in the source article. As an example, an estimated cloze % of 51.3 should yield a grade level of 9:

$$9 = (-27.4004 * (51.3/100)) + 23.06395$$

10.5 Danielson-Bryan 1

Danielson-Bryan 1 (206) is meant for student materials and calculates the grade level of a document based on sentence length and character count.

DB1 was designed to be a faster method (compared to Farr-Jenkins-Paterson) of calculating readability scores on UNIVAC 1105 computers. Rather than counting syllables, DB1 counts characters and punctuation, which was simpler to analyze on computers of that era. This performance benefit is more of a historical reference now; modern computers can calculate syllable counts optimally.

This test was trained against 383 selections from the 1950 edition of McCall-Crabbs *Standard Test Lessons in Reading*. For validation, the derived formula had an $r^2 = 0.575$, similar to Farr-Jenkins-Paterson's $r^2 = 0.553$. This regression equation is as follows:

Danielson-Bryan 1 formula

$$G = 1.0364 * (RP/W) + .0194 * (RP/S) - .6059$$

Where:

G	Grade level
W	Number of words (see below)
RP	Number of characters[a]
S	Number of sentences

[a] In this context, characters are letters, numbers, and punctuation (except for sentence-ending punctuation)

In the original article, the statistic of "number of spaces between words" is suggested, rather than explicitly saying "number of words." Logically, this should be the number of words minus 1. However, in the article's example, number of spaces between words is the same as number of words (even when there are dashes connecting words). It is therefore assumed that the authors' intention was to count the number of words using the following logic:

- Count the number of spaces.
- Treat dashes connecting words as spaces.
- Add 1, considering how there would be one less space than words.

For this reason, the *number of words* statistic is a more accurate description of what the authors intended.

10.6 Danielson-Bryan 2

Danielson-Bryan 2 (206) calculates an index score of a document based on sentence length and character/punctuation count. It is a variant of Flesch Reading Ease, with the difference being that it uses character and punctuation counts instead of syllable counts. Being a variant of Flesch, its scores range from 0–100 (the higher the score, the easier to read), with average documents being within 60–70.

DB2 was designed to be a faster method (compared to Farr-Jenkins-Paterson) of calculating readability scores on UNIVAC 1105 computers. Rather than counting syllables, DB2 counts characters and punctuation, which was simpler to analyze on computers of that era. This performance benefit is more of a historical reference now; modern computers can calculate syllable counts optimally.

This test was trained against 376 passages from *McCall-Crabbs Standard Test Lessons in Reading* (1926 ed.). For validation, the derived formula had an $r^2 = 0.575$, similar to Farr-Jenkins-Paterson's $r^2 = 0.553$. This regression equation is as follows:

Danielson-Bryan 2 formula

$$I = 131.059 - 10.364 * (RP/W) - .194 * (RP/S)$$

Where:

I	Flesch index score
W	Number of words[a]
RP	Number of characters[b]
S	Number of sentences

[a] Refer to Danielson-Bryan 1 for an explanation of why number of words is used, instead of number of spaces between words.
[b] In this context, characters are letters, numbers, and punctuation (except for sentence-ending punctuation)

Table 10.4: Danielson-Bryan 2 (Flesch Reading Ease) Index Table

Flesch Score	Description
90–100	Very easy, third-grade level
80–89	Easy, fourth-grade level
70–79	Fairly easy, fifth-grade level
60–69	Standard, sixth-grade level
50–59	Fairly difficult, junior high school level
30–49	Difficult, high school level
0–29	Very difficult, college level

10.7 Degrees of Reading Power (GE)

The Degrees of Reading Power (grade equivalent) (Carver 303–16) test calculates the difficulty level of a document by performing the DRP test and then converting its units score into a grade level.

This test was trained on the same 330 110-word passages used to build the Bormuth test, along with an additional 30 samples. These additional samples were 100-word passages tested by students from the graduate library at the University of Maryland (305–06). Using these samples (and their cloze and Rauding scores), Carver built a table to convert between DRP units and grade levels (see below). Next, he compared the DRP-GE results against DRP units, Rauding scale scores, and cloze scores. The DRP-GE (grade equivalency) results strongly correlated with these other scores:

- DRP: $r = 0.98$
- Rauding Scale-GE: $r = 0.84$
- cloze: $r = -0.81$

To calculate the grade equivalency, first calculate the DRP difficulty of a document. Then, look up the DRP score from the following table to find its respective grade level:

DRP-Difficulty	DRP-GE	DRP-Difficulty	DRP-GE
0–39	1.5	59	6.8
40	1.6	60	7.3
41	1.7	61	7.8
42	1.7	62	8.3
43	1.8	63	8.8
44	2.0	64	9.4
45	2.1	65	10.0
46	2.3	66	10.6
47	2.5	67	11.2
48	2.8	68	11.8
49	3.0	69	12.5
50	3.3	70	13.1
51	3.6	71	13.7
52	3.9	72	14.4
53	4.3	73	15.0
54	4.7	74	15.7
55	5.1	75	16.4
56	5.5	76	17.1
57	5.9	77	17.9
58	6.3	78–100	18.0

10.8 Degrees of Reading Power

The Degrees of Reading Power (Kibby 416–27; Carver 303–16) formula calculates the difficulty level of a document in terms of DRP units. (These units range from 0 [easy] to 100 [difficult]).

The DRP difficulty score of a document is used to match it to a student based on their DRP ability score. DRP ability tests are manual tests administered to students to gauge their reading levels.

This test is based on the Bormuth Cloze test and uses the same criteria for its calculation. Basically, this test uses the Bormuth formula to calculate a predicted cloze score, converts it to percentage format, and then inverts it to arrive at the units score.

Degrees of Reading Power formula

$$\begin{aligned} U =& 100 - 100 * (.886593 - .083640 * (R/W) + \\ & .161911 * (D/W)^3 - .021401 * (W/S) + \\ & .000577 * (W/S)^2 - .000005 * (W/S)^3) \end{aligned}$$

Where:

U	Units score
R	Number of characters
W	Number of words
D	Number of familiar Dale-Chall words
S	Number of sentences

10.9 Easy Listening Formula

The Easy Listening Formula is designed for "listenability" and is meant for radio and television broadcasts. Fang (65) stated that broadcasted news generally has much shorter sentences compared to print media. Therefore, he created the ELF test to focus solely on complex-word density, not using sentence length as a factor. That is, rather than looking at a sentence's word count, the density of syllables in a sentence is used instead.

This test was trained on 36 television scripts from the following:

- ABC
- CBS
- NBC
- Local newscasts from the Los Angeles network stations KABC-TV, KNXT, and KNBC

To compare listenability to readability, the study was also trained on 36 newspaper samples from the following:

- *The New York Times*
- *The Wall Street Journal*
- *The Christian Science Monitor*
- *The Los Angeles Times*
- *The Chicago Tribune*
- *The St. Louis Post-Dispatch*

The newscasts yielded the following ELF ranges:

- ABC: 9.9–12.0
- CBS: 9.6–11.9
- NBC: 8.7–10.7
- KNXT: 11.5–13.1
- KNBC: 10.2–12.3
- KABC-TV: 6.4–9.0

While the newspaper samples yielded the following:

- *The New York Times*: 16.8–18.2
- *The Wall Street Journal*: 13.8–15.1
- *The Christian Science Monitor*: 10.3–13.1
- *The Los Angeles Times*: 14.7–18.0
- *The Chicago Tribune*: 14.2–18.6
- *The St. Louis Post-Dispatch*: 11.3–17.1

Note the stark difference between the printed and televised-material scores. This indicates a much higher complex-word density in printed news versus televised news.

To further validate the test, Fang compared Flesch Reading Ease scores against ELF and found a strong correlation of $r = 0.96$.

The ELF score of a sentence is calculated by counting the number of syllables above one for each word. For example:

> In any sentence, count each syllable above one per word.

The above sentence contains four polysyllabic words: any (2), sentence (2), above (2), and syllable (3). In this case, *any*, *sentence*, and *above* each have 1 syllable above 1, and *syllable* has 2 syllables above one. Adding these values will yield a score of 5.

This process is repeated for every sentence, and then these scores are averaged. A simpler method is shown below:

Easy Listening formula

$$G = (B - W)/S$$

Where:

G	Grade level
W	Number of words
B	Number of syllables
S	Number of sentences

10.10 McAlpine EFLAW

The McAlpine EFLAW© formula (McAlpine) is used to determine the ease of reading English text for ESL/EFL (English as a Second/Foreign Language) readers. This test calculates the index score of a document based on sentence length and number of miniwords (i.e., words that contain three or less letters).

Where most other formulas penalize longer words, this test does the opposite, penalizing the use of miniwords. This is because it is designed for ESL readers, as McAlpine explains: "Miniwords are confusing because they have many meanings and are often a sign of wordiness or idioms. . . . these flaws bamboozle EFL readers."

To improve a score, McAlpine recommends lowering the use of miniwords and shortening all sentences to 20 words or less. She also recommends avoiding the following:

- double negatives
- negative questions (e.g., "Are you *not* currently an Ohio resident?")
- idioms/clichés (e.g., "tip of the iceberg")
- helper (auxiliary) verbs (e.g., "can," "may")
- abstract nouns (e.g., "peace," "tension")
- use of pronouns that may be unclear (e.g., "Gabi visited her mother after *she* was released from the hospital.")

McAlpine EFLAW formula

$$I = (W + T)/S$$

Where:

I	The index score
W	Number of words
T	Number of miniwords
S	Number of sentences

Table 10.8: McAlpine EFLAW Index Table

EFLAW Score	Description
1–20	Very easy to understand
21–25	Quite easy to understand
26–29	Mildly difficult
30+	Very confusing

10.11 Farr-Jenkins-Paterson

The Farr-Jenkins-Paterson readability formula (333–37) calculates the Flesch difficulty level of a document based on sentence length and number of monosyllabic words.

Farr-Jenkins-Paterson was designed as a simpler way to calculate a Flesch score because it uses monosyllabic word counting instead of counting all syllables.

This test was trained on 360 100-word samples, producing the following equation:

Farr-Jenkins-Paterson formula

$$I = -31.517 - (1.015 * (W/S)) + (1.599 * ((M/W) * 100))$$

Where:

I	Flesch index score
W	Number of words
M	Number of monosyllabic words
S	Number of sentences

For validation, Flesch and FJP scores from the aforementioned samples were compared, with a significant correlation of $r = 0.93$.

10.12 Flesch-Kincaid

The Flesch-Kincaid readability formula is Kincaid et al.'s modified version of Flesch Reading Ease (Derivation 14) that was recalculated "to be more suitable for Navy use" (11); hence, it is designed for technical documents and manuals. In addition to being a recalculation, this variation of Flesch returns a grade-level result, rather than an index score. This test calculates the grade level of a document based on sentence length and syllable count.

This test was trained against reading-test results from 531 U.S. Navy personnel from four Naval technical training schools. Using these results, the following multiple regression equation was derived:

Flesch-Kincaid formula

$$G = (11.8 * (B/W)) + (.39 * (W/S)) - 15.59$$

A simplified variation was also provided:

$$G = (12 * (B/W)) + (.4 * (W/S)) - 16$$

Where:

G	Grade level
W	Number of words
B	Number of syllables
S	Number of sentences

Initially, this test was named New Reading Ease Formula (14), but was later referred to as Flesch-Kincaid (CRES 19).

Note that this test sounds out numerals' digits when syllabizing. For example, "1918" will be counted as four syllables ("one"-"nine"-"one"-"eight"). This behavior can be adjusted from the Options dialog if you prefer a different numeral syllabizing method.

> Note
> The simplified version of this formula uses a lower precision calculation. It was originally offered for making a manual calculation of this test easier. With the modern availability of computers, it is recommended to use the higher-precision formula.

10.13 Flesch-Kincaid (Simplified)

Refer to sec. 10.12.

10.14 Flesch Reading Ease

The Flesch Reading Ease readability formula calculates an index score of a document based on sentence length and number of syllables.

Flesch Reading Ease (*Art of Readable Writing* 213; *Plain English* 23) is best suited for school textbooks and technical manuals. It is a standard used by many U.S. government agencies, including the U.S. Department of Defense. Scores range from 0–100 (the higher the score, the easier to read) and average documents should be within the range of 60–70.

Flesch Reading Ease formula

$$I = 206.835 - (84.6 * (B/W)) - (1.015 * (W/S))$$

Where:

I	Flesch index score
W	Number of words
B	Number of syllables
S	Number of sentences

Table 10.12: Flesch Reading Ease Table

Flesch Score	Description
90–100	Very easy
80–89	Easy
70–79	Fairly easy
60–69	Standard (plain English)
50–59	Fairly difficult
30–49	Difficult
0–29	Very difficult

Note
Note that this test treats numerals as monosyllabic words by default. This behavior can be changed from the Options dialog.

Along with the raw score, a chart is also available to visualize the score's meaning:

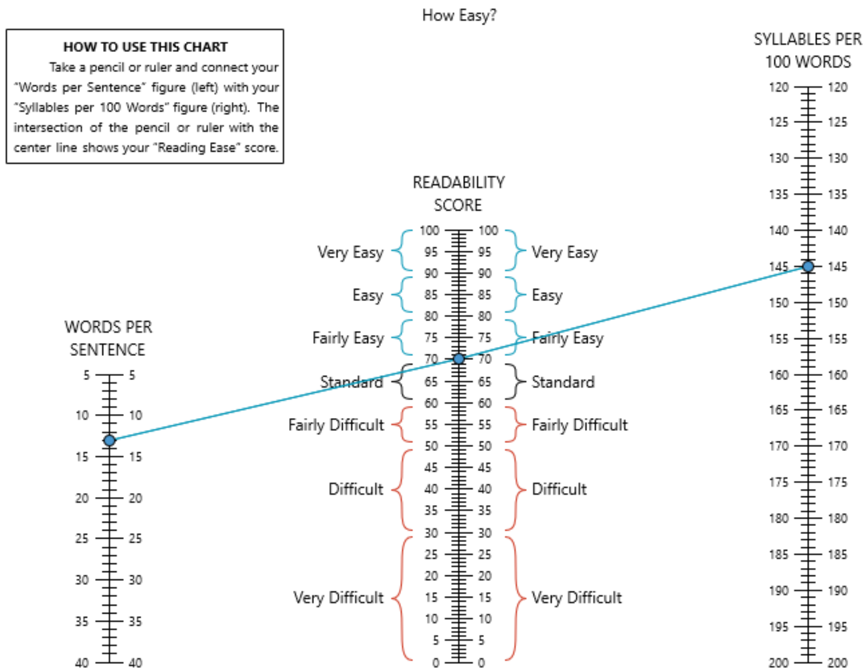


Figure 10.1: Flesch chart

This chart consists of three "rulers." The middle ruler displays the Flesch score as a red point. On both sides of this ruler are brackets showing the reading levels. The bracket that the point falls into will indicate the reading level of the document. The rulers on the left and right of the score ruler will display the factors used to calculate the score. A straight line is drawn between these points to demonstrate how the factors calculated the score.

In the above chart, we can see that the document's average words per sentence is 13 and its syllables per 100 words is 146. This yields a score of 71, which indicates that the document is "fairly easy" to read.

10.15 FORCAST

The FORCAST readability formula (Caylor et al. 13–17) was devised for assessing U.S. army technical manuals and forms. It is the only test not designed for running narrative, so it is best suited for multiple-choice quizzes, applications, entrance forms, etc. This test calculates the grade level of a document based on its number of monosyllabic words.

This formula was trained against U.S. Army "job reading material" (13), using the criterion of cloze passages where readers scored around 35% comprehension. (Most other tests usually use the 50% criterion.) The researchers initially tested regression equations using 15 variables for their new formula, but noted the single factor of monosyllabic words yielded a "sufficiently high" (15) correlation of $r = 0.86$ against the mean cloze scores. Based on this, the following formula was derived:

FORCAST formula

$$G = 20 - (M/10)$$

Where:

G	Grade level
M	Number of monosyllabic words (per 150-words)

A follow-up study for U.S. Air Force regulation materials was later administered (Hooke et al. 13–21). This study included 900 Air Force personnel across 13 bases and used the 40% cloze criterion. The researchers noted during the study that reviewers' estimated grade levels for regulation documents were generally higher than actual FORCAST scores. Based on this, the authors recommended more extensive use of this test for Air Force materials.

Note that FORCAST results may be slightly different from other tests because it does not take sentence length into account. If your document is structured mostly with tables and lists, then expect some variance between the FORCAST grade level and other tests' grade levels.

This test is designed for a 150-word sample, but normalization can be used to analyze entire documents.

Note
The name FORCAST is a pseudo acronym for its authors: J. Patrick FORd, John S. CAylor, and Thomas G. STicht.

10.16 Fry Graph

Fry (“Saves Time” 513–16, 575–78; “Clarifications, Validity, and Extension” 242–52) is a graphical test for English documents. It calculates a document’s grade level from its average number of sentences and syllables per hundred words. These averages are plotted onto a chart where their intersection determines the reading level of the content.

The original version of this graph included grade levels 1–12 and college (“Saves Time” 513–516; *Elementary Reading Instruction* 217). It was trained against samples from the following books:

- *Light in the Forest*
- *Of Mice and Men*
- *The Pearl*
- *Shane*
- *Death be Not Proud*
- *The Moon is Down*
- *To Kill a Mockingbird*
- *A Tale of Two Cities*
- *Silas Marner*
- *Act One*

The graph’s results were compared to other tests, yielding the following correlations:

- Dale-Chall: $r = 0.94$
- Flesch Reading Ease: $r = 0.96$
- Botel: $r = 0.78$
- SRA: $r = 0.98$

In 1977, Fry extended the graph to include grades 13–17 by extrapolating the averages from the preceding 3 years on the graph (“Clarifications, Validity, and Extension” 251).

The Fry graph is designed for both primary and secondary-age reading materials. Below is an example of a Fry graph:

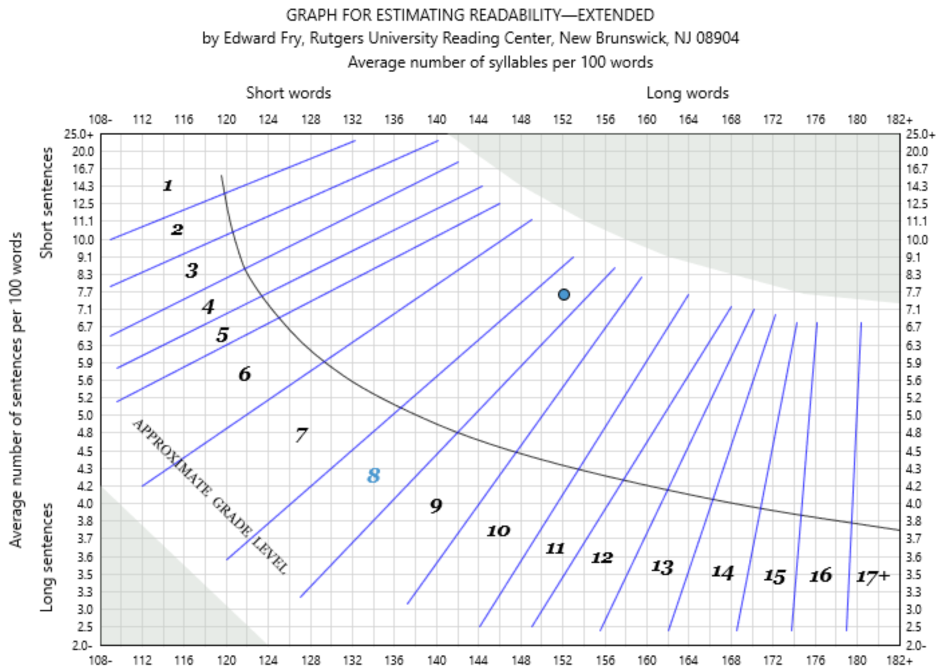


Figure 10.2: Fry graph

Note the following aspects of this graph:

- Grade levels from 1 to 17+ are represented by bands along the middle. If a score falls within one of these bands, then that will be the document's grade level.
- A curved line running through the middle of the grade-level bands. This represents the "smoothed mean of the sample passages. If you plot a large number of passages with a wide range, they will tend to fall somewhere near the line" (243).
- Shaded areas in the bottom-left and top-right corners. These represent the long-sentences and long-words regions, respectively. If a document's score falls into either of these regions, then one of these factors is making the document too difficult to fit into a grade level.

The Fry graph instructions are:

- Extract a 100-word passage from the selection. If the material is long, take subsamples from the beginning, middle, and end.
- Count the number of sentences in each passage. Count a half sentence as .5.
- Count the number of syllables in each passage.
- Find the point on the chart.
- If the sample's syllable or sentence count is too low or high for it to be plotted, then adjust that factor so that it can fit onto the chart. For example, if a sample has 187 syllables per 100 words, then this will need to be adjusted to 182.
- Repeat this process for each sample, and then average the samples.

This test is designed for 100-word samples, with 3 samples being recommended for best results. (Normalization can also be used to analyze larger samples or entire documents.)

Note that numerals are fully syllabized for this test. For example, *1945* will be counted as four syllables (*one-nine-four-five*).

10.17 Gunning Fog

The Gunning Fog readability formula (39–41) calculates the grade level of a document based on its number of sentences and complex words.

Complex words are words that contain three or more syllables, with the exception of:

- Proper nouns.
- Words made three syllables by adding *ed* or *es* (e.g., *created* or *trespasses*).
- Compound words composed of simpler words. *Readability Studio* considers hyphenated words to be compound words (e.g., *horse-power* or *and/or*). (Note that forward slashes (/) are treated like hyphens in this context.)

This test can optionally count independent clauses as separate sentences, as *The Technique of Clear Writing* instructs. It should be noted, however, that this can cause misleadingly low scores for documents that contain run-on sentences. Consider the following:

> The Meldorf® statistical library is available in C++ source code–a low-level programming language–and in the form of a COM library–a Microsoft® technology for inter-program communication; the COM library can be called from numerous languages, such as: Visual Basic, C#, and Delphi.

If you treat independent clauses as separate sentences, then this will be seen as six small sentences and produce a Fog score of 10.2. If you count sentences by only looking for ., ?, or !, then this will be counted as a single 44-word sentence and produce a Fog score of 19. To penalize run-on sentences (which is recommended), then leave the option Count independent clauses as separate sentences unchecked for this test.

Gunning Fog formula

$$G = .4 * (W/S + ((F/W) * 100))$$

Where:

G	Grade level
W	Number of words
F	Number of complex words
S	Number of sentences

10.18 Harris-Jacobson Wide Range

The Harris-Jacobson Wide Range Readability Formula test (19–30) calculates the grade level of a document based on sentence length and number of unique unfamiliar words. Words are unfamiliar if they do not appear on a list of common words that are known to most second-grade students. The following are also considered familiar:

- Regular verb forms of any common word (i.e., *ing*, *es*, *s*, *ed*, and *ies* endings). Irregular forms are not familiar.
- Regular plural and possessive forms of any common word (i.e., *s*, *es*, and *ies* endings). Irregular forms are not familiar.
- Adjectival or adverbial forms of common words. This includes forms with endings such as *ly*, *est*, *er*, and *ily*.
- Hyphenated words if both parts of the word are common words. For example, *apple-tree* and *and/or* would be familiar. (Note that forward slashes (/) are treated like hyphens in this context.) Although this rule is not explicitly stated in *Basic Reading Vocabularies*, it is recommended because both Spache and New Dale-Chall apply it.
- All proper nouns.

Note that numerals are entirely excluded from this test and are considered neither familiar nor unfamiliar.

The Wide Range Readability Formula is based on previous Harris-Jacobson formulas designed for preprimer through eighth-grade reader-level materials. The Wide Range variation was trained on a new series of basal readers (30). This series included:

Primary Levels	Secondary Levels
1 preprimer	1 fourth reader
1 primer	1 fifth reader
1 first reader	1 sixth reader
2 second readers	1 seventh reader
2 third readers	1 eighth reader

Using these new basal readers as the test's criterion, a new regression equation and expanded word list were built for the Wide Range formula. This new formula could significantly correlate samples from these books to their reported reading levels. Further, to predict readability scores for materials beyond the eighth-grade level, the lookup table used by the test (see below) was extrapolated to include secondary and adult-level materials.

> Note
> This test treats headers and subheaders as full sentences, but excludes lists and tables. The default text exclusion method will be overridden for this specific test.

Harris-Jacobson is calculated from the following steps:

- Obtain the V1 score: (Number of unique unfamiliar words / Number of words) * 100.
- Obtain the V2 score: Number of words / Number of sentences.
- Multiply V1 by .245.
- Multiply the V2 score by .160.
- Add together the result of step 3, step 4, and .642 to achieve the predicted raw score.
- Round off the predicted raw score to one decimal place.
- Find the predicted raw score in the following table to acquire the final readability score:

Predicted Raw Score	Readability Score	Predicted Raw Score	Readability Score
1.1	1.0	4.6	4.8
1.2	1.0	4.7	5.0
1.3	1.0	4.8	5.2
1.4	1.1	4.9	5.4
1.5	1.2	5.0	5.5
1.6	1.3	5.1	5.7
1.7	1.4	5.2	5.9
1.8	1.5	5.3	6.0
1.9	1.7	5.4	6.2
2.0	1.8	5.5	6.4
2.1	1.9	5.6	6.5
2.2	2.0	5.7	6.7
2.3	2.1	5.8	6.9
2.4	2.2	5.9	7.1
2.5	2.3	6.0	7.3
2.6	2.4	6.1	7.5
2.7	2.6	6.2	7.7
2.8	2.7	6.3	7.9
2.9	2.8	6.4	8.1
3.0	2.9	6.5	8.3
3.1	3.1	6.6	8.5[a]
3.2	3.2	6.7	8.7
3.3	3.3	6.8	8.9
3.4	3.4	6.9	9.1
3.5	3.5	7.0	9.2
3.6	3.6	7.1	9.4
3.7	3.7	7.2	9.6
3.8	3.8	7.3	9.8
3.9	3.9	7.4	10.1
4.0	4.0	7.5	10.3
4.1	4.1	7.6	10.5
4.2	4.3	7.7	10.7
4.3	4.5	7.8	10.9
4.4	4.6	7.9	11.1
4.5	4.7	8.0	11.3

[a] Readability scores above 8.5 were derived by extrapolation (21)

Because this formula is based on the usage of familiar words (rather than syllable or letter counts), it is often regarded as a more accurate test for younger readers.

10.19 Lix

The Läsbarhetsindex (Lix) readability formula (Björnsson 480–97) was designed for documents of any Western European language. It calculates a document's index score based on sentence length and number of long words (i.e., words containing seven or more characters). Viewing long words as those with seven or more characters is what differentiates it from similar tests (which use six or more characters). This difference is what allows it to be a more balanced "yardstick" that can be applied to numerous languages.

The philosophy behind using one formula for multiple languages is that, at their core, most languages are similar in difficulty. This approach argues that a language having various idiosyncrasies and longer words is not why it scores at more difficult levels. The reason languages score higher is because of how they are used (i.e., word choice, writing style) and presented (e.g., longer paragraphs and sentences, fewer images). This differs from the reasoning behind other tests based on adjusted English formulas. Those tests are designed under the assumption that non-English languages are inherently more difficult (because of comparatively longer words). For example, Gilliam-Peña-Mountain and the German reinterpretation of Lix follow this logic of adjusting English formulas for non-English text. Lix, however, chooses to use the same formula to gauge multiple languages. It views documents that score higher to be a result of writing style and presentation, not the language itself.

This test was originally designed for Danish, English, French, German, and Finnish and trained on "thousands of books and reading passages" (481) during 1968, 1969, 1974, 1975, and 1979. During the 1974 English study, the formula was tested with:

- 20 children's books
- 30 fiction books
- 30 non-fiction books
- 20 technical books

Other language studies were conducted later using newspaper articles as their training material. These languages included Danish, English, Finnish, French, German, Italian, Norwegian, Portuguese, Russian, Spanish, and Swedish. (Anderson conducted a separate study with Greek material in 1981.) The researchers' findings noted the following average scores across the languages:

- Swedish (47), Norwegian (48), and Danish (51) were the lowest scoring (i.e., easiest to read)
- ...followed by English (52), French (55), and German (59)
- ...then Italian (65), Spanish (67), and Portuguese (70)
- ...and Finnish (72) and Russian (65) being the most difficult

The researchers conjectured these differences had more to do with use of language than with the languages themselves (495).

The index score of Lix can be summarized as follows:

Table 10.15: Lix Index Table

Lix Score	Description
0–29	Very easy (books for children)
30–39	Easy (fiction)
40–49	Average (factual prose)

Lix Score	Description
50–59	Difficult (technical literature)
60 and above	Very difficult

Note that Lix also includes a chart to visualize the meaning of its index score:

Figure 10.3: Lix gauge

Lix formula

$$I = W/S + 100 * (X/W)$$

Where:

I	Lix index score
W	Number of words
X	Number of long words (7+ characters)
S	Number of sentences

As a final note, Anderson contributed a grade-level conversion table for Lix index scores (Anderson, "Lix and Rix: Variations of a Little-Known Readability Index" 494) as follows:

Table 10.17: Lix Grade-level Conversion Table

Lix Score	Equivalent Grade Level
56+	College
52–55	12
48–51	11
44–47	10
40–43	9
36–39	8
32–35	7
28–31	6
24–27	5
20–23	4
15–19	3
10–14	2
Below 10	1

10.20 Modified SMOG

The Modified SMOG test (L. L. Smith 129–32) calculates the grade level of a document based on sentence length and number of difficult (i.e., 3 or more syllable) words and is designed for primary and secondary-age materials.

Modified SMOG differs from SMOG in the following ways:

- It uses the unique number of difficult words (rather than the total).
- It uses a smaller constant in its formula.
- The score is rounded, rather than floored.

Because of these adjustments, it will generally produce scores closer to other tests than SMOG does—in particular, the Fry graph (129).

The author's decision not to count repeated difficult words was a major factor in making this test more appropriate for primary readers. As he explains, in a particular pre-primer book "the word 'kangaroo' appeared 52 times . . . it was the only polysyllabic word in the book. To count the word more than once would have inflated the readability score to an unrealistic level" (130).

The steps for calculating Modified SMOG are:

- Take 3 ten-sentence samples.
- Count the number of unique words containing 3 or more syllables.
- Calculate the square root of this count and round it.
- If this square root is 4 or more, then add 1. Otherwise, the square root should remain the same. This will be the grade level score.

Note that numerals are fully syllabized (i.e., sounded out) for this test, so the program overrides your numeral syllabication setting when calculating it.

This test requires a 10-sentence sample, with 3 samples being recommended for best results. (Normalization can also be used to analyze larger samples or entire documents.)

10.21 New Automated Readability Index

The New Automated Readability Index is Kincaid et al.'s modified version of ARI (Derivation 14) that was recalculated "to be more suitable for Navy use" (11); hence, it is designed for technical documents and manuals. This test calculates the grade level of a document based on sentence length and character count.

This test was trained against reading-test results from 531 U.S. Navy personnel from four Naval technical training schools. Using these results, the following multiple regression equation was derived:

New Automated Readability Index formula

$$G = (5.84 * (RP/W)) + (.37 * (W/S)) - 26.01$$

Where:

G	Grade level
W	Number of words
RP	Number of strokes[a]
S	Number of sentences

[a] Strokes are characters and punctuation (except for sentence-terminating punctuation).

A simplified variation was also provided:

$$G = (6 * (RP/W)) + (.4 * (W/S)) - 27.4$$

Note
The simplified version of this formula uses lower precision. It was originally offered to make manual calculations of this test easier. With the modern availability of computers, it is recommended to use the higher-precision formula.

10.22 New Automated Readability Index (Simplified)

Refer to sec. 10.21.

10.23 New Dale-Chall

The New Dale-Chall readability test (Chall and Dale, *Manual for the New Dale-Chall Readability Formula* 1–7) calculates the grade level of a document based on sentence length and number of unfamiliar words. Words are unfamiliar if they do not appear on a list of 3,000 common words that are known to most 4th grade students. The following are also considered familiar:

- Regular plural and possessive forms of common words (i.e., *s*, *es*, and *ies* endings). Irregular forms (e.g., *oxen*) are not familiar.
- Adjectival and verb forms of common words with the following endings: *d*, *ed*, *ied*, *ing*, *s*, *es*, *ies*, *r*, *er*, *est*, *ier*, *iest*.
- Compound and hyphenated words if both parts of the word are common words. For example, *long-haired* and *and/or* would be familiar. Note that forward slashes (/) are also treated as compound-word punctuation.
- The first instance of any proper noun—all subsequent occurrences are considered familiar.
- All numerals.

> Note
> This test treats headers and subheaders as full sentences, but excludes lists and tables. The default text exclusion method will be overridden for this specific test, although this behavior can be customized from the Options dialog.

The initial version of this test (Dale and Chall) was a multiple regression trained against 376 passages from McCall-Crabbs *Standard Test Lessons in Reading*. After comparing the results of this formula against other tests (such as Flesch and Lorge), it was validated against manually judged reading levels from other materials. These materials included “fifty-passages of health-education materials” (18) with a correlation of $r = 0.92$. Also included were “78 passages on foreign affairs from current-events magazines, government pamphlets, and newspapers” (18) with a correlation of $r = 0.90$.

The New Dale-Chall test replaces this regression equation by looking up the factors’ intersection in a table and returning a grade range (e.g., *4* or *7–8*). (This grade range is a notable difference from other tests that use floating-point precision grade levels.) It also includes an updated version of the 3,000 common words.

Because this test is based on the usage of familiar words (rather than syllable or letter counts), it is often regarded as a more accurate test for younger readers.

10.24 New Farr-Jenkins-Paterson

The New Farr-Jenkins-Paterson readability formula calculates the grade level of a document based on sentence length and number of monosyllabic words.

New Farr-Jenkins-Paterson is Kincaid et al.'s recalculated version of the FJP test (Derivation 17) that was recalculated "to be more suitable for Navy use" (11); hence, it is designed for technical documents and manuals. In addition to being a recalculation, this variation of FJP (i.e., a Flesch-like test) returns a grade-level result, rather than an index score.

This test was trained against reading-test results from 531 U.S. Navy personnel from four Naval technical training schools. Using these results, the following multiple regression equation was derived:

New Farr-Jenkins-Paterson formula

$$G = 22.05 + (.387 * (W/S)) - (.307 * ((M/W) * 100))$$

Where:

G	Grade level
W	Number of words
M	Number of monosyllabic words
S	Number of sentences

10.25 New Fog Count

The New Fog Count readability formula is Kincaid et al.'s modified version of the Gunning Fog Index (14) that was recalculated "to be more suitable for Navy use" (11); hence, it is designed for technical documents and manuals. This test calculates the grade level of a document based on sentence length and number of words containing three or more syllables.

This test was trained against reading-test results from 531 U.S. Navy personnel from four Naval technical training schools. Using these results, the following multiple regression equation was derived:

New Fog Count formula

$$G = (((Z + (3 * F))/S) - 3)/2$$

Where:

G	Grade level
F	Number of complex words
Z	Number of easy words
S	Number of sentences

Note
This test uses the same easy-word and sentence counting methods as the Gunning Fog test.

10.26 (Powers-Sumner-Kearl) Dale-Chall

The (Powers-Sumner-Kearl) Dale-Chall readability formula (99–105) calculates the grade level of a document based on sentence length and number of unfamiliar Dale-Chall words.

The goal of Powers, Sumner, and Kearl was to develop updated versions of four popular readability formulas: Dale-Chall, Flesch, Gunning Fog, and Farr-Jenkins-Paterson. These updated formulas would be based on the 1950 edition of the McCall-Crabbs test criterion. By using this revised test material, Powers, Sumner, and Kearl aimed to “modernize the formulas by taking advantage of the more recently administered tests” and to “establish formulas which are derived from identical materials” (100).

The training materials used to build the regression formulas were 383 prose passages from the 1950 edition of McCall-Crabbs’s *Standard Test Lessons in Reading*. For validation, the formulas were tested against 113 samples from “various publications” (100). Their results (shown below) implied somewhat strong correlations for their Dale-Chall and Flesch formulas and mild correlations for their FJP and Gunning Fog formulas:

- recalculated Dale-Chall: $r^2 = 0.5092$
- recalculated Flesch: $r^2 = 0.4034$
- recalculated Gunning Fog: $r^2 = 0.3440$
- recalculated Farr-Jenkins-Paterson: $r^2 = 0.3407$

The recalculated Dale-Chall formula is as follows:

(PSK) Dale-Chall formula

$$\begin{aligned} G = &3.2672 + (.0596 * (W/S)) + \\ &(.1155 * ((UDC/W) * 100)) \end{aligned}$$

Where:

G	Grade level
S	Number of sentences
UDC	Number of unfamiliar Dale-Chall words
W	Number of words

Although this is an update of the 1948 Dale-Chall formula, the 1995 New Dale-Chall test is recommended instead. This formula does not apply as much weight to the sentence length factor as New Dale-Chall does and comparatively yields lower results for difficult (i.e., upper-secondary education) documents. Therefore, this test should only be used for primary and lower-secondary educational documents. It is also only recommended if floating-point precision for the results is required, as New Dale-Chall uses grade ranges instead.

10.27 (Powers-Sumner-Kearl) Farr-Jenkins-Paterson

The (Powers-Sumner-Kearl) Farr-Jenkins-Paterson readability formula (99–105) calculates the grade level of a document based on sentence length and number of monosyllabic words.

The goal of Powers, Sumner, and Kearl was to develop updated versions of four popular readability formulas: Dale-Chall, Flesch, Gunning Fog, and Farr-Jenkins-Paterson. These updated formulas would be based on the 1950 edition of the McCall-Crabbs test criterion. By using this revised test material, Powers, Sumner, and Kearl aimed to "modernize the formulas by taking advantage of the more recently administered tests" and to "establish formulas which are derived from identical materials" (100).

The training materials used to build the regression formulas were 383 prose passages from the 1950 edition of McCall-Crabbs's *Standard Test Lessons in Reading*. For validation, the formulas were tested against 113 samples from "various publications" (100). Their results (shown below) implied somewhat strong correlations for their Dale-Chall and Flesch formulas and mild correlations for their FJP and Gunning Fog formulas:

- recalculated Dale-Chall: $r^2 = 0.5092$
- recalculated Flesch: $r^2 = 0.4034$
- recalculated Gunning Fog: $r^2 = 0.3440$
- recalculated Farr-Jenkins-Paterson: $r^2 = 0.3407$

The recalculated Farr-Jenkins-Paterson formula is as follows:

(PSK) Farr-Jenkins-Paterson formula

$$G = 8.4335 + (.0923 * (W/S)) - (.0648 * ((M/W) * 100))$$

Where:

G	Grade level
W	Number of words
M	Number of monosyllabic words
S	Number of sentences

10.28 (Powers-Sumner-Kearl) Flesch

The (Powers-Sumner-Kearl) Flesch readability formula (99–105) calculates the grade level of a document based on sentence length and number of syllables.

The goal of Powers, Sumner, and Kearl was to develop updated versions of four popular readability formulas: Dale-Chall, Flesch, Gunning Fog, and Farr-Jenkins-Paterson. These updated formulas would be based on the 1950 edition of the McCall-Crabbs test criterion. By using this revised test material, Powers, Sumner, and Kearl aimed to "modernize the formulas by taking advantage of the more recently administered tests" and to "establish formulas which are derived from identical materials" (100).

The training materials used to build the regression formulas were 383 prose passages from the 1950 edition of McCall-Crabbs's *Standard Test Lessons in Reading*. For validation, the formulas were tested against 113 samples from "various publications" (100). Their results (shown below) implied somewhat strong correlations for their Dale-Chall and Flesch formulas and mild correlations for their FJP and Gunning Fog formulas:

- recalculated Dale-Chall: $r^2 = 0.5092$
- recalculated Flesch: $r^2 = 0.4034$
- recalculated Gunning Fog: $r^2 = 0.3440$
- recalculated Farr-Jenkins-Paterson: $r^2 = 0.3407$

The recalculated Flesch formula is as follows:

(PSK) Flesch formula

$$G = (ASL * .0778) + (SY * .0455) - 2.2029$$

Where:

G	Grade level
ASL	Average sentence length
SY	Number of syllables per 100 words

Note

Note that this test treats numerals as monosyllabic words, in conjunction with the original Flesch test.

10.29 (Powers-Sumner-Kearl) Gunning Fog

The (Powers-Sumner-Kearl) Gunning Fog readability formula (99–105) calculates the grade level of a document based on sentence length and number of syllables.

The goal of Powers, Sumner, and Kearl was to develop updated versions of four popular readability formulas: Dale-Chall, Flesch, Gunning Fog, and Farr-Jenkins-Paterson. These updated formulas would be based on the 1950 edition of the McCall-Crabbs test criterion. By using this revised test material, Powers, Sumner, and Kearl aimed to "modernize the formulas by taking advantage of the more recently administered tests" and to "establish formulas which are derived from identical materials" (100).

The training materials used to build the regression formulas were 383 prose passages from the 1950 edition of McCall-Crabbs's *Standard Test Lessons in Reading*. For validation, the formulas were tested against 113 samples from "various publications" (100). Their results (shown below) implied somewhat strong correlations for their Dale-Chall and Flesch formulas and mild correlations for their FJP and Gunning Fog formulas:

- recalculated Dale-Chall: $r^2 = 0.5092$
- recalculated Flesch: $r^2 = 0.4034$
- recalculated Gunning Fog: $r^2 = 0.3440$
- recalculated Farr-Jenkins-Paterson: $r^2 = 0.3407$

The recalculated Gunning Fog formula is as follows:

(PSK) Gunning Fog formula

$$\begin{aligned} G =& 3.0680 + (.0877 * (W/S)) + \\ & (.0984 * ((F/W) * 100)) \end{aligned}$$

Where:

G	Grade level
W	Number of words
F	Number of complex words
S	Number of sentences

Note
This test uses the same easy-word and sentence counting methods as the Gunning Fog test.

10.30 Raygor Estimate Graph

The Raygor Estimate graph (259–63) is a graphical test for English documents. It calculates a document's grade level from its average number of sentences and long (6+ character) words per hundred words. These averages are plotted onto a chart where their intersection determines the reading level of the content.

Note that this graph is very similar to the Fry graph. The intent of this test was to provide a quicker and simpler way to plot a readability level, assuming that counting characters is easier than syllables. To confirm this, a secondary study (Baldwin and Kaufman 148–53) conducted an experiment involving 155 students from the University of Tulsa. The students performed Fry and ELF tests on 6 100-word passages. Afterwards, an independent *t* test was performed to gauge the difference in time spent manually performing the two readability tests. What the researchers found was a significant difference, where Raygor took an average of 250.26 seconds to complete versus 323.45 seconds for Fry.

In another experiment, Baldwin and Kaufman compared the correlations between Fry and Raygor scores. They collected three 100-word samples from 100 books from the Learning Resource Center at The University of Tulsa. With these books, the "Levels of difficulty in the sample ranged from upper elementary to professional (beyond the college level)" (149). The researchers found a significant Kendall rank order correlation of 0.875 between scores from the two tests.

This graph is primarily used in secondary education to help classify teaching materials and books into their appropriate reading groups. Below is an example of a Raygor graph:

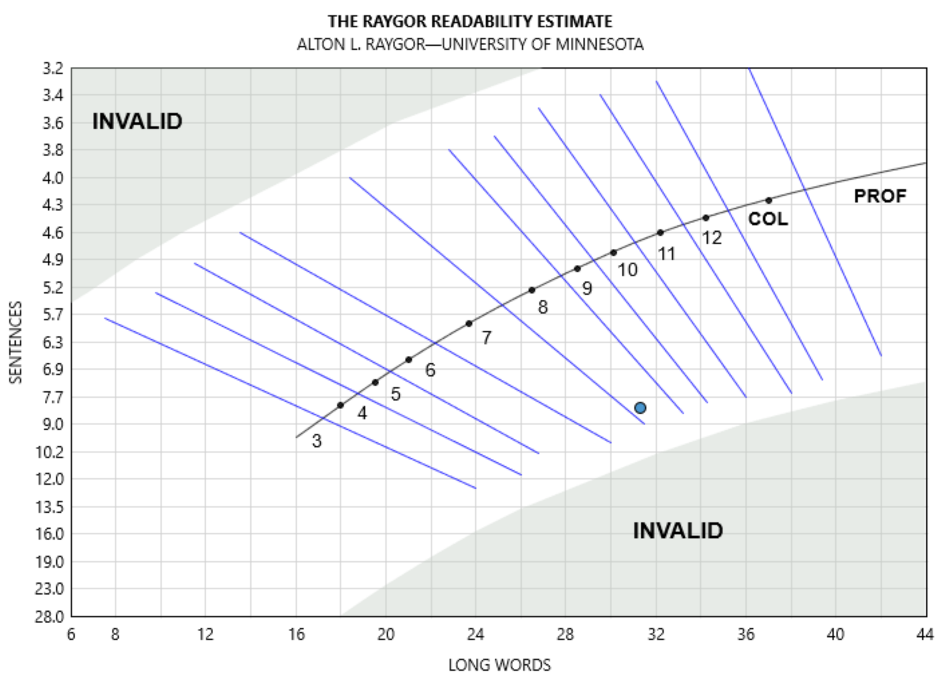


Figure 10.4: Raygor Estimate graph

The Raygor Estimate instructions are:

- Extract a 100-word passage from the selection. If the material is long, take subsamples from the beginning, middle, and end. Numerals should be skipped.
- Count the number of sentences, estimated to the nearest tenth.
- Count the number of words that are six or more letters.
- Find the point on the chart.
- If the sample's character or sentence count is too low or high for it to be plotted, then adjust that factor so that it can fit onto the chart. For example, if a sample has 30 sentences per 100 words, then this will need to be adjusted to 28.
- Repeat this process for each sample, and then average the samples.

This test is designed for 100-word samples, with three samples being recommended for best results. (Normalization can also be used to analyze larger samples or entire documents.)

10.31 Rix

The Rate Index (Rix) readability formula ("Lix and Rix" 490–96) was designed for documents of any Western European language. It calculates a document's index score based on sentence length and number of long words (i.e., words containing seven or more characters). It is based on the Lix readability formula (Björnsson 480–97) and follows the same philosophy of using a unified formula for multiple languages.

"Rate" in this context refers to the rate of long words throughout the document. Comparing against other tests that traditionally recommend fixed sample sizes (e.g., SMOG), Anderson noted that Rix could be applied to any length of text without normalization. As he explained, "the concept of rate has the advantage of being applied over any number of sentences, for it is a running average" ("Lix and Rix" 495).

Regarding the factors used for this test, Anderson defined a sentence "as a sequence of words terminated by a full-stop (period), question or exclamation mark, colon, or semicolon" (495). In other words, sentences units should be used for the calculation. (As for Lix, Björnsson did not define how sentences should end, so it is recommended to use the traditional *.*, *?*, and *!* markers for that test.)

As for defining long words, Anderson recommended "excluding hyphens, punctuation marks, and brackets" (495) when determining a word's length. Although Björnsson did not explicitly state this for Lix, it is a reasonable assumption that was his original intention.

Prior to creating this formula, Anderson conducted a validation study for Lix, using a set of 36 prose passages. These 150-word passages were originally scored from a cloze study (Miller and E. Coleman 851). Later, they were scored in another study involving judged-rankings and word-recollection tests (Aquino 346–56). (Both portions of Aquino's study used 14 *Southwest Regional Laboratory for Educational Research and Development* employees.) Anderson analyzed the Lix scores against these other measurements and found significant correlations, as shown below:

- against the Miller and Coleman's cloze scores, $r = 0.89$
- against word-recall scores, $r = 0.82$
- against the judges' rankings, $r = 0.88$

After validating Lix, Anderson created Rix and compared the two tests' results. He found that they almost perfectly correlated ($r = 0.99$). Afterwards, he validated Rix against the aforementioned 36-passage studies and found similarly strong correlations.

Rix formula

$$I = X/U$$

Where:

I	Rix index score
X	Number of long words (7+ characters)
U	Number of sentence units

Table 10.25: Rix Grade-level Conversion Table

Rix Index Score	Grade Level
7.2 and above	College
6.2 and above	12
5.3 and above	11
4.5 and above	10
3.7 and above	9
3.0 and above	8
2.4 and above	7
1.8 and above	6
1.3 and above	5
.8 and above	4
.5 and above	3
.2 and above	2
Below .2	1

10.32 SMOG

The SMOG readability formula (McLaughlin 639–46) calculates the grade level of a document based on complex (i.e., 3 or more syllable) word density and is designed for secondary-age readers.

SMOG attempts to predict "full comprehension" (L. L. Smith 129), whereas most other formulas predict 50%–75% comprehension. Because of this, SMOG will generally produce higher scores comparatively (usually 1–2 grade levels higher).

This test was trained on 390 passages from the 1961 edition of *McCall-Crabbs Standard Test Lessons in Reading*. The following regression equation was derived with a correlation of $r = 0.985$:

SMOG formula

$$G = 1.0430 * \sqrt{C} + 3.1291$$

Where:

G	Grade level
C	Number of complex (3+ syllables) words from 30 sentences

A simplified variation is also available (McLaughlin 643):

$$G = FLOOR(\sqrt{C}) + 3$$

$FLOOR$ refers to rounding the result of $\sqrt{C}$ down to the closest perfect square.

Note
The simplified version of this formula uses lower precision. It was originally offered to make manual calculations of this test easier.

This test requires 3 10-sentence samples, but normalization can be used to analyze entire documents.

Numerals are fully syllabized (i.e., sounded out) for this test, so the program overrides your numeral syllabication setting when calculating it.

Note that this test is colloquially referred to as an acronym for "Simple Measure of Gobbledygook." The origin of its name, however, is actually a nod to Robert Gunning's Fog index.

10.33 SMOG (Simplified)

Refer to sec. 10.32.

10.34 Spache Revised

The Spache Revised test (*Good Reading* 195–207) calculates the grade level of a document based on sentence length and number of unique unfamiliar words. This test uses a list of 1,041 common words that are known to younger readers (4^{th} grade and below). Any word that does not appear on this list is considered unfamiliar. The following are also considered familiar:

- Regular verb forms of any common word (i.e., *ing*, *es*, *s*, *ed*, and *ies* endings). Irregular forms are not familiar.
- Regular plural and possessive forms of any common word (i.e., *s*, *es*, and *ies* endings). Irregular forms are not familiar.
- Adjectival or adverbial forms of common words. This includes forms with endings such as *ly*, *est*, *er*, and *ily*.
- Derivatives of common words where the function changes (e.g., an adjective changed to function as a noun) are not familiar unless explicitly on the common word list. For example, *brave* is a common word, but *bravery* would not be considered familiar.
- Hyphenated words if both parts of the word are common words. For example, *apple-tree* and *and/or* would be familiar. Note that forward slashes (/) are treated like hyphens in this context.
- All proper nouns.
- All numerals.

Spache Revised is generally used for primary-age (i.e., Kindergarten to 7^{th} grade) readers to help classify school textbooks and literature. This differs from New Dale-Chall, which is better meant for secondary age readers.

Because this formula uses familiar words (rather than syllable or letter counts), it is often regarded as a more accurate test for younger readers.

Note that *Readability Studio* uses the revised 1974 version of Spache.

Spache Revised formula

$$G = (.121 * (W/S)) + (.082 * UUS) + .659$$

Where:

G	Grade level
S	Number of sentences
W	Number of words
UUS	Number of unique unfamiliar words per 100 words (see note below)

Note
Spache recommends not counting the same unfamiliar word more than once if it appears in subsequent 100-word samples (200). For this reason, it is recommended to follow this practice when analyzing an entire document.

10.35 Stocker's Catholic Supplement

Stocker's Catholic supplement (87–89) is an extension to the Dale-Chall word list designed for Catholic students. Based on a 1971 study of 4th-grade Catholic students[1], this list represents over 200 words (not on the standard Dale-Chall list) familiar to 80% of them.

This list is used along with the Dale-Chall list and offers more accurate scoring with materials meant for Catholic schools.

To view this list, select the Readability tab, click the Word Lists button, and select Stocker's Catholic Supplement from the menu.

To include this list as part of your New Dale-Chall scoring, select the Tools tab and click the Options button to display the Options dialog. Next, expand Readability Scores, select Test Options, and then check Include Stocker's Catholic supplement. After selecting this option, all New Dale-Chall tests performed in new projects will include this supplement.

[1] The students involved in this study were "6,743 randomly selected Catholic fourth-graders attending 136 parochial schools throughout the country" (87–88).

10.36 Wheeler-Smith

The Wheeler-Smith (397–99) test is designed for primary-age materials (Kindergarten to 4^{th} grade). This test calculates the grade level of a document based on unit length and number of polysyllabic (2+ syllable) words. It should be noted that this differs from most other tests that use 3+ syllable words as a difficulty factor. The researchers selected 2+ syllable words because primary readers will have more difficulty with them than secondary and adult readers.

Although not explicitly stated in the article, the authors' definition of "polysyllabic" as being 2+ syllable can be inferred from the example[2] that they provided:

> The Woodsman saw the seven sisters come into their bedroom. He saw them put on the shoes in which they danced. He saw them put on their party dresses.
> When they were all dressed, the oldest sister opened a door in the floor near her bed.
> "Sisters, we are ready," she said.
> They walked down the secret steps.
> The Woodsman put on his magic cap and followed them down the steps.
> Thanks to his magic cap, no one was able to see him.
> Soon they came to a wonderful wood.
> Some of the trees had blue leaves.
> Some had glass leaves. Others had leaves of gold.

The authors reported 20 polysyllabic words in the above sample. *Wonderful* is the only 3+ syllable word in this sample, whereas there are 18 2+ syllable words. It may be inferred that the authors were considering *their* as 2 syllables, which in that case would yield 20 2+ syllable (i.e., "polysyllabic") words.

In regards to unit (i.e., sentence) length, the researchers noted that sentence lengths strongly correlated with the manually-rated grade levels. Because of this, sentence length was given a stronger weight in the formula (compared to previous formulas).

For the building and validation of this formula, nine basic reading series were used.

Finally, the authors offered recommendations while reviewing a reported score. First, if a score appears to be higher than expected, then you should exclude proper nouns and rerun the test. Second, they recommended that when assigning a book for independent reading, assign it one level below the reported score. In other words, if a book scores at the 4^{th} grade level, then it should be assigned to 3^{rd} grade readers if they are reading it independently.

Wheeler-Smith formula

$$I = (W/U) * (P/W) * 10$$

[2] This sample was taken from *Down Singing River*, Betts-Welch, American Book Company.

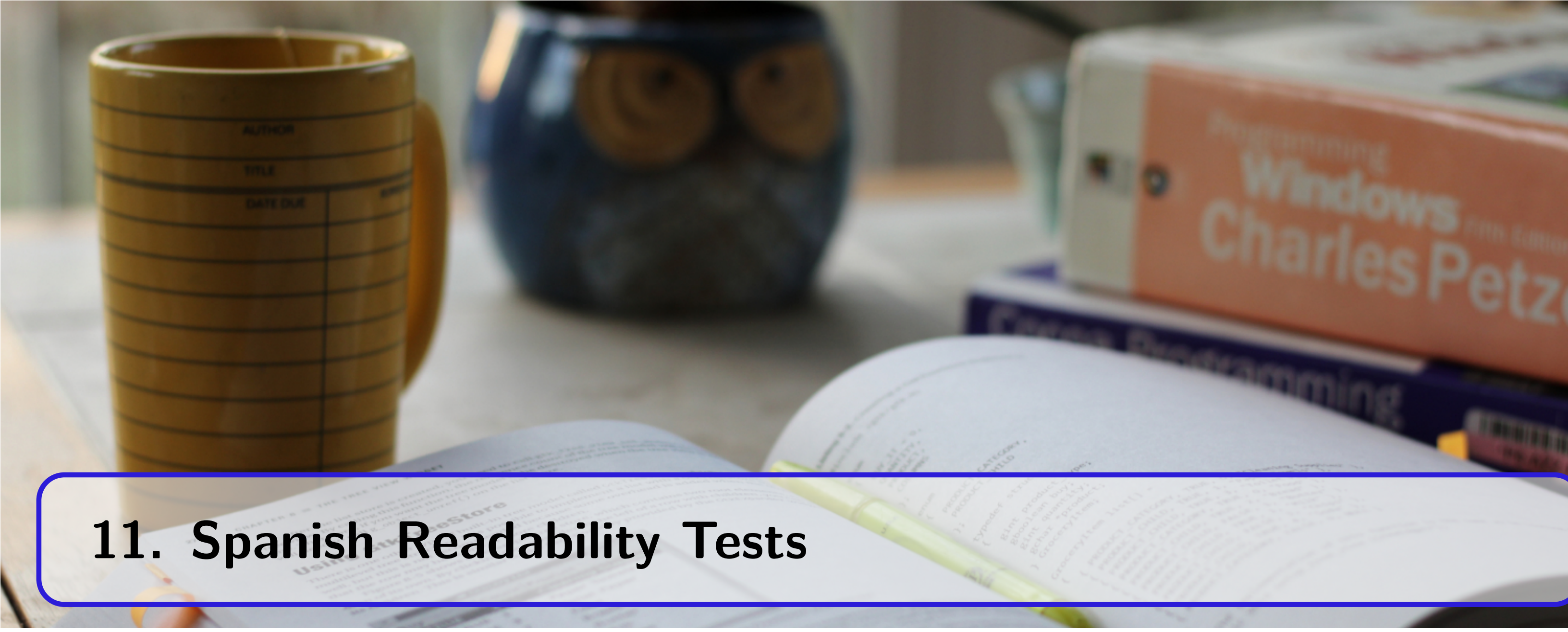
Where:

I	Index value
P	Number of polysyllabic (2+ syllable) words
W	Number of words
U	Number of units

Then, find the index value in the following chart to acquire the final readability score:

Table 10.29: Wheeler-Smith Grade-level Conversion Table

Index Score	Grade Level
4.0–8.0	Primer (Kindergarten)
8.1–11.5	First grade
11.6–19.0	Second grade
19.1–26.5	Third grade
26.6–34.5	Fourth grade



11. Spanish Readability Tests

11.1 Crawford

The Crawford test is designed for "elementary level materials in Spanish" (4). This test calculates the grade level of a document based on sentence length and syllable count.

The test was derived from a multiple regression equation trained against 10 basal readers series collected from the United States, Latin America, and Spain. (These series represented grade levels 1–6 and contained 789 100-word passages in total.) Across these reader series, Crawford noted somewhat significant correlations for syllables per 100 words (an average of $r = 0.4152$) and a consistently stronger correlation for sentences per 100 words (an average of $r = 0.6395$). Using these findings, the following equation was built:

Crawford formula

$$G = (S * -.205) + (B * .049) - 3.407$$

Where:

G	Grade level
S	Number of sentences per 100 words
B	Number of syllables per 100 words

A chart visualizing this test's score is also available:

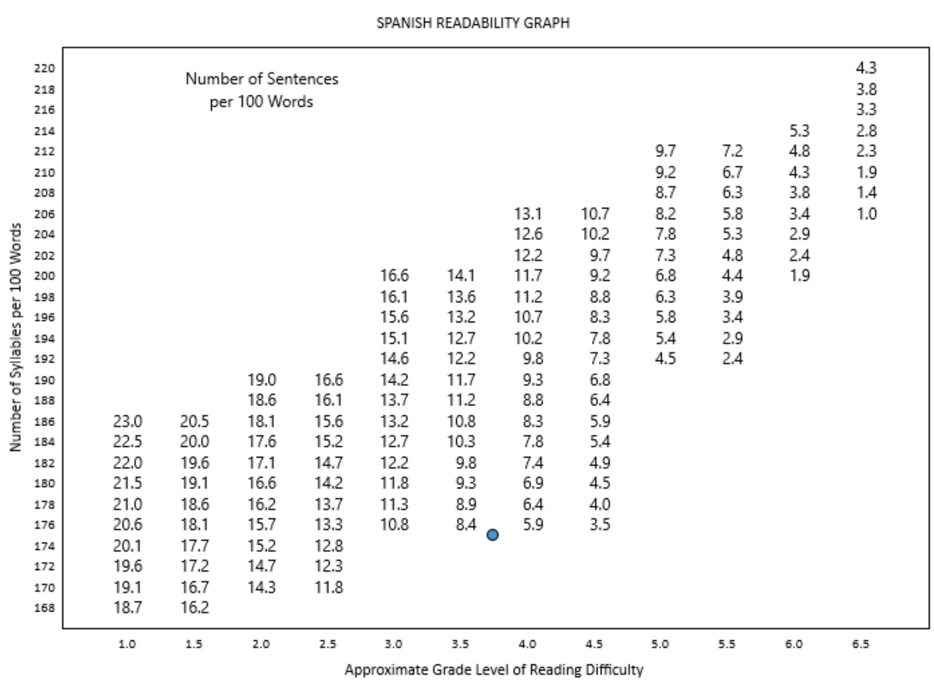


Figure 11.1: Crawford chart

In this chart, the y-axis and interior (i.e., plot area) represent the factors, and the x-axis represents the final score. Specifically, the y-axis represents the sample's syllable count, and the interior area represents the sample's sentence count. After finding the intersection of these two factors, the position on the x-axis respective to this intersection yields the grade-level score.

Note that this differs from other graphs which use the x- and y-axes to represent the factors and their intersection to represent the score. Instead, this chart displays the sentence-count values as a series of steps across the plot area.

For example, in the above graph the number of syllables is 175. The number of sentences is approximately 7 (between 8.4 and 5.9 [on the plot, respective to 175 on the y-axis]). After finding the intersection of these factors, the point's x-axis value is 3.7. Hence, 3.7 is the sample's grade-level score.

This test is designed for 100-word samples and uses the same statistics as the Fry test. (Normalization can also be used to analyze larger samples or entire documents.)

11.2 FRASE Graph

FRASE (Fry Readability Adaptation for Spanish Evaluation) (Vari-Cartier 141–48) is a graphical test for Spanish text. This test calculates a document's language level from its average number of sentences and syllables per hundred words. These averages are plotted onto a chart where their intersection determines the reading level of the content.

The test uses methodologies borrowed from the Fry graph, such as using the same factors (sentence length and syllable count). However, it also includes adjustments which differentiate it. One adjustment is that the x-axis begins at 182 (Fry begins at 108)—this is used to account for Spanish text having higher syllable counts compared to English. The second adjustment is that it uses reading levels ("Beginning," "Intermediate," "Advanced Intermediate," and "Advanced"), rather than grade levels. By using reading levels, documents can be classified more broadly into reading groups, which makes this test ideal for SSL (Spanish as a second language) readers.

The FRASE graph is designed for most text and can be used for both primary and secondary-age reading materials, as well as SSL materials. It was trained on eight textbooks representing the four levels of difficulty:

- *Nueva Vista*
- *Español: A Descubrirlo*
- *Entre Nosotros*
- *Vista Hispánica*
- *Civilización y Cultura*
- *Multivista Cultura*
- *Español: Curso Primero*
- *Perspectivas*

These textbooks "covered a variety of reading topics which dealt with family life, entertainment, customs, the Hispanic people, history, and politics" (143). 300–350 word excerpts were randomly selected from each book, scored on a FRASE graph, and then objectively and subjectively validated.

These validations involved correlating the FRASE scores against "teacher judgments, Spaulding formula ratings, cloze test scores, and informal multiple choice tests" (144). (The "teacher judgments" validation involved 16 secondary-level Spanish teachers ranking the samples into the levels of beginner, intermediate, advanced intermediate, and advanced.) The comparisons were significantly correlated, with the following results:

- cloze tests: $r = -0.93$
- Spaulding scores: $r = 0.91$
- teacher judgments: $r = 0.95$
- multiple choice tests: $r = -0.97$

Below is an example of a FRASE graph:

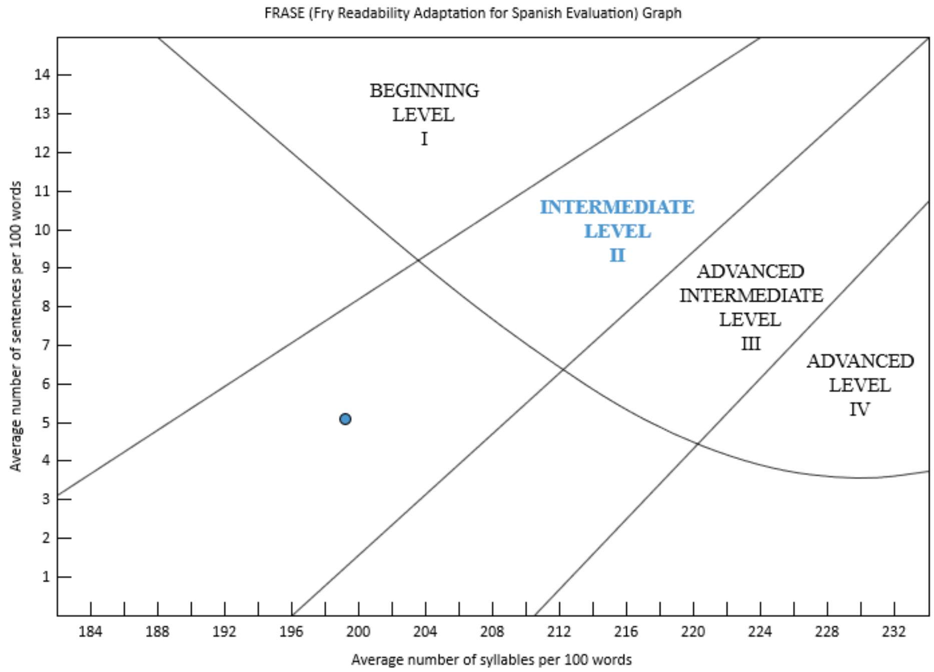


Figure 11.2: FRASE graph

The FRASE graph procedure is:

- Extract a 100-word passage from the selection. If the material is long, take subsamples from the beginning, middle, and end.
- Count the number of sentences in each passage. Count a half sentence as .5.
- Count the number of syllables in each passage.
- Find the intersecting point on the plot. The quadrant where the point lies indicates the reading level of the document.

This test is designed for 100-word samples, with 3 samples being recommended for best results. (Normalization can also be used to analyze larger samples or entire documents.)

11.3 Gilliam-Peña-Mountain Graph

Gilliam-Peña-Mountain (426–30) is a graphical test for Spanish text based on the Fry graph ("Clarifications, Validity, and Extension" 242–52). It is identical to the Fry graph, except that its X (syllable-count) axis begins at 175 (instead of 108). This is necessary to account for Spanish text having a much higher syllable count than its English counterpart. As the authors stated, "Spanish syllable counts, however, were so high that they threw many of the samples outside the bands or off the graph entirely" (427).

This test calculates the grade level from the average number of sentences and syllables per hundred words. These averages are plotted onto a chart where their intersection determines the reading level of the content.

This variation of Fry was trained on samples from the following textbooks:

- *¡Vas bien!*
- *¿Ya te vas?*
- *¡Hasta luego!*
- *Nuestros amigos*
- *Senda 1*
- *Trabaja y aprende*
- *La ciudad*
- *Del campo al pueblo*
- *Aventuras maravillosas*
- *Otros amigos, otras culturas*
- *Senda 3*
- *Conozcamos a Puerto Rico*
- *Por tierras vecinas*

It was also trained on the following juvenile trade books:

- *Aquí viene el ponchado*
- *Osito*
- *El caso del forastero hambriento*
- *Leonardo el león y Ramón el ratón*
- *Jorge el curioso*
- *Teresita y las orugas*
- *Danielito y los dinosaurios*
- *Johnny Lost (Juanito Perdito)*
- *El niño que no creía en la primavera*

Syllable and sentence statistics were calculated from these samples. Initially, the researchers noted that most of these statistics could not be plotted onto a Fry graph, because of the high syllable count. They deduced that a constant would need to be subtracted from this syllable count to be able to plot it on the graphs. Next, they determined where on the graph these materials should lie based on either the grade level provided by the publisher or by plotting an English translation of the material onto a graph. By comparing the differences between the actual and expected syllable counts, it was determined that subtracting 67 from the Spanish syllable counts would adjust the score to appear on the graph where it was expected to be.

The Gilliam-Peña-Mountain graph is designed for both primary and secondary-age reading materials. Below is an example of a Gilliam-Peña-Mountain graph:

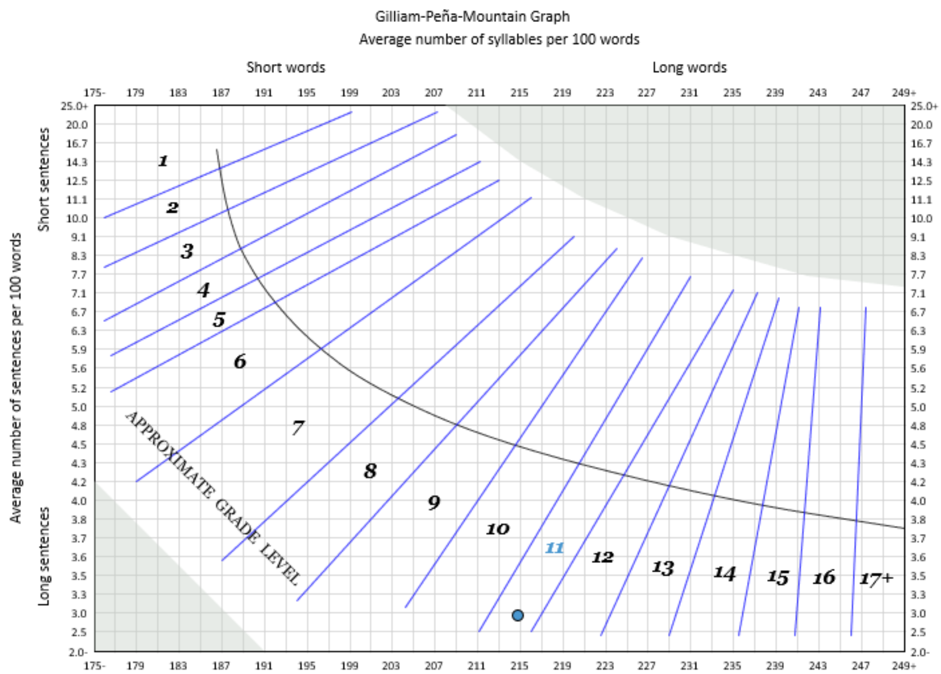


Figure 11.3: Gilliam-Peña-Mountain graph

The Gilliam-Peña-Mountain graph formula is:

- Extract a 100-word passage from the selection. If the material is long, take subsamples from the beginning, middle, and end.
- Count the number of sentences in each passage. Count a half sentence as .5.
- Find the point on the chart.
- Count the number of syllables in each passage and subtract 67 to adjust for the higher syllabic nature of Spanish text. However, note that *Readability Studio* shifts the syllable-count axis by 67, rather than adjusting the actual syllable count. This approach preserves the syllable count, while still being able to plot the score on the graph.
- If the sample's syllable or sentence count is too low or high for it to be plotted, then adjust that factor so that it can fit onto the chart. For example, if a sample has 261 syllables per 100 words, then this will need to be adjusted to 249.
- Repeat this process for each sample, and then average the samples.

This test is designed for 100-word samples, with 3 samples being recommended for best results. (Normalization can also be used to analyze larger samples or entire documents.)

Note that numerals are fully syllabized for this test. For example, "1945" will be counted as nine syllables ("u-no"-"nu-e-ve"-"cua-tro"-"cin-co").

11.4 INFLESZ Scale

The INFLESZ Scale is a Spanish test derived from earlier adaptations of Flesch Reading Ease (*Art of Readable Writing* 213; *Plain English* 23). It builds on Szigriszt's *Escala de Nivel de Perspicuidad*, which retained the original Flesch calculation, but adjusted its classifications for Spanish-language texts. While Szigriszt's Scale adjusted the difficulty levels for Spanish text, it relied on a relatively small and non-random corpus for validation. INFLESZ preserves Szigriszt's score, but realigned the scale boundaries and included fewer categories. This realignment was validated on a larger, more systematically sampled corpus. Hence, this revised scale is better suited for assessing the readability of a wide range of Spanish-language texts. (The authors' stated focus was on health-related materials.)

The INFLESZ Scale was validated using 210 Spanish-language publications drawn from three categories:

- *Publicaciones de quiosco* (newsstand publications), including national, regional, and sports newspapers, popular magazines, and comics
- *Textos escolares* (primary and secondary school textbooks) from major Spanish publishers
- *Revistas científicas* (scientific and medical journals aimed at health professionals)

From each publication, three random text fragments of at least 500 words were selected, yielding a total of 630 analyzed fragments.

Szigriszt index formula

$$I = 206.835 - (W/S) - (62.3 * (B/W))$$

Where:

I	Szigriszt index score
W	Number of words
B	Number of syllables
S	Number of sentences

Table 11.3: INFLESZ Scale Table

INFLESZ Score	Description
81–100	*Muy fácil* (Very easy)
66–80	*Bastante fácil* (Fairly easy)
56–65	*Normal* (Normal)
41–55	*Algo difícil* (Fairly difficult)
0–40	*Muy difícil* (Very difficult)

The INFLESZ category boundaries are based on the article's instructions that *Muy Fácil* applies to scores "puntuaciones superiores a 80" [scores greater than 80] (Barrio-Cantalejo et al. 140). This implies a strict greater-than cutoff for the upper boundary, rather than an inclusive threshold. As a result, the INFLESZ cutoffs follow a different boundary logic than Flesch Reading Ease, which treats category limits as inclusive.

Below is an example of the INFLESZ Scale:

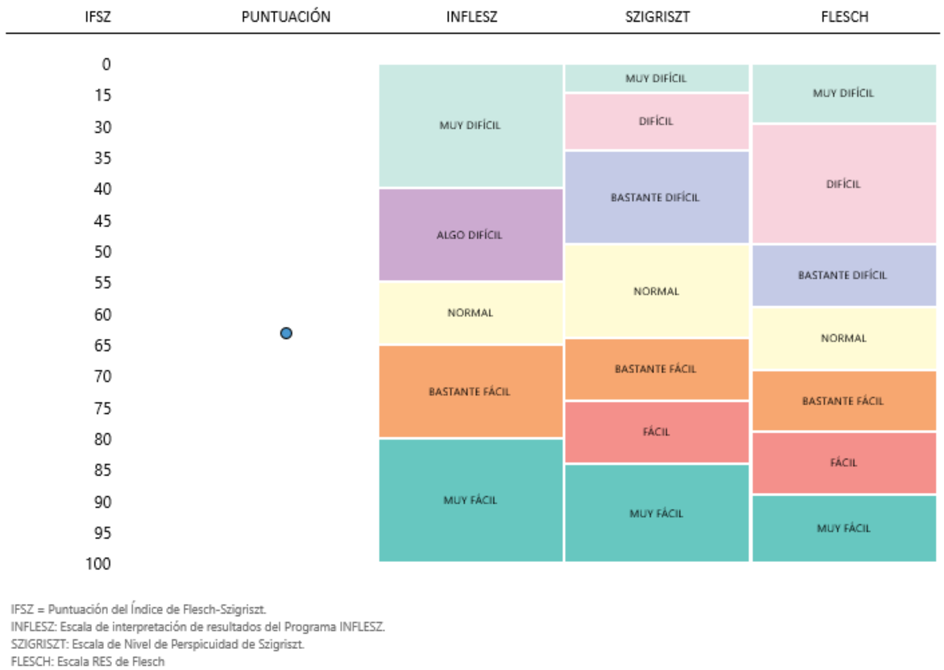


Figure 11.4: INFLESZ Scale

This chart consists of vertically aligned scales. The left scale, INFLESZ, presents which difficulty level the score falls into. (This is the primary focus of interpretation.) To the right are the original Szigriszt and Flesch Reading Ease scales, respectively. (These are only included for comparison.)

11.5 Lix

Refer to sec. 10.19.

11.6 Rix

Refer to sec. 10.31.

11.7 SOL

The SOL readability formula (Contreras et al. 21–29) calculates the grade level of a document based on complex-word density.

SOL is a modified version of SMOG (McLaughlin 639–46) that was recalibrated for Spanish text. This test calculates the SMOG score of a Spanish document, and then adjusts the score to take into account the comparably higher complex-word counts found in Spanish text. This adjustment is similar to what Gilliam-Peña-Mountain (Gilliam et al. 426–30) uses for plotting Spanish text onto the Fry graph.

This test was trained on multiple blocks of 30-sentence samples from the following English, Spanish, and French materials:

- *The Lancet* (Spanish and English)
- *El País* (Spanish)
- *Le Monde* (French)
- *International Herald Tribune* (English)
- *Don Quixote* (Spanish, English, and French)
- *The Bible* (Spanish, English, and French)
- *Snow White and the Seven Dwarfs* (Spanish, English, and French)
- *The Three Bears* (Spanish, English, and French)
- *The Wolf and the Seven Little Kids* (Spanish, English, and French)
- *The Three Little Pigs* (Spanish, English, and French)
- Story books (a set of story books) (Spanish, English, and French)

SMOG scores for these samples (including available translations) were collected. Next, the scores between each language for every set of material were correlated. The researchers found significantly higher SMOG scores for French compared to English, and higher SMOG scores for Spanish compared to French. To adjust for these differences, the following regression equation was built to convert a Spanish SMOG calculation to English ($r^2 = 0.60$):

SOL formula

$$G = (1.0430 * \sqrt{C} + 3.1291) * .74 - 2.51$$

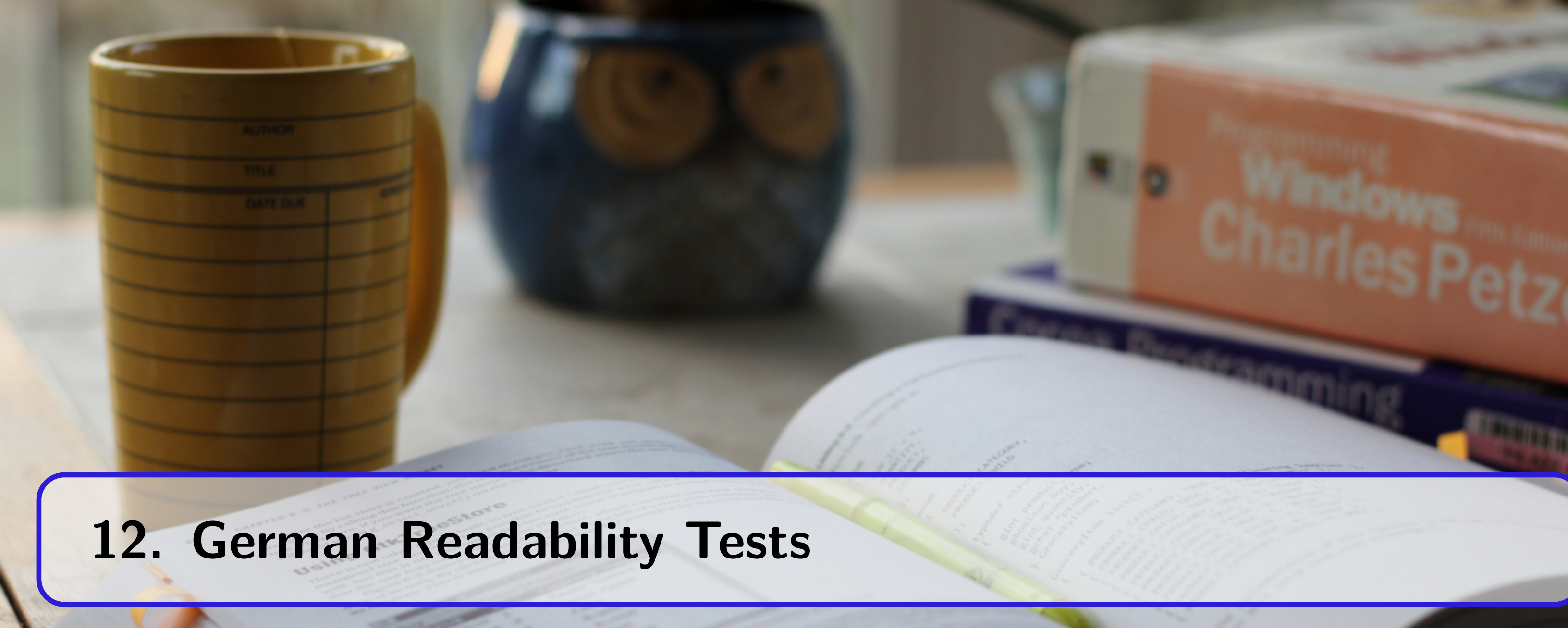
Where:

G	Grade level
C	Number of complex words (3+ syllables) from 30 sentences

Like SMOG, this test fully syllabizes numerals (i.e., sounds them out) , so the program overrides your numeral syllabication setting for this test. It also excludes lists and tables, but includes headers and footers (Contreras et al. 23). (SMOG does not explicitly follow this rule.)

Note that this test uses the high-precision version of the SMOG formula. This is recommended by the authors to avoid any rounding issues when converting scores between Spanish and English (24).

This test requires 3 ten-sentence samples, but normalization can be used to analyze larger samples or entire documents.



12. German Readability Tests

12.1 Amstad

The Amstad test is a recalculation of Flesch Reading Ease for German text (78–81, 50). Its formula is adjusted from the original Flesch to take into account German's comparatively longer words. It also uses a lower weighting for the sentence-length factor, placing most of the influence on word length.

The factors used for this test are syllables per word ("*Silben pro Wort*") and words per sentence ("*Wörter pro Satz*") (81). (These are the same factors used for Flesch Reading Ease).

Amstad is best suited for school textbooks and technical manuals. Scores range from 0–100 (the higher the score, the easier to read) and average documents should be within the range of 60–70.

Amstad formula

$$I = 180 - (W/S) - (58.5 * (B/W))$$

Where:

I	Amstad (i.e., Flesch) index score
W	Number of words
B	Number of syllables
S	Number of sentences

Amstad offered these interpretations for scoring:

Table 12.2: Amstad Conversion Table

Reading Ease Score	Description of Style	Typical Magazine
90-100	Very easy	comics
80-89	Easy	pulp-fiction
70-79	Fairly easy	slick-fiction
60-69	Standard	digest
50-59	Fairly difficult	quality
30-49	Difficult	academic
0-29	Very difficult	scientific

Note
Like Flesch Reading Ease, this test treats numerals as monosyllabic words by default. This behavior can be changed from the Options dialog.

12.2 Lix

Refer to sec. 10.19.

12.3 Lix (German children's literature)

This test is a German variation of Läsbarhetsindex (Lix) designed for children's literature. It uses the same index-score formula as Lix, but uses different scales for assigning the difficulty and grade-level scores. This adjustment was developed by Renström (152–153) and Bamberger & Vanecek (64, 187) for use with German literature ranging from 1st to 8th grade. (Renström developed the adjusted difficulty gauge; Bamberger & Vanecek developed the grade-level conversion.)

Note that this test differs from the original Lix approach of using the same scoring for any document, regardless of language. Instead, this test is designed specifically for German materials, adjusting the grade-level score and classification of the index value.

Regarding the index score, the following variation of the Lix gauge is used to classify the difficulty (Renström 153; Bamberger and Vanecek 65; Schulz 53):

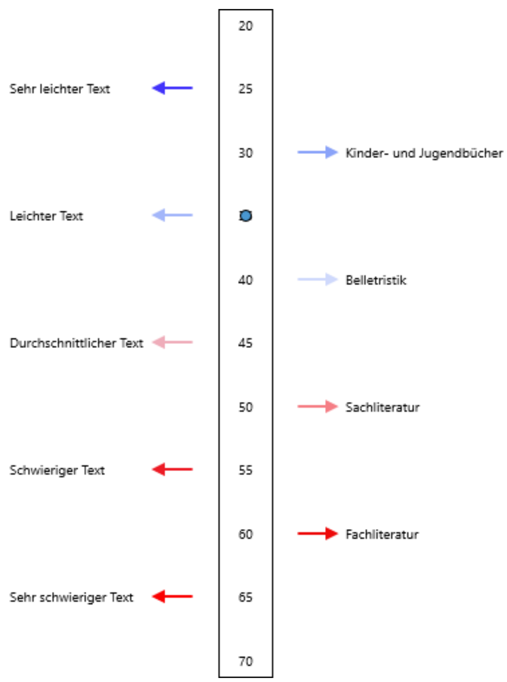


Figure 12.1: German Lix gauge (children's literature)

This version adjusts the scaling of the score, assuming that German text is inherently more difficult than other languages. It also includes additional levels of difficulty (e.g., "children and young adult").

To convert the Lix index score to a grade, this test uses the following table:

Table 12.3: Lix (German children's literature) Conversion Table

Index Score	Grade
0-23	1st grade
24-26	2nd grade
27-29	3rd grade
30-31	4th grade
32-33	5th grade
34-35	6th grade
36-37	7th grade
38+	8th grade

12.4 Lix (German technical literature)

This test is a German variation of Läsbarhetsindex (Lix) designed for technical literature. It uses the same index-score formula as Lix, but uses different scales for assigning the difficulty and grade-level scores. This adjustment was developed by Renström (152–153) and Bamberger & Vanecek (64, 187) for use with German literature ranging from 3rd to 15th grade. (Renström developed the adjusted difficulty gauge; Bamberger & Vanecek developed the grade-level conversion.)

Note that this test differs from the original Lix approach of using the same scoring for any document, regardless of language. Instead, this test is designed specifically for German materials, adjusting the grade-level score and classification of the index value.

Regarding the index score, the following variation of the Lix gauge is used to classify the difficulty (Renström 153; Bamberger and Vanecek 65; Schulz 53):

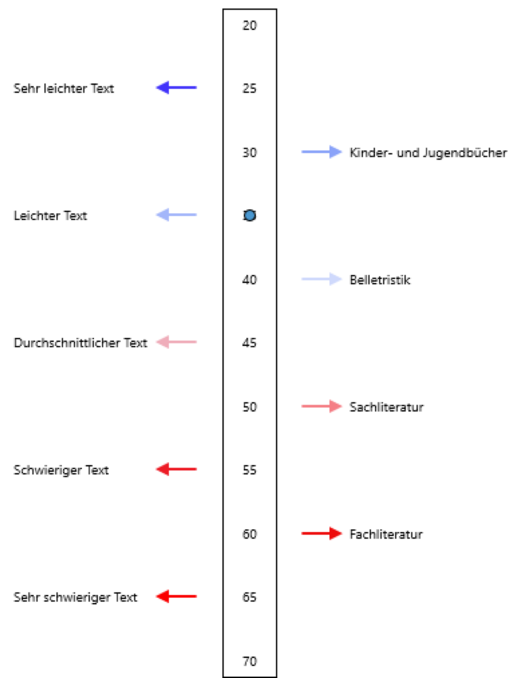


Figure 12.2: German Lix gauge (technical literature)

This version adjusts the scaling of the score, assuming that German text is inherently more difficult than other languages. It also includes additional levels of difficulty (e.g., "children and young adult").

To convert the Lix index score to a grade, this test uses the following table:

Table 12.4: Lix (German technical literature) Conversion Table

Index Score	Grade
0-30	3rd grade
31-33	4th grade
34-37	5th grade
38-40	6th grade
41-43	7th grade
44-47	8th grade
48-50	9th grade
51-53	10th grade
54-56	11th grade
57-59	12th grade
60-63	13th grade
64-69	14th grade
70+	15th grade

12.5 Neue Wiener Sachtextformel 1

Neue Wiener Sachtextformel 1 (1.nWS) (Bamberger and Vanecek 83, 187) is used for evaluating German nonfiction (specifically, 6th-10th grade materials). This test returns a grade-level score based on word length/complexity and sentence length.

Neue Wiener Sachtextformel 1 formula

$$
\begin{aligned}
G = & 0.1935 * ((C/W) * 100) + 0.1672 * (W/S) + \\
& 0.1297 * ((X/W) * 100) - 0.0327 * ((M/W) * 100) - 0.875
\end{aligned}
$$

Where:

G	Grade level
C	Number of 3+ syllable[a] words
X	Number of 7+ character[b] words
M	Number of monosyllabic[c] words
S	Number of sentences
W	Number of words

[a] *"drei- und mehrsilbigen"*
[b] *"mehr als sechs buchstaben"*
[c] *"einsilbigen"*

12.6 Neue Wiener Sachtextformel 2

Neue Wiener Sachtextformel 2 (2.nWS) was developed by Bamberger & Vanecek for evaluating German nonfiction (specifically, very light materials up to 5th grade) (84, 187). This test returns a grade-level score based on word length/complexity and sentence length.

Neue Wiener Sachtextformel 2 formula

$$\begin{aligned} G = & 0.2007 * ((C/W) * 100) + 0.1682 * (W/S) + \\ & 0.1373 * ((X/W) * 100) - 2.779 \end{aligned}$$

Where:

G	Grade level
C	Number of 3+ syllable[a] words
X	Number of 7+ character[b] words
S	Number of sentences
W	Number of words

[a] *"drei- und mehrsilbigen"*
[b] *"mehr als sechs buchstaben"*

12.7 Neue Wiener Sachtextformel 3

Neue Wiener Sachtextformel 3 (3.nWS) was developed by Bamberger & Vanecek for evaluating German nonfiction (specifically, very light materials up to 5th grade) (84, 187). This test returns a grade-level score based on word complexity and sentence length.

Neue Wiener Sachtextformel 3 formula

$$G = 0.2963 * ((C/W) * 100) + 0.1905 * (W/S) - 1.1144$$

Where:

G	Grade level
C	Number of 3+ syllable[a] words
S	Number of sentences
W	Number of words

[a] *"drei- und mehrsilbigen"*

12.8 Quadratwurzelverfahren

Quadratwurzelverfahren ("Square root method") is an alternative method for calculating SMOG (Bamberger-Vanecek) (78). Where SMOG (Bamberger-Vanecek) uses a 30-sentence sample, Quadratwurzelverfahren uses a 100-word sample; otherwise, they produce similar results.

Quadratwurzelverfahren formula

$$G = \sqrt{((C * (100/W))/(S * (100/W)) * 30)} - 2$$

Where:

G	Grade level
C	Number of 3+ syllable[a] words
S	Number of sentences
W	Number of words

[a] *"drei- und mehrsilbigen"*

Note

Based on the examples from Bamberger-Vanecek (78), the grade-level score should be truncated (or "floored") to one-point precision, not rounded.

12.9 Rix

Refer to sec. 10.31.

12.10 Rix (German fiction)

This test is a German variation of Rix designed for fiction books. It uses a modified index-score formula from Rix, as well as different logic for assigning a grade-level score. This test was developed by Bamberger & Vanecek (64, 187) for use with German fiction ranging from 1st to 11th grade.

Rix (German fiction) formula

$$I = ((100 * (X/W))/(U * (100/W))) * 10$$

Where:

I	Rix index score
W	Number of words
X	Number of long words (7+ characters)
U	Number of sentence units

To convert the Rix index score to a grade, this test uses the following table:

Table 12.6: Rix Index Score (German Fiction) to Grade Level Conversion Table

Index Score	Grade
0-13.5	1st grade
13.51-17	2nd grade
17.1-20.5	3rd grade
20.51-24	4th grade
24.1-27.5	5th grade
27.51-31	6th grade
31.1-34.5	7th grade
34.51-38	8th grade
38.1-41.5	9th grade
41.51-45	10th grade
45.1+	11th grade

12.11 Rix (German non-fiction)

This test is a German variation of Rix designed for non-fiction books. It uses a modified index-score formula from Rix, as well as different logic for assigning a grade-level score. This test was developed by Bamberger & Vanecek (64, 187) for use with German non-fiction ranging from 4^{th} to 14^{th} grade.

Rix (German non-fiction) formula

$$I = ((100 * (X/W))/(U * (100/W))) * 10$$

Where:

I	Rix index score
W	Number of words
X	Number of long words (7+ characters)
U	Number of sentence units

To convert the Rix index score to a grade, this test uses the following table:

Table 12.8: Rix Index Score (German Non-fiction) to Grade Level Conversion Table

Index Score	Grade
0-26	4^{th} grade
26.1-32	5^{th} grade
32.1-38	6^{th} grade
38.1-45	7^{th} grade
45.1-52	8^{th} grade
52.1-57	9^{th} grade
57.1-66	10^{th} grade
66.1-75	11^{th} grade
75.1-84	12^{th} grade
84.1-100	13^{th} grade
100+	14^{th} grade

12.12 Schwartz German Readability Graph

The Schwartz German Readability Graph (1–21) is a derivative of Fry's readability graph and is designed for instructional textbooks for primary and secondary students. Like Fry, it relies on sentence length and word complexity (i.e., syllable counts) to determine a document's difficulty.

As an elementary teacher, Schwartz was familiar with English tests and had found them "extremely helpful in classifying texts according to levels and fitting a student with proper reading material" (3). Later, as a high school German teacher, she was often asked if there were any German readability tests.

After corresponding with Prof. George Klare and Dr. Richard Bamberger, she confirmed that only two German tests existed (as well as some unpublished cloze procedure experiments). She then interviewed West German publishers to see how they classified school textbooks. They informed her that their process was to compare student dictionaries against their materials to gauge difficulty, as well as use large print and illustrations.

Schwartz then set out to create a German readability test, given the lack of such a tool at the time. She initially experimented with word frequencies as a factor, but decided that sentence length and word complexity were easier to calculate when computer assistance wasn't available. She also determined that a graphical test—such as Fry—was easier to use by classroom teachers than regression formulas. Hence, Schwartz created a new graph similar to Fry, but adjusted it for the West German basal readers that she used in her study.

When designing the graph, Schwartz adjusted the syllable-count axis to take into account German words' tendency to have higher syllable counts. As she noted, "It is commonplace that the average German word contains more syllables than the average English one" (14). In her samples, she noted that English averaged 25 syllables per 100 words, while German averaged 37 (14).

Another difference from the Fry graph is that some grade bands cover two grades (whereas each band on a Fry graph represents a unique grade).

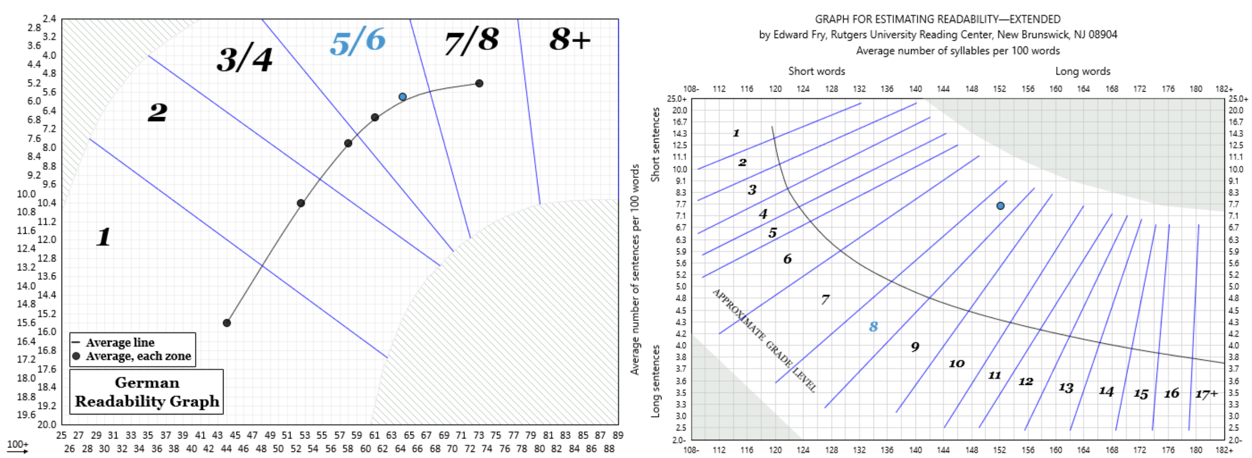


Figure 12.3: Comparison of bands between Schwartz and Fry graphs, respectively

As Schwartz noted:

> Furthermore, it might be of interest to the American reader to note that it used to be quite common in Germany to have certain reader levels combined in one volume: usually third and fourth, fifth and sixth, seventh and eighth, sometimes even fifth through eighth, and that often there was no distinction made inside the book itself as to in which grade a particular text had to be read. (5–6)

In the study, between 15–21 random 100-word samples were taken from 2nd–8th grade textbooks and between 11–15 100-word samples from 1st grade textbooks. The following textbooks were used:

Grade 1:

- *Bunte Lesewelt*
- *Bei uns daheim*
- *Unsere neue Fibel*
- *Bunte Welt*
- *Mein erstes Buch zum Anschauen, Zeichnen, Lesen und Schreiben*

Grade 2:

- *Mein Lesebuch für das 2. Schuljahr*
- *Lesebuch für das 2. Schuljahr der Grundschule*
- *schwarz auf weiß, Primarstufe, Text 2*
- *Deutsches Lesebuch für die Grundschule für das zweite Schuljahr*
- *auswahl, Lesebuch für Schulen, Grundschule 2. Schuljahr*
- *Lesebuch für die Volksschulen, 2. Schuljahr*

Grade 3:

- *Mein Lesebuch für das 3. Schuljahr*
- *Lesebuch für das 3. Schuljahr der Grundschule*
- *Deutsches Lesebuch für das dritte und vierte Schuljahr*
- *schwarz auf weiß, ein neues Lesebuch- Drittes Schuljahr*

Grade 4:

- *Mein Lesebuch für das 4. Schuljahr*
- *Lesebuch für das 4. Schuljahr der Grundschule*
- *Geschichten Berichte Gedichte*
- *schwarz auf weiß, ein neues Lesebuch- Viertes Schuljahr*
- *Deutsches Lesebuch, Viertes Schuljahr*

Grades 5/6:

- *Kein schöner Land*
- *Bayerishes Lesebuch für das 5. und 6. Schuljahr*
- *Lesebuch für die Volksschulen*

Grades 7/8:

- *Bayerisches Lesebuch für das siebte und achte Schuljahr*
- *Lesebuch für die Volksschulen, Vierter Band. 7. und 8. Schuljahr*
- *Lesebuch. C 8 (8. Schuljahr)*

Schwartz noted the following validation results:

- Grade 1: 83% of the 70 samples fell in the grade 1 band on the graph, 91% falling in either the grade 1 or 2 bands.
- Grade 2: 30.5% of the 105 samples fell in grade 2, although 82% fell in the 1, 2, or 3/4 bands.
- Grades 3 and 4: 29% of the 164 samples fell in the 3/4 band, although 77% fell within the 2, 3/4 , or 5/6 bands.
- Grades 5 and 6: 40% of the 72 samples fell in the 5/6 band, while 81% fell in the 3/4, 5/6, or 7/8 bands.

Altogether, of the 411 samples for grade levels 1–6, 75% of them either fell into the proper band on the graph, or within one grade level.

When counting the sentences and syllables in a document, Schwartz had the following recommendations:

- Exclude headers ("headlines")
- Treat numerals as one syllable
- Speech set off by a colon mid-sentence should be counted as a separate sentence

Regarding the last point, it could be inferred that independent clauses set off by colons, semicolons, and em dashes should also be counted as separate sentences. Because of this, the program uses sentence units when counting sentences. Numerals are also counted as one syllable by the program, regardless of the project's settings.

Plotting and interpreting the Schwartz graph is the same as other Fry-like graphs. Once you have found the average number of syllables and sentences per 100-words, plot the intersection of these two statistics on the graph. The band that the point falls in will represent the grade-level score of the sample.

> Note
> An item of interest when interpreting this graph is the x-axis. Although the labels read *25*, *26*, etc., they actually represent *125*, *126*, etc. The *100+* with the arrow underneath it (to the left) indicates that the axis labels should have 100 added to them. This is done to make the labels more compact.
> Although this is an unusual display, this is how the graph is drawn in the original article. Therefore, to preserve fidelity with the source material, it is recommended to use this same style.

12.13 SMOG (Bamberger-Vanecek)

SMOG (Bamberger-Vanecek) (78) is a German variation of SMOG designed for secondary-age readers. This test calculates the grade level of a document based on complex-word density.

Note that numerals are fully syllabized (i.e., sounded out) for this test, so the program overrides your numeral syllabication setting when calculating it.

SMOG (Bamberger-Vanecek) formula

$$G = \sqrt{C} - 2$$

Where:

G	Grade level
C	Number of 3+ syllable[a] words from 30 sentences

[a] *"drei- und mehrsilbigen"*

This test requires a 10-sentence sample, with 3 samples being recommended for best results. (Normalization can also be used to analyze larger samples or entire documents.)

Note
Based on the examples from Bamberger-Vanecek (78), the grade-level score should be truncated to one-point precision, not rounded.

12.14 Wheeler-Smith (Bamberger-Vanecek)

Wheeler-Smith (Bamberger-Vanecek) (77, 186) is a German variation of Wheeler-Smith designed for both primary and secondary-age reading material. It calculates the grade level of a document based on unit length and number of complex words.

Wheeler-Smith (Bamberger-Vanecek) formula

$$I = ((W/U) * ((C/W) * 100))/10$$

Where:

I	Index value
C	Number of 3+ syllable[a] words
W	Number of words
U	Number of units

[a] *"drei- und mehrsilbigen"*

Then, find the index value in the following chart to acquire the final readability score:

Table 12.9: Wheeler-Smith (Bamberger-Vanecek) Conversion Table

Grade	Score
First grade	2.5–6
Second grade	6.1–9
Third grade	9.1–12
Fourth grade	12.1–16
Fifth grade	16.1–20
Sixth grade	20.1–24
Seventh grade	24.1–29
Eighth grade	29.1–34
Ninth grade	34.1–38
Tenth grade	38.1–42

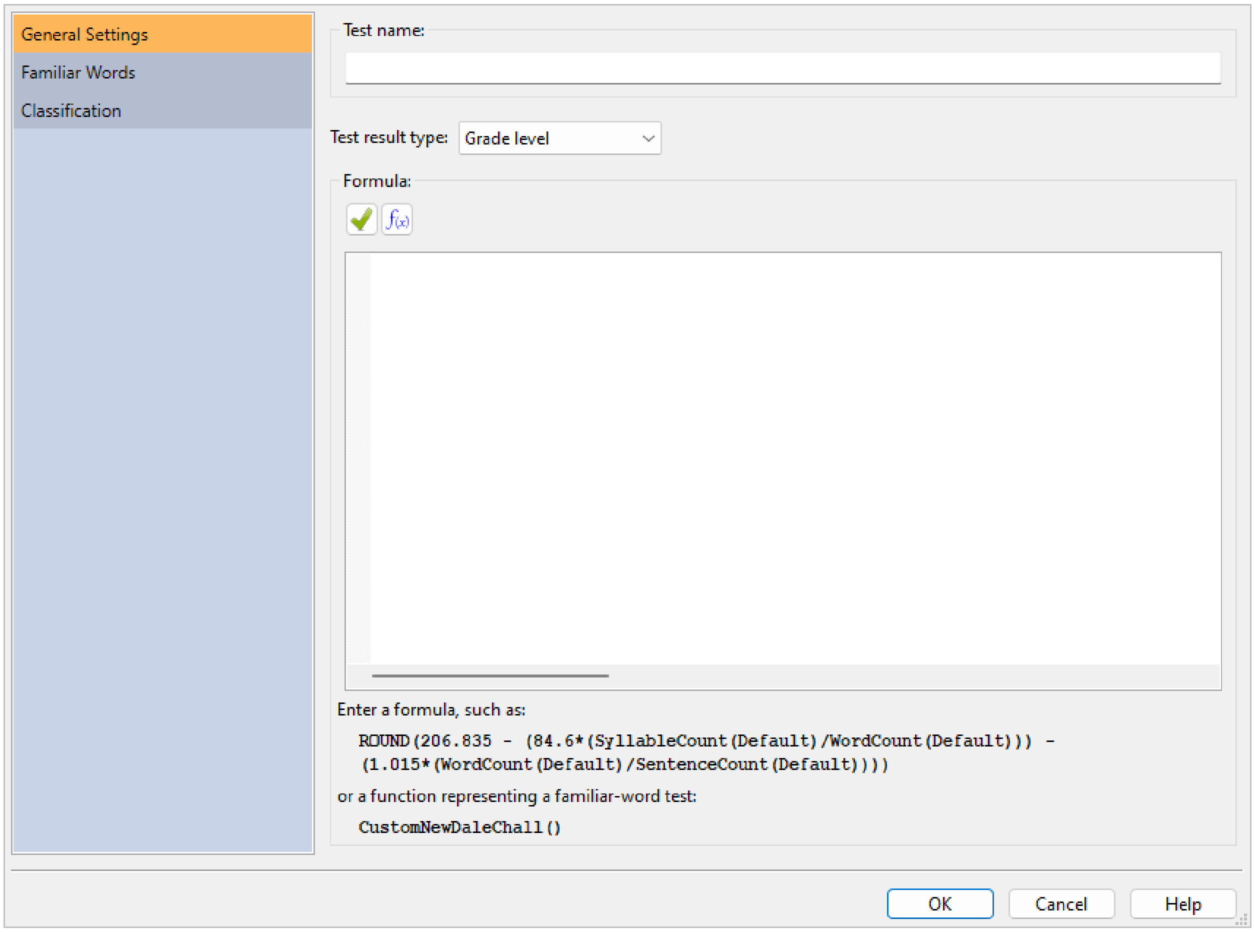
Note
Although Bamberger and Vanecek (77, 186) do not specify using units instead of sentences, the use of units is assumed to match Wheeler-Smith (398).
Based on the examples from Bamberger and Vanecek (77, 186), index values should be truncated to one-point precision, not rounded.

13. Custom Tests & Bundles

13.1 Creating a Custom Test

Along with offering standard tests, such as New Dale-Chall and Spache Revised, *Readability Studio* also includes the ability to create your own tests. Many customizable features are offered for new tests, such as being able to select your own familiar-word list and which formula to use.

To create a custom test, go to the Readability tab, click the arrow beneath the Custom Tests button and select Add Test... from the menu. At this point, the Add Custom Test dialog will be displayed:



In the General Settings section, enter a name for this test (e.g., "NDC Computer") in the Test name field. This will be the name of the test as it appears in your results and on the menu.

Next, enter the test's formula in the Formula field. Essentially, there are two types of formulas: predefined familiar-word formulas and custom-built formulas. A predefined familiar-word formula will use a standard test's formula along with your own familiar-word criteria. This is useful for expanding a standard test (such as New Dale-Chall) by including your own word list. The available predefined formulas are:

- `CustomNewDaleChall()`: this will use the New Dale-Chall formula.
- `CustomSpache()`: this will use the Spache Revised formula.
- `CustomHarrisJacobson()`: this will use the Harris Jacobson formula.

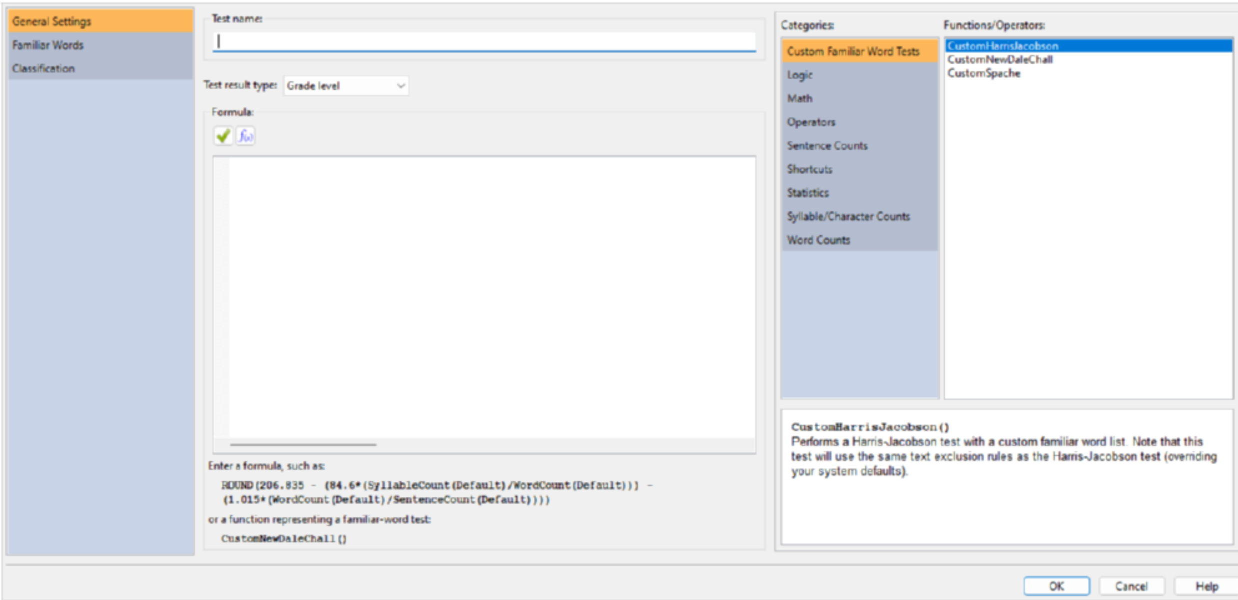
The second type of formula is a custom-built formula. For example, the formula for Lix is:

```
SevenCharacterPlusWordCount() / SentenceCount()
```

To create a new formula based on Lix that treats 6+ character words as difficult, enter this:

```
SixCharacterPlusWordCount() / SentenceCount()
```

To view all available functions, click the fx button. This will display a function browser on the right side of the dialog:



The left side of this browser shows the function categories, while the right side shows the functions from the selected category. Beneath, a description of the selected function will be shown. Note that double-clicking a function on the right side will insert it into the formula.

Refer to sec. 13.2 for a list of available functions.

After entering your formula, you will need to specify what type of test this will be from the Test result type combobox. This will determine which column the test's score will appear in the Readability Scores window. The following types of tests are available:

- Grade level
- Index value
- Predicted cloze score

If your test is using any of the predefined familiar-word formulas (`CustomNewDaleChall()`, `CustomSpache()`, `CustomHarrisJacobson()`) or using the functions `UnfamiliarWordCount()` or `UniqueUnfamiliarWordCount()`, then you will need to define how familiar words are detected. Expand Familiar Words to view two sections: Word Lists and Proper Nouns & Numerals.

If you want to include a list of your own familiar words, then check the option Include custom familiar word list. Next, enter the path to the familiar-word file in the File containing familiar words field. This file must be a text file and each word must be separated by a space, tab, comma, semicolon, or new line. An example file would look like this:

```
apple orange pear
```

or like this:

```
apple
orange
pear
```

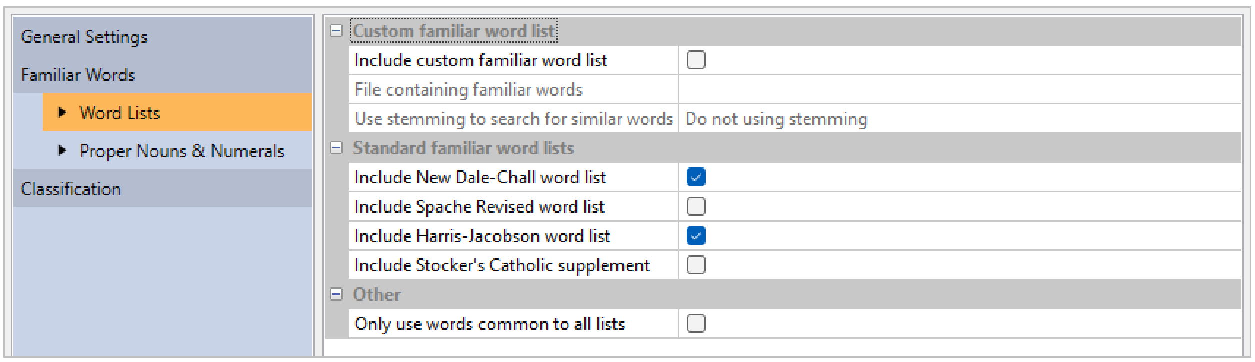
After entering the file path, the words from the file will be shown in the familiar words grid.

Next, you can specify which type of stemming (if any) to use. Stemming compares words based on their roots and is useful for finding words that appear in different forms. This comparison is performed when words from a document are being matched against your familiar-word list. For example, let us say that you have the word *open* in your list and have English stemming selected. When you analyze a document, words such as *opened* and *opening* will be considered familiar, even if they are not explicitly in your word list.

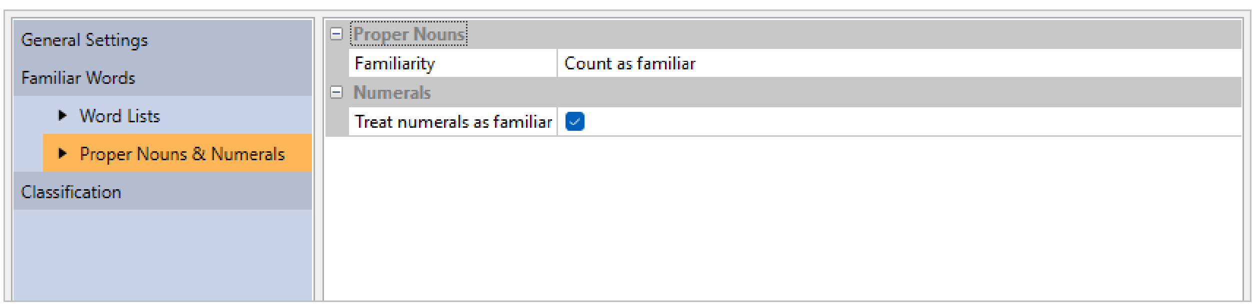
> Note
> The application implements 3.0.1 of the *Snowball* standard for its stemming algorithm.

If you are not using stemming, then it is recommended that you enter each inflected form of your words (e.g., possessive, plural) in your familiar-word list. For example, you would need *open*, *opens*, *opening*, etc. explicitly in your list to cover each form of *open*.

Along with your own word list, you can also use the New Dale-Chall, Spache, and Harris-Jacobson word lists. This enables you to make your word list an appendage to any of these standard tests, therefore creating your own extended version of these tests. This behavior can be toggled with the options shown below:

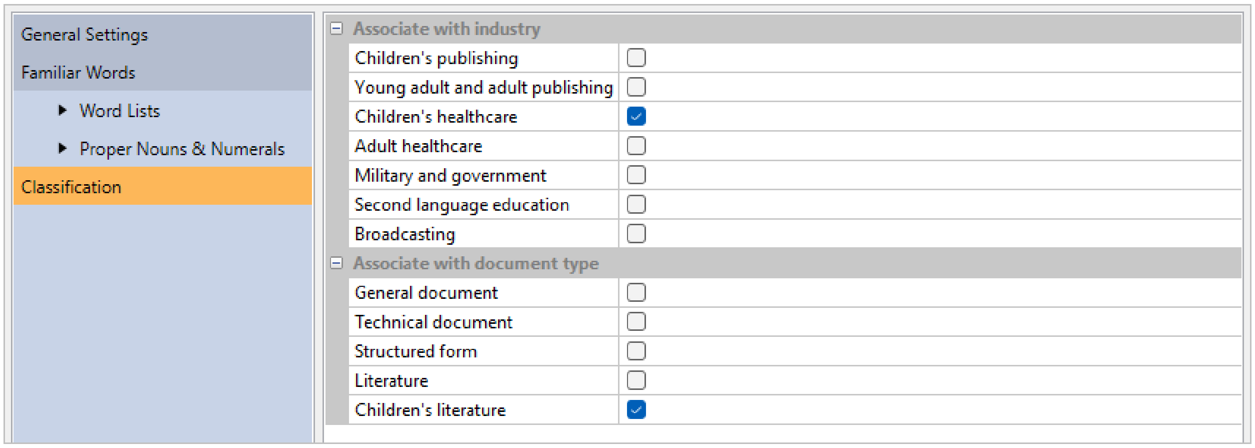


Standard familiar-word tests normally consider numbers and proper nouns as familiar. This is the default behavior for a custom test as well, but these can be toggled on the Proper Nouns and Numbers sections (shown below):



Finally, you can specify how this test interacts with the New Project wizard by associating it with document types and industries. Click the Classification icon to display these options. Then select all the documents and industries that this test should relate to.

For example, select the Children's literature and Children's healthcare check boxes (as shown below) and click OK.



Now create a new project and select Children's healthcare as the industry type. Once the program loads the document, your custom test will be automatically included in the results.

13.2 Custom Test Functions

The following functions and operators are available for use in custom test formulas.

Table 13.1: Operators

Operator	Description
*	Multiplication.
/	Division.
%	Modulus: Divides two values and returns the remainder.
+	Addition.
-	Subtraction.
^	Exponentiation; the number in front of ^ is the base, the number after it is the power to raise it to.
**	Exponentiation (this is an alias for ^)
=	Equals.
<	Less than.
>	Greater than.
<>	Not equal to.
!=	Not equal to (this is an alias for <>)
>=	Greater than or equal to.
<=	Less than or equal to.
&	Logical conjunction (AND).
|	Logical alternative (OR).
( )	Groups sub-expressions, overriding the order of operations.

For operators, the order of precedence is:

Operator	Description
( )	Instructions in parentheses are executed first.
^	Exponentiation.
*, /, and %	Multiplication, division, and modulus.
+ and -	Addition and subtraction.

For example, the following:

$$5 + 5 + 5/2$$

Will yield `12.5`. `5/2` is executed first, then added to the other fives. However, by using parentheses:

$$(5 + 5 + 5)/2$$

You can override it so that the additions happen first (resulting in `15`), followed by the division (finally yielding `7.5`). Likewise, `(2+5)^2` will yield 49 (7 squared), while `2+5^2` will yield 27 (5 squared, plus 2).

Table 13.3: Logic Functions

Function	Description
AND(Value1, Value2, ...)	Returns true if all conditions are true.
IFS(Condition1, Value1, Condition2, Value2, ...)	Checks up to twelve conditions, returning the value corresponding to the first met condition. This is shorthand for multiple nested `IF` commands, providing better readability. Will accept 1–12 condition/value pairs. NaN will be returned if all conditions are false.
IF(Condition, ValueIfTrue, ValueIfFalse)	If *Condition* is true (non-zero), then *ValueIfTrue* is returned; otherwise, *ValueIfFalse* is returned.
NOT(Value)	Returns the logical negation of *Value*.
OR(Value1, Value2, ...)	Returns true if any condition is true.

Table 13.4: Math Functions

Function	Description
ABS(Number)	Absolute value of *Number*.
CEIL(Number)	Smallest integer not less than *Number*. `CEIL(-3.2)` = -3 `CEIL(3.2)` = 4
CLAMP(Number, Start, End)	Constrains *Number* within the range of *Start* and *End*.
EXP(Number)	Euler to the power of *Number*.
FAC(Number)	Returns the factorial of *Number*. The factorial of *Number* is equal to 1*2*3*...* *Number*
FACT(Number)	Alias for `FAC()`
FLOOR(Number)	Returns the largest integer not greater than *Number*. `FLOOR(-3.2)` = -4 `FLOOR(3.2)` = 3
LN(Number)	Natural logarithm of *Number* (base Euler).
LOG10(Number)	Common logarithm of *Number* (base 10).
MIN(Value1, Value2, ...)	Returns the lowest value from a specified range of values.
MAX(Value1, Value2, ...)	Returns the highest value from a specified range of values.
MOD(Number, Divisor)	Returns the remainder after *Number* is divided by *Divisor*. The result has the same sign as divisor.
POW(Base, Exponent)	Raises *Base* to any power. For fractional exponents, *Base* must be greater than 0.
POWER(Base, Exponent)	Alias for `POW()`.
RAND()	Generates a random floating point number within the range of 0 and 1.
ROUND(Number, NumDigits)	*Number* rounded to *NumDigits* decimal places. If *NumDigits* is negative, then *Number* is rounded to the left of the decimal point. (*NumDigits* is optional and defaults to zero.) `ROUND(-11.6, 0)` = 12 `ROUND(-11.6)` = 12 `ROUND(1.5, 0)` = 2 `ROUND(1.55, 1)` = 1.6 `ROUND(3.1415, 3)` = 3.142 `ROUND(-50.55, -2)` = -100
SIGN(Number)	Returns the sign of *Number*. Returns `1` if *Number* is positive, zero (`0`) if *Number* is `0`, and `-1` if *Number* is negative.
SQRT(Number)	Square root of *Number*.
TRUNC(Number)	Discards the fractional part of *Number*. `TRUNC(-3.2)` = -3 `TRUNC(3.2)` = 3

Table 13.5: Statistical Functions

Function	Description
AVERAGE(value1,value2,...)	Returns the average of a specified range of values.
SUM(value1,value2,...)	Returns the sum of a specified range of values.

Table 13.6: Custom Familiar Word Test Functions

Function	Description
CustomHarrisJacobson()	Performs a Harris-Jacobson test with a custom familiar word list. Note that this test will use the same text exclusion rules as Harris-Jacobson (overriding your system defaults).
CustomNewDaleChall()	Performs a New Dale-Chall test with a custom familiar word list. Note that this test will use the same text exclusion rules as New Dale-Chall (overriding your system defaults).
CustomSpache()	Performs a Spache Revised test with a custom familiar word list.

Table 13.7: Document Statistics Functions

Function	Description
CharacterCount()	Returns the number of characters (i.e., letters and numbers). This function takes an argument specifying which text exclusion method to use: `Default`, `DaleChall`[a], or `HarrisJacobson`[b].
CharacterPlusPunctuationCount()	Returns the number of characters (i.e., letters and numbers) and punctuation. Note that sentence-ending punctuation is not included in this count.
IndependentClauseCount()	Returns the number of units/independent clauses. A unit is a section of text ending with a period, exclamation, question mark, interrobang, colon, semicolon, or dash.
SentenceCount()	Returns the number of sentences. This function takes an argument specifying which text exclusion or sentence-counting method to use: `Default`, `DaleChall`, `HarrisJacobson`, or `GunningFog`[c].
SyllableCount()	Returns the number of syllables. This function takes an argument specifying which numeral syllabizing method to use: `Default`, `NumeralsFullySyllabized`, or `NumeralsAreOneSyllable`.

[a] Uses the Dale-Chall text-exclusion method.
[b] Uses the Harris-Jacobson text-exclusion method.
[c] Uses the Gunning Fog sentence-counting method.
This will be either traditional sentence counting or unit counting.

Table 13.8: Word Statistics Functions

Function	Description
FamiliarDaleChallWordCount()	Returns the number of familiar New Dale-Chall words.
FamiliarHarrisJacobsonWordCount()	Returns the number of familiar Harris-Jacobson words.
FamiliarSpacheWordCount()	Returns the number of familiar Spache words.
FamiliarWordCount()	Returns the number of familiar words (based on a custom list) from the document.
HardFogWordCount()	Returns the number of difficult Gunning Fog words.

Function	Description
MiniWordCount()	Returns the number of miniwords (refer to sec. 10.10).
NumeralCount()	Returns the number of numerals.
OneSyllableWordCount()	Returns the number of monosyllabic words.
ProperNounCount()	Returns the number of proper nouns.
SevenCharacterPlusWordCount()	Returns the number of words consisting of seven or more characters.
SixCharacterPlusWordCount()	Returns the number of words consisting of six or more characters.
ThreeSyllablePlusWordCount()	Returns the number of words consisting of three or more syllables. This function takes an argument specifying which numeral syllabizing method to use: `Default` or `NumeralsFullySyllabized`.
UnfamiliarDaleChallWordCount()	Returns the number of unfamiliar New Dale-Chall words.
UnfamiliarHarrisJacobsonWordCount()	Returns the number of unfamiliar Harris-Jacobson words.
UnfamiliarSpacheWordCount()	Returns the number of unfamiliar Spache words.
UnfamiliarWordCount()	Returns the number of unfamiliar words (based on a custom list) from the document.
UniqueOneSyllableWordCount()	Returns the number of unique monosyllabic words.
UniqueSixCharacterPlusWordCount()	Returns the number of unique words consisting of six or more characters.
UniqueUnfamiliarDaleChallWordCount()	Returns the number of unique unfamiliar New Dale-Chall words.
UniqueUnfamiliarHarrisJacobsonWordCount()	Returns the number of unique unfamiliar Harris-Jacobson words.
UniqueUnfamiliarSpacheWordCount()	Returns the number of unique unfamiliar Spache words.
UniqueUnfamiliarWordCount()	Returns the number of unique unfamiliar words (from a custom list).
UniqueThreeSyllablePlusWordCount()	Returns the number of unique words consisting of three or more syllables. This function takes an argument specifying which numeral syllabizing method to use: `Default` or `NumeralsFullySyllabized`.
UniqueWordCount()	Returns the number of unique words.
WordCount()	Returns the number of words. This function takes an argument specifying which text exclusion method to use: `Default`, `DaleChall`, or `HarrisJacobson`.

> Note
> New Dale-Chall and Harris-Jacobson familiar word functions will take their respective test's text exclusion options into account.

Table 13.9: Shortcuts

Shortcut	Description
B	Shortcut for SyllableCount(Default).
C	Shortcut for ThreeSyllablePlusWordCount(Default).
D	Shortcut for FamiliarDaleChallWordCount().
F	Shortcut for HardFogWordCount().
L	Shortcut for SixCharacterPlusWordCount().
M	Shortcut for OneSyllableWordCount().
R	Shortcut for CharacterCount(Default).
RP	Shortcut for CharacterPlusPunctuationCount().
S	Shortcut for SentenceCount(Default).
T	Shortcut for MiniWordCount().
U	Shortcut for IndependentClauseCount().
UDC	Shortcut for UnfamiliarDaleChallWordCount().
UUS	Shortcut for UniqueUnfamiliarSpacheWordCount().
W	Shortcut for WordCount(Default).
X	Shortcut for SevenCharacterPlusWordCount().

Comments

Comments can be embedded within an expression to clarify its intent. C/C++ style comments are supported, which provide:

- multi-line comments (text within a pair of /* and */).
- single-line comments (everything after a // until the end of the current line).

For example:

```
/* Returns a value between 1-3, representing the usage
   of buzz words in the document.*/                          ①
IF(// Review the % of buzz words in the document.
   // Over 15% is unacceptable                               ②
   (FamiliarWordCount()/WordCount())*100 > 15, 3 /* Unacceptable */,
   // Over 5% should be reviewed
   IF((FamiliarWordCount()/WordCount())*100 > 5, 2,
   // 5% or less is acceptable
   1 /* Acceptable */) )
```

① Multi-line comment
② Single-line comment

13.3 Custom Test Dialog

13.3.1 General Settings

From the Readability tab, click the Custom Test button and then select Add Test... to display the Add Custom Test dialog. This dialog is used to edit or create a custom test.

Test name: If creating a new test, enter its name into this field.

Test type: Select from this list the type of result this test returns. This will determine which column the test's score will appear in the Readability Scores window. The following types of tests are available:

- Grade level
- Index value
- Predicted cloze score

Formula

fx Click this button to display the Function Browser dialog. In this dialog, you can browse the available functions and insert them into your formula. Refer to sec. 13.2 for a list of available functions.

✓ Click this button to validate the formula. You will be alerted if your formula contains any syntax errors.

Formula: Enter the test's formula into this field. As you type, a list of available functions will appear beneath the cursor. To select a function from this list, first use the arrow keys on your keyboard to navigate through the list. Once you have selected the desired function, hit the Spacebar on your keyboard to insert it into your formula. You can also insert functions from the Function Browser.

Word Lists

13.3.1.1 Custom familiar word list

Include custom familiar word list: Check this option to use your own word list to determine if a word is familiar.

File containing familiar words: Enter the path to the familiar-word list into this field. This list must be a text file where each word is separated by a space, tab, comma, semicolon, or new line. An example file would look like this:

```
computer install program videogame
```

or like this:

```
computer,install,program,videogame
```

or like this:

```
computer
install
program
videogame
```

Use stemming to search for similar words: Select from this list the stemming method (if any) to use when comparing your familiar words with the words in a document.

Stemming compares words based on their root. This is useful for finding words that appear in different forms from how they are entered in your word list. For example, let us say that you enter the word *installation* in your familiar word file. If you select English as your stemming type, then occurrences of the words *installation*, *install*, *installed*, *installs*, etc. found in your document will be considered familiar. This is because each of these words shares the same root, *install*. The benefit of this is that you do not need to enter every form of every word into your familiar word file.

If you are not stemming, then it is recommended to enter each inflected form of your words (e.g., possessive, plural) in your word list.

Standard familiar word lists

Include New Dale-Chall familiar word list: Check this option to use the New Dale-Chall word list (along with any other selected word lists) to determine if a word is familiar.

Include Spache Revised familiar word list: Check this option to use the Spache Revised word list (along with any other selected word lists) to determine if a word is familiar.

Include Harris-Jacobson familiar word list: Check this option to use the Harris-Jacobson word list (along with any other selected word lists) to determine if a word is familiar.

Include Stocker's Catholic supplement: Check this option to include Stocker's supplementary word list for Catholic students (along with any other selected word lists) to determine if a word is familiar.

Other

Only use words common to all lists: Check this option to consider words familiar only if they appear on your custom list *and* other selected lists.

> Note
> This option is only available if you are including two or more word lists.

> Warning
> This option is only recommended for when you want to find words that appear within a union of your word list and a standard list(s) (e.g., Spache).

13.3.2 Proper Nouns & Numerals

Proper Nouns

Familiarity

Count as unfamiliar: Select this option to consider all proper nouns (that do not appear on any familiar-word list) as unfamiliar.

Count as familiar: Select this option to consider all proper nouns as familiar.

Count only first occurrence of each as unfamiliar: Select this option to consider only the first occurrence of each proper noun (that does not appear on any familiar-word list) as unfamiliar.

Numerals

Treat numerals as familiar: Check this to also consider numeric words as familiar.

Classification

Associate with document type: Select the document types that you want this test to be associated with when creating a new project. This will only take effect if the project wizard selects tests for you based on document type.

Associate with industry: Select the industries that you want this test to be associated with when creating a new project. This will only take effect if the project wizard selects tests for you based on industry or field.

13.4 Adding a Custom-word Test

From the Readability tab, click the Custom Test button and then select Add Custom [Test]... to display the Add Custom [Test] dialog. The following tests available are: New Dale-Chall, Spache Revised, and Harris-Jacobson. This dialog is used to create a custom word test and is a simplified version of the standard Custom Test dialog.

Test name: Enter the name of the new test (e.g., "Harris-Jacobson (Social Studies)") into this field.

File containing familiar words: Enter the path to the familiar-word list into this field. This list must be a text file where each word is separated by a space, tab, comma, semicolon, or new line. An example file would look like this:

```
computer install program videogame
```

or like this:

```
computer,install,program,videogame
```

or like this:

```
computer
install
program
videogame
```

13.5 Test Bundles

Test bundles enable you to organize tests into collections that you can quickly add to a project. Rather than selecting numerous tests from the menus one-by-one, you can simply select a test bundle from the menu. From there, all tests within the bundle will be added to the project.

To access the test bundles, select the Readability tab and click the Bundles button. A menu will appear which contains all the available bundles, as well as options to add, edit, and remove them.

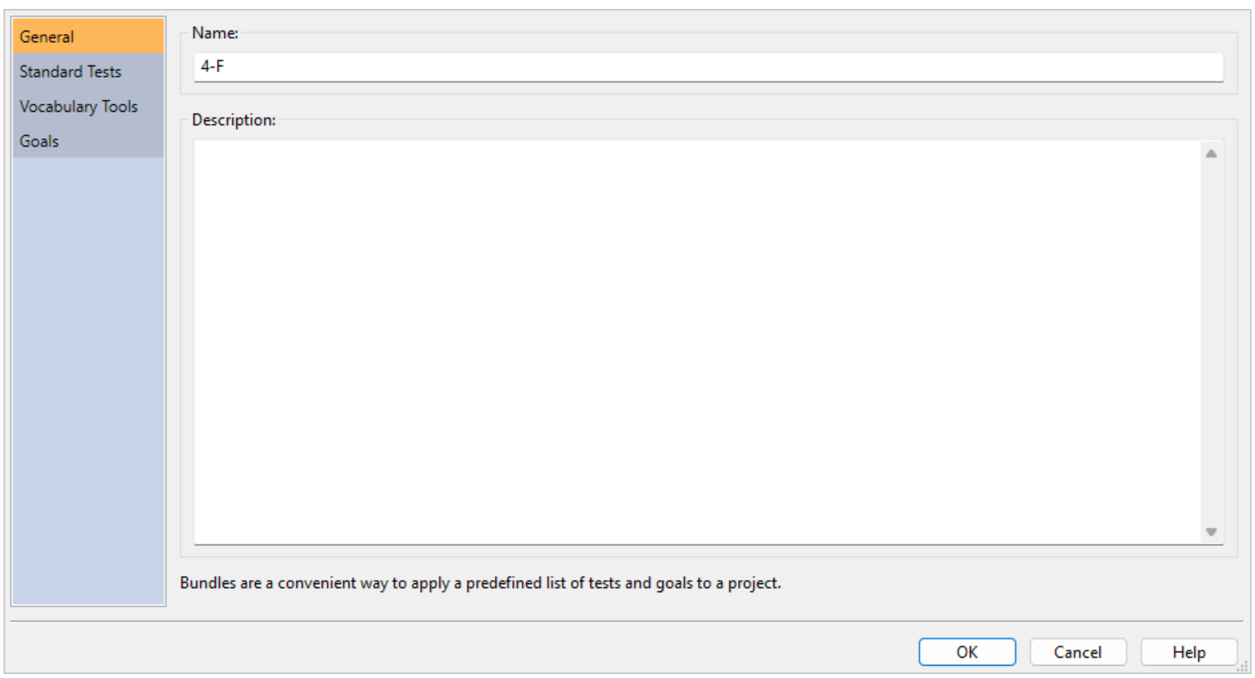
By default, the industry and document-type bundles used in the New Project wizard will be available. This includes:

- General document
- Technical document
- Structured form
- Literature
- Children's literature
- Children's publishing
- Young adult and adult publishing
- Children's healthcare
- Adult healthcare
- Military and government
- Second language education
- Broadcasting

More specific bundles based on research articles are also included, such as *Patient Consent Forms* (Grundner 9–10) and *Kincaid's Tests for Navy-Personnel Materials*. (Some of these bundles may include goals, if recommended in the original article.)

Select any of these bundles from the menu and all the tests within it will be added to the active project. Note that when you add a bundle to a project, all other tests currently in the project will be removed. This ensures that only the tests relevant to the bundle will be in the project after it is applied.

You can also create your own bundles. To do this, select Add... from the Bundles menu to open the Add Test Bundle dialog:



On the General page of this dialog, you will first enter a name for the bundle. (This name is how it will appear in the Bundles menu.) Next, you will select all the tests that you wish to include in this bundle. (Tests are available on the Standard Tests, Custom Tests, and Vocabulary Tools pages.)

Along with selecting tests, you can also specify the range of values that a document should score within for any given test. First, on the Goals page, click the add button to include a new goal. Next, double-click in the Test column to select the test. Finally, click in either the Recommended minimum score or Recommended maximum score column and enter the acceptable score range. Now, when this bundle is applied to a project, the document's pass/fail status will be displayed in the Goals section.

Beneath the test goals grid is another grid for creating statistics goals as well. For example, you can add a goal such as specifying the maximum average sentence length for the document.

Once you are done, click the OK button to add the bundle to the menu.

After this bundle is added, select it from the menu and all the tests in the bundle will be added to the active project.

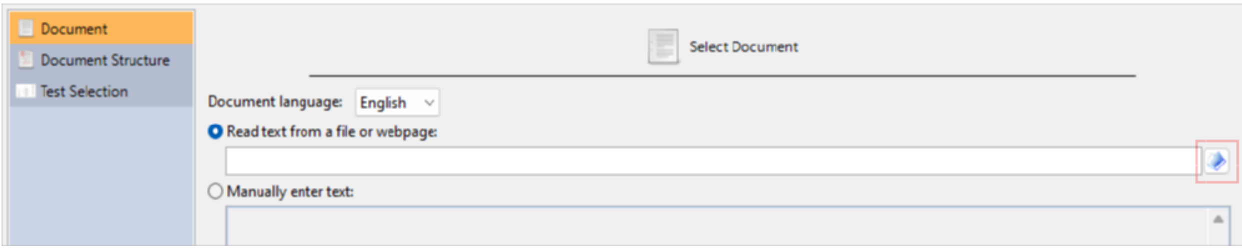
V Examples

14. Getting Started

Within this chapter, we will step through some examples demonstrating how to analyze, review, and improve a document. After that, we will continue with examples showing how to analyze a webpage and how to review difficult sentences.

14.1 Analyzing a Book

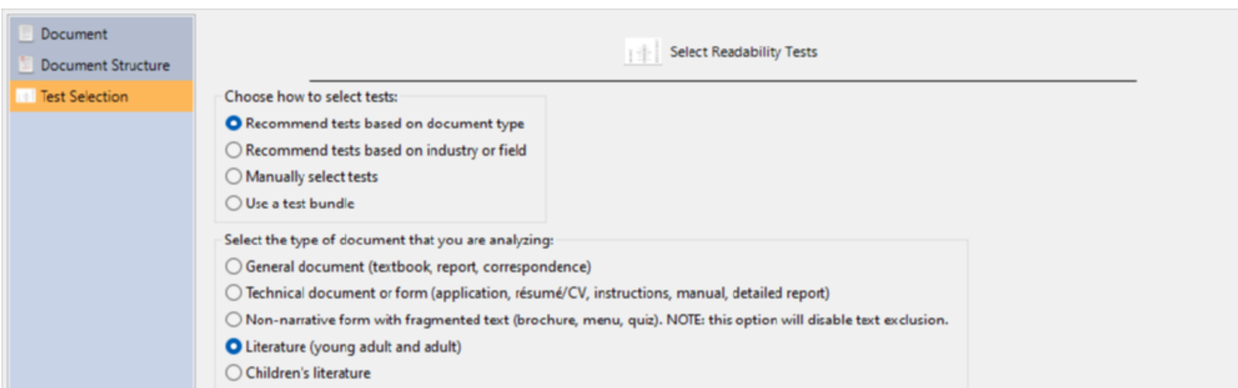
In this example, we will analyze a literary novel to help determine which age group it is most appropriate for (in terms of readability, not content). The first step is to open the New Project wizard and choose the book. Click the New button on the Home tab to open the New Project wizard. First, select English as the document language (if this option is available). Then, select the option Read text from a file or webpage. Click the button next to the file path field to browse for it:



This will open the Select Document to Analyze dialog. Choose the file that you want to analyze and click the Open button to accept.

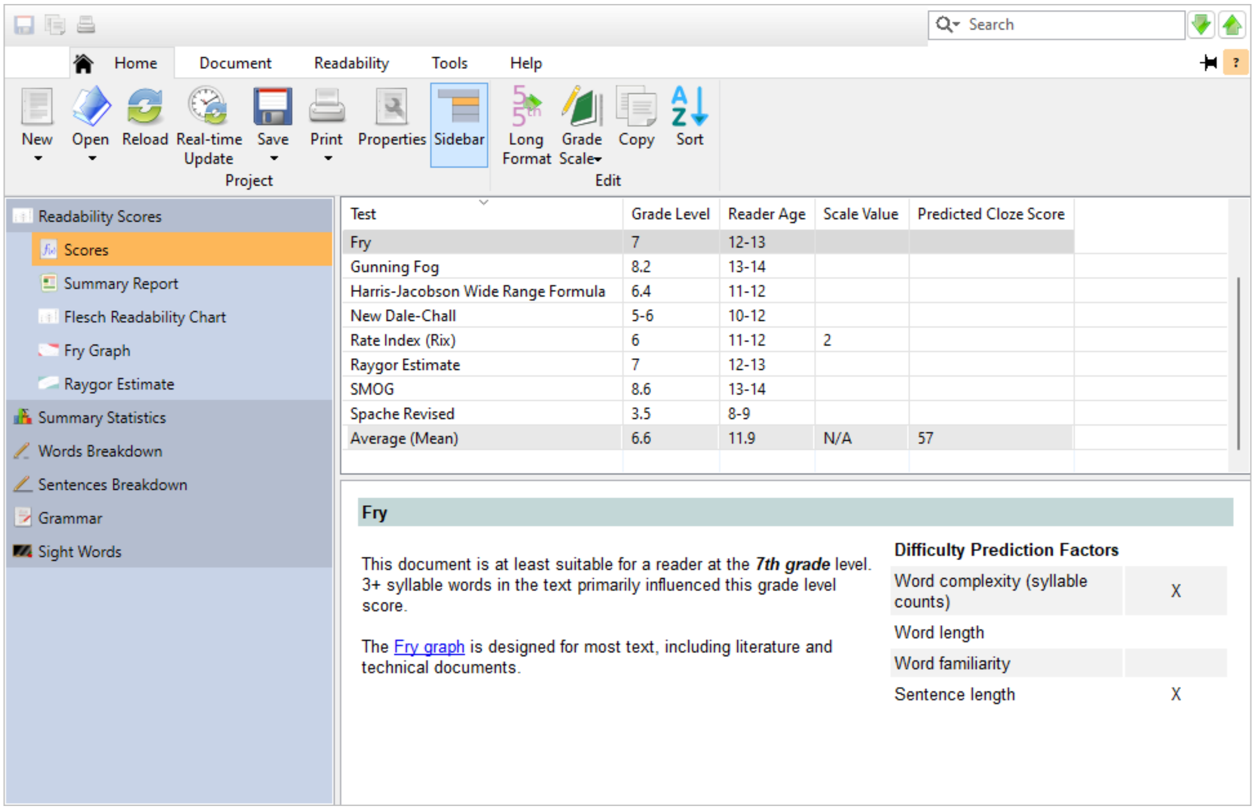
You will then return to the Document page. Click the Forward > button twice to proceed to the Test Selection dialog.

Because this is a literary work, select the Recommend tests based on document type. Then select the Literature (young adult and adult) option (because we believe that this book is meant for older children), as shown below:



Finally, click the Finish button to begin analyzing the project.

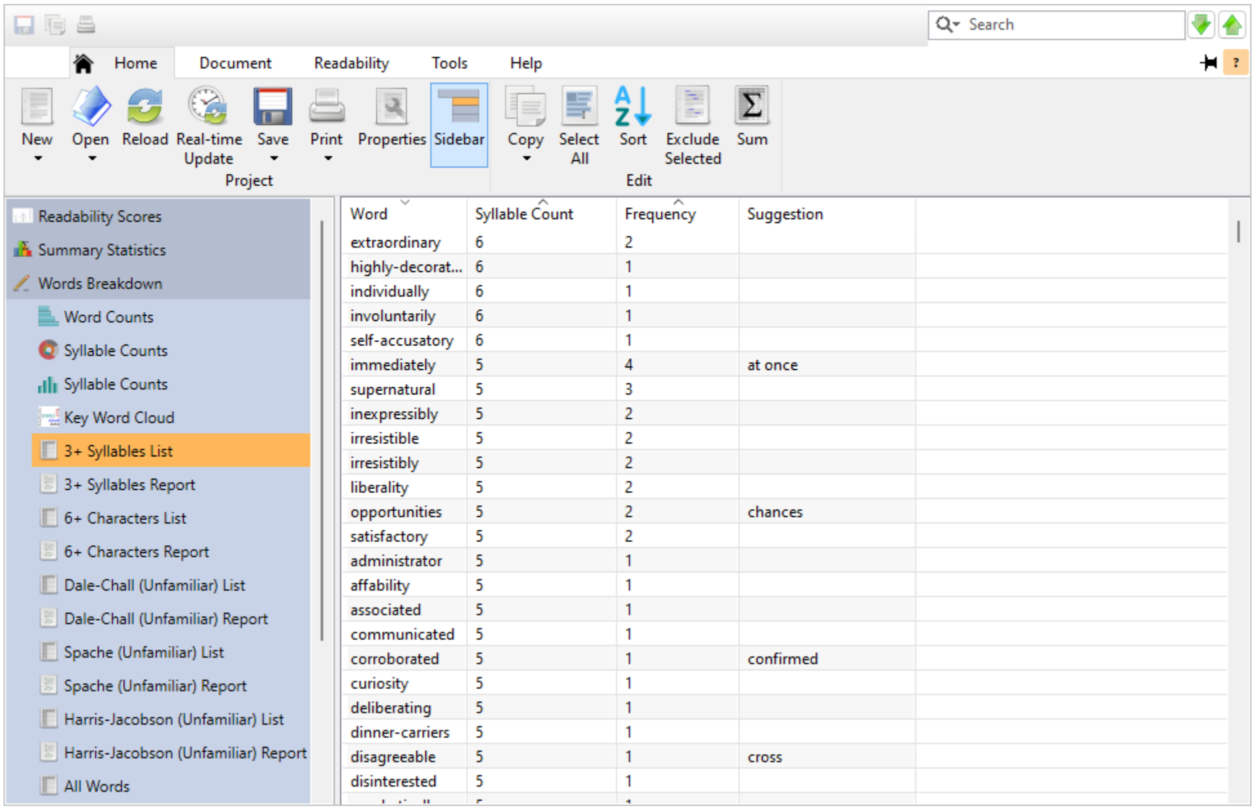
At this point we can review the test scores of the book to help determine its reading level. Click the Readability Scores icon on the project sidebar. As we can see below, this book (Charles Dickens's *A Christmas Carol*) appears to be readable for at least readers in the 10–13 age (or above) range. This implies that it may be too difficult for younger readers due to longer sentences and/or difficult words.



To help understand why this book may be difficult for younger readers, click on the Words Breakdown icon on the project sidebar to display various lists of difficult words. Select the 3+ syllable words subitem to view the list of complex words.

Next, we will sort this list, going from highest to lowest syllable count. Click the Sort button on the ribbon and select Sort Descending from the button's menu. On the Sort Columns Descending dialog, double-click on the first row in the grid and select the column Syllable Count. Then double-click on the row below that and select Frequency. These options will tell the program to sort the words from highest to lowest syllable count, and in the case of any ties to sort by frequency. Click OK to sort the list.

Now the words will be sorted by syllable count and then by frequency. As illustrated below, there are many words containing five or more syllables, which can be very difficult for most readers of any age.



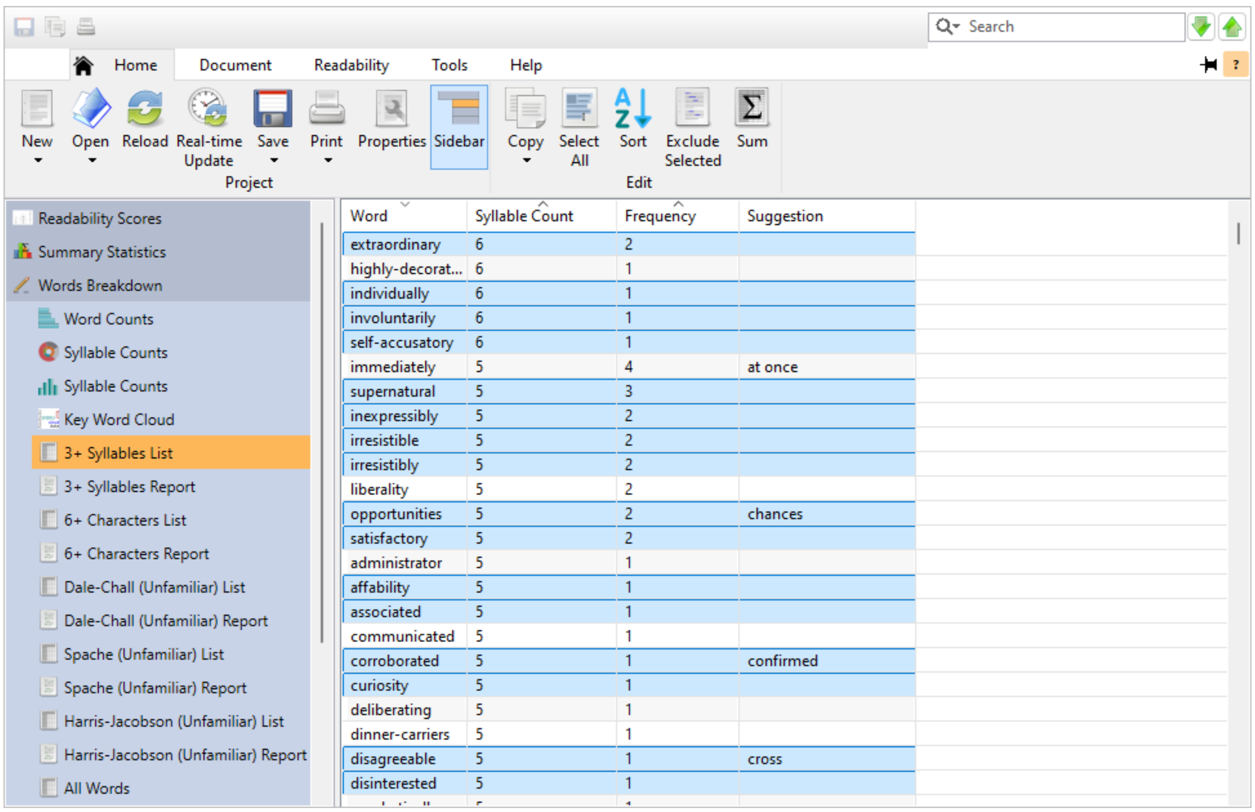
If you are an educator assigning this book to your students, you can export the more difficult words to a report. You can hand this report out to your students and ask them to define these terms before reading the book. This will help prepare them for the book and make it easier and more enjoyable to read.

To do this, click the Save button on the Home tab and use the Save As... dialog to save this list to either an HTML or text file. You can also do this with any of the other difficult word lists, such as the Spache unfamiliar words list. Refer to sec. 14.2 for an example.

14.2 Improving Students' Vocabulary

In this example, we will save a list of the more complex words from a book. If you are an educator, you can send this list to your students and ask them to define these words prior to reading the book. This can be a great way to prepare them for future readings, as well as helping them build their overall vocabulary.

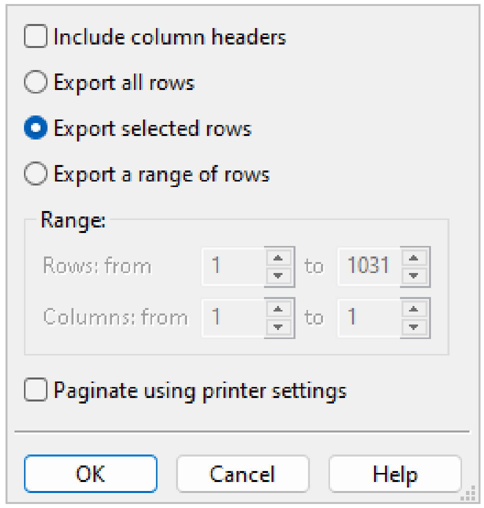
Continuing from the previous example (see sec. 14.1), select the top item in the hard word list. Now scroll down to the last word that has five syllables, hold down the Shift (⊞, 🐧) or ⇧ () key, and select this word. Now you will have all the words with five or more syllables selected. If there are any words that you want to add or remove from your selection, hold down the Ctrl (⊞, 🐧) or ⌘ () key and click on them. Your word list will look like this:



Now that you have selected the words that you want to save, click the Save button on the Home tab. From the menu that appears, select Export 3+ Syllables List....

First, you will be prompted to save as either HTML or text (refer to sec. 7.5 for a discussion of these formats). After specifying the format, the Save As dialog will appear. Select where you want to save the file and click Save.

Next, the List Export Options dialog will appear. (By default, all the words will be exported.) To export just the words that you have highlighted, select the option Export selected rows. Also, because we are only interested in saving the Word column—not the Syllable Count or Frequency Count columns—enter 1 in the Columns: from and to: fields. Finally, uncheck the option Include column headers because we are only exporting one column.



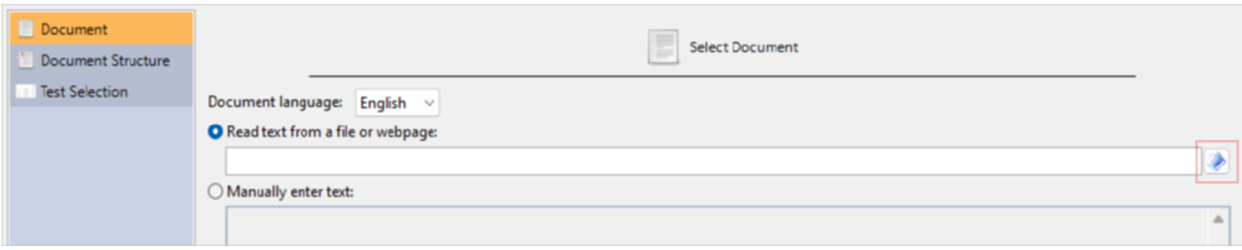
Click OK to export the word list to distribute to your students to help prepare them for their reading assignments.

> Note
> This example book is available by going to the Help tab of the ribbon, clicking the Example Documents button, and selecting *A Christmas Carol*.

14.3 Analyzing a Work in Progress

In this example, we will analyze recipes that we are working on for a middle-school home economics class. Our target audience will be seventh-eighth graders and we will use *Readability Studio* to meet their reading level.

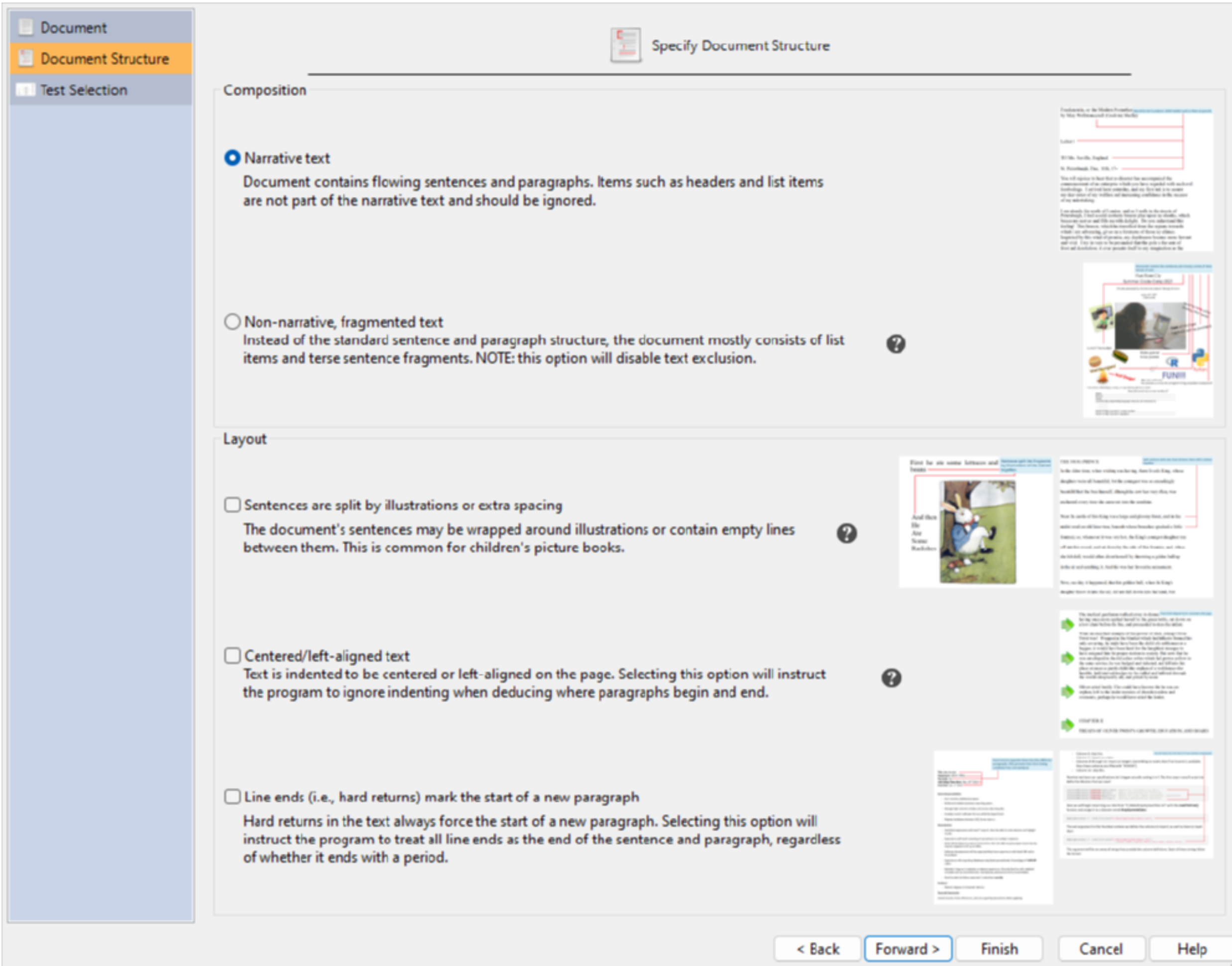
The first step is to open the New Project wizard and select the recipe file. Click the New button on the Home tab to open the New Project wizard. Next, select English as the document language (if this option is available). Then, select the option Read text from a file or webpage. Click the button next to the file path field to browse for it:



This will open the Select Document to Analyze dialog. Choose the file that you want to analyze and click the Open button to accept.

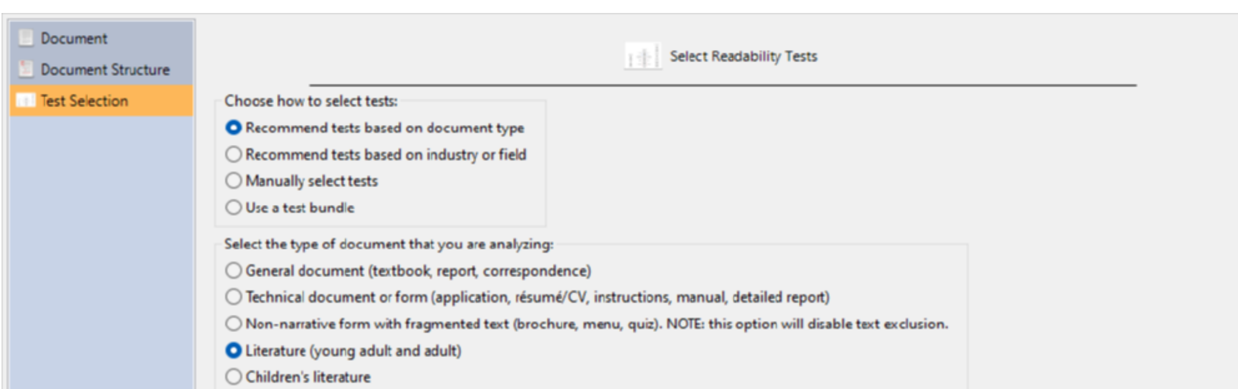
> **Note**
> The original and revised versions of this recipe are available from the Example Documents menu. From the Help tab, click the Example Documents button and then select either "Cocoa Desserts" or "Cocoa Desserts (Revised)" from the menu. When prompted about how to open the document, select Create a new project.

You will then return to the Document page. Click the Forward > button to proceed to the Document Structure page, as shown below:

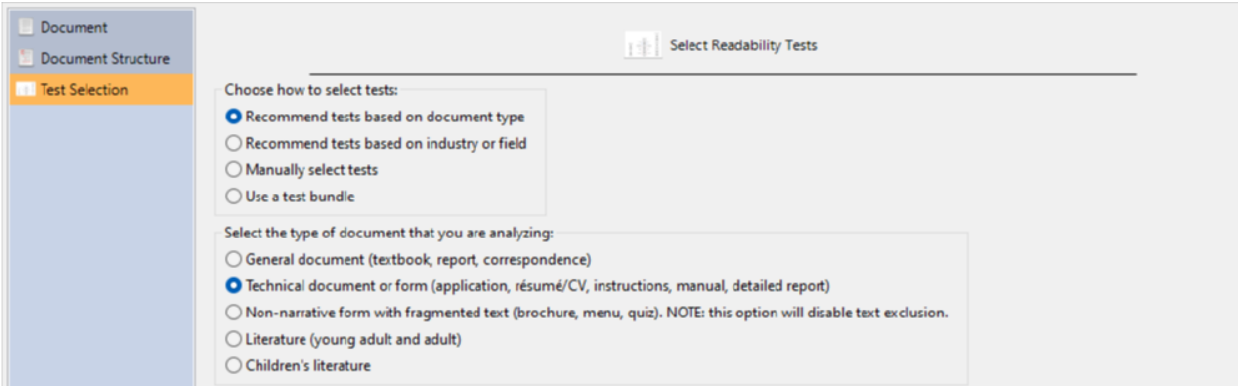


On this page, we will specify how this document is formatted. The first option, Narrative text, is meant for documents that consist of the standard sentence and paragraph structure. The second option, Non-narrative, fragmented text, is meant for documents that contain very few sentences and mostly consist of sentence fragments. This document contains narrative text, so select Narrative text.

Our recipe file does not have any special formatting (e.g., text centering), so keep all options in the Layout section unchecked. Now that we have specified the structure of the document, click the Forward > button to proceed to the Test Selection page, as shown below:



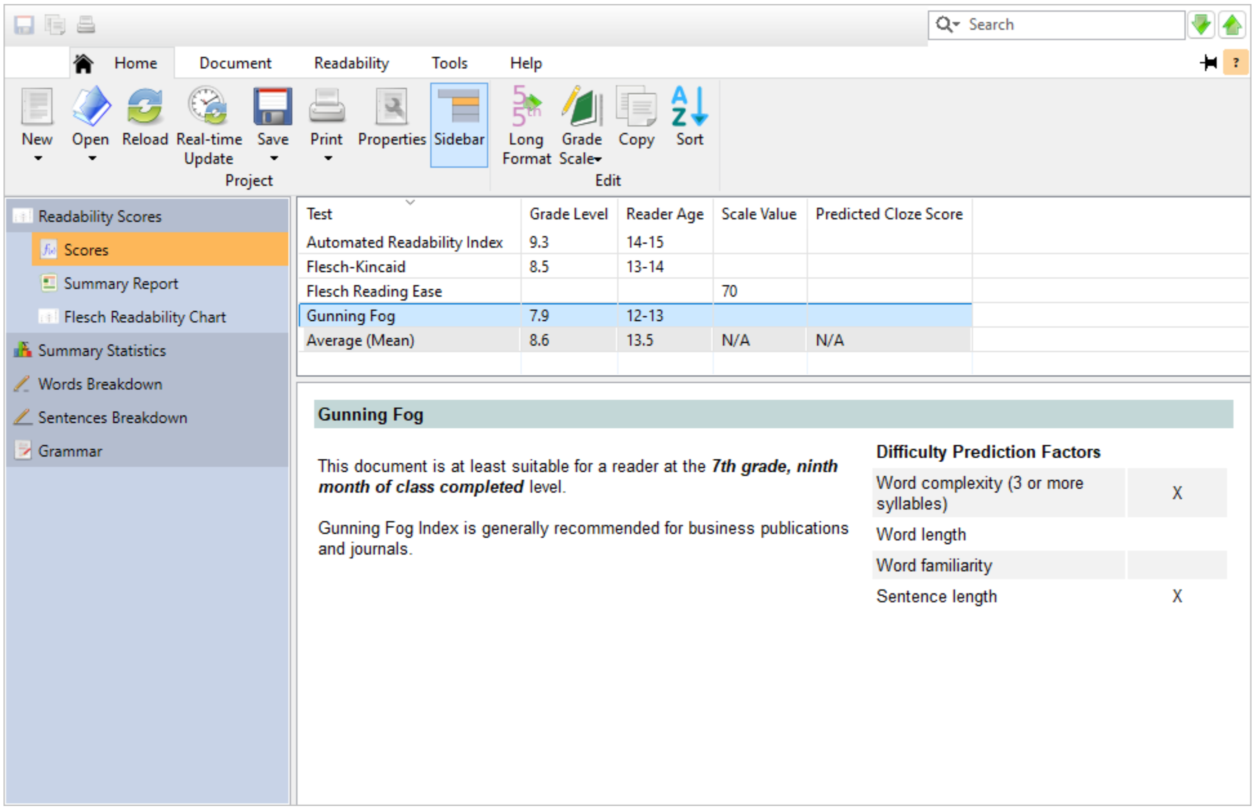
This file does not belong to any particular industry, but we do know that it is an instructional document. Select the Recommend tests based on document type:



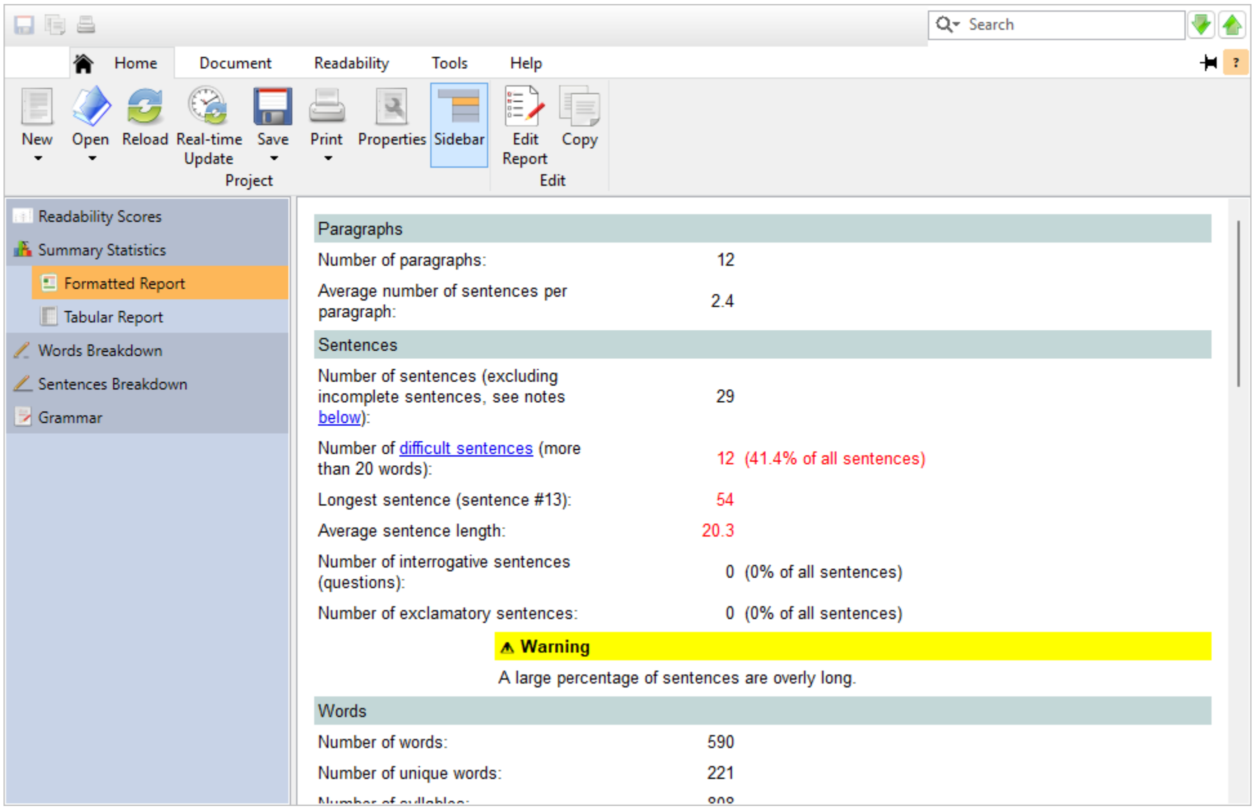
Because this is an instructional document containing small narrative blocks, we will consider this to be a technical document. Select Technical document or form and click the Finish button to begin analyzing the project.

If you are excluding incomplete sentences (the default), then a warning will be displayed. Because this file contains numerous incomplete sentences, the program will ask you if excluding them is what you truly intend. In this case, these sentences are just list items that we should ignore, so click Continue excluding incomplete sentences.

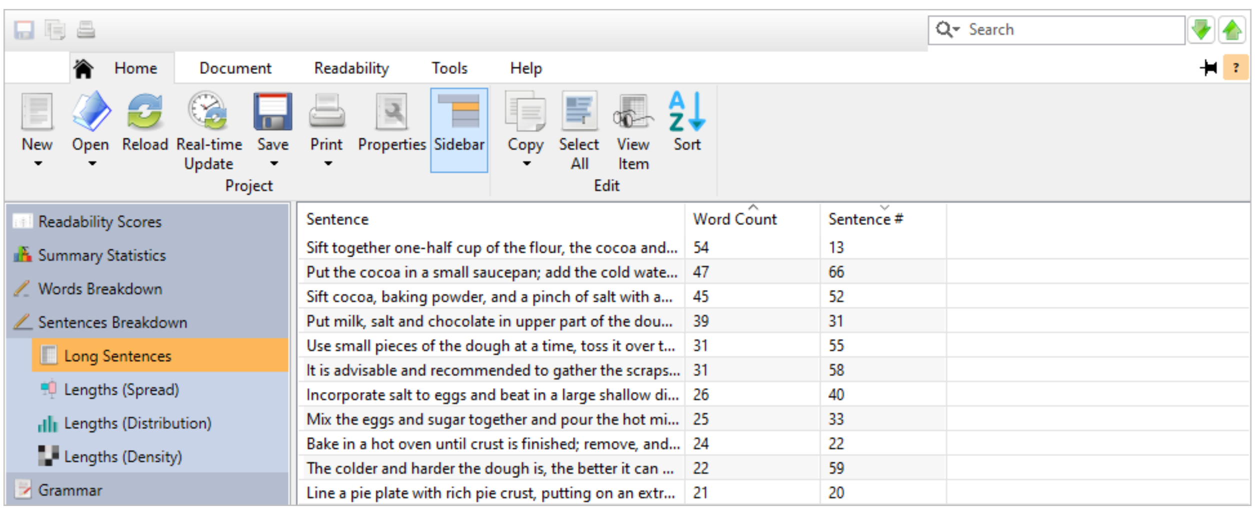
At this point, we can review the test scores of the recipe to determine which reading age it is appropriate for. Click on the Readability Scores icon on the project sidebar to display the scores. As we can see below, it appears to be readable for at least readers in the upper eighth grade. This is higher than our target audience and we may need to make some improvements to this recipe to make it easier to read.



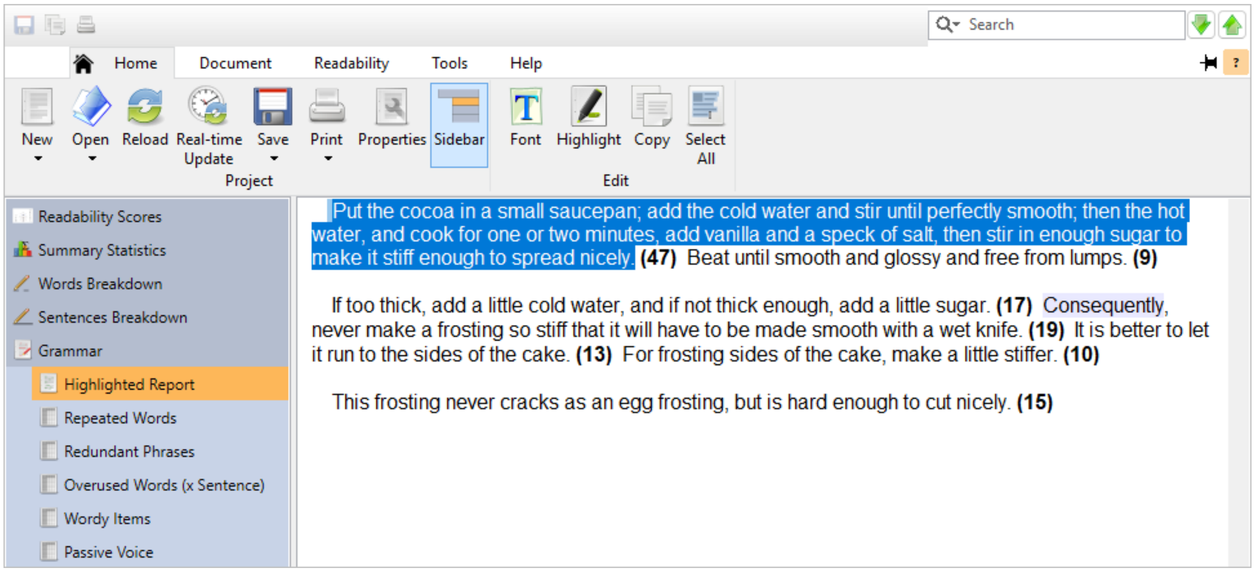
To help understand why this recipe may be difficult for younger readers, click on the Summary Statistics icon to review the statistics and see if the program has any advice for us. In this section, any statistics that may be problematic will be highlighted in red, along with a warning being displayed. As we can see below, there are a lot of long sentences (with the longest being 54 words long):



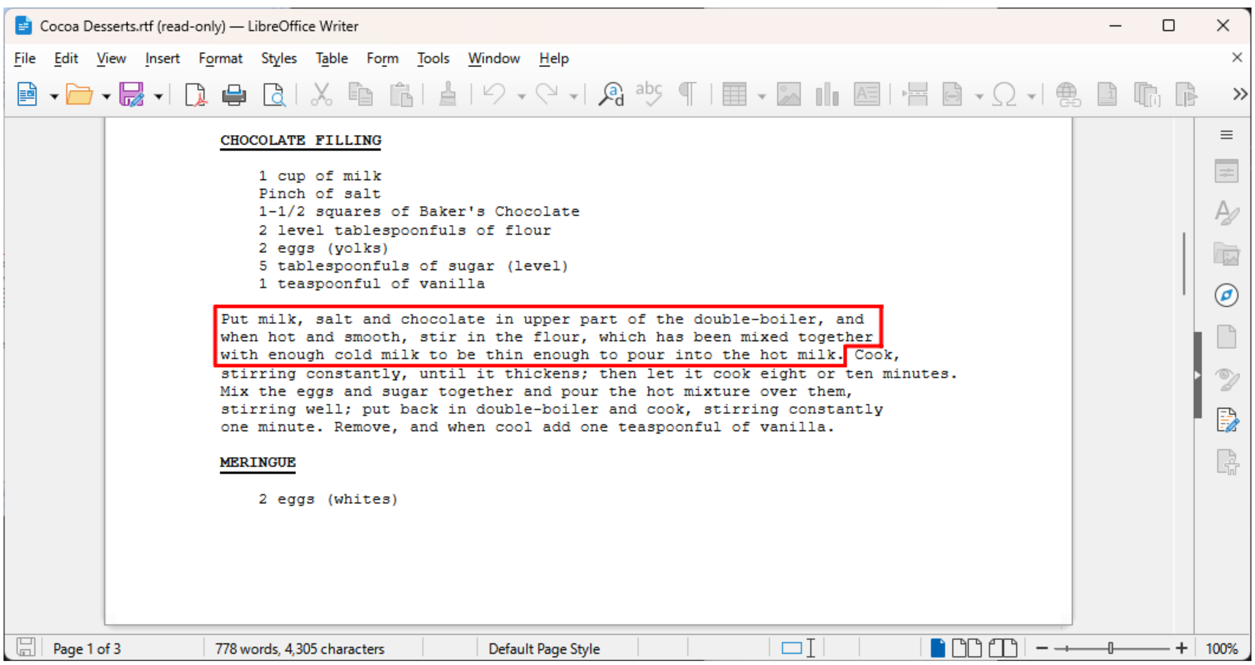
Click on the difficult sentences link in this section to go to the Long Sentences subitem of the Sentences Breakdown section. This window will display a list of all the overly-long sentences in our recipes. As we can see, there are a few rather long sentences that are adversely affecting our scores.



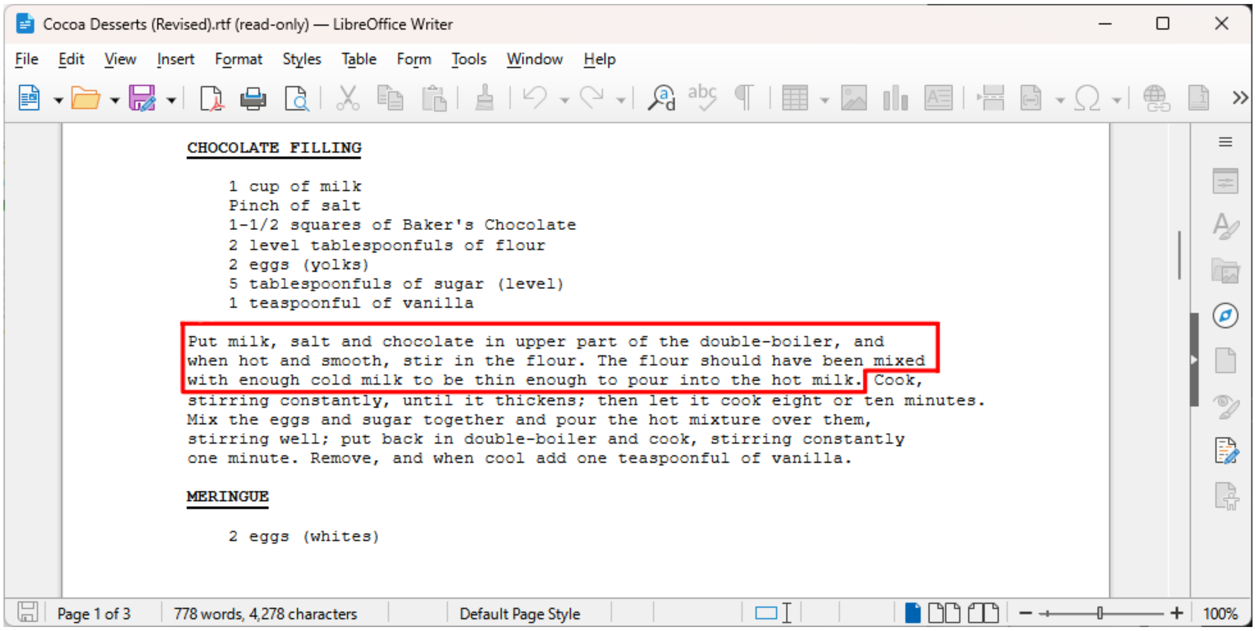
Long sentences can seem unfocused and difficult to read, and it is recommended to shorten or split them. To view any of these sentences in its original context, double-click on it in this list. The program will switch to the Highlighted Report page with this sentence highlighted:



Now that we have found the difficult sentences, let's split them up so that they are easier to read. From the Document tab, click the Edit Document button to open the recipes file for editing. Find one of the long sentences and review it. For example, consider this sentence:



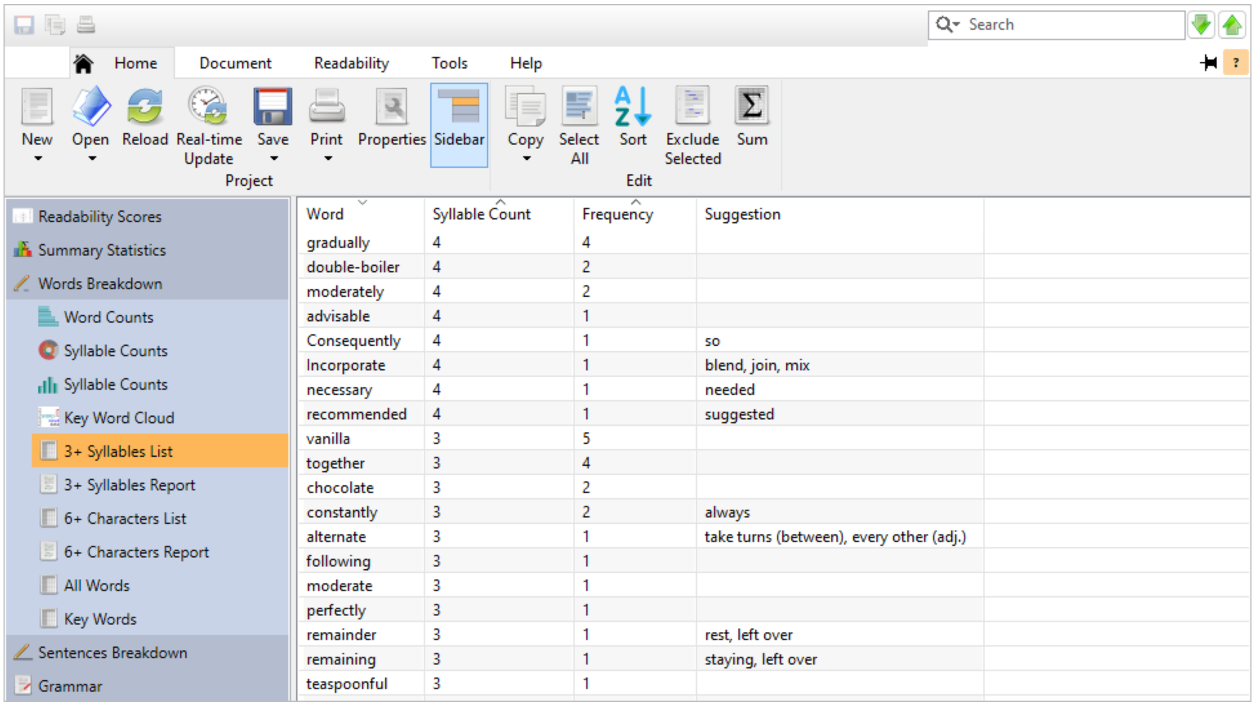
Not only is it long, but it is a little confusing. It is unclear what was mixed with the cold milk because there are so many steps in this sentence. To make it easier to follow, split this sentence into two and reword it:



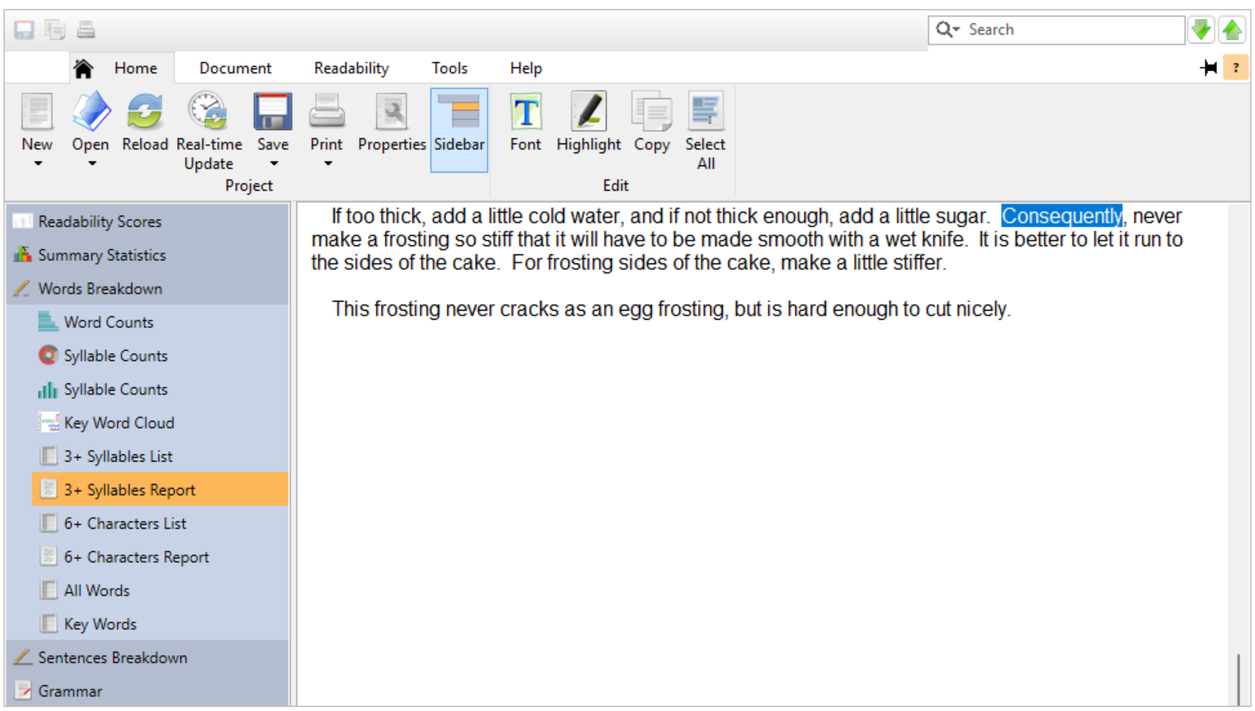
Continue this process for as many of the long sentences as you can. Sentences in the 20-word range are normally acceptable, but sentences over 30 words should be revised.

Other useful tools for shortening sentences in *Readability Studio* are the other pages in the Grammar section. Click on the Grammar icon on the project sidebar. Click on the Repeated Words page and note how the phrase *and and* is in the file (obviously a typo). Next, click on the Redundant Phrases page. Note how the phrase *mixed together* is redundant and the program is suggesting that it be replaced with *mixed*. A Wordy Items and Clichés page may also appear, but there do not seem to be any of these in this file. By fixing these items, we can shorten sentences and improve the overall document.

Although there are not many difficult words in this file, it may be worthwhile to at least review them. Click on the Words Breakdown icon on the project sidebar. This will display various lists of difficult words. Select the 3+ syllable words page and click on the Syllable Count column header twice to sort the words into descending order. As illustrated below, there are a few words containing four syllables, which can be difficult for most readers.



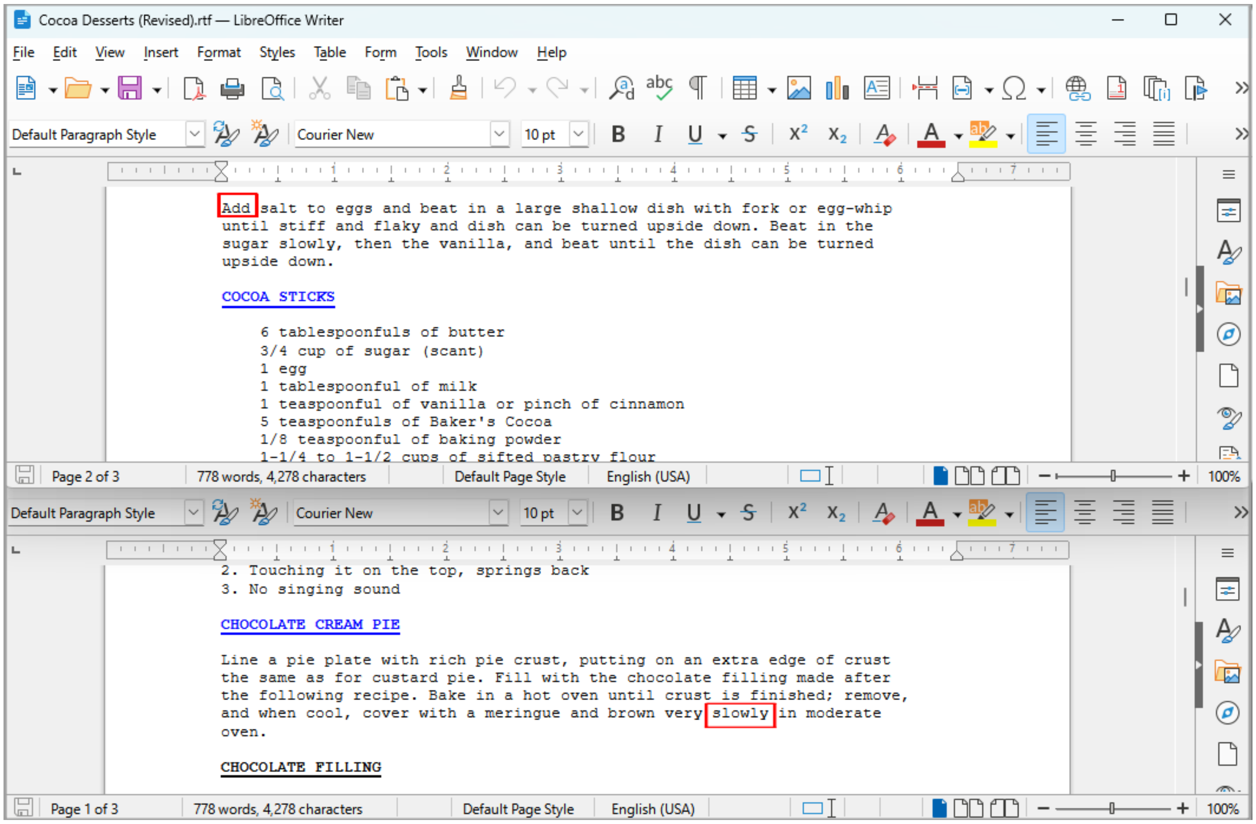
To view these words in their original context, double-click on any word in this list. The program will switch to the 3+ syllable words page, as illustrated below:



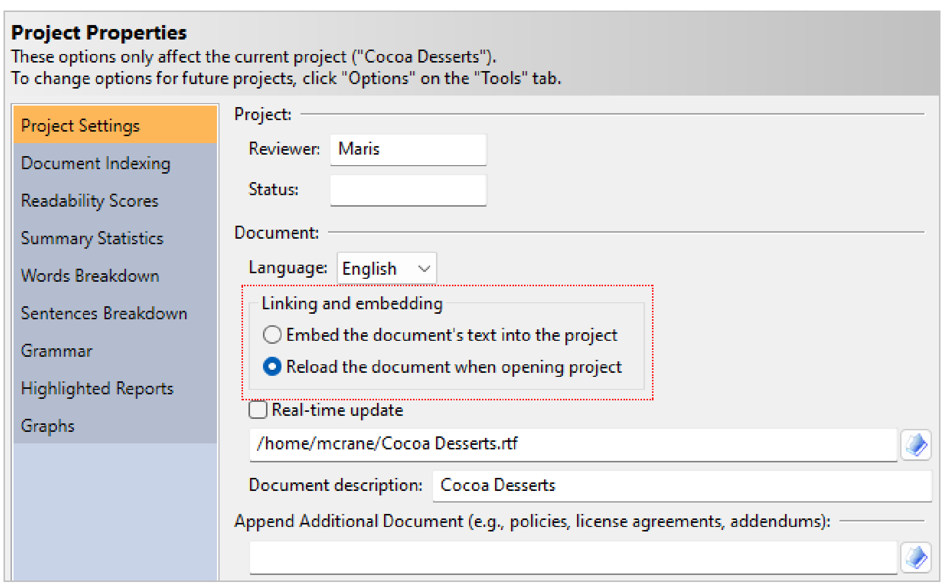
You can click on any of the other pages to view other difficult words in their original context, such as 6+ character words or unfamiliar New Dale-Chall words.

To continue simplifying this recipe, we will replace some of these difficult words. Words such as *incorporate*, *consequently*, and *gradually* could also be replaced with shorter, simpler synonyms.

As illustrated below, we have replaced a few words, such as substituting *incorporate* with *add* and *gradually* with *slowly*.



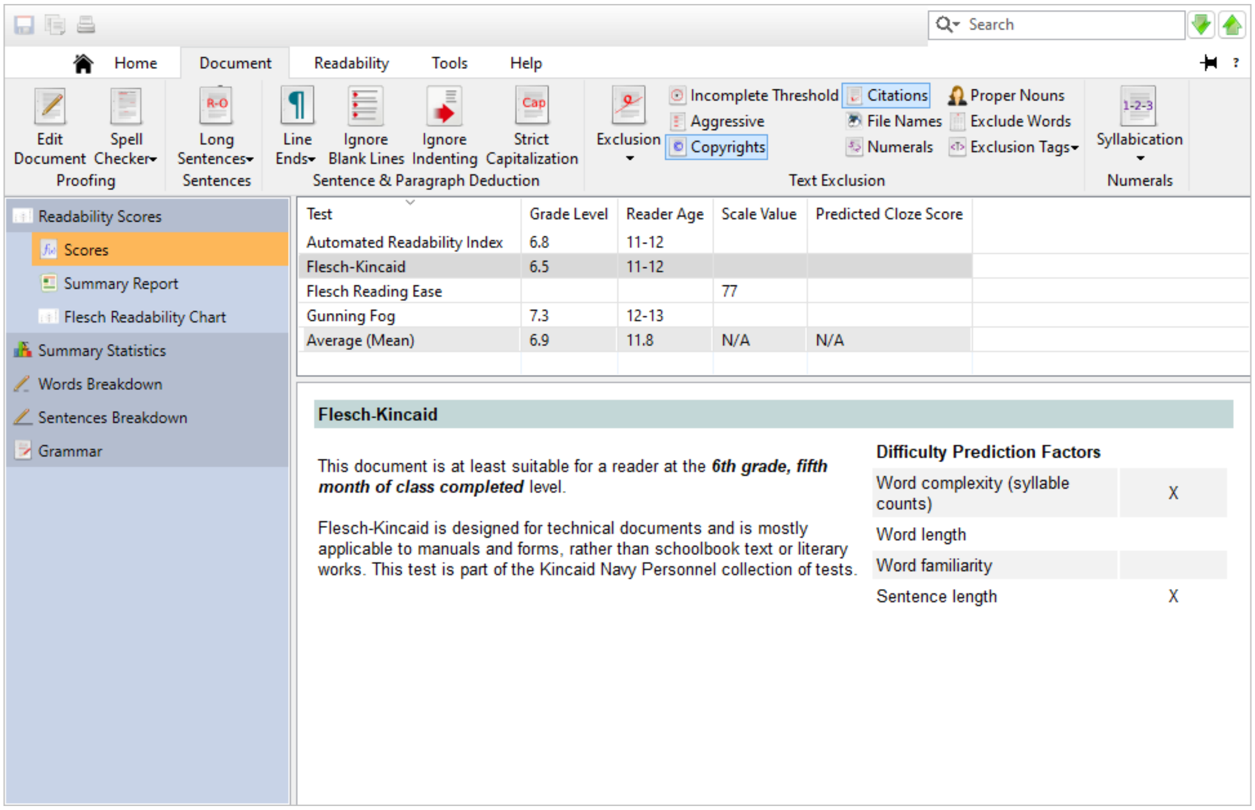
After we finish reworking the original document, we will want to analyze it again to see if it now meets our target audience. Because this project is already linked to the recipe document, we can reload it (rather than starting all over with a new project). To view this connection, click the Properties button on the Home tab to open the Project Properties dialog. Select the Project Settings icon and note the option Reload the source document when opening project:



Also of note is the Real-time update checkbox. If this is checked, then the program will automatically reload the project when the document is edited. This is discussed in a later example (see sec. 16.2); for now, leave this unchecked.

Close this dialog and click the Reload button on the Home tab.

The program will now reload the recipe document, meaning that all the improvements that we just made will be imported into the project. As we can see below, our grade level scores now meet our target audience and this recipe will be easier and more enjoyable to read for our home economics class.



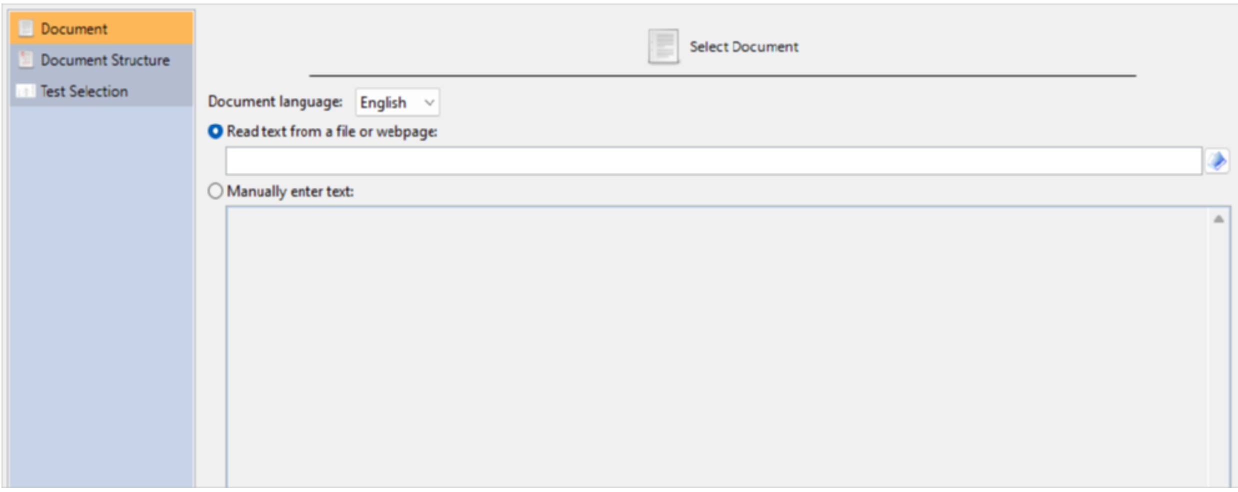
💡 Tip

You can click the Reload button at any time to reanalyze the source document as you edit it externally. Likewise, pressing Real-time Update will tell the program to reload automatically as the document is saved from another program. Finally, whenever you open this project, it will automatically reload the source document (in this case, the recipe file).

14.4 Analyzing a Webpage

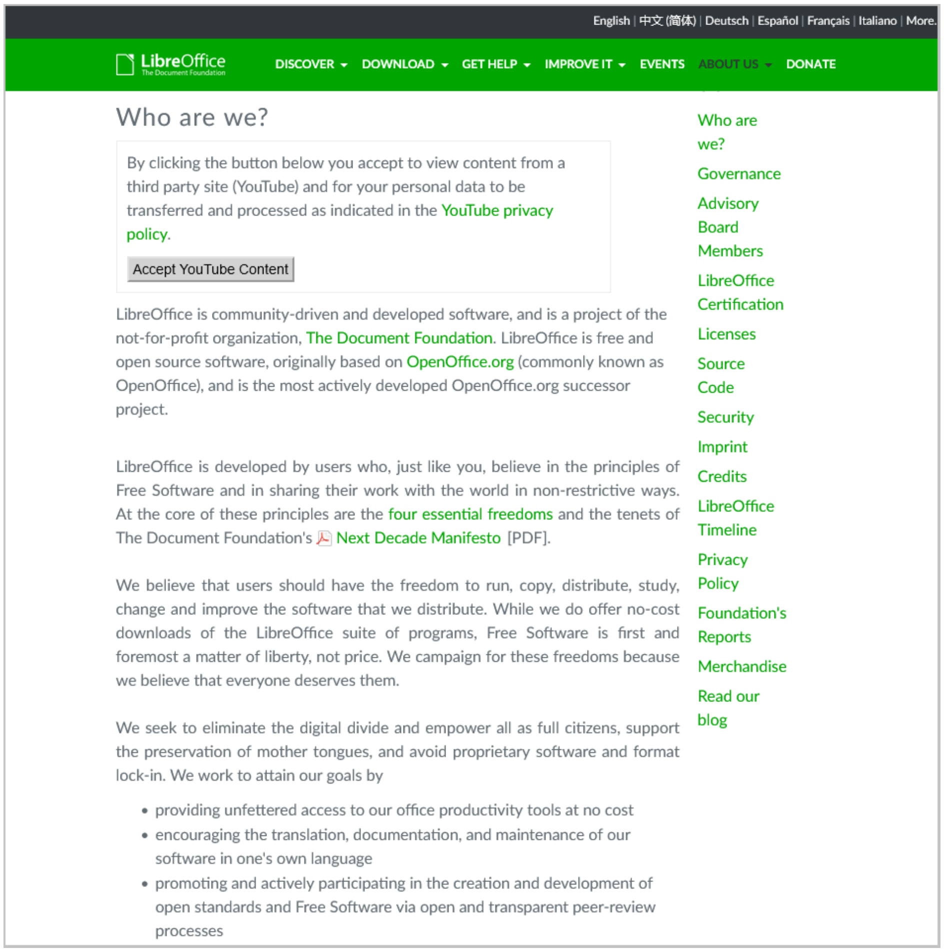
In this example, we will create a project that reads and analyzes a webpage. Although you can save a webpage from your web browser and load the HTML file, *Readability Studio* can also directly connect to a webpage.

First, click on the New button (on the Home tab). The New Project wizard dialog will appear, as shown below:

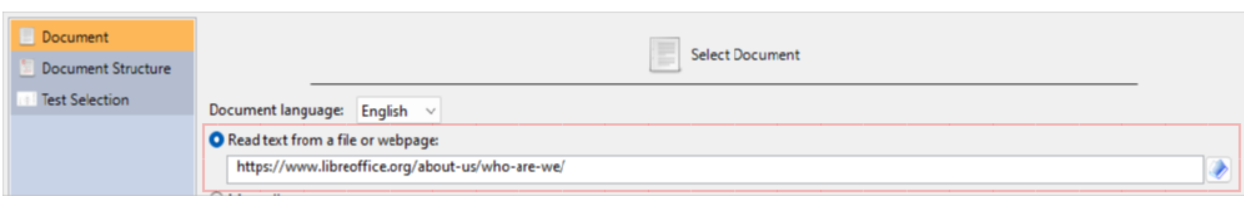


First, select English as the document language.

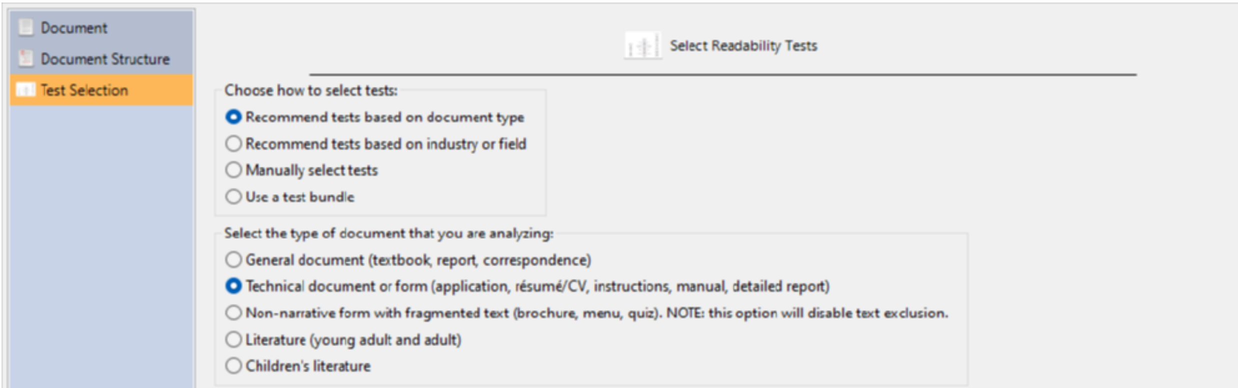
Next, open any web browser, go to the webpage, and copy its URL (i.e., address) onto the clipboard.



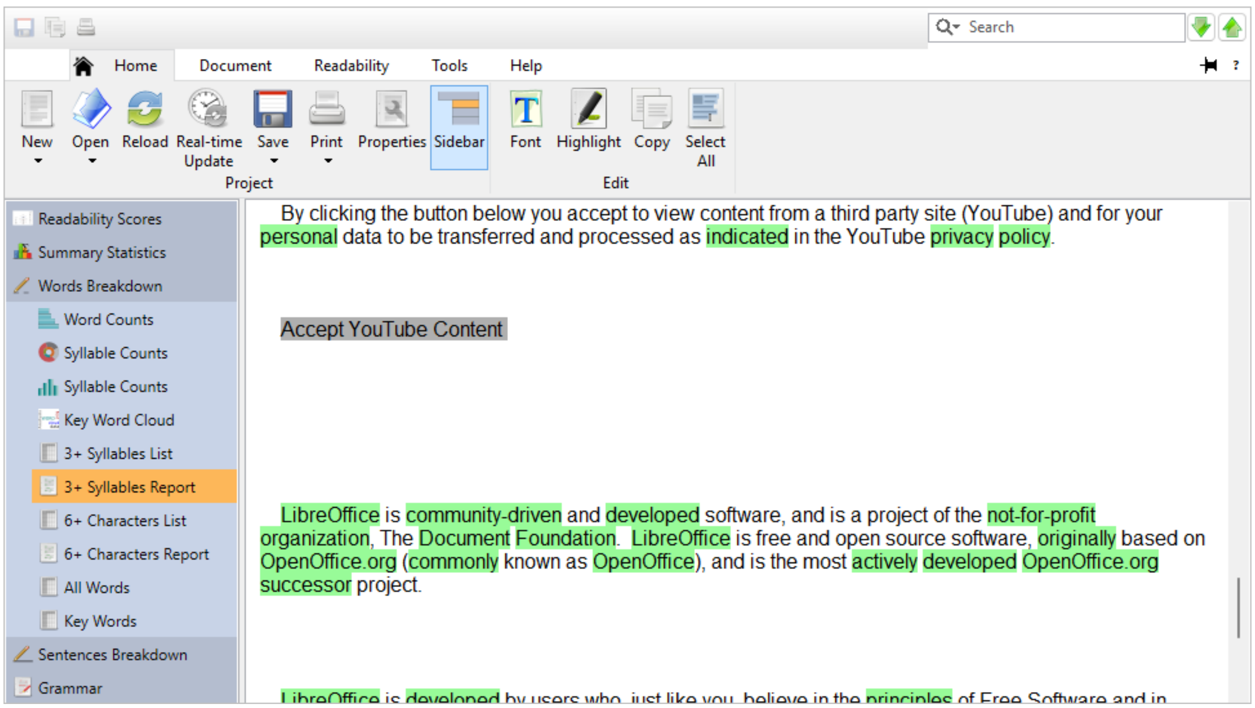
Now, go back to the project wizard and select Read text from a file or webpage. Paste the URL into the file-path field, as illustrated below.



After specifying the URL, choose to analyze the page by document type and then select Technical document. Most webpages would fall under the category of either general or technical document. Because this page discusses office software, we will select technical.

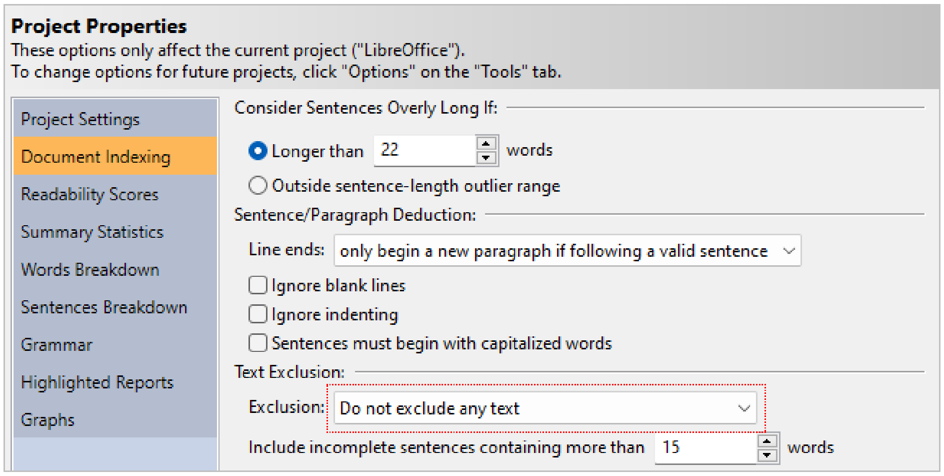


Next, accept the rest of the defaults on the New Project wizard by clicking the Finish button. Once the project is finished loading, click on the Words Breakdown icon on the sidebar and select 3+ Syllable Words.



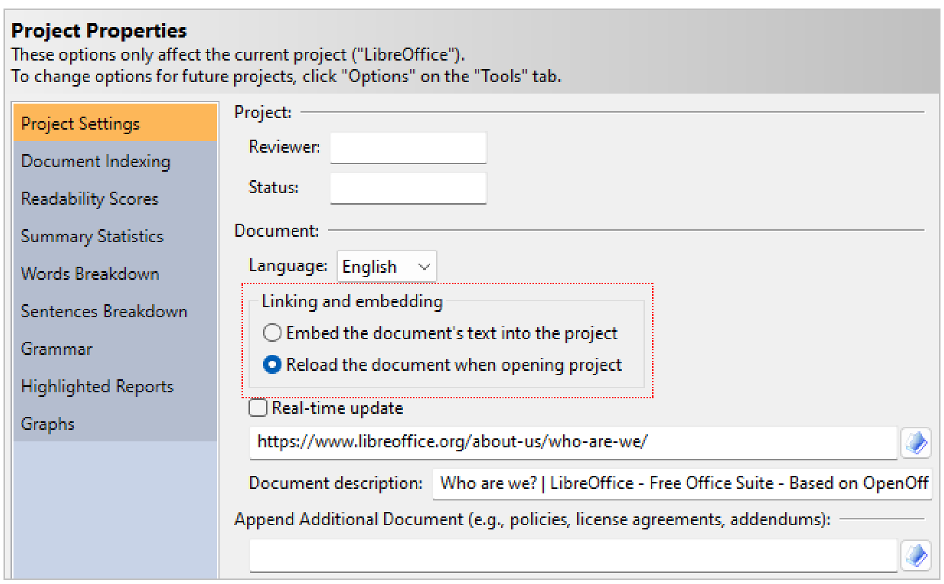
Note how some text is highlighted gray. These are incomplete sentences, and because *Readability Studio*, by default, ignores incomplete sentences, nothing highlighted gray is factored into the analysis. That is, nothing highlighted gray here counts towards the overall sentence, paragraph, and word totals and are also not included in any readability test calculation. Aside from non-narrative forms, it is normally recommended to ignore incomplete sentences. This is especially true for webpages, because often HTML tables and lists are used on websites to display menus—items that are not normally part of the page's narrative.

If you wish to include incomplete sentences in the analysis, click the Properties button on the Home tab to display the Project Properties dialog. Next, click the Document Indexing icon, and select the Do not exclude any text option.



Note that this option is also available on the Options dialog (from the Tools tab) if you wish to change this behavior for all future projects.

One last item to discuss about loading a webpage is whether you want to reanalyze it later as it changes. By default, the project will directly link to the webpage, although you can change this to embed a snapshot of the page into the project. To toggle this behavior, click the Properties button on the Home tab to display the Project Properties dialog. Then click the Project Settings icon. Here, we can see that Reload the source document when opening project is selected.



By having this option selected, you will reload the page's content whenever you reopen the project (or at any time by pressing F5). If you prefer to embed the current version of this page into the project, then select Embed the source document's original text in the project file. Refer to sec. 8.2 for more information about these options.

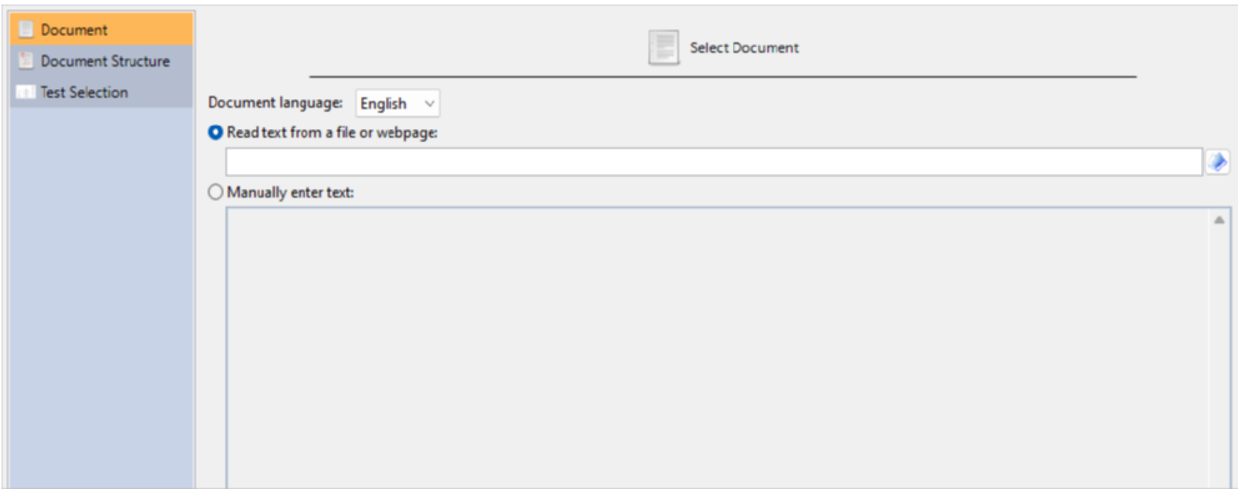
Note that webpages requiring authentication (i.e., requiring a user name and password) are not currently supported. To work around this, log into the webpage from any browser, save the page as a local HTML file, and then load that into a new project.

14.5 Searching for Overly Long Sentences

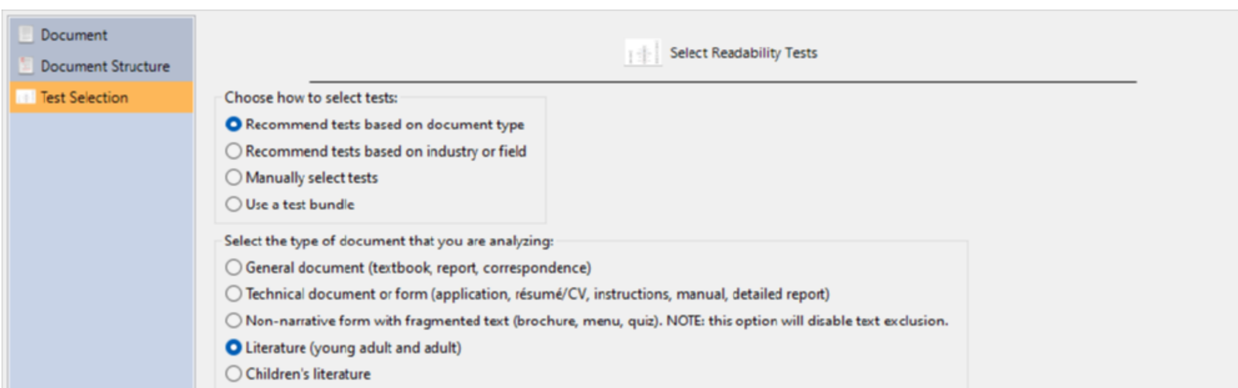
In this example, we will review a document for overly-long sentences. This example will demonstrate the available methods for defining an overly-long sentence. In addition, it will also demonstrate how to search for and review these sentences.

The first step is to open the example file "Features." From the Help tab, click the Example Documents button and then select "Features" from the menu. Next, when prompted about how to open the document, select Create a new project.

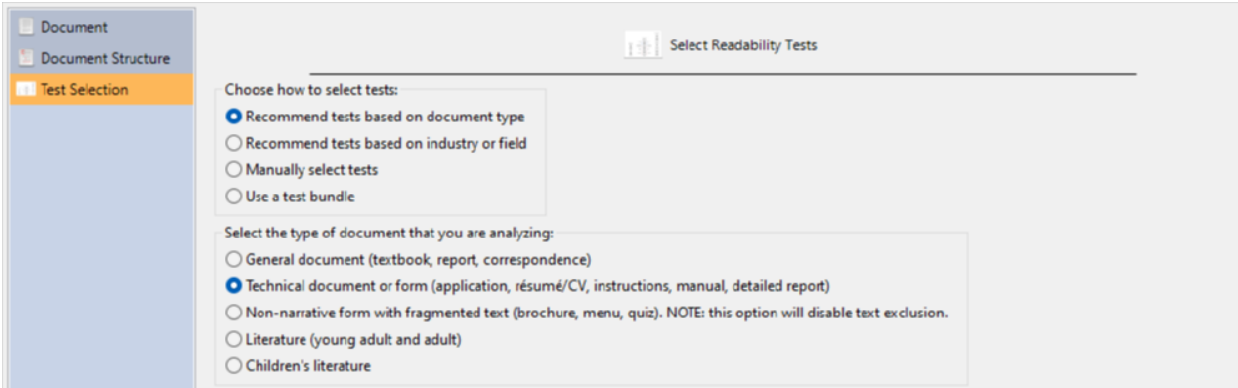
You will then be presented with the New Project wizard dialog, as shown below:



Select English as the document language (if this option is available) and click the Forward > button until you reach the Test Selection page, as shown below:



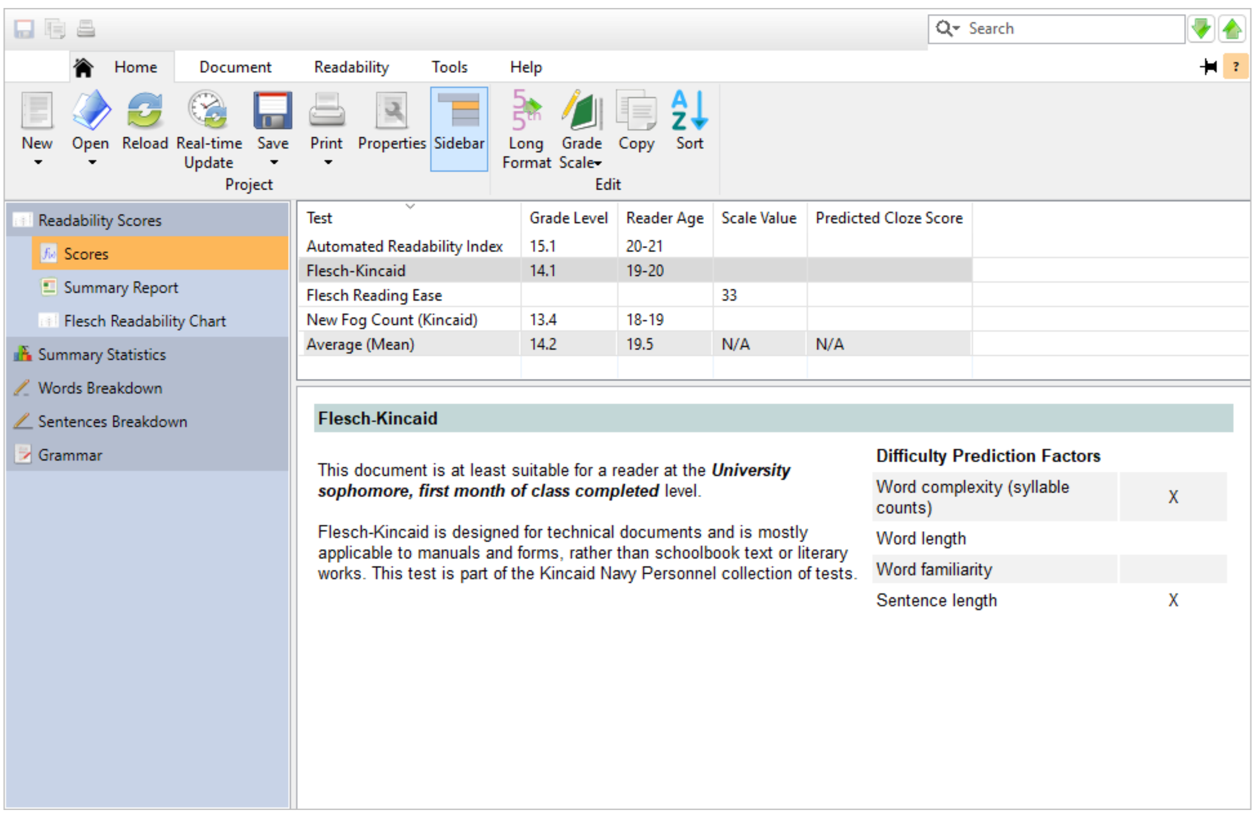
Select the Recommend tests based on document type and select the Technical document option, as shown below:



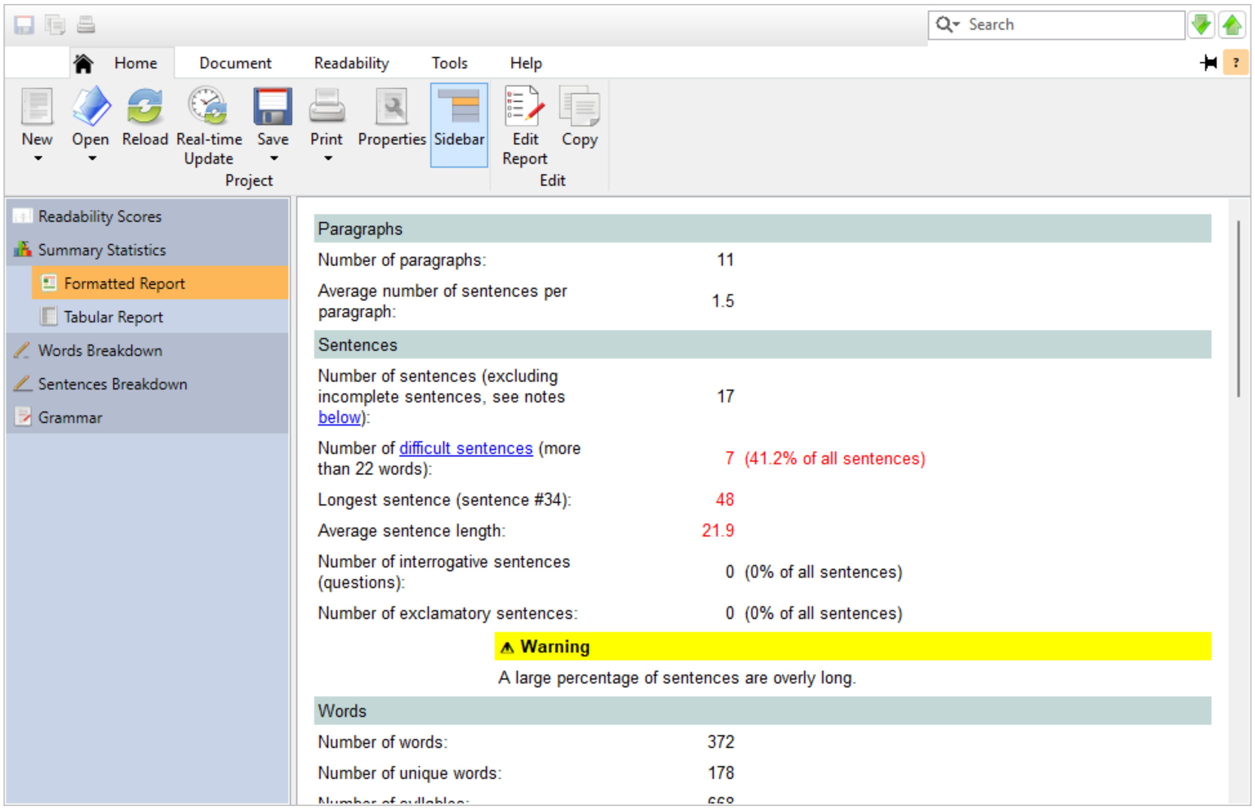
Click the Finish button to begin analyzing the project.

If you are excluding incomplete sentences (the default), then a warning will be displayed. Because this file contains numerous incomplete sentences, the program will ask you if excluding them is what you truly intend. In this case, these sentences are just list items that we should ignore, so click Continue excluding incomplete sentences.

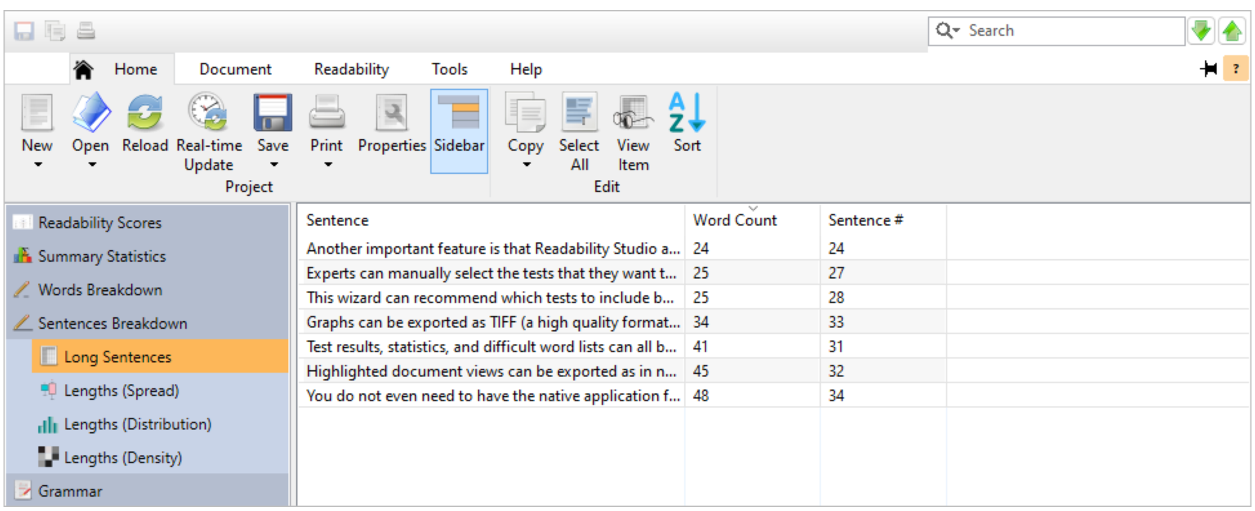
Once the project is loaded, the readability scores will be presented. As we can see below, this document scores high within the collegiate range.



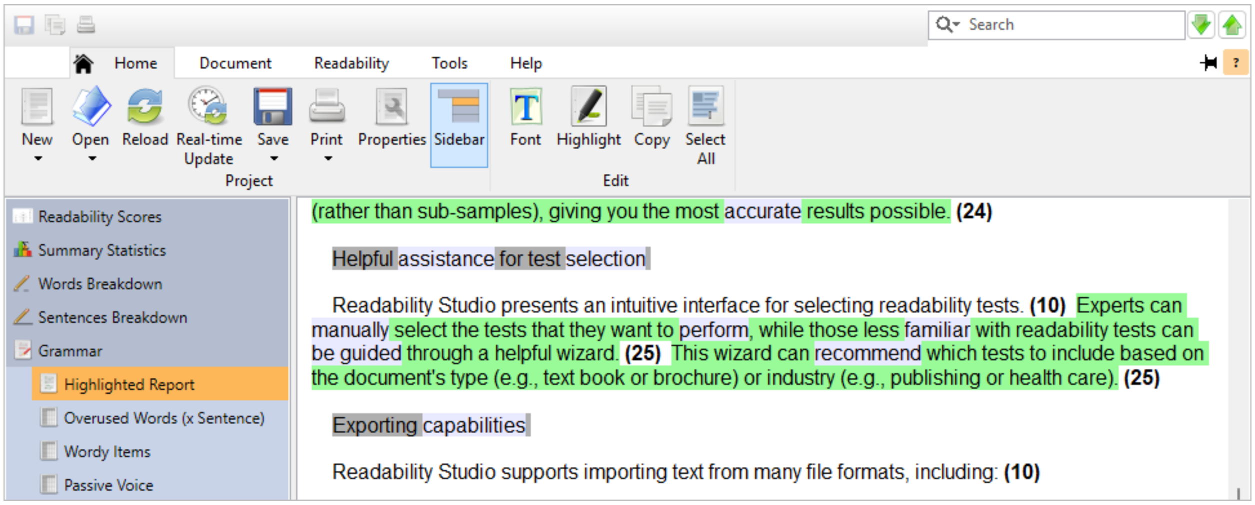
This document should be an easy-to-read list of software features intended for a broad range of readers. These scores are a bit too high, so now we will explore the results to understand why it is so difficult to read. Select the Summary Statistics icon in the project sidebar to view the document's statistics.



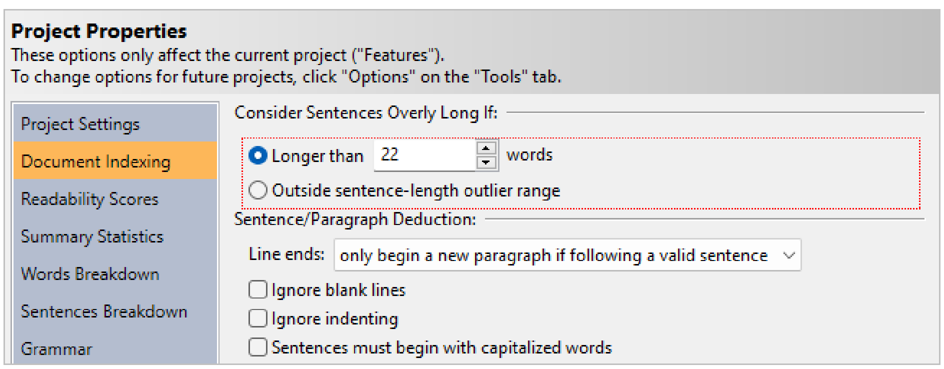
As we can see, there are some very long sentences in this document and they are the biggest factor in our high scores. To view a list of these sentences, click the difficult sentences link. You can also click the Sentences Breakdown icon in the sidebar and select the Long Sentences page.



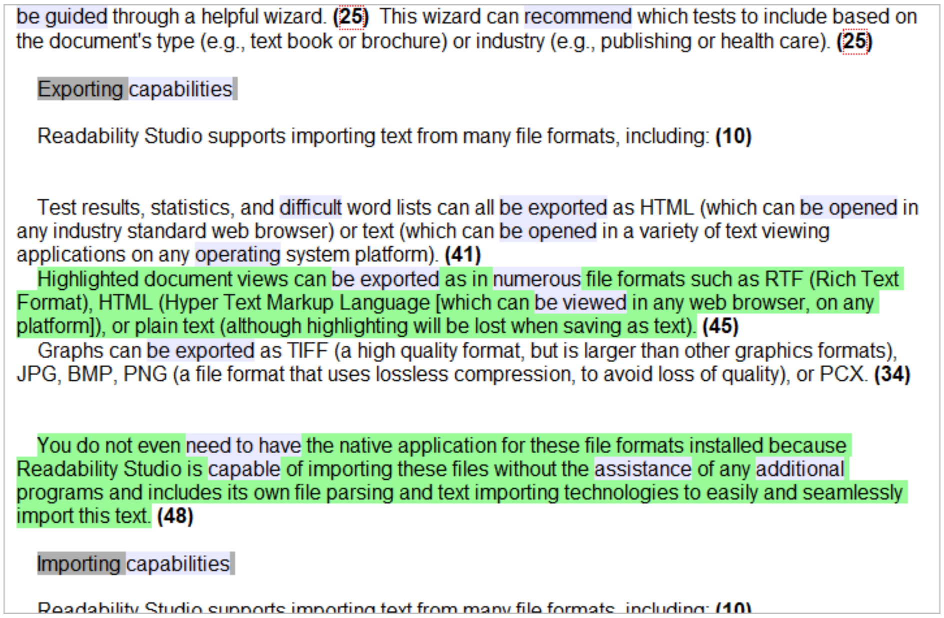
This list can be sorted alphabetically, by word count, or by location. To view any sentence in its original context, double-click on it. This will take you to the Grammar section. In this window, all the overly-long sentences and wordy items are highlighted and each sentence is followed by its respective word count (in parentheses).



By default, the program considers sentences overly long if they contain more than 22 words (CRES 19). As we can see, there are a few sentences that are barely over 22 words that are not problematic. However, there are a few clustered sentences which exceed 40 words. At this point, we will want to temporarily filter out the smaller of these long sentences and search for the “worst offenders.” From the Home tab, click the Properties button and then select Analysis. As shown below, we have a couple of different ways to determine if a sentence is overly long:



The first option is to define a specific word count as the long sentence boundary. Enter *25* where it says Longer than [22] words and click the OK button. The program will now reanalyze the file, classifying any sentences longer than 25 (instead of 22) as overly long. As shown below, sentences that are between 23–25 words are no longer being considered overly long:


be guided through a helpful wizard. (25) This wizard can recommend which tests to include based on the document's type (e.g., text book or brochure) or industry (e.g., publishing or health care). (25)

Exporting capabilities

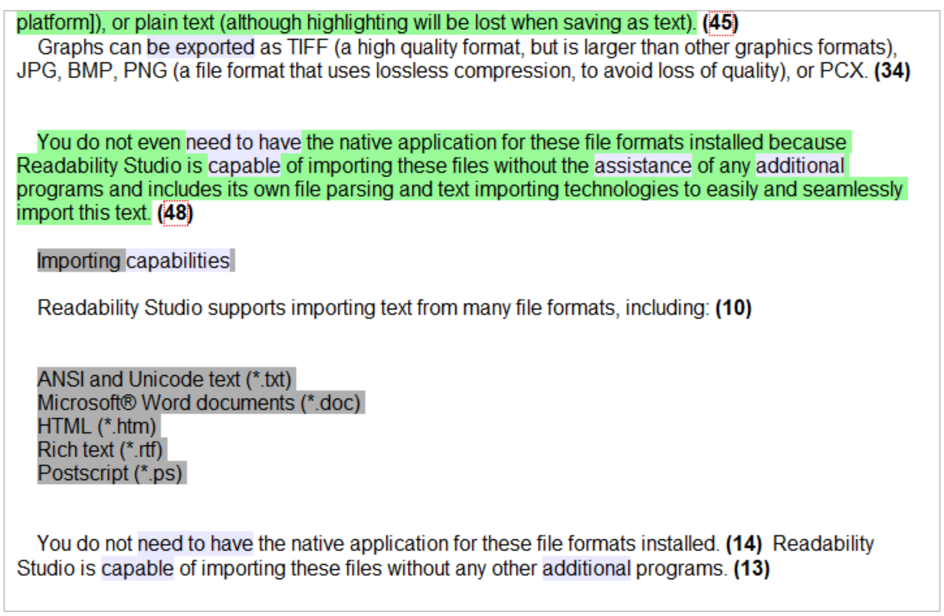
Readability Studio supports importing text from many file formats, including: **(10)**

Test results, statistics, and difficult word lists can all be exported as HTML (which can be opened in any industry standard web browser) or text (which can be opened in a variety of text viewing applications on any operating system platform). **(41)**
Highlighted document views can be exported as in numerous file formats such as RTF (Rich Text Format), HTML (Hyper Text Markup Language [which can be viewed in any web browser, on any platform]), or plain text (although highlighting will be lost when saving as text). **(45)**
Graphs can be exported as TIFF (a high quality format, but is larger than other graphics formats), JPG, BMP, PNG (a file format that uses lossless compression, to avoid loss of quality), or PCX. **(34)**

You do not even need to have the native application for these file formats installed because Readability Studio is capable of importing these files without the assistance of any additional programs and includes its own file parsing and text importing technologies to easily and seamlessly import this text. **(48)**

Importing capabilities

The second option is to consider sentences overly long if they are comparatively longer than the other sentences in the document. From the Home tab, click the Properties button and then select Analysis. Select the option Outside sentence-length outlier range and click the OK button. At this point, the boundary for overly-long sentences will be calculated based on the outlier range of the sentence lengths. After reanalyzing the file, note how there are only two highlighted sentences. One sentence contains 45 words and the other 48 words. These are the two "worst offenders" and should be either reworded or split into small sentences.


platform]), or plain text (although highlighting will be lost when saving as text). (45)
Graphs can be exported as TIFF (a high quality format, but is larger than other graphics formats), JPG, BMP, PNG (a file format that uses lossless compression, to avoid loss of quality), or PCX. **(34)**

You do not even need to have the native application for these file formats installed because Readability Studio is capable of importing these files without the assistance of any additional programs and includes its own file parsing and text importing technologies to easily and seamlessly import this text. (48)

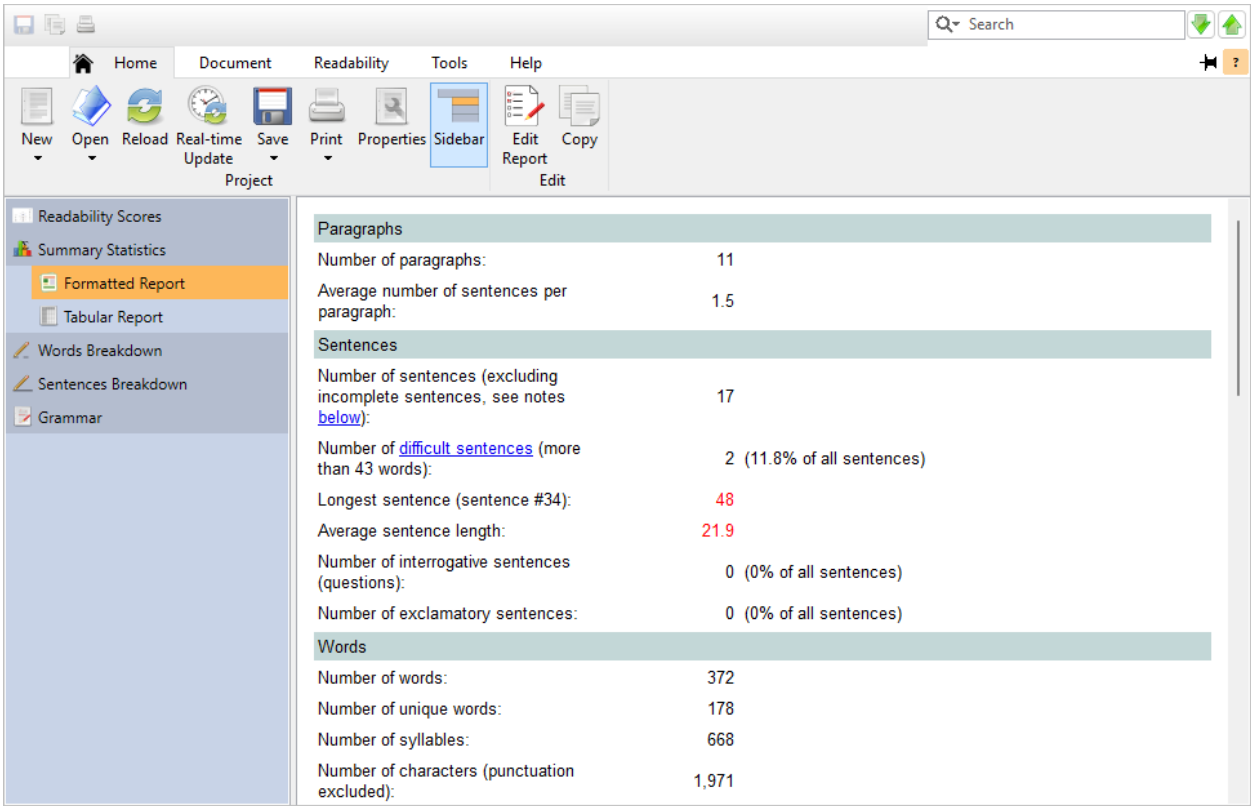
Importing capabilities

Readability Studio supports importing text from many file formats, including: **(10)**

ANSI and Unicode text (*.txt)
Microsoft® Word documents (*.doc)
HTML (*.htm)
Rich text (*.rtf)
Postscript (*.ps)

You do not need to have the native application for these file formats installed. **(14)** Readability Studio is capable of importing these files without any other additional programs. **(13)**

To see what the outlier range is, click on the Summary Statistics icon in the project sidebar. Here we can see that any sentence more than 43 words is considered overly long.



This option is only recommended for situations where most of the sentences are an acceptable length, but a few (very) long sentences skew the average. This option is useful for finding these extreme sentences.

14.6 Finding Difficult Sections

Along with searching for overly-long sentences, we should also review the distribution of longer sentences. If there are sections of a document where longer sentences seem to clump together, then these areas may be more difficult to read. These sections may need to be rewritten or restructured so that we don't have entire paragraphs that readers will struggle with.

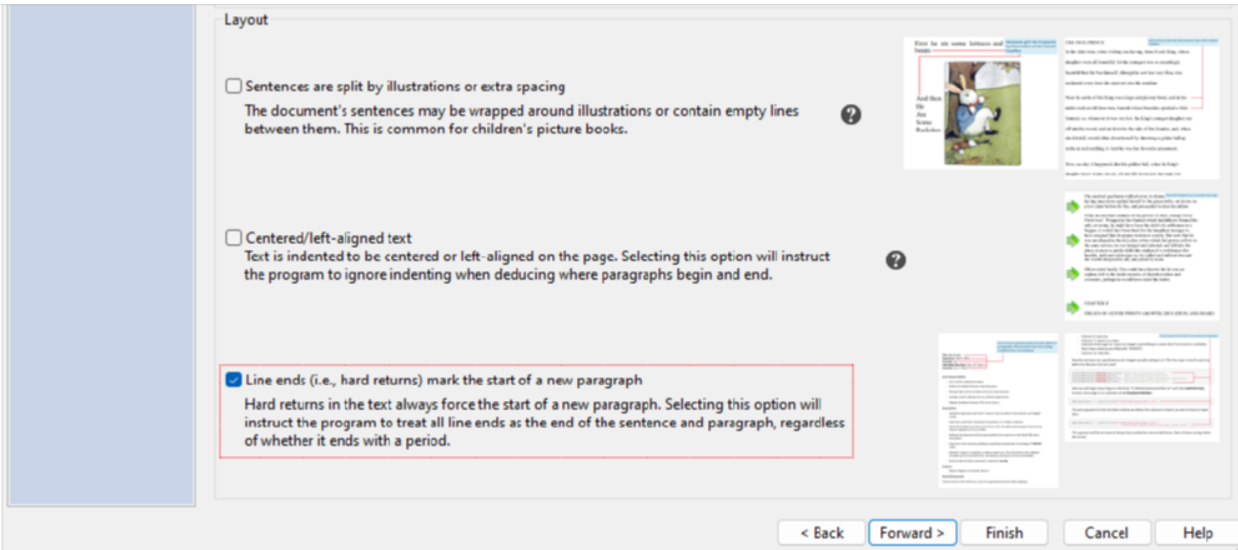
In a previous example (sec. 14.5), we used a box plot to review the dispersion of sentence lengths (and to find outliers). Although this is useful, these graphs aggregate their data and do not preserve their order. In other words, they lump sentence lengths together and don't show them in the order that they appear in the document. To compare the sentences' lengths in relation to where they are in the document, we will need to use a heatmap.

A heatmap is a graph which displays items as color-filled squares, where the color represents an item's value in relation to other items. To construct a heatmap, the following is performed:

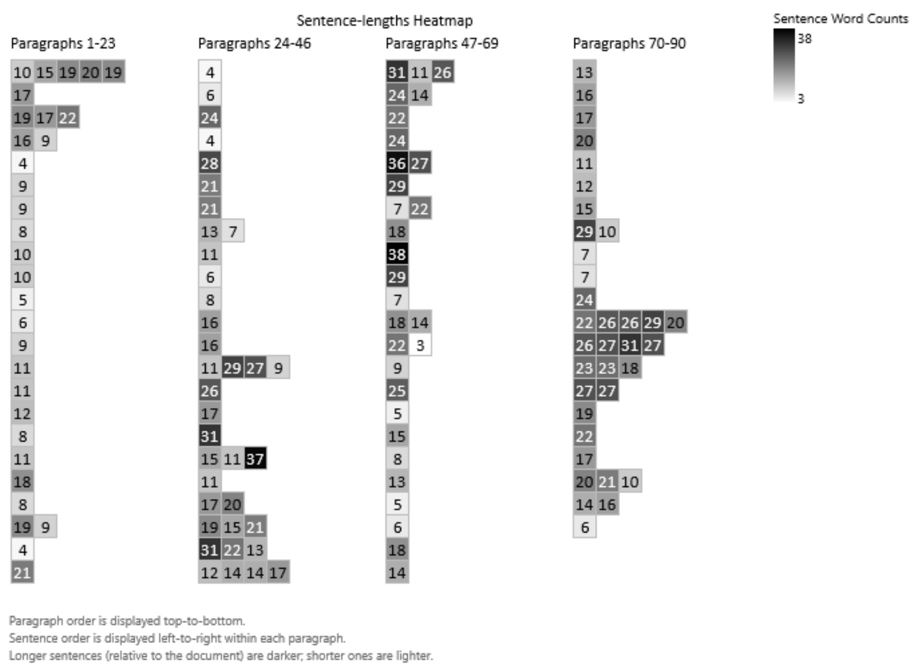
- The highest and lowest values from a range of data are found.
- Two colors are selected to represent the high and low values from the data.
- A color gradient between these colors is built, representing the full range of the data.
- A square for each data point is drawn in a rectangular grid. These data point squares are drawn in the same order as they appear in the data, going from left-to-right and top-to-bottom.
- Each square is filled with the color from the gradient where the value falls. For example, if a gradient goes from white (lowest value) to black (highest value) and a data point is the highest value, then its square will be black. Likewise, mid-value points will be displayed as gray.
- Optionally, categorized data can split into separate heatmaps, where each line represents a different group. (All groups still use a common color gradient scale.)

For our purpose, sentences will be shown in the order that they appear in the document. This will enable us to not only find longer sentences, but also see where numerous difficult sentences clump together.

Let's begin by opening a mid-sized technical document. Select the Help tab of the ribbon, click the Example Documents button, and select "Importing and Exporting Fixed-Width Data with R" from the menu. On the following prompt, tell the program to create a new project. When the New Project Wizard appears, select the Document Structure page. This example document contains lines of programming of code separated by hard returns. Because of this, we will want to force each of these lines to be a new paragraph. (Doing so will help with excluding them.) Check the option Line ends and click Finish:

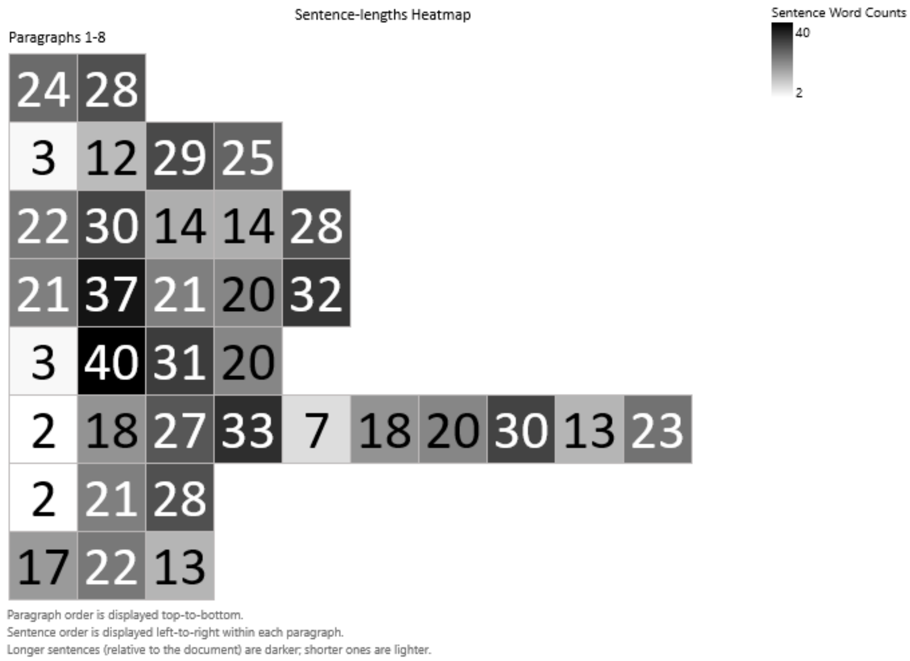


Once the project is created, select Sentences Breakdown on the sidebar, and click on the Lengths (Density) subitem. Here we will see a heatmap showing the sentences going from left-to-right, where their colors represent the length. Additionally, if the document contains 500 or less paragraphs, then the sentences will be grouped by paragraphs. In this example, the heatmap will show four columns, each containing ~25 paragraphs. These paragraphs go from top-to-bottom, then left-to-right across the columns. In each paragraph, its sentences are shown left-to-right.



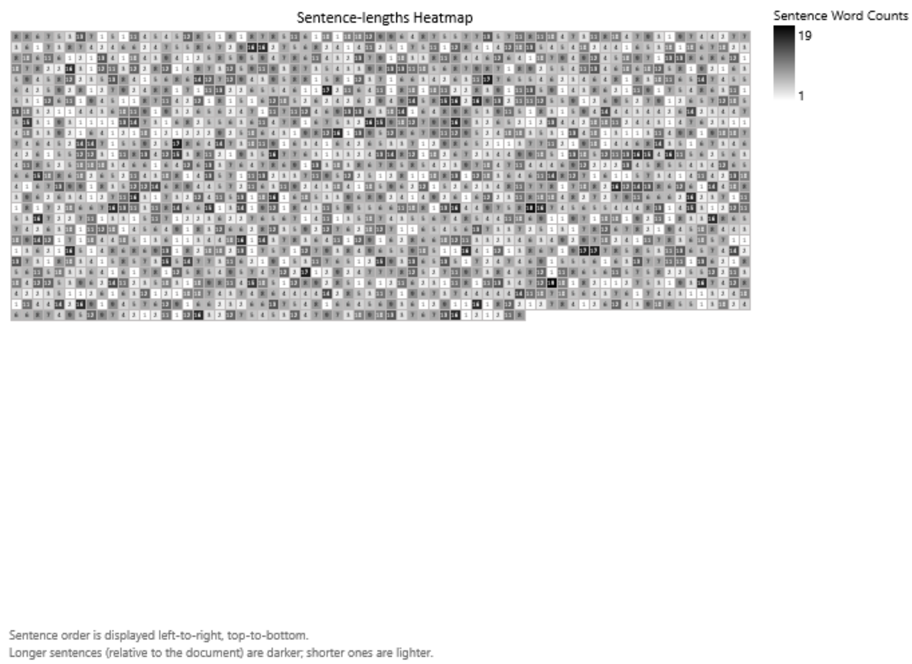
In the above heatmap, we can see that the first half of the document contains mostly one-word sentences (most of which are medium length). In most of the second half of the document there are a few more longer sentences, although the paragraphs do remain within 1–2 sentences. In the last ten paragraphs, however, we do see longer paragraphs that also happen to contain longer sentences. In this case, we may want to review the ending section of the document.

Close this project and return to the Help tab of the ribbon. Click the Example Documents button again and select "Instructional Disclaimer." Create a new project with the defaults and select the heatmap when the project opens.



In the above graph, we can see that most paragraphs begin with a short intro sentence, but are then followed by rather long sentences. (Some sentences being as high as 40 words.) Also of interest is a paragraph containing ten sentences—which will be difficult to read through.

Finally, let's view a novel. Close this project and return to the Help tab of the ribbon. Click the Example Documents button again and select *A Christmas Carol*. Create a new project with the defaults and select the heatmap when the project opens.



Because this document contains more than 500 paragraphs, grouping will not be used. Instead, the sentences will appear in a single heatmap, where all sentences are displayed left-to-right, top-to-bottom. In the above graph, we can see that the novel's sentences range from 1–19 words in length. Also, the sentence lengths are rather evenly distributed. In other words, the longer sentences are not clumping together within any particular section.

14.7 Reviewing a Flyer

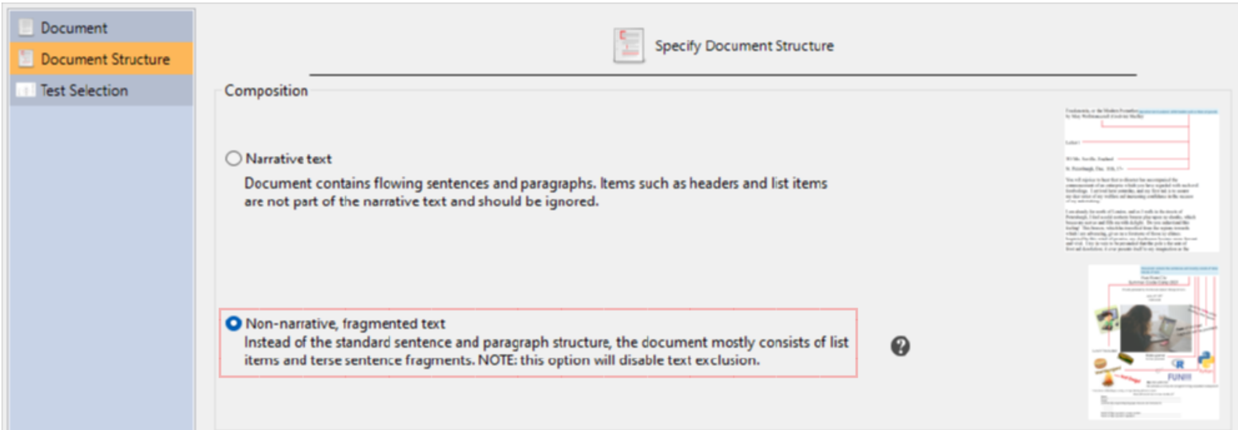
In this example, we will analyze a flyer for a summer program being offered by a local software company. This flyer consists of short text boxes designed to grab the attention of young adults interested in learning computer programming. Of interest is that this document contains a few sentences, but the bulk of its content is simple text boxes.

Almost all readability formulas use sentence length as a factor. Because of this, short text blocks like these are traditionally excluded and only the full sentences are analyzed. For this document, however, we prefer to analyze everything since so much of its content is disjointed text boxes.

First, let's view the document. From the Help tab on the ribbon, click the Example Documents button, and select "Summer Code Camp." Next, you will be prompted to either create a new project or view the document; select View document. Once the flyer is opened, note how it mostly contains simple text boxes:

Rather than excluding these and only analyzing the few remaining sentences, we will analyze everything. Also, we will only use a test that is designed to not rely on sentence lengths.

Close this document and return to *Readability Studio*. Click the Example Documents button and again select "Summer Code Camp." Next, select Create a new project. When the New Project Wizard appears, select the Document Structure page and select Non-narrative, fragmented text:



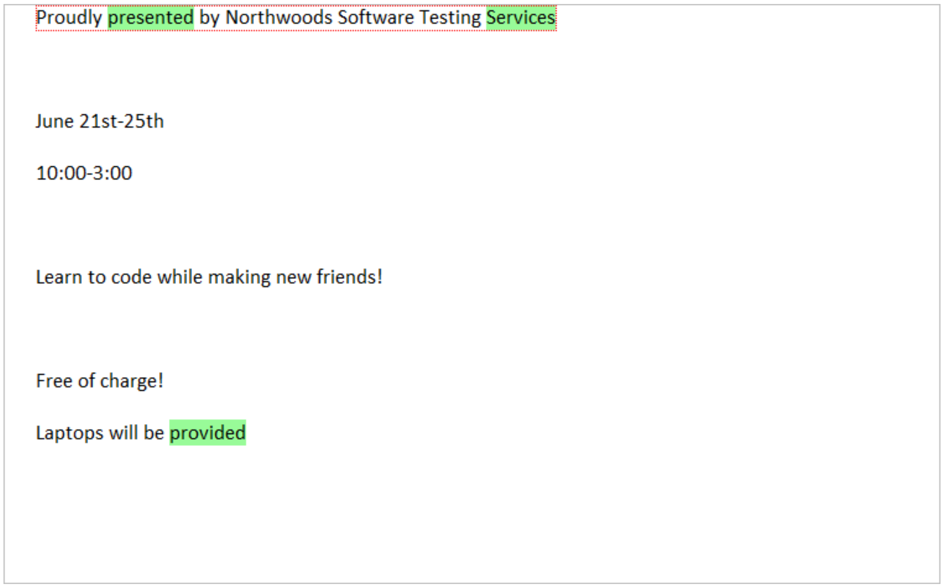
Note
Going to the Test Selection page, selecting Recommended test selection based on document type, and selecting Non-narrative form will have the same effect.

Finally, click the Finish button to create the project. When the project appears, note that only the FORCAST test was performed because it does not use sentence lengths. Also, viewing any highlighted report will show that all text was included in the analysis.

We will have a somewhat high FORCAST score, so we will want to review the lengthy words and see if we can make any improvements.

An interesting aspect of this flyer is that it has two audiences: children and their parents. The eye-catching text boxes in the upper half of the flyer are targeted for children, while the lower half requires parental involvement. With this in mind, we will review both halves of the flyer from these perspectives.

Select Words Breakdown on the sidebar and then select 3+ Syllables Report. Here we can see some phrases that could be simplified:



Proudly presented by Northwoods Software Testing Services

June 21st-25th

10:00-3:00

Learn to code while making new friends!

Free of charge!

Laptops will be provided

For example, we could remove *Proudly presented* or replace it with *Offered*. Scrolling down further:

No previous computer programming experience required!

(vegan) Hot Dogs!

*Your choice of hamburger, corn dog, or vegan hot dog and one (1) snack

Please fill out and return no later than May 25th

Name:

School:

Grade:

The highlighted line above could be rewritten to something like *New to programming? No Problem!*.

Scrolling to the bottom, we can review the second half of this flyer. The content here is meant for parents and their children to fill out together.

Please fill out and return no later than May 25th

Name:

School:

Grade:

List three (3) programming languages that you are interested in:

Parent or legal guardian's contact number:

Parent or legal guardian's signature:

Although there are some longer words here, the responsible adult intended to sign this should be able to understand the content.

Refer to sec. 14.8 and sec. 15.5 for a continuation of this example where the flyer is revised and has an addendum added to it.

14.8 High-level Review of a Disclaimer

In this example, we will perform a quick review of the concepts, word complexity, and paragraph complexity of a disclaimer.

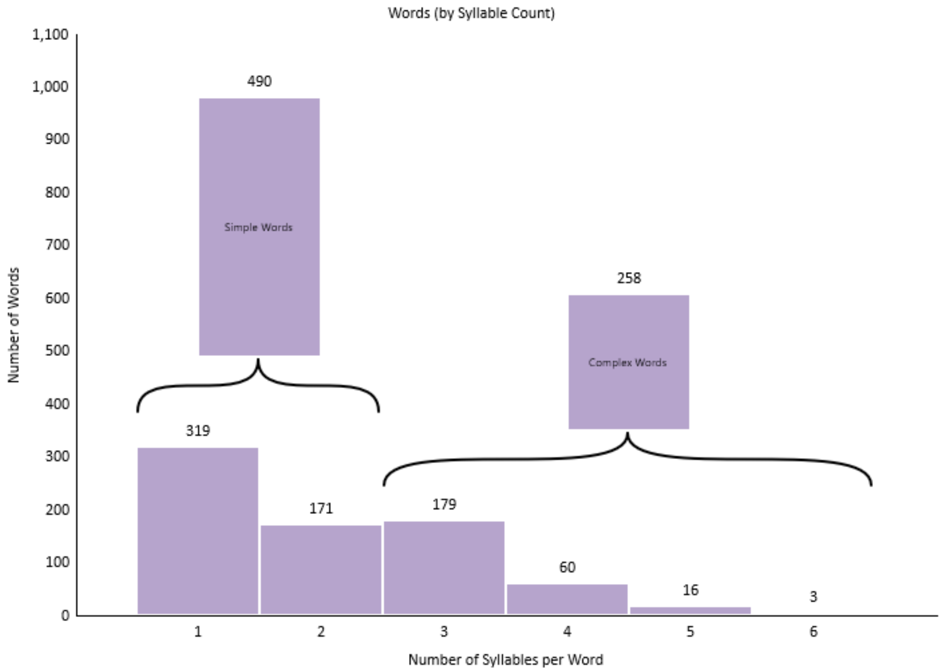
Continuing from the "Reviewing a Flyer" example (refer to sec. 14.7), our fictitious software company has been purchased by a private equity firm. After the purchase, this firm has decided to make some changes to the Summer Code Camp program. In particular, they have created a disclaimer that parents must agree to before their children can participate. In a later example, we will explore in more detail the difficult aspects of this disclaimer. For now, however, we will first create summary graphs to give us an idea of what we are dealing with.

From the Help tab on the ribbon, click the Example Documents button, and select "Instructional Disclaimer." Next, select Create a new project. When the New Project Wizard appears, leave the defaults and click Finish.

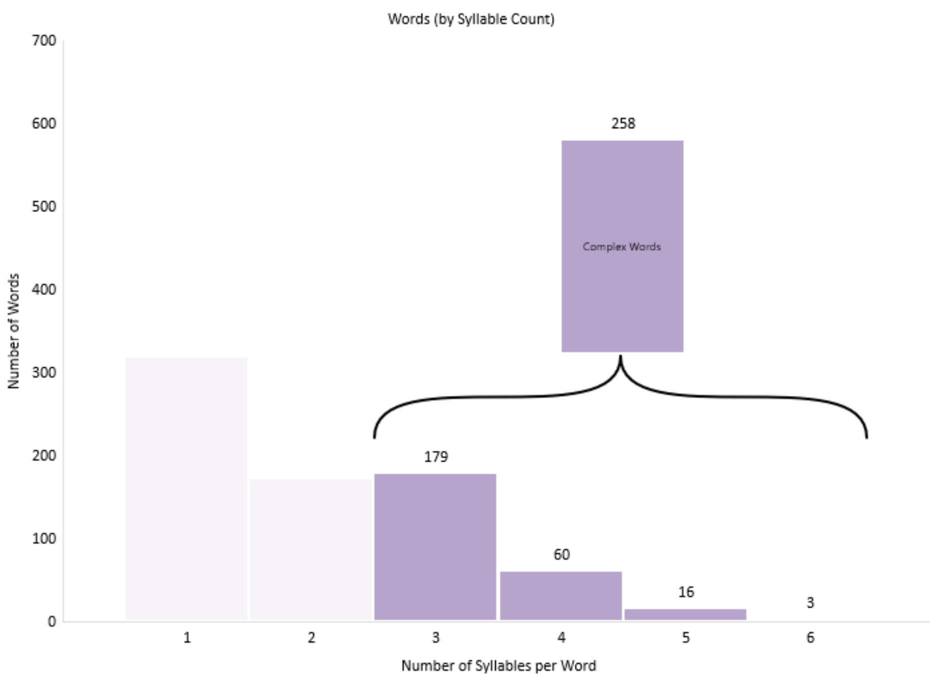
First, let's review the concepts of this document to get a broad idea of what it is about. Click Words Breakdown on the sidebar and then select Key Word Cloud. This word cloud shows the document's words, after they have been combined (via stemming) and common words (e.g., articles) have been removed. The size of the words is relative to their frequencies. As an example, the words "Company" and "Customer" appear much more frequently than any other word:

From this, we can glean that this document is about a customer and company relationship. Looking further, we may notice other words such as "claims," "responsibilities," "legal," and "liable." From here, we could deduce that this may be a legally-binding agreement (and a bit adversarial).

Next, click Syllable Counts on the sidebar (the one with the histogram icon). In these histograms, we want to see a right-tailed distribution, where the bin sizes go down as you go from one-syllable words to more complex ones. Although that is the pattern we see here, it has an alarming number of complex words — approximately a third of the document.

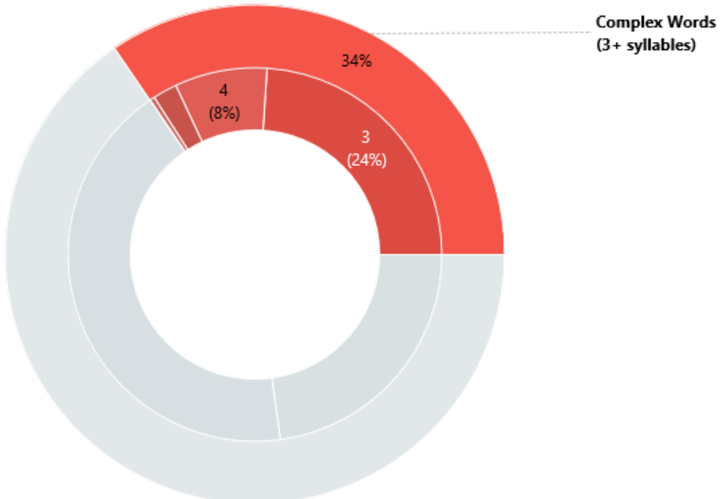


To draw attention to these groups of words, press the Showcase on the ribbon. Now the cumulative bar group and labels will only be shown for these bars:



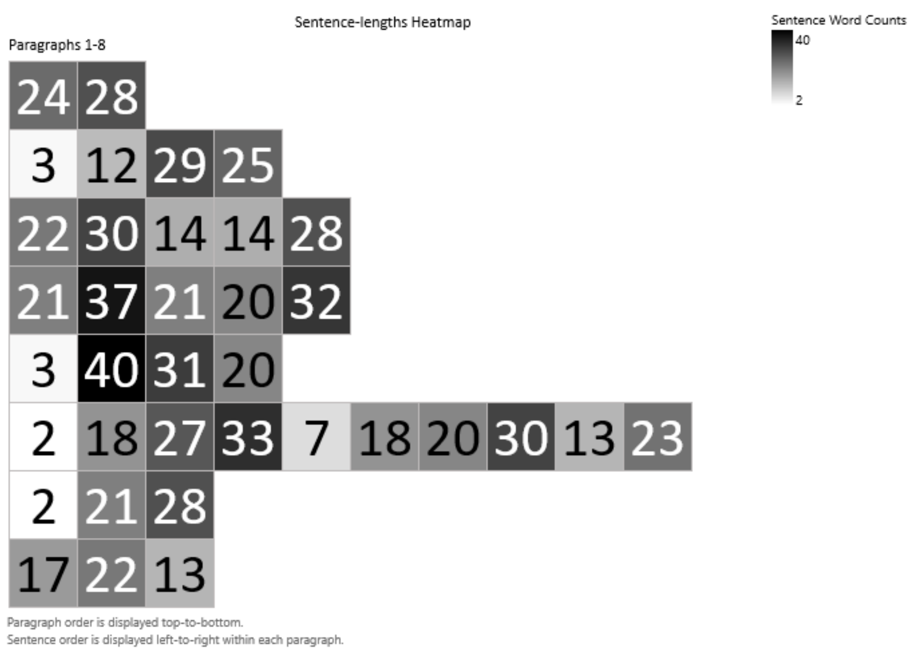
If using a solid color for the bars, then the one- and two-syllable word bars will also be translucent. By “ghosting” these bars (Tufte 24–25), the complex-word bars will receive the focus.

For another way to visualize complexity, click the Syllable Counts on the sidebar (the one with the pie icon). Here we can see how much of the document consists of complex words by looking at the outer ring. We can further see a breakdown of these complex words in the inner ring, where almost a quarter of the document is three-syllable words:



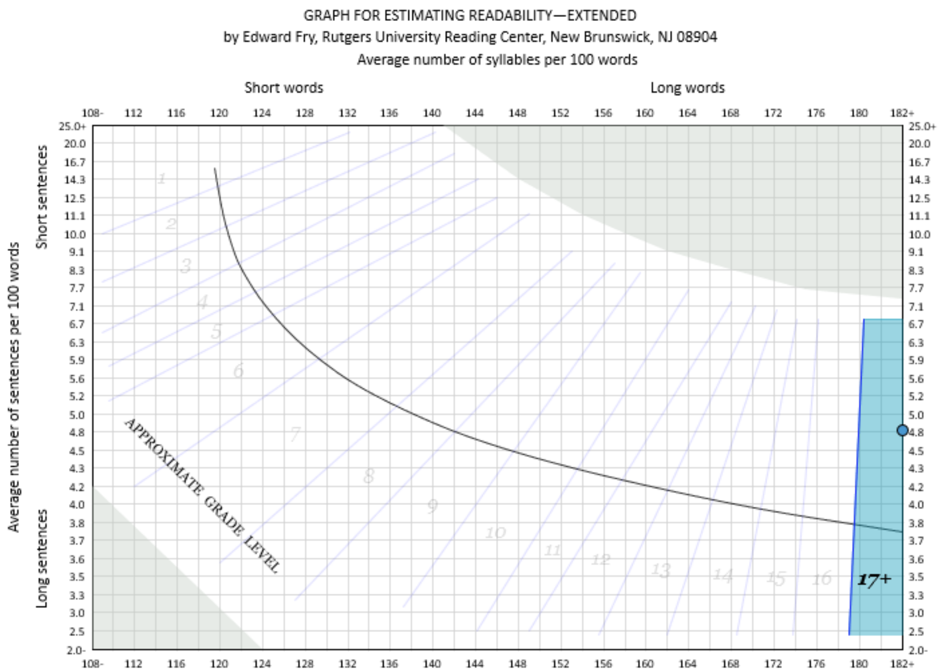
Note that because we previously pressed Showcase, the simple-word slices are translucent and their labels are hidden.

Next, let's review a visual layout of the sentence lengths. Click Sentences Breakdown on the sidebar and then select Lengths (Density). Going from top-to-bottom, this graphic displays a row for each paragraph. Across each row is a box for each sentence in the respective paragraph. For each box, its length is shown as a number at its center. Also, the darker the box, the longer its length is compared to the rest of the sentences in the document. In this example, we can see that the sixth paragraph has ten sentences in it:



There are also a few lengthy sentences in this paragraph, judging from the numbers and dark colors across the row.

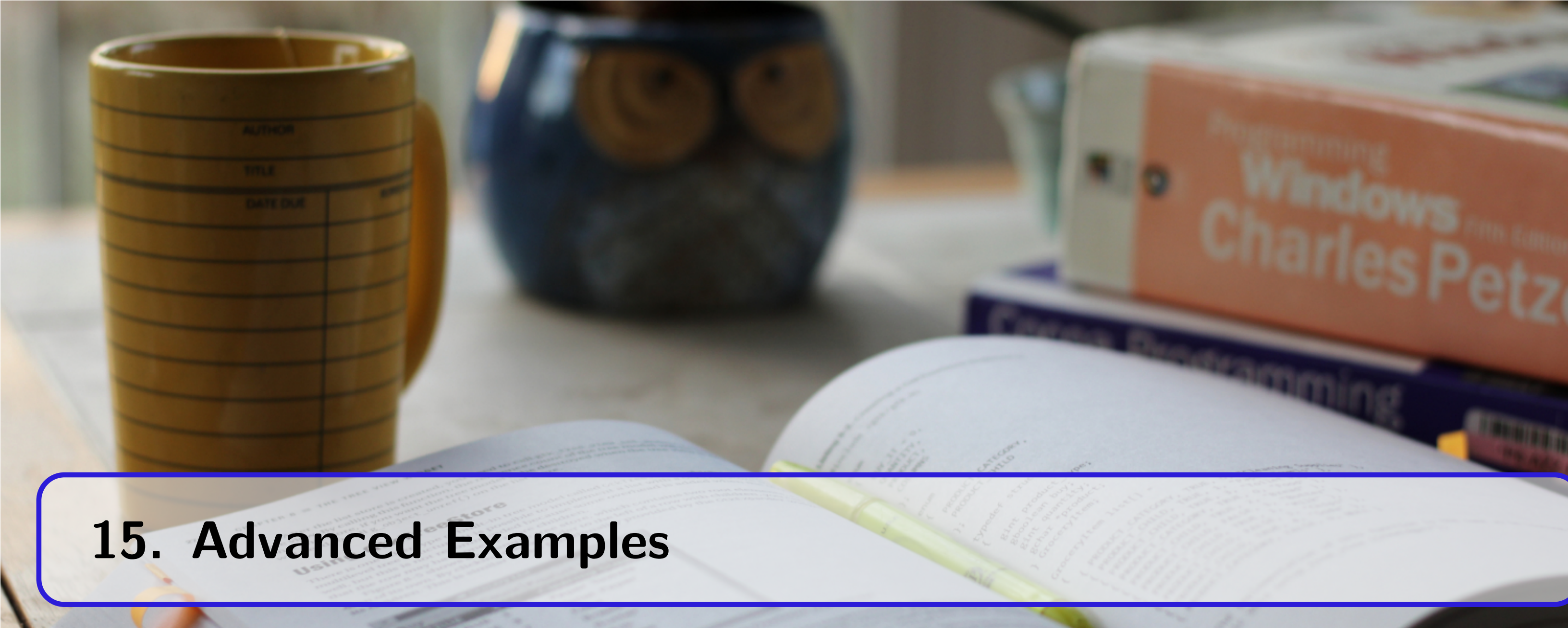
Finally, let's review how word and sentence complexity affects readability scoring. From the Readability tab, click Secondary > Fry to add a Fry test. Click on the Readability Scores icon on the project sidebar and select Fry Graph.



Here, we can see that the document is scoring at *17+*. Long words are the main contributing factor for this, as the score point is pulled to the far right. Long sentences, however, have a much lesser effect given the vertical position of the point.

Something to note is that because we enabled showcasing earlier, the section with the score will be more prominent. Likewise, the other areas and labels have been "ghosted," drawing attention away from them. If you wish to see this graph as it would normally appear, then toggle the Showcase button on the ribbon.

Refer to sec. 15.5 for a continuation of this example. There, this document will be explored in further detail and used as an addendum to the original flyer from sec. 14.7.



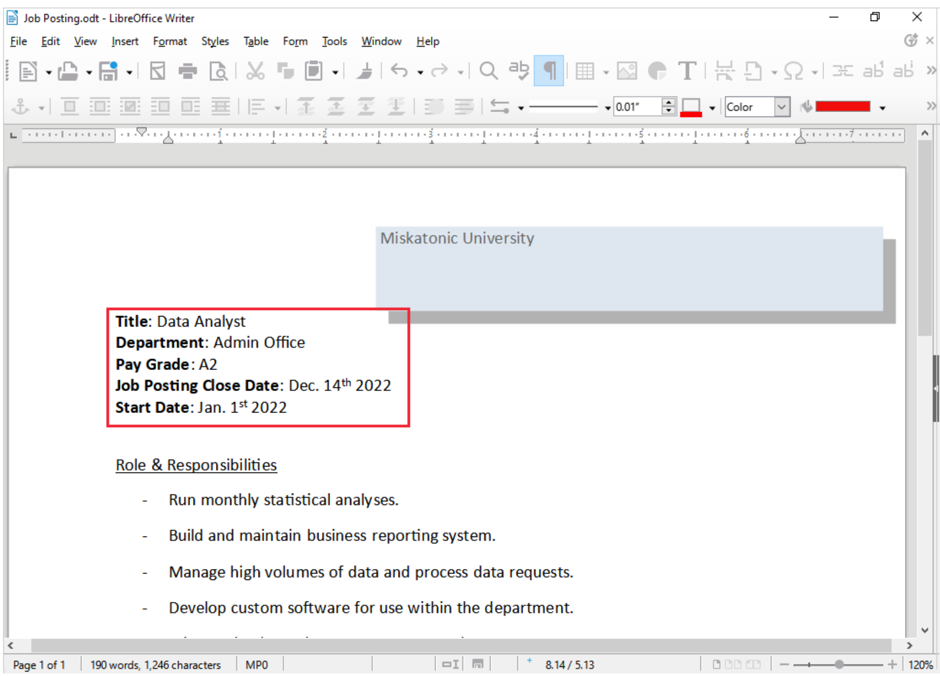
15. Advanced Examples

Now that we have covered the basics of the program, we will discuss import options, text exclusion, and including addendums. After that, we will continue with showing how to create test bundles.

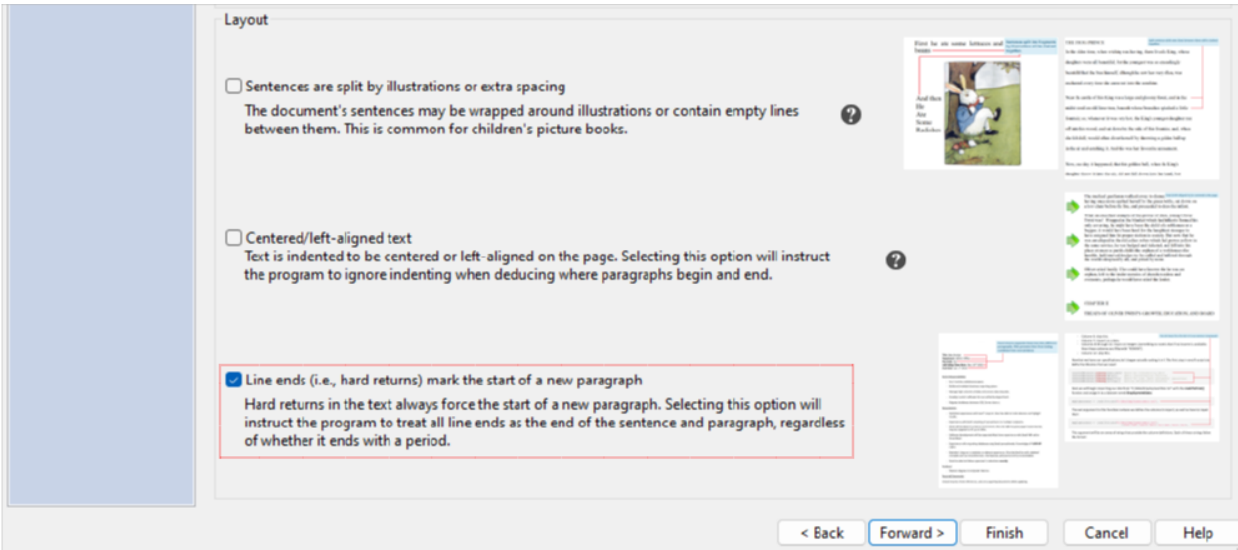
15.1 Preserving Hard Returns

In this example, we will import a document which contains short lines of text that are separated by hard returns. These lines of text are not separated by blank lines, and are neither indented nor start with bullet points (the way list items often appear). The default for the program is to chain text lines such as this into a single sentence. In the case of this document, however, we will want to treat each line as a separate paragraph.

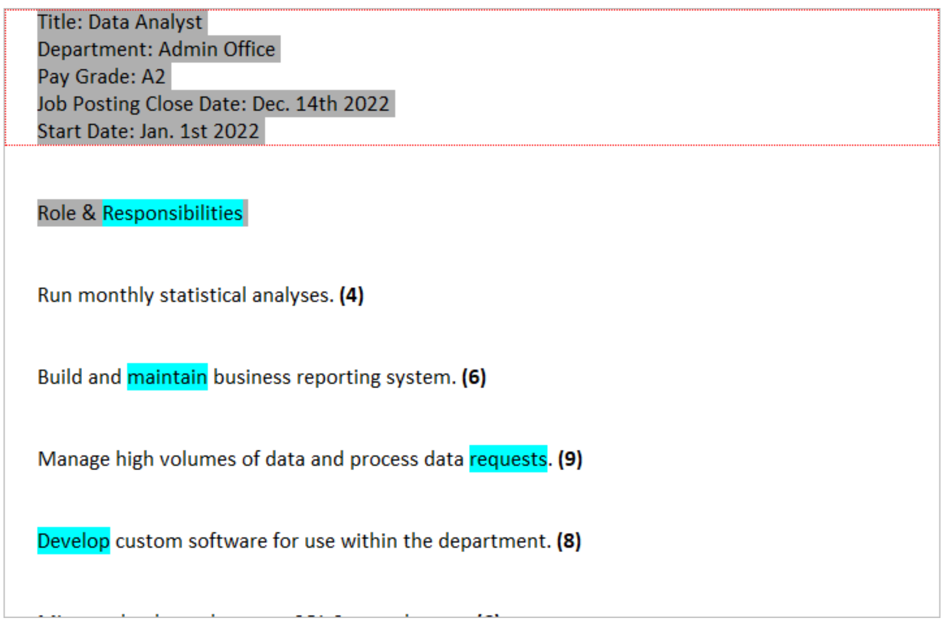
First, let us view the document. From the Help tab on the ribbon, click the Example Documents button, and select "Job Posting." Next, you will be prompted to either create a new project or view the document; select View document. Once the document is opened, note the text in the upper left corner. Here we have short descriptions about a job posting that we will want to preserve as separate lines when we import:



Close this document and return to *Readability Studio*. Click the Example Documents button and again select "Job Posting." Next, select Create a new project. When the New Project Wizard appears, select the Document Structure page and check Line ends (i.e., hard returns) mark the start of a new paragraph:



Finally, click the Finish button to create the project. When the project appears, select Grammar on the sidebar and select Highlighted Report. Note how the job descriptors at the top have been preserved as separate lines:



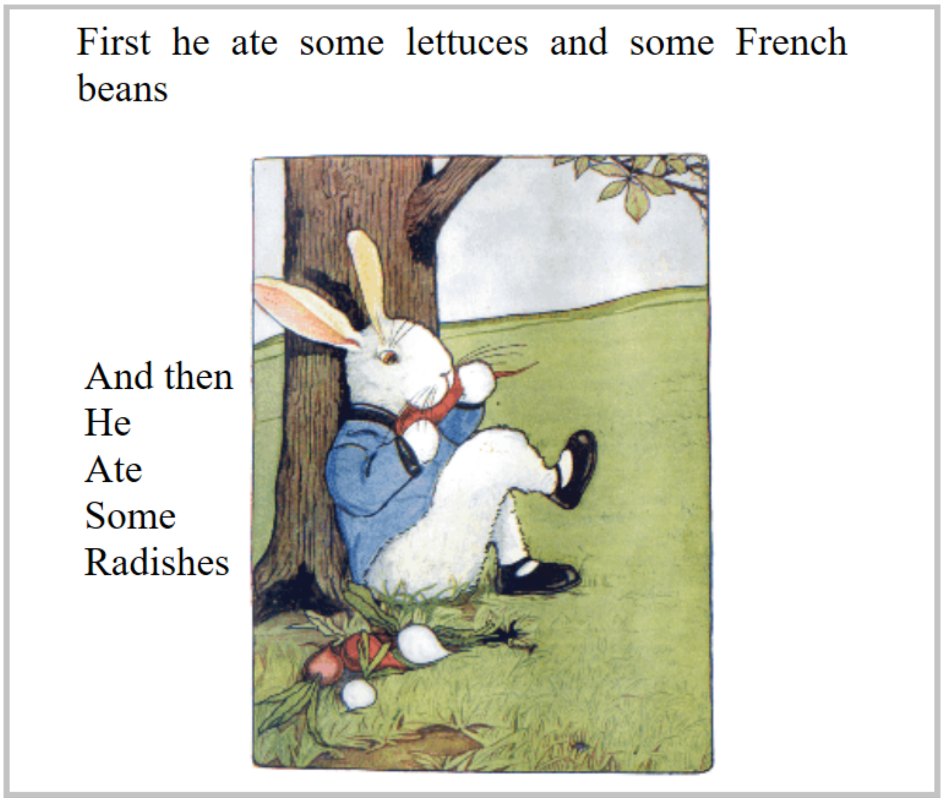
Because these are terse, unterminated lines of text, they will also be excluded from our analyses.

At this point, we can add tests and begin reviewing this (rather dubious) job posting.

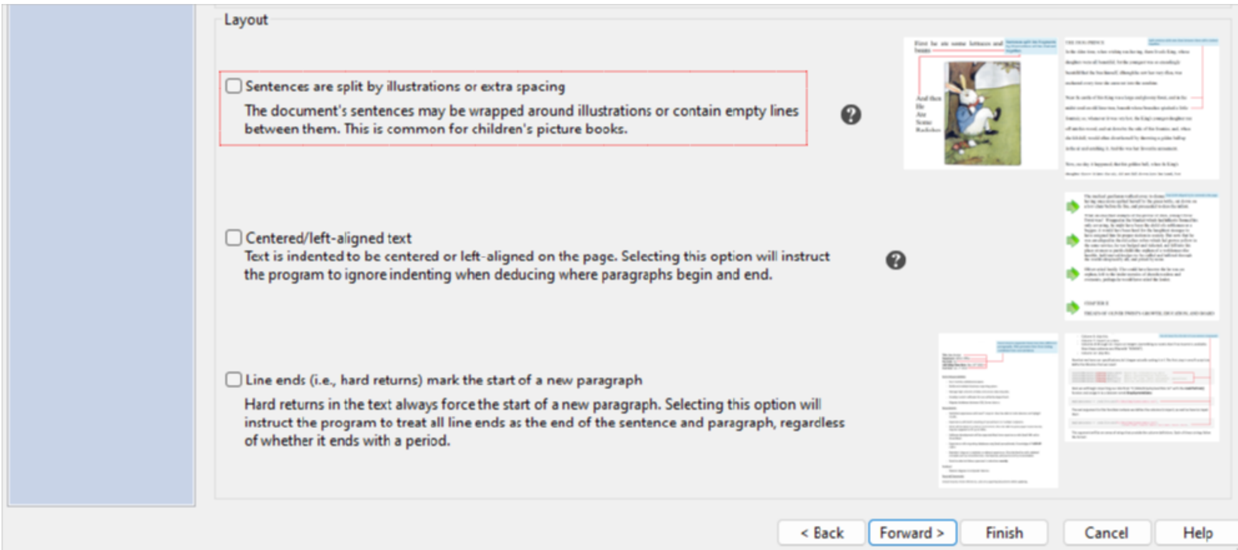
15.2 Sentences Broken by Blank Lines

In this example, we will import a picture book which has unusual sentence formatting. In the document we will be looking at, some of the sentences are broken apart by hard returns and blank lines so that they can wrap around images.

From the Help tab, click the Example Documents button and then select *The Tale of Peter Rabbit*. When prompted about what to do with the file, select View document. As shown below, we can see a sentence that is partially on top of an image, and the rest of it is split across multiple lines to fit next to the image.

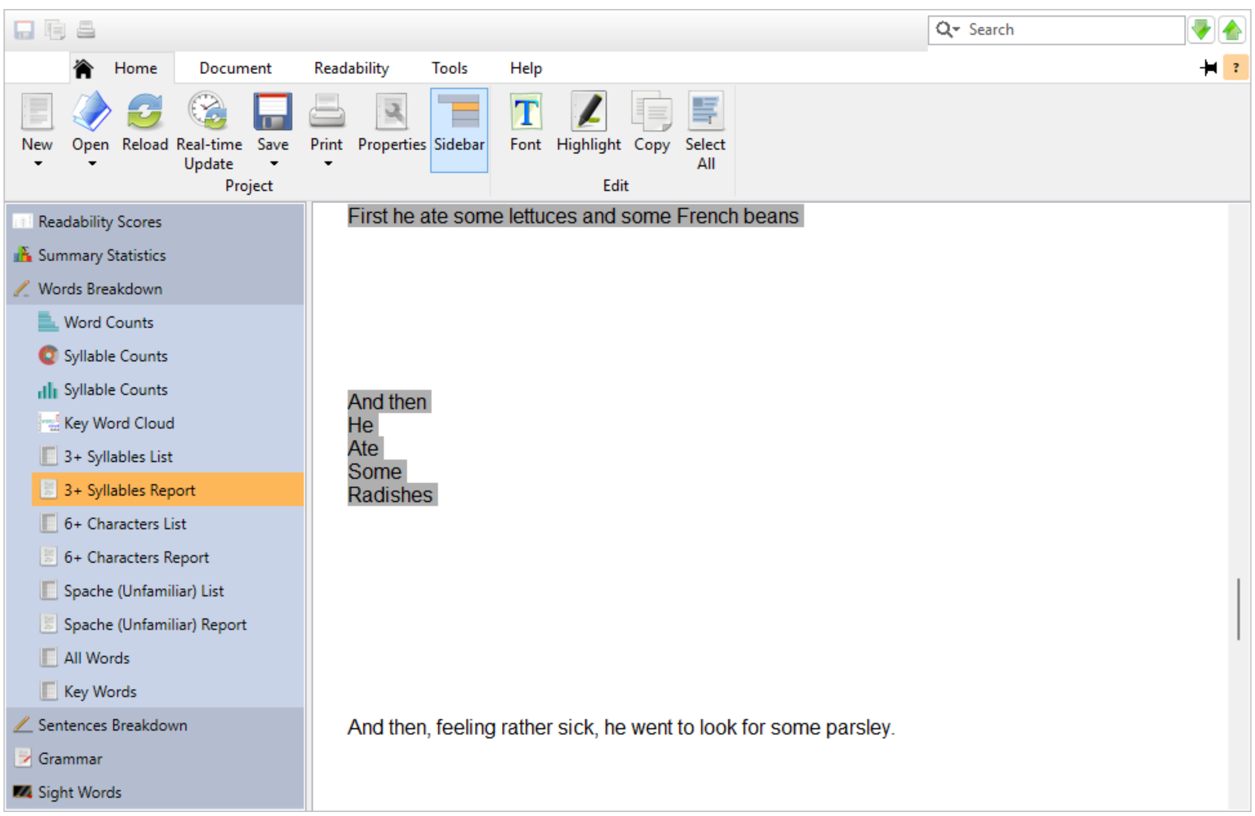


Close this document and return to the Help tab. Click the Example Documents button and again select *The Tale of Peter Rabbit*. This time, select Create a new project. From the project wizard, select the Document Structure page and note the option Sentences are split. Selecting this will instruct the program to ignore hard returns and blank lines and combine these fragments into full sentences:



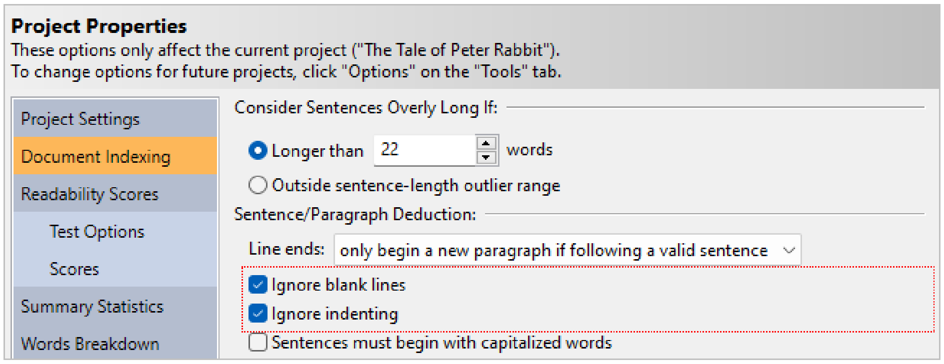
For this example, though, let's leave this unchecked to see what happens. Next, select the Test Selection page and select Children's literature. Click Finish to create the project.

When the project appears, click on the Words Breakdown icon on the project sidebar and select 3+ Syllable Report. Note how some sentences have been split into incomplete fragments (as it appeared in the book):

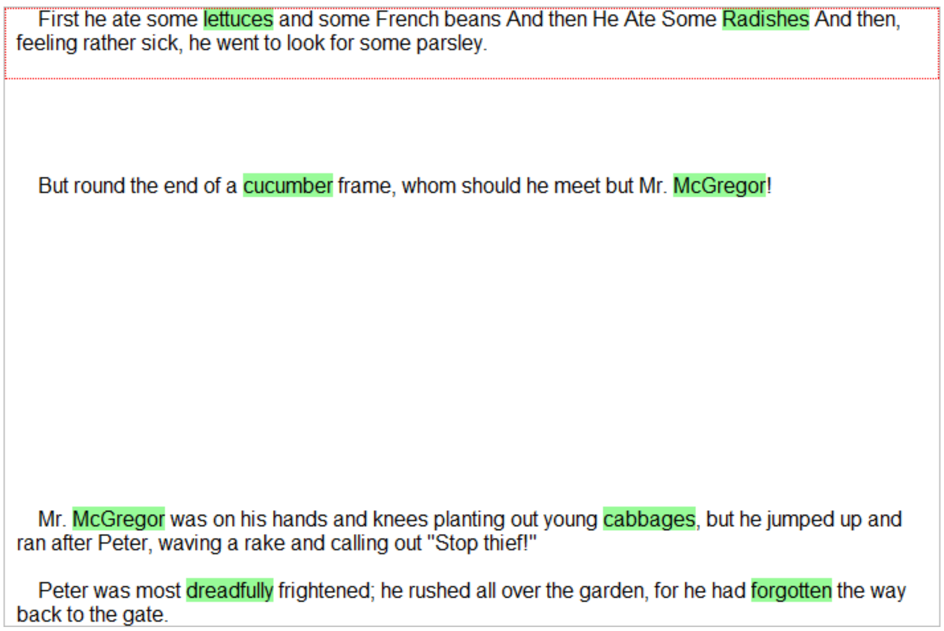


To correct this, we will change how sentences are deduced for this project. Click the Properties button on the Home tab and then select Document Indexing. First, ensure that Line ends is set to only begin a new paragraph if following a valid sentence (this should be the default). Then check the options

Ignore blank lines and Ignore indenting and click OK to accept. (Note that we also selected Ignore indenting because this file contains some indented text.)



At this point, the project will be reanalyzed and these sentence fragments will be chained together:


First he ate some lettuces and some French beans And then He Ate Some Radishes And then, feeling rather sick, he went to look for some parsley.

But round the end of a cucumber frame, whom should he meet but Mr. McGregor!

Mr. McGregor was on his hands and knees planting out young cabbages, but he jumped up and ran after Peter, waving a rake and calling out "Stop thief!"

Peter was most dreadfully frightened; he rushed all over the garden, for he had forgotten the way back to the gate.

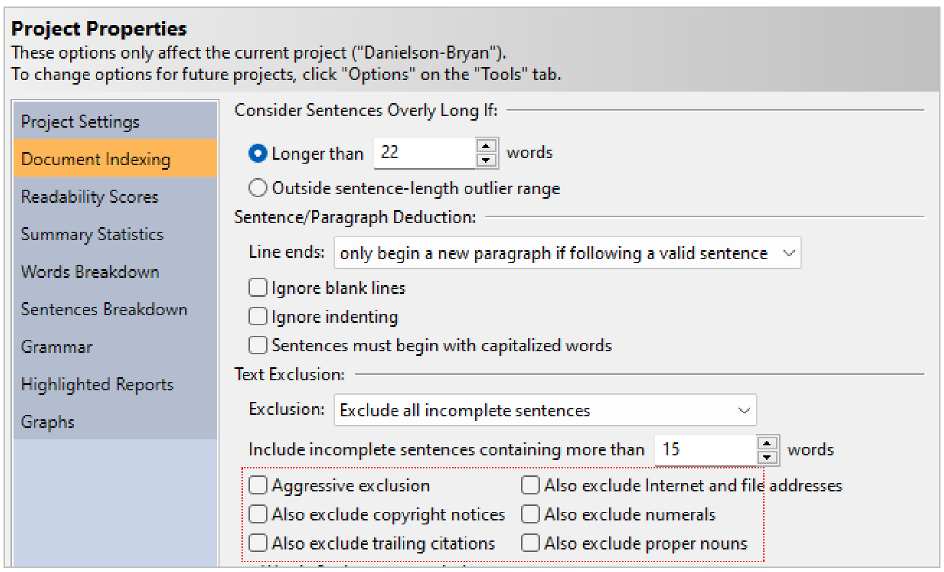
15.3 Excluding Text

When reviewing a document, it is usually preferable to ignore certain items, such as headers and citations (refer to sec. 20.1 for more information). In this example, we will discuss the various methods for excluding these items.

The first step is to open the example file "Danielson-Bryan." From the Help tab, click the Example Documents button and then select "Danielson-Bryan" from the menu. (When prompted about how to open the document, select Create a new project.) When the New Project wizard appears, leave the defaults and click Finish. A warning about an incomplete sentence being included because of its length will appear.

For now click Close; we will review this sentence later in this example.

Once the project is loaded, on the Home tab click the Properties button under the Project section. On the Properties dialog, click Analysis on the sidebar and select Document Indexing. Uncheck all the options shown below and click OK:



Next, click Grammar on the project's sidebar and then click Highlighted Report. In this window, any text that is being excluded will be highlighted gray. For example, note that the header *Danielson-Bryan 1 Test* is excluded, as shown below:

Danielson-Bryan 1 Test

Danielson-Bryan 1 (206) is meant for student materials and calculates the grade level of a document based on sentence length and character count. **(23)**

DB1 was designed to be a faster method (compared to Farr-Jenkins-Paterson) of calculating readability scores on UNIVAC 1105 computers. **(19)** Rather than counting syllables, DB1 counts characters and punctuation, which was simpler to analyze on computers of that era. **(19)** This performance benefit is more of an historical reference now; most modern computers and software will calculate syllable counts optimally. **(20)**

The Danielson-Bryan 1 Formula

G =1.0364*(RP/W) +.0194*(RP/S) -.6059

Scroll midway down the document and note how another header, an equation, and a table are also excluded. However, a couple of items in this block are being included that shouldn't be:

Where: **(1)**

G Grade level

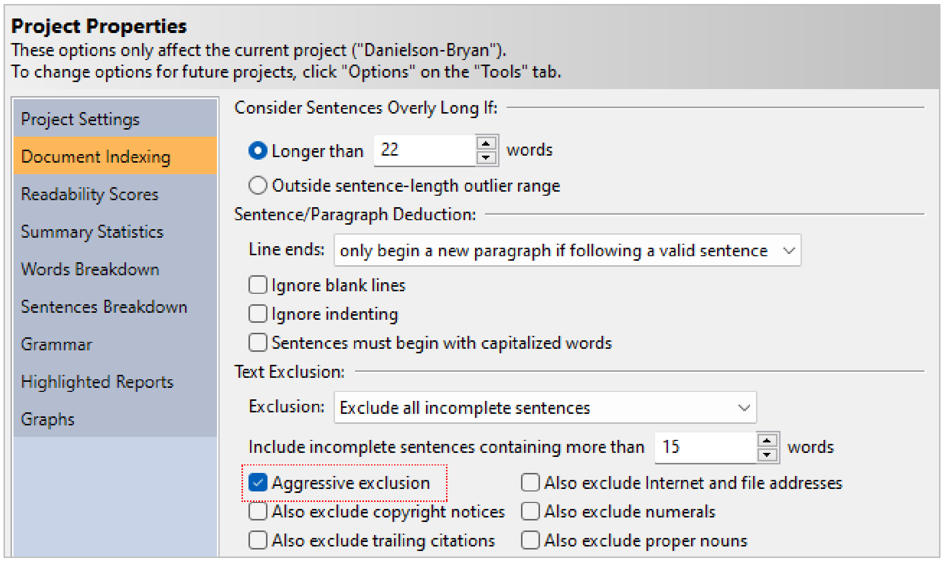
W Number of words*

RP Number of characters, in this context being letters, numbers, and punctuation (except for sentence-ending punctuation, such as periods) **(19)**

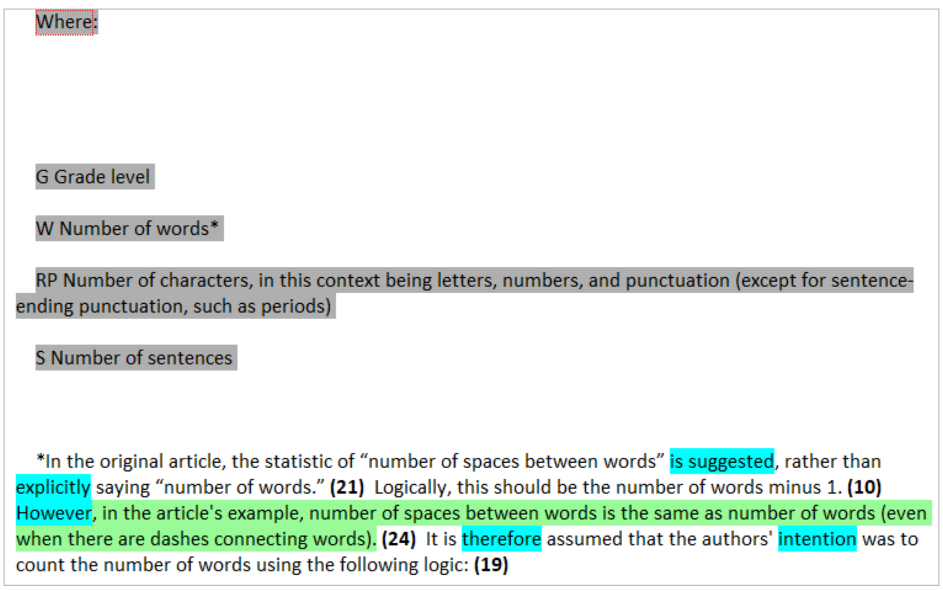
S Number of sentences

*In the original article, the statistic of "number of spaces between words" is suggested, rather than explicitly saying "number of words." **(21)** Logically, this should be the number of words minus 1. **(10)** However, in the article's example, number of spaces between words is the same as number of words (even when there are dashes connecting words). **(24)** It is therefore assumed that the authors' intention was to count the number of words using the following logic: **(19)**

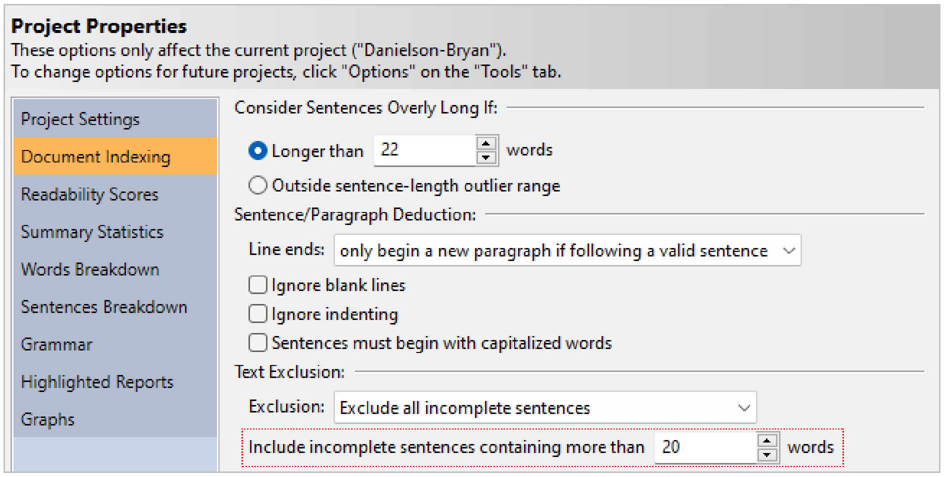
The first is the sentence *Where:*. Sentences that end with a semicolon and are followed by a list will normally be considered valid. In this case though, we will want to exclude this sentence too. Open the Properties dialog again, click Analysis and select Document Indexing. Check the option Aggressively deduce lists, as shown below:



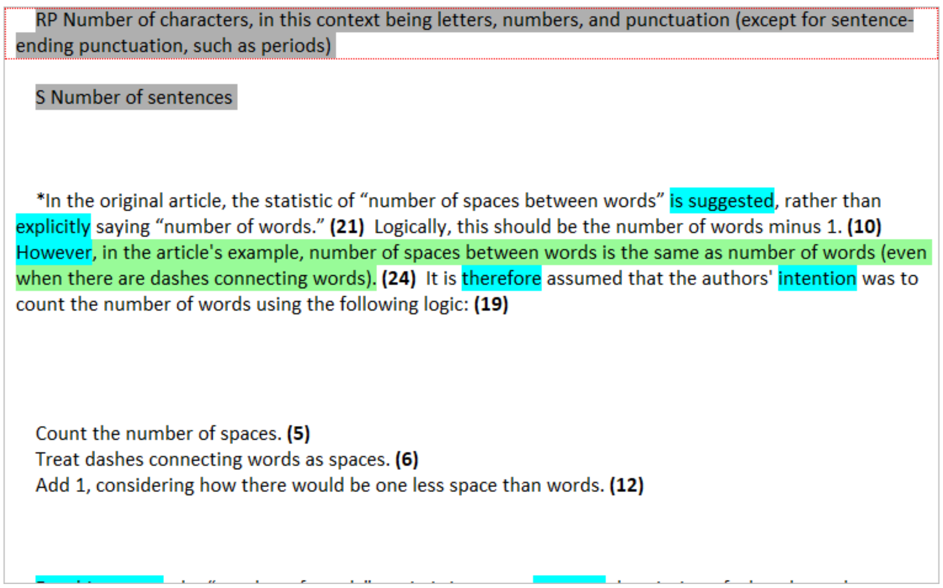
Checking this option will exclude any valid sentence (containing less than 10 words) that is inside of a block of incomplete sentences. Click OK and note how *Where:* is now excluded:



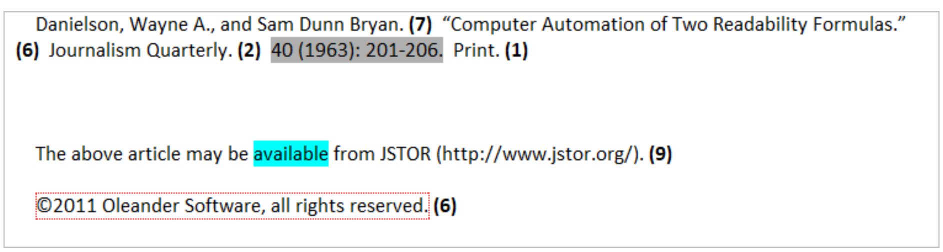
The next item to note is the sentence *Number of characters....* This is really an item in a table that is missing a period, so we should be excluding it. However, currently it is being included because it contains more than 15 words. To adjust this, go back to the Document Indexing options for this project. Change the value for Include incomplete sentences containing more than [15] words to a higher value (e.g., 20) and click OK:



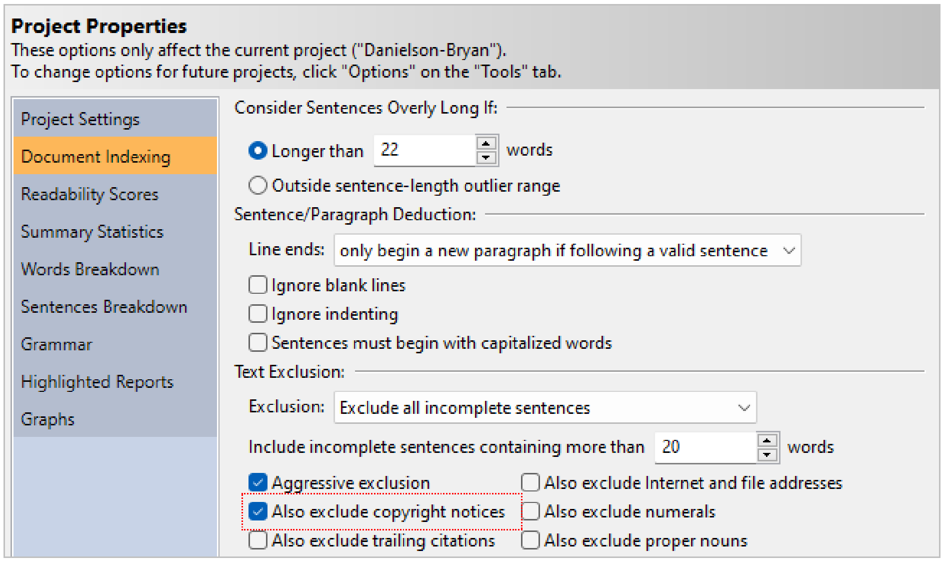
Note how the *Number of characters...* sentence is now excluded:



Now scroll to the bottom of the report window and note the copyright notice:

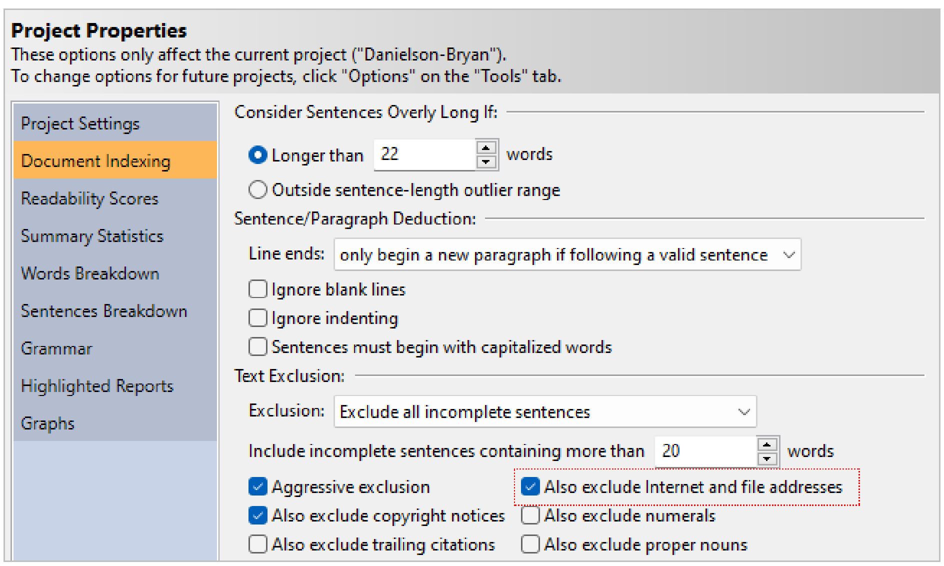


Go to the Document Indexing options, check Also exclude copyright notices, and click OK:

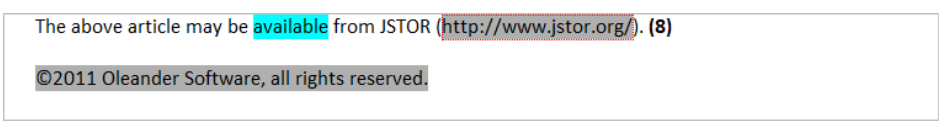


Now this copyright notice will be excluded from the analysis:

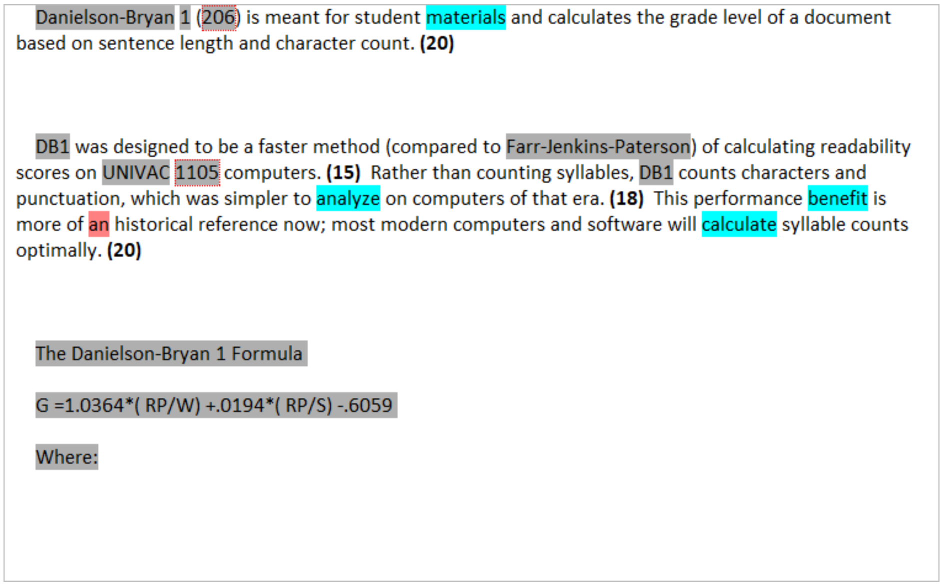
Along with incomplete sentences, specific types of words can also be excluded. For example, at the bottom of this window note the hyperlink *http://www.jstor.org/*. To exclude this, go to the Document Indexing options, check Also exclude Internet and file addresses, and click OK:



Note how *http://www.jstor.org/* is now excluded:



Other items, such as proper nouns and numbers can also be excluded. Go to the Document Indexing options, check Also exclude numerals and Also exclude proper nouns, and click OK. Then scroll to the top of the report window. First note how numbers, such as the page citation and computer series, are now excluded:


Danielson-Bryan 1 (206) is meant for student materials and calculates the grade level of a document based on sentence length and character count. **(20)**

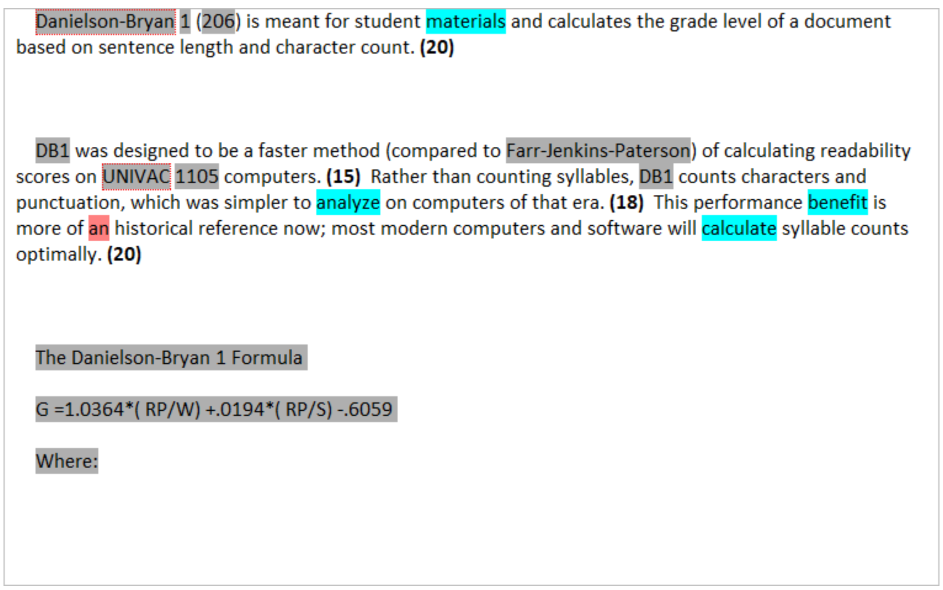
DB1 was designed to be a faster method (compared to Farr-Jenkins-Paterson) of calculating readability scores on UNIVAC 1105 computers. **(15)** Rather than counting syllables, DB1 counts characters and punctuation, which was simpler to analyze on computers of that era. **(18)** This performance benefit is more of an historical reference now; most modern computers and software will calculate syllable counts optimally. **(20)**

The Danielson-Bryan 1 Formula

G =1.0364*(RP/W) +.0194*(RP/S) -.6059

Where:

Also note how proper nouns, such as *Danielson-Bryan* and *UNIVAC*, are now excluded:


Danielson-Bryan 1 (206) is meant for student materials and calculates the grade level of a document based on sentence length and character count. **(20)**

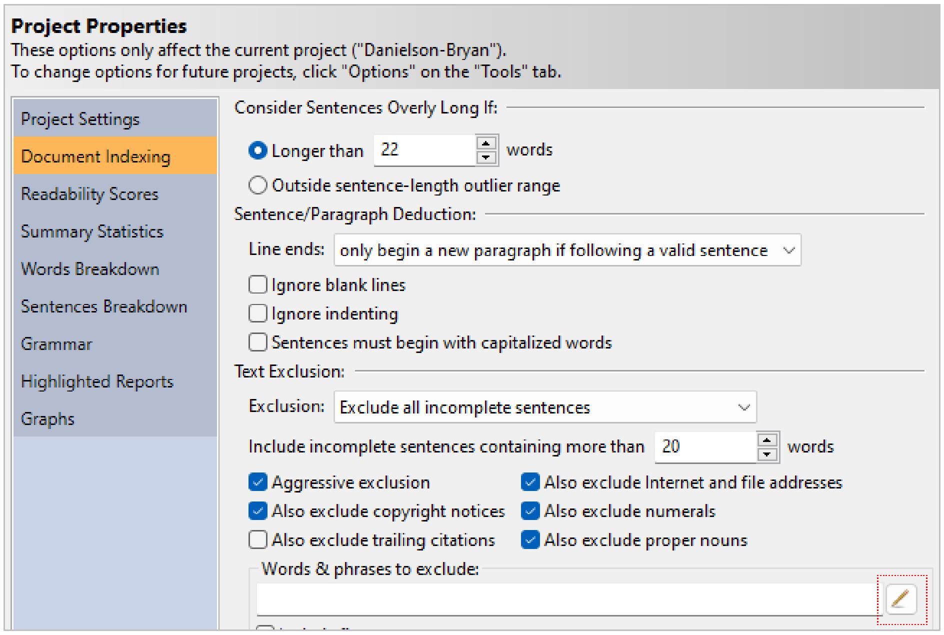
DB1 was designed to be a faster method (compared to Farr-Jenkins-Paterson) of calculating readability scores on UNIVAC 1105 computers. **(15)** Rather than counting syllables, DB1 counts characters and punctuation, which was simpler to analyze on computers of that era. **(18)** This performance benefit is more of an historical reference now; most modern computers and software will calculate syllable counts optimally. **(20)**

The Danielson-Bryan 1 Formula

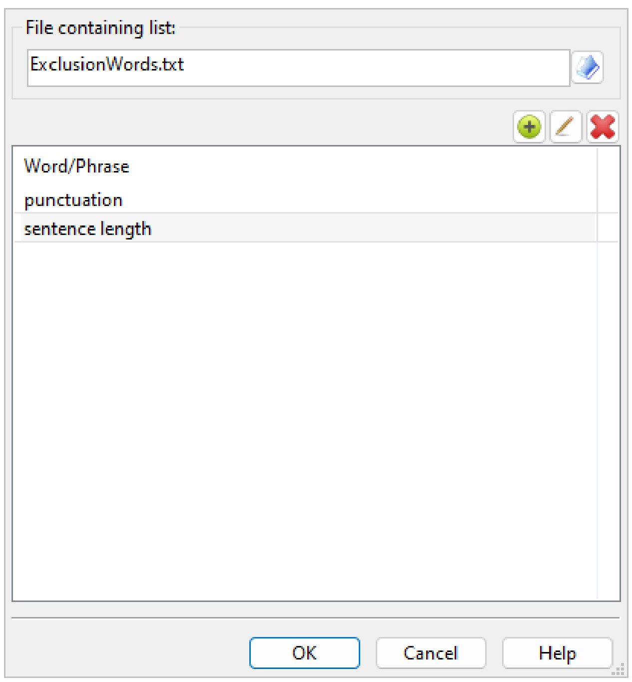
G =1.0364*(RP/W) +.0194*(RP/S) -.6059

Where:

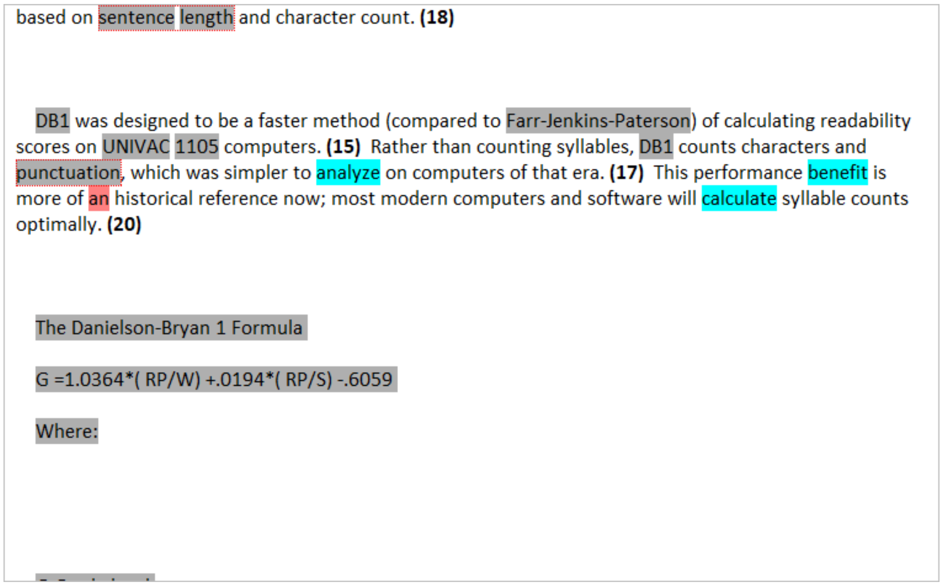
You can also exclude specific words and phrases. From the Document Indexing options, click the edit button in the Words & phrases to exclude section:



In the Edit Phrase List dialog, either double-click the grid or click the add item button to begin adding words or phrases to exclude. In one line enter *punctuation*, and in another line enter *sentence length*:



Click OK and all instances of *punctuation* and *sentence length* are now excluded:


based on sentence length and character count. **(18)**

DB1 was designed to be a faster method (compared to Farr-Jenkins-Paterson) of calculating readability scores on UNIVAC 1105 computers. **(15)** Rather than counting syllables, DB1 counts characters and punctuation, which was simpler to analyze on computers of that era. **(17)** This performance benefit is more of an historical reference now; most modern computers and software will calculate syllable counts optimally. **(20)**

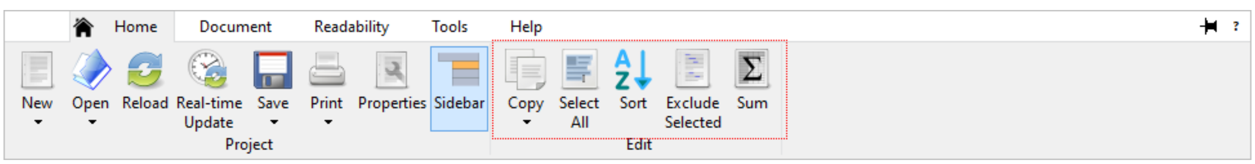
The Danielson-Bryan 1 Formula

G =1.0364*(RP/W) +.0194*(RP/S) -.6059

Where:

Separate instances of *sentence* and *length* will still be included, they will only be excluded if they appear together to form the phrase *sentence length*. Refer to sec. 8.3 for more on how custom word/phrase exclusion works.

Along with editing the word list from the Properties dialog, you can also add words directly from the results. Select Words Breakdown on the sidebar and select any list under this section (e.g., 3+ Syllable List). In this list window, select any words that you wish to exclude and click Exclude Selected on the Home tab of the ribbon:



The selected words will be added to your exclusion list and the project will then be updated.

Note that when excluding individual words, these words are treated as if they are not in the document at all. This means that excluding words will affect the sentences that they appear in. For example, by excluding *Danielson-Bryan*, *1*, and *206*, the first sentence will have a word count of 20 instead of 23.

Another section of text that could be excluded is the citation at the bottom. Go to the Document Indexing options, check Also exclude trailing citations and click OK. Scroll to the bottom of the document and note how everything below the header *Citations* is now excluded:


Citations

Danielson, Wayne A., and Sam Dunn Bryan. "Computer Automation of Two Readability Formulas." Journalism Quarterly. 40 (1963): 201-206. Print.

The above article may be available from JSTOR (http://www.jstor.org/).

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At this point, all that remains in the analysis is the actual prose—all "noise" from the document has been filtered out.

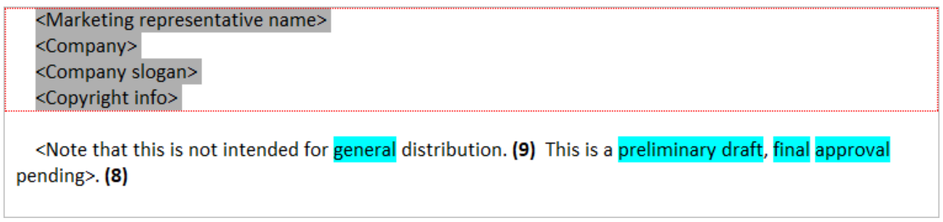
> **Tip**
> All the options discussed here are also available on the Document tab of the ribbon.

15.4 Excluding Placeholders and Blocks of Text

Continuing from the previous example, we will now discuss how to exclude specific blocks of text.

The first step is to open the example file "Press Release." From the Help tab, click the Example Documents button and then select "Press Release" from the menu. (When prompted about how to open the document, select Create a new project.) When the New Project wizard appears, leave the defaults and click Finish.

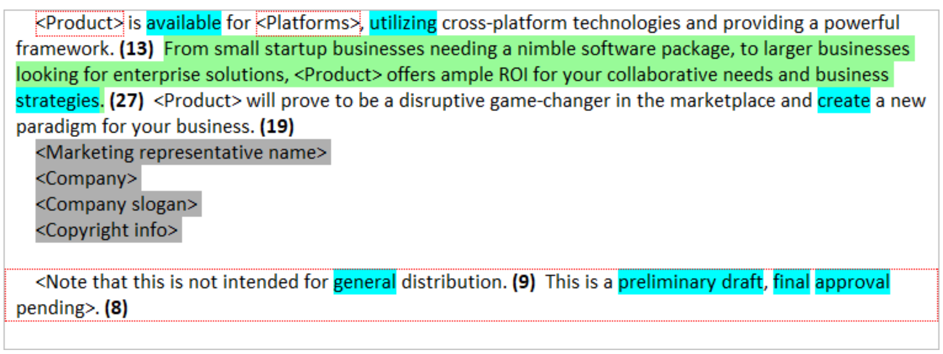
Once the project is loaded, click Grammar on the project's sidebar and then click Highlighted Report. In this window, any text that is being excluded will be highlighted gray. For example, an indented list is being excluded, as shown below:



<Marketing representative name>
<Company>
<Company slogan>
<Copyright info>

<Note that this is not intended for general distribution. **(9)** This is a preliminary draft, final approval pending>. **(8)**

However, note that there are a few placeholders in this file, such as *<Product>*:



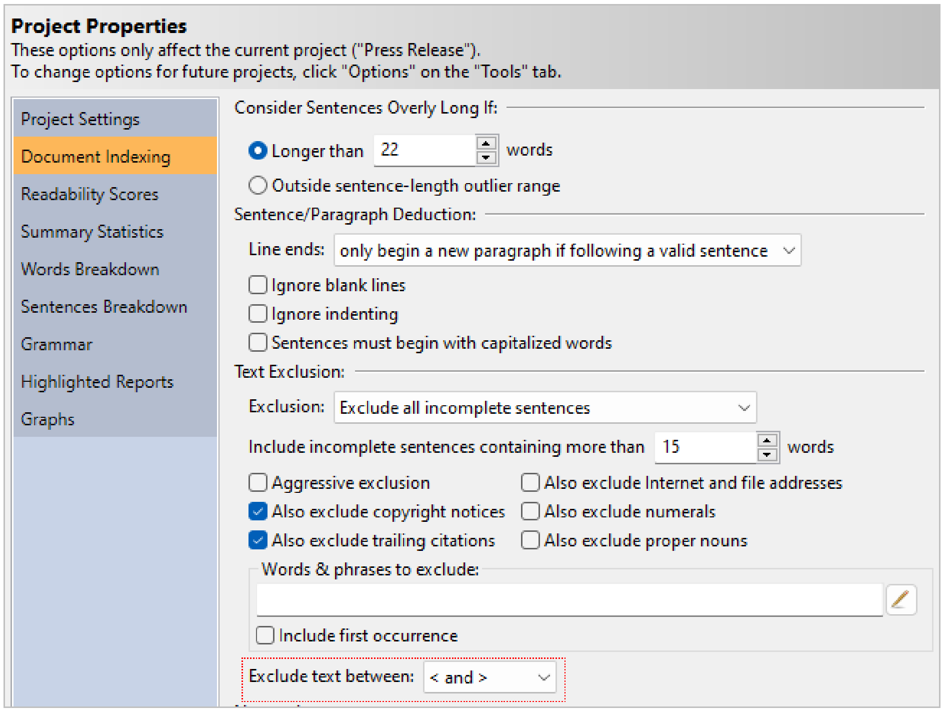
<Product> is available for <Platforms>, utilizing cross-platform technologies and providing a powerful framework. **(13)** From small startup businesses needing a nimble software package, to larger businesses looking for enterprise solutions, <Product> offers ample ROI for your collaborative needs and business strategies. **(27)** <Product> will prove to be a disruptive game-changer in the marketplace and create a new paradigm for your business. **(19)**

<Marketing representative name>
<Company>
<Company slogan>
<Copyright info>

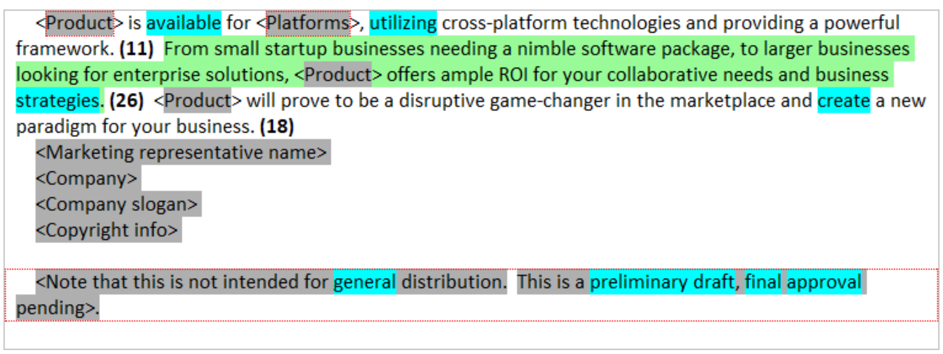
<Note that this is not intended for general distribution. **(9)** This is a preliminary draft, final approval pending>. **(8)**

We will want to exclude these items because they will not be in the final draft. However, because these are valid words and sentences (i.e., they are not headers, list items, etc.), then the standard exclusion features will not remove them. Also, using a list of exclusion words may be too aggressive. For example, adding *product* to an exclusion list will remove the *<Product>*, but it will also exclude any other instance of *product*. Also, the last paragraph is valid, but because it is just a comment we will want to exclude this also. For these situations, we will need to use tagged-block exclusion.

Open the Properties dialog and select Document Indexing (Home > Properties > Document Indexing). Set the option Exclude text between to < and >, as shown below:



Selecting this option will exclude text in between pairs of angle brackets. Click OK and note how the placeholders and final paragraph are now excluded:

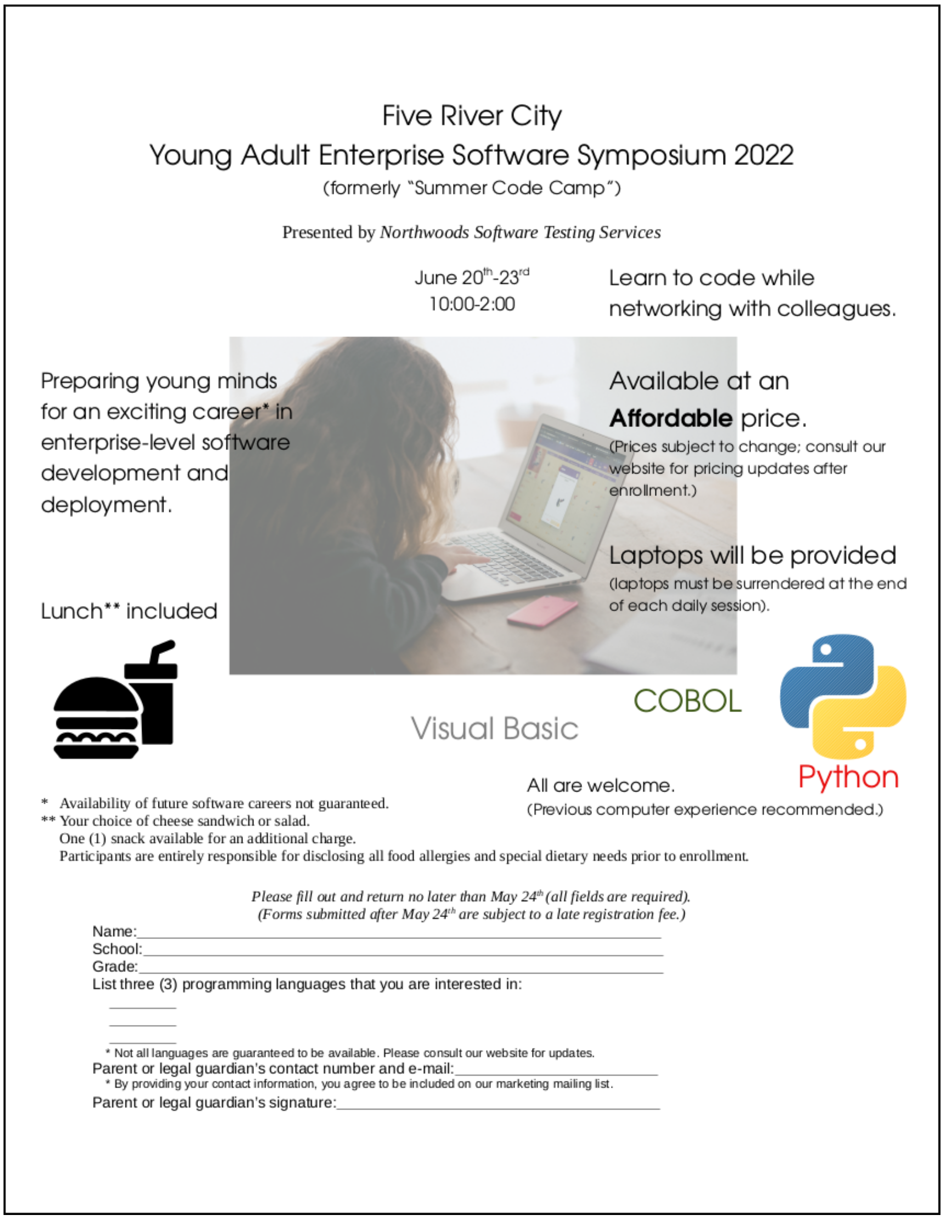


💡 Tip
All the options discussed here are also offered on the Document tab of the ribbon.

15.5 Including an Addendum

Continuing from the "Reviewing a Flyer" (sec. 14.7) and "High-level Review of a Disclaimer" (sec. 14.8) examples, our fictitious private equity firm now plans to append a disclaimer to the flyer.

Along with policy changes to the program, they have redesigned the program's flyer to appear more professional. Let's first take a look at the updated flyer. From the Help tab on the ribbon, click the Example Documents button, and select "YA Enterprise Software Symposium." Next, you will be prompted to either create a new project or view the document; select View document. Once the document is opened, note how the flyer's short text boxes have been replaced by full sentences:


Five River City
Young Adult Enterprise Software Symposium 2022
(formerly "Summer Code Camp")

Presented by *Northwoods Software Testing Services*

June 20th-23rd
10:00-2:00

Learn to code while networking with colleagues.

Preparing young minds for an exciting career* in enterprise-level software development and deployment.

Available at an **Affordable** price.
(Prices subject to change; consult our website for pricing updates after enrollment.)

Laptops will be provided
(laptops must be surrendered at the end of each daily session).

Lunch** included

COBOL

Visual Basic

Python

All are welcome.
(Previous computer experience recommended.)

* Availability of future software careers not guaranteed.
** Your choice of cheese sandwich or salad.
One (1) snack available for an additional charge.
Participants are entirely responsible for disclosing all food allergies and special dietary needs prior to enrollment.

Please fill out and return no later than May 24th (all fields are required).
(Forms submitted after May 24th are subject to a late registration fee.)

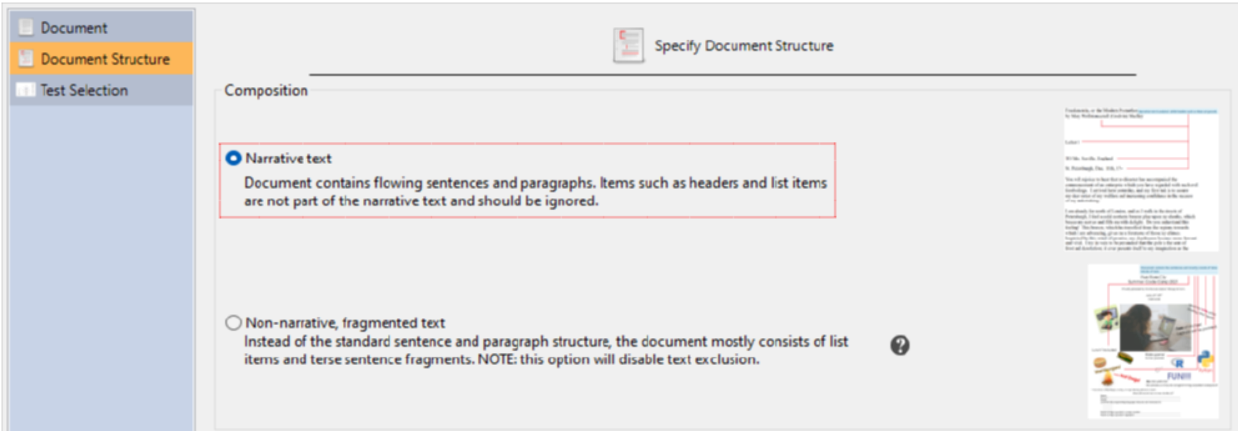
Name:__________
School:__________
Grade:__________
List three (3) programming languages that you are interested in:

______
______
______

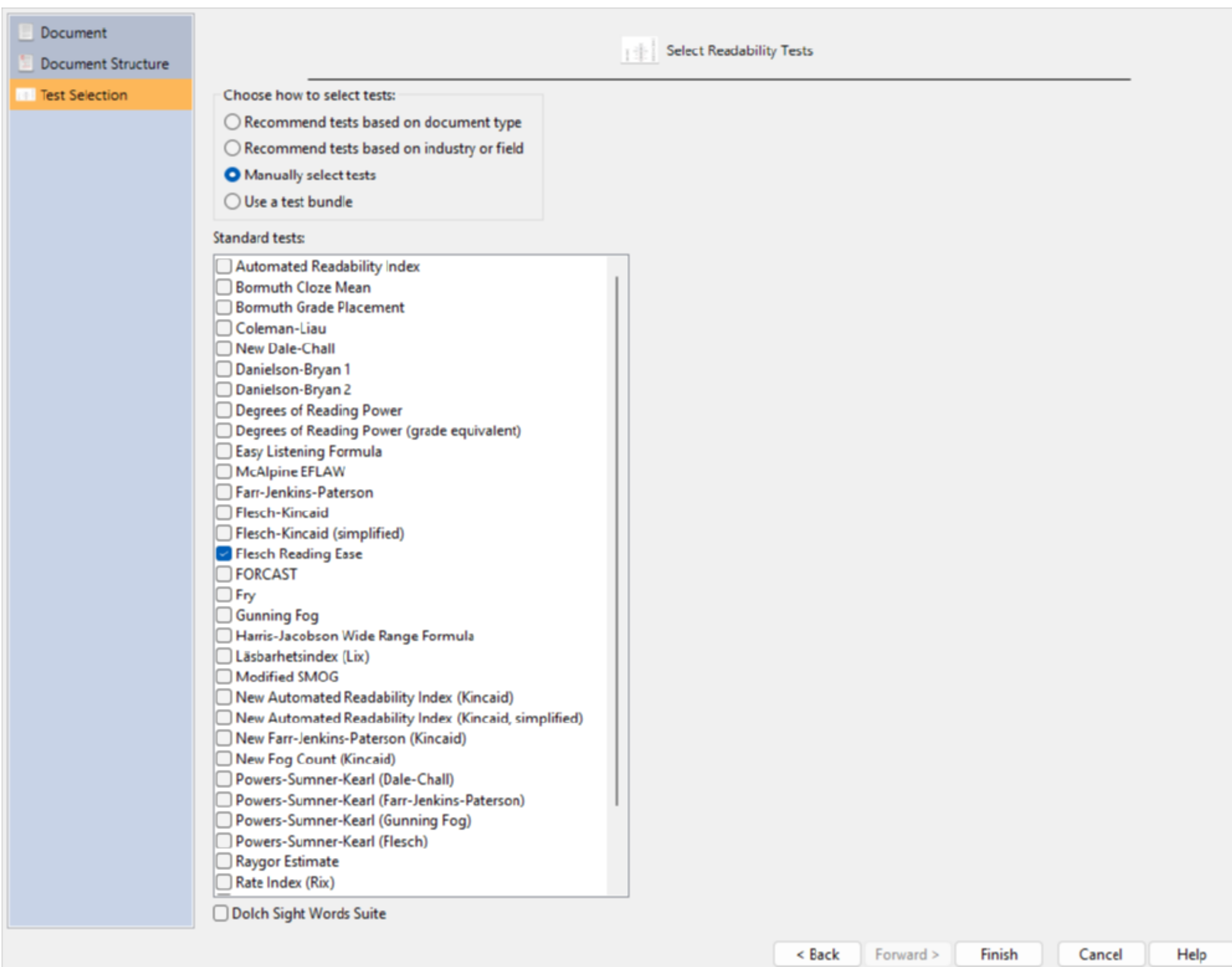
* Not all languages are guaranteed to be available. Please consult our website for updates.
Parent or legal guardian's contact number and e-mail:__________
* By providing your contact information, you agree to be included on our marketing mailing list.
Parent or legal guardian's signature:__________

This means that when we import this document, we will specify that it is a narrative document and apply text exclusion. (This is in contrast to the earlier flyer, where we specified it as non-narrative, fragmented text.)

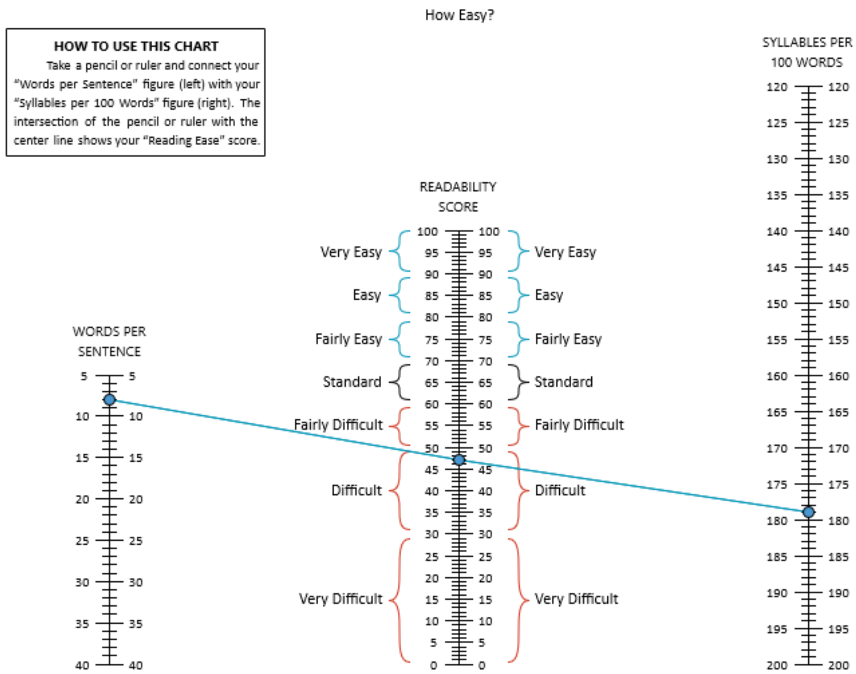
Close this document and return to *Readability Studio*. Click the Example Documents button and again select "YA Enterprise Software Symposium." Next, select Create a new project. When the New Project Wizard appears, select the Document Structure page and select Narrative text:



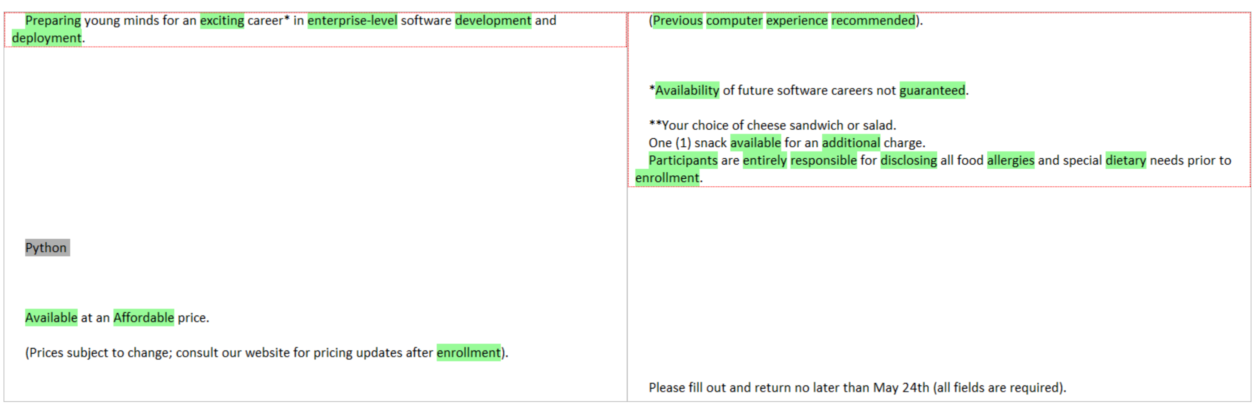
Next, select the Test Selection page and select Manually select tests. In the grid below, check Flesch Reading Ease:



Finally, click the Finish button to create the project. When the project appears, select Flesch Reading Ease under the Scores section in the sidebar.



Note that this new flyer is difficult, so let's review it. Click Words Breakdown on the sidebar and then select 3+ Syllables Report. As we scroll through this window, note the highlighted difficult words:


Preparing young minds for an exciting career* in enterprise-level software development and deployment.

Python

Available at an Affordable price.

(Prices subject to change; consult our website for pricing updates after enrollment).

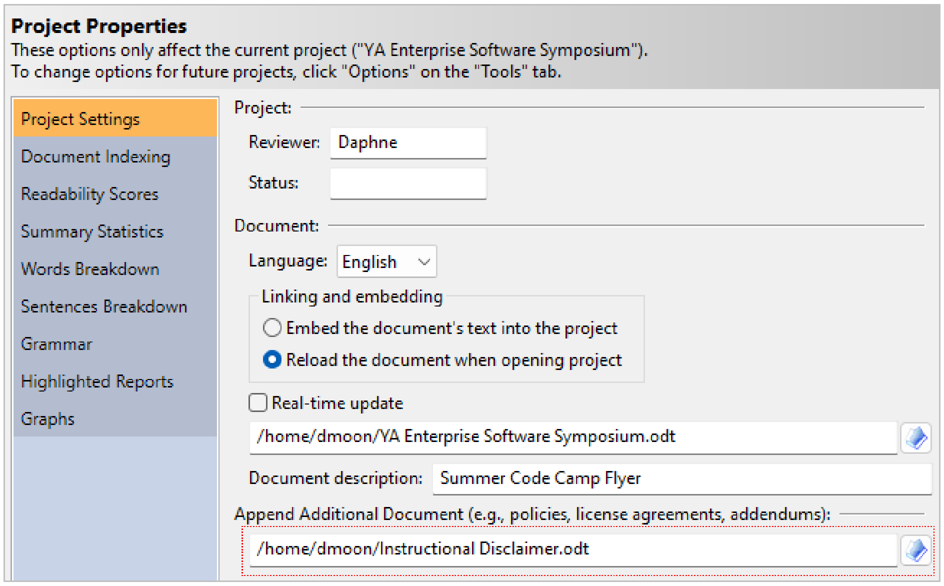
(Previous computer experience recommended).

*Availability of future software careers not guaranteed.

**Your choice of cheese sandwich or salad.
One (1) snack available for an additional charge.
Participants are entirely responsible for disclosing all food allergies and special dietary needs prior to enrollment.

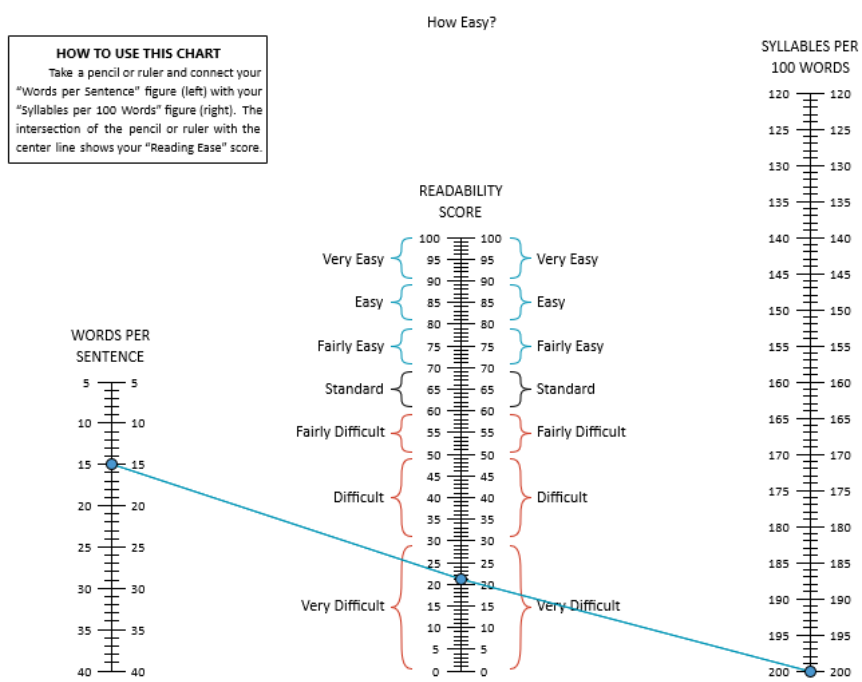
Please fill out and return no later than May 24th (all fields are required).

As we can see, this document could use some improving. Before we can do that, though, the new owners of our software company have also requested that a disclaimer be appended to all sign-up forms such as this. Rather than running a separate analysis on this disclaimer, we would like to append this document to our flyer and analyze them together. To do this, go to the Home tab of the ribbon and select Properties. On the Properties dialog, select Project Settings on the sidebar. Next, under the Append Additional Document section, click the folder button and navigate to the *examples* folder from where *Readability Studio* is installed. Select the file "Instructional Disclaimer.odt":



💡 Tip
You can also enter *@[EXAMPLESDIR]/Instructional Disclaimer.odt* into the Append Additional Document field and the program will find the file in the examples folder for you.

Click OK to accept. When the project finishes reloading, select Flesch Reading Ease under the Scores section again.



Note how the document is now very difficult. Click Words Breakdown on the sidebar and then select 3+ Syllables Report. As we scroll to where our disclaimer was appended, note the numerous difficult words:

Any instruction, advice, or products provided by Company do not include any sort of warranty or guarantee of Customer's perceived quality. Although Company has provided quality instruction in the past, this performance being represented is historical; past performance is not a reliable indicator of current or future results and Customer's expectations of quality is in no way guaranteed. The quality of instruction may fluctuate and Company cannot be held responsible for lower level of quality that Customer may experience. Customer hereby releases Company and Owners from any perceived obligation or expectations related to Customer's subjective measurements of instructional quality. Customer also releases Company and Owners from any legal responsibilities as a result of interaction with Company's products, services, advice, or third-party catering services (e.g., allergic reactions, inaccurate educational materials, Customer "disappointment").

Claims of Service. Although Company attempts to provide the highest level of quality for its products and services, Company may not be held liable for Customer's perceived expectations resulting from Company's marketing materials or communications via third parties (e.g., customer "word of mouth"). Customer acknowledges that claims made by Company in promotional materials are for the sole purpose of marketing and may or may not reflect the actual product or service supplied by Company. Exaggerated claims are intended for promotional purposes only and Company is no way responsible for delivering implied products or services.

Next, click Grammar on the sidebar and then select Highlighted Report. Scrolling to the disclaimer portion of the document, note the numerous, meandering sentences:

quality. **(20)** Customer also releases Company and Owners from any legal responsibilities as a result of interaction with Company's products, services, advice, or third-party catering services (e.g., allergic reactions, inaccurate educational materials, Customer "disappointment"). **(32)**

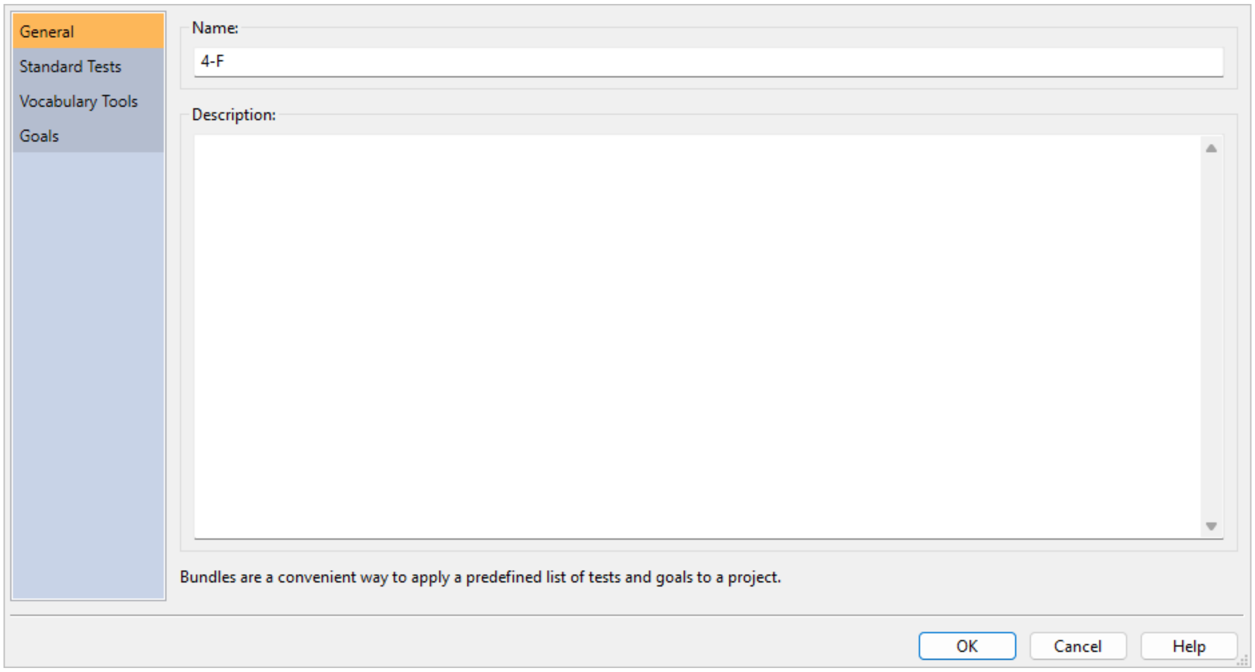
Claims of Service. **(3)** Although Company attempts to provide the highest level of quality for its products and services, Company may not be held liable for Customer's perceived expectations resulting from Company's marketing materials or communications via third parties (e.g., customer "word of mouth"). **(40)** Customer acknowledges that claims made by Company in promotional materials are for the sole purpose of marketing and may or may not reflect the actual product or service supplied by Company. **(31)** Exaggerated claims are intended for promotional purposes only and Company is no way responsible for delivering implied products or services. **(20)**

Customer Feedback. **(2)** Customer agrees to supply any concerns or complaints about Company's products and services exclusively to Company or Owners. **(18)** Customer may not discuss their experience with Company's products or services to any third party if Customer's comments may be construed as negative, derogatory, obstructive, or inflammatory. **(27)** Customer agrees that any disparaging comments or criticism about Company expressed to any third party by Customer will be grounds for legal action, where Customer will be responsible for punitive and compensatory damages.

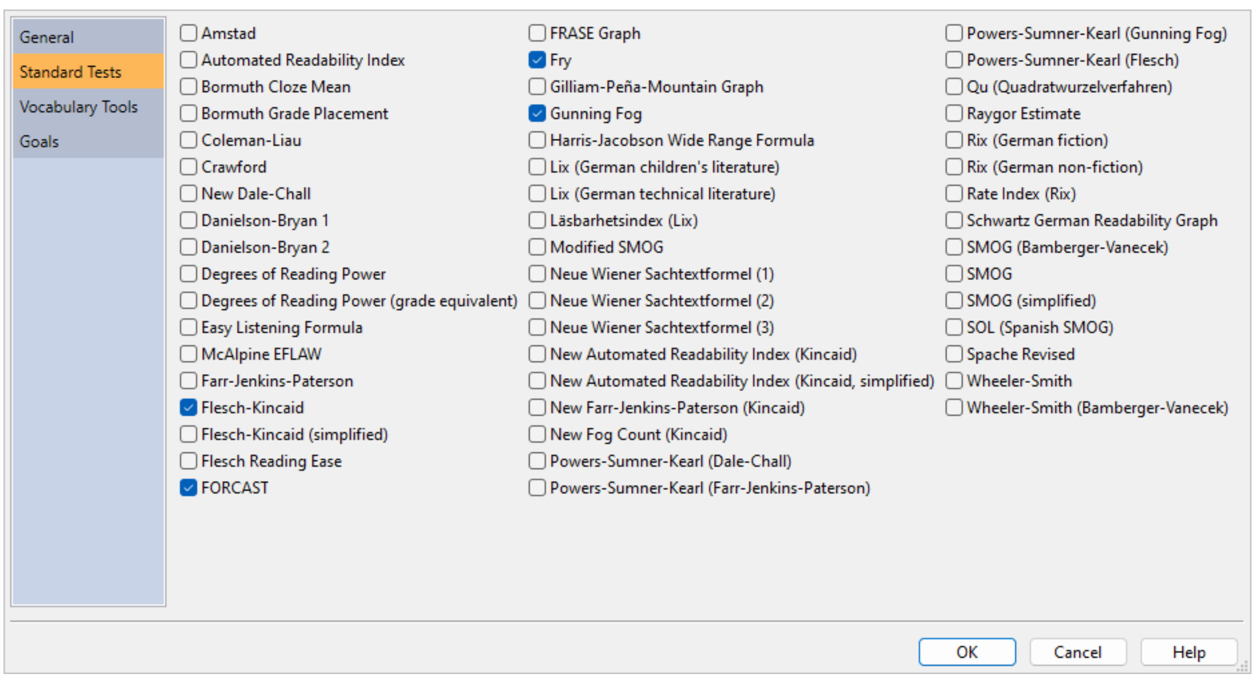
At this point, we will have quite of a bit of improvements to make to both the flyer and disclaimer. (And parents may want to start looking for a different summer camp.)

15.6 Creating a Test Bundle

Test bundles are a convenient way to add multiple tests to a project at once. In this example, we will create a test bundle named "4F" which includes the Fry, FORCAST, Fog, and Flesch-Kincaid tests. The first step is to select the Readability tab, click the Bundles button, and select Add... from the menu. This will open the Add Test Bundle dialog:

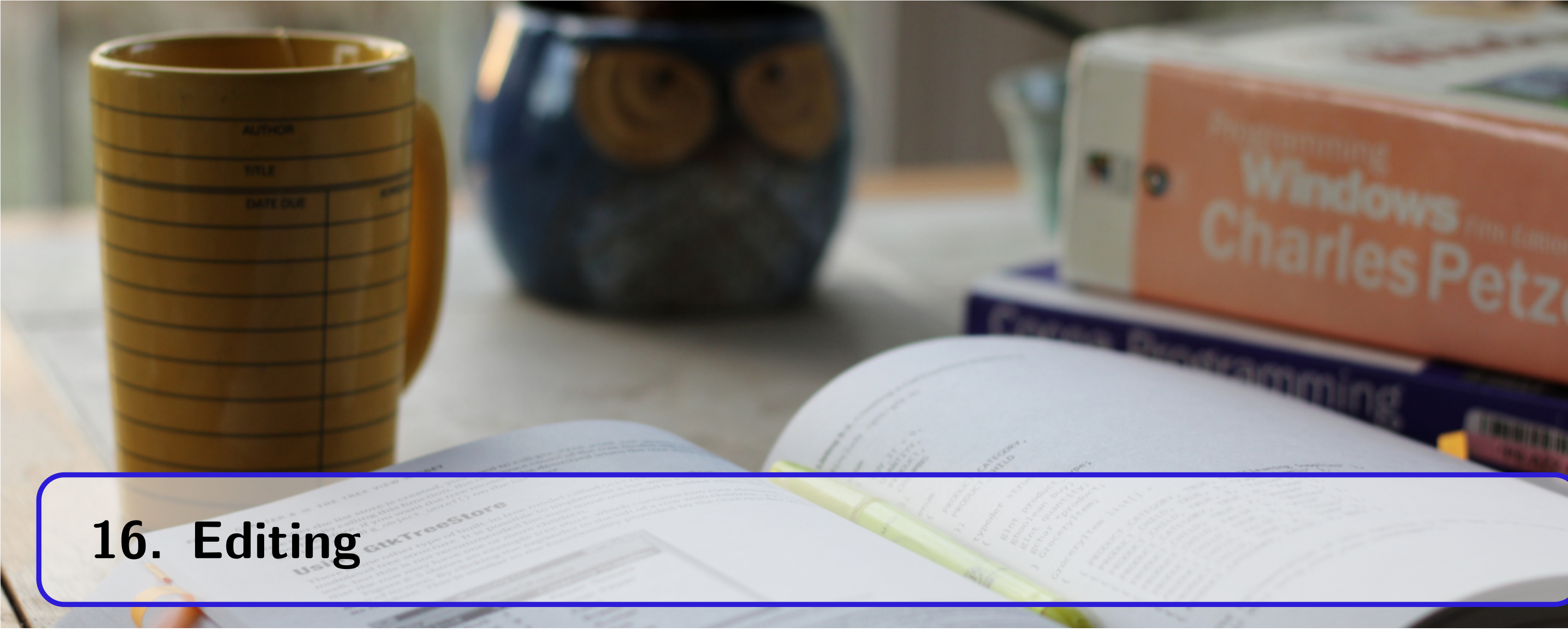


Enter the name "4F" into the Bundle Name text field. Next, select the Standard Tests page and check the Fry, FORCAST, Fog, and Flesch-Kincaid tests, as shown below:



Click the OK button to accept.

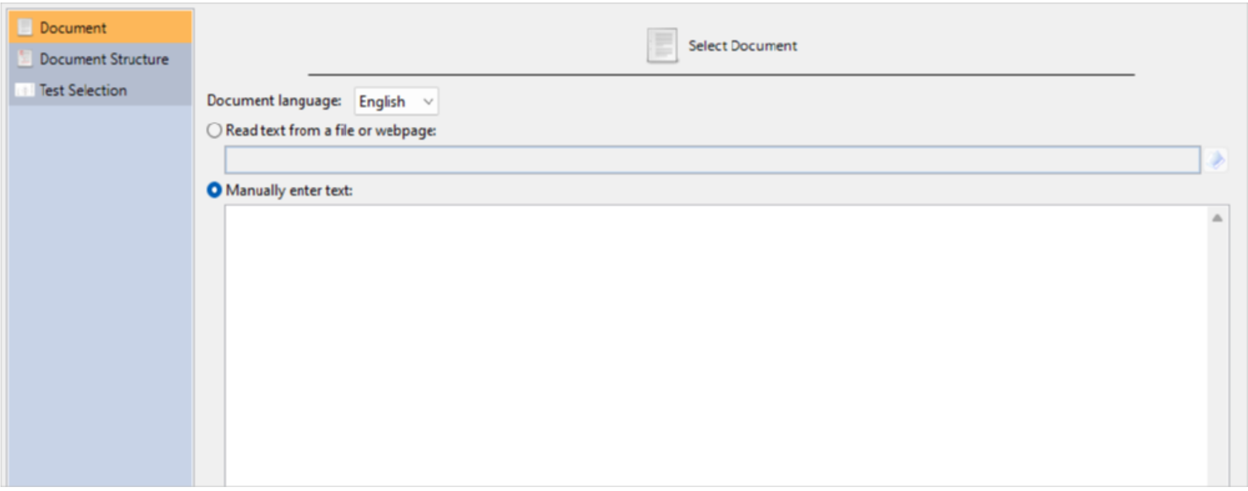
To use the bundle, first open any project. Next, select the Readability tab, click the Bundles button, and select 4F from the menu. Now the program will replace all tests within the project with the tests from the “4F” bundle.



16. Editing

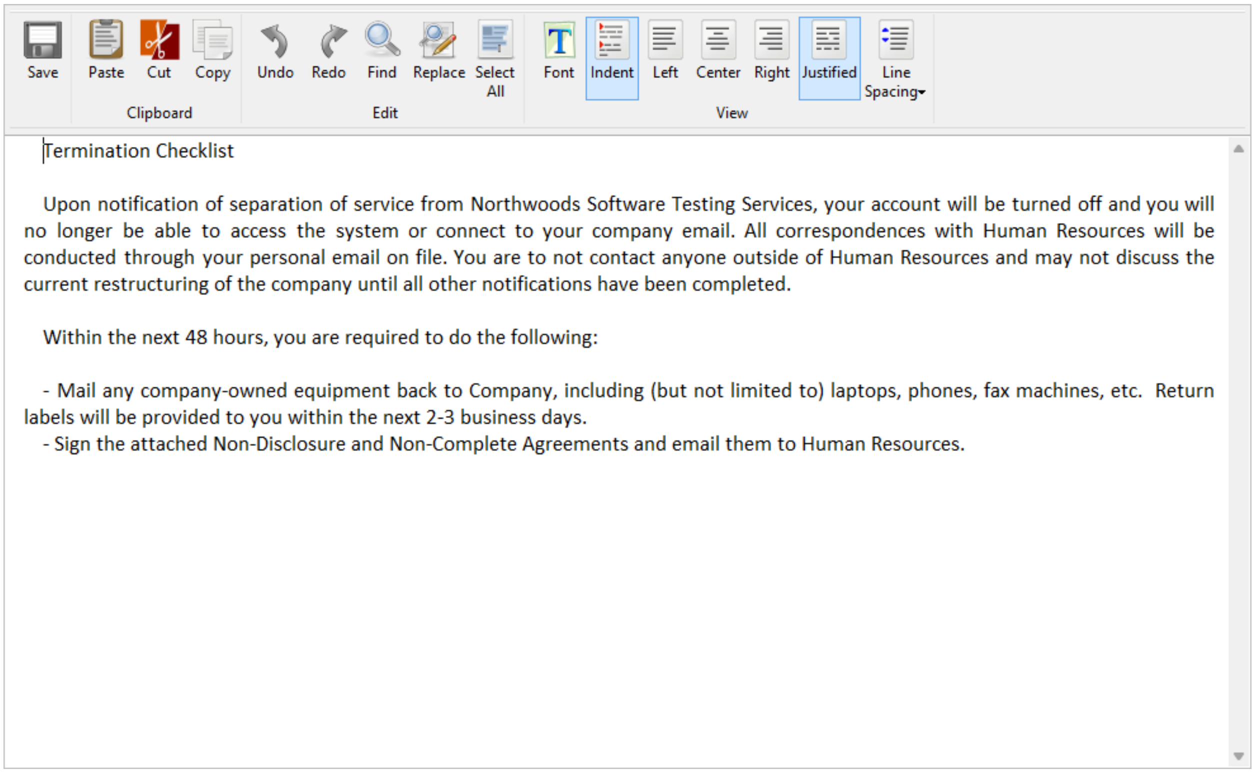
16.1 Editing within Readability Studio

In this example, we will write and edit a document from within *Readability Studio* itself. Click the New button on the Home tab to open the New Project wizard. Next, select the Manually enter text option.

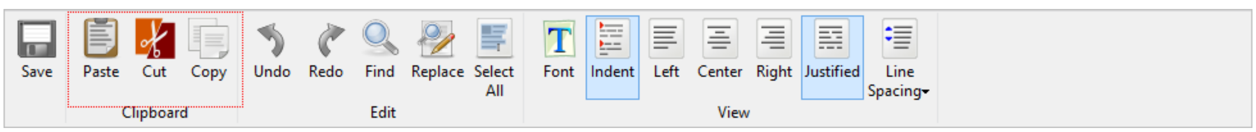


You can either type or paste text into this text box. In this example, we will leave it blank and edit it later. For brevity, we will use the defaults for document structure and test selections. (These can always be changed later from the Document and Readability tabs on the ribbon.) Click the Finish button to create the project.

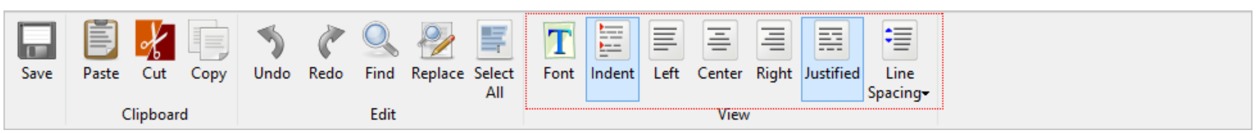
Because the project does not have any content yet, only empty scores and statistics will be shown. To add content, select the Document tab on the ribbon and click the Edit Document button. This will show the Edit Embedded Document dialog, where we will write our document.



Along the ribbon of this dialog are various editing features, such as find & replace, clipboard operations, and undo/redo.

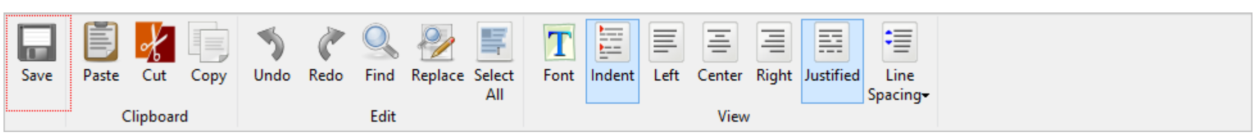


In the View section of the ribbon are options for changing the appearance of the editor.



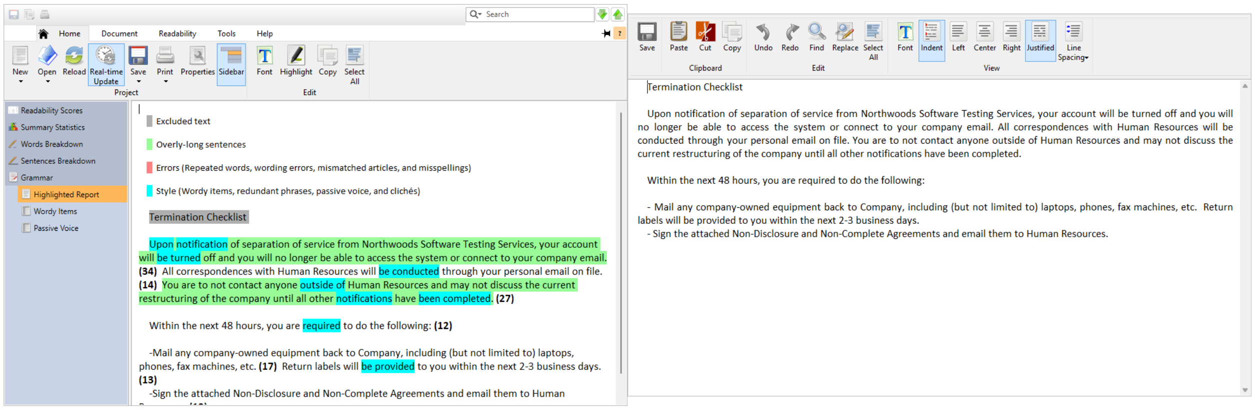
For example, we can toggle line spacing after line breaks by selecting Line Spacing > Display space after hard returns.

As you edit the document, you can click the Save button on the editor:



at any time to send your edits back to the project. The project will then analyze your changes and reload its results.

By placing the project window and editor side-by-side, you can review any issues while you edit.



As you edit, you can also change the parsing options or add tests. With the project window selected, select the Readability tab on the ribbon and click Secondary › Flesch Reading Ease. A Flesch test will now be available and can be viewed by selecting Readability Scores on the sidebar. Likewise, various parsing features (e.g., line-end handling, text exclusion) can be changed on the Document tab on the ribbon.

You can also search for issues in the editor by double-clicking on items in the project's lists. Click Words Breakdown on the sidebar and select any list in this section (e.g., 3+ Syllable List). Double-click any word in this list and it will become selected in the editor. (Clicking F3 on the keyboard will search for the next instance.)

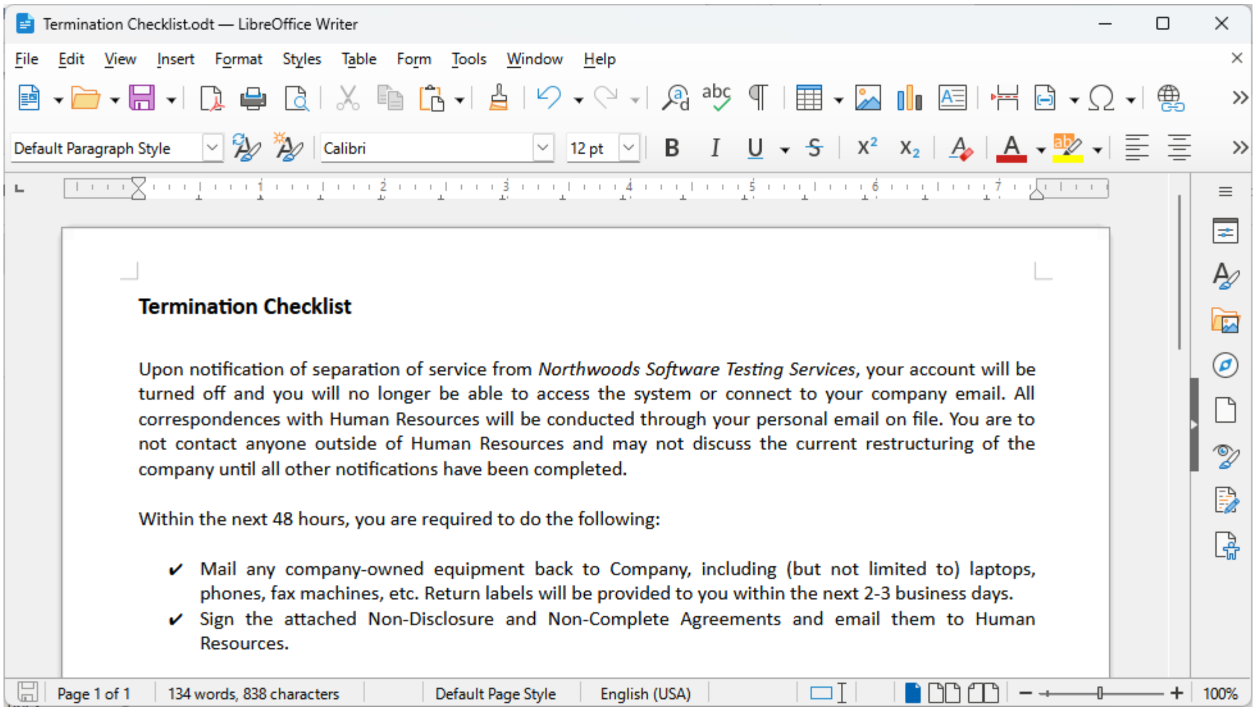
> **Note**
> If the editor is not open, then double-clicking a list item will search for it in the respective text window within the project.

Once you are finished editing and ready to export your document, copy it and paste it into another program. For example, click Select All and Copy, then paste into a program such as *Microsoft Word*, *Outlook*, *Pages*, *LibreOffice*, etc.

In the next example (see sec. 16.2), we will edit our document from another program and view the changes in *Readability Studio* in real-time.

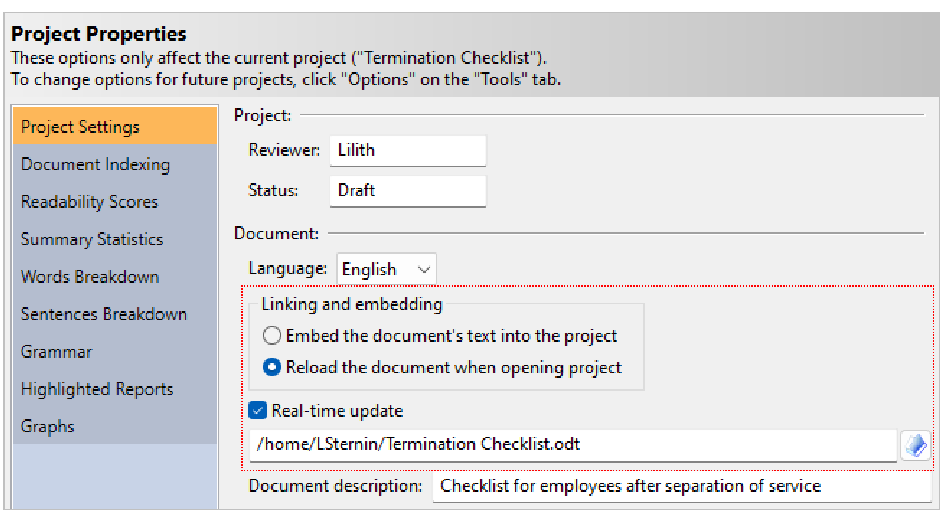
16.2 Editing from Other Programs

Continuing from the previous example (see sec. 16.1), we have copied the text from the editor into another program. (*LibreOffice* was used in this example, but any other editor will work as well.)



From *LibreOffice*, save the document and move the program over to the side of the screen. Next, go back to the project in *Readability Studio*. From the *Home* tab on the ribbon, click the Properties button. On the *Project Properties* dialog, select *Project Settings* on the sidebar. Then, under *Linking and embedding*, select *Reload the document when opening project*.

Because we created this project with entered text, it is not linked to a document. Under *Linking and embedding* is a text box where you enter a document path. Click the button next to this box and select the file that you just saved from *LibreOffice*. Also, check *Real-time update*, which will be discussed later.

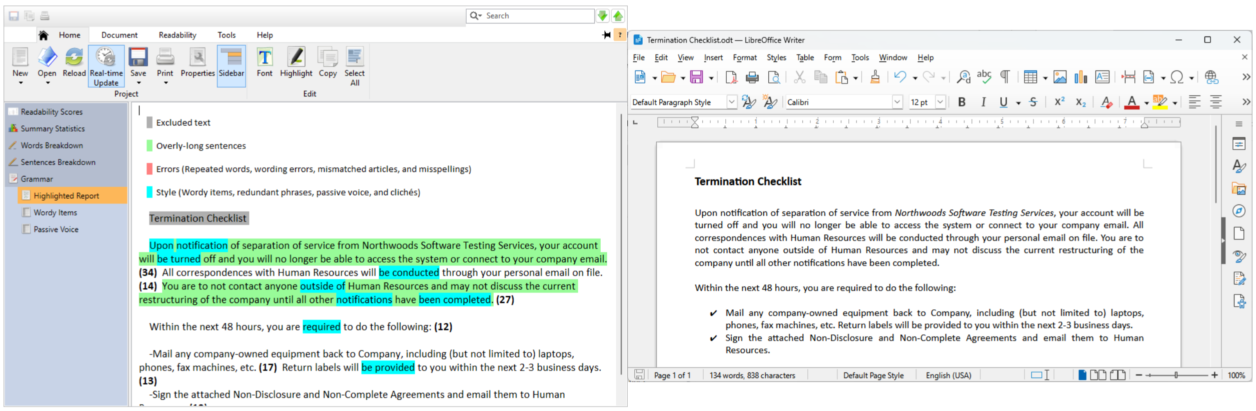


Click OK to accept these changes.

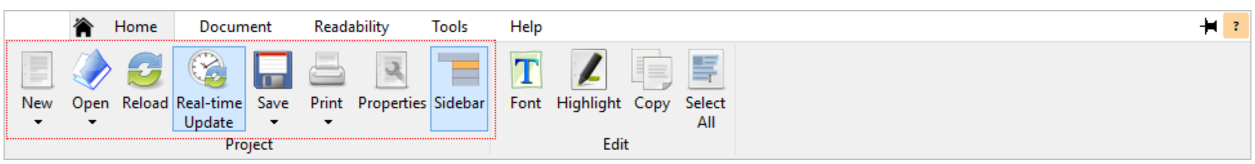
The project will link to the external document and analyze that, rather than the embedded text that we were editing before. Also, because we selected *Real-time update*, the project will reanalyze the document as we edit it in *LibreOffice*. The program will check every few seconds to see if the document's modified time has changed; if it has, then the project will reload.

> **Note**
> You will need to save from your external editor before *Readability Studio* can detect any changes.

By placing the two programs side-by-side, you can review any issues while you edit.



The real-time update option is also available on the ribbon and can be toggled at any time:



If this option is unpressed, then the project will only update when you explicitly click the Reload button.

Once you are finished editing, you can close the other program and continue reviewing the results in *Readability Studio*.

> **Tip**
> To reopen the document in your default editor, click Edit Document from the ribbon's Document tab.

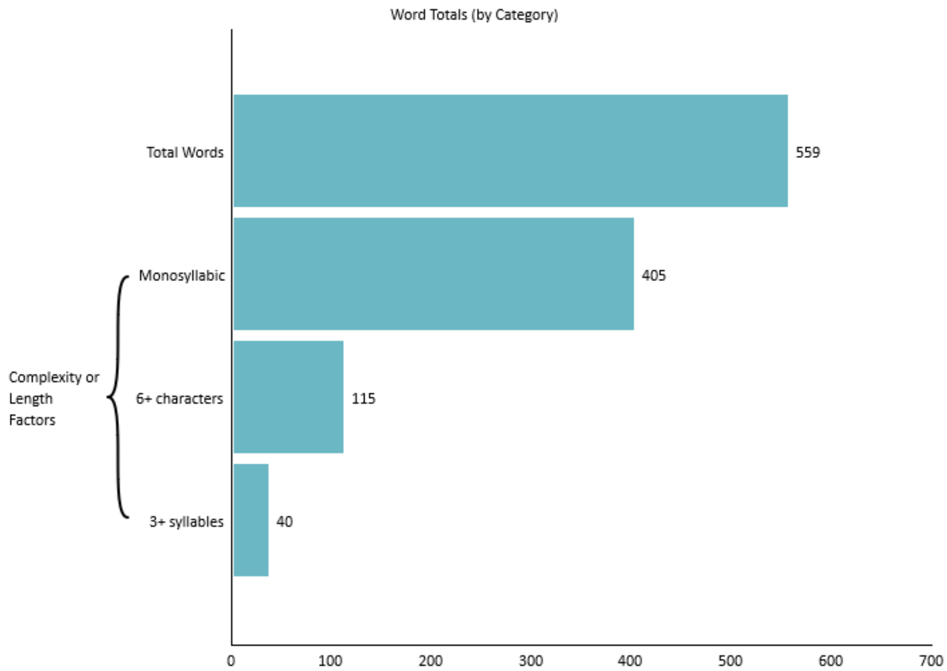
17. Customizing Graphs

17.1 Customizing Bars

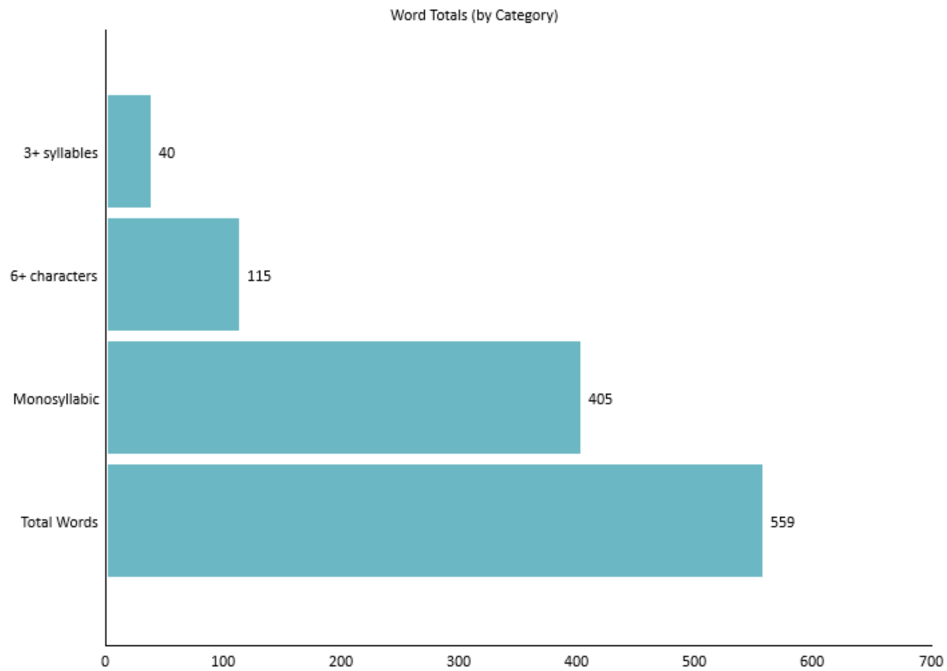
In this example, we will discuss the various ways to change the appearance of bars in a bar chart or histogram.

The first step is to open the example file “Chocolate Eclairs.” From the Help tab on the ribbon, click the Example Documents button and then select “Chocolate Eclairs” from the menu. (When prompted about how to open the document, select Create a new project.) When the New Project wizard appears, leave the defaults and click Finish.

Once the project has finished loading, select Words Breakdown on the sidebar, then click Word Counts beneath that. A bar chart showing a breakdown of word categories will be displayed:



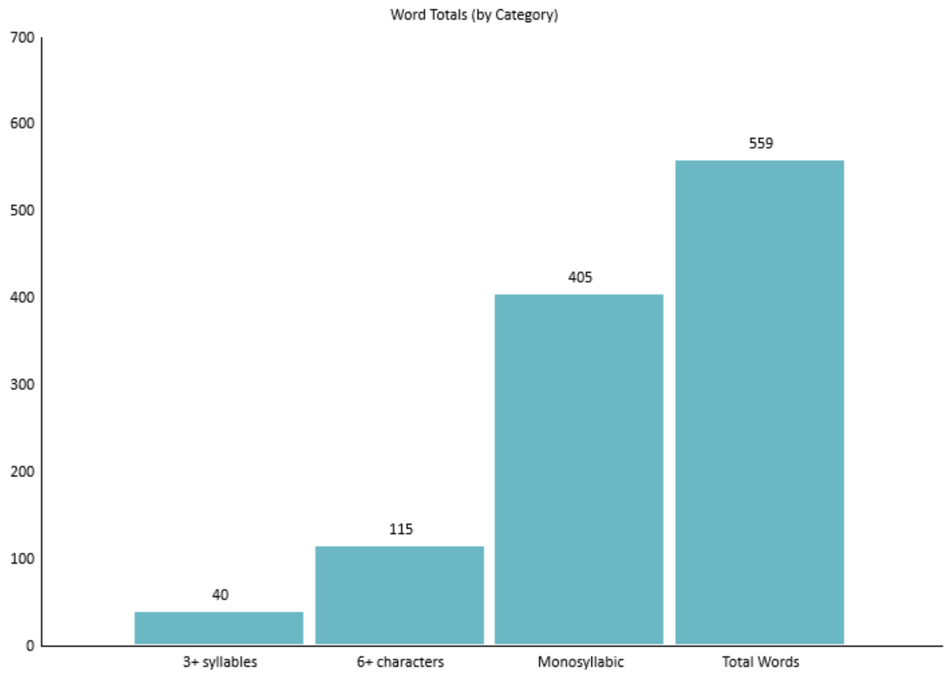
Now we will customize the appearance of this graph. One change that we can make is to reorder the columns in this bar chart. On the Home tab, click the Sort button under the Edit section. Select Sort Descending from the menu and note that the bars are now shown from largest to smallest from the origin:



Note
When changing the order of the bars, any axis brackets surrounding them will be removed.

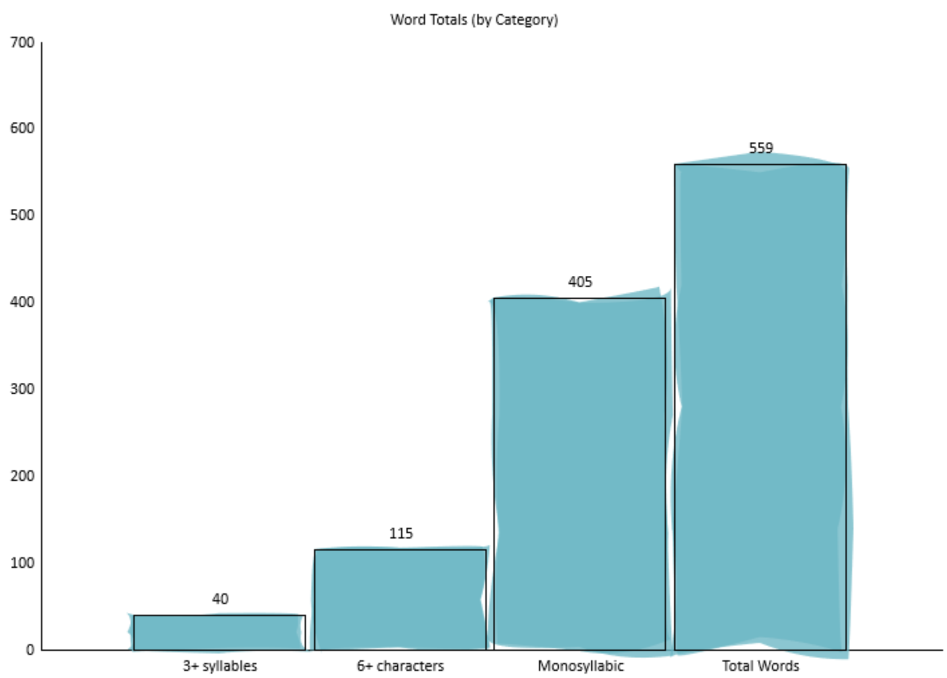
Next, we can change the direction of the bars. Click the Orientation button in the Edit section. Select Vertical from the menu.

Now the bars will be arranged vertically:



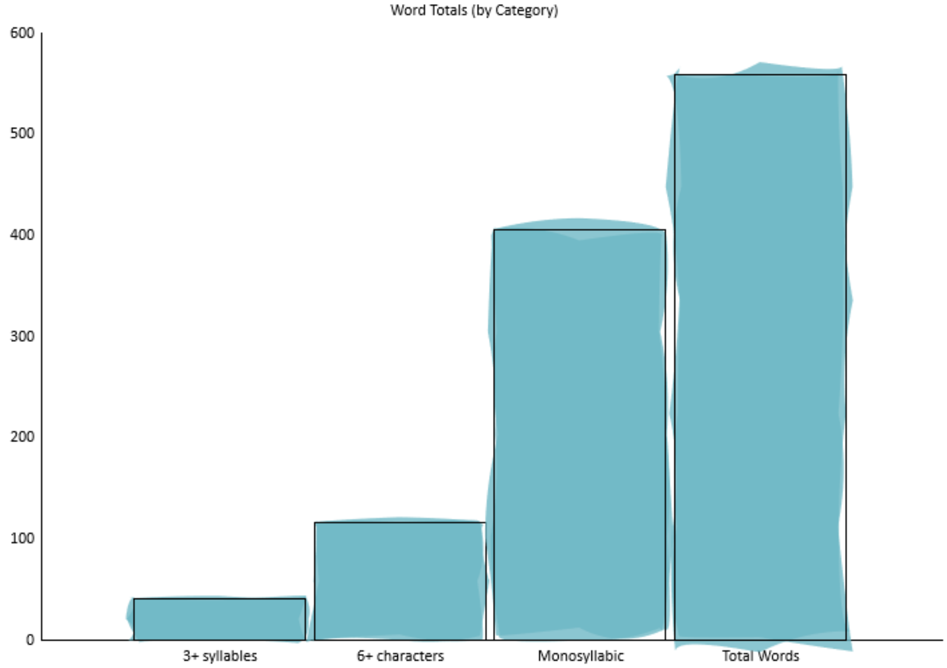
Finally, let's change how the bars are drawn. Click the Bar Style button in the Edit section and select Thick Watercolor.

The bars will now have a watercolor paint effect, rather than the default solid look:



Images and shapes can also be used to draw our bars, as demonstrated in a later example.

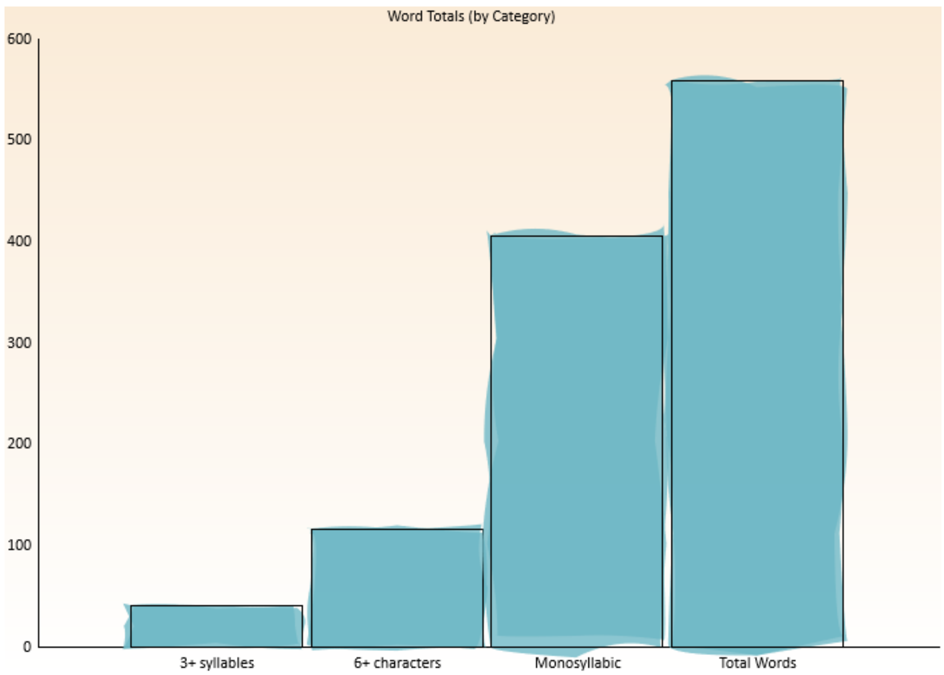
Finally, let's remove the labels to simplify the graph. Click the Labels button on the ribbon so that it is unpressed. Now the bars will be drawn without labels above them:



17.2 Customizing Backgrounds

Continuing from our previous example, let's now change the background of the graph. On the Home tab, click Background › Background Color › Color... . When the color selection dialog appears, select a new color (e.g., antique white) and click OK. After selecting a color, check the option Fade from the same menu.

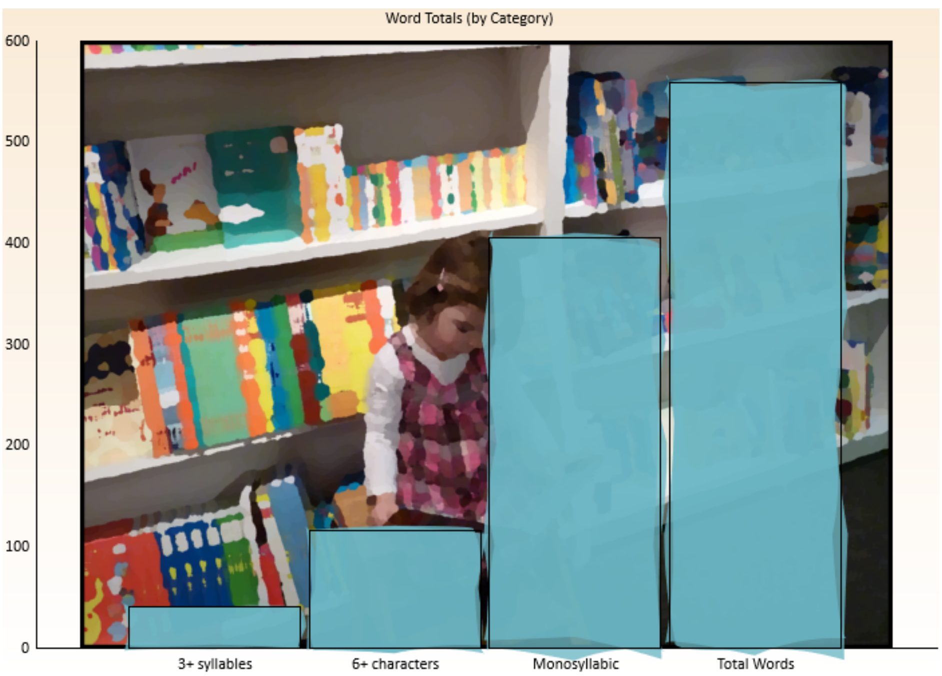
Now, the graph will have an antique white background with a gradient effect:



Next, click the Background button again and select Plot Image › Image... . When the image selection dialog appears, choose an image and click OK.

This image will now be shown as the background for your graphs:

To go along with the watercolor bars, we can apply an oil painting effect to the image. Click the Background button and select Plot Image > Effects > Oil Painting.



17.3 Adding Watermarks and Logos

Next, let's put a watermark across our graph. On the Home tab, click Print › Watermark... Enter the following into the Watermark dialog and click OK:

INTERNAL USE ONLY

Processed on @DATE@

Note how this label is now lightly written across the graph:

> Tip
> Entering `@DATE@` in the watermark field will show the current date when the graph is rendered. Refer to sec. 7.8 for other watermark options.

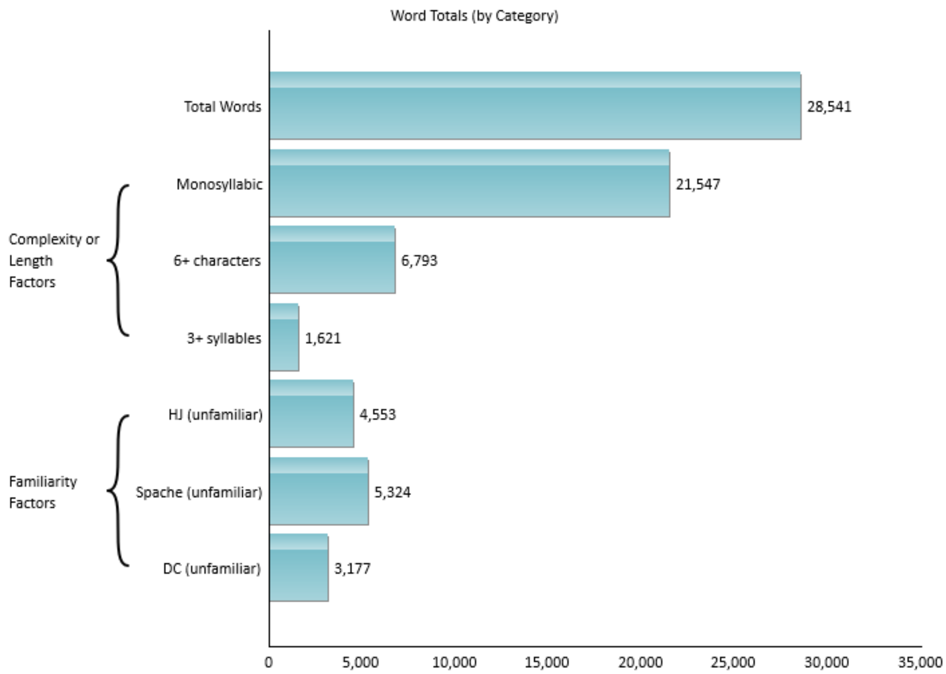
> Tip
> The watermark will be embedded in your project; this will ensure that it is portable when sharing with others.

Now let's put a company logo on our graph. Click the Logo button in the Edit section. Select an image and click OK.

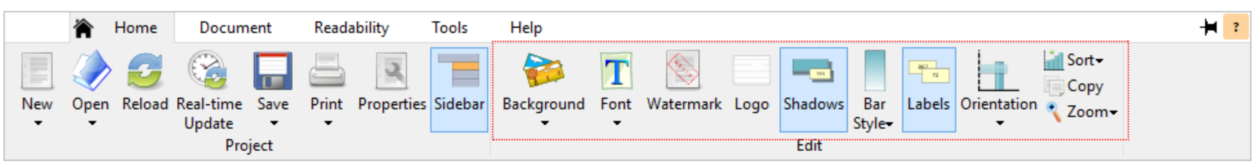
Now this image will be shown in the bottom right corner of the graph:

17.4 Using Stipples

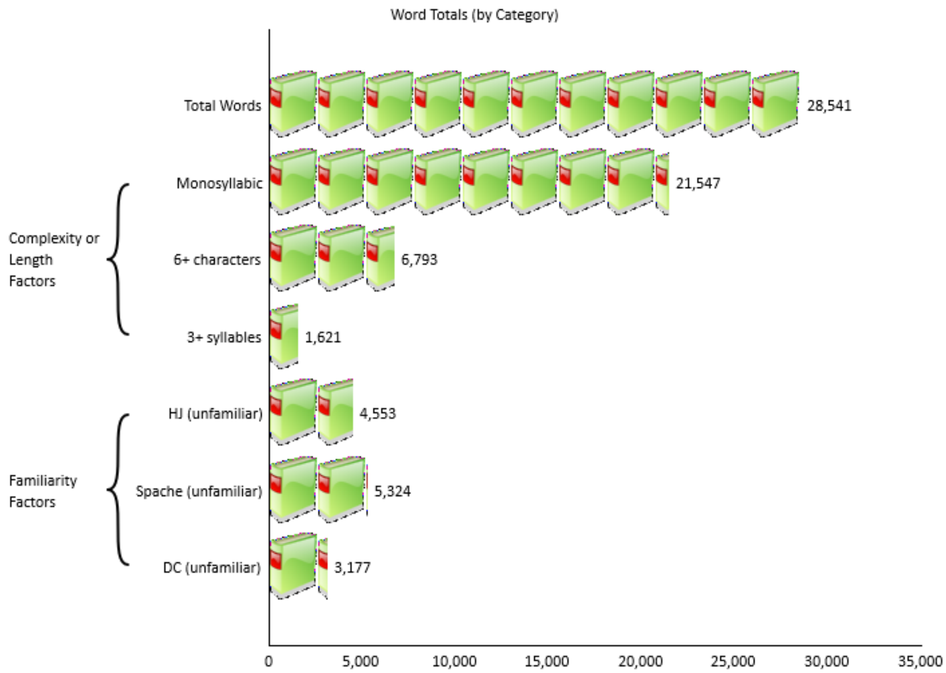
In this example, we will use stipples for our bar charts. Basically, a stipple enables us to use stacked images or shapes to draw our bars. First, create or open any standard project and select Words Breakdown on the sidebar. Next, select the Word Counts subitem and note how the bars will be filled rectangles:



On the Home tab of the ribbon, click the Bar Style button and select Select stipple image... from the menu:



After selecting an image, the bars will be drawn as a repeating pattern of this image:



> **Tip**
> Enabling the Shadows option on the ribbon will draw a shadow under each image.

Note that images of any size can be used. If the image is larger than the bars' width, then it will be scaled down to fit. Images that are smaller will remain their original size (rather than being upscaled) to preserve their quality.

Along with images, a set of predefined shapes are also available to choose from. To select one, click the Bar Style button again and select Select stipple shape... from the menu. Select a shape from the list (e.g., Car or Newspaper), and this shape will then be used for the bar pattern:

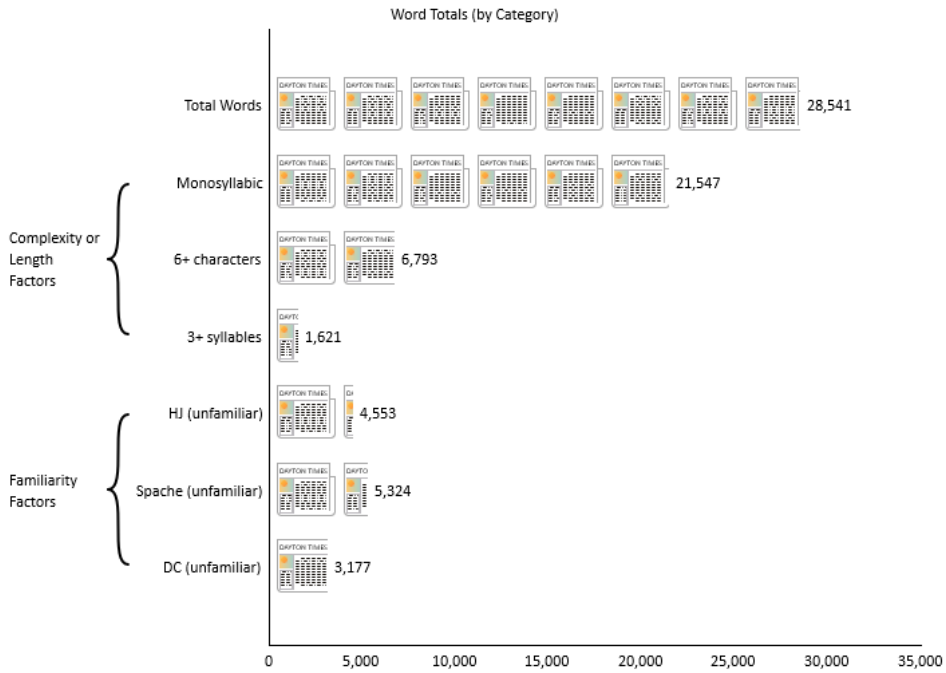


Figure 17.1: Newspapers used to draw bars

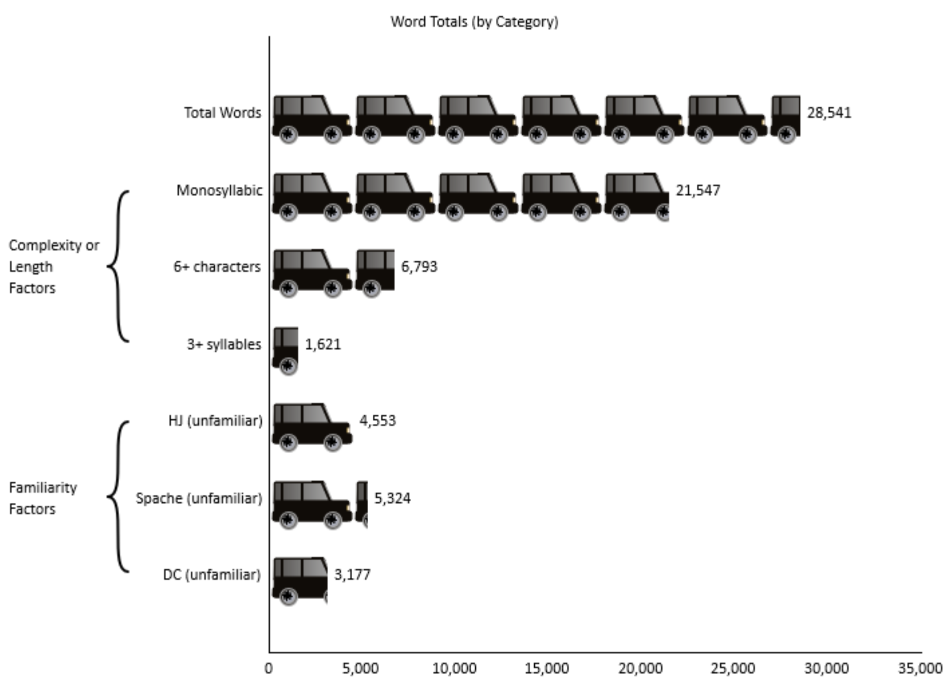


Figure 17.2: Cars (specifically, a 2006 Scion xB) used to draw bars

These features can also be applied to box plots and histograms. With any of these graph types selected, perform the same operation from the ribbon to draw with a stipple.

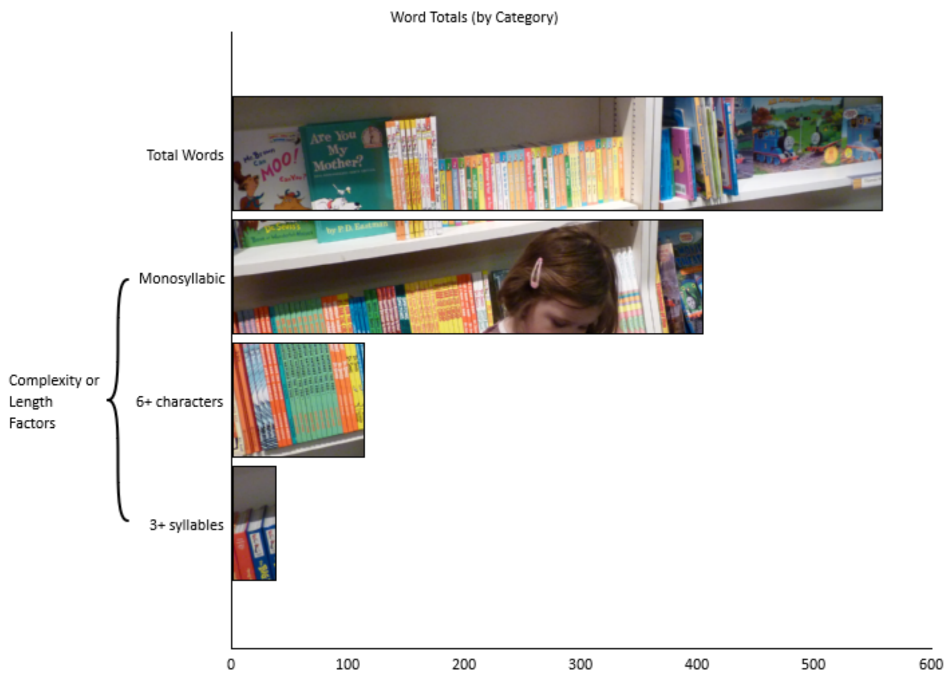
17.5 Selective Colorization

In this example, we will create a bar chart with a selective colorization effect. This technique is where the background is black & white and various focal points (i.e., the bars) are in color.

From the Help tab on the ribbon, click the Example Documents button and then select "Chocolate Eclairs" from the menu. (When prompted about how to open the document, select Create a new project.) When the New Project wizard appears, leave the defaults and click Finish.

Once the project has finished loading, select Words Breakdown on the sidebar, then click Word Counts beneath that. A bar chart showing a breakdown of word categories will be displayed. First, let's remove the labels to simplify the graph. On the Home tab of the ribbon, click the Labels button so that is it unpressed.

Next, we will use an image to be displayed across all the bars. On the Home tab, click Bar Style > Common Image. When the image selection dialog appears, select an image and click OK. Now, the bars will be filled with the image:



Next, click the Background button and select Plot Image > Image.... When the image selection dialog appears, choose the same image and click OK. We will also need to adjust how this image fits into the plot area. Our goal is to have the background image line up with the same image drawn across the bars. Click the Background button and select Plot Image > Fit > Crop & Center.

This image will now be shown as the background for the graph and bars:

As it is now, you can only see the bars' outlines. To make the bars pop, we will change the background to be black & white and faded. First, click the Background button and select Plot Image › Opacity... . On the Set Opacity dialog, select 100 and click OK. Next, click the Background button again and select Plot Image › Effect › Grayscale.

Now, we should see a selective colorization effect on our bar chart:

17.6 A Retro Appearance

In this example, we will create a basic-looking graph, with a more "retro" appearance. We will do this by removing all adornments and effects, and by applying washed-out colors and monospace fonts.

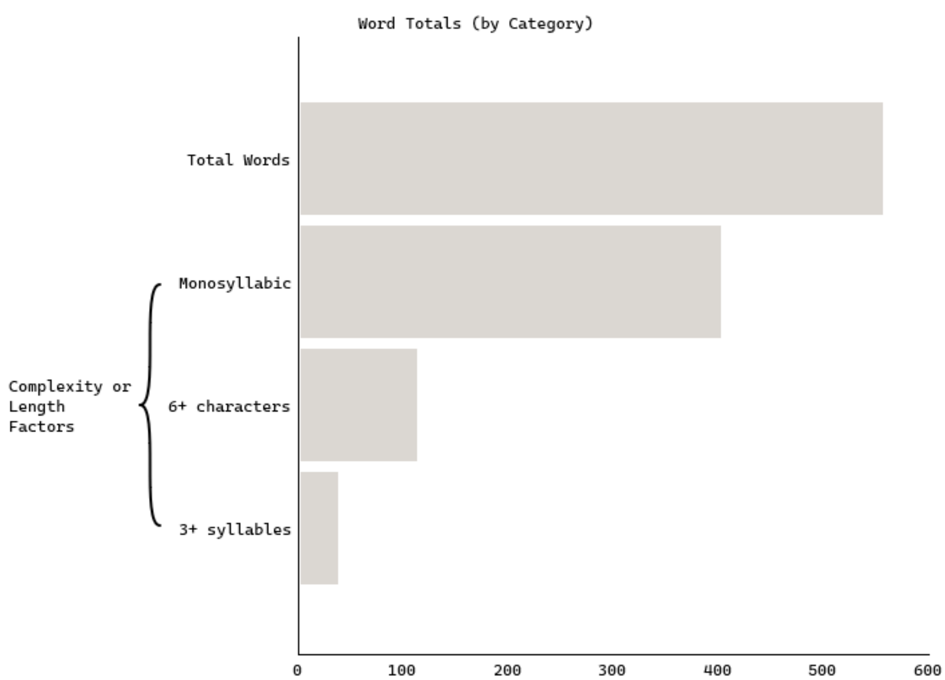
From the Help tab on the ribbon, click the Example Documents button and then select "Chocolate Eclairs" from the menu. (When prompted about how to open the document, select Create a new project.) When the New Project wizard appears, leave the defaults and click Finish.

Once the project has finished loading, select Words Breakdown on the sidebar, then click Word Counts beneath that. First, let's remove the labels to simplify the graph. On the Home tab of the ribbon, click the Labels button so that is it unpressed.

Next, we will remove any visual effect from the bars. On the Home tab, click Bar Style › Solid. Then, click Bar Style › Color... and select a light gray.

Next, click the Font button and select X Axis.... When the font selection dialog appears, choose a monospace font (e.g., Consolas or Courier New) and click OK. Do the same for Font › Y Axis... and Font › Top Titles.... Doing this will give a typewriter look to the graph's text.

Once we are done, our bar chart will have a simpler, "retro" look:



17.7 A Dark-mode Appearance

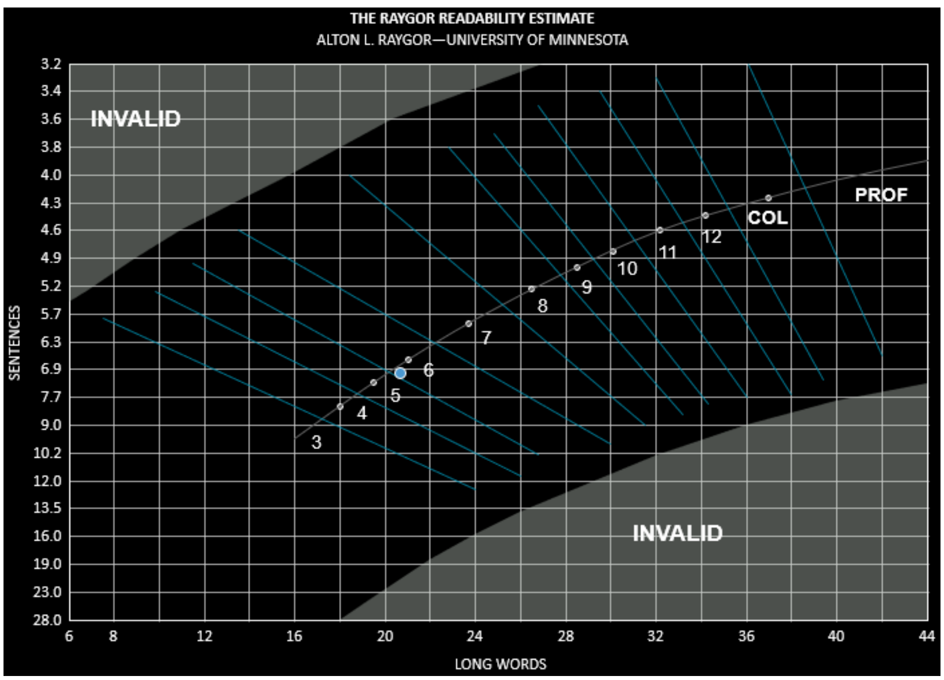
In this example, we will create a graph with a high-contrast, “dark mode” appearance.

From the Help tab on the ribbon, click the Example Documents button and then select “Chocolate Eclairs” from the menu. (When prompted about how to open the document, select Create a new project.) When the New Project wizard appears, leave the defaults and click Finish.

Once the project has finished loading, select the Readability tab on the ribbon. Click Secondary > Raygor Estimate to add a Raygor test to the project. Next, click Readability Scores on the sidebar, then click Raygor Estimate beneath that.

By default, graph backgrounds are white and most labels are black. If you change the background to black, then the program will automatically change the labels to white to provide contrast. To do this, select the Home tab on the ribbon and click Background > Background color > Color. From the color picker, select pure black and click OK.

Now our graphs will have a dark, high-contrast appearance:



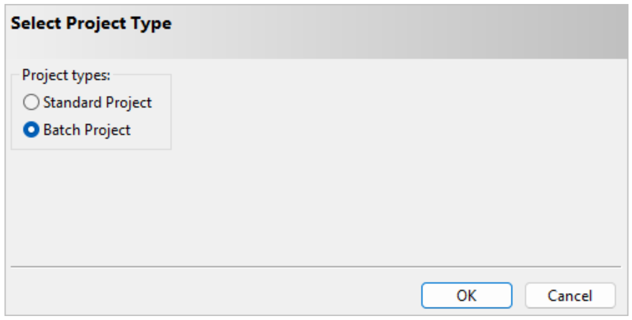
> **Note**
> Printing with dark backgrounds is not recommended, as it will consume a lot of ink.

18. Batch Projects

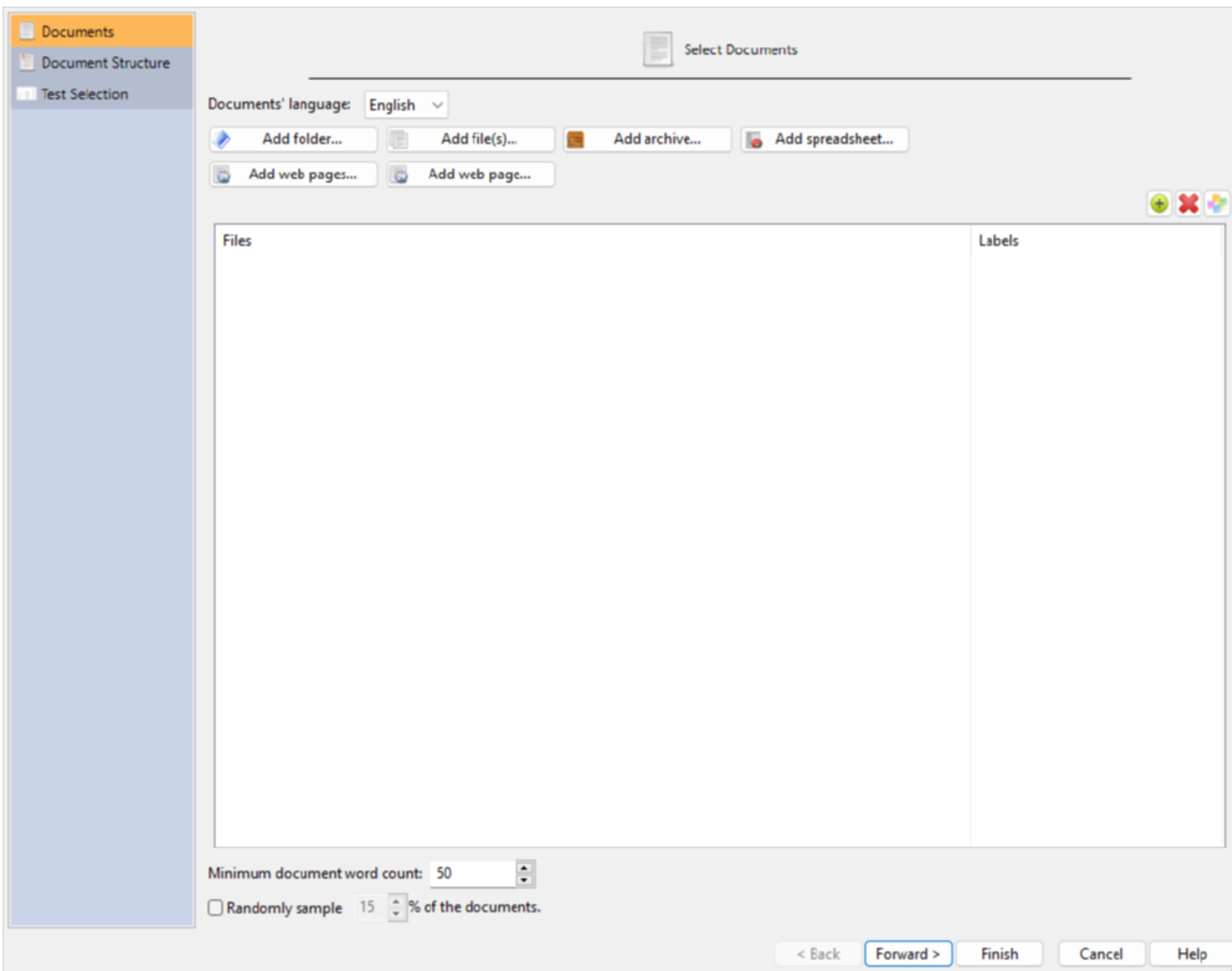
Within this chapter, we will step through examples showing how to analyze batches of documents. This will involve gathering documents from folders, grouping them, and then processing them all at once. We will also discuss the batches' aggregated results and how to interpret them.

18.1 Reviewing a Collection of Documents

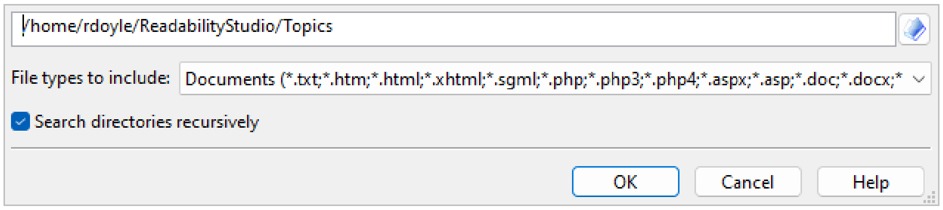
To determine the readability of a document collection, you will need to create a new batch project. First, click the New button on the Home tab to open the Select Project Type dialog, as shown below:



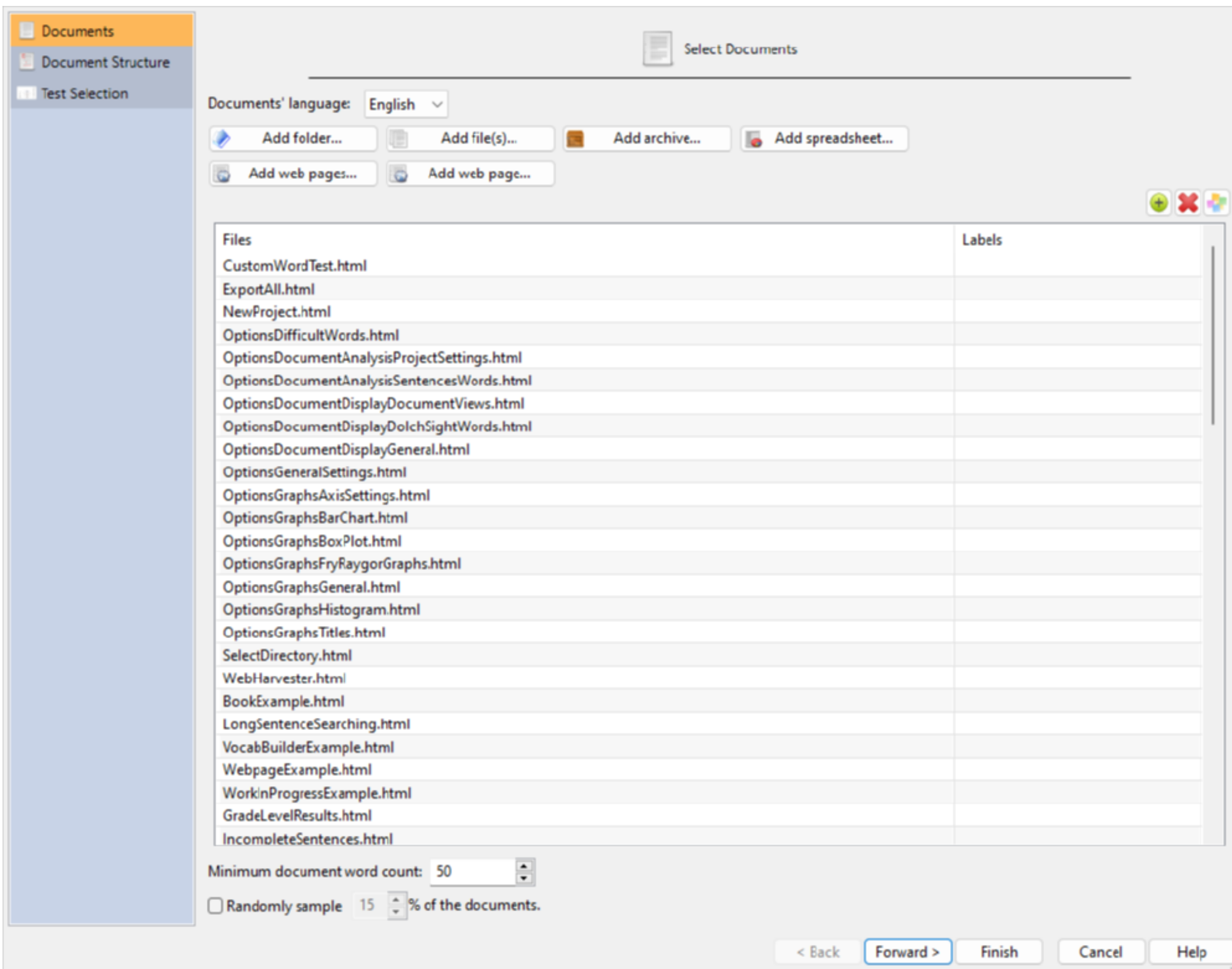
Select Batch Project and click the OK button to proceed to the New Project wizard dialog:



Select English as the document language. Next, you can add files from a folder, website, spreadsheet, or archive to the project. We will be reviewing files from a folder, so click the Add folder... button. You will be presented with the Select Directory dialog:

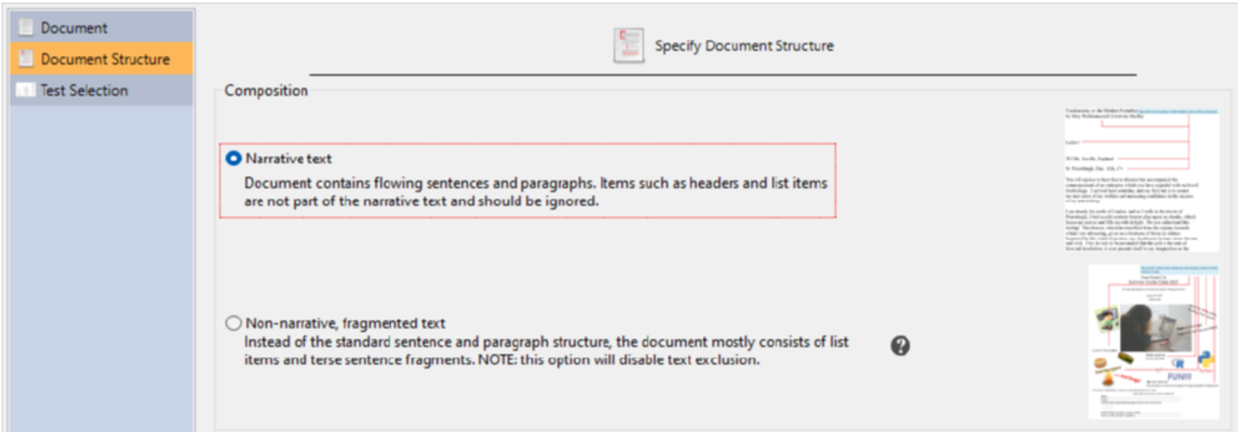


Enter the path to your document folder in the top field. We will want to search for files inside of this folder and its subfolders, so leave Search directories recursively checked. Click the OK button to accept, and you will then be prompted about how to label the documents. Leave the defaults and click OK and the program will begin its file search. After the program is finished searching for documents, you will return to the New Project Wizard. Note how all the files that were found are now included in the file list:

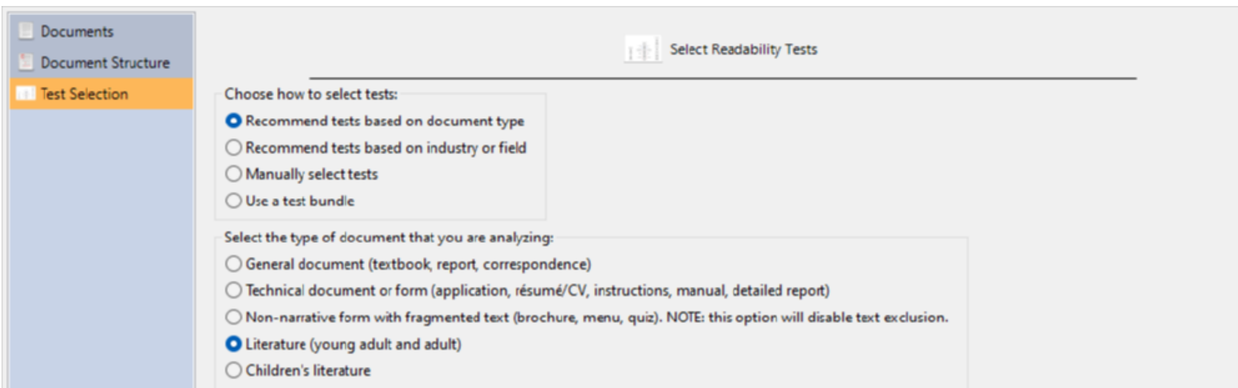


To add more files from other folders, repeat the previous step.

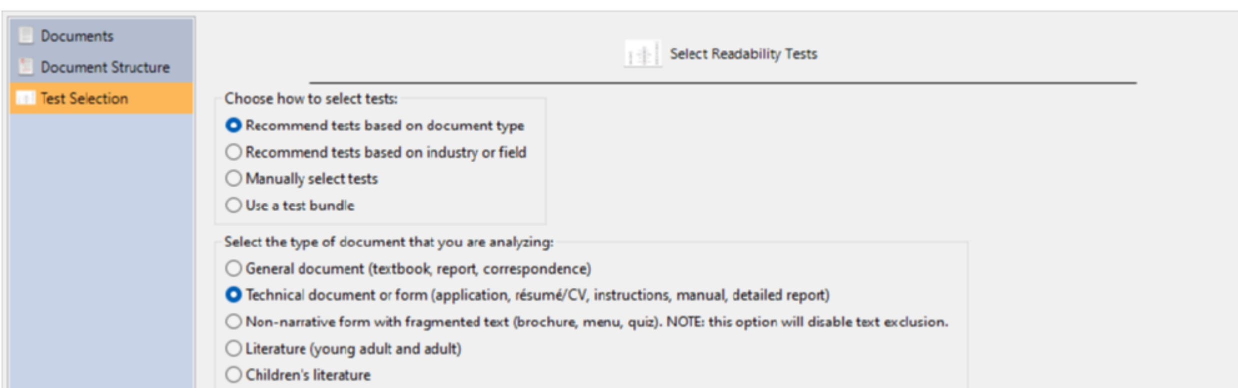
We will analyze all the documents in this collection, so just ignore the % of documents to randomly sample option. Click the Forward > button to continue to the Document Structure page, as shown below:



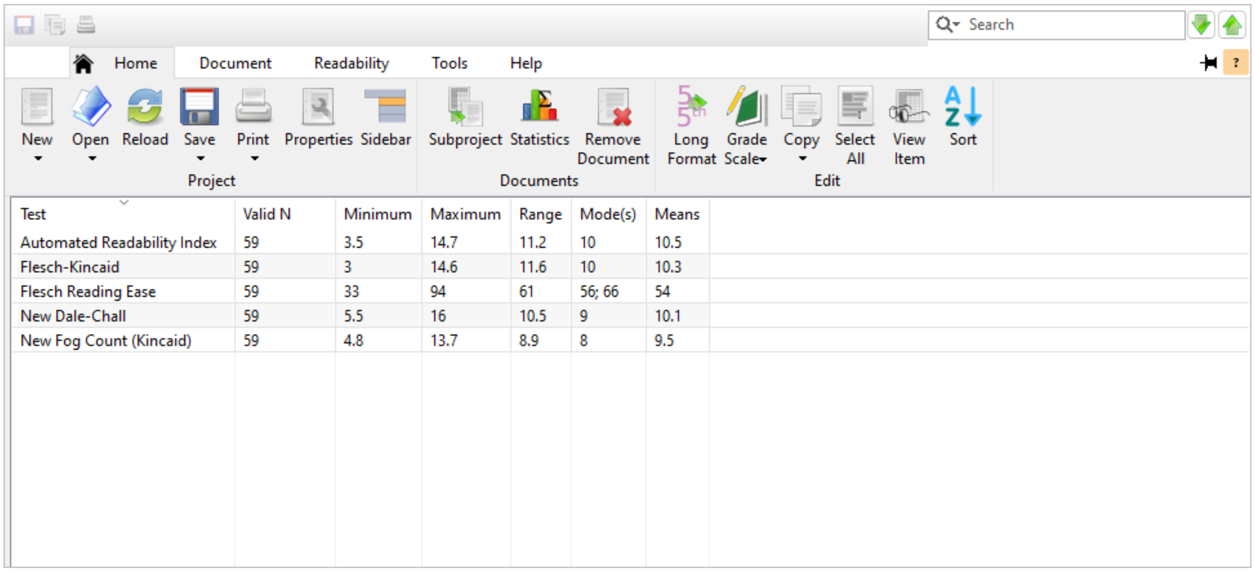
In this dialog, you will specify the format that best describes your documents. We are analyzing standard, narrative documents, so select Narrative text. After selecting your document structure, click the Forward > button to continue to the Test Selection page, as shown below:



On this page, you will select how you want *Readability Studio* to choose the readability tests to perform. Select the Recommend tests based on document type:



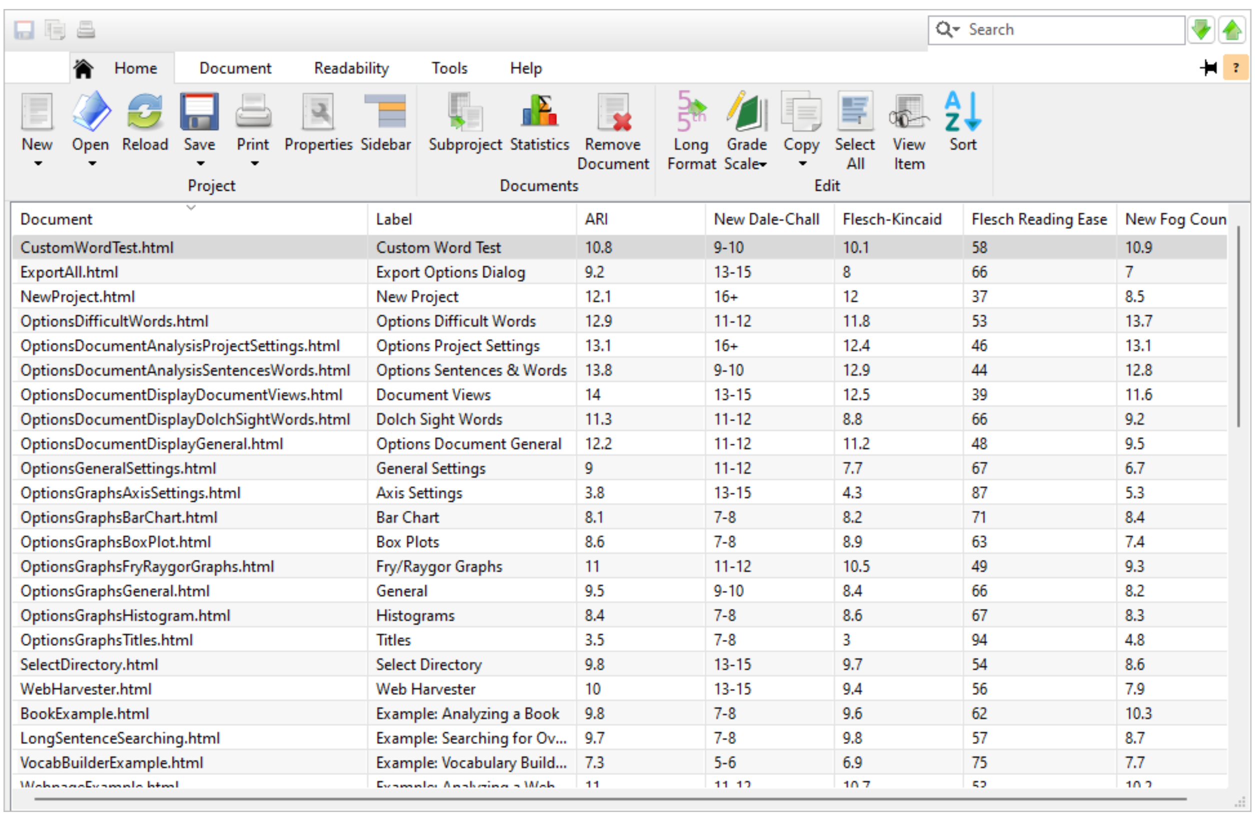
We will be analyzing technical documents in this example, so select Technical document or form. Click the Finish button and the project will be created and will appear like this:



Test	Valid N	Minimum	Maximum	Range	Mode(s)	Means
Automated Readability Index	59	3.5	14.7	11.2	10	10.5
Flesch-Kincaid	59	3	14.6	11.6	10	10.3
Flesch Reading Ease	59	33	94	61	56; 66	54
New Dale-Chall	59	5.5	16	10.5	9	10.1
New Fog Count (Kincaid)	59	4.8	13.7	8.9	8	9.5

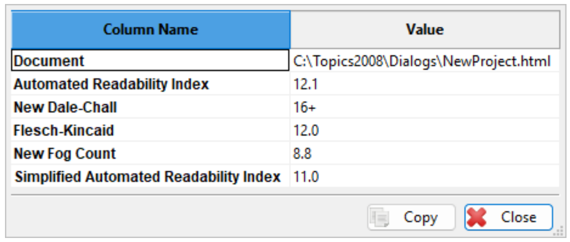
The first items presented to us are the aggregated statistics for the test scores. This will give us a general overview of the collection's readability level. By looking at the Means column, we can see that the average reading levels of our documents are in the high-school range. Also, the mean Flesch score is 54, which is somewhat difficult. The next column to review is the Maximum column. Here we can see that some of the higher test scores (e.g., 14.6) are into the collegiate level. From this, we can summarize that this collection is a little too difficult to read and that some documents need revising.

To view a breakdown of all the documents and their respective test scores, click on the Raw Scores subitem. Here we can sort any of the test columns to see which documents scored the highest.



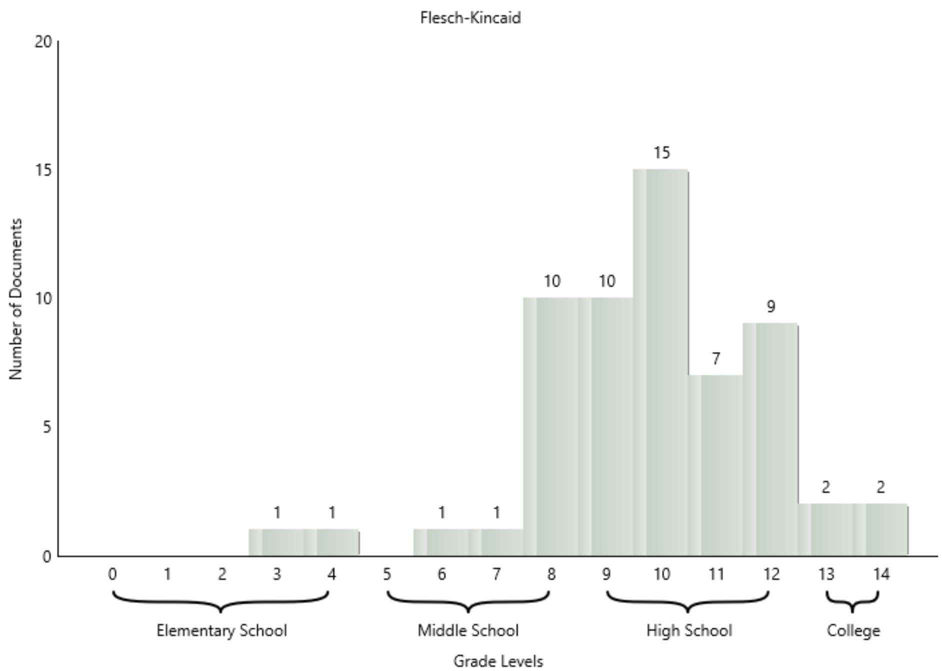
Document	Label	ARI	New Dale-Chall	Flesch-Kincaid	Flesch Reading Ease	New Fog Coun
CustomWordTest.html	Custom Word Test	10.8	9-10	10.1	58	10.9
ExportAll.html	Export Options Dialog	9.2	13-15	8	66	7
NewProject.html	New Project	12.1	16+	12	37	8.5
OptionsDifficultWords.html	Options Difficult Words	12.9	11-12	11.8	53	13.7
OptionsDocumentAnalysisProjectSettings.html	Options Project Settings	13.1	16+	12.4	46	13.1
OptionsDocumentAnalysisSentencesWords.html	Options Sentences & Words	13.8	9-10	12.9	44	12.8
OptionsDocumentDisplayDocumentViews.html	Document Views	14	13-15	12.5	39	11.6
OptionsDocumentDisplayDolchSightWords.html	Dolch Sight Words	11.3	11-12	8.8	66	9.2
OptionsDocumentDisplayGeneral.html	Options Document General	12.2	11-12	11.2	48	9.5
OptionsGeneralSettings.html	General Settings	9	11-12	7.7	67	6.7
OptionsGraphsAxisSettings.html	Axis Settings	3.8	13-15	4.3	87	5.3
OptionsGraphsBarChart.html	Bar Chart	8.1	7-8	8.2	71	8.4
OptionsGraphsBoxPlot.html	Box Plots	8.6	7-8	8.9	63	7.4
OptionsGraphsFryRaygorGraphs.html	Fry/Raygor Graphs	11	11-12	10.5	49	9.3
OptionsGraphsGeneral.html	General	9.5	9-10	8.4	66	8.2
OptionsGraphsHistogram.html	Histograms	8.4	7-8	8.6	67	8.3
OptionsGraphsTitles.html	Titles	3.5	7-8	3	94	4.8
SelectDirectory.html	Select Directory	9.8	13-15	9.7	54	8.6
WebHarvester.html	Web Harvester	10	13-15	9.4	56	7.9
BookExample.html	Example: Analyzing a Book	9.8	7-8	9.6	62	10.3
LongSentenceSearching.html	Example: Searching for Ov...	9.7	7-8	9.8	57	8.7
VocabBuilderExample.html	Example: Vocabulary Build...	7.3	5-6	6.9	75	7.7
WebpageExample.html	Example: Analyzing a Web	11	11-12	10.7	53	10.2

To view any document's scores in a vertical layout, double-click on its row in this list to display the View Item dialog:



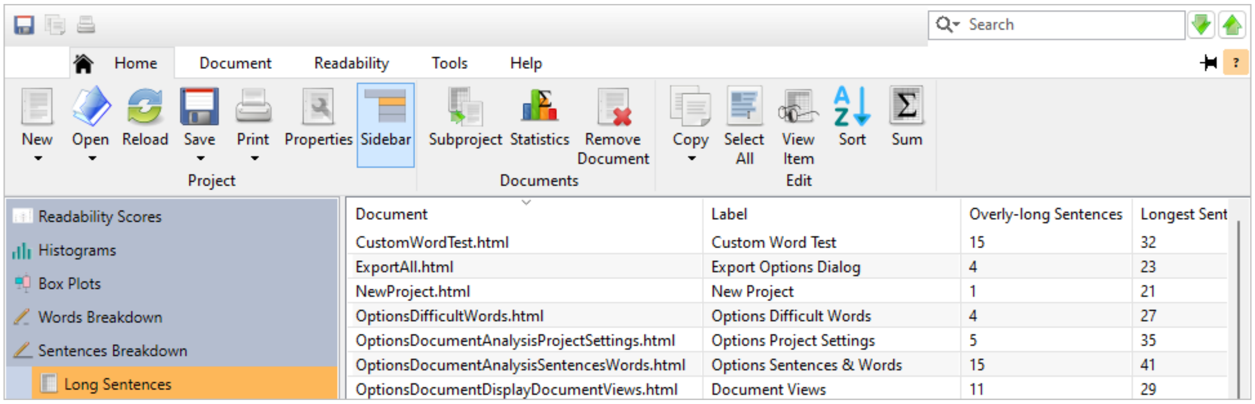
Column Name	Value
Document	C:\Topics2008\Dialogs\NewProject.html
Automated Readability Index	12.1
New Dale-Chall	16+
Flesch-Kincaid	12.0
New Fog Count	8.8
Simplified Automated Readability Index	11.0

The distribution of these raw scores can also be graphically viewed. Click on the Score Box Plots icon on the project sidebar to view the box plots, and click on the Histograms icon to view the histograms. Box plots show the spread of each test's scores, whereas histograms show a test's scores divided into groups. For example, below is a histogram showing the Flesch-Kincaid scores:



Judging from this graph, we have quite a few difficult documents in this batch. There are 11 files in the 12–13 range, which is not a comfortable reading level. Refer to sec. 5.2 and sec. 5.3 for further information.

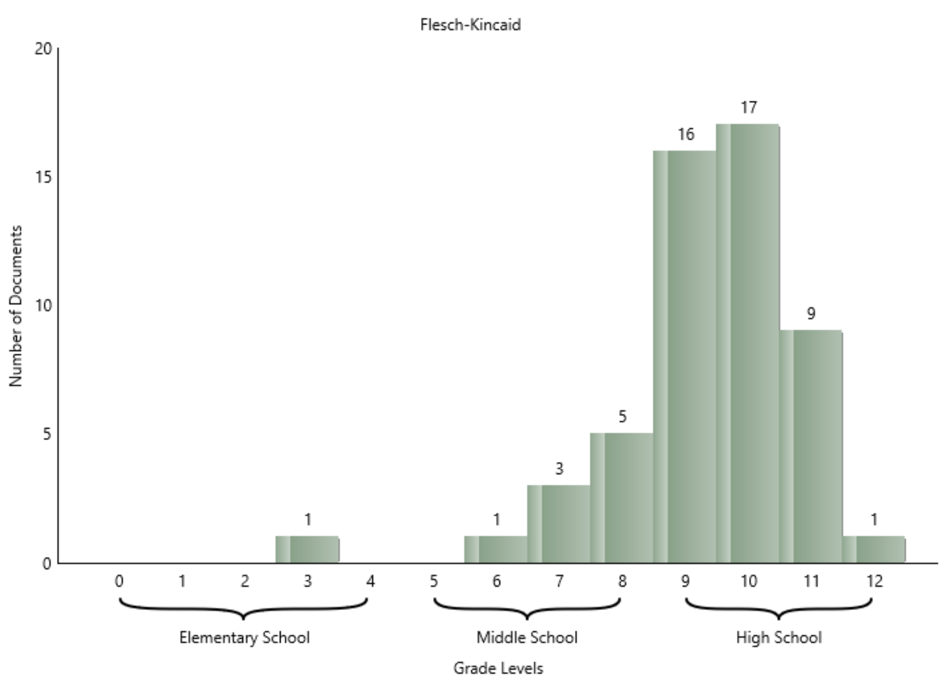
Now that we know which documents are scoring highly, we can begin revising them. The next step is to review any grammar issues that may be contributing to overly-long sentences. Click on the Grammar icon to display any grammar issues detected in the documents. First select the Long Sentences subitem, as shown below:



Here we can see which documents have the most overly-long sentences, as well as where the longest sentences are. Once we have found a document of interest, we may want to review it in greater detail. Select a document in this list and then click the Subproject button on the Home tab. This will create a standard project, where we can review the long sentences individually. We can also view the document with these sentences highlighted in their original context. Note that standard projects can be created this way from any list which includes the document names.

Other tabs in the Grammar section are also available to help improve your documents. If any of these grammatical issues are encountered, then lists for wordy items, clichés, repeated words, lowercased sentences, and conjunction-starting sentences will be included. For the purpose of shortening sentences, the Repeated Words and Wordy Items pages should be reviewed.

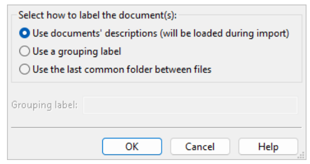
Once some of the documents have been revised, you can reload the batch project with the F5 button on your keyboard. As we can see in this updated Flesch-Kincaid histogram, there is already some improvement in this batch. There are no longer any documents scoring above the 12^{th} grade, and only 10 documents remain in the 10–12 grade range.



18.2 Adding Descriptive Labels to Documents

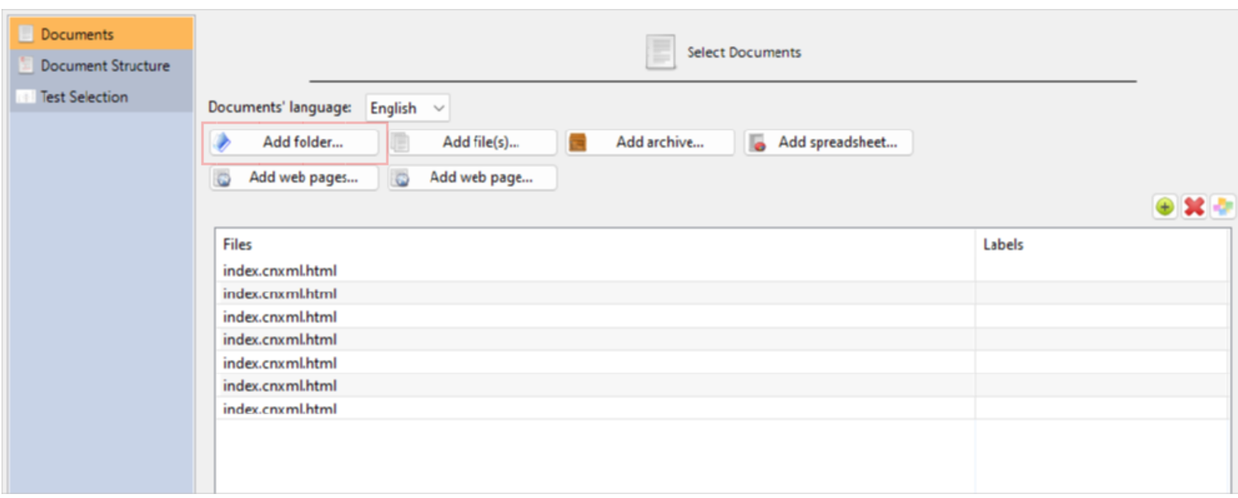
Sometimes documents that we receive do not have descriptive names that are easily recognizable. An example of this is downloaded webpages that may have names such as “index.html” or “2A67B2H.html.” In this example, we will show how to include descriptive labels next to our documents in a batch so that we can distinguish between them.

On the Home tab of the ribbon, click the New button. When prompted for which type of project to create, select Batch Project. Once the New Project Wizard appears, click the Add folder... button. On the Select Directory dialog, enter the folder path containing your documents and then click OK. Now we will be shown the Select Labeling dialog.



On this dialog, we can either specify a single label for all documents or use the documents’ metadata to create unique descriptions for each one. For this example, we are wanting to include unique, descriptive labels for each document, so leave Use documents’ descriptions selected and click OK.

Now, the program will load the files from the folder that we had selected. For example, below is a list of files where each one has the same name:



Because we had selected Use documents’ descriptions for our labeling, then the titles from the documents will be extracted when the analysis begins. From there, these titles will be shown as descriptive labels next to each document in the results. Click Finish to begin the analysis, and once the project appears, select Raw Scores under Readability Scores:

Document	Label	SMOG
index.cnxml.html	Introduction	7.6
index.cnxml.html	Definitions of Statistics, Probabilit...	7.6
index.cnxml.html	Data, Sampling, and Variation in ...	8.1
index.cnxml.html	Frequency, Frequency Tables, and...	8.1
index.cnxml.html	Experimental Design and Ethics	8.1
index.cnxml.html	Data Collection Experiment	7.6
index.cnxml.html	Sampling Experiment	8.1

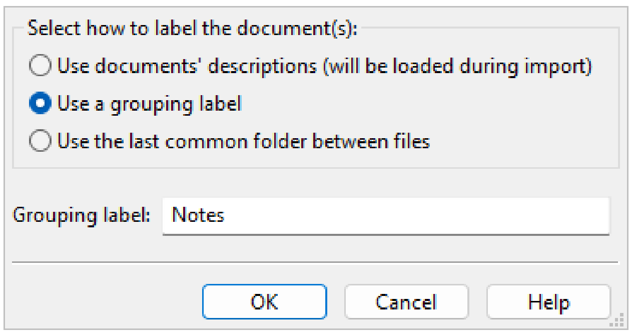
Note how there is a Label column next to the Document column. Despite the documents all having the same name, we can use this column to help us differentiate between them.

18.3 Grouping Documents in a Batch

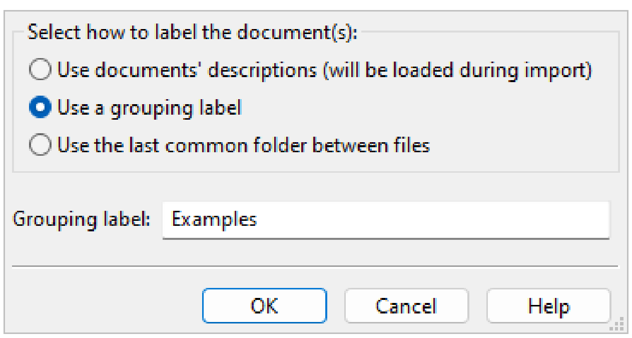
Sometimes when creating a batch, we may want to group the documents within it. For example, clusters of documents within the batch may be from different websites, authors, or studies. By adding group labels to these clusters, we can compare them in the results. In this example, we will show how to add group labels to a batch of documents and how to view them in the results.

On the Home tab of the ribbon, click the New button. When prompted for which type of project to create, select Batch Project. When the New Project Wizard appears, click the Add folder... button. On the Select Directory dialog, enter the folder path containing your documents and then click OK. Now we will be shown the Select Labeling dialog.

On this dialog, we can either specify a single label for all documents or use the documents' metadata to create unique descriptions for each one. For this example, we are wanting to group this folders' documents together, so select Use a grouping label, enter a label (e.g., *Notes*) and click OK.



Repeat this process with another folder, but enter a different grouping label. For example, we could select a folder of help examples and apply the label *Examples* to it:



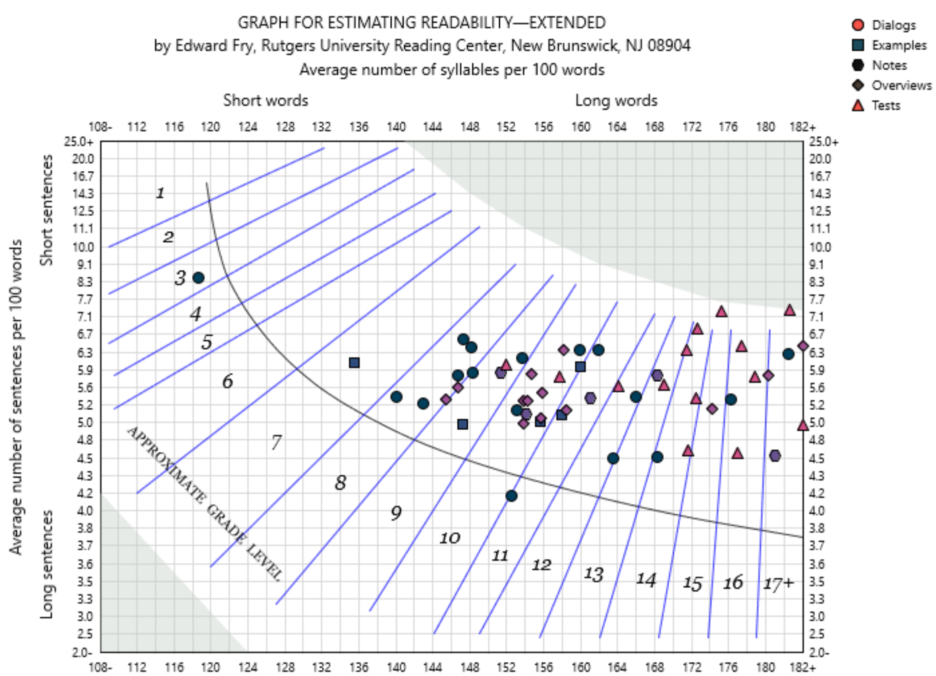
> Note
> You can apply a group label to all selected documents by clicking the group button above the list.

After adding your documents, click Finish. When the project appears, select Raw Scores under Readability Scores. Here we can see a list of documents, along with their group labels next to them:

Document	Label	Flesch Reading Ease	Fry
IncompleteSentences.html	Notes	33	17
SMOG.html	Tests	34	17
Features.html	Overviews	36	17
NewProject.html	Dialogs	37	17
Spache.html	Tests	37	16
IntroductoryOverview.html	Overviews	38	17
OptionsDocumentDisplayDocumentViews.html	Dialogs	39	16
OptionsAndSettingsOverview.html	Overviews	40	15
ColemanLiau.html	Tests	41	Text is too diffi...
SARI.html	Tests	41	16
PSK.html	Tests	42	16
FORCAST.html	Tests	42	14
OptionsDocumentAnalysisSentencesWords.html	Dialogs	44	14
OptionsDocumentAnalysisProjectSettings.html	Dialogs	46	12
DaleChall.html	Tests	46	14
FleschKincaid.html	Tests	46	14
Raygor.html	Tests	47	Text is too diffi...
Syllables.html	Notes	47	13
OptionsDocumentDisplayGeneral.html	Dialogs	48	12
Fry.html	Tests	48	14
NewFogCount.html	Tests	50	12
ProperNounsInWordList.html	Notes	52	11
OptionsDifficultWords.html	Dialogs	53	11

Here, we can sort by group or scores and see how the groups compare. In the above example, we can see that the *Tests* group is scoring poorly.

Grouping is also shown in the readability graphs. For example, if we included a Fry graph, then it may appear as such:



Here, we can see the *Tests* group again scoring rather high, as much of this group's documents are clustering in the higher grade ranges.

19. Creating Custom Tests

With custom tests, you can create enhanced versions of existing tests like New Dale-Chall that include your own word list. You can also create entirely new tests with your own formulas, and design them to return either grade-level, predicted cloze, or index scores.

In this chapter, we will step through examples showing how to create these types of tests.

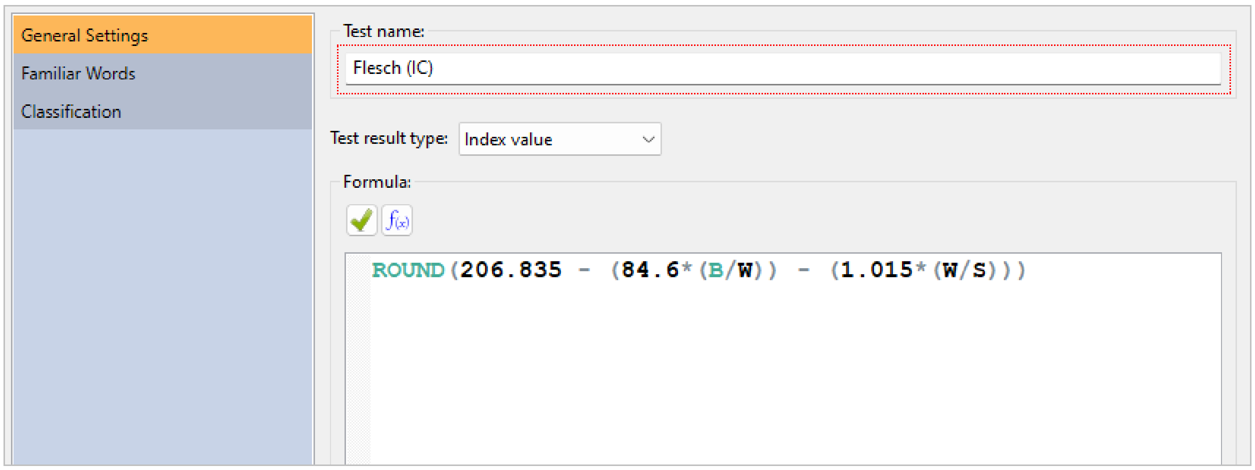
19.1 Creating a Modified Flesch Test

In this example, we will create a custom test. Custom tests are useful for either creating an entirely new formula or a new one based on an existing formula.

For this example, we will use the Flesch formula for our basis. We will take the standard Flesch formula and substitute its sentence count factor with independent clause count. This will effectively create a formula which does not penalize documents for containing numerous compound sentences.

The first step to creating a new test is to click the Custom Tests button on the Readability tab and select Add Test Based on... from the menu. This will present a dialog containing formula-based tests; select Flesch Reading Ease and click OK.

Because this test will use independent clauses, change the name to "Flesch (IC)":



Next, we will change the formula. The formula for Flesch is defined as:

$$I = 206.835 - (84.6 * (B/W)) - (1.015 * (W/S))$$

Where:

I	Flesch index score
W	Number of words
B	Number of syllables
S	Number of sentences

Formulas can either be typed into the Formula field, or you can use the Function Browser. For example:

```
206.835 - (84.6*(SyllableCount()/WordCount())) -
    (1.015*(WordCount()/SentenceCount()))
```

There are shortcuts available for various functions, so this formula can also be simplified to this:

```
206.835 - (84.6*(B/W)) - (1.015*(W/S))
```

By default, all formula results will be returned with floating-point precision. For this particular formula, we will prefer to simply round the result. To do this, we will add the ROUND function to our formula (with 0 as the precision width):

```
ROUND(206.835 - (84.6*(B/W)) - (1.015*(W/S)), 0)
```

This formula will yield the same results as the Flesch test. For this example, we will have our new test count independent clauses instead of sentences. To do this, change SentenceCount() (or S) to IndependentClauseCount() (or U). For example:

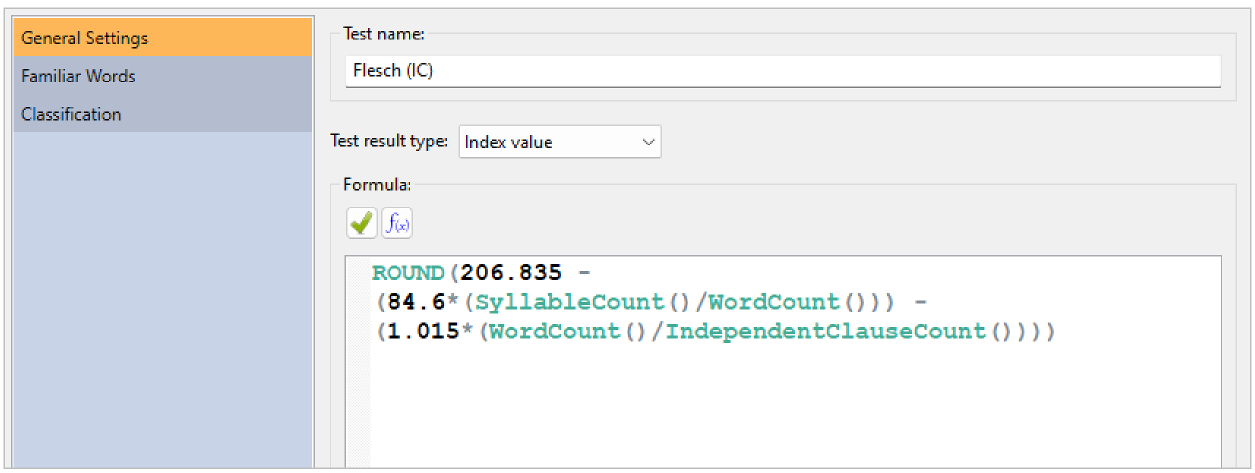
```
ROUND(206.835 - (84.6*(B/W)) - (1.015*(W/U)), 0)
```

Or like this:

```
ROUND(206.835 - (84.6*(SyllableCount()/WordCount())) -
    (1.015*(WordCount()/IndependentClauseCount())), 0)
```

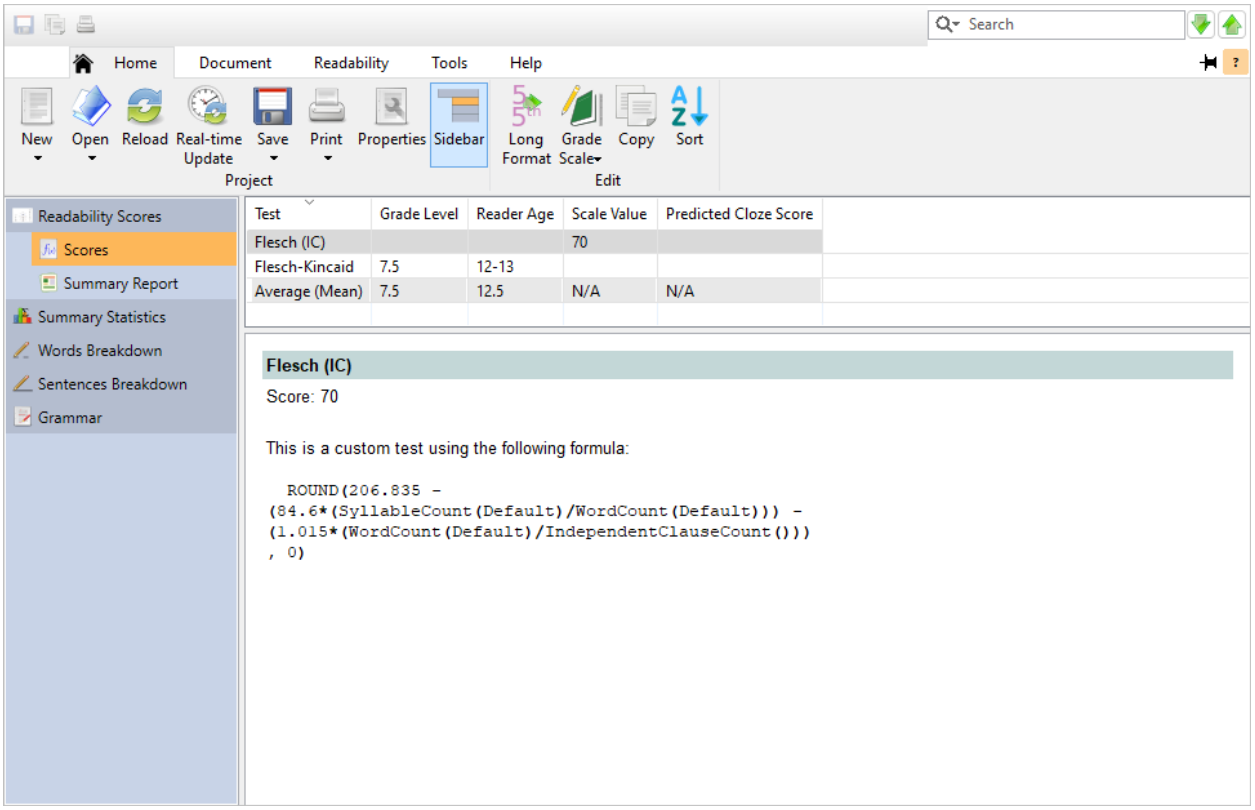
The final step is to specify the type of test that this is. Flesch Reading Ease is an index test, so verify that the test result type is set to Index value. This will instruct the program to place the result into the Scale Value column (see below).

Once finished, the dialog will look like this:



Click the OK button and our new test will be added to the system.

To add this test to any project, first select the Readability tab. Next, click the Custom Tests button and then select Flesch (IC) from the menu. Select the Readability Scores icon on the project pane and note that Flesch (IC) has been added to the project's results:

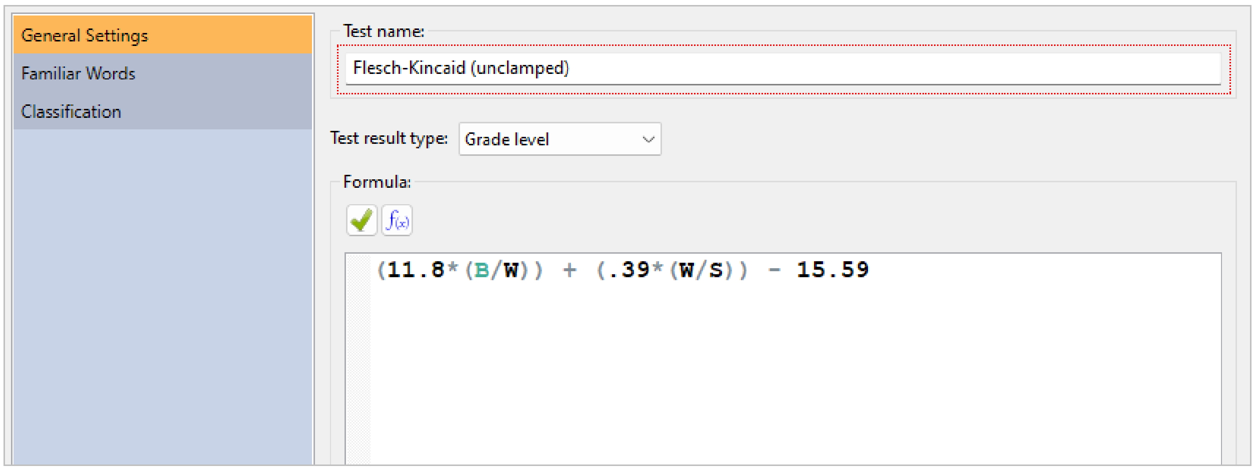

Search
Home
Document
Readability
Tools
Help
New
Open
Reload
Real-time Update
Save
Print
Properties
Sidebar
Project
Long Format
Grade Scale
Copy
Sort
Edit
Readability Scores
Scores
Summary Report
Summary Statistics
Words Breakdown
Sentences Breakdown
Grammar
Test
Grade Level
Reader Age
Scale Value
Predicted Cloze Score
Flesch (IC)
70
Flesch-Kincaid
7.5
12-13
Average (Mean)
7.5
12.5
N/A
N/A
Flesch (IC)
Score: 70
This is a custom test using the following formula:
ROUND(206.835 -
(84.6*(SyllableCount(Default)/WordCount(Default))) -
(1.015*(WordCount(Default)/IndependentClauseCount()))
, 0)

19.2 Creating an Unclamped Flesch-Kincaid

As described in sec. 1.2, regression-based tests are clamped to the reasonable limit of 19^{th} grade (i.e., PhD). If you wish to override this, however, you can create a copy of a test and not set an upper limit for its score.

In this example, we will create a custom Flesch-Kincaid test and leave its formula unrestricted, allowing for scores such as 27 (i.e., 27^{th} grade).

First, select the Readability tab of the ribbon. Next, click the Custom button and select Add Custom Test Based on.... Select Flesch-Kincaid from the list, which will create a custom test based on Flesch-Kincaid. When the Edit Custom Test dialog appears, enter the name "Flesch-Kincaid (unclamped)":



We will leave the formula as-is, so click OK to accept.

From here, we can add this test via the New Project Wizard by selecting Manually select tests and selecting it in the Custom tests group box. We can also add it to an existing project from the Custom button on the Readability tab.

When adding this test to documents with extreme sentence lengths or difficult-word counts, we may see Flesch-Kincaid (unclamped) scores going beyond 19.

> Tip
> If you do want to clamp a test to a given range, wrap it in a `CLAMP` function call. For example, `CLAMP(formula, 0, 4)` will clamp the result of *formula* within 0–4 (i.e., Kindergarten to 4^{th} grade).

19.3 Creating a Custom New Dale-Chall Test

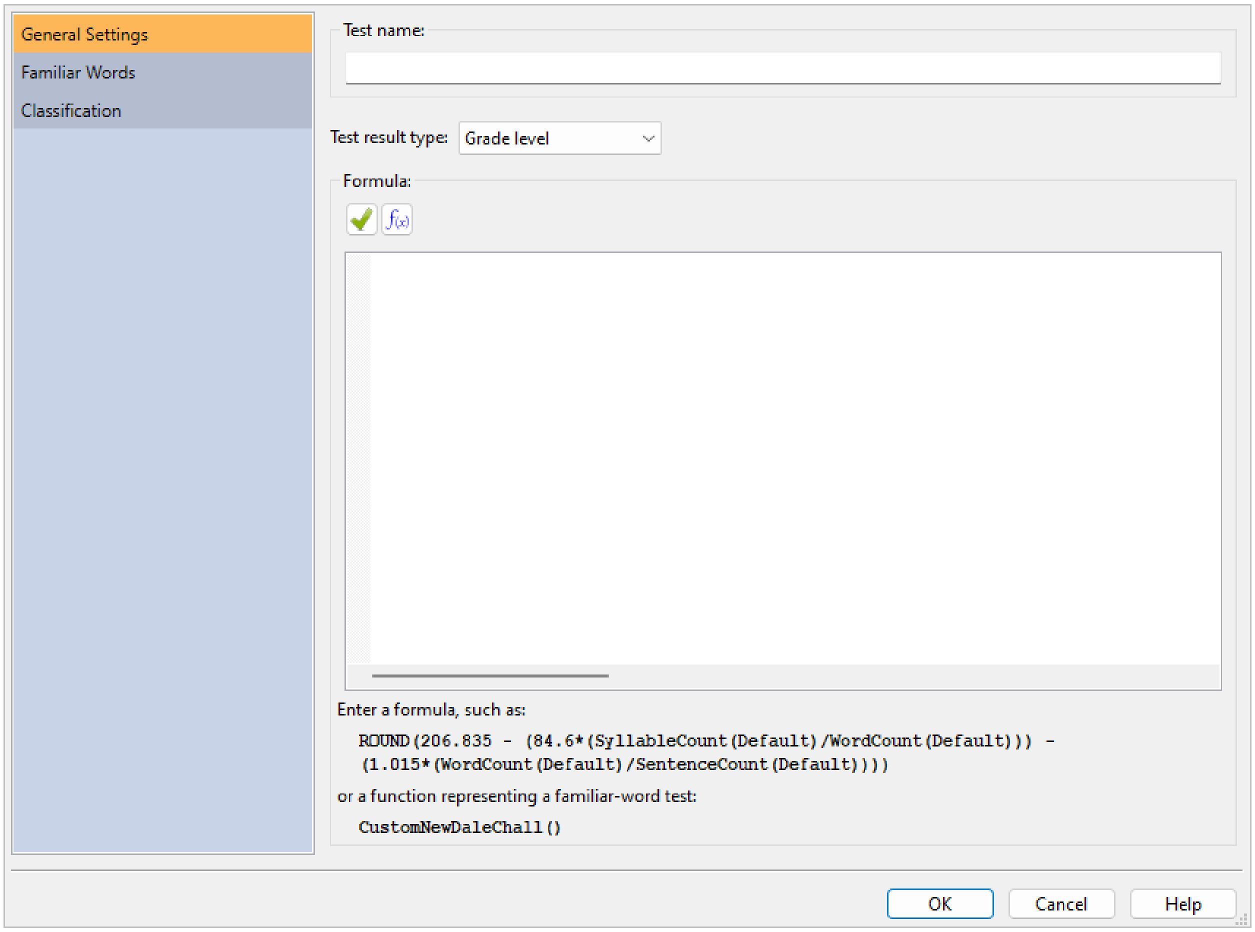
In this example, we will create a custom New Dale-Chall test specifically designed for recipes for a group of young students. Although the standard New Dale-Chall list is quite extensive, we may wish to append some words that we know are familiar to our kitchen-savvy students.

The first step is to create a list of baking-specific words that our readers are familiar with that are not already on the standard New Dale-Chall list. This can be any text file where the words are separated by spaces, tabs, commas, or semicolons. For this example, we will use this list:

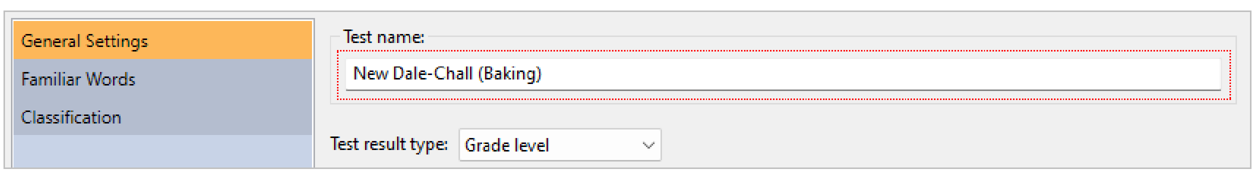
```
beaten, custard, eclair, éclair, flaky, glaze, glace, glossy, grate, meringue, pastry, pinch, platter, prepare, recipe, saucepan, sift, skewer, teaspoon, teaspoonful, thicken, tumbler, unbeaten, vanilla, whisk, scrape, tablespoon, tablespoonful
```

The next step is to create a custom test and connect our word list to it. For a custom New Dale-Chall, the simplest way is to click the Custom Tests button on the Readability tab and select Add Custom “New Dale-Chall”... from the menu. This will present the Add Custom Word Test dialog. From this dialog, we would enter the name of the new test (e.g., “New Dale-Chall (Baking)”) and file path to our word list, and then the program would add this test. However, for this example we will use the more advanced Add Custom Test dialog.

Click the Custom Tests button on the Readability tab and select Add... from the menu. This will present the Add Custom Test dialog, as shown below:



Because this will be a New Dale-Chall test meant for baking, we will enter the name “New Dale-Chall (Baking)”:



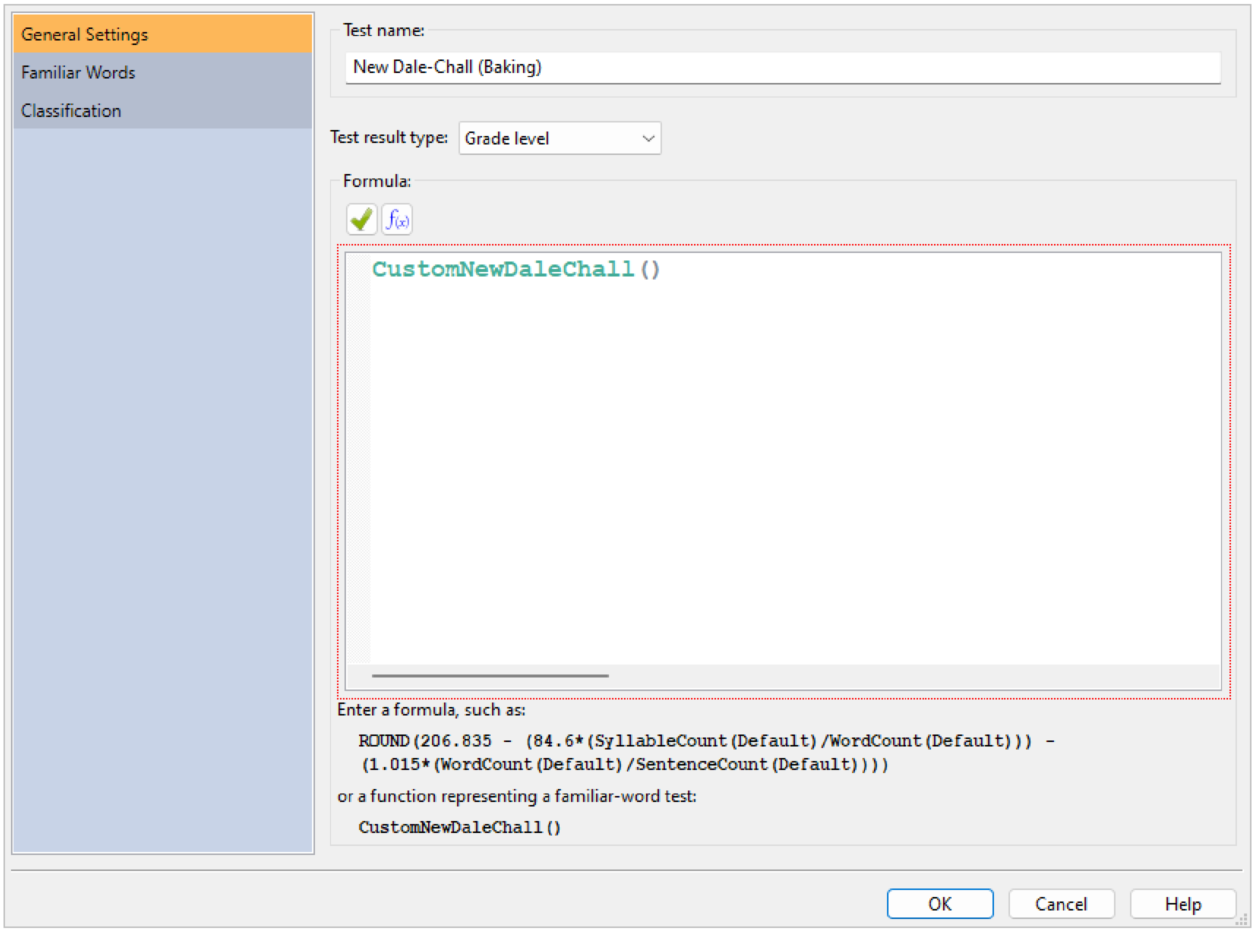
Next, we will enter the formula. The formula for a custom New Dale-Chall test is:

```
CustomNewDaleChall()
```

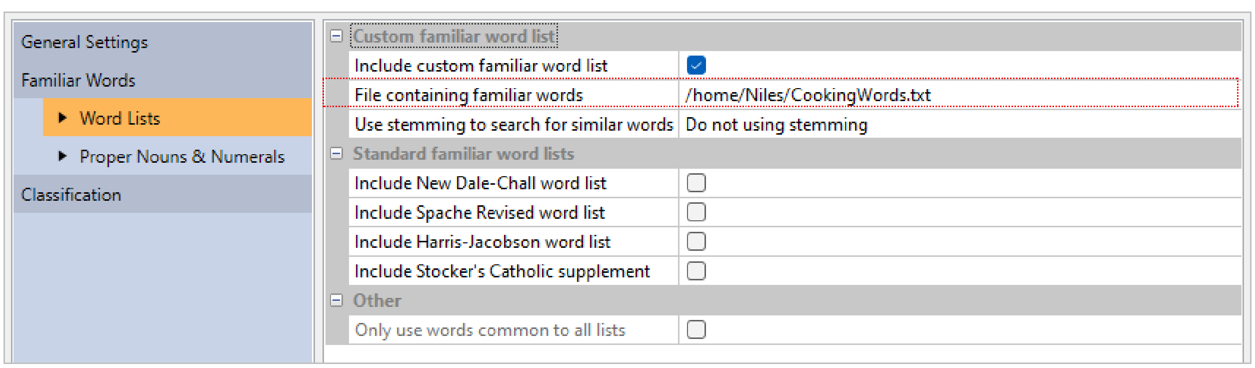
This formula will instruct the program to use the standard New Dale-Chall formula with our familiar-word criteria, which we will define later.

The next step is to specify the type of test that this is in the Test result type field. By default, new tests will be defined as grade-level tests, which means that the result will be truncated to be within the 0 (Kindergarten) to 19 (doctorate) range. New Dale-Chall is a grade-level test, so we will leave this option set to the default.

Once finished, the dialog will look like this:



Now we will need to define what a familiar word is, so click the Familiar Words icon. Because we will be appending our own words to the test, check the option Include custom familiar word list. Next, enter the file path to the custom-word file in the File containing familiar words field. You can also click the button next to this field to select and edit your word list file:



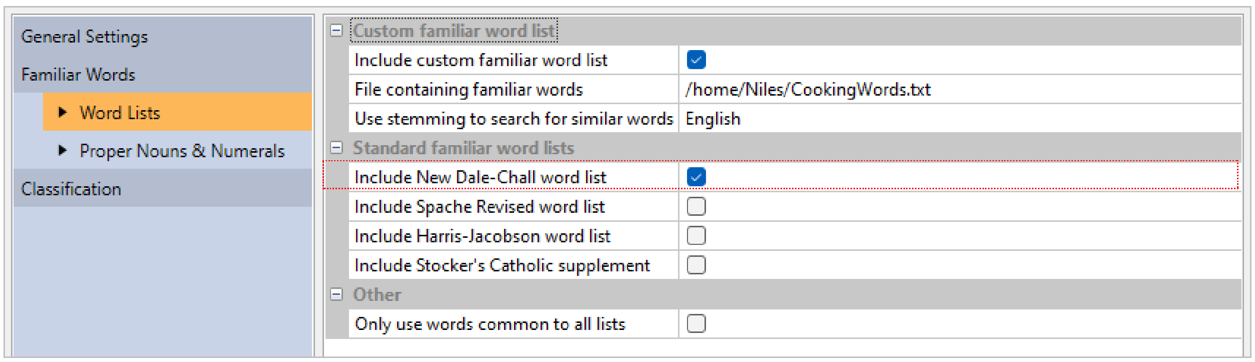
Once the file has been selected, the words from that file will be loaded.

There are two ways that our custom words can be compared to words in the recipe that we will be analyzing. The first is to simply compare words between our list and the recipe exactly as they appear. This is accomplished by setting Use stemming to search for similar words to Do not use stemming. The advantage of this is that we will have precise control over which variations of our words will be counted as familiar. For example, we could add *glaze* and *glazes* to our list and only these variations of *glaze* would be counted as familiar. This way, other variations (e.g., *glazy*) will remain unfamiliar. The disadvantage of this method is that we must explicitly add every variation of *glaze* to our list that we want to be familiar. For larger lists, this can be tedious and error-prone.

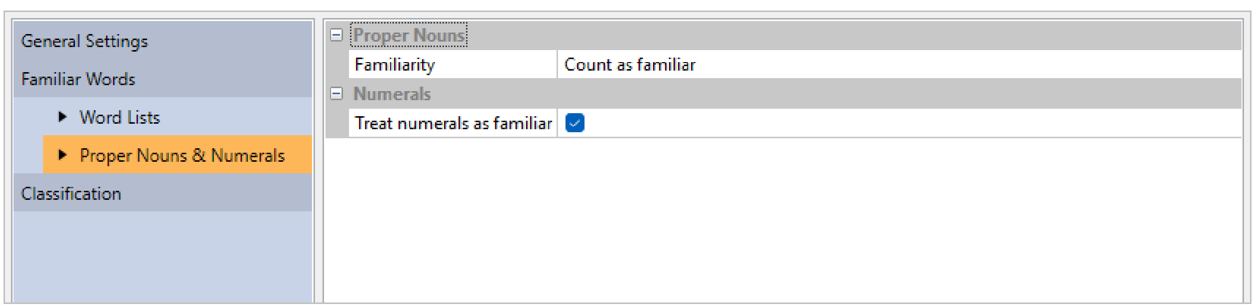
The other option is to use stemming. Stemming will compare the roots of the words between our list and our recipe, rather than the words themselves. For example, say that we have English stemming selected and the word *glaze* is on our list. In this case, *glaze* will be stemmed to *glaze*. As our recipe file is analyzed, words such as *glazed*, *glazes*, and *glazing* will also be stemmed to *glaze*. Because these words all share the same root as a word from our list, they will all be counted as familiar. The advantage of this approach is that we do not need to add every possible variation of every word on our list. The disadvantage is that we will lose precise control over which word variations should be familiar or not.

To keep our word list simple, we will use English stemming. Select English as the stemmer.

Next, we will define which other word lists will be included in our test. Because this is a New Dale-Chall test, check the option Include New Dale-Chall word list:



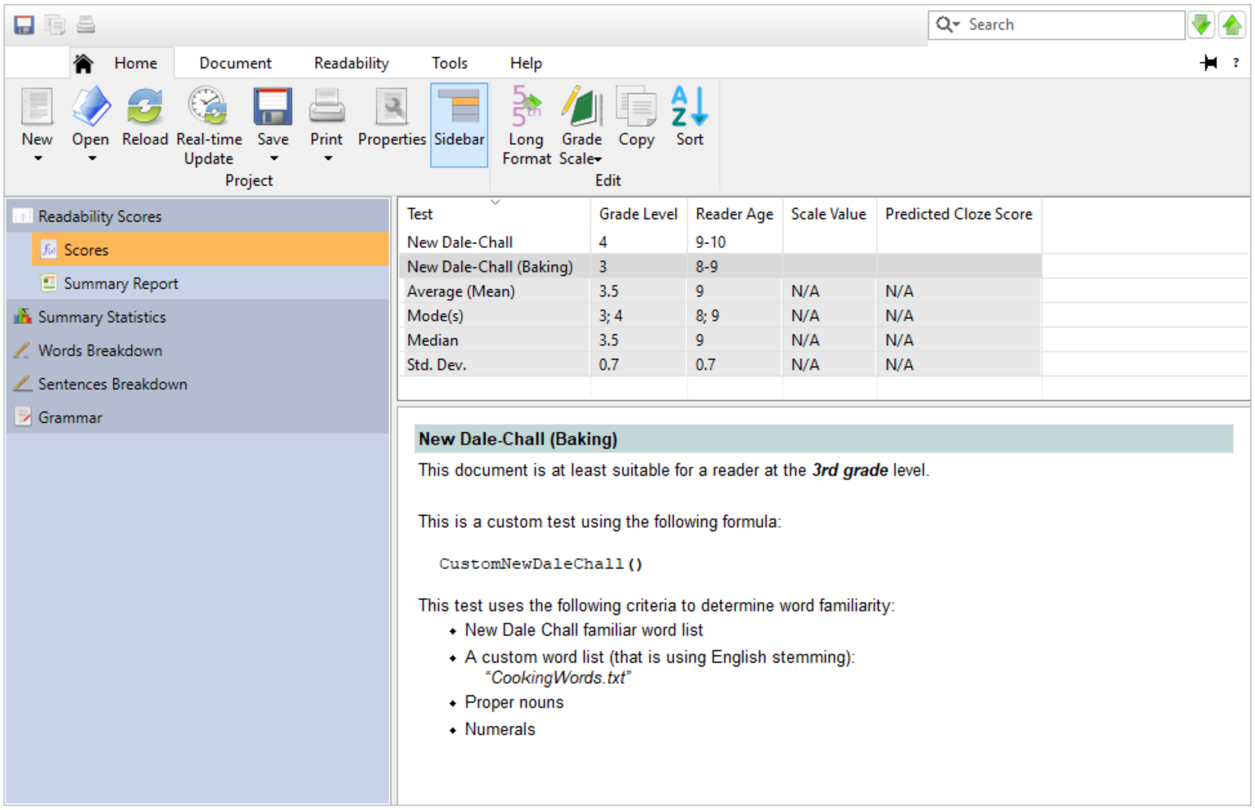
The remaining logic that we must define is how to count proper nouns and numerals. Select the Proper Nouns & Numerals page:



For this example, we will treat all proper nouns and numerals as familiar. This is the default, so leave these options unchanged.

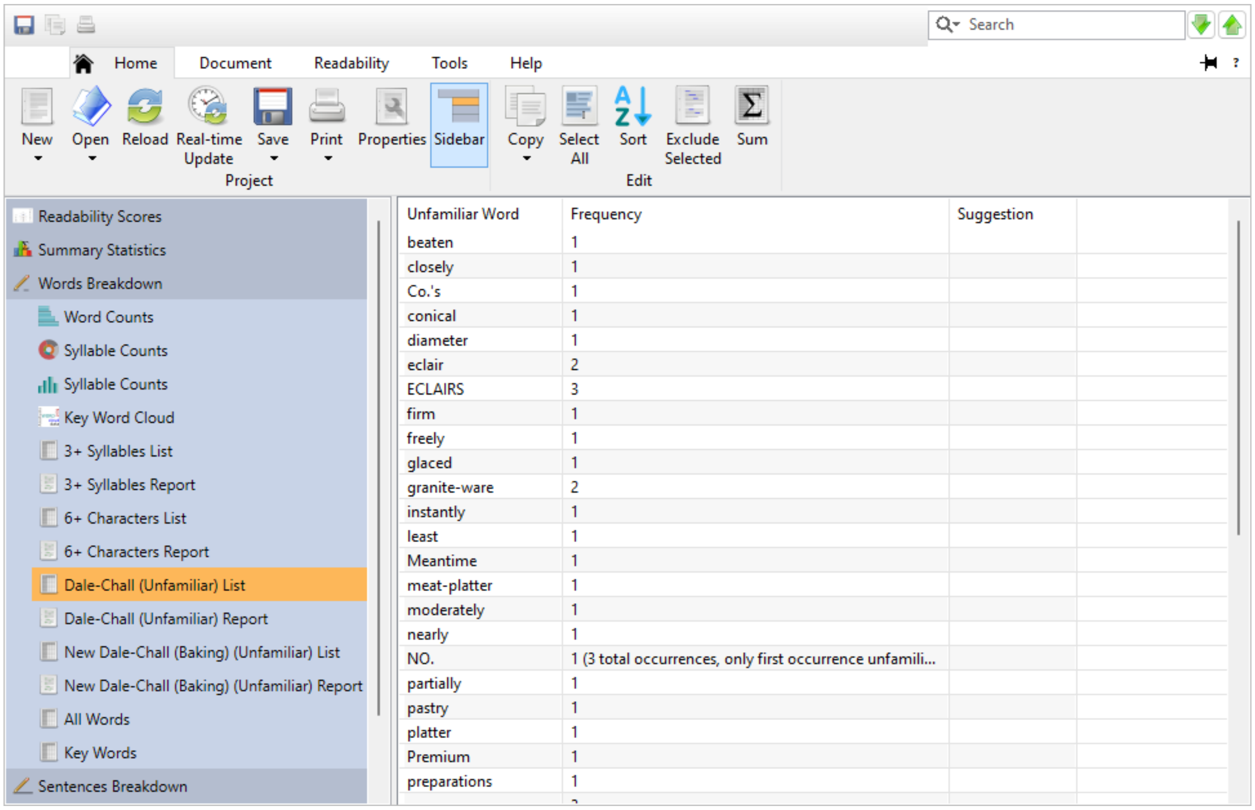
Click the OK button and our new test will be added to the system. At this point, the formula and all our settings will be validated. Because our proper-noun options differ from the standard New Dale-Chall test, the program will prompt to confirm this. We are intending to treat all proper nouns as familiar and do not wish to use the same settings as New Dale-Chall, so click Do not adjust. Our new test, New Dale-Chall (Baking), has been added to the program.

To add this test to any project, first select the Readability tab. Next, click the Custom Tests button and then select New Dale-Chall (Baking). Select the Readability Scores icon on the project pane and note that New Dale-Chall (Baking) has been added to the project's results:

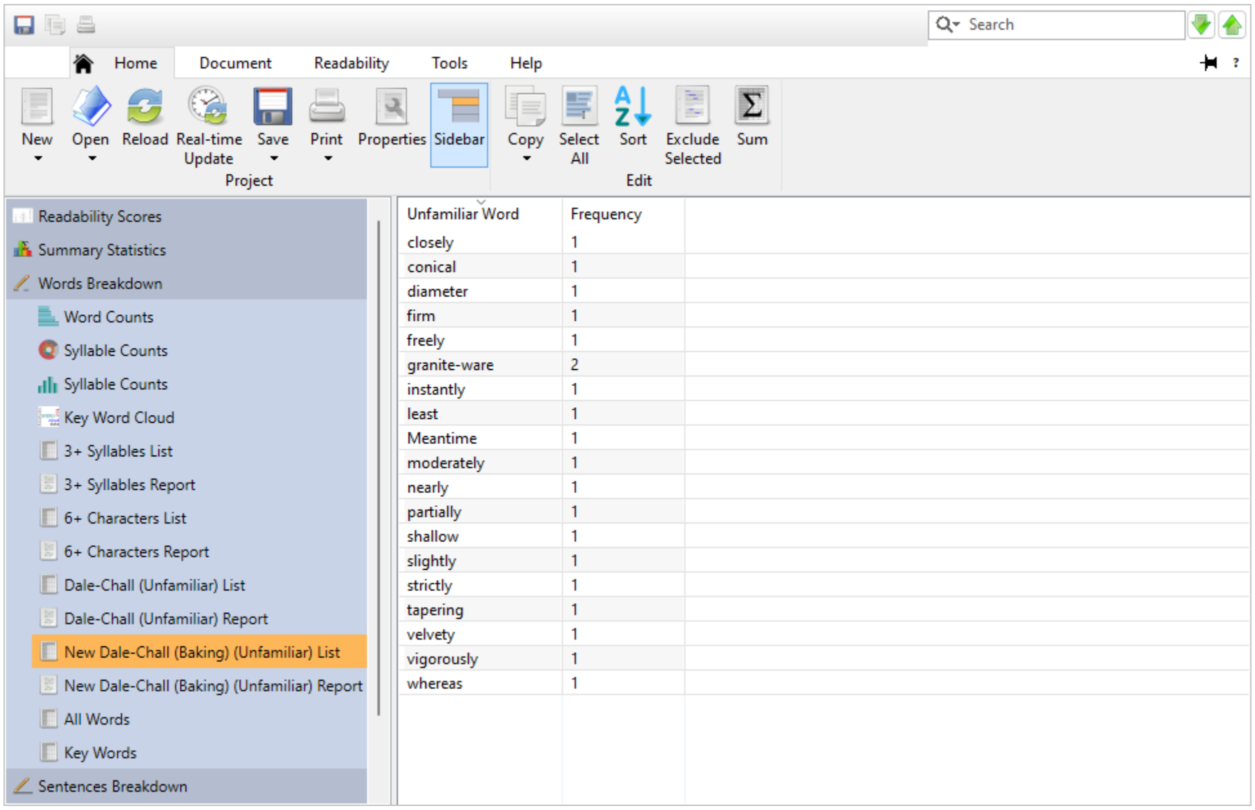


Note how the result for our custom test is lower than the standard New Dale-Chall test. This is because our test is using an expanded list of known words and will consider fewer words unfamiliar than the New Dale-Chall test will.

Now that the test has been added, we can review some of its output. Click on the Words Breakdown icon on the project pane. As shown below, there are numerous unfamiliar New Dale-Chall words in this recipe:

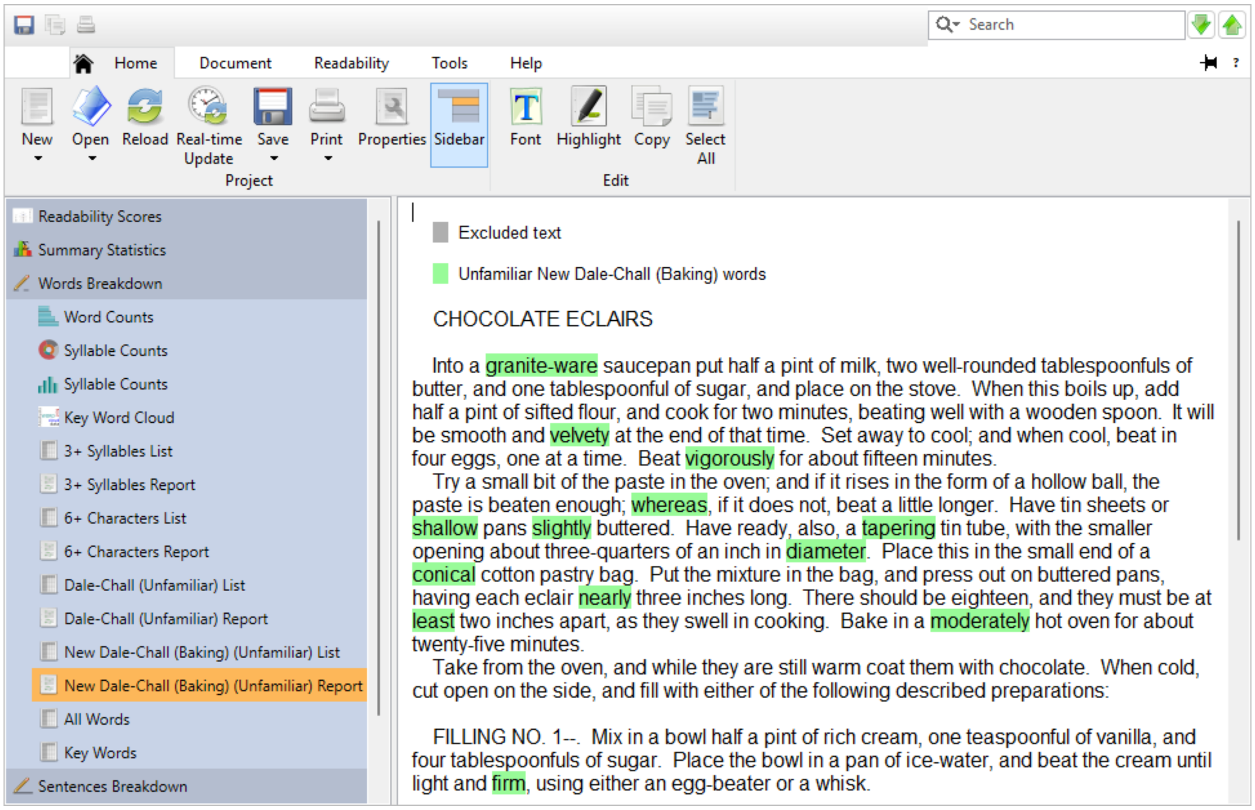


However, because our custom test is including some extra words not on the New Dale-Chall list, our test will have fewer unfamiliar words. Select the New Dale-Chall (Baking) subitem to review the unfamiliar words from the recipe relative to our custom test:



Note how there are far fewer unfamiliar New Dale-Chall (Baking) words when compared to the standard New Dale-Chall output.

We can also review these words in their original context. Click on the Highlighted Reports icon on the project pane, and then select the New Dale-Chall (Baking) page. Here we can see exactly where all the unfamiliar New Dale-Chall (Baking) words are in our recipe.



At this point, we have a couple of options. The first is to replace some of these unfamiliar words so that our students can better understand the recipe. If the reading level of this recipe was higher, this would be recommended. However, because this recipe already has a low reading level and contains few unfamiliar words, this may not be necessary.

The other option is to export these unfamiliar words and use that output to review these new words with our students. Not only will this ease the students' comprehension of the recipe, but it will serve as a helpful vocabulary-building exercise.

> **Tip**
> The recipe used in this example is available from the Example Documents menu. From the Help tab, click the Example Documents button, and then select "Chocolate Eclairs." When prompted about how to open the document, select Create a new project.

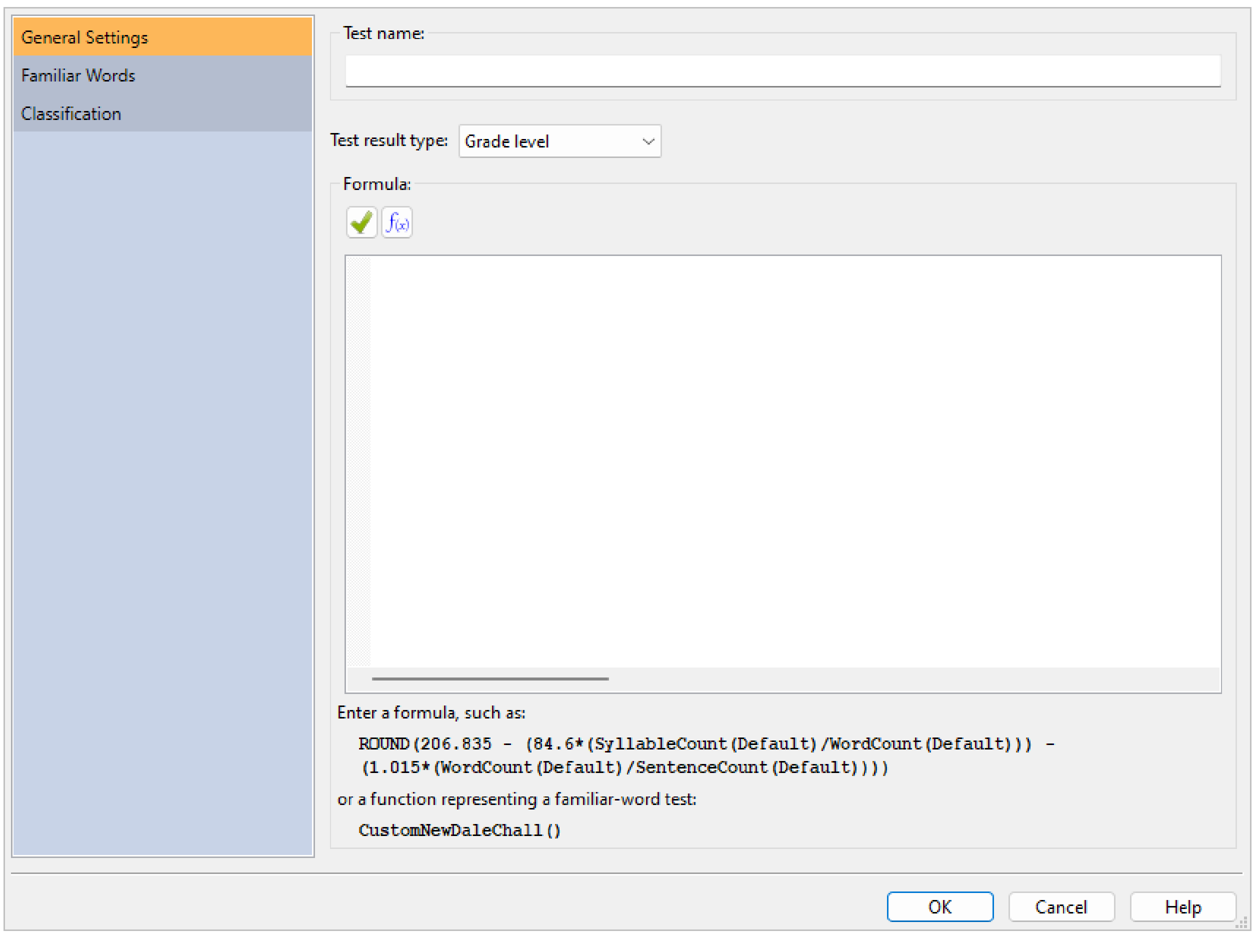
19.4 Creating a Custom Index Test

In this example, we will create a test designed for classifying a document within a custom scale. We will make a list of "buzzwords" that are commonly overused in marketing documents, then create a test to see how much a given document uses them. Also, rather than calculating the conventional grade-level or predicted cloze score, our test will return a value using our own scale.

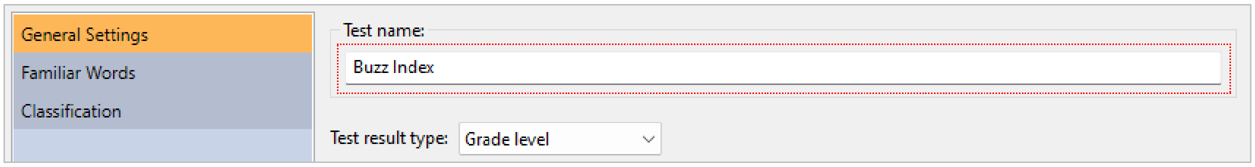
The first step is to create a list of "buzzwords." This can be any text file where the words are separated by spaces, tabs, commas, or semicolons. For this example, we will use this list:

```
agile, collaborate, cross-platform, disrupt, empower, enterprise, framework,
game-change, groundbreaking, holistic, impact, infrastructure, innovate,
integrate, lean, leverage, marketplace, next-generation, nimble, paradigm,
platform, powerful, proactive, revolution, ROI, seamless, solution, startup,
strategy, streamline, suite, sustainable, synergy, technology, utilize,
value-add, visibility, workflow
```

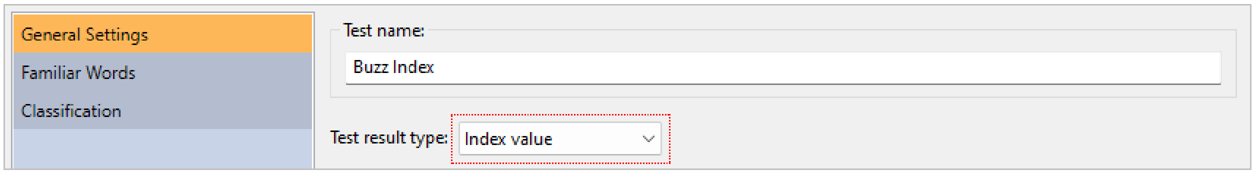
The next step is to create a custom test and connect our word list to it. Click the Custom Tests button on the Readability tab and select Add... from the menu. This will present the Add Custom Test dialog, as shown below:



Enter "Buzz Index" as the test name:

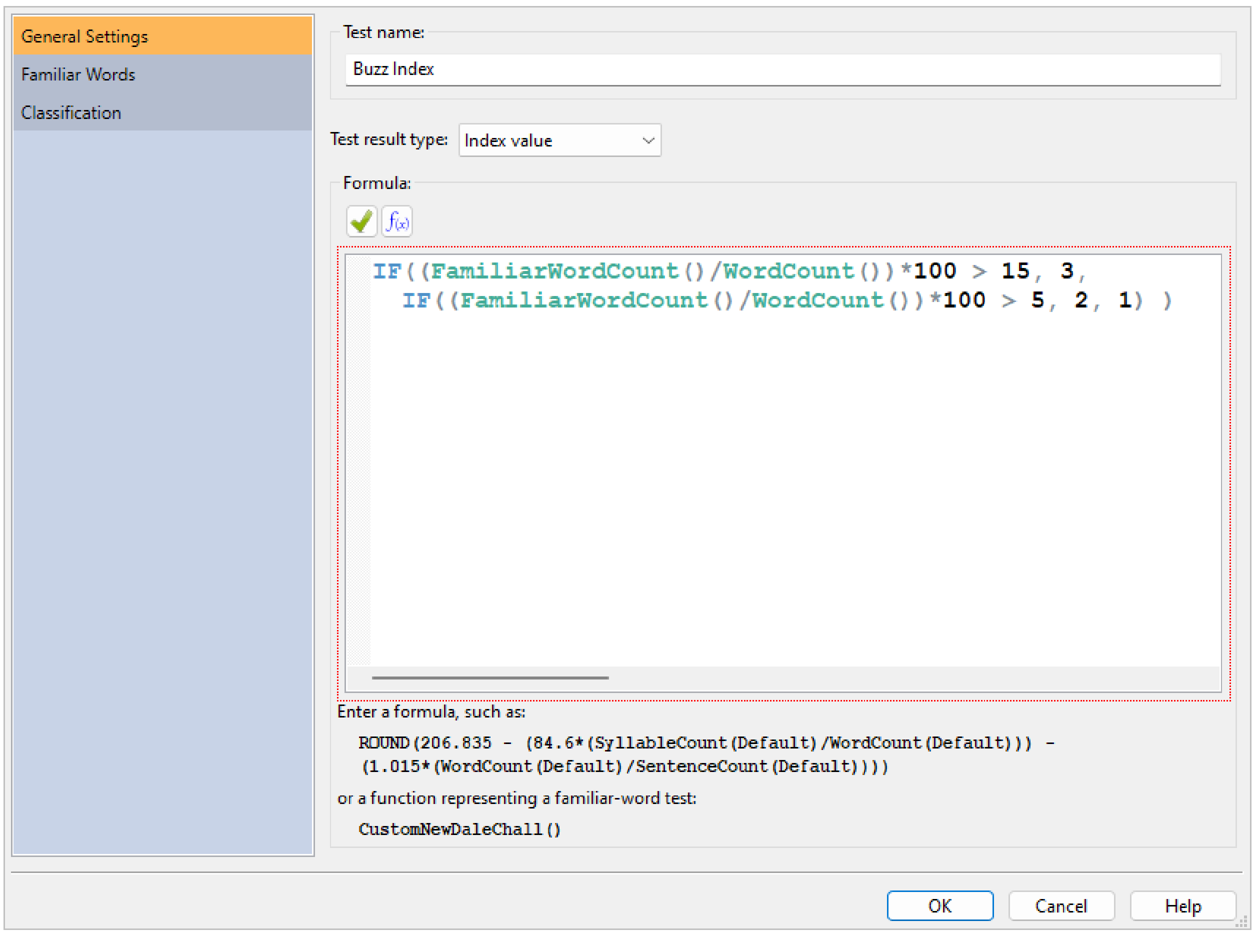


Because this test will use our own special type of scoring, set Test result type to Index value:



Next, we will enter the formula. Enter the following into the Formula field:

```
IF((FamiliarWordCount()/WordCount())*100 > 15, 3,
   IF((FamiliarWordCount()/WordCount())*100 > 5, 2, 1) )
```



Basically, this formula classifies a file based on its percentage of buzz words. First, it calculates this percentage in this line:

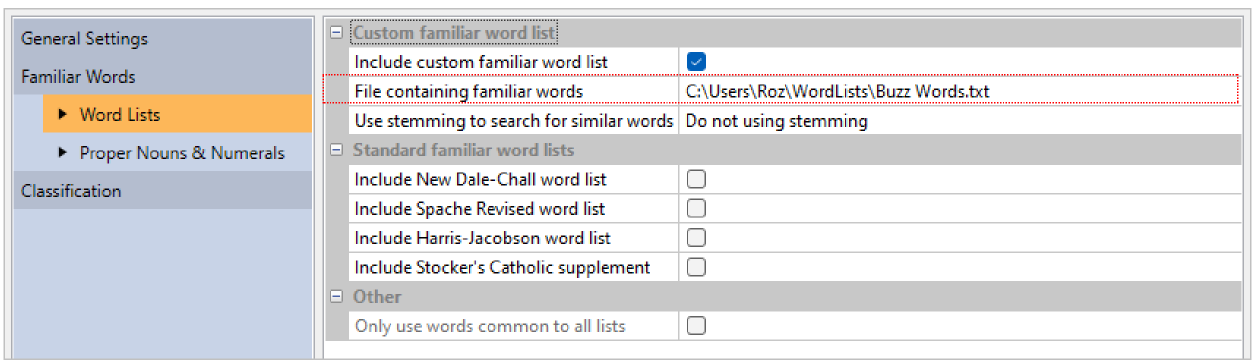
```
(FamiliarWordCount()/WordCount())*100
```

`FamiliarWordCount()` represents the number of words in the document that are found from our buzzword list. Then this count is divided by the total number of words in the document (`WordCount()`). This will yield a floating-point value (e.g., *.17*), so we multiply that by 100 to change it to an integer (e.g., *17*).

After getting the buzzword percentage, we will then classify the file within a scale of 1–3. Three means that there is an unacceptably high percentage of buzzwords, two means a high percentage, and one means an acceptable percentage. To do this classification, we will use the ternary function `IF`.

The first parameter of `IF` is the expression to evaluate (in this case, the buzzword percentage). The second parameter is the value to return if the expression is true. The third parameter is what to return if the expression is false. In our formula, the first `IF` statement checks if the percentage is more than 15%—if it is, then it returns level 3. If the percentage is less than 15, then the third parameter (i.e., the second `IF`) is executed. This second `IF` statement verifies whether the buzzword percentage is more than 5%—if it is, then it returns level 2. If the percentage is less than 5, then the formula finally returns level 1.

Now we will need to define the words to search for in our document, so click the Familiar Words icon. Because we will be using our own words, check the option Include custom familiar word list. Next, enter the file path to the custom-word file in the File containing familiar words field. You can also click the button next to this field to select and edit your word list file:

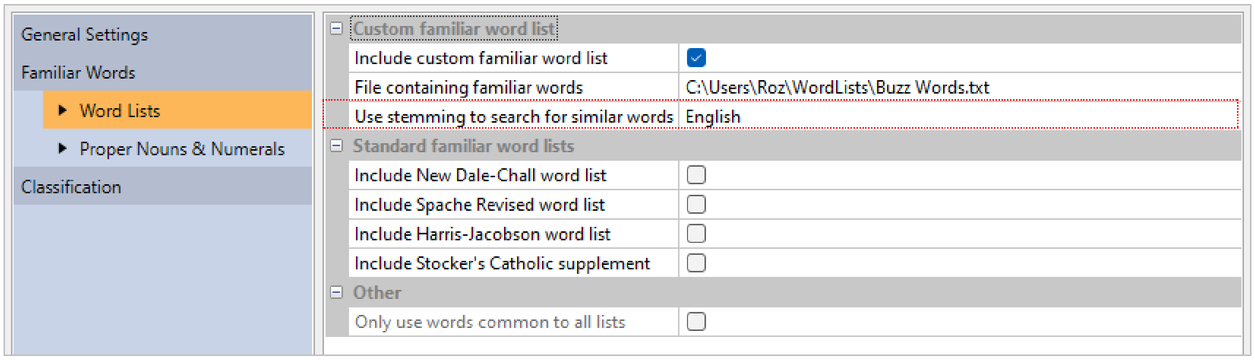


Once the file has been selected, the words from that file will be loaded.

There are two ways that our custom words can be compared to words in the press release that we will be analyzing. The first is to simply compare words between our list and the recipe exactly as they appear. This is accomplished by setting Use stemming to search for similar words to Do not use stemming. The advantage of this is that we will have precise control over which variations of our words will be counted as familiar. For example, we could add *glaze* and *glazes* to our list and only these variations of *glaze* would be counted as familiar. This way, other variations (e.g., *innovates*) will remain unfamiliar. The disadvantage of this method is that we must explicitly add every variation of *innovate* to our list that we want to be familiar. For larger lists, this can be tedious and error prone.

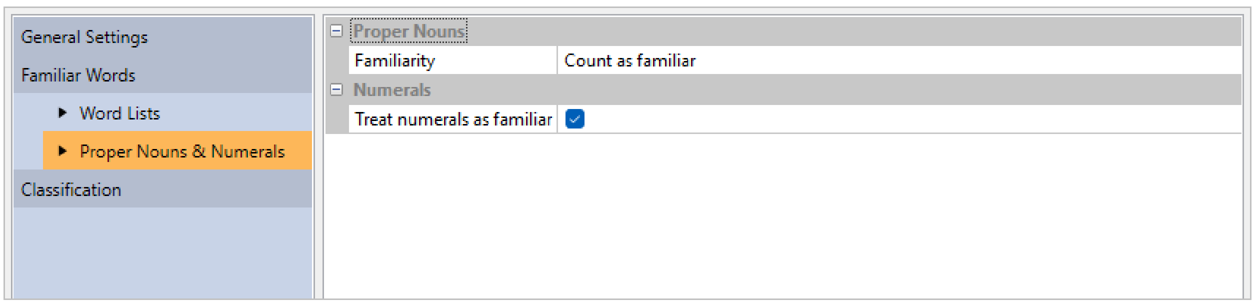
The other option is to use stemming. Stemming will compare the roots of the words between our list and our recipe, rather than the words themselves. For example, say that we have English stemming selected and the word *innovate* is on our list. In this case, *innovate* will be stemmed to *innovate*. As our file is analyzed, words such as *innovates*, *innovated*, and *innovating* will also be stemmed to *innovate*. Because these words all share the same root as a word from our list, they will all be counted as familiar. The advantage of this approach is that we do not need to add every possible variation of every word on our list. The disadvantage is that we will lose precise control over which word variations should be familiar or not.

To keep our word list simple, we will use English stemming. Select English as the stemmer:

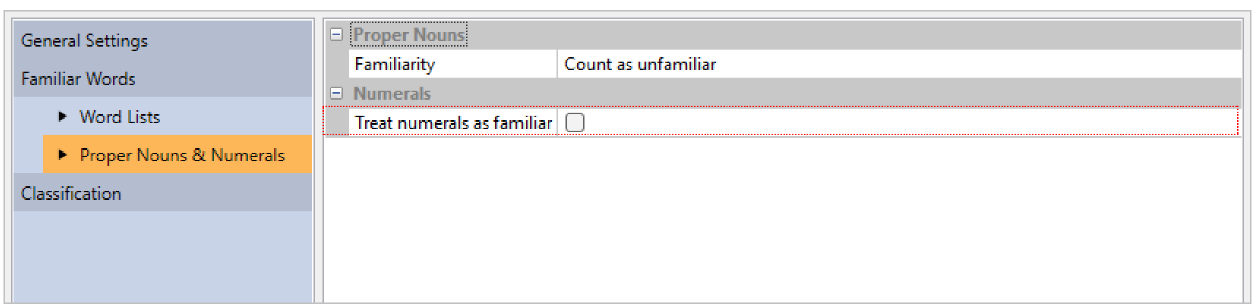


We will not be using any other words for our test, so leave the other word lists unchecked.

The remaining logic that we must define is how to handle proper nouns and numerals. Select the Proper Nouns & Numerals page:



For this test, we are only looking for words explicitly from our list. This means that we will need to ignore proper nouns and numbers. Set Proper Nouns/Familiarity to Count as unfamiliar and uncheck Treat numerals as familiar:

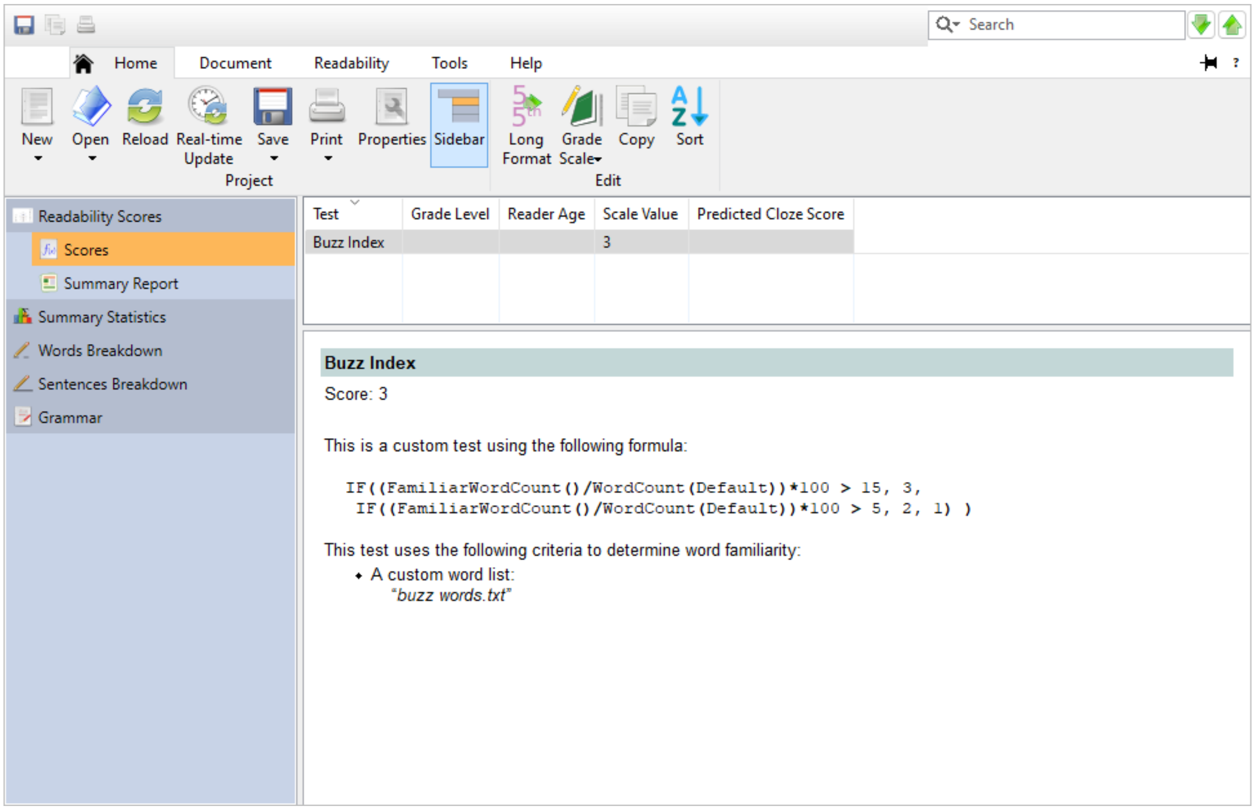


Click the OK button and our new test will be added to the system.

To add this test to any project, first select the Readability tab. Next, click the Custom Tests button and then select Buzz Index. Select the Readability Scores icon on the project pane and note that Buzz Index has been added to the project's results:

To demonstrate this, click the Help tab of the ribbon. Then, click the Examples button and select Press Release from the menu. Choose the defaults in the standard New Project wizard to create a new project from this document.

Once the file is processed, click the Highlighted Reports icon on the sidebar to view the document. After doing a quick reading of this press release, there may be some concern about the author's enthusiasm for marketing speak. To confirm this, select the Readability tab on the ribbon, click Custom Tests and select Buzz Index. Next, click the Readability Scores icon on the sidebar and select Scores underneath that. As we had feared, the document is scoring at the unacceptable level of 3:



At this point, we will ask the author to revise this document until the score is at a more acceptable level. Because we are using a familiar-word test's logic in reverse for this test, we are actually interested in lowering the number of familiar (not unfamiliar) words from the document. To show which familiar (i.e., buzzwords) are being used, select the Highlighted Reports icon on the sidebar and select Buzz Index (unfamiliar) beneath that. The words that are *not* highlighted are the buzzwords that the author will need to limit. To export this window, click the Home tab of the ribbon, click the Save button, and select Export Highlighted Buzz Index (unfamiliar). The author can then use this output to see which sections need revising in this press release.

VI Notes

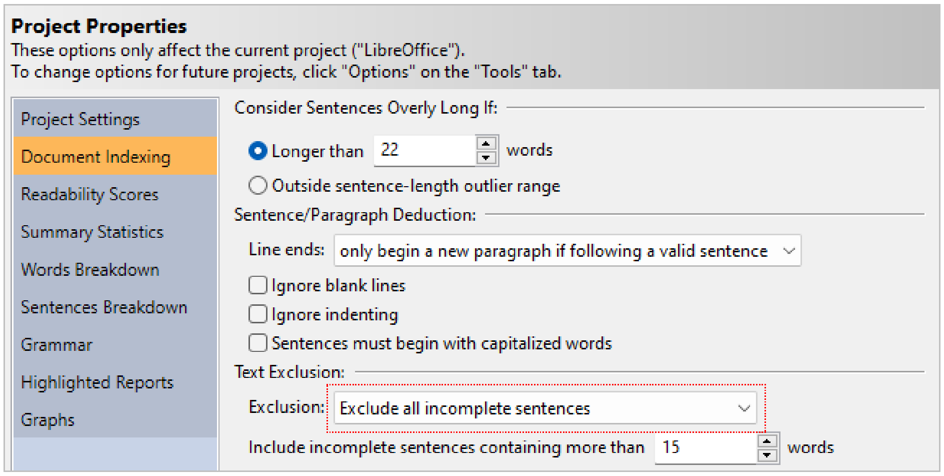
20. Analysis Notes

20.1 How Text is Excluded

Most experts recommend ignoring items such as chapter titles, list items, and table of contents when calculating reading levels. One reason that these items should be ignored is because they falsely lower the average sentence length (ASL). For example, chapter titles are usually just a couple of words. If you treat every section and chapter title as a regular sentence, then these short "sentences" will dramatically lower the ASL. In turn, this will falsely lower most of your test scores. The other reason to ignore these items is because they are not usually part of the main content, but rather dividers or sectional markers. These items are read, but it is the narrative content that readers may struggle with. Therefore, it is the content that should be our main focus, and titles and list items should just be considered "noise."

These items, by default, are ignored for all newly created projects. That is, any word inside an incomplete sentence is not factored into any of the statistics or tests. Incomplete sentences are not factored into the overall sentence and paragraph counts, which in turn will prevent them from skewing the readability scores.

To toggle this behavior for an existing project, click the Properties button the Home tab to display the Project Properties dialog. Next, click the Document Indexing icon. Here you can select either Do not exclude any text, Exclude all incomplete sentences, or Exclude all incomplete sentences, except headings, as illustrated below.



If you are excluding incomplete sentences, note that any non-terminated sentence longer than 15 words will still be counted as a regular sentence. If a sentence is longer than a certain length and is missing its terminating punctuation, then the missing period is likely just a mistake. To accommodate for these errors, the program will consider these sentences valid. To customize this behavior, enter a different minimum sentence length in the Include incomplete sentences containing more than [15] words option (pictured above).

Note that these options are also available on the Options dialog (on the Tools tab) to change this behavior for all future projects.

Although ignoring these items is generally recommended, there is one exception: non-narrative (i.e., structured) forms. Files such as résumés and application forms often contain many terse and incomplete sentences and very little narrative text. In these situations, you should include incomplete sentences and use tests that do not emphasize sentence length, such as FORCAST. When creating a new project, there are two ways to implicitly set these options. The first way is to select non-narrative, fragmented text when specifying the document structure. The other way is to select non-narrative form when specifying the document type. By selecting either of these, the program will use the appropriate options and tests for these types of files.

Additional Items

Although not traditionally mentioned in readability articles, other items may also be excluded within *Readability Studio*. These include:

- copyright notices
- trailing citations
- internet and file addresses
- specific words or phrases

Refer to sec. 8.3 for a detailed explanation of these options.

20.2 How Sentences are Counted

Readability Studio considers periods, question marks, exclamation points, and interabangs (!?) as the end of a sentence, although there are a few exceptions.

The first exception is periods that are part of an abbreviation, acronym, or initial. For example, the following is seen as only one sentence:

> Jane B. Doe has worked at A.C.M.E., inc. for seven years.

Another exception is the last sentence inside of a quote. If this sentence is followed by a lowercase word, then it will not end until the next sentence terminating punctuation is encountered. For example, the following is seen as two sentences:

> “Where did it go? I can’t find it anywhere!” he said.

Because *he* is lowercase, the *!* will not be considered the end of the sentence and *he said* will be part of the preceding sentence. However, if *He* is capitalized, then the *!* will be seen as the end of a sentence and the above will be counted as three sentences.

There is also a special rule for ellipses. If ellipses are followed by a capitalized word, then it will be considered the end of the sentence. Otherwise, it is simply seen as punctuation and does not terminate the sentence. The ellipses symbol (…) and dot sequences (...) are both recognized.

One last special case is when a colon or semicolon is found at the end of a line. In the case of a colon, it will be seen as the end of a sentence because it is assumed that this sentence is leading into a list. For a semicolon, it will be seen as the end of a lengthy list item that should be considered a valid sentence. If a colon or semicolon is found in the middle of a sentence, then it will not end that sentence.

> **Note**
> If Aggressively exclude is enabled, then semicolons will never mark the end of a sentence. This is useful for when your document contains programmer code that you wish to exclude.

By default, lowercase words can be considered the start of a new sentence (with the exception mentioned above). Although it is poor grammar to not capitalize the first word of a sentence, this writing style is often found in some literature. This behavior can be toggled using the option Sentences must begin with capitalized words.

Finally, East Asian sentence-ending punctuation is always treated as the end of a sentence. This includes the ideographic full stop (。), its half-width variant (｡), the full-width question mark (？), and exclamation mark (！). Unlike Western periods, these characters do not require special handling for abbreviations or spacing because East Asian languages do not use them in those contexts.

20.3 Line (Sentence) Chaining

Line chaining (or sentence chaining) is a sentence deduction method where lines of text are joined until a *.*, *?*, or *!* is found. In other words, multiple lines of text can be joined into a single sentence. This is normally preferred behavior when text is formatted to a specific width, such as the following:

> WARNING:
> Before cleaning the blade (or performing
> any general maintenance), be sure to turn
> OFF the mower. Failure to comply may
> result in injury.

Although this is five lines of text, we would expect this to be seen as three sentences. By chaining lines 2, 3, and part of 4 into one sentence (from *Before* to *mower.*), we properly see the sentences (regardless of the line breaks).

However, this can be a problem if your document contains headers, footers, and list items. Consider this example:

> COCOA CAKE
> The following ingredients are required:
> - 1/2 a cup of butter
> - 3/4 a cup of milk
> - 1 cup of sugar
> - 6 level tablespoonfuls of Baker's Cocoa
> - 3 eggs
> - 2 level teaspoonfuls of baking powder
> - 1 teaspoonful of vanilla
> - 1–1/2 or 2 cups of sifted pastry flour
>
> Cream the butter, stir in the sugar gradually, add the unbeaten eggs, and beat all together until very creamy.

In this situation, the header would be joined with the first sentence. Additionally, all the list items would be combined into one large sentence, and could even be joined with the sentence following them. This can greatly skew your sentence lengths and lead to erroneous results.

Readability Studio uses a more intelligent form of line chaining when deducing sentences. By default, the program will only chain lines together if all the following criteria are met:

- A line is not terminated by a *.*, *?*, *!*, or *:*
- There are no blank lines following the line
- The next line of text is not tabbed over or bulleted

In the above example, the program will detect the blank line between the header and the first sentence and not chain them together. Instead, the header will be seen as an incomplete sentence. The program will also detect how the list items are tabbed over and not chain them together. Instead, each list item will be seen as an incomplete sentence. Any line that begins with a tab or two (or more) spaces will be considered as a tabbed line. Lines that begin with hyphens, dashes, or numbers followed by a period will be considered bulleted and will also not be chained to the proceeding sentence.

As we can see, the program helps avoid incorrect line chaining. Just as important, it will chain lines correctly when it is actually necessary. Consider the following:

> WARNING:
> Before cleaning the blade (or performing
> any general maintenance), be sure to turn
> OFF the mower. Failure to comply may
> result in injury.

The first line, *WARNING:*, will be seen as a single sentence because the line ends with a colon. Although colons are not normally treated as a sentence terminating character, an exception is made when it is encountered at the end of a line. The next line, *Before cleaning the blade (or performing,* does not end with a sentence terminating character. There are no empty lines after this line, and the next line is not bulleted or tabbed; therefore, this line will be chained to the next line. This chaining will continue until an empty line or a sentence terminating character is encountered. In this case, all these lines will be joined into one sentence:

> Before cleaning the blade (or performing
> any general maintenance), be sure to turn
> OFF the mower.

Note that even if a header does not precede an empty line or end with a sentence terminating character, line chaining will be intelligently avoided if the following sentence is tabbed or spaced over. The first sentence of a new paragraph is normally spaced over, so this is expected behavior. In this example, the following sentence is tabbed over and the program will know not to chain these lines:

> WARNING
>   Before cleaning the blade...

A caveat with line chaining is when a header is not terminated and not followed by an empty line, and the following sentence is not tabbed over. For example:

> WARNING
> Before cleaning the blade...

In this case, these two lines will be erroneously joined into one sentence. To completely disable all possible line chaining, select the option Line ends always begin a new paragraph. Note that this option is only recommended if all line endings in your document represent new paragraphs. If your document contains line ends to control the width of the text, then selecting this option will cause incorrect sentence deduction.

20.4 Controlling How Sentences are Deduced

How sentences are structured can vary from document to document. Because of this, *Readability Studio* offers options to control how sentences are parsed and analyzed. These features are available under the Sentence deduction section of the Document Indexing page. To view these features, select Options from the Tools tab and then click on the Document Indexing icon. Under the Sentence deduction section, there are two options: Ignore blank lines when determining the end of a sentence and Sentences must begin with capitalized words.

Tools › Options › Document Indexing › Sentence deduction

The option Sentences must begin with capitalized words controls whether sentences must begin with capitalized words. Selecting this option will make *Readability Studio* more grammatically strict when counting sentences; however, it is recommended to leave this option unselected for "loosely written" documents.

The option Ignore blank lines is used for controlling how multi-line sentences with blank lines between them are seen. For example:

> Chapter 1
>
> It was the best days of our lives.

If Ignore blank lines is unselected (the default), then the above example is seen as two sentences. *Chapter 1* is seen as an incomplete sentence because there is a blank line between it and the next sentence. Then *It was the best days of our lives.* is seen as the second sentence. This behavior is recommended for most documents because it helps remove headers and list items from the analysis.

There are special situations where you would need multi-line sentence fragments to be chained together. For example, picture books that wrap sentences around images may cause sentences to be split into fragments with blank lines between them. Refer to sec. 15.2 for an example.

20.5 How Syllables are Counted

Most software that counts syllables employs one of two methods. The first is a dictionary-based system that includes a list of common words and their respective syllable counts. As the software counts a document's syllables, each word is looked up in a dictionary to retrieve its syllable count. The disadvantage of this method is that if a word is not found, then the program must prompt you to enter the syllable count.

The second syllable-counting method is to use an algorithm. These algorithms generally consist of counting vowel blocks[1] in a word to determine the syllable count. The advantage of this is that it will work with any word, without user interaction. The disadvantage is that these algorithms are traditionally less accurate than a dictionary approach. To be truly effective, these algorithms must consider special cases, such as:

- Situations where vowel blocks split into two syllables. For example, the *ui* in *fruit* is one syllable, but in *intuitive* it splits into two syllables.

[1] An exception for silent *e* is made for English syllabizing.

- Situations where special consonant blocks may split into two syllables. For example, in the name *McCoy*, the prefix is really a separate syllable.

Readability Studio uses an algorithmic approach which takes these linguistic nuances into account.

Different algorithms should be used for different languages. The rules for vowel-blocks, silent *e*'s, and number counting can greatly vary between languages. For example, *85* would be two syllables in English (*eight five*), whereas in Spanish[2] it would be four syllables (*o-cho cin-co*). To account for this, *Readability Studio* uses the proper syllabizer based on the language that you specify for your project.

Finally, for readability analysis, syllables should be counted on a word-by-word basis and not use synalepha. Synalepha is the blending of word sounds together often found in poetry. Thus, it is generally not suggested to analyze text which has a poetic meter.

How Numbers are Syllabized

Numbers are generally syllabized using one of two approaches:

- Sound out each digit of the number. For example, *87* would be three syllables (*eight se-ven*).
- Count the entire number as one syllable.

By default, *Readability Studio* counts numbers as monosyllabic words; however, this behavior can be toggled from the Document Indexing options. Note that some tests may override these settings if they require a specific method. For example, Fry will always sound out each numeric digit when syllabizing.

How Symbols are Syllabized

The following symbols are syllabized as follows:
= 1 syllable ("pound")
£ = 1 syllable ("pound" or "quid") if at the beginning of the word
± = 3 syllables ("plus mi-nus") if at the beginning of the word
$ = 2 syllables ("do-llar") if at the beginning of the word
€ = 2 syllables ("eu-ro") if at the beginning of the word
¢ = 1 syllable ("cent") if at the end of the word
% = 2 syllables ("per-cent") if at the end of the word
° = 2 syllables ("de-grees") if at the end of the word
¼ = 2 syllables ("one fourth") if at the end of the word
½ = 2 syllables ("one half") if at the end of the word
¾ = 2 syllables ("three fourths") if at the end of the word

20.6 Repeated Word Exceptions

Generally, repeated words are a typo; however, there are a few exceptions. For example:

[2] For Spanish text, a specialized version of the Cuayáhuitl (412–15) algorithm is used.

> The report is due tomorrow. I'm concerned *that that* report won't be ready in time. The last time that I spoke to our boss, she said that she *had had* it with overdue reports.

Although awkwardly worded, the duplicate *that* and *had* are grammatically correct. To account for this, *Readability Studio* ignores certain repeated words, such as:

- ha ha
- had had
- no no
- that that
- *die die* (German article)
- *der der* (German article)
- *das das* (German article)
- *sie sie* (German pronoun)

Words separated by punctuation are also ignored when deducing repeated words. For example:

> I need a list of new features. Each new *feature, feature* by feature, is required to be documented.

These will not be seen as repeated words because of the comma.

20.7 Article Mismatching

There are two special cases for article-mismatch detection: single letters and acronyms. For single letters, *Readability Studio* will sound out the letter when determining whether the article in front of it is correct. For example:

> All the students did well, except for one that received *a 'F'*.

In this example, *a 'F'* will be considered incorrect because the program will sound out the letter *F* as *ef*. An *ef* sound requires the article *an* to be grammatically correct, so this will be considered wrong.

Similar logic is applied to acronyms that probably would have each letter sounded out. This deduction is made if an acronym contains no vowels. For example:

> *A SDLC* meeting should be held after the product is launched.

The program will sound out the letters in *SDLC* because it has no vowels, resulting in *es dee el see*. Because of the *es* sound, the article *an* is required. Therefore, the program will mark this as an article mismatch.

Note that if an acronym contains a vowel, then the program will assume that it is pronounced as a regular word. For example:

> Run *a FORCAST* test if your document doesn't have a regular sentence structure.

Rather than treating the word *FORCAST* as if it begins with an *ef* sound, it sounds it out like *forecast*. Therefore, the article *a* is considered correct.

20.8 Non-narrative/Fragmented Text

Most readability formulas are designed for narrative text, where a work consists of flowing sentences and paragraphs. For these formulas, the average sentence length (ASL) is an important factor.

Items such as headers and list items can adversely affect the accuracy of the ASL. If they are chained to the following sentence, then they will inflate the ASL. If they are treated as full sentences, then they will falsely lower the ASL. Because of this, most experts recommend removing these and only analyzing the narrative sections of a document.

By default, *Readability Studio* follows this advice. However, there are situations where this rule does not apply.

For files consisting mostly of short blocks of text (e.g., quizzes, flyers, or menus) instead of flowing narrative, this presents a problem. Sentence lengths will be abnormally short and skew most test results. Also, if incomplete sentences are being excluded, then most of the document will not even be analyzed.

To avoid these issues, it is recommended to include incomplete sentences in the analysis and to only use the FORCAST test. When creating a new project, there are two ways to implicitly set these options. The first way is to select non-narrative, fragmented text when specifying the document structure. The other way is to select non-narrative form when specifying the document type.

20.9 Joining Hyphenated Words

Some published documents format each line of text to a specific width to fit on the page. In doing so, longer words at the end of some lines may be split and hyphenated. For example:

> COBOL is an acronym for COmmon *Bus-iness* Oriented Language. It is commonly used for calculating monetary transactions and generating reports. For example, COBOL can be found in the backend of many Point-of-Sale systems.

Note how *Business* and *Point-of-Sale* are split across multiple lines.

When *Readability Studio* encounters a hyphenated word at the end of a line, it will perform these steps:

- Both parts of the word will be joined.
- The hyphen(s) will be removed and if what remains is a known word, then that will be used. Otherwise,...
- The hyphen(s) will be kept as part of the word.

With the above example, *Bus-iness* will be recognized as *Business*. In contrast, *Point-of-Sale* will remain as a hyphenated compound word because *PointofSale* is not a known word.

20.10 Special Characters

When a *%* is preceded by a space, the program will check if the character before the space is a number. If it is, then the *%* will be joined with the preceding word; otherwise, it will be treated as the start of a new word. Consider the following:

> What *%* of viewers enjoyed the film *Prometheus*?
> From my independent survey, over *95 %* of respondents said that they did.

In the above example, the first *%* will be counted as a word. The second one will be combined to create the word *95 %*.

20.11 Typographical Ligatures

Ligatures are glyphs that combine two (or more) letters. For example, *æ* is a combination of the letters *a* and *e*. Some ligatures represent unique characters that are part of languages (e.g., *œ* in French). Other times, they are stylistic substitutions used to improve typography. An example of this is *fi*; in some fonts, the dot of the *i* may overlap the hood of the *f*. To prevent this, a special ligature replaces any *fi* sequences for a more refined appearance.

In the case of typographic ligatures, *Readability Studio* will substitute them with their constituent characters. This allows for more accurate analyses and word searching. These substitutions include:

- f f
- f i
- f l

- f f i
- f f l
- f t
- s t

20.12 Combined Accents

Diacritical markers are accents combined with letters to produce a new letter. For example, *e* combined with combining acute accent (U+0301) will create an "e with acute" (*é*). In some files, these accented characters may be stored as two-characters sequences, rather than a single character. For example, the file will contain the sequence *e* + U+0301 rather than the precomposed character *é*. While most text editors display them identically, searching for *é* will fail because the file contains two distinct code points.

During import, *Readability Studio* will substitute these sequences with their proper characters. This will ensure accurate results and text searching. All diacritical combinations for European letters are supported, including the following accents:

- grave
- acute
- circumflex
- diaeresis/umlauts/trema
- dot above
- ring
- caron
- short solidus
- long solidus
- tilde
- retroflex hook below
- cedilla

20.13 Loanwords

Words and phrases from other languages that are commonly used in another are referred to as loanwords (or borrowed words). For example, schadenfreude, doppelgänger, and armada are often heard in the English lexicon, despite their German and Spanish origins. In the context of readability, these words are counted and syllabized just as any other native word. For example:

> That first sip of Mokka Java coffee while relaxing at Winans with a good Edward Tufte book...*la dolce vita*.

In the above example, *la dolce vita* will be counted as part of the sentence and syllabized using the English syllabizer's rules.

This also applies to dual-role characters, such as Greek letters, which are often used as mathematical symbols. In this context, analyses will treat individual Greek letters as monosyllabic words. On the other hand, contiguous Greek letters will be seen as a word and syllabized as such.

Readability Studio can process most European letters (and Japanese, see below), meaning that it will count the letters of foreign words as expected. (This applies to all words, not just loanwords.) In terms of syllable counting, it will use the rules defined by the project. For example, a Spanish project will use the Spanish syllabizer for all words.

As an example:

> "привет," Ivan said to his trusty localization engineer.

In an English project, *привет* will be counted as two syllables. Despite being a Russian word, *Readability Studio* will recognize the vowels *и* and *е* in it. From there, the English syllable-counting algorithm will calculate this as being two syllables.

In regards to Japanese script, *Readability Studio* supports modern Japanese orthography (as defined in the Unicode Basic Multilingual Plane [BMP]). This includes:

- Hiragana
- Katakana (including the half-width and phonetic extensions for Ainu)
- Jōyō/Jinmeiyō Kanji

When a Japanese word is encountered, each character is counted as one syllable, except for small kana (e.g., ュ, ョ, ッ). (Small kana modifies the preceding character and are not counted as separate syllables.) For example:

> Check the option "ドキュメンテーション" to install the documentation.

Here, ドキュメンテーション will be counted as an eight-syllable word (the ュ and ョ will not count as syllables).

> **Note**
> Note on word separation: this feature is intended for individual Japanese words appearing within other languages. Because Japanese does not use spaces between words, larger blocks of Japanese text will be treated as a single "word" with an extremely high syllable count. If your documents contain sections of Japanese, it is recommended to exclude them using the Exclude text between feature.

20.14 Link Lists (HTML Importer)

When importing HTML documents, three or more consecutive hyperlinks not separated by text will be treated as a link list. This will involve laying out each link on a separate line, with all of them tabbed over. This will prevent what would usually be a footer containing links from being seen as a sentence. For example:

> Our Products
>
> Readability Studio
>
> Our Corporate Mission
>
> Inspire through software | Align clients' global strategies | Make the world a better place | Elevate world consciousness

Because there isn't any text in between the links at the bottom (only | separating them), it is assumed that each link is a separate unit and not part of a sentence. When the document is imported, it will be processed as such:

> Our Products
>
> Readability Studio
>
> Our Corporate Mission
>
> Inspire through software |
> Align clients' global strategies |
> Make the world a better place |
> Elevate world consciousness

Note that the link list must also start from a new line. In other words, consecutive links that are proceeded by content on the same line will remain part of the text. For example:

> The recommended tools for continuous integration are: ccp-check, Microsoft PREFast, clang-tidy, GCC.

This content will be imported as it is shown above (i.e., a full sentence).

20.15 Option Changes that Require Reanalysis

When changing the options of a project (either from the ribbon or Project Properties dialog), most changes can be applied without the program needing to analyze the document again. For example, changing graph options or adding a test will process immediately, without the program needing to refresh the rest of the results. This is helpful when working with large documents or batch projects that may take some time to process.

There are some options that when changed will require a project to reload. These are listed below:

- Any changes made from the Document tab of the ribbon. This includes all options related to document analysis, spell checking, and editing the source document(s) (if documents or text are embedded in the project).
- Adding a Dolch section.
- Adding a custom test.

When any of the aforementioned changes are made, the project will reanalyze the source document(s) and recalculate all its results.

Note that you can always reload a project manually by clicking the Reload button on the Home tab of the ribbon. This is useful when you are editing the source document from another program and want to refresh the results in *Readability Studio*.

21. Scoring Notes

21.1 Grade Level Results

Most readability tests that return a grade level include a mantissa (i.e., decimal value), which represents the month of the grade. For example, *1.6* would be the sixth month of first grade (Carver 306–07). However, this only applies to tests that are computed from a mathematical formula (generally, a regression equation). Tests that use different methods—such as lookup tables or graphs—usually only return a grade level as an integer.

For example, New Dale-Chall looks up your unfamiliar word and sentence counts from a table and returns a grade range. This will yield results such as *7–8*, meaning that a document falls within the seventh to eighth grade reading level.

Lix and Rix use a similar method. These tests calculate an index score and then look up the grade level from a table using this value.

Finally, graphical tests (e.g., Fry) only return integral grade levels because the bands on these graphs only represent the grade, not the grade and month.

21.2 Grade Scales

Grade-level scores are traditionally displayed in U.S. (K–12+) format, but can be converted to other scales. The following grade scales are available to choose from:

Table 21.1: United States of America

Grade Levels[a,b]
Kindergarten (grade 0)
Grades 1–8
High School (grades 9–12)
College (grades 13–16)
Graduate School (grades 17–18)
Doctorate (grade 19+)

a www.ed.gov/k-12reforms
b www.education.ohio.gov/Topics/Learning-in-Ohio

Table 21.2: England and Wales

Key Stages[a,b]
Key stage 1: Ages 5–7 (Years 1–2)
Key stage 2: Ages 7–11 (Years 3–6)
Key stage 3: Ages 11–14 (Years 7–9)
Key stage 4: Ages 14–16 (Years 10–11)
Sixth Form (2 years)
College (4 years)
Graduate School (2 years)
Doctorate

a www.curriculum.qca.org.uk/index.aspx
b www.old.accac.org.uk/index.php

Table 21.3: Canadian Provinces & Territories

Grade Levels	Grade Levels
K–12+ (Alberta)[a]	K–12+ (Nova Scotia)[h]
Elementary (Kindergarten–grade 6)	Elementary (Primary–grade 6)
Junior High (grades 7–9)	Junior High (grades 7–9)
Senior High (grades 10–12)	Senior High (grades 10–12)
College (grades 13–16)	College (grades 13–16)
Graduate School (grades 17–18)	Graduate School (grades 17–18)
Doctorate (grade 19+)	Doctorate (grade 19+)
K–12+ (British Columbia)[b]/Yukon[c]	K–12+ (Nunavut)[i]
Elementary (Kindergarten–grade 7)	Kindergarten (grade 0)
Junior Secondary (grades 8–10)	Grades 1–12
Senior Secondary (grades 11–12)	College (grades 13–16)
College (grades 13–16)	Graduate School (grades 17–18)
Graduate School (grades 17–18)	Doctorate (grade 19+)
Doctorate (grade 19+)	
K–12+ (Manitoba)[d]	K–12+ (Ontario)[j]
Early (Kindergarten–grade 4)	Elementary (grades 1–8)
Middle (grades 5–8)	Grades 9–12
Senior (grades 9–12)	College (grades 13–16)
College (grades 13–16)	Graduate School (grades 17–18)
Graduate School (grades 17–18)	Doctorate (grade 19+)
Doctorate (grade 19+)	
K–12+ (New Brunswick)[e]	K–12+ (Prince Edward Island)[k]
Elementary (Kindergarten–grade 5)	Kindergarten
Middle School (grades 6–8)	Elementary (grades 1–6)
High School (grades 9–12)	Intermediate School (grades 7–9)
College (grades 13–16)	Senior High (grades 10–12)
Graduate School (grades 17–18)	College (grades 13–16)
Doctorate (grade 19+)	Graduate School (grades 17–18)
	Doctorate (grade 19+)

Table 21.3: Canadian Provinces & Territories *(continued)*

Grade Levels	Grade Levels
K–12+ (Newfoundland & Labrador)[f]	K–12+ (Saskatchewan)[l]
Kindergarten	Kindergarten
Primary (grades 1–3)	Elementary Level (grades 1–5)
Elementary (grades 4–6)	Middle Level (grades 6–9)
Intermediate (grades 7–9)	Secondary Level (grades 10–12)
Senior High (Level I–III)	College (grades 13–16)
College (grades 13–16)	Graduate School (grades 17–18)
Graduate School (grades 17–18)	Doctorate (grade 19+)
Doctorate (grade 19+)	
K–12+ (Northwest Territories)[g]	Quebec[m]
Primary (Kindergarten–grade 3)	Maternelle (Kindergarten)
Intermediate (grades 4–6)	École Primaire (grades 1–6)
Junior Secondary (grades 7–9)	École Secondaire (grades 7–11)
Senior Secondary (grades 10–12)	CEGEP (two years of university prep)
College (grades 13–16)	University (3 years for Bachelor’s degree)
Graduate School (grades 17–18)	Graduate degree (2 years)
Doctorate (grade 19+)	Doctoral degree

[a] www.education.alberta.ca/media/832568/guidetoed.pdf
[b] www.bced.gov.bc.ca/reporting/glossary.php
[c] www.education.gov.yk.ca/psb/curriculum.html
[d] www.edu.gov.mb.ca/ks4/schools/gts.html
[e] www.gnb.ca/0000/anglophone-e.asp
[f] www.ed.gov.nl.ca/edu/sp/pcdbgl.htm
[g] www.ece.gov.nt.ca/Divisions/kindergarten_g12/curriculum/Elementary_Junior_Secondary_School_Handbook/Elementary_Junior_Secondary_School_Handbook.htm
[h] www.ednet.ns.ca/pdfdocs/psp/psp_03_04_full.pdf
[i] www.ntanu.ca/assets/docs/Handout-Nunavut%20Approved%20Teaching%20Resources%20Version%20(5).pdf
[j] www.edu.gov.on.ca/eng/educationFacts.html
[k] www.edu.pe.ca/curriculum/default.asp
[l] www.sasked.gov.sk.ca/branches/curr/index.shtml
[m] www.ibe.unesco.org/en/access-by-country/europe-and-north-america/canada/curricular-resources.html

21.3 Cloze Scores and Readability Formulas

A cloze procedure is a test given to a student to measure their comprehension ability. This test consists of the following steps:

- Take a 250 word passage from a document.
- Delete every fifth word and replace it with a blank of equal length. There should be a total of 50 blanks when finished.
- Have the student fill in as many of the blanks as he/she can.
- Count the number of blanks that he/she filled in correctly. Traditionally, synonyms are not accepted, but this can be at the discretion of the administrator. Misspellings are acceptable.
- Multiply the number of correct answers by two. This is the percentage of correct answers and will range from 0 (zero comprehension) to 100 (total comprehension).

Cloze procedures can also be used to gauge how easily a passage of text can be comprehended. In this context, a cloze procedure is given to a group of students, and the mean (average) of their scores is the level of comprehensibility for the passage. Sixty percent and higher are considered to be acceptable levels of comprehensibility. Scores lower than 60% indicate that the passage may be too difficult to be comprehensible and should be rewritten.

Cloze scores such as these are the criteria used to validate most readability formulas. Generally, researchers use the passages and results from published cloze procedures as their test data when creating a formula. This process usually involves multiple regression with the cloze scores of the passages as the dependent variable and factors such as sentence length and syllable count as the independent variables. This analysis will determine which factors most strongly affect the cloze scores. After finding the most effective factors, these factors are then incorporated into a formula. Through experimentation, the formula is adjusted to produce results that strongly correlate with the original cloze scores. The stronger this correlation is, the more accurate the formula will be in predicting the readability of new text samples.

Some readability formulas produce predicted cloze scores (rather than grade levels). A predicted cloze score represents an estimation of a passage's mean score if it were made into a cloze test and given to a group of students. This can be useful when selecting passages to use for cloze tests.

21.4 Proper Nouns in the Difficult Word Lists

When reviewing your difficult word lists, you may notice that some proper nouns are displayed with additional information. This is to help discern how many times the given word is considered unfamiliar in the document (for certain tests).

Certain tests, such as Spache Revised, treat all proper nouns as familiar, even if they do not appear on the test's familiar-word list. However, if the same word occurs in the document as non-proper, then that instance will be considered unfamiliar if it is not on the test's familiar-word list.

Consider the following example:

> The *Daily Examiner* reported today that a tax examiner is reviewing the company's records. The examiner will have his full report by the end of the week.

The list of unfamiliar Harris-Jacobson words will report the following:

```
Examiner    2 (3 total occurrences, 1 proper and familiar, 2 non-proper and unfamiliar)
```

This means that *Examiner* in the phrase *Daily Examiner* is proper and Harris-Jacobson treated it as familiar. Although the word *examiner* is not a familiar Harris-Jacobson word, a reader will recognize that it is proper. From there, he/she can deduce that it is the name of a specific person, place, or thing (e.g., a newspaper). Therefore, it is somewhat familiar to the reader, even if he/she does not know the word.

However, the other two instances of *examiner* are regular nouns. Because *examiner* is not on the Harris-Jacobson familiar-word list, these two instances are considered unfamiliar. When Harris-Jacobson is computed, a frequency count of 2 (not 3) will be used for this particular word when tallying the number of unfamiliar words.

If a word appears only in proper form throughout the document, then it will not appear on various unfamiliar lists (e.g., Spache). A proper noun will never appear on these lists unless there are conflicting non-proper instances of the same word.

Note that this also applies to any custom tests that treat proper nouns as familiar.

21.5 Adding Words to Familiar-Word Tests

Although both Dale-Chall and Spache were updated in recent years, some may find that these tests lack contemporary words known by young readers. For example, some computer terms that 4^{th} graders are fluent in are not considered familiar by these tests. Proper nouns such as *Microsoft* and *MacBook* will be considered familiar, but some non-proper computer words (e.g., *megabyte*) will not.

Readability Studio offers custom tests as a solution for extending these tests. You can create a new test with New Dale-Chall, Spache, or Harris-Jacobson as a template, and append your own word list to it. When you perform this custom test, any word on the standard word list or your word list will be considered familiar. Refer to sec. 13.1 for more information.

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[1] https://www.edisonohio.edu/
[2] https://www.wright.edu/

A. Keyboard Shortcuts

Table A.1: Application

To do this	Press
Open the help	F1
Enable verbose logging	Hold down Shift (⊞, 🐧) or ⇧ () while the program loads

Table A.2: Projects

To do this	Press
Show/hide the sidebar	F2
Show/hide the ribbon	F4
Reload the project	F5
Create a new project	Ctrl + V (Windows, Linux) or ⌘ + V (macOS)
Save the project	Ctrl + S (Windows, Linux) or ⌘ + S (macOS)
Print the active window	Ctrl + P (Windows, Linux) or ⌘ + P (macOS)
Remove the selected test	Ctrl + Del (Windows, Linux) or ⌘ + ⌫ (macOS)
Edit/view the selected document	F6
Create subproject (batch projects only)	Ctrl + → (Windows, Linux) or ⌘ + → (macOS)

Table A.3: Graphs

To do this	Press
Print	Ctrl + P (Windows, Linux) or ⌘ + P (macOS)
Copy	Ctrl + C (Windows, Linux) or ⌘ + C (macOS)
Zoom in	+
Zoom out	-
Reset zoom	*

Table A.4: Lists

To do this	Press
Print	Ctrl+P (⊞, 🐧) or ⌘+P ()
Copy selection	Ctrl+C (⊞, 🐧) or ⌘+C ()
Copy first column of selection	Ctrl+Shift+C (⊞, 🐧) or ⌘+⇧+C ()
Copy column from selection (column corresponds to the number)	Ctrl+Numpad [1–9] (⊞, 🐧) or ⌘+Numpad [1–9] ()
Find	Ctrl+F (⊞, 🐧) or ⌘+F ()
Find next	F3
Select all	Ctrl+A (⊞, 🐧) or ⌘+A ()
Go to first item	Ctrl+↑ (⊞, 🐧) or ⌘+↑ ()
Go to last item	Ctrl+↓ (⊞, 🐧) or ⌘+↓ ()
Remove selected item(s)	Del (⊞, 🐧) or ⌫ ()
Edit selected item	F2
Accept edited item	Enter, Tab, ↑, ↓ (or mouse click outside the edit box)
Cancel editing item	Esc
Add an item	Ctrl+Ins (⊞, 🐧) or ⌘+Ins ()
Group selected items (batch project wizard file list)	Ctrl+G (⊞, 🐧) or ⌘+G ()

Table A.5: Script Editor

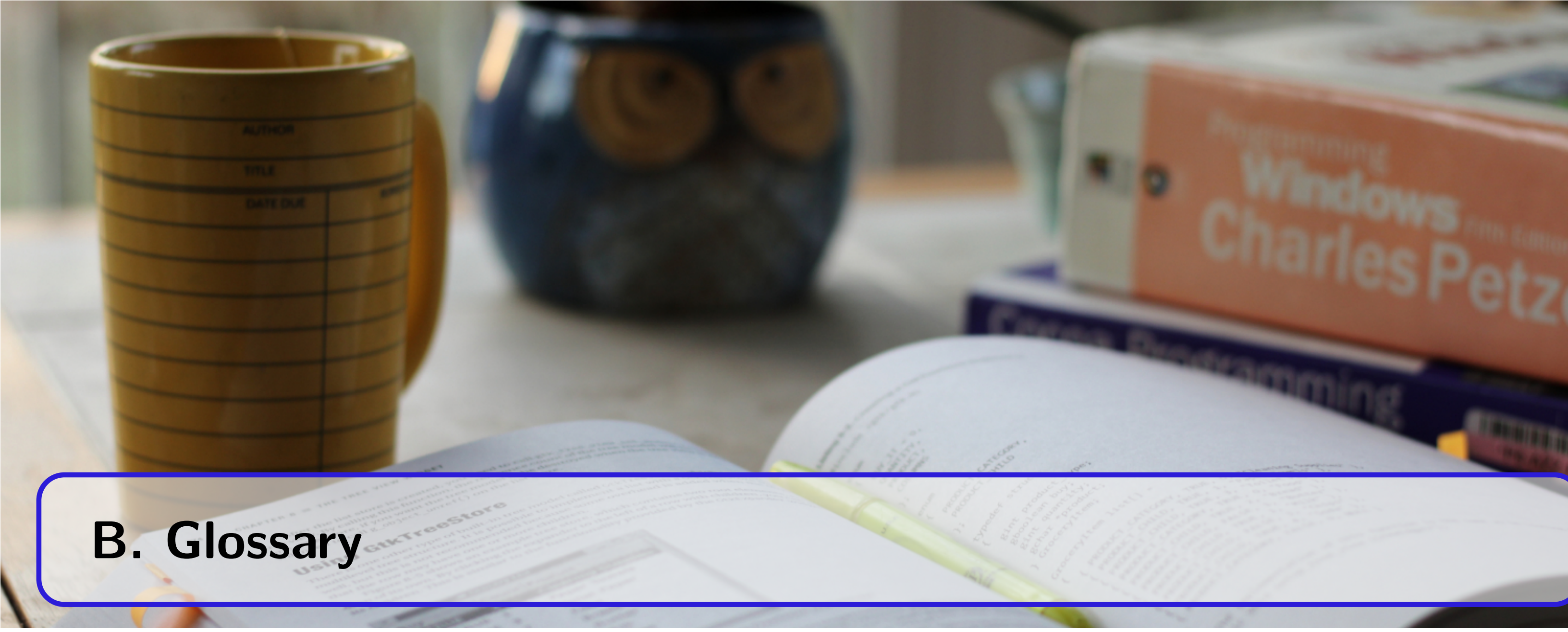
To do this	Press
Create a new script	Ctrl+Shift+N (⊞, 🐧) or ⌘+⇧+N ()
Open a script	Ctrl+Shift+O (⊞, 🐧) or ⌘+⇧+O ()
Save the script	Ctrl+Shift+S (⊞, 🐧) or ⌘+⇧+S ()
Run selection (or full script if no selection)	F5
Stop running	Shift+F5 (⊞, 🐧) or ⇧+F5 ()
Cut selected text	Ctrl+X (⊞, 🐧) or ⌘+X ()
Paste text	Ctrl+V (⊞, 🐧) or ⌘+V ()
Copy selected text	Ctrl+C (⊞, 🐧) or ⌘+C ()
Undo previous edit	Ctrl+Z (⊞, 🐧) or ⌘+Z ()
Redo previous edit	Ctrl+Y (⊞, 🐧) or ⌘+Y ()
Find	Ctrl+F (⊞, 🐧) or ⌘+F ()
Find next	F3
Replace	Ctrl+H (⊞, 🐧) or ⌘+H ()
Select all	Ctrl+A (⊞, 🐧) or ⌘+A ()
Duplicate current line	Ctrl+D (⊞, 🐧) or ⌘+D ()

Table A.6: Embedded Document Editor

To do this	Press
Save back to the project	Ctrl+S (,) or ⌘+S ()
Cut selected text	Ctrl+X (,) or ⌘+X ()
Paste text	Ctrl+V (,) or ⌘+V ()
Copy selected text	Ctrl+C (,) or ⌘+C ()
Undo previous edit	Ctrl+Z (,) or ⌘+Z ()
Redo previous edit	Ctrl+Y (,) or ⌘+Y ()
Insert date	Ctrl+; (,) or ⌘+; ()
Insert time	Ctrl+Shift+; (,) or ⌘+⇧+; ()
Find	Ctrl+F (,) or ⌘+F ()
Find next	F3
Replace	Ctrl+H (,) or ⌘+H ()
Select all	Ctrl+A (,) or ⌘+A ()

Table A.7: Web Harvester

To do this	Press
Load links from HTML content	Ctrl+K (Windows) or ⌘+K (Apple)



B. Glossary

B.1 Readability & Grammar Terms

ASL
: Average sentence length.

cloze test
: A cloze procedure is a test given to readers to measure their comprehension ability. Refer to sec. 21.3 for an extended explanation.

diacritic
: Accents combined with letters to produce a new letter.

interrobang
: The combination of a question mark and exclamation point at the end of a sentence. Interrobangs are shown as either two separate characters (*?!* or *!?*) or a single character (‽).

ligature
: A glyph that combines two (or more) letters.

polysyllabic
: Technically speaking, this means any word containing more than 1 syllable. In the case of some tests (e.g., Wheeler-Smith), this term is used in this context. Other tests' sources, however, tend to use this term to describe words of 3 or more syllables. Because of this inconsistency, it is recommended to avoid using this term and instead state explicitly the intention of a given test.

synalepha
: The blending of word sounds together often found in poetry.

unit
: A section of text ending with a period, exclamation mark, question mark, interrobang, colon, semicolon, or dash. Lengthy, complex sentences generally contain more than one unit.

B.2 Statistical Terms

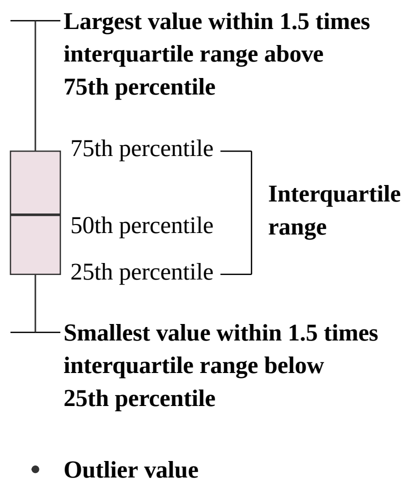
box plots

Box plots are used to display the distribution of data, where:

- The middle point is the median.
- The box is the 25th-75th percentile range.
- The whiskers are the non-outlier range.

Outlier values that do not fall within these areas are plotted outside the whiskers.

> Note
> *Readability Studio* uses the Tukey hinges method for calculating quartiles.



(a) Adapted from DeCicco ("Exploring ggplot2 boxplots")

Figure B.1: Box plot layout

IQR

Interquartile range. Refer to fig. B.1.

kurtosis
: The peakedness of a distribution. Zero indicates a normal distribution, a negative value indicates a flat distribution, and a positive value indicates a sharp curve.

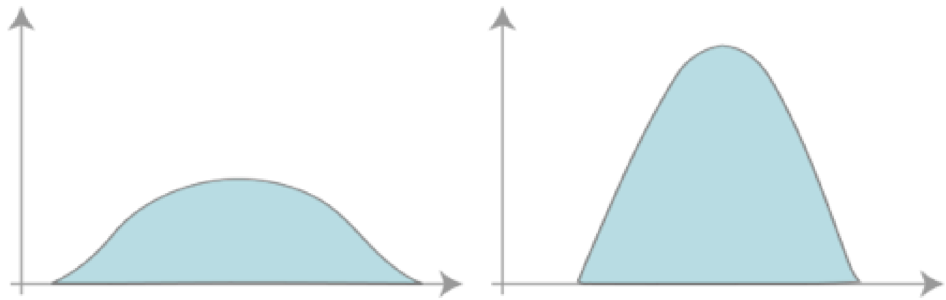


Figure B.2: Negative and positive kurtosis

lower quartile
: The value at the 25% percentile. This is the value separating the lower quarter of the values from the rest of the values.

maximum
: The highest value in a range of data.

mean
: The average value of a range of data. This is calculated by adding all the values and dividing the total by N.

median
: The middle point separating the lower and upper halves of a range of data. This is calculated by sorting the values and taking the value in the middle if the number of values is odd. If the number of values is even, then the mean of the two middle points is taken.

minimum
: The lowest value in a range of data.

mode
: The most frequently occurring value in a range of data. Note that a range can have multiple modes if two or more unique values appear the same number of times.

normalization

The adjustment of values from different scales into a common scale.
For example, say that a test requires a 100-word sample, but your document is 200 words and contains 22 sentences. Rather than taking a 100-word sample, the full document's measurements can be normalized and used instead. In this case, we need to normalize 200 words to be 100 words, which we can do by multiplying it by 0.5.

$$200 * 0.5 = 100$$

Next, we multiply the 22 sentences by the same factor of 0.5 and arrive at 11.

$$22 * 0.5 = 11$$

Now that the document's measurements have been normalized to the required sample's scale, we can use these values (100 words and 11 sentences) with the test.

outlier

An extreme (high or low) value that is numerically distant from the rest of the data.

overplotting

When visualizing points along a single dimension (e.g., the y-axis of a box plot), points with the same score will overlap each other. Because of this, it becomes difficult to fully see the distribution of the data. Considering the following box plot:

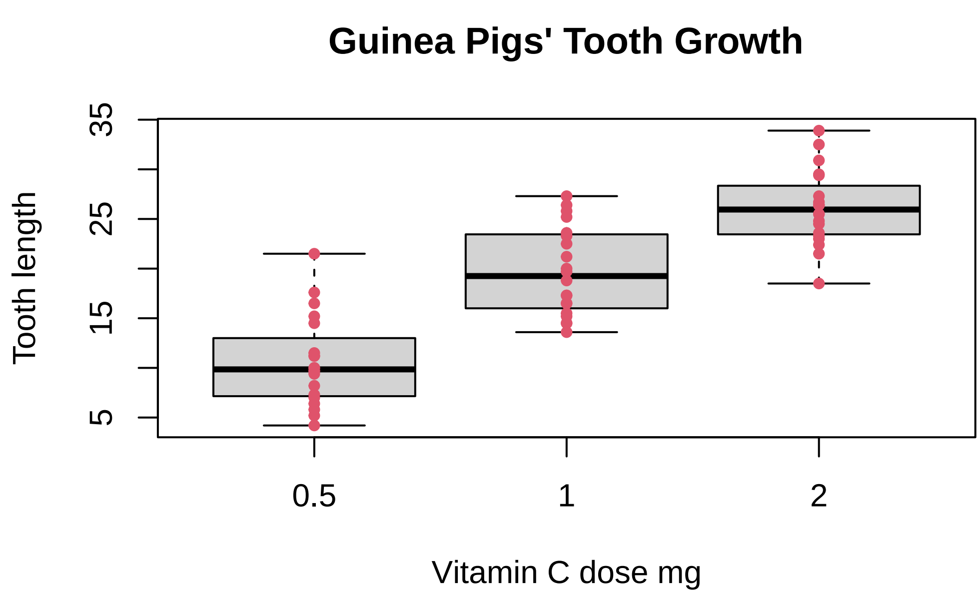


Figure B.3: Box plot with no jittering

Although we can interpret the quartile ranges, it is not obvious how the data clumps overall.

To correct this, a technique called jittering is used to adjust the x-axis values of overlapping points. Traditionally, jittering uses randomness to move the points' position around both sides of the axis, as such:

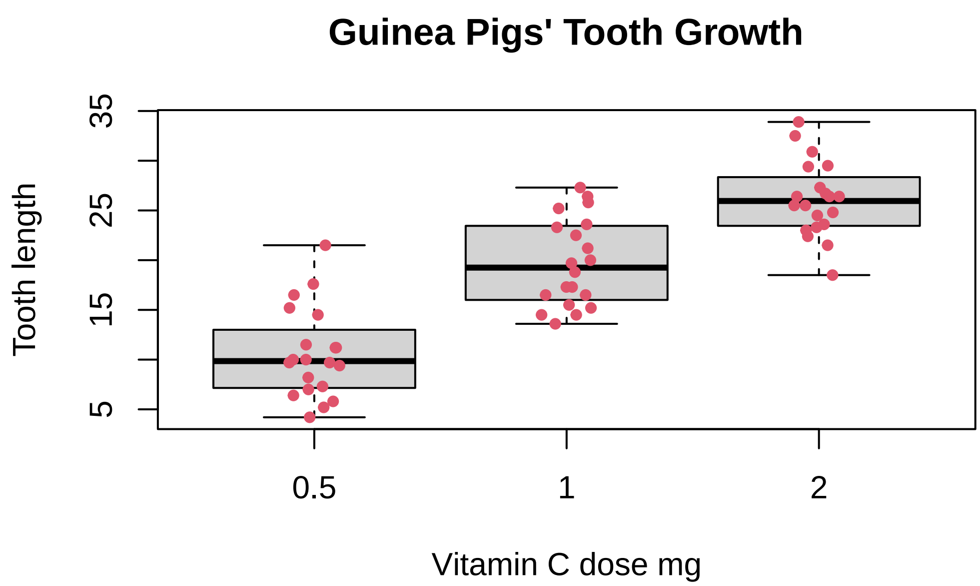


Figure B.4: Box plot with random jittering

Because of this randomness, some points may still overlap. To improve this, a newer technique known as beeswarm jittering instead adjusts points the minimum distance from the axis. This ensures that each point is seen and more tightly packed (offering a more accurate view of how the data are distributed). An example of this is as follows:

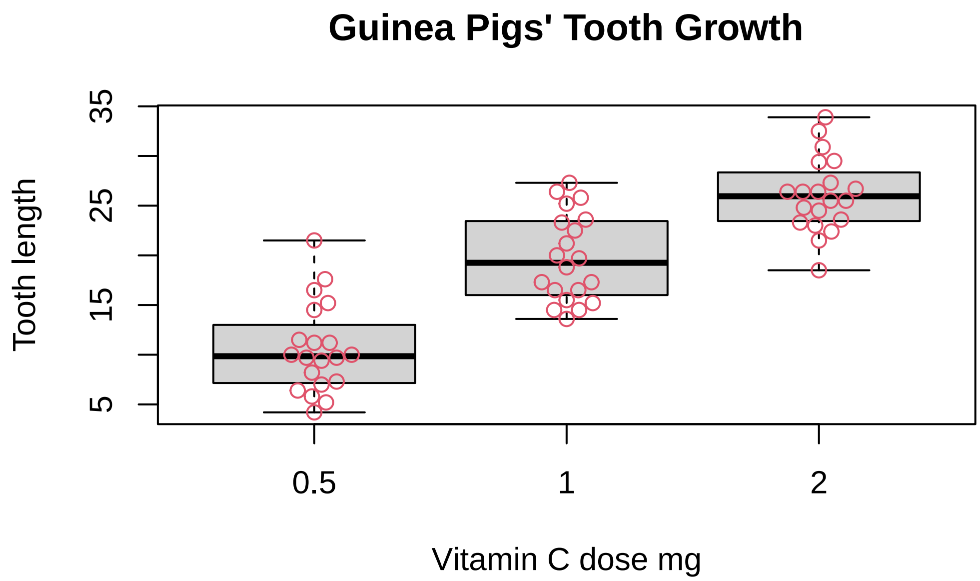


Figure B.5: Box plot with beeswarm jittering

r

The correlation between two factors (e.g., sentence length vs. reading level). *r* values range from -1.0 to 1.0, where -1.0 is a perfect negative correlation and 1.0 is a perfect positive correlation.

r^2

The proportion of variation in a dependent variable explained by the independent variable(s). r^2 values range from 0.0 to 1.0, where 0.0 indicates no explanatory power and 1.0 indicates complete explanation.

range

The distance between the lowest and highest values in a range of data.

skewness
: The asymmetry of the probability distribution. A zero skew indicates a symmetrical balance in the distribution. A negative skew indicates that the left side of the distribution is longer and most of the values are concentrated on the right. A positive skew indicates that the right side of the distribution is longer and most of the values are concentrated on the left.

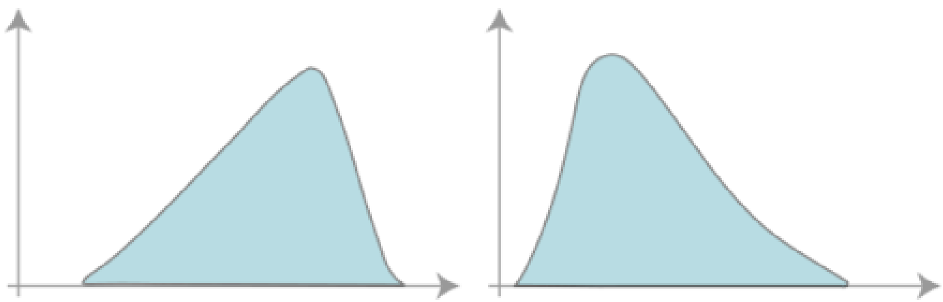


Figure B.6: Negative and positive skewness

standard deviation
: The variability (or spread) of the values. A low standard deviation indicates that most values are close to the data's mean. In contrast, a high standard deviation indicates values that stray far from the mean.

upper quartile
: The value at the 75% percentile. This is the value separating the upper quarter of the values from the rest of the values.

valid N
: The number of valid values in a range of data.

variance
: Another measurement of variability, which is standard deviation squared.

References

Amstad, Toni. “Wie Verstandlich Sind Unsere Zeitungen?” 1978. Universitat Zurich, Dissertation.

Anderson, Jonathan. “Analysing the Readability of English and Non-English Texts in the Classroom with Lix”. Annual Meeting of the Australian Reading Association, Aug. 1981.

———. “Lix and Rix: Variations of a Little-Known Readability Index”. *Journal of Reading*, vol. 26, no. 6, 1983, pp. 490–96.

Aquino, Milagros R. “The Validity of the Miller-Coleman Readability Scale”. *Reading Research Quarterly*, vol. 4, no. 3, Spring 1969, pp. 342–57.

Baldwin, R. Scott, and Rhonda K. Kaufman. “A Concurrent Validity Study of the Raygor Readability Estimate”. *Journal of Reading*, vol. 23, no. 2, 1979, pp. 148–53.

Bamberger, Richard, and Erich Vanecek. *Lesen-Verstehen-Lernen-Schreiben: Die Schwierigkeitsstufen von Texten in deutscher Sprache*. Jugend und Volk/Diesterweg/Sauerländer, 1984.

Barrio-Cantalejo, Inés María, et al. “Validación de la Escala INFLESZ para evaluar la legibilidad de los textos dirigidos a pacientes”. *Anales del Sistema Sanitario de Navarra*, vol. 31, no. 2, 2008, pp. 135–52.

Björnsson, C.H. “Readability of Newspapers in 11 Languages”. *Reading Research Quarterly*, vol. 18, no. 4, 1983, pp. 480–97.

Björnsson, C.H., and Birgit Hård af Segerstad. *Lix på Franska och tio andra Språk*. Pedagogiskt centrum Stockholms skolförvaltning, 1979.

Bormuth, John R. *Development of Readability Analyses*. United States Department of Health, Education, & Welfare, 1969.

———. "Readability: A New Approach". *Reading Research Quarterly*, vol. 1, no. 3, Spring 1966, pp. 79–132.

Carver, Ronald P. "Measuring Readability Using DPR Units". *Journal of Reading*, vol. 27, no. 4, 1985, pp. 303–16.

Caylor, John S., Thomas G. Sticht, Lynn C. Fox, and J. Patrick Ford. *Methodologies for Determining Reading Requirements of Military Occupational Specialties*. United States Office of the Chief of Research / Development, 1973.

Chall, Jeanne S., and Edgar Dale. *Manual for the New Dale-Chall Readability Formula*. Brookline Books, 1995.

———. *Readability Revisited: The New Dale-Chall Readability Formula*. Brookline Books, 1995.

Coleman, Meri, and T.L. Liau. "A computer readability formula designed for machine scoring". *Journal of Applied Psychology*, vol. 60, no. 2, 1975, pp. 283–84.

Contreras, Alfonso, Rafael Garcí a-Alonso, Marta Echenique, and Fedora Daye-Contreras. "The SOL Formulas for Converting SMOG Readability Scores Between Health Education Materials Written in Spanish, English, and French". *Journal of Health Communications*, vol. 4, 1999, pp. 21–29.

Crawford, Alan. "A Spanish Language Fry-Type Readability Procedure: Elementary Level". *Bilingual Education Paper Series*, vol. 7, no. 8, 1984, pp. 1–17.

Cuayáhuitl, Heriberto. "A Syllabification Algorithm for Spanish". *Lecture Notes in Computer Science*, vol. 2945, 2004, pp. 412–15.

Dale, Edgar, and Jeanne S. Chall. "A Formula for Predicting Readability". *Educational Research Bulletin*, vol. 27, no. 1, 1948, pp. 11–20 , 28.

Danielson, Wayne A., and Sam Dunn Bryan. "Computer Automation of Two Readability Formulas". *Journalism Quarterly*, vol. 40, 1963, pp. 201–06.

DeCicco, Laura. “Exploring ggplot2 Boxplots - Defining Limits and Adjusting Style”, 10 Aug. 2018. Accessed 28 Aug. 2020.

Dolch, E.W. “A Basic Sight Vocabulary”. *Elementary School Journal*, vol. 36, no. 6, 1936, pp. 456–60.

Fang, Irving. “The ‘Easy Listening Formula’ ”. *Journal of Broadcasting*, vol. 10, no. 1, 1966, pp. 63–68.

Farr, James N., James J. Jenkins, and Donald G. Paterson. “Simplification of the Flesch Reading Ease Formula”. *Journal of Applied Psychology*, vol. 35, no. 5, 1951, pp. 333–37.

Flesch, Rudolf Franz. *How to Write Plain English: A Book for Lawyers and Consumers*. Harper & Row, 1979.

———. *The Art of Readable Writing*. Harper & Row, 1949.

Fry, Edward. “A Readability Formula That Saves Time”. *Journal of Reading*, vol. 11, no. 7, 1968, pp. 513–16 , 575–78.

———. *Elementary Reading Instruction*. McGraw-Hill, 1977.

———. “Fry’s Readability Graph: Clarifications, Validity, and Extension to Level 17”. *Journal of Reading*, vol. 21, no. 3, 1977, pp. 242–52.

Gilliam, Bettye, Sylvia C. Peña, and Lee Mountain. “The Fry Graph Applied to Spanish Readability”. *Reading Teacher*, vol. 33, 1980, pp. 426–30.

Grundner, T. M. “How to Make Consent Forms More Readable”. *IRB: Ethics & Human Research*, vol. 3, no. 7, 1981, pp. 9–10.

Gunning, Robert. *The Technique of Clear Writing*. Rev. ed., McGraw-Hill, 1968.

Harris, A. J., and M. D. Jacobson. *Basic Reading Vocabularies*. Macmillan, 1982.

Hooke, Lydia R., Philip J. DeLeo, and Capt. Sharon L. Slaughter. *Readability of Air Force Publications: A Criterion Referenced Evaluation. Final Report*. United States Air Force Human Resources Lab, 1979.

Kibby, Michael W. “Test Review: The Degrees of Reading Power”. *Journal of Reading*, vol. 24, no. 5, 1981, pp. 416–27.

Kincaid, J. Peter, James A. Aagard, and John W. O'Hara. *Development and Test of a Computer Readability Editing System (CRES). Final Report, June 1978 through December 1979*. United States Naval Training Analysis / Evaluation Group, 1980.

Kincaid, J. Peter, Jr. Lt. Robert P. Fishburne, Richard L. Rogers, and Brad S. Chissom. *Derivation of New Readability Formulas (Automated Readability Index, Fog Count and Flesch Reading Ease Formula) for Navy Enlisted Personnel*. United States Naval Technical Training Command, 1975.

Kincaid, J. Peter, et al. *Use of the Automated Readability Index for Evaluating Peer-Prepared Material for Use in Adult Reading Education*. United States Department of Health, Education, & Welfare / Office of Education, 1972.

McAlpine, Rachel. "From Plain English to Global English", 2006. Accessed 22 Aug. 2010.

McCall, W.A., and L.M. Crabbs. *Standard Test Lessons in Reading*. Columbia U Teachers College, 1926.

———. *Standard Test Lessons in Reading*. Rev. ed., Columbia U Teachers College, 1950.

———. *Standard Test Lessons in Reading*. Rev. ed., Columbia U Teachers College, 1961.

McLaughlin, Harry. "SMOG grading - a new readability formula". *Journal of Reading*, vol. 12, no. 8, 1969, pp. 639–46.

Miller, G.R., and E.B. Coleman. "A Set of Thirty-Six Prose Passages Calibrated for Complexity". *Journal of Verbal Learning and Verbal Behavior*, vol. 6, 1967, pp. 851–54.

Powers, R.D., W.A. Sumner, and B.E. Kearl. "A recalculation of four adult readability formulas". *Journal of Educational Psychology*, vol. 49, no. 2, 1958, pp. 99–105.

Raygor, Alton L. "The Raygor Readability Estimate: A Quick and Easy Way to Determine Difficulty". *Reading: Theory, Research, and Practice. Twenty-Sixth Yearbook of the National Reading Conference*. Proc. of National Reading Conference, Dec. 1977, pp. 259–63.

Renström, Jan. "Texte und Textbehandlung im Deutschunterricht". *Die Unterrichtspraxis / Teaching German*, vol. 13, no. 2, Autumn 1980, pp. 150–55.

Schulz, Renate A. "Literature and Readability: Bridging the Gap in Foreign Language Reading". *The Modern Language Journal*, vol. 65, no. 1, Spring 1981, pp. 43–53.

Schwartz, Roswitha E. W. “An Exploratory Effort to Design a Readability Graph for German Material”. 1975. State U of New York at Albany, Reading Department, Unpublished study.

Smith, E. A., and R. J. Senter. *Automated Readability Index*. United States Aerospace Medical Research Laboratorties, 1967.

Smith, Lawrence L. “Using a Modified SMOG in Primary and Intermediate Grades”. *Reading Horizons*, vol. 24, no. 2, 1984, pp. 129–32.

Spache, George. “A new readability formula for primary-grade reading materials”. *The Elementary School Journal*, vol. 53, no. 7, 1953, pp. 410–13.

———. *Good Reading for Poor Readers*. Garrard Publishing Company, 1974.

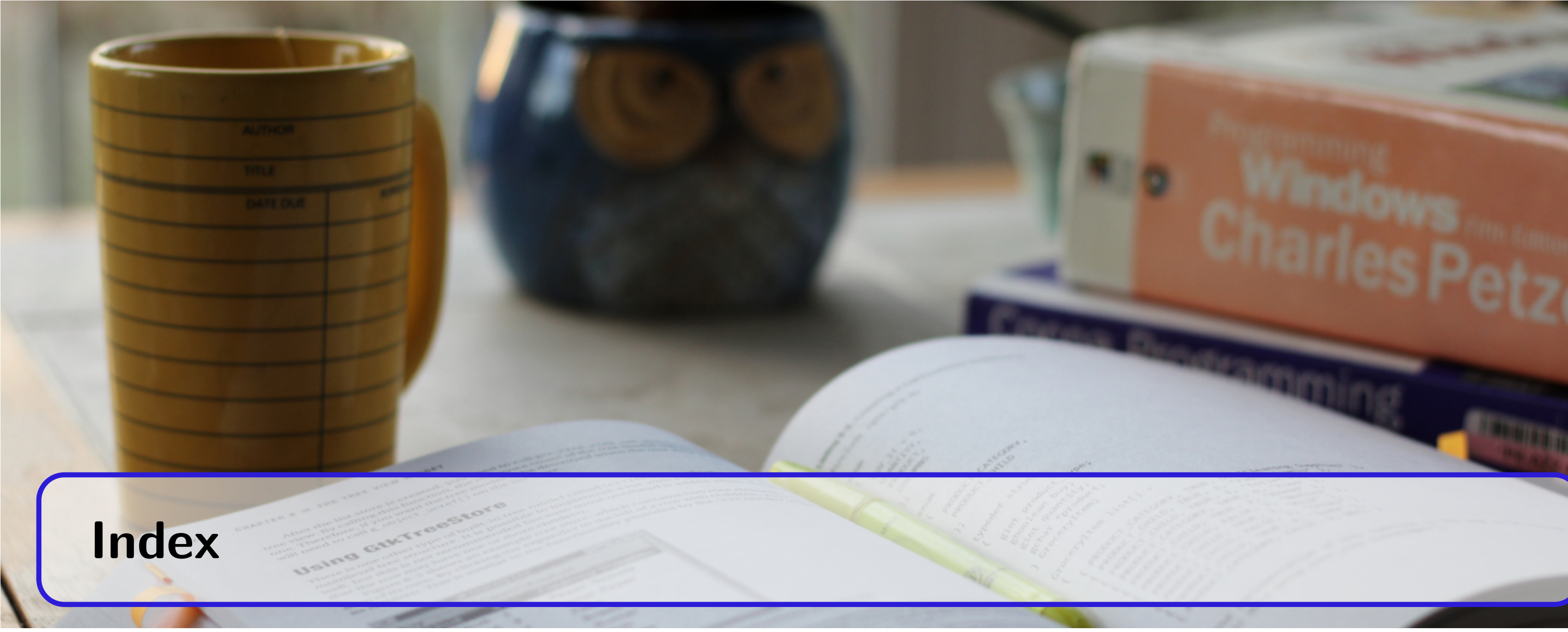
Stocker, Leonard P. “Increasing the Precision of the Dale-Chall Readability Formula”. *Reading Improvement*, vol. 8, no. 3, Winter 1971, pp. 87–89.

Tufte, Edward. *Seeing with Fresh Eyes*. Graphics P LLC, 2020.

Vari-Cartier, Patricia. “Development and Validation of a New Instrument to Assess the Readability of Spanish Prose”. *The Modern Language Journal*, vol. 65, no. 2, 1981, pp. 141–48.

Wheeler, Lester R., and Edwin H. Smith. “A Practical Readability Formula for the Classroom Teacher in Elementary Grades”. *Elementary English*, vol. 31, 1954, pp. 397–99.

Wolinski, John T., and Maurice W. Bozman. “Using the Raygor Readability Estimate for Social Studies”. *The Social Studies*, vol. 74, no. 5, 1983, pp. 215–19.



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Blake Madden is the Director of Institutional Research at Edison State Community College. He earned a Bachelor of Arts in English from the University of Dayton and has researched readability formulas for more than twenty years. Before entering academia, he spent eighteen years working in the statistical software industry, primarily in localization engineering.

Colophon

This book was typeset using LaTeX and built with open-source tools.

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